

Honeywell
Aerospace Electronics Systems
P.O. Box 97001
Redmond, Washington 98073-9701
U.S.A.
(CAGE 97896)

TO HOLDERS OF COMPONENT MAINTENANCE MANUAL, PUB. NO. 012-0342-001,
UNIVERSAL FLIGHT DATA RECORDER (UFDR) (PART NO. 980-4100-XXXX AND
980-4101-001, -002)

REVISION NO. 19 DATED 22 DEC 2006

HIGHLIGHTS

This revision is a full replacement of Pub. No. 012-0342-001, dated Jul 16/03. All revised pages have a new publication date, as identified in the List of Effective Pages. Technical changes in text are identified by a revision bar opposite the change. Changes to the Illustrated Parts List do not have revision bars.

Please replace your copy of the manual with the attached revision. Retain the binder and tab dividers.

Page	Description of Change
T-1, T-2	Revised to show the revision date of this manual. Added the Honeywell address.
SBL-1, SBL-2	Added details from the following Service Bulletins: 980-4100D/B-31-2 (012-0342-102), with Revision 1; 980-4100D/B-31-5 (012-0342-105), with Revision 5; 980-4100D/B-31-11 (012-0342-111), with Revision 6; 980-4100DBA-31-14 (012-0342-114), with Revision 2; 980-4100D/B-31-15 (012-0342-115), with Revision 5; 980-4100ABD-31-19 (012-0342-119), Revision 1.
LEP-1 thru LEP-8	Revised to show where changes are made in the manual.
TC-1 thru TC-14	Revised to show the location of data in the manual.
INTRO-1 thru INTRO-8	Added new INTRODUCTION section.
3	Revised Figure 2, Accessory Codes.
102	Revised subparagraph I.(1)(c).
167 thru 174	Revised Figure 122, Chassis and Tape Recorder Assembly Schematic to the latest revision. Combined sheets 4 and 5.
177, 178	Revised Figure 123, Filter Schematic to the latest revision.
179 thru 186	Revised Figure 124, Motherboard Schematic to the latest revision.
187 thru 194	Revised Figure 125, Power Supply, A1, Schematic to the latest revision.
195 thru 198, 198.1/198.2	Revised Figure 126, Read/Write Interface, A2, Schematic to the latest revision.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Page	Description of Change
198.3 thru 198.14	Revised Figure 127, Transport Interface Assembly, A3, Schematic to the latest revision.
198.15 thru 198.24	Revised Figure 128, Recording System Controller, A4, Schematic to the latest revision.
198.25 thru 198.30	Revised Figure 129, Airframe Wiring Interface, A8, Schematic to the latest revision.
198.53	Added Note to paragraph 9.F.
402	No revision bar. Caution regarding vapor degreaser removed at paragraph 3.A.
508	Converted lbs. to oz. in Figure 506.
719	Converted lbs. to oz. in paragraph 7.Q. (7) in three places.
1001 thru 1098, 1098.1 thru 1098.52	Updated ILLUSTRATED PARTS LIST with latest parts data.

Honeywell
Aerospace Electronics Systems
P.O. Box 97001
Redmond, Washington 98073-9701
U.S.A.

Component Maintenance Manual with illustrated parts list

Universal Flight Data Recorder (UFDR) Part No.

980-4100-AXXS	980-4100-DXXS
980-4100-AXXN	980-4100-DXXN
980-4100-AXUS	980-4100-DXUS
980-4100-AXUN	980-4100-DXUN
980-4100-BXUS	980-4101-001
980-4100-BXUN	980-4101-002

This document contains technical data and is subject to U.S. export regulations.

These commodities, technology, or software were exported from the United States in accordance with the export administration regulations. Diversion contrary to U.S. law is prohibited.

31-30-32

Title Page T-1

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

PROPRIETARY NOTICE

This document and the information disclosed herein are proprietary data of Honeywell. Neither this document nor the information contained herein shall be used, reproduced, or disclosed to others without the written authorization of Honeywell, except to the extent required for installation or maintenance of the recipient's equipment.

NOTICE - FREEDOM OF INFORMATION ACT (5 USC 552) AND DISCLOSURE OF CONFIDENTIAL INFORMATION GENERALLY (18 USC 1905)

This document is being furnished in confidence by Honeywell. The information disclosed herein fall within exemption (b) (4) of 5 USC 552 and the prohibitions of 18 USC 1905.

S2006

Honeywell is a U.S. registered trademark of Honeywell. All other marks are owned by their respective companies.

31-30-32

Title Page T-2

Copyright 2006 Honeywell

Revised 22 Dec 2006

All Rights Reserved

01 Dec 1980

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

RECORD OF REVISIONS

For each revision, put the revised pages in your manual and discard the superseded pages. Write the revision number and date, date put in manual, and the incorporator's initials in the applicable columns on the Record of Revisions. The initial H shows Honeywell is the incorporator.

Revision Number	Revision Date	Date Put In Manual	By	Revision Number	Revision Date	Date Put In Manual	By
1	Jun 15/81	Jun 15/81	H	19	22 Dec 2006	22 Dec 2006	H
2	Sep 30/81	Sep 30/81	H				
3	Apr 30/82	Apr 30/82	H				
4	Sep 30/82	Sep 30/82	H				
5	Mar 30/84	Mar 30/84	H				
6	Jun 21/84	Jun 21/84	H				
7	Apr 11/86	Apr 11/86	H				
8	Not Available	Not Available	H				
9	Jul 28/89	Jul 28/89	H				
10	Jan 15/93	Jan 15/93	H				
11	Jul 14/95	Jul 14/95	H				
12	Jun 28/96	Jun 28/96	H				
13	Nov 15/00	Nov 15/00	H				
14	Apr 12/01	Apr 12/01	H				
15	Oct 15/02	Oct 15/02	H				
16	Oct 28/02	Oct 28/02	H				
17	Jun 23/03	Jun 23/03	H				
18	Jul 16/03	Jul 16/03	H				

31-30-32

Page RR-1
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

31-30-32

Esta hoja fue impresa desde la BTv y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

RECORD OF TEMPORARY REVISIONS

Instructions on each page of a temporary revision tell you where to put the pages in your manual. Remove temporary revision pages only when discard instructions are given. For each temporary revision, put the applicable data in the record columns on this page.

Temporary Revision Number	Temporary Revision Date	Temporary Revision Status	Date Put in Manual	By*	Date Removed from Manual	By*
31-1	Jun 03/83	Inactive	Jun 21/84	H		
31-2	Oct 14/83	Inactive	Jun 21/84	H		
31-3	Nov 04/83	Inactive	Jun 21/84	H		
31-4	Nov 22/83	Inactive	Jun 21/84	H		
31-5	Feb 15/85	Inactive	Apr 11/86	H		
31-6	Jul 19/85	Inactive	Apr 11/86	H		
31-7	Jul 15/86	Inactive	Jul 28/89	H		
31-8	Dec 23/87	Inactive	Jul 28/89	H		
31-9	Jan 20/88	Inactive	Jul 28/89	H		
31-10	May 13/88	Inactive	Jul 28/89	H		
31-11	May 24/88	Inactive	Jul 28/89	H		
31-12	May 30/89	Inactive	Jul 28/89	H		
31-13	Nov 17/89	Inactive	Jan 15/93	H		
31-14	Jun 08/90	Inactive	Jan 15/93	H		

* The initial H in this column shows Honeywell has done this task.

** Temporary revisions are incorporated in the manual by normal revision.

31-30-32

Page RTR-1
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Temporary Revision Number	Temporary Revision Date	Temporary Revision Status	Date Put in Manual	By*	Date Removed from Manual	By*
31-15	Aug 03/90	Inactive	Jan 15/93	H		
31-16	Mar 25/91	Inactive	Jan 15/93	H		
31-17	Mar 23/92	Inactive	Jan 15/93	H		
31-18	Apr 22/92	Inactive	Jan 15/93	H		
31-19	Nov 02/92	Inactive	Jan 15/93	H		
31-20	Mar 10/93	Inactive	Jul 14/95	H		
31-21	Jul 02/93	Inactive	Jul 14/95	H		
31-22	Aug 13/93	Inactive	Jul 14/95	H		
31-23	Mar 17/94	Inactive	Jul 14/95	H		
31-24	Mar 23/94	Inactive	Jul 14/95	H		
31-25	Sep 08/94	Inactive	Jul 14/95	H		
31-26	Dec 10/94	Inactive	Jul 14/95	H		
31-27	Mar 17/95	Inactive	Jul 14/95	H		
31-28	Nov 14/95	Inactive	Jun 28/96	H		
31-29	Feb 14/96	Inactive	Jun 28/96	H		
31-30	May 14/96	Inactive	Jun 28/96	H		
31-31	Aug 20/96	Inactive	Nov 15/00	H		
31-32	Feb 06/97	Inactive	Nov 15/00	H		

31-30-32

Page RTR-2
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTv y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Temporary Revision Number	Temporary Revision Date	Temporary Revision Status	Date Put in Manual	By*	Date Removed from Manual	By*
31-33	Jul 01/97	Inactive	Nov 15/00	H		
31-34	Dec 16/97	Inactive	Nov 15/00	H		
31-35	Feb 10/98	Inactive	Nov 15/00	H		
31-36	Dec 15/98	Inactive	Nov 15/00	H		
31-37	Oct 15/99	Inactive	Nov 15/00	H		
31-38	Dec 16/99	Inactive	Nov 15/00	H		
31-39	Jan 31/00	Inactive	Nov 15/00	H		
31-40	Apr 30/01	Inactive	Oct 28/02	H		
31-41	Jun 20/01	Inactive	Oct 28/02	H		
31-42	Aug 19/02	Inactive	Oct 28/02	H		
31-43	Sep 24/03	Inactive	22 Dec 2006	H		
31-44	Oct 31/03	Inactive	22 Dec 2006	H		
31-45	May 28/04	Inactive	22 Dec 2006	H		
31-46	Dec 22/04	Inactive	22 Dec 2006	H		
31-47	Feb 3/05	Inactive	22 Dec 2006	H		
31-48	Jun 10/05	Inactive	22 Dec 2006	H		
31-49	Sep 16/05	Inactive	22 Dec 2006	H		

31-30-32

Page RTR-3
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

31-30-32

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

SERVICE BULLETIN LIST

Service Bulletin	Date Incorporated	Description
980-4100D/B-31-1 (012-0342-101)	May 17/82	MOD 4 - Improves stability of various electronics circuits. Replaces kapton drive belt with dacron/neoprene drive belt.
980-4100D/B-31-2 (012-0342-102)	Sep 15/82	MOD 5 - Improves service life by changing peripheral belt to a thicker belt and makes other mechanical changes in the transport assembly to accommodate the thicker belt.
980-4100D/B-31-3 (012-0342-103) with Revision 1	Jun 24/83	MOD 6 - Factory incorporated on serial number 2000 and subsequent. Replaces vendor sensitive part (A3U2) with more reliable part.
980-4100D/B-31-4 (012-0342-104)	Oct 31/83	MOD 8 - Requires inspection of serial numbers: 1688-1690, 2016-2021, 2024-2028, to determine proper installation of read/write heads.
980-4100D/B-31-5 (012-0342-105) with Revision 5	22 Dec 2006	MODS 11 and 15 - Provides for the use of mylar tape as an alternative to Kapton tape. Replaces the aluminum capstan and P-belt rollers with stainless steel versions.
980-4100D/B-31-6 (012-0342-106)	Jul 28/89	MOD 13 - Enhances UFDR compatibility with Flight Data Acquisition Units (FDAUs) which are equipped with mechanical reset annunciators. With Mod incorporated, the maintenance status signal from the UFDR is inhibited to the FDAU during self-test.
980-4100D/B-31-7 (012-0342-107)	Jul 28/89	MOD 14 - Improves compatibility of UFDR with Flight Data Acquisition Units (FDAUs) equipped with mechanical reset type annunciators when the recorder is used as part of an ARINC 573 AIDS. Mfg on SN 4225 and on.
980-4100D/B-31-8 (012-0342-108)	Jul 28/89	MOD 15 - Reduces the generation of debris which originates from drag idlers and capstan on the Tape Transport Assembly. Mfg on SN 4170 and on.
980-4100D/B-31-9 (012-0342-109)	Jul 28/89	MOD 16 - Prevents possible mechanical clearance problems between Power Supply support assembly and circuit traces on Power Supply. Mfg on SN 3951 and on.
980-4100D/B-31-10 (012-0342-110)	Jul 28/89	MOD 18 - Mod must be performed on all affected units to ensure that they will fit in all aircraft installations. Applies to UFDRs with unwired pins installed in upper rear connector section. Mfg on SN 3998 and on.
980-4100D/B-31-11 (012-0342-111) with Revision 6	22 Dec 2006	MOD 22 - Provides improvements to the Tape Transport Assembly following field and factory evaluation of Mod 11 changes. Mfg on SN 6000 and on.

Date Incorporated is the date of the CMM revision in which Service Bulletin information was incorporated.

31-30-32

Page SBL-1
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Service Bulletin	Date Incorporated	Description
980-4100D/B-31-12 (012-0342-112)	Jul 28/89	MOD 24 - Prevent UFDR from entering a fault condition in Recording System Controller, which is exhibited by the presence of all 1's recorded on tape. Mfg on SN 5453 and on.
980-4100D/B-31-13 (012-0342-113)	Jul 28/89	MOD 25 - Provides improvements on Tape Transport Assembly which allow better control of the tape path, prevent possible bearing contamination, and help prevent tape deck debris. Mod 5 must be incorporated prior to incorporating this Mod.
980-4100DBA-31-14 (012-0342-114) with Revision 2	22 Dec 2006	Combination of Mod 11 and 15 per SB 5 and Mod 22 per SB 11 in order to modify units at the same time to incorporate 3 mods.
980-4100D/B-31-15 (012-0342-115) with Revision 5	22 Dec 2006	Converts B and D Models to A Models - Implements firmware changes to Recording System Controller required in Airbus A320, A330 and A340 installations.
980-4100ABD-31-16 (012-0342-116)	Jun 15/93	MOD 26 - Allows for substitution of 8251A PCI for 8251 and 9551 PCIs. Mfg on SN 8500 and on.
980-4100ABD-31-17 (012-0342-117)	Jan 15/93	MOD 27 - To prevent interference between thermal enclosure liner and flywheel.
980-4100ABD-31-18 (012-0342-118)	Jul 14/95	Converts magnetic tape UFDR to Solid State UFDR, PN 980-4120-().
980-4100ABD-31-19 (012-0342-119) Revision 1	Jun 26/96 No Effect	MOD 31 - Implement firmware change correcting reporting of faults at power up.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

LIST OF EFFECTIVE PAGES

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
Title		TC-11	* 22 Dec 2006
T-1	* 22 Dec 2006	TC-12	* 22 Dec 2006
T-2	* 22 Dec 2006	TC-13	* 22 Dec 2006
Record of Revisions		TC-14	* 22 Dec 2006
RR-1	* 22 Dec 2006	Introduction	
RR-2	* 22 Dec 2006	INTRO-1	* 22 Dec 2006
Record of Temporary Revisions		INTRO-2	* 22 Dec 2006
RTR-1	* 22 Dec 2006	INTRO-3	* 22 Dec 2006
RTR-2	* 22 Dec 2006	INTRO-4	* 22 Dec 2006
RTR-3	* 22 Dec 2006	INTRO-5	* 22 Dec 2006
RTR-4	* 22 Dec 2006	INTRO-6	* 22 Dec 2006
Service Bulletin List		INTRO-7	* 22 Dec 2006
SBL-1	* 22 Dec 2006	INTRO-8	* 22 Dec 2006
SBL-2	* 22 Dec 2006	Description and Operation	
List of Effective Pages		1	* 22 Dec 2006
LEP-1	* 22 Dec 2006	2	* 22 Dec 2006
LEP-2	* 22 Dec 2006	3	* 22 Dec 2006
LEP-3	* 22 Dec 2006	4	* 22 Dec 2006
LEP-4	* 22 Dec 2006	5	* 22 Dec 2006
LEP-5	* 22 Dec 2006	6	* 22 Dec 2006
LEP-6	* 22 Dec 2006	7	* 22 Dec 2006
LEP-7	* 22 Dec 2006	8	* 22 Dec 2006
LEP-8	* 22 Dec 2006	9	* 22 Dec 2006
Table of Contents		10	* 22 Dec 2006
TC-1	* 22 Dec 2006	11	* 22 Dec 2006
TC-2	* 22 Dec 2006	12	* 22 Dec 2006
TC-3	* 22 Dec 2006	13	* 22 Dec 2006
TC-4	* 22 Dec 2006	14	* 22 Dec 2006
TC-5	* 22 Dec 2006	15	* 22 Dec 2006
TC-6	* 22 Dec 2006	16	* 22 Dec 2006
TC-7	* 22 Dec 2006	17	* 22 Dec 2006
TC-8	* 22 Dec 2006	18	* 22 Dec 2006
TC-9	* 22 Dec 2006	19	* 22 Dec 2006
TC-10	* 22 Dec 2006	20	* 22 Dec 2006

* indicates a changed or added page.

F indicates a foldout page.

31-30-32

Page LEP-1
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
21	* 22 Dec 2006	118	* 22 Dec 2006
22	* 22 Dec 2006	119	* 22 Dec 2006
23	* 22 Dec 2006	120	* 22 Dec 2006
24	* 22 Dec 2006	121	* 22 Dec 2006
25	* 22 Dec 2006	122	* 22 Dec 2006
26	* 22 Dec 2006	123	* 22 Dec 2006
27	* 22 Dec 2006	124	* 22 Dec 2006
28	* 22 Dec 2006	125	* 22 Dec 2006
29	* 22 Dec 2006	126	* 22 Dec 2006
30	* 22 Dec 2006	127	* 22 Dec 2006
31	* 22 Dec 2006	128	* 22 Dec 2006
32	* 22 Dec 2006	129	* 22 Dec 2006
33	* 22 Dec 2006	130	* 22 Dec 2006
34	* 22 Dec 2006	131	* 22 Dec 2006
35	* 22 Dec 2006	132	* 22 Dec 2006
36	* 22 Dec 2006	133	* 22 Dec 2006
37	* 22 Dec 2006	134	* 22 Dec 2006
38	* 22 Dec 2006	135	* 22 Dec 2006
Testing and Troubleshooting		136	* 22 Dec 2006
101	* 22 Dec 2006	137	* 22 Dec 2006
102	* 22 Dec 2006	138	* 22 Dec 2006
103	* 22 Dec 2006	139	* 22 Dec 2006
104	* 22 Dec 2006	140	* 22 Dec 2006
105	* 22 Dec 2006	141	* 22 Dec 2006
106	* 22 Dec 2006	142	* 22 Dec 2006
107	* 22 Dec 2006	143	* 22 Dec 2006
108	* 22 Dec 2006	144	* 22 Dec 2006
109	* 22 Dec 2006	145	* 22 Dec 2006
110	* 22 Dec 2006	146	* 22 Dec 2006
111	* 22 Dec 2006	147	* 22 Dec 2006
112	* 22 Dec 2006	148	* 22 Dec 2006
113	* 22 Dec 2006	149	* 22 Dec 2006
114	* 22 Dec 2006	150	* 22 Dec 2006
115	* 22 Dec 2006	151	* 22 Dec 2006
116	* 22 Dec 2006	152	* 22 Dec 2006
117	* 22 Dec 2006	153	* 22 Dec 2006

* indicates a changed or added page.

F indicates a foldout page.

31-30-32

Page LEP-2
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTv y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
154	* 22 Dec 2006	198.4	* 22 Dec 2006
155	* 22 Dec 2006	F 198.5/198.6	* 22 Dec 2006
156	* 22 Dec 2006	F 198.7/198.8	* 22 Dec 2006
157	* 22 Dec 2006	F 198.9/198.10	* 22 Dec 2006
158	* 22 Dec 2006	F 198.11/198.12	* 22 Dec 2006
159	* 22 Dec 2006	F 198.13/198.14	* 22 Dec 2006
160	* 22 Dec 2006	198.15	* 22 Dec 2006
161	* 22 Dec 2006	198.16	* 22 Dec 2006
162	* 22 Dec 2006	F 198.17/198.18	* 22 Dec 2006
163	* 22 Dec 2006	F 198.19/198.20	* 22 Dec 2006
164	* 22 Dec 2006	F 198.21/198.22	* 22 Dec 2006
165	* 22 Dec 2006	F 198.23/198.24	* 22 Dec 2006
166	* 22 Dec 2006	198.25	* 22 Dec 2006
167	* 22 Dec 2006	198.26	* 22 Dec 2006
168	* 22 Dec 2006	F 198.27/198.28	* 22 Dec 2006
F 169/170	* 22 Dec 2006	F 198.29/198.30	* 22 Dec 2006
F 171/172	* 22 Dec 2006	198.31	* 22 Dec 2006
F 173/174	* 22 Dec 2006	198.32	* 22 Dec 2006
175	* 22 Dec 2006	198.33	* 22 Dec 2006
176	* 22 Dec 2006	198.34	* 22 Dec 2006
F 177/178	* 22 Dec 2006	198.35	* 22 Dec 2006
179	* 22 Dec 2006	198.36	* 22 Dec 2006
180	* 22 Dec 2006	198.37	* 22 Dec 2006
F 181/182	* 22 Dec 2006	198.38	* 22 Dec 2006
F 183/184	* 22 Dec 2006	198.39	* 22 Dec 2006
F 185/186	* 22 Dec 2006	198.40	* 22 Dec 2006
187	* 22 Dec 2006	198.41	* 22 Dec 2006
188	* 22 Dec 2006	198.42	* 22 Dec 2006
F 189/190	* 22 Dec 2006	198.43	* 22 Dec 2006
F 191/192	* 22 Dec 2006	198.44	* 22 Dec 2006
F 193/194	* 22 Dec 2006	198.45	* 22 Dec 2006
195	* 22 Dec 2006	198.46	* 22 Dec 2006
196	* 22 Dec 2006	198.47	* 22 Dec 2006
F 197/198	* 22 Dec 2006	198.48	* 22 Dec 2006
F 198.1/198.2	* 22 Dec 2006	198.49	* 22 Dec 2006
198.3	* 22 Dec 2006	198.50	* 22 Dec 2006

* indicates a changed or added page.

F indicates a foldout page.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
198.51	* 22 Dec 2006	Check	
198.52	* 22 Dec 2006	501	* 22 Dec 2006
198.53	* 22 Dec 2006	502	* 22 Dec 2006
198.54	* 22 Dec 2006	503	* 22 Dec 2006
198.55	* 22 Dec 2006	504	* 22 Dec 2006
198.56	* 22 Dec 2006	505	* 22 Dec 2006
Atlas Test Specifications		506	* 22 Dec 2006
201	* 22 Dec 2006	507	* 22 Dec 2006
202	* 22 Dec 2006	508	* 22 Dec 2006
Disassembly		509	* 22 Dec 2006
301	* 22 Dec 2006	510	* 22 Dec 2006
302	* 22 Dec 2006	511	* 22 Dec 2006
303	* 22 Dec 2006	512	* 22 Dec 2006
304	* 22 Dec 2006	513	* 22 Dec 2006
305	* 22 Dec 2006	514	* 22 Dec 2006
306	* 22 Dec 2006	Repair	
307	* 22 Dec 2006	601	* 22 Dec 2006
308	* 22 Dec 2006	602	* 22 Dec 2006
309	* 22 Dec 2006	603	* 22 Dec 2006
310	* 22 Dec 2006	604	* 22 Dec 2006
311	* 22 Dec 2006	605	* 22 Dec 2006
312	* 22 Dec 2006	606	* 22 Dec 2006
313	* 22 Dec 2006	607	* 22 Dec 2006
314	* 22 Dec 2006	608	* 22 Dec 2006
315	* 22 Dec 2006	609	* 22 Dec 2006
316	* 22 Dec 2006	610	* 22 Dec 2006
317	* 22 Dec 2006	611	* 22 Dec 2006
318	* 22 Dec 2006	612	* 22 Dec 2006
Cleaning		613	* 22 Dec 2006
401	* 22 Dec 2006	614	* 22 Dec 2006
402	* 22 Dec 2006	615	* 22 Dec 2006
403	* 22 Dec 2006	616	* 22 Dec 2006
404	* 22 Dec 2006	617	* 22 Dec 2006
405	* 22 Dec 2006	618	* 22 Dec 2006
406	* 22 Dec 2006	619	* 22 Dec 2006

* indicates a changed or added page.

F indicates a foldout page.

31-30-32

Page LEP-4
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
620	* 22 Dec 2006	721	* 22 Dec 2006
621	* 22 Dec 2006	722	* 22 Dec 2006
622	* 22 Dec 2006	723	* 22 Dec 2006
623	* 22 Dec 2006	724	* 22 Dec 2006
624	* 22 Dec 2006	725	* 22 Dec 2006
625	* 22 Dec 2006	726	* 22 Dec 2006
626	* 22 Dec 2006	727	* 22 Dec 2006
627	* 22 Dec 2006	728	* 22 Dec 2006
628	* 22 Dec 2006	729	* 22 Dec 2006
629	* 22 Dec 2006	730	* 22 Dec 2006
630	* 22 Dec 2006	731	* 22 Dec 2006
631	* 22 Dec 2006	732	* 22 Dec 2006
632	* 22 Dec 2006	733	* 22 Dec 2006
633	* 22 Dec 2006	734	* 22 Dec 2006
634	* 22 Dec 2006	735	* 22 Dec 2006
Assembly (Including Storage)		736	* 22 Dec 2006
701	* 22 Dec 2006	737	* 22 Dec 2006
702	* 22 Dec 2006	738	* 22 Dec 2006
703	* 22 Dec 2006	739	* 22 Dec 2006
704	* 22 Dec 2006	740	* 22 Dec 2006
705	* 22 Dec 2006	741	* 22 Dec 2006
706	* 22 Dec 2006	742	* 22 Dec 2006
707	* 22 Dec 2006	743	* 22 Dec 2006
708	* 22 Dec 2006	744	* 22 Dec 2006
709	* 22 Dec 2006	745	* 22 Dec 2006
710	* 22 Dec 2006	746	* 22 Dec 2006
711	* 22 Dec 2006	747	* 22 Dec 2006
712	* 22 Dec 2006	748	* 22 Dec 2006
713	* 22 Dec 2006	749	* 22 Dec 2006
714	* 22 Dec 2006	750	* 22 Dec 2006
715	* 22 Dec 2006	751	* 22 Dec 2006
716	* 22 Dec 2006	752	* 22 Dec 2006
717	* 22 Dec 2006	753	* 22 Dec 2006
718	* 22 Dec 2006	754	* 22 Dec 2006
719	* 22 Dec 2006	755	* 22 Dec 2006
720	* 22 Dec 2006	756	* 22 Dec 2006

* indicates a changed or added page.

F indicates a foldout page.

31-30-32

Page LEP-5
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
757	* 22 Dec 2006	1013	* 22 Dec 2006
758	* 22 Dec 2006	1014	* 22 Dec 2006
759	* 22 Dec 2006	1015	* 22 Dec 2006
760	* 22 Dec 2006	1016	* 22 Dec 2006
761	* 22 Dec 2006	1017	* 22 Dec 2006
762	* 22 Dec 2006	1018	* 22 Dec 2006
Fits and Clearances		1019	* 22 Dec 2006
801	* 22 Dec 2006	1020	* 22 Dec 2006
802	* 22 Dec 2006	1021	* 22 Dec 2006
803	* 22 Dec 2006	1022	* 22 Dec 2006
804	* 22 Dec 2006	1023	* 22 Dec 2006
Special Tools, Fixtures, and Equipment		1024	* 22 Dec 2006
901	* 22 Dec 2006	1025	* 22 Dec 2006
902	* 22 Dec 2006	1026	* 22 Dec 2006
903	* 22 Dec 2006	1027	* 22 Dec 2006
904	* 22 Dec 2006	1028	* 22 Dec 2006
905	* 22 Dec 2006	1029	* 22 Dec 2006
906	* 22 Dec 2006	1030	* 22 Dec 2006
F 907/908	* 22 Dec 2006	1031	* 22 Dec 2006
909	* 22 Dec 2006	1032	* 22 Dec 2006
910	* 22 Dec 2006	1033	* 22 Dec 2006
911	* 22 Dec 2006	1034	* 22 Dec 2006
912	* 22 Dec 2006	F 1035/1036	* 22 Dec 2006
Illustrated Parts List		1037	* 22 Dec 2006
1001	* 22 Dec 2006	1038	* 22 Dec 2006
1002	* 22 Dec 2006	1039	* 22 Dec 2006
1003	* 22 Dec 2006	1040	* 22 Dec 2006
1004	* 22 Dec 2006	1041	* 22 Dec 2006
1005	* 22 Dec 2006	1042	* 22 Dec 2006
1006	* 22 Dec 2006	1043	* 22 Dec 2006
1007	* 22 Dec 2006	1044	* 22 Dec 2006
1008	* 22 Dec 2006	1045	* 22 Dec 2006
1009	* 22 Dec 2006	1046	* 22 Dec 2006
1010	* 22 Dec 2006	1047	* 22 Dec 2006
1011	* 22 Dec 2006	1048	* 22 Dec 2006
1012	* 22 Dec 2006	1049	* 22 Dec 2006

* indicates a changed or added page.

F indicates a foldout page.

31-30-32

Page LEP-6
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
1050	* 22 Dec 2006	1087	* 22 Dec 2006
1051	* 22 Dec 2006	1088	* 22 Dec 2006
1052	* 22 Dec 2006	1089	* 22 Dec 2006
1053	* 22 Dec 2006	1090	* 22 Dec 2006
1054	* 22 Dec 2006	F 1091/1092	* 22 Dec 2006
1055	* 22 Dec 2006	1093	* 22 Dec 2006
1056	* 22 Dec 2006	1094	* 22 Dec 2006
1057	* 22 Dec 2006	1095	* 22 Dec 2006
1058	* 22 Dec 2006	1096	* 22 Dec 2006
1059	* 22 Dec 2006	1097	* 22 Dec 2006
1060	* 22 Dec 2006	1098	* 22 Dec 2006
1061	* 22 Dec 2006	F 1098.1/1098.2	* 22 Dec 2006
1062	* 22 Dec 2006	F 1098.3/1098.4	* 22 Dec 2006
1063	* 22 Dec 2006	1098.5	* 22 Dec 2006
1064	* 22 Dec 2006	1098.6	* 22 Dec 2006
1065	* 22 Dec 2006	1098.7	* 22 Dec 2006
1066	* 22 Dec 2006	1098.8	* 22 Dec 2006
1067	* 22 Dec 2006	1098.9	* 22 Dec 2006
1068	* 22 Dec 2006	1098.10	* 22 Dec 2006
1069	* 22 Dec 2006	1098.11	* 22 Dec 2006
1070	* 22 Dec 2006	1098.12	* 22 Dec 2006
1071	* 22 Dec 2006	1098.13	* 22 Dec 2006
1072	* 22 Dec 2006	1098.14	* 22 Dec 2006
1073	* 22 Dec 2006	1098.15	* 22 Dec 2006
1074	* 22 Dec 2006	1098.16	* 22 Dec 2006
1075	* 22 Dec 2006	1098.17	* 22 Dec 2006
1076	* 22 Dec 2006	1098.18	* 22 Dec 2006
1077	* 22 Dec 2006	1098.19	* 22 Dec 2006
1078	* 22 Dec 2006	1098.20	* 22 Dec 2006
1079	* 22 Dec 2006	F 1098.21/1098.22	* 22 Dec 2006
1080	* 22 Dec 2006	F 1098.23/1098.24	* 22 Dec 2006
1081	* 22 Dec 2006	1098.25	* 22 Dec 2006
1082	* 22 Dec 2006	1098.26	* 22 Dec 2006
1083	* 22 Dec 2006	1098.27	* 22 Dec 2006
1084	* 22 Dec 2006	1098.28	* 22 Dec 2006
F 1085/1086	* 22 Dec 2006	1098.29	* 22 Dec 2006

* indicates a changed or added page.

F indicates a foldout page.

31-30-32

Page LEP-7
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subheading and Page</u>	<u>Date</u>	<u>Subheading and Page</u>	<u>Date</u>
1098.30	*	22 Dec 2006	
1098.31	*	22 Dec 2006	
1098.32	*	22 Dec 2006	
1098.33	*	22 Dec 2006	
1098.34	*	22 Dec 2006	
1098.35	*	22 Dec 2006	
1098.36	*	22 Dec 2006	
1098.37	*	22 Dec 2006	
1098.38	*	22 Dec 2006	
1098.39	*	22 Dec 2006	
1098.40	*	22 Dec 2006	
1098.41	*	22 Dec 2006	
1098.42	*	22 Dec 2006	
F 1098.43/1098.44	*	22 Dec 2006	
1098.45	*	22 Dec 2006	
1098.46	*	22 Dec 2006	
1098.47	*	22 Dec 2006	
1098.48	*	22 Dec 2006	
1098.49	*	22 Dec 2006	
1098.50	*	22 Dec 2006	
1098.51	*	22 Dec 2006	
1098.52	*	22 Dec 2006	

* indicates a changed or added page.

F indicates a foldout page.

31-30-32

Page LEP-8
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

TABLE OF CONTENTS

<u>Subject</u>	<u>Page</u>
INTRODUCTION	INTRO-1
1. Proprietary, Export, and Precautionary Data	INTRO-1
A. Proprietary Notice	INTRO-1
B. Export Notice	INTRO-1
C. Special Precautions	INTRO-1
2. Content Data	INTRO-2
A. How to Use This Manual	INTRO-2
B. Verification	INTRO-3
C. Symbols	INTRO-4
D. Weights and Measurements	INTRO-4
E. Acronyms and Abbreviations	INTRO-5
DESCRIPTION AND OPERATION	1
1. General	1
A. Physical Description	1
B. Dust Cover Assembly	2
C. Product Identification	2
D. Leading Particulars	2
E. Tape Transport	2
F. Thermal Enclosure	5
G. Recording Concept	5
H. Modes of Operation	6
I. Fast Playback Feature	8
J. Failure Warning and Maintenance Status Monitoring	8
K. Input Data Forms	10
2. Block Diagram Analysis of Recorder Operation	10
3. Theory of Operation	13
A. General	13
B. Recording System Controller, A4	14
C. Overview of Recording System Activity	18
D. Transport Interface Assembly, A3	21
E. Airframe Wiring Interface Assembly, A8	27
F. Read/Write Interface Assembly, A2	30
G. Power Supply Assembly, A1	36

31-30-32

Page TC-1
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
TESTING AND TROUBLESHOOTING	101
1. General	101
A. Test Equipment Required	101
B. Power Requirements	101
C. Test Conditions	101
D. Flight Data Transcription Requirements	101
E. Test Procedure Requirements	101
F. ATE Users	102
G. Readout and Test Unit (RTU) Users	102
H. Acceptance Test/Transcription Unit (ATU) Users	102
I. Underwater Locating Device (ULD) Test	102
2. Functional Test Procedure	104
A. Insulation Resistance Check	104
B. Initial Setup	104
C. Nonflight Recording Modes Operating Tests	106
D. Flight Recording Mode Operating Test	107
E. CHECKSTROKE Stability Test	108
F. Write-to-Tape Test Procedure	110
G. Playback Tests	111
H. Gain Margin Test	116
I. System Status Indicator	117
J. Abnormal Power Condition Tests	118
K. Data Output Signal Level Test	119
L. End of Functional Test	119
3. Power Supply, A1, Calibration Procedure	120
4. Read/Write Interface, A2, Calibration Procedure	123
5. Recording System Controller, A4, Calibration Procedure	125
A. Setup Procedure	125
B. Check Procedure	125
C. Calibration Procedure	125
6. Troubleshooting (UFDR Tester Users)	127
A. General	127
B. Built-In Test Equipment (BITE)	127
C. Causes of STATUS Indicator ON	128
D. Causes of Software Detected Fault Condition	128

31-30-32

Page TC-2
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
E. Causes of Data Process Fault 1: CHEKIN \Not= CHEKWR	129
F. Causes of Data Process Fault 2: RAM buffer contains all 1's or 0's	129
G. Causes of Data Read/Write Fault Condition: CHEKRD \Not= CHEKWR	130
H. Causes of Lock-Up Circuit Detected Fault	131
I. Causes of Track Select Fault Condition: A3U13-13 Set LOW and/or OPORT-2, Bit 6, set HIGH	132
J. Fault Logic Diagrams	132
7. UFDR RTU Test Procedure	154
A. Test Equipment Required	154
B. Equipment Setup	154
C. Return to Service Information	155
D. Transport Test	156
E. Manual Acceptance Test	159
F. Diagnostic Mode	159
8. Troubleshooting (RTU Users)	159
A. General	159
B. Fault Logic Diagram	160
9. ATU Test Procedure	198.45
A. Inspection of UFDR	198.45
B. Base Software Installation	198.45
C. ATP Software Installation	198.45
D. Install the ADRAS for Windows Software	198.49
E. Configuring ATEasy for the ATU	198.52
F. Checking UFDR Insulation Resistance	198.53
G. UFDR ATU Test Setup	198.53
H. Tape Transport Test	198.54
ATLAS TEST SPECIFICATION	201
1. Introduction	201
2. General	201
DISASSEMBLY	301
1. General	301
2. General Techniques	301
3. Recorder, Universal Flight Data	301
4. Power Supply A1	304
5. Read/Write Interface Assembly, A2	305

31-30-32

Page TC-3
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
6. Transport Interface Assembly, A3	305
7. Recording System Controller Assembly, A4	305
8. Airframe Wiring Interface Assembly, A8	305
9. Power Supply Assembly, A1A1	305
10. Tape Recorder Assembly	305
11. Chassis Assembly	306
A. Filter Removal	306
B. Elapsed Time Indicator (ETI) Removal	306
C. Connector J1 Removal	306
D. Motherboard Removal	307
E. Shockmount Removal	308
F. Connector J2 Removal	308
G. LED Indicator Removal	308
H. Connector J5 Removal	308
I. Tywrap Clamp Removal	308
12. Tape Transport Assembly	308
A. Tape Reel and Tape Reel Pin Removal	308
B. Motor Drive (Flat) Belt Removal	309
C. Drive Pulley Bearing Removal	309
D. Tape Guide Roller and Roller Pin Removal	309
E. Drive Belt Roller and Roller Pin Removal	309
F. Magnetic Head Removal	310
G. EOT/BOT Sensors Removal	310
H. Distribution Board Removal	310
I. Peripheral Belt Roller and Roller Pin Removal.	311
13. Motor Drive Assembly	311
14. Distribution Board	312
15. Motor Drive Assembly	312
16. Tape Transport Assembly	312
A. Tape Reel Removal	312
B. Disassembly of Reel Hub Assembly	313
C. Tape Reel Pin Removal	313
D. Flat Belt and Tape Transport Pulley Removal	313
E. Tape Transport Pulley Bearing Removal	313
F. Tape Guide Roller and Pin Removal	313

31-30-32

Page TC-4
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
G. Disassembly of Drive Belt Roller Guide Assembly	314
H. Magnetic Head Removal	314
I. EOT/BOT Sensors Removal	314
J. Distribution Board Removal	315
K. Disassembly of P-Belt Pulley Assembly	315
17. Power Supply, A1	315
18. Chassis Assembly	316
A. EMI Filter Removal	316
B. EMI Filter and Diode Removal	316
C. Elapsed Time Indicator (ETI) Removal	316
D. Connector J1 Removal	316
E. Motherboard Removal	316
F. Shockmount Removal	317
G. Connector J2 Removal	318
H. LED Indicator Assembly Removal	318
I. Connector J5 Removal	318
J. Tywrap Base Removal	318
K. Card Guide Removal	318
CLEANING	401
1. General	401
2. Cleaning Materials	401
3. Cleaning Procedures	401
A. Printed Wiring Board Assemblies	402
B. Connectors	402
C. Machined Metal Parts or Assemblies	402
D. Miscellaneous Metal Parts	403
E. Nonmetallic Parts	403
F. Bearings	403
G. Tape Transport Cleaning Procedure	403
H. Underwater Locating Device (ULD)	405
I. Underwater Acoustic Beacon (UAB)	406
CHECK	501
1. General	501
A. Component Physical Examination	501
B. Component Electrical Examination	501

31-30-32

Page TC-5
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
2. Specific Checks	501
3. Perpendicularity Check	503
A. General	503
B. Procedure	503
4. Height Check	505
A. General	505
B. Procedure	505
5. Preload Check	506
A. General	506
B. Procedure	507
6. End Play Check	508
A. General	508
B. Procedure	508
7. Track Location Check	509
A. General	509
B. Procedure	509
8. Tape Stacking, P-Belt Running Location, and Tape Reversing Time Checks	512
A. General	512
B. Procedure	512
9. Tape Tracking Check	513
A. General	513
B. Procedure	513
REPAIR	601
1. General	601
2. PWB Repair Procedure	603
3. Conformal Coating Removal	603
4. Installation of Replacement Components	603
5. Fits and Clearance Requirements	604
6. Procedure for Applying Conformal Coating	604
7. Reflective Tape Installation	604
8. Degaussing Procedure	604
9. Bearing Information	606
10. Bearing Preload Information	608
11. Bearing Handling Information	608
12. Finished Metallic Surfaces	609

31-30-32

Page TC-6
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
13. Thermal Enclosure Repair	609
A. Determining Extent of Delamination	609
B. Repair Procedure, Liner and Insulator Replacement	609
C. Repair Procedure, Liner Separation From Insulator	611
14. Field Rework of UFDR Head Assemblies	612
A. Inspection Procedure	613
B. Magnetic Head Assembly Relap Procedure	615
C. Supplemental Lapping Procedure	616
15. Filter Assembly Repair Procedure	617
A. Test Equipment Required	617
B. Repair Procedure	617
C. Test Procedure	618
16. Tape Reel Loading Procedure	622
A. Equipment Requirements	622
B. Material Requirements	622
C. Tape Winder Preparation	623
D. Tape Loading Procedure	623
17. Power Supply Cable Clamp Installation	626
18. Stepper Motor PN 360-0082-002/004 Bearing Replacement	627
A. Materials Required	627
B. Disassembly	627
C. Clean and Check	628
D. Assembly	628
19. Preventive Maintenance Information	628
20. Power Supply (A1A1) -002 or less to -003 Modification	631
21. Stepper Motor PN 360-0082-001/003 Bearing Replacement	631
A. Materials Required	631
B. Disassembly	631
C. Clean and Check	632
D. Assembly	632
ASSEMBLY (INCLUDING STORAGE)	701
1. General	701
2. General Techniques	701
3. Assembly Materials	701

31-30-32

Page TC-7
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
4. Chassis Assembly	705
A. Assembly of Chassis	705
B. Card Guide Installation	705
C. Shockmount Installation	705
D. Tywrap Base Installation	705
E. Connector J5 Installation	705
F. Connector J2 Installation	706
G. LED Indicator Assembly Installation	706
H. Elapsed Time Indicator (ETI) Installation	706
I. EMI Filter Installation	707
J. Connector J1 Installation	707
K. Assembly of Motherboard PWB	707
L. Motherboard PWB Installation	707
5. Power Supply, A1	708
6. Tape Transport Assembly	709
7. Tape Transport Assembly	714
8. Motor Drive Assembly	730
9. Distribution Board	732
10. Motor Drive Assembly	733
11. Tape Transport Assembly	735
12. Chassis Assembly	740
13. Tape Recorder Assembly	743
14. Tape Recorder Assembly	744
A. Tape Transport Assembly Installation	744
B. Flat Belt Slip Torque Check	745
C. Motor Drive Belt Tracking Check	746
D. Magnetic Tape and P-belt Installation	746
E. Drive Torque Test	751
F. Bulk Tape Erasure	751
G. Install Top Plate	751
H. Install Top Cover	751
15. Power Supply Assembly, A1A1	752
16. Airframe Wiring Interface, A8	754
17. Recording System Controller, A4	754
18. Transport Interface Assembly, A3	755

31-30-32

Page TC-8
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Subject</u>	<u>Page</u>
19. Read/Write Interface, A2	755
20. Power Supply, A1	756
21. Recorder, Universal Flight Data	757
22. Final Confidence Test	760
23. Storage After Assembly	760
A. Temporary Storage	760
B. Extended Storage	760
C. Peripheral Belt Storage	761
FITS AND CLEARANCES	801
SPECIAL TOOLS, FIXTURES, AND EQUIPMENT	901
ILLUSTRATED PARTS LIST	1001
1. Overview	1001
2. Contents of the IPL	1001
3. List of Suppliers	1005
4. Equipment Designator Index	1009
5. Numerical Index	1015
6. Parts List	1033

31-30-32

Page TC-9
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

List of Illustrations

Figure		Page
Figure Intro-1.	Symbols	INTRO-4
Figure 1.	Recorder	1
Figure 2.	Accessory Codes	3
Figure 3.	Leading Particulars	3
Figure 4.	Tape Velocity Versus Displacement Graph	6
Figure 5.	Modes of Operation	7
Figure 6.	Harvard Bi-Phase and NRZ Codes	10
Figure 7.	UFDR Functional Block Diagram	11
Figure 8.	UFDR Recorder Connectors	12
Figure 9.	Upper Fields Address Decode	15
Figure 10.	Interrupt Timing Diagram	19
Figure 11.	External Equipment Control Data Signals	22
Figure 12.	Timing Diagram for Transition Bit Generation	26
Figure 13.	Timing Diagram for Clock Bit Generation	26
Figure 14.	Timing Diagram for Data Bit Generation	26
Figure 15.	Timing Diagram for Bi-Phase Bit Generation	27
Figure 16.	Relay Interconnection Diagram	29
Figure 17.	Summary of Relay Conditions	30
Figure 18.	Read/Write Circuit Simplified Diagram	31
Figure 19.	Logic Signals for Track Select	32
Figure 20.	Typical Input/Output Waveforms	34
Figure 21.	Comparator U10 Waveforms	35
Figure 22.	Typical Stepper Motor	38
Figure 23.	Pattern of Stepper Motor Signal Phases	38
Figure 101.	Functional Tester Control Panel	105
Figure 102.	Acceptable Waveforms for Playback Signals Modulation Envelope	109
Figure 103.	Unacceptable Waveforms for Playback Signals Modulation Envelope	109
Figure 104.	Scope Indication of 1's and 0's	114
Figure 105.	Data Output Signal Waveform	120
Figure 106.	Power Supply Assembly A1A1 Resistor Location	121
Figure 107.	Read/Write Interface, A2., Resistor Location	124
Figure 108.	Recorder System Controller, A4, Resistor Location	126
Figure 109.	Assembly Level Fault Logic Diagram	133
Figure 110.	Recording System Controller Assembly, A4, Fault Logic Diagram	139

31-30-32

Page TC-10
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Figure		Page
Figure 111.	Read/Write Interface Assembly, A2, Fault Logic Diagram	144
Figure 112.	Transport Interface Assembly, A3, Fault Logic Diagram	147
Figure 113.	Aircraft Wiring Interface Assembly, A8, Fault Logic Diagram	150
Figure 114.	Power Supply Assembly, A1, Fault Logic Diagram	151
Figure 115.	Tape Reproducer Assembly, Fault Logic Diagram	153
Figure 116.	UFDR Test Setup	154
Figure 117.	Top Level Menu	156
Figure 118.	Transport Test Warning Screen	157
Figure 119.	UFDR Transport Test Header Screen	157
Figure 120.	Transport Test Header Screen with Data	158
Figure 121.	UFDR Fault Logic Diagram	160
Figure 122.	Chassis and Tape Recorder Assembly Schematic	167
Figure 123.	Filter Schematic	175
Figure 124.	Motherboard Schematic	179
Figure 125.	Power Supply, A1, Schematic	187
Figure 126.	Read/Write Interface, A2, Schematic	195
Figure 127.	Transport Interface Assembly, A3, Schematic	198.3
Figure 128.	Recording System Controller, A4, Schematic	198.15
Figure 129.	Airframe Wiring Interface, A8, Schematic	198.25
Figure 130.	Integrated Circuit Diagram	198.31
Figure 131.	Initial Instructions Dialog	198.46
Figure 132.	ATU Configuration Select Dialog	198.46
Figure 133.	Software Choice Dialog	198.47
Figure 134.	ATEasy Directory Dialog	198.48
Figure 135.	Change Disk Dialog	198.48
Figure 136.	SSFDR/SSUFDR/UFDR ATP Program Group	198.48
Figure 137.	ATU Model Software List	198.50
Figure 138.	Run Dialog	198.51
Figure 139.	Setup Initiation	198.51
Figure 140.	Enter Disk Window	198.52
Figure 141.	Defaults Dialog Window	198.53
Figure 142.	Equipment Setup	198.54
Figure 143.	Select Profile Dialog	198.55
Figure 144.	Return to Service Test Selection	198.55
Figure 301.	ULD Disassembly	302

31-30-32

Page TC-11
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Figure</u>		<u>Page</u>
Figure 401.	Cleaning Materials	401
Figure 501.	Perpendicularity Check	504
Figure 502.	Perpendicularity Check Requirements	504
Figure 503.	Tape Guide Roller Height Check	506
Figure 504.	Hub Assembly Height Check	506
Figure 505.	Preload Check	507
Figure 506.	Preload Requirements	508
Figure 507.	End Play Requirements	508
Figure 508.	Track Wiring Check Pattern	511
Figure 509.	Track Spacing Pattern	512
Figure 601.	Repair Materials	601
Figure 602.	Reflective Tape Locations	605
Figure 603.	Bearing Lubrication Information	606
Figure 604.	Tape Transport Mounting Hole Pattern	611
Figure 605.	Liner Hole Pattern	612
Figure 606.	Head Assembly Measurement	614
Figure 607.	Magnetic Head Pole Tip Lapping Setup	615
Figure 608.	Tip Depth and Tilt Final Measurements	616
Figure 609.	Filter Assembly Test Setup	619
Figure 610.	Test Fixture Test Switch Positions	620
Figure 611.	Tape Winder	623
Figure 612.	BOT/EOT Windows, Mylar Tape	624
Figure 613.	BOT/EOT Windows, Kapton Tape	625
Figure 614.	Cable Clamp Installation	627
Figure 615.	Stepper Motor, PN 360-0082-002/004	629
Figure 616.	Preventive Maintenance Items	630
Figure 617.	Stepper Motor, PN 360-0082-001/003	633
Figure 701.	Assembly Materials	702
Figure 702.	Installation of LED Indicator Assembly	706
Figure 703.	Foam Tape Installation on EMI Shield	709
Figure 704.	Reel Assembly	713
Figure 705.	Cable Assembly	715
Figure 706.	Head and Shim Installation	716
Figure 707.	Reel Hub Assembly (70, 70A)	718
Figure 708.	Reel Hub Assembly (70B, 70C)	719

31-30-32

Page TC-12
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Figure</u>		<u>Page</u>
Figure 709.	Shim Requirements, Reel Hub Assembly (70B, 70C)	719
Figure 710.	Height Check, Reel Hub Assembly	720
Figure 711.	Reel Hub Heights	720
Figure 712.	Drive Belt Roller Guide Assembly	722
Figure 713.	Height Check, Peripheral Belt Roller Pin	723
Figure 714.	Peripheral Belt Pulley Assembly	723
Figure 715.	Preliminary Eccentric Roller Pin Alignment	724
Figure 716.	Tape Guide Roller Assembly (140)	726
Figure 717.	Tape Guide Roller Assembly (440, 440A)	727
Figure 718.	Tape Guide Roller Assembly (485, 485B)	728
Figure 719.	Height Check, Tape Guide Roller Assemblies	729
Figure 720.	Tape Guide Heights	729
Figure 721.	Roller Guide Assembly	731
Figure 722.	Stepper Motor Assembly	731
Figure 723.	Motor Cable Assembly	732
Figure 724.	Wire Chart, Distribution Board	733
Figure 725.	Tape Guide Roller Assembly	736
Figure 726.	Reel Assembly	737
Figure 727.	Belt, Tape, and Wire Layout	739
Figure 728.	Belt Installation (1, 1A)	743
Figure 729.	Flat Belt Installation (1B and on)	745
Figure 730.	Watch Paper Installation (Tape Cleaning)	746
Figure 731.	Peripheral Belt Force at Elongation Measurement Limits	748
Figure 732.	Peripheral Belt Installation on Idler Pulleys	748
Figure 733.	Peripheral Belt Installation on Capstan Pulley	749
Figure 734.	Peripheral Belt Installation on Loaded Reel	749
Figure 735.	Tape Loading Fixture Mounting Location	750
Figure 736.	Sealing Connector	752
Figure 737.	Installation of AR3 (220B, 220C, IPL Figure 7)	753
Figure 738.	Wire Chart, Recording System Controller, A4	755
Figure 739.	Foam Tape Installation on Shield	758
Figure 740.	Underwater Locating Device (ULD)	759
Figure 801.	Fits and Clearances	801
Figure 901.	Special Tools, Fixture, and Equipment	901
Figure 902.	Locating Jig, UFDR Environmental Enclosure Repair	905

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>Figure</u>		<u>Page</u>
Figure 903.	Belt Test Fixture	906
Figure 904.	UFDR Motor Mover Schematic	907
Figure 905.	UFDR Motor Mover Parts List	909
IPL Figure 1.	Recorder, Universal Flight Data	1035
IPL Figure 2.	Power Supply Assembly (A1).	1048
IPL Figure 3.	Read/Write Interface Assembly (A2)	1052
IPL Figure 4.	Transport Interface Assembly (A3).	1056
IPL Figure 5.	Recording System Controller Assembly (A4).	1062
IPL Figure 6.	Airframe Wiring Interface Assembly (A8).	1076
IPL Figure 7.	Power Supply Assembly (A1A1)	1080
IPL Figure 8.	Tape Recorder Assembly	1084
IPL Figure 9.	Chassis Assembly	1091
IPL Figure 10.	Tape Transport Assembly.	1098.1
IPL Figure 11.	Motor Drive Assembly	1098.10
IPL Figure 12.	Distribution Board.	1098.12
IPL Figure 13.	Motor Drive Assembly	1098.15
IPL Figure 14.	Tape Transport Assembly.	1098.21
IPL Figure 15.	Power Supply Assembly (A1).	1098.38
IPL Figure 16.	Chassis Assembly	1098.43
IPL Figure 17.	Filter	1098.50

List of Tables

<u>Table</u>		<u>Page</u>
Table Intro-1.	Related Publications	INTRO-3

31-30-32

Page TC-14
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

INTRODUCTION

1. Proprietary, Export, and Precautionary Data

A. Proprietary Notice

- (1) This document and the information disclosed herein are proprietary data of Honeywell. Neither this document nor the information contained herein shall be used, reproduced, or disclosed to others without the written authorization of Honeywell, except to the extent required for installation or maintenance of the recipient's equipment. FREEDOM OF INFORMATION ACT (5 USC 552) AND DISCLOSURE OF CONFIDENTIAL INFORMATION GENERALLY (18 USC 1905).
- (2) This document is being furnished in confidence by Honeywell. The information disclosed herein fall within exemption (b) (4) of 5 USC 552 and the prohibitions of 18 USC 1905. Copyright 2006 Honeywell. All Rights Reserved.
- (3) Honeywell is a U.S. registered trademark of Honeywell. All other marks are owned by their respective companies.

B. Export Notice

- (1) This document contains unrestricted technical data and is being exported under license exception TSU/OTS in accordance with EAR Section 740.13(a).
- (2) These commodities, technology, or software were exported from the United States in accordance with the export administration regulation. Diversion contrary to U.S. law is prohibited. ECCN: 7E994 Schedule B#4901.99.0050

C. Special Precautions

- (1) Warnings, cautions, and notes in this manual give the data that follows:
 - A WARNING is an operation or maintenance procedure or condition that, if not obeyed, can cause injury or death.
 - A CAUTION is an operation or maintenance procedure or condition that, if not obeyed, can cause damage to the equipment.
 - A NOTE gives data to make the work easier or gives directions to go to a procedure.
- (2) All personnel who operate equipment and do maintenance specified in this manual must know and obey the safety precautions. The warning and cautions that follow apply to all parts of this manual.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

WARNING: BEFORE YOU USE A MATERIAL, REFER TO THE MANUFACTURES' MATERIAL SAFETY DATA SHEETS FOR SAFETY INFORMATION. SOME MATERIALS CAN BE DANGEROUS.

CAUTION: DO NOT USE MATERIALS THAT ARE NOT EQUIVALENT TO MATERIALS SPECIFIED BY HONEYWELL. MATERIALS THAT ARE NOT EQUIVALENT CAN CAUSE DAMAGE TO THE EQUIPMENT AND CAN VOID THE WARRANTY.

CAUTION: THE UNIVERSAL FLIGHT DATA RECORDER (UFDR) CONTAINS ITEMS THAT ARE ELECTROSTATIC DISCHARGE SENSITIVE (ESDS). IN THE IPL OF THIS MANUAL, THESE ITEMS ARE IDENTIFIED AS ESDS. IF YOU DO NOT OBEY THE NECESSARY CONTROLS, A FAILURE OR UNSATISFACTORY OPERATION OF THE UNIT CAN OCCUR FROM ELECTROSTATIC DISCHARGE. USE APPROVED INDUSTRY PRECAUTIONS TO KEEP THE RISK OF DAMAGE TO A MINIMUM WHEN YOU TOUCH, REMOVE, OR INSERT PARTS OR ASSEMBLIES.

2. Content Data

A. How to Use This Manual

- (1) The instructions in this manual give the data necessary to do all recommended maintenance functions to put the UFDR in serviceable condition. Standard maintenance procedures that technicians are thought to know are not given in this manual.
- (2) Refer to the table of contents to see which subheadings are included in this manual. The table of contents identifies those subheadings that are not applicable or require no special instructions.
- (3) We recommend that the tests in TESTING AND TROUBLESHOOTING be done before the unit is disassembled. These tests can tell the condition of the UFDR or most probable cause of any malfunction. Should any malfunction occur, repair as necessary.
- (4) To decrease the length of sentences and titles, complete part numbers are not always shown in this manual. A piece of the part number can show applicability to a specific assembly or component on illustrations as well as in text and tables. Complete part numbers are always given in the parts list subheading. Refer to the introduction in the ILLUSTRATED PARTS LIST (IPL) to find how to use that subheading.
- (5) Related publications that are referred to in this manual are identified in Table Intro-1.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Table Intro-1. Related Publications

Publication	Publication No.	ATA No.
The United States Government Printing Office (GPO) Style Manual 2000		
Abbreviations for Use on Drawings and in Text*	ASME Y14.38-1999 (Formerly ASME Y1.1-1989)	
Graphic Symbols for Electrical and Electronics Diagrams*	ANSI Y32.2 (1975)	
Standard Letter Symbols for Units of Measurement*	ANSI/IEEE Standard (Std) 260 (1978)	
Graphic Symbols for Logic Functions*	ANSI/IEEE STD 91 (1984)	
NOTES: 1. You can order a Honeywell publication from Honeywell as follows: Telephone No.: 800-601-3099 (U.S.A.) or 602-365-3099 (International) Fax No.: 602-822-7272 E-mail: cas-publications-distribution@honeywell.com 2. *Available from the American National Standards Institute, New York, NY		

B. Verification

- (1) Verification of these technical instructions is done by performance or by simulation of the necessary procedures. Checks of the manual by the engineering staff make sure the instructions and description data agree with the applicable engineering specifications and drawings and are accurate and sufficient. The level of verification for this manual is shown in the list that follows.

Subheading	Level of Verification
Testing and Troubleshooting	By performance, 10 Sep 1997
Disassembly	By performance, 3 Nov 1992
Assembly	By performance, 3 Nov 1992

- (2) Honeywell will revise this manual as necessary to give current data. The sources for data supplied in this manual include engineering drawings and change orders released as of 1 Apr 2006.

31-30-32

Page INTRO-3
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

C. Symbols

- (1) The symbol in Figure Intro-1 may be used to identify static sensitive (ESDS) devices.

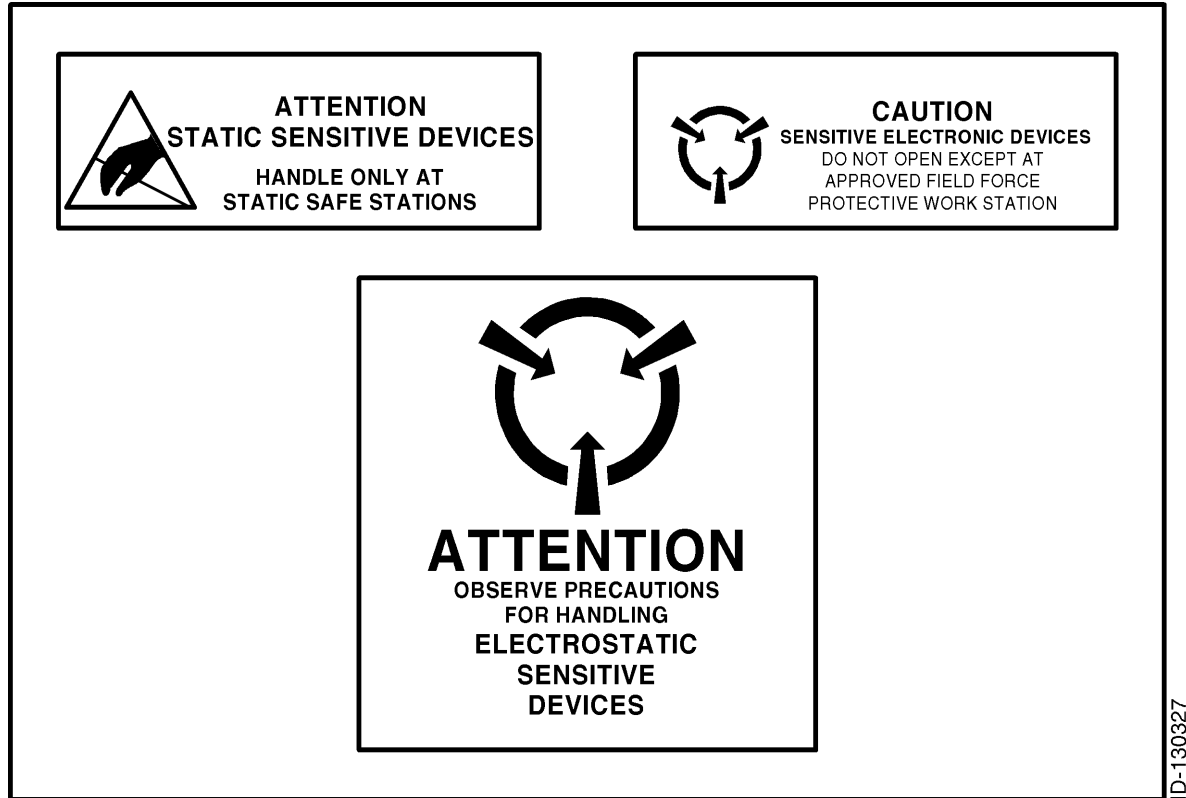


Figure Intro-1. Symbols

D. Weights and Measurements

- (1) All weights and measurements are in U.S. values.
- (2) The letter symbols for units of measurement are the same as shown in the GPO Style Manual and in ANSI/IEEE Std 260.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

E. Acronyms and Abbreviations

- (1) The acronyms and abbreviations that follow help the reader identify terms and definitions used by Honeywell.
- (2) The letter symbols for units of measurement are the same as shown in the GPO Style Manual and in ANSI/IEEE Std 260.

Term	Definition
ANSI	American National Standards Institute
ARINC	Aeronautical Radio Incorporated
ASME	American Society of Mechanical Engineers
ATA	Air Transport Association
ATE	Acceptance Test Equipment
ATLAS	Abbreviated Test Language for Avionics Systems
AWG	American Wire Gage
BCD	Binary-Coded Decimal
BITE	Built In Test Equipment
Biφ	Biphase
BOT	Beginning of Tape
CHEKIN	Check In
CHEKRD	Check Read
CHEKWR	Check Write
CHK	Check
CMM	component maintenance manual
C-MOS	Complementary Metal Oxide Semiconductor
CPU	Central Processor Unit
D-A	Digital to Analog
DAU (DAS)	Data Acquisition Unit (Data Acquisition System)
DFDR	Digital Flight Data Recorder
DMM	Digital Multimeter
DVM	Digital Voltmeter
EAROM	Erasable Read Only Memory
EOT	End of Tape
ESD	Electrostatic Sensitive Device
ESDS	electrostatic discharge sensitive
EXT	External

31-30-32

Page INTRO-5
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Term	Definition
FDAU	Flight Data Acquisition Unit
FDR	Flight Data Recorder
FLT	Flight
FWD	Forward
GPO	Government Printing Office
GPS	Ground Playback Station
Hz	Hertz
IC	Integrated Circuit
IEEE	Institute of Electrical and Electronics Engineers
INCR	Increment
IPL	Illustrated Parts List
IPORT	Import Port
IPS	Inches Per Second
IRG	Inter-Record-Gap
KHz	Kilohertz
LED	Light Emitting Diode
LSB	Least Significant Bit
MSB	Most Significant Bit
NAND	Inverted AND Gate
NOR	Inverted OR Gate
NRZ	Non-Return to Zero
OPORT	Output Port
PCI	Programmable Communications Interface
P/N or PN	Part number
P/PA	Preambles and Postambles
PWB (PCB)	Printed Wiring Board (Printed Circuit Board)
RAM	Random Access Memory
REC	Record
REV	Reverse
ROM	Read Only Memory
RSC	Recording System Controller
SN	Serial Number
STC	System Timing Controller

31-30-32

Page INTRO-6
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Term	Definition
Std	standard
TI	Transport Interface
TP	Test Point
TTL	Transistor-Transistor Logic
UAB	Underwater Acoustic Beacon
UFDR	Universal Flight Data Recorder
ULD	Underwater Locating Device
UUT	Unit Under Test
VAC	Volts Alternating Current
VCO	Voltage Controlled Oscillator
VDC	Volts Direct Current
XOR	Exclusive OR Gate

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

DESCRIPTION AND OPERATION

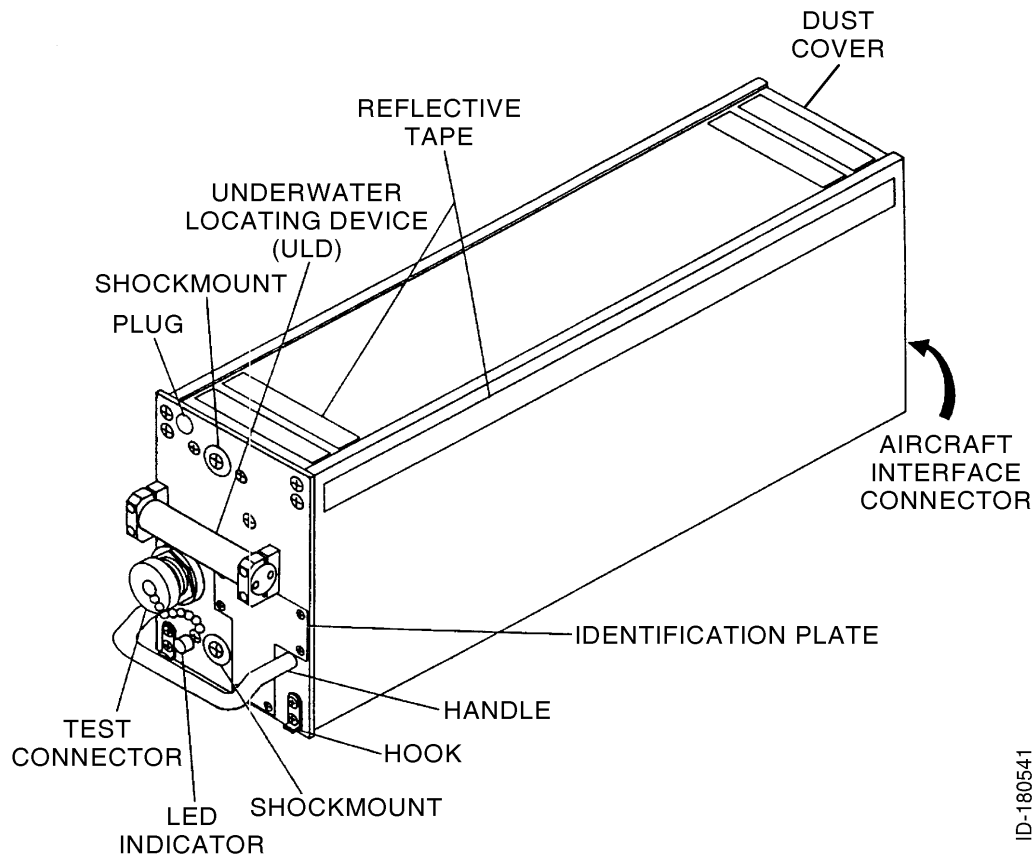
1. General

The Universal Flight Data Recorder (UFDR) is manufactured by Honeywell. It is a high performance 25 hour crash protected recorder which is designed to be interchangeable with all older ARINC 573/717 Digital Flight Data Recorders (DFDR). The basic configuration functions as a DFDR only and is intended for use with a remote Flight Data Acquisition Unit (FDAU) for all ARINC 573 and/or 717 applications.

Paragraph A provides a summary of the assemblies used in ARINC 573/717 configurations. Recorders which are factory configured with para. A(1) through A(8) function as a DFDR recorder only. The ULD (Para. A(9)) is an optional device that can be attached to any UFDR configuration as the mounting capability is available on all recorder front panels.

A. Physical Description

The recorder is packaged in a standard ARINC 404 1/2 ATR long unit. It is intended to be mounted in a standard aircraft rack with no external provisions for shock mounts. Figure 1 illustrates the packaged recorder. The unit is modular in construction and in its basic configuration, consists of the following:



ID-180541

Recorder
Figure 1

31-30-32

Page 1
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (1) Dust cover.
- (2) Front panel/Chassis assembly.
- (3) Transport/environmental enclosure with stepper drive motor.
- (4) Power Supply, A1.
- (5) Read/Write Interface, A2.
- (6) Transport Interface Assembly, A3.
- (7) Recording System Controller, A4.
- (8) Airframe Wiring Interface, A8.
- (9) Underwater Locating Device (if required).

B. Dust Cover Assembly

The dust cover assembly is a stainless steel box which encloses the entire recorder except for the aircraft interface connector and front panel. It provides protection during handling of the unit and, in case of an accident involving fire, increases thermal protection by deflecting flames from the protective enclosure. To assist in recovering the recorder in event of an accident, the cover is painted international orange with large black letters painted on the left and right sides. In addition, reflective strips are added on the left side, top, right side, bottom, and back of the cover to further aid recovery. Two different lettering nomenclatures are used. In the standard version, the lettering on both sides is in English and reads: **DIGITAL FLIGHT DATA RECORDER**.

In the two language version, the lettering on one side reads in English: **FLIGHT RECORDER - DO NOT OPEN**. On the other side, it reads in French: **REGISTREUR DE VOL NE PAS OUVRIR**. The lettering in this version is fire survival material.

C. Product Identification

All UFDR recorders are basically a DFDR recorder and differ only in accessory configuration. All units bear the same model number: 980-4100 plus a four letter accessory code. Figure 2 illustrates and explains the use of the accessory codes.

D. Leading Particulars

Figure 3 lists the UFDR leading particulars.

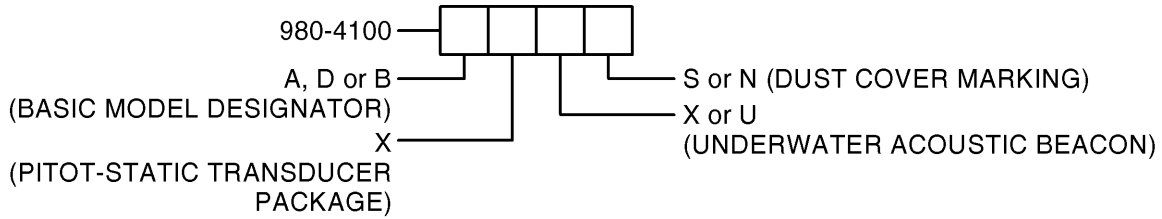
E. Tape Transport

The tape transport is fully contained within a crash proof thermal enclosure which supports the external drive motor. The transport is a co-planar, peripheral belt drive device which contains 388 feet of 1/4 inch magnetic tape, a reel drive belt, two reel hubs, a capstan/fly-wheel, two belt idlers, five tape guides, EOT/BOT sensors, two four-channel read/write heads, two four-channel erase heads, a distribution board, wiring harness, the primary drive belt and intumescent feedthroughs. The primary drive belt passes through a hole in the wall of the enclosure to connect the drive motor with the capstan/fly-wheel. A second hole in the wall is used to pass wiring for the heads and tape sensors. The two holes are lined with intumescent feedthroughs which char and plug the holes in event of fire.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



D - ARINC 573/717 configuration

B - BAC configuration (at present same as D)

A - Airbus configuration (Firmware changes to RSC)

X - No transducer package

S - Standard case

N - Non-standard case (case with two language fireproof lettering)

X - No Underwater Locating Device (ULD) fitted

U - ULD fitted at factory

Accessory Codes
Figure 2

PHYSICAL CHARACTERISTICS

Size:	1/2 ATR long.
Length:	20.092 inches maximum (minus handle).
Width:	5.031 inches maximum.
Height:	7.839 inches maximum.
Weight:	28 pounds maximum, ARINC 573/717 configuration.
Case:	Stainless steel, international orange with standard or two language nomenclature.

ELECTRICAL CHARACTERISTICS

Power:	115 VAC, 400 Hz, single phase. 35 watts, ARINC 573/717 configuration.
Connector, J1:	DPX2MA57P00P34-0001, mates with DPX2MA57S57S33-0001.
Connector, J2:	KPT07A22-55S, mates with KPSE06F22-55P.

Leading Particulars
Figure 3 (Sheet 1 of 2)

31-30-32

Page 3
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

RECORDER CHARACTERISTICS

Capacity:	25 hours at 768 bits per second.
Tape Speed:	
Write Mode:	Incremental 5 ips (0.41 ips-net).
Read Mode:	Continuous 5 ips.
Tape Type:	1/4 inch magnetic (Mylar or Kapton).
Tape Length:	388 feet
Tape Format:	Saturation level block biphase with preambles, postambles and inter-record-gaps (IRG). Data is recorded sequentially on 8 successive tracks.
IRG:	0.06 inch.
Packing Density:	2246 BPI.
Transport Type:	Co-planar reel to reel, belt drive, single motor.
Motor Type:	7.5 degree stepper (700 steps/second).
EOT/BOT sensing:	Clear tape leaders (4.00 inch) in conjunction with optical devices.
Tracks:	8-track interleaved single channel.
Heads:	2 interleaved 4-channel read/write heads and 2 interleaved 4-channel erase heads.

ENVIRONMENTAL CHARACTERISTICS

Altitude:	-1000 feet to +50,000 feet.
Temperature:	-55 to +70°C.
Humidity:	0 to 95% relative at approximately 32°C.

Leading Particulars Figure 3 (Sheet 2 of 2)

The externally mounted drive motor is a 7.5 degree stepper motor which operates in a semi-slewing mode to drive the capstan/fly-wheel. The capstan/fly-wheel in turn drives an elastomer belt which passes over the two reel hubs at their periphery and the two belt idlers. Differential belt tension and consequently, differential belt speeds are produced throughout the belt path as the belt passes these four successive load points. The result is a slightly different speed at the periphery of each of the two reel hubs. Thus, the tape drive mechanism not only produces tape motion but also tape tension. Clear tape leaders are utilized at both ends of the tape in conjunction with optical sensors to effect motor reversing and track incrementing. The two combination read/write four-channel heads and two four-channel erase heads produce eight sequential tracks of recording.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

F. Thermal Enclosure

The thermal enclosure consists of two halves of a hardened steel outer shell with Min-K fire insulation and fiber glass liner. A gasket is used between the two halves and intumescent material lines the belt and wire entry holes. The thermal enclosure is mounted to the front panel/chassis assembly by four vibration isolators.

G. Recording Concept

The tape transport is designed to move tape in either of two kinds of motor operation. These are:

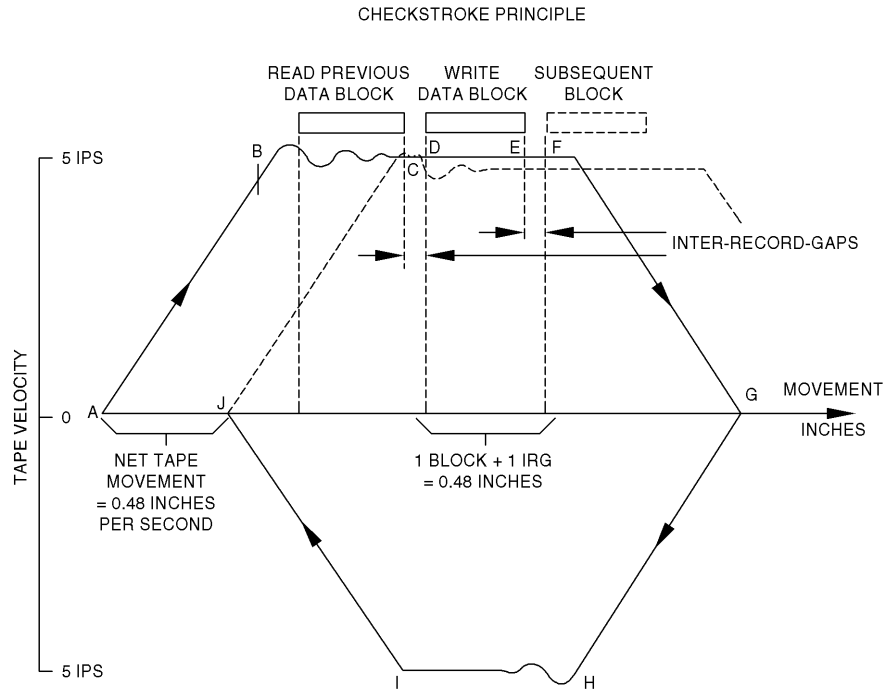
- with the motor running at a continuous speed of 5 ips and
- with the motor running incrementally at 5 ips.

Data is written on the tape at the rate of 11.4 K-bits per second; consequently, since the tape is moving at 5 ips only approximately 0.36 inches of tape is required to record a one second segment of data. In order to record 25 hours of flight data on one 388 foot length of tape, the tape is backed up after the recording of each one second segment, incremented with a 0.06 inch inter-record-gap (IRG) and recorded sequentially on successive tracks. To facilitate this type of recording technique, the tape transport drive motor must be started and stopped at known fixed intervals. In addition, timing must be provided to allow the tape transport drive motor to accelerate up to rated speed (5 ips), record the segment of data and decelerate to a stopped condition. Following the stopped condition, the motor must be energized to move the tape in the reverse direction and again timing must be provided to allow the motor to come up to speed, move the tape back and decelerate to stop at a precise position to assure 0.41 inches of net tape movement per second. Since the tape is backed up and repositioned following the recording of each segment of data, it becomes convenient to read the segment of data just recorded and thereby perform a recording system check. This advanced method of verifying the validity of recorded data and proper system CHECKSTROKE®, insert an IRG, record the next data segment and decelerate to a stopped condition. Figure 4 depicts the operation of the recording concept and the UFDR CHECKSTROKE® principal.

The sequence of events is as follows:

- (A) start the motor forward and generate the acceleration ramp;
- (B) enable the playback circuit, change to steady speed and read the previously recorded data segment;
- (C) disable playback, transfer control and enable the write circuit;
- (D) write data to tape;
- (E) finish writing data to tape and leave head on for erase;
- (F) start a deceleration ramp;
- (G) stop, disable write, check end of tape sensors, if end of tape not sensed, start motor in reverse and generate acceleration ramp;
- (H) change speed to steady state;
- (I) start reverse deceleration ramp;
- (J) stop.

If end of tape is sensed in step G, steps H, I and J are not performed. Instead, the processor causes the track select circuit to signal a track increment.



Tape Velocity Versus Displacement Graph
Figure 4

H. Modes of Operation

(1) Flight Recording

Nominally there are only two modes of operation: **Flight recording** and **non-flight recording**. In the flight recording mode, input data is obtained from an external FDAU and the recorder is permanently wired for ARINC 573/717 compatibility.

(2) Non-Flight Recording Modes

There are five non-flight recording modes; all of them are externally controlled by the application of commands to the external equipment connector mounted on the front panel. Four binary control lines are available to communicate the mode of operation to the microprocessor. Since the microprocessor is programmed to check these lines at regular intervals, the signals for a normal flight recording mode are also transmitted by the lines. When ground equipment is connected to the external connector, the hold (standby) mode must be selected first before any other non-flight recording mode can be selected. This step functions as a safety precaution to safeguard against extraneous mode changes. Also, returning to the flight record mode always takes precedence over other modes; consequently, even if flight record mode is not specifically selected, it is implemented on removal of ground equipment from the connector. Figure 5 lists the modes of operation and logic signals which must be present on the binary control lines for each mode to be implemented. A description of each non-flight recording mode is given in the following paragraphs:

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

FUNCTION SELECTED	LOGIC SIGNALS:			
	E	C	B	A
Flight Recording	X	1	1	1
Run Forward	X	1	0	1
Run Reverse	X	1	1	0
Rewind (Seek BOT)	X	1	0	0
Fast Forward (Seek EOT)	X	0	1	1
Write Tone Marker	X	0	1	0
Increment Track	0	0	0	0
Hold (Standby) (No track increment)	1	0	0	0

Modes of Operation
Figure 5

- (a) The **Run Forward Mode** moves the tape continuously at 5 ips in a playback mode. The direction is appropriate to the track in use at the time the command is given. Playback data starting from T - 25 hours may be obtained.
- (b) The **Run Reverse Mode** backs tape up in a direction opposite to that applicable to the selected track. This mode is used typically for reading small amounts of playback data, for example from: T - 5 minutes to T - 0 minutes.
- (c) In both of the **Run Forward** and **Run Reverse** modes, tracks are automatically incremented or decremented when tape ends are sensed, and tape movement continues on the next track.
- (d) The **Rewind and Fast Forward Modes** cause the tape to move at 5 ips to BOT and EOT, respectively, and stop. No track change occurs.
- (e) When the **Write Tone Marker Mode** is selected, the transport runs continuously at 5 ips in the normal direction and a 2.5 KHz data signal is written on the current position of the tape on the track in use at the time the mode is selected. The duration of the tone is the duration of the mode. It is normally used to mark the point from which any ground controlled mode is initiated and sensing the written tone marker provides the means of repositioning the tape at the end of ground operations.
- (f) The **Increment Track Mode** is implemented by the test set only. None of the other ground equipment units have controls to implement this mode. The UFDR recorder must be placed in the Hold Mode for this mode to be accepted.
- (g) The **Hold or Standby Mode** overrides all other non-flight recording modes and must be selected to allow manual control of track change software instructions. The Hold Mode is continuously active when other modes are absent. No other functions are executed in the Hold mode. The tape transport does not move and any incoming data is not processed.

31-30-32

Page 7
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

I. Fast Playback Feature

All UFDR recorders are designed so that data may be copied from the recorder by either a Ground Playback Station (GPS) or a portable copy recorder. When data is copied by the GPS, the recorder is removed from the aircraft and interfaced with the GPS. Data is then extracted one track at a time in a serial data stream. The rate is 11.4 kilobits per second and 25 hours of recorder data can be extracted in approximately 2 hours. When data is copied by the portable copy recorder on the aircraft, all 8 tracks are copied in parallel in one pass from BOT to EOT. A total of 30 minutes is required to make the copy: 15 minutes for the copy recording and 15 minutes to move the tape from its current position to BOT and subsequently from EOT to the original position. This enables the UFDR to continue recording from the same point on the tape. The copying process does not cause any erasure of the recorded data.

In both methods of fast playback, the tape is moved continuously at 5 ips and control of the tape movement is through commands applied via the external connector. Both methods also require the writing of a tone marker onto the tape. In the GPS method, the tone marker indicates when the 25 hours of data has been transcribed. In the portable copy recorder method, the tone marker indicates when the tape has been returned to its original position.

In the flight data recorder, the input of each of 8 pairs of differential playback preamplifiers is permanently wired to the winding of each read/write head. Also, the output of each pair of differential playback preamplifiers is permanently wired to the external equipment connector; thus, the output of each pair of preamplifiers is readily available at the connector. In addition, playback signals from each of the playback preamplifiers are routed through an analog multiplexer to a separate single playback channel. The output of this channel is used in the CHECKSTROKE® function and is available at the external connector. The channel may be switched to receive the playback from the amplifiers of any of the read/write head windings.

J. Failure Warning and Maintenance Status Monitoring

The design of the UFDR recorder includes features which provide for the monitoring of conditions which affect the recording system. This self-test failure monitoring system in ARINC 573/717 DFDR versions is designed to provide two kinds of status reports. The first is called a system status report and provides a failure warning indication to flight crew personnel during flight recording operations. The second is called a maintenance status report and provides equipment failure diagnostic information to ground test equipment and illuminates the BITE indicator on the front panel of the recorder. The maintenance status report is produced by a built-in-test equipment (BITE) feature. The ARINC 573/717 system status report failure warning circuit is activated by any one of the following conditions:

- Input data loss (input data RAM buffer not filled within 1.25 seconds)
- The condition of the recorder being in any operating mode other than flight recording
- Loss of 115 VAC input power (system status relay power)
- The occurrence of a BITE fault (OPORT-2, Bit 6 set HIGH, software detected fault).

When a system status report condition occurs, a logic zero signal is transmitted via OPORT-2, BIT 4 to the system status relay located on the A8 card.

A major function of the BITE feature, which is available in all UFDR configurations, is the verification of data circuit integrity. Data circuit integrity is provided by the operation of the UFDR CHECKSTROKE® feature. In this feature, three data samples are taken for

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

comparison. The first sample is called CHEKIN (checkin) and is a sample of a segment of data at the time it is received. The second data sample is called CHEKWR (checkwrite) and is a sample of the same segment of data at the time it is transmitted to the write circuits. The third data sample is called CHEKRD (checkread) and is a sample of the same segment of data at the time it is played back from the tape. Thus, when the CHECKSTROKE® feature finds that CHEKIN equals CHEKWR and CHEKWR equals CHEKRD, the integrity of the following functions is inherently verified:

- Microprocessor, RAM, or ROM and support IC's.
- Write drivers and write head current.
- Tape motion (tape not broken).
- Read/write circuits.
- Generated power supply voltages.
- Playback amplifiers.
- Data receiver/transmitter circuits.

In addition to monitoring data processing circuits for data circuit integrity, the BITE feature also monitors the following for faulty operation:

- EOT/BOT sensors.
- Track select.
- CPU (for lock-up condition).
- Data Validity (hardware malfunctions as indicated by all zeros or all ones in the RAM buffer).

If the CHECKSTROKE® features find that CHEKRD does not equal CHEKWR, a data bit fault status flag is set in the RAM buffer. When 8 consecutive read after write errors are detected, a maintenance status fault signal is transmitted via OPORT-2, bit 6 (logic state 1 signal) to an NAND circuit. Other software monitored conditions which also transmit a maintenance fault signal via OPORT-2, bit 6 are: CHEKIN does not equal CHEKWR (data process fault 1), track memory data not equal to track select data, wrong sensor ON or both sensors ON, and data invalid as indicated by RAM buffer containing all zeros or all ones (data process fault 2). The NAND circuit which receives the maintenance fault signal from the output of OPORT-2, bit 6, also receives a maintenance fault signal from the recording system lock-up detector circuit in the event it does not receive a retrigger signal within a 5 second interval. The retrigger signal is produced by the CPU when it executes a read to IPORT-2. The third signal sent to the NAND circuit is a track select fail signal produced by the track select memory circuit A3U13. The output of the NAND circuit is sent to the maintenance status relay on the A8 card to activate it when any of the monitored conditions signal a maintenance fault.

In addition to activating the maintenance status relay, certain of the monitored conditions also provide fault signals to test points. These are data process fault 1 (CHEKIN does not equal CHEKWR) to test point TP6 via OPORT-1, bit 1; data process fault 2 (data validity, RAM contains all zeros or all ones) to TP5 via OPORT-1, bit 0; and data check error (CHEKRD does not equal CHEKWR) to the external connector (J2-L), via OPORT-2, bit 5.

Honeywell

COMPONENT MAINTENANCE MANUAL

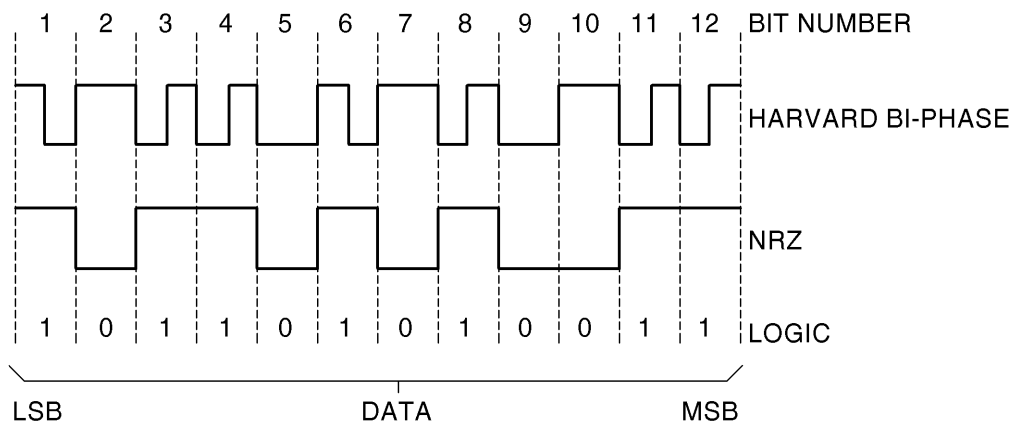
Universal Flight Data Recorder / Part No. 980-4100/980-4101

K. Input Data Forms

All DFDR UFDR recorder configurations are designed for the receipt of input data which meet the format requirements of ARINC 573. Specifically, DFDR UFDR recorder configurations provide for the reception of data which originate from a remote Flight Data Acquisition Unit (FDAU). Output signals from this type of FDAU have the following characteristics:

- Input word structure is Harvard Bi-Phase coding.
- Data rate is 64 12 bit words per second (768 bits per second).
- Words are formatted into four frames of 64 words each.
- The first word of each subframe is a sync word.

The illustration in Figure 6 shows a comparison of Harvard Bi-Phase coding and digital NRZ coding. Harvard Bi-Phase coding carries information in the form of phase transitions. Whether a phase transition occurs within each bit cell determines the character of the cell. Also, the Harvard Bi-Phase code provides self-clocking information because the start of each cell gives a phase transition.



Harvard Bi-Phase and NRZ Codes
Figure 6

ID-180543

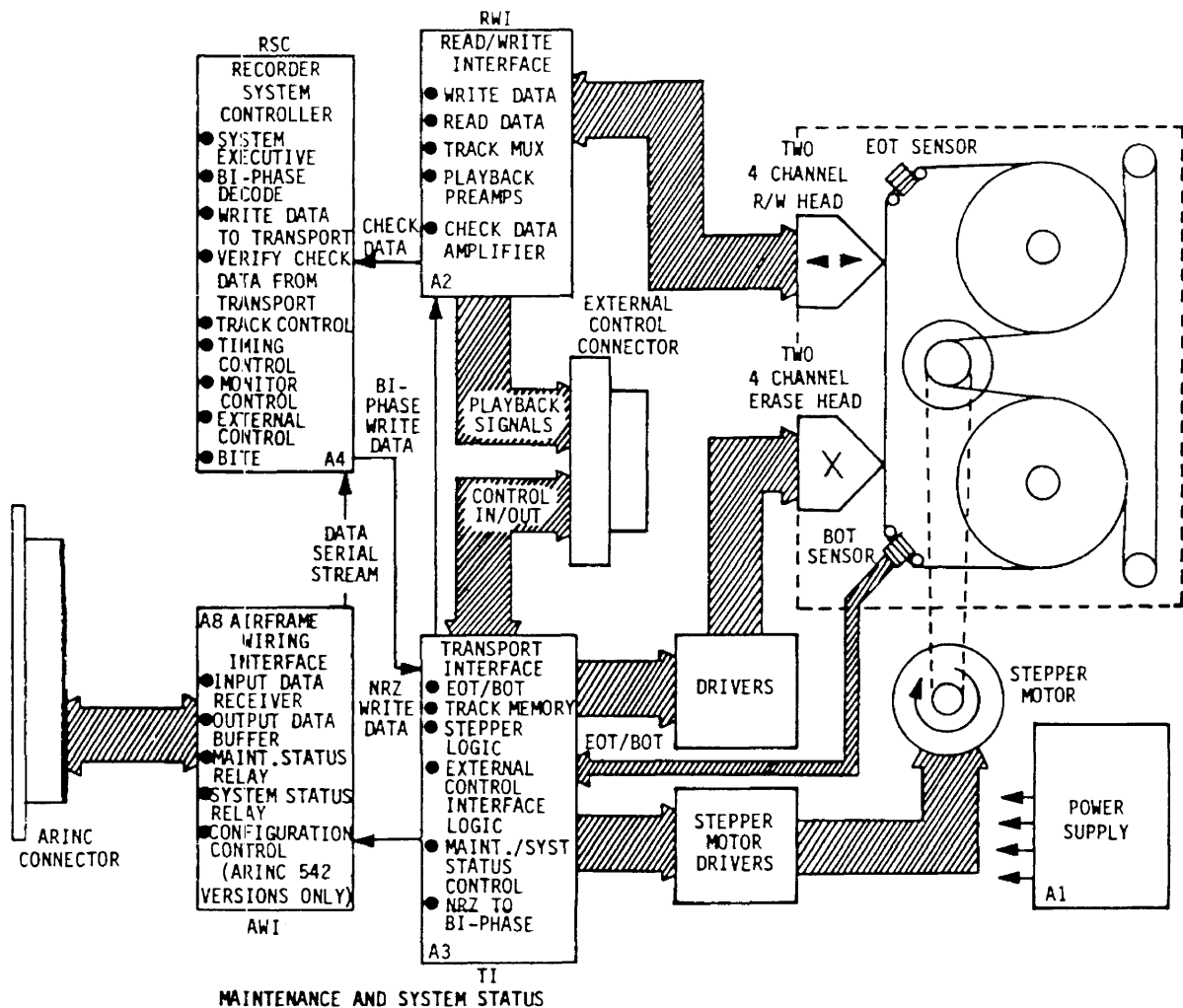
2. Block Diagram Analysis of Recorder Operation

Figure 7 is a general block diagram for all UFDR configurations. The large arrows between blocks indicate that more than one wire is used for the movement of signals. The input data are always applied through the block marked Airframe Wiring Interface (AWI) which corresponds to assembly A8. The card contains an input data receiver, an output data buffer, a maintenance status relay and a system status relay. Output data is transformer coupled Harvard Bi-Phase in a serial data stream. The TTL bi-phase data signal is fed to the next block, the Recording System Controller (RSC). Other signals also applied to the Airframe Wiring Interface are the BITE signal from the transport interface and the system status signal, also from the circuits of the transport interface block. UFDR Recorder connectors and their functions are listed in Figure 8.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



UFDR Functional Block Diagram
Figure 7

The block marked Recording System Controller (RSC) represents the A4 assembly. It contains the Central Processor Unit (CPU), a clock generator, a System Timing Controller (STC), a Programmable Communications Interface (PCI), a bi-phase demodulator circuit, address buffers, address decoders, RAM buffers, a Read Only Memory (ROM) which contains the program of instructions and other microprocessor signal handling circuits. All system functions are executed by the program of instructions (software) stored in the ROM. When bi-phase data is received by the RSC, it is demodulated and applied to the PCI serial data input. From the PCI, the data is processed under the direction of the CPU to RAM buffer storage. Data from the RAM buffers are fed once per second via the PCI serial output section to the write circuits of the Read/Write Interface via the Transport Interface. In addition, the RSC provides stepper motor timing control signals, track select control signals, data CHECKSTROKE® functions, BITE monitoring functions and external equipment control signals.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CONNECTOR	FUNCTION
J1A and J1B	Provides interconnection between UFDR recorder and external signal and power sources. This aircraft interface connector is located at rear of recorder.
J2	External ground equipment connector, located on front panel.
J3	Erase head/sensor connector, located on motherboard.
J4	Motor connector, located on motherboard.
J5	Head connector, located on chassis which connects transport wiring to motherboard wiring.
XA1	Receptacle for Power Supply, A1.
XA2	Receptacle for Read/Write Interface, A2.
XA3	Receptacle for Transport Interface, A3.
XA4	Receptacle for Recording System Controller, A4.
XA8	Receptacle for Airframe Wiring, A8

UFDR Recorder Connectors
Figure 8

The block marked Transport Interface (TI) represents the A3 assembly. It contains the input and output ports which enable the CPU to communicate instructions to be executed by the implementing circuits and to receive reports on operating conditions from monitor circuits. Instructions to be implemented which are transmitted by the circuits of this block are the stepper motor logic signals, the track select signals, BITE monitor signals, system status signals, and external equipment control signals. Also transmitted by the circuits of the TI block are the data signals after they have been reconverted to bi-phase for writing to tape. Monitor circuit signals which are transmitted via the circuits of this block to the CPU are the EOT/BOT tape condition signals, memory of the track select signals, track increment signal and signals from the external connector which indicate the ground mode of operation for the externally connected ground equipment or the selection of the flight record mode when the ground equipment is disconnected. The TI also transmits motor command signals to the four windings of the stepper motor via the block marked stepper motor drivers. These signals cause the tape drive motor to step forward or backward depending on the sequence of the actuation of the windings.

The block marked Read/Write Interface represents the A2 assembly. It contains 8 pairs of differential playback preamplifiers, 2 analog multiplexers, 8 pairs of differential write drivers and a separate single playback channel which converts the bi-phase signals to TTL signal level. These circuits perform a two fold function: (1) They process data received from the RSC via the Transport Interface into write current signals and (2) They handle playback signals for copy recording and read after write verification for the BITE function. When serial bi-phase data is received by the Read/Write Interface, it is gated into one of eight pairs of write driver circuits by the track select signals and written to tape by the selected read/write head. The two four-channel R/W heads are enclosed by a dotted block which includes the tape transport mechanism.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

3. Theory of Operation

The following paragraphs contain a description of each of the modules (PWB assemblies) which are used in the DFDR UFDR Recorder. The descriptions are preceded by a general description of recorder operation.

A. General

Data to be recorded are obtained from the output of a FDAU. Data inputted to the recording system is continuous bi-phase data in accordance with the ARINC 573 format which must be converted to a TTL level digital format and demodulated. To convert the serial bi-phase data to TTL logic level data (coded 2-state 1 and 0) format, a bi-phase converter circuit located on the Recording System Controller card, A4, is used. The converted TTL logic level data is then passed in 8-bit bytes to one of two RAM buffers where it is stored until the RAM buffer is full. A full RAM buffer contains 96 bytes or 768 bits of data. When the RAM buffer is full, an 8-bit preamble and an 8-bit postamble is added to identify the beginning and the end of each 768 bit data block. For recording purposes, each data block represents a nominal one second recording segment. When a recording segment is in one RAM buffer, continuing input data is stored in the other RAM buffer while the recorder simultaneously processes the first segment to tape via the read/write electronics.

In addition to providing normal flight data recording capabilities, the UFDR design also incorporates a CHECKSTROKE® feature which provides verification of the integrity of the recorded data as part of a built-in test equipment (BITE) system. The BITE system also provides fault indications for tape motion, EOT/BOT, track select, input data loss and system lock-up. The recorder is designed to sense the principal modes of operation. These are: (1) The flight recording mode, (2) Non-flight recording modes and (3) A standby or hold mode. The flight recording mode is the only valid in-flight mode. The non-flight recording modes are used for ground functions such as maintenance and copy recording. Non-flight recording modes are accessed when the ground equipment is connected to the external equipment connector, J2. The hold mode is a transitional mode which overrides all other externally operated modes. It must be selected before the non-flight recording modes can be selected. Sensing of the mode of operation is software directed and occurs after the software has implemented the initialization process. To determine the mode of operation, the software directs the CPU to read the data at input port 2 (IPORT-2). Input data bits D0, D1 and D2 at IPORT-2 are set to a logic state HIGH by the action of pull-up resistors connected to the input lines if no external equipment is connected to the external equipment connector, J2.

Initialization is the process of making the various circuits of the recorder ready to perform their function. It occurs on power-up and whenever a condition warranting a reset condition is detected. To start the initialization process, an RC network consisting of A4R1/C2 senses +5 VDC when a logic LOW reset signal from one of three sources is applied to the junction of A4R1/C2 and the RESIN terminal of A4U2, the clock generator. These sources are the +5 VDC under-voltage monitor, A4U17; the line voltage under-voltage monitor, A1U3; and the lock-up detector circuit, A3U19A/B. These three signal sources are open collector wire OR'ed to the junction cited above; thus, when a logic LOW is applied to the junction, a slow rise signal is presented to an internal circuit of A4U2. A4U2 then applies a logic HIGH, reset signal to the reset terminal of the CPU, A4U1, and the PCI, A4U5. In addition, the logic HIGH reset signal is sent via A4U12D, a hex inverter, to the output ports: OPORT-0, A3U2; OPORT-1, A3U3; and OPORT-2, A3U14. Also, a logic LOW reset signal is applied to the stepper motor memory

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

latch circuit, A3U10. This logic LOW forces the four outputs of A3U10 to an all logic LOW condition. When the CPU is reset, it suspends all normal internal operations and clears itself for a beginning operation. It then fetches from the ROM, A4U8, software provided initialization instructions. These instructions direct the CPU to initialize the RAM, I/O ports, STC and PCI. On completion of the initialization process, mode selection is made by the CPU reading IPORT-2 and the recorder commences recording activity.

B. Recording System Controller, A4 (see Figure 128)

(1) General

This card functions as a microprocessor control center to control the recording system. It contains the 8-bit microprocessor (CPU), U1; a clock generator, U2; a bus driver, U3; a system timing controller (STC), U4; a programmable communications interface (PCI), U5; a RAM, U6 and U7; a ROM, U8; an address buffer, U9 and U10; an address decoder, U11; an interrupt select gate, U14; a bi-phase converter circuit, U13 and U15B; an under-voltage monitor, U17; and other circuit components such as NAND gates and HEX inverters.

(2) CPU

The CPU, U1, is the heart of the system; its principal functions are to provide system timing, to furnish address(es) of the various memory devices and to read and write 8-bit bytes of data from and onto the data bus lines. Time signals are provided to the CPU by the clock generator, U2, which provides phase 1, phase 2 and a READY signal. The phase 1 and phase 2 signals function as internal CPU timing signals while the READY signal functions to permit the CPU to enter a wait state and allow external circuits to respond to requests from the CPU. The READY input signal synchronizes the CPU with the external circuits. Other input signals to the CPU are the INTERRUPT signal from the interrupt select gate, U14, and the RESET signal from the clock generator, U2. Output signals from the CPU are addresses, data (data are also input to the CPU on the same lines), and a SYNC signal to the clock generator, U2. The SYNC signal indicates the start of each CPU cycle and the presence of CPU status information on the data bus.

The CPU address lines are 16 bits wide and are divided into two principal fields: an upper field of 5 bits and a lower field of 11 bits. In the upper field, the three top bits; A15, A14 and A13 contain the addresses of 8 single-address devices. The table in Figure 9 contains the addresses and the devices addressed. The other two bits of this field function as enables for the stage, U11, which decodes the addresses of this field. The lower field contains the multiple addresses used by the ROM (10 or 11 bits) and the RAM (8 bits). One bit of this field (A0) is used in conjunction with a decoded address bit from the address decoder, U11, of the upper field to form a two bit 2 address format for the PCI, U5, and the STC, U4.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

ADDRESS BITS			DEVICE ADDRESSED
A15	A14	A13	
0	0	0	ROM (address field 0000H through 07FFH)
0	0	1	Not used
0	1	0	-DS-0 (IPORT-0/OPORT-0)
0	1	1	-DS-1 (IPORT-1/OPORT-1)
1	0	0	RAM (address field 4000H through 40FFH)
1	0	1	-DS-2 (IPORT-2)
1	1	0	PCI (addresses 6000H and 6001HL)
1	1	1	STC (addresses 7000H and 7001H)

Upper Fields Address Decode
Figure 9

(3) Address Driver

The 16-bit address field of the CPU is sent to an address driver stage consisting of U9 and U10. The output of U9 and U10 is placed on 16 lines of the address bus. In addition to driving the address bus, U9 and U10 also function as a buffer. Connected to appropriate address bus lines are devices to be addressed by multiple addresses and the upper field address decoder, U11.

(4) Clock Generator

The clock generator, U2, consists of a crystal oscillator and a divide-by-nine circuit. The crystal frequency of 18.8744 MHz is converted to the system operating frequency of 2.097 MHz by the divide-by-nine circuit. In addition to providing the phase 1, phase 2, and READY signals to the CPU, U2 also provides a TTL version phase 2 clock signal to the STC, U4, and the PCI, U5. Also, the clock generator contains the RESTART circuit which was described in paragraph 3A. One other signal also produced by the clock generator is the status strobe (-STSTB). At the beginning of each CPU machine cycle, the CPU issues status information on its data bus. This information pertains to the kind of action that is to take place during the machine cycle. The clock generator produces the -STSTB signal by AND'ing the SYNC signal from the CPU with the phase 1 signal during time T2 of the CPU machine cycle. Thus, the -STSTB signal occurs at the start of each machine cycle at the earliest time that status data is stable on the data bus. The -STSTB signal is sent directly to the data bus driver, U3. The -STSTB signal ensures proper system timing during the execution of all of the CPU machine cycles.

(5) Bus Driver

U3 is a bidirectional 8-bit data bus driver with three output states: logic HIGH, logic LOW and high impedance. It is designed to buffer the CPU from memory and input/output devices and drive the data bus lines. In addition to the -STSTB signal, other control signals which are applied to U3 are the data bus input strobe (DBIN), the hold input from

31-30-32

Page 15
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

the CPU (HLDA) and the write input strobe (-WR). All input control signals are active HIGH and all output control signals are active LOW. The principal function of U3 is to time the read and write operation of the data bus. It transfers data into and out of the RAM, ROM, PCI, and STC on this card. Through the card edge connectors, data is moved to and from the input/output ports of the Transport Interface card, A3.

(6) RAM

Temporary storage of data received into the system through the PCI or of data produced by the system is provided by the RAM. The RAM consists of two static CMOS 256 x 4 RAM chips, U6 and U7, connected to form a 256 x 8 memory array. Before data can be read out of or written into the RAM, appropriate control signals must be applied to the RAM. These are the chip select signal (-CS1), the chip enable signal (-STR) and the write enable signal (-WR). When these three signals are applied to the RAM in a logic LOW state, data may be written into the RAM. When the -CS1 and the -STR are in the logic LOW state and the -WR is in a logic HIGH state, data may be read from the RAM. The RAM is powered from a supply labeled VCMOS. This supply is located on the Transport Interface card, A3, and is designed to ensure that the RAM is powered for at least 200 milliseconds after a system power loss (See paragraph 3D(7)).

(7) ROM

U8 contains the program memory which controls the operation of the recording system. It is a 16,384 bit bi-polar programmable read only memory (ROM). The program is located with 2,048 x 8 bit addressable bytes. To access the memory requires a logic level LOW on the CE3 and CS1 terminals. The CE3 terminal is activated LOW by the address decoder, U11, and the CS1 terminal is activated LOW when the bus driver, U3, transmits a MEMR signal.

(8) Bi-Phase Converter

The bi-phase converter circuit consists of U13A, B, D, U15B and counter number 1 of the STC. Counter number 1 is programmed by an instruction stored in the ROM and applied by the CPU to function as a one-shot with a time constant which is 3/4 of a bit cell period. Level detection of the bi-phase input signals is accomplished by U13A and U13B. Output signals from U13B function to synchronize the triggering of the STC one-shot with the clocking of the positive edge triggered D flip-flop, U15B. As bi-phase transition signals are applied from U13B to the clock terminal of U15B and the gate terminal of the one-shot, converted TTL logic level signals are produced at the Q terminal of U15B. These signals are applied to the Receive Data (RxD) terminal of the PCI, U5. Also, a clock rate signal from U13D is applied to the Receive Clock Input (-RxC) terminal of the PCI to control the serial data input transfer rate.

(9) Programmable Communications Interface

U5, the Programmable Communications Interface (PCI), is a universal synchronous asynchronous transmitter-receiver. In this application, it is designed to receive asynchronous serial data inputs, format them into 8-bit bytes of data and parallel transmit them via the data bus lines to the CPU. During a write operation, the CPU parallel transmits 8-bit bytes of data to the PCI. The PCI, on receipt of the data, converts it to serial format and transmits it synchronously to the write electronics. The frequency of the

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

data transmitted is controlled by the -TXCLK signal received from A3U5A. This signal, in turn, is produced when the divide-by-two flip-flop circuits: A3U4B and A3U5A scale the 4XTXCLK signal produced by the STC, U4 (see paragraph 3D(8)). The receipt and transfer of data can occur simultaneously in the PCI. The PCI contains a receiver data buffer, a receiver section, a transmit data buffer, a transmitter section, a command register, a status register, a control logic section and a mode control section. Data control, operation and format options are all controlled in the PCI by commands received from the CPU.

Control signals which must be applied to the PCI are as follows: A two bit address field received from U11 and U10; A memory read (MEMR) or a memory write (MEMW) signal from U3 and a clock signal from U3. Other control signals which must be present at appropriate times for the PCI to function properly are: The data terminal ready (-DTR) signal; The receiver ready (-RxRDY) signal; The transmitter ready (TxRDY) signal; The clear-to-send (-CTS) signal; The transmitter clock (-TxC) signal; and the transmitter empty (-TxE) signal. The -DTR, -RxRDY and TxRDY signals are PCI generated signals which are sent to the PCI via U14 during PCI operations. The -CTS and -TxC signals are received by the PCI during data transmission to the write electronics. The TxE signal is sent by the PCI when the PCI transmitter section has transmitted its data.

(10) System Timing Controller

U4, the System Timing Controller (STC) is a large scale integrated circuit which contains five sections of 16-bit general purpose counters. The mode of operation and function of all five sections is completely programmable by data applied to the STC via the data bus lines. In this application, the function of each counter is programmed by instructions transmitted to the STC by the CPU. The function of each counter is as follows: Counter number 1 is programmed to function as a non-retriggerable delayed one-shot, counting to a terminal count (TC). When it reaches TC, it reloads itself and repeats the cycle. The output from this counter facilitates the conversion of bi-phase data to TTL logic level data by producing a pulse that has a time duration of 3/4 of a data bit cell period. Counter number 2 is programmed to function as a transition counter. It counts the number of transitions in the data stream during a CHEKWR and CHEKRD operation (see paragraphs 1H and 1K) and thus facilitates operation of the CHECKSTROKE\R feature. Counter number 3 is programmed to function as a stepper motor timing counter that determines the interval of motor stepping operations. Counter number 4 is programmed to function as a frequency scaler. It produces the input clock signal for the data reformat circuitry in the Transport Interface card, A3, at four times the data write frequency (4 x 11.398 KHz). During external equipment operation, the 4XTXCLK frequency is 10 KHz (4 x 2500 Hz).

(11) Under-Voltage Monitor

U17 is an under-voltage monitor (supply voltage level detector) which is designed to cause a system RESTART signal to be generated whenever the Vcc voltage goes out of range. When the Vcc goes below 4.5 VDC, the device output which is an open collector TTL compatible signal, pulls the RESTART line to a logic LOW. The RESTART line remains LOW until the Vcc rises above 4.7 VDC. At that point the output of the device goes HIGH. This allows the output of the reset circuit of U2 to go HIGH and clear the reset condition throughout the system.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

(12) Interrupt Circuit Gate

U14 is an AND/OR logic inverting gate. It consists of three logic AND gate sections OR'ed together and inverted. Each of the three AND sections is used to provide an interrupt signal to the CPU. These interrupt signals are the stepper motor terminal count (C3OUT) interrupt, the receive ready (RxRDY) interrupt which indicates to the CPU that 8-bits of serial input data are ready for transfer to the CPU and the transmitter ready (TxRDY) interrupt. This interrupt indicates to the CPU that the data received from the CPU has been transferred to the write electronics. When the CPU acknowledges an interrupt signal from U14, it uses the polling technique to sequentially accept interrupt signals. The sequence of acceptance in this equipment is as follows: (1) C3OUT; (2) TxRDY; and (3) -RxRDY. If all of the input signals to U14 are logic state LOW, the output signal to U2B is a logic state HIGH and no interrupt is signaled to the CPU. Interrupt signals for TxRDY and RxRDY are transmitted to U14 by the PCI, U5. Motor interrupt signals (C3OUT) are transmitted by the STC, U4. When the PCI transmits a HIGH for a TxRDY interrupt, it must also transmit a data terminal ready (-DTR) signal which is inverted to a HIGH by U2C before U14 can AND the two signals to produce the interrupt signal. When the STC, U4, transmits a HIGH for a motor interrupt (C3OUT) signal, the input circuit of U14 requires that the signal be AND'ed with a signal (-C3OUT) that confirms that the motor is enabled. This signal is inverted to a HIGH by U2A. When the PCI, U5, transmits a HIGH for RxRDY, it is AND'ed by a section of U14 that has its other terminals permanently held HIGH by +5 VDC applied through R18.

C. Overview of Recording System Activity

After the recording system has been initialized and the mode of operation selected (for this discussion the mode of operation is assumed to be flight recording), data to be recorded is processed by the recorder. This data is received through a bi-phase line receiver located on the airframe wiring interface card, A8, (which is discussed in paragraph 3E(1)) and sent to the bi-phase converter circuit on the A4 card. Output signals from the bi-phase converter circuit are applied to the RxD terminal of the PCI. In the PCI, two registers are used to interface with the CPU. These are the PCI data register, PCIDAT, and the PCI control register, PCICTL. Through the PCIDAT the CPU directs the byte timing rate (interrupt timing), the routing of received data to RAM buffer A or B and the writing of data from RAM buffer A or B to tape via the read/write card, A2. Through the PCICTL, the CPU controls the operation of the PCI by loading control words into it. Thus, when the PCI receives an 8-bit byte of data, it generates an interrupt signal to the CPU by placing a logic HIGH on its receiver ready (-RxRDY) output terminal. This logic HIGH is sent to U14 where it is AND'ed with two other signals which are held at a logic HIGH permanently. Thus, an interrupt request is presented to the CPU by U14 which indicates that there is an 8-bit byte of data in the PCI receiver ready to be sent to a RAM buffer. When the CPU acknowledges the interrupt, it causes the PCI to parallel transfer the byte of data to one of the RAM buffers selected by the CPU. This procedure continues until the CPU counter indicates that the buffer is filled with 96 bytes of data or time out has occurred indicating a loss of data. U14 also receives interrupt signals from two other sources. These are the transmitter ready (TxRDY) signal from the PCI and the motor terminal count signal (-C3OUT) produced by the STC when the tape is in motion.

The TxRDY interrupt signal is used during a write cycle routine. When 8 bits of data have been serially shifted out of the PCI to the write electronics, the PCI sets the TxRDY line to U14 to a

Honeywell

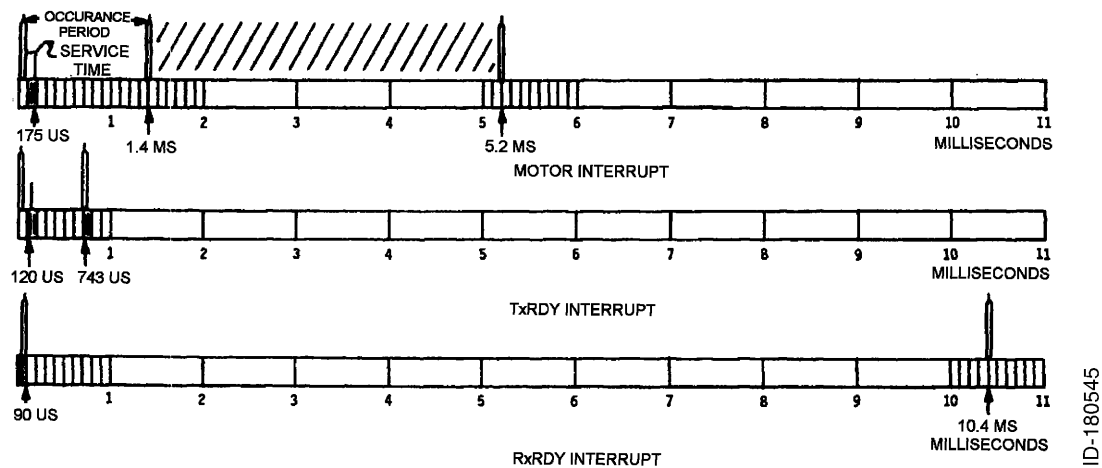
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

logic HIGH. At the input to U14, this logic HIGH is AND'ed with the data terminal ready (-DTR) signal which previously had been transmitted to the interrupt gate by the PCI on initialization. When the CPU acknowledges the interrupt, it causes the RAM buffer to parallel transfer another 8-bit byte of data to the PCI transmitter buffer register. This procedure continues until all of the data in a RAM buffer has been transferred. As the data is being transferred, the CPU counts the bytes, including the preamble and the postamble, and as the last byte is being processed, causes the PCI to disable the TxRDY line and sets the TxEMPTY ENABLE bit at OPORT-0.

In addition to writing data to tape with a preamble and a postamble the write operation routine is designed to verify the data block previously written (read after write) on the tape. It calls another subroutine to increment the track selection codes, if necessary. It pre-erases three inches of tape if the selected end of tape sensor is on. It verifies the recorded data by generating and comparing a data checksum for a BITE maintenance status. It controls tape motion, generates IRG timing and lastly, calls for the wait-for-buffer-to-fill (RDWAIT) subroutine.

Figure 10 is a timing diagram which shows the interval of the three interrupts. When the CPU scans U14 for interrupts, it uses the polling technique and polls the interrupts in the following order: (1) MOTOR, (2) TxRDY and (3) RxRDY. Figure 10 shows the service time for each interrupt to be 175 microseconds, 120 microseconds and 90 microseconds, respectively. Therefore, the total service time required for interrupts is 385 microseconds. Since the most frequent interrupt (TxRDY) occurs every 743 microseconds, another interrupt does not occur until 388 microseconds after all three interrupts have been serviced.



Interrupt Timing Diagram
Figure 10

To implement the write instruction routine, the CPU first transmits an instruction word to the STC and initializes its own stack pointers for motor phase and duration instructions. In the STC, the CPU instruction initializes the operation of counter 3 and causes the STC to transmit a signal to U14 and A3U10. The signal at U14 acts as the MOTOR interrupt signal and the

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

signal at A3U10 acts as a clock signal for moving the motor phase signals to the motor drive circuits. The CPU, on acknowledging the MOTOR interrupt signal, passes the motor phase signals on the data bus to OPORT-1 from which they are moved via A3U10 to the motor drive circuits. The CPU then waits for the counter to reach a count of 32 which indicates that the motor is up to speed. When the motor is up to speed, the CPU sets flags in the motor status (MOTSTA) register, transmits a read enable bit (bit-0, logic state 1) to OPORT-0 and an execute word to the STC. The data bit at OPORT-0 enables the playback of data previously written on the tape and the word sent to the STC enables STC counter 2. STC counter 2 then generates a 16-bit word for the CHECKSUM operation. In the meantime, the CPU waits for a motor step count of 88 to occur. When a motor step count of 88 does occur, the CPU disarms STC counter 2, stores the 16-bit CHEKRD word in the RAM register and transmits a read disable bit (bit 0, logic state 1) to the OPORT-0. This completes the read-after-write operation. On completion of this operation, the CPU then performs three error checking operations. The first error check test performed is a comparison of CHEKIN with CHEKWR. If the error check finds that CHEKIN does not equal CHEKWR (data process fault 1), bit 6 of OPORT-2 is set to a logic state 1. In addition, a fault indication is signaled by OPORT-1, bit 1, to test point (TP6). The second error check performed is a comparison of CHEKRD and CHEKWR checksums. If the two do not compare, a read-after-write error is indicated and a logic state 1 is shifted in the RAM register, CHEKER. If the two do compare, a logic state 0 is shifted in this register. In addition, the signal generated is transmitted to the external connector, J2-L, via OPORT-2, bit 5. The third error check test performed is a test of the read-after-write error weighting register, CHEKER. If the serially accumulated bits indicate 8 consecutive CHEKER errors, a maintenance status fault signal is transmitted via OPORT-2, bit 6 (logic state 1).

When a motor step count of 92 occurs, the CPU sets the write data byte counter in the RAM buffer to zero, enables counter 2 in the STC for a checksum accumulation, transmits the preamble byte to a PCI-DAT register, transmits a data terminal ready and an enable signal to the PCICTL register, transmits a write enable bit (bit 1, logic state 1) to OPORT-0 and stores the OPORT-0 image in the RAM register. The PCI then transmits the TxRDY interrupt signal to the CPU. The CPU, on acknowledging the TxRDY interrupt signal, executes the transfer of data from the RAM buffer through the PCI to the write electronics. It also causes the write-in-progress flag, -DS-0, to be set to logic state 0. From motor step 92 through motor step 140, data is written to tape at a rate of 11.4 kilobits per second. When motor step 140 is reached, the CPU disarms STC counter 2, stores the 16-bit CHEKWR word in the RAM register, transmits a write disable bit (bit 1, logic state 1) to OPORT-0, disarms STC counter 3 (the motor step counter), causes the PCI to transmit TxEMPTY and calls for a motor STOP subroutine. When the CPU fetches the motor STOP subroutine, it transmits to OPORT-1 a data word which sets the motor phase output bits (BIT 4, 5, 6 and 7) for a motor stop condition. It then waits for the motor to step count from 144 to 176. At a count of 176, the CPU disarms STC counters 2 and 3, sets bit 2 of OPORT-1 to a logic state 1 (-C3OUT disabled) and enters a do nothing loop for 20 milliseconds to wait for motor inertia to subside. At the end of the wait period, the motor drive hold signal is transmitted via OPORT-1, bit 1 (set to logic state 1). Following the motor stop subroutine, the CPU must not fetch and execute a motor backup subroutine (CSTROK) for the next read-after-write operation and the subsequent writing of the next segment of input data to tape. To do this, the CPU reads the address of the current track and the BOT/EOT sensors. If the selected sensor is not indicating a track change, the CPU tests the LSB of the current track address to determine motor direction. On determining motor direction, the CPU sends a step control word to the STC and

a motor phase control word to OPORT-1 to implement the motor start routine. The CPU then waits for the motor step counter of the STC to reach a count of 88. When a count of 88 is reached, the motor has repositioned the tape. The CPU then calls for the stop subroutine to stop the tape drive motor. When the step subroutine is completed, the system is ready to repeat the write instruction routine.

D. Transport Interface Assembly, A3 (See Figure 127)

(1) General

The principal circuits of this card are the input/output ports: IPORT-0, U1; OPORT-0, U2; OPORT-1, U3; OPORT-2, U14; IPORT-2, U15; an erasable read only memory (EAROM), U13; a motor phase memory latch, U10; a lock-up detect circuit, U19A and B; and a circuit designed to reformat the TTL level data back into the bi-phase data format; U4A/B, U5A/B, U6A/B, U7A and U12B/C. In addition, several other kinds of basic logic elements are used to perform routine functions.

(2) Input/Output Ports

The input/output ports enable the CPU to communicate with implementing circuits and to receive data on operating conditions. IPORT-0 receives and transmits to the CPU memory data relating to the track selected and the status of the EOT and BOT sensors. IPORT-2 receives and transmits to the CPU data relating to the mode of operation selected, i.e., flight recording mode or non-flight recording modes. Of the 8 bits in the logic field of OPORT-0, 3 bits are used to transmit track select data from the CPU to the track select memory, U13. Two other bits are used also in relation to track select operations. Of these two bits, one is used to signal track select failure and the other is used to initiate the track select write cycle. The remaining three bits of the OPORT-0 logic field signal are -Read Enable, -Write Enable and -TxEMPTY Enable.

Of the 8 bits in the logic field of OPORT-1, 6 bits transmit data relating to stepper motor operation and 2 bits transmit signals to two test points relating to data processing faults. Of the 6 bit field, bits 4, 5, 6, and 7 transmit motor drive phase signals, bit 3 transmits the motor hold signal and bit 2 transmits the -C3OUT, interrupt signal. Bit 0 transmits to TP5 the status of data process fault 2. Bit 1 transmits to TP6 the status of data process fault 1 (see paragraph 1K for additional information).

Of the 8 bits in the logic field of OPORT-2, one bit is not used. Of the 7 remaining bits, 3 bits (bits 0, 1 and 2) are used by the CPU to transmit control data signals to external equipment via the external equipment connector, J2. Figure 11 gives the output codes used and their meaning. Of the three remaining bits, bit 4 transmits the systems status fault signal to the system status relay on the A8 card. Bit 5 transmits the data check error signal (CHEKRD does not equal CHEKWR) to the external equipment connector, J2-L. Bit 6 transmits the maintenance status fault signal to the maintenance status relay and the system status relay on the A8 card. Bit 7 transmits the ERASE ENABLE signal to the erase head decode-driver (located on the tape recorder distribution board assembly).

In all of the input and output ports, data applied to the D input terminals is latched to the Q output terminals on the positive transition (logic LOW to logic HIGH) of the signal applied to the clock terminal. In IPORT-0, data is latched to the Q output terminals for reading by the CPU when the CPU directs the address decoder, A4U11, to transmit the -DS-0 signal and the bus driver, A4U3, to transmit the -MPU RD signal. When these

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

signals are applied to the A3 card, they are NOR'ed together by U7C which applies a positive transition signal to the clock terminal of IPORT-0. When the clock signal transitions negative (HIGH to LOW), U9A, a hex inverter, applies a positive transition to the output control enable terminal of IPORT-0. This HIGH, which does not affect the internal functioning of the port, places the Q output terminals in a high impedance state. IPORT-2 is clocked in a similar manner. A4U11 transmits the -DS-2 signal and A4U3 the -MPU RD signal. These signals are NOR'ed by U16A which applies the positive transitioning signal to the clock terminal of IPORT-2, U15. When the clock signal transitions negative, the transition is inverted to a positive transition by hex inverter, U11F, and is applied to the output control enable terminal of IPORT-2 and to U19A of the lock-up detect circuit.

ECO STATUS CODE			UFDR STATUS INDICATED
C	B	A	
0	0	0	FLIGHT RECORDING MODE
0	0	1	BOT SENSOR ACTIVATED
0	1	0	EOT SENSOR ACTIVATED
0	1	1	FAST FWD MODE (SEEKING EOT)
1	0	0	REWIND MODE (SEEKING BOT)
1	0	1	RUN FORWARD MODE
1	1	0	RUN REVERSE MODE
1	1	1	STANDBY (HOLD)

NOTE: POSTIVE LOGIC

External Equipment Control Data Signals
Figure 11

The output ports are clocked also in a similar manner except that the signals NOR'ed are the -MPU WR signal with one of the following: -DS-0, -DS-1 or -DS-2 for OPORT-0, OPORT-1 and OPORT-2, respectively. These signals must each be a logic state HIGH for their respective -NOR circuit to produce the required positive transition clock signal to clock their respective output port. The -NOR circuit for each output port is as follows: OPORT-0, U7B and U8B; OPORT-1, U7D and U8A; and OPORT-2, U16A and U11F. In addition, the output ports employ another feature which is different from the input ports. The three output ports are enabled or disabled by the -MPU RESET signal which is applied to the control enable terminal of each simultaneously through a set/reset flip-flop latch. The set/reset latch consists of U8C and U8D. When a -MPU RESET signal is transmitted to the A3 card, it is inverted to a logic HIGH by hex inverter, U9B, and applied to the set terminal, pin 8 of U8C. This causes a logic LOW to appear at pin 10 of U8C and a logic HIGH at pin 13 of U8D. The HIGH disables the output ports by placing the output Q terminals of each in a high impedance state. The LOW from pin 10 of U8C is

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

applied to the reset terminal of the motor phase memory latch, U10, and causes the output of U10 to be forced to an all LOW logic state condition. Subsequently, when the CPU reads IPORT-0, by directing the appropriate signals to clock IPORT-0, the clock signal is applied also to the reset terminal, pin 12, U8D, of the set/reset latch. Thus, the latch is reset following the removal of the -MPU RESET condition.

(3) Motor Phase Memory Latch

U10, the motor phase memory latch, is a quad edge triggered D flip-flop with common clock and reset inputs. Data on the D input terminals are loaded into the latch on the LOW to HIGH transition of the clock pulse. All Q output terminals are forced to a logic state LOW independent of the clock and data terminals when the reset terminal (MR) is at a logic LOW. In this application, output signals are taken from the latch at the Q (true) terminals and sent to the stepper motor drive circuit on the power supply card, A1.

(4) Lock-Up Detect Circuit

U19A and U19B are monostable multivibrators (one-shot) circuits. The characteristic of a one-shot circuit is to have only one stable state (monostable). The circuit is designed to stay in this stable state until it is triggered. When it is triggered, it switches to the other quasistable state where it stays for a fixed period of time as determined by the RC time constant of externally connected components. After the fixed period of time, it returns to the monostable state. In this application, U19A is designed to have its quasistable state extended by the application of a trigger pulse before the quasistable state is terminated. Thus, as long as the trigger pulses occur at intervals of time less than the time established by the RC components, the one-shot (U19A) remains in the quasistable state. If, however, a trigger pulse does not occur within the proper time frame, U19A switches states. When it does, the output of its Q terminal goes to a logic LOW and signals a -LOCK-UP FAILURE to an NAND circuit (U12A). At the same time, the $\bar{Q}$ terminal of U19A triggers U19B. When U19B is triggered, it produces at its $\bar{Q}$ terminal a negative pulse (logic LOW) of 1 millisecond in time duration which functions as the system RESTART signal. The trigger pulses for U19A are the CPU directed enable signals applied to IPORT-2, U15. If the CPU does not read IPORT-2 at least every 5 seconds, U19A signals a -LOCK-UP FAILURE.

(5) Track Select

U13, the erasable read only memory (EAROM) is a non-volatile 4-bit data latch designed to retain the data latched into it even in the absence of power. The data applied to its data input terminals comes from OPORT-0 and consists of three bits which identify the track selected and a fourth bit which indicates whether a read-after-write test of track select codes compares with originally applied data codes. Data is written to the EAROM when the CPU causes OPORT-0, bit 4 to apply a logic state LOW to the SAVE terminal of U13. The read-after-write test is performed when the CPU reads the track select codes at the output of U13 via IPORT-0. If the read-after-write test indicates a fault condition, the fourth bit (bit 3, track select failure) transmits a logic state LOW to U13. This signal causes the CPU to rewrite the track select data codes in an attempt to clear the fault condition. U13 requires 30 milliseconds to complete a write cycle. If rewrite of the track select data codes does not clear the fault condition, U13 transmits a logic state LOW to an OR circuit, which, in turn, transmits a fault signal to the maintenance status relay of the A8 card.

31-30-32

Page 23
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

When A3U2 is replaced by a part other than part number 321-0213-006, the circuit may produce spurious save signals when a power-down condition occurs. When these spurious save signals are produced by A3U2, A3U13 may respond to these erroneous signals and, consequently, on subsequent power-up, signal an erroneous track change.

Therefore, to prevent this from occurring, Transport Interface Assembly, Part number 722-2441-004 and subsequent, are configured with additional circuit components which are used to prevent A3U13 from responding to spurious save signals on power-down.

These circuit components are Q1, Q2, CR4 and CR5 and associated circuit resistors. These components function in the following manner: Under normal conditions, Q1 is conducting and Q2 is cut off. When a power-down condition occurs, the -RESTART signal is applied to the base terminal of Q1, causing it to go to cut-off. When Q1 goes to cut-off, Q2 goes to conduction and applies ground to the Vss terminal of U13 and the save terminal of U13 through CR5. U13, being thus disabled, is unable to respond to spurious signals which U2 may transmit at this time.

(6) Maintenance Status NAND Circuit

The maintenance status NAND circuit consists of U6D, U12A and U11F. In addition to the track select fail signal, a signal from U19A of the lock-up detect circuit and a signal from OPORT-2, bit 6, for data process faults 1 and 2 are applied to the OR circuit. If all three inputs to U12A are logic state HIGH, U11F transmits a logic state HIGH through buffer, U20B, to the maintenance status relay circuit of the A8 card. If any of the three inputs goes to a logic state LOW, U11F transmits through the buffer, U20B, a logic state LOW. This LOW indicates a fault condition.

(7) RAM VCMOS Circuit

The circuit consisting of CR1, CR2, CR3, R11 and C3 functions to ensure that at least 3 VDC is supplied to the RAM, A4U6/U7, for at least 200 milliseconds after primary power is lost. The VCMOS supply is established by the charge on capacitor, C3.

(8) NRZ to Bi-Phase Reformat Circuit

Processed data is transmitted to this card by the PCI, A4U5, of the Recording System Controller card in the NRZ (TTL level) data format. Before this data can be written to tape by the write electronics of the Read/Write card, A2, it must be reconstructed into the bi-phase format. The circuitry on this card which performs this function consists principally of the following circuit components: U4A, a D edge triggered flip-flop which functions as the control element for the circuitry; U4B and U5A, both edge triggered flip-flops, each functioning as a divide-by-two flip-flop; U5B, another edge triggered flip-flop which functions as the bi-phase data output circuit; U6B, the clock bit gate; U12B, the data bit gate; and U12C, the bi-phase bit gate. Input signals required to operate the circuitry are:

- TXDATA
- 4XTXCLK
- TX EMPTY
- -WR ENABLE
- TX EMPTY ENABLE.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Output signals from the circuitry are:

- Bi-phase data
- TXCLK
- Clear to Send (-CTS)
- -WR ENABLE.

Before reading the operation of the circuitry, study Figure 6. Notice that the bit cells for NRZ (TTL level) coded signals are a series of logic state HIGHs and LOWs: whereas, in Harvard bi-phase coded signals, the character of a bit cell is determined by the presence or absence of a phase transition within each bit cell. The function of this circuitry is to put this transition into the appropriate input data bit cell, and thus convert them to Harvard bi-phase format.

Operation of the circuitry is initiated when the CPU applies a -WR ENABLE signal through OPORT-0, bit 1, and hex inverter, U9E, to the set (SD) terminal of flip-flop, U4A. The logic HIGH causes U4A to produce a logic HIGH at its Q output terminal. This logic HIGH is applied to NAND gate, U6A, where it is AND'ed with the same logic HIGH from U9E that was applied to the SD terminal of U4A. The resulting logic LOW output from U6A is sent from the card through pin A17 of the edge connector as the -CTS signal and inverted to a logic HIGH by hex inverter, U9F. The logic HIGH from U9F is applied to the clear (CD) terminals of the divide-by-two flip-flops; U4B and U5A and the bi-phase output flip-flop, U5B. Thus cleared, the circuitry is ready to reformat NRZ data. Incoming input NRZ data is applied to one of the three input terminals of the data bit gate, U12B. Simultaneously, the 4XTXCLK signal from the STC, A4U4, on the Recording System Controller card, is applied to the clock terminal of the first divide-by-two flip-flop, U4B, and to a -NOR gate, U7A. Applied to the other input terminal of U7A is the -2XTXCLK signal from the -Q terminal of U4B.

Referring now to the timing diagram shown in Figure 12, it can be seen that when the 4XTXCLK signal and the complement of the 2XTXCLK (-2XTXCLK) signal are a logic state LOW at the input of U7A, the output produces a signal called the transition bit. This bit is sent to two places:

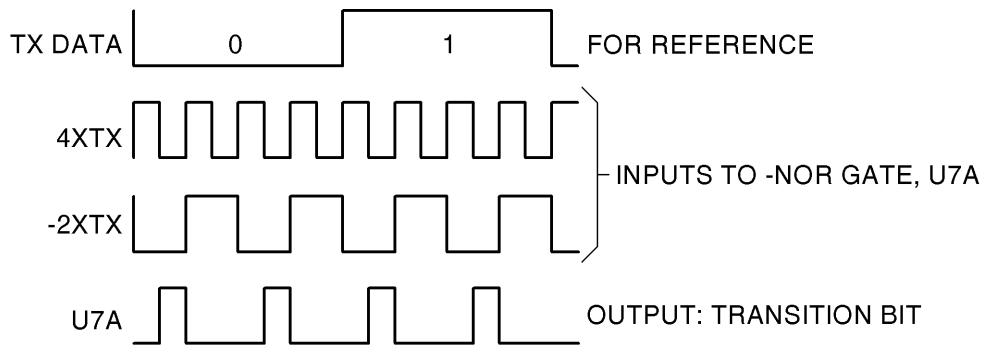
- U12B, the data bit gate
- U6B, the clock bit gate.

At the clock bit gate, U6B, the transition bit is NOR'ed with the TXCLK signal from the Q terminal of U5A, the second divide-by-two flip-flop, and produces at the output of U6B, the clock bit (see Figure 13). At the data bit gate, U12B, the transition bit is NOR'ed with the complement of the TXCLK signal (-TXCLK) and the NRZ data bit signals. As can be seen by examining the timing diagram in Figure 14, the data bit gate, U12B, produces a data bit whenever the three input signals are in a logic state HIGH. At U12C, the output from the clock bit gate, U6B, and the data bit gate, U12B, is AND'ed to produce a signal called the bi-phase clock bit (see Figure 15). This signal is used to clock the output of the bi-phase output flip-flop, U5B, which produces the reformatted output signals with the bi-phase transitions occurring in the proper data bit cells.

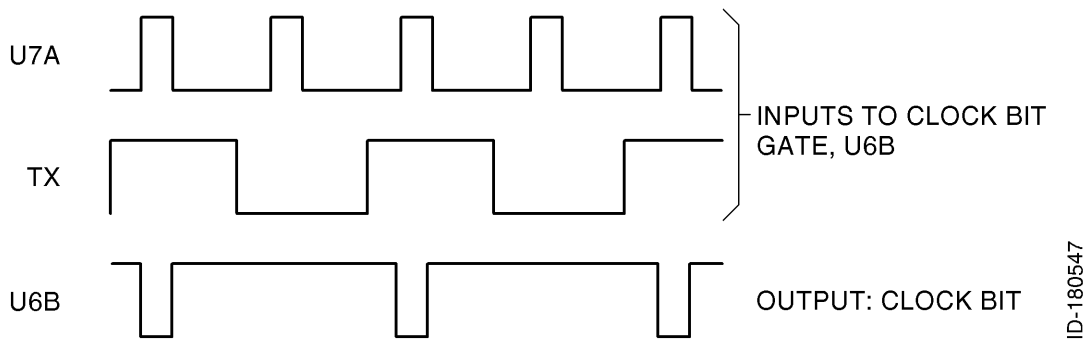
Honeywell

COMPONENT MAINTENANCE MANUAL

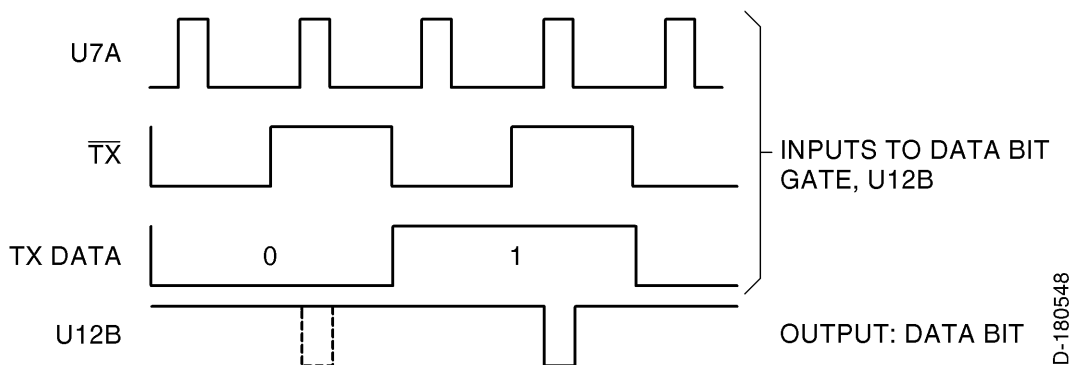
Universal Flight Data Recorder / Part No. 980-4100/980-4101



Timing Diagram for Transition Bit Generation
Figure 12



Timing Diagram for Clock Bit Generation
Figure 13



Timing Diagram for Data Bit Generation
Figure 14

31-30-32

Page 26
22 Dec 2006

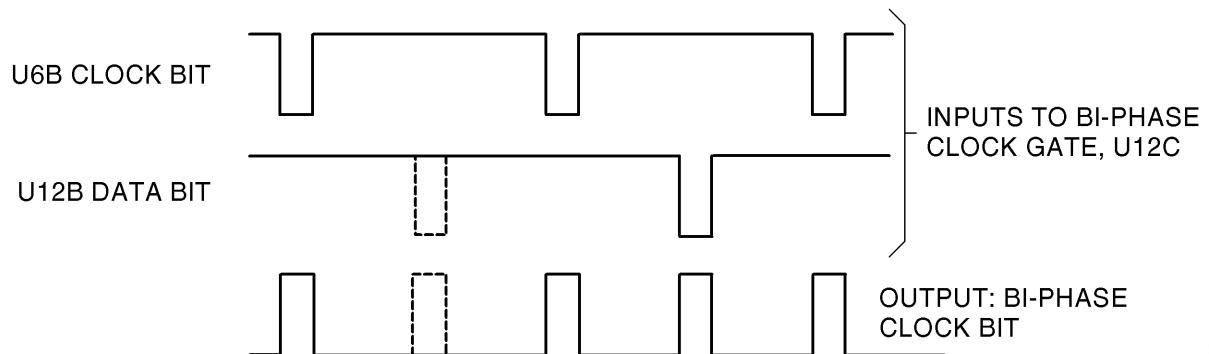
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180549

Timing Diagram for Bi-Phase Bit Generation
Figure 15

When the PCI, A4U5, completes the transmission of the last byte of NRZ data to this circuitry, it activates the -TX EMPTY signal which is applied to the clock terminal of U4A. At the beginning of the transmission of the last byte, the CPU under the direction of the software determines that all of the data including the postamble has been transferred. It then causes the PCI to disable the TxRDY interrupt and sets the TX EMPTY ENABLE bit at OPORT-0 to a logic state LOW. This LOW is applied to the data terminal of U4A and is clocked to the output Q terminal of the flip-flop by the -TX EMPTY signal. From the output Q terminal, the LOW is applied to the clear (CD) terminal of flip-flops U4B, U5A and U5B and functions to shut off the reformat circuitry until the next 96 bytes of NRZ data are ready to be written to tape.

E. Airframe Wiring Interface Assembly, A8 (see Figure 129)

Data signals from a remotely located FDAU are inputted to the recorder via this card.

(1) Input/Output Circuit

Signals received from the FDAU are applied to a multiplexer, U1, through T1, Q1 and Q2. The signals are bi-polar, bi-phase data signals which are converted to TTL logic level bi-phase signals through T1, Q1 and Q2. T1 transformer couples the bi-polar signal to its secondary which is center tapped to ground; thus, when each end of the secondary swings positive with respect to ground, Q1 or Q2 conducts to saturation. When each end swings negative with respect to ground, Q1 or Q2 cuts off. Since each transistor is connected to +5 VDC through a pull-up resistor, R7 and R6, the output of each is a TTL logic level signal. These signals are applied to U1. U1 is a data selector multiplexer circuit which is designed to pass the FDAU converted TTL signal.

At the multiplexer, signal selection is controlled by the application of a signal to the select terminal (pin 1). Since the recorder is operating as a DFDR (ARINC 573/717), a logic HIGH (+5 VDC) is permanently applied to the select terminal through R12. If the recorder is operating as an FDR (ARINC 542), the operation of configuration detection latching relay, K101, applies a ground (logic LOW) to the select terminal of U1. This pulls the terminal LOW by dropping the +5 VDC across R12 and signal data from the DAS is selected. Signal data from U1 is applied next to the R/S latch consisting of U5A and U5B.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

This latch functions to improve the shape of the bi-phase data signals and facilitates reconverting the bi-phase signals to bi-polar format.

TTL level bi-phase signals from the latch are sent through pin A28 of P1, the connector, XA8, to the Receiving System Controller, A4. Signals from the latch are applied also to one of two terminals on each of two two-input terminal positive NAND drivers; U2A and U2B. One the other input terminal of U2A and U2B is an enable/disable signal received from U5D. U5D functions to disable U2A and U2B when a fault condition is detected by the BITE maintenance status system. When the BITE maintenance status system detects a fault condition, the Transport Interface card, A3, transmits to this card a logic level LOW signal. This signal is received through pin A9 of connector XA8, and applied to a hex inverter, U4D. U4D inverts the logic level LOW to a HIGH and applies it to pin 12 of U5D. When U5D receives a logic level HIGH, it produces a logic level LOW at its output terminal providing that its other input terminal is at a logic level HIGH, also. This terminal is held permanently HIGH by +5 VDC applied to it through R12. The LOW from U5D disables U2A and U2B by causing the output of both to go to a logic HIGH state.

When U2A and U2B are enabled, complementary bi-phase data signals are produced at the output terminals, pins 5 and 3. Since pins 5 and 3 are connected to opposite ends of the primary of T2 and the primary is center tapped to +5 VDC through R11, a bi-polar bi-phase data signal is developed across the primary as polarity of the signals from U2A and U2B change. These bi-polar bi-phase data signals are then coupled to the secondary of T2 which transmits signals via pins B28 and B29 of connector XA8 to the ground equipment connector, J2, on the front panel of the recorder.

(2) Fault Indicator Circuits

When U4D produces a logic level HIGH, indicating the BITE maintenance system has detected a fault condition, a HIGH is sent to U5D, U3A, U4B, U4C and U4E. U3A is a current driver which activates K1, the DFDR maintenance status relay, when it produces a logic level LOW at its output terminal. U3A produces a logic level LOW at its output terminal when both of its input terminals are HIGH. Since one of its output terminals is permanently connected to +5 VDC through R9, it is always at a logic HIGH. Therefore, when U4A transmits a logic HIGH to U3A, a logic LOW is applied to K1, causing it to activate. When K1 is activated, Airplane Indicator Power, which may be +28 VDC in some aircraft, is removed from the cockpit fault indicator because a set of normally closed contacts of K1 open when K1 is activated.

U4B and U4C are open collector drivers connected in parallel to drive the BITE indicator on the front panel of the recorder. The BITE indicator is a LED with its cathode connected through pin B25 of connector XA8 and R13 to the open collector drivers. The anode of the LED is connected to +12 VDC. When the logic output of the drivers is LOW, the BITE indicator is illuminated. Thus, the +12 VDC on pin B25 when the BITE indicator is OFF, is pulled down to approximately +10 VDC when the indicator is ON.

U4E is an open collector driver also and functions to indicate a BITE fault condition to ground equipment (the UFDR Functional Tester, for example) which may be connected to J2.

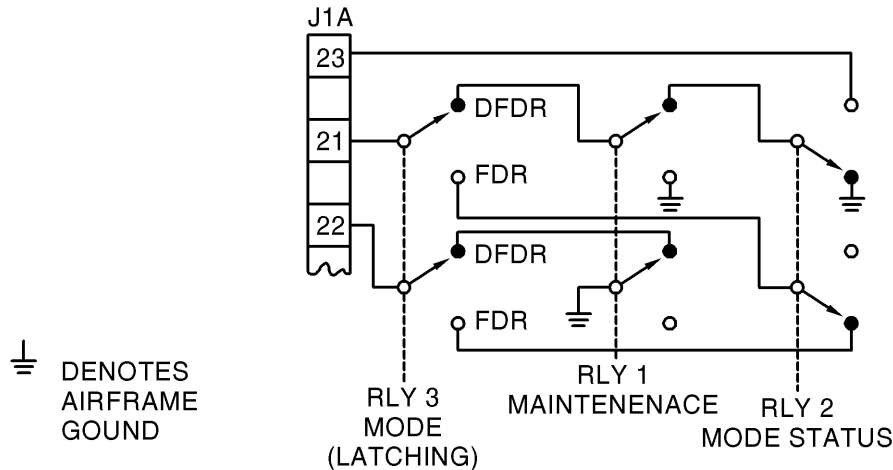
Figure 16 is a schematic diagram of the interconnections of K1 with K2. K2 functions as the System Status relay. Indicator Power received via J1A-23 is routed through contacts

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

of K1 and K2 and applied via J1A-21 to the cockpit fault indicator. A BITE Maintenance Status fault condition removes indicator power from the cockpit fault indicator circuit by activating K1 while a System Status fault condition removes indicator power by deactivating K2. K2 is normal activated.



LATCHED MODE	CONDITION	PIN 21	PIN 22	RELAY STATES
DFDR	FLIGHT RECORDING	+27.5 V	GND	RLY 2 ENERGIZED
DFDR	SYSTEM STATUS FAULT	GND	GND	NEITHER ENERGIZED
DFDR	MAINT. STATUS (BITE) FAULT	GND	O.C.	RLY 1 ENERGIZED
DFDR	NO POWER APPLIED	GND	GND	NEITHER ENERGIZED
FDR	FLIGHT RECORDING	OPEN	OPEN	RLY 2 ENERGIZED
FDR	SYSTEM STATUS FAULT	SHORTED	SHORTED	NEITHER ENERGIZED
FDR	NO POWER APPLIED	SHORTED	SHORTED	NEITHER ENERGIZED

ID-180550

Relay Interconnection Diagram
Figure 16

(3) System Status Fault Condition

When a System Status fault condition is detected (see paragraph 1K(1) through (4)), a logic level LOW signal is transmitted to this card by OPORT-2, bit 4, from the Transport Interface card, A3. This logic LOW is applied to U5C which is a two input terminal NAND gate connected to function as a -OR gate. Thus, when a logic level LOW is applied to either input terminal, a logic level HIGH is produced at the output terminal. This HIGH is converted to a LOW by U4F and applied to U3B. U3B is the current driver used to drive relay K2. Normally, U3B produces a logic level LOW when the system status conditions are satisfactory. This logic level LOW normally activates K2. However, when a system status fault condition is detected, U4F applies a LOW to U3B which sends a logic HIGH to K2 to deactivate it. Figure 17 contains a table which summarizes BITE Maintenance Status and System Status fault signal circuit configurations.

31-30-32

Page 29
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

RELAY STATES	CONDITIONS	INDICATION AT	
		J1A-21	J1A-22
K1 not activated K2 activated	Normal flight recording	Indicator Power	Ground
K1 not activated K2 not activated	System Status fault (no power applied)	Ground	Ground
K1 activated K2 activated	BITE Maintenance Status Fault	Ground	Open Circuit
K1 activated K2 not activated	BITE Maintenance Status and System Status Fault	Ground	Open Circuit

Summary of Relay Conditions
Figure 17

F. Read/Write Interface Assembly, A2 (see Figure 126)

This card contains the electronics which perform the functions of writing data to or reading (playback) data from the tape. It contains 2 write drivers: U1 and U2 with a bank of write current setting resistors: U3 and U4, a write current reference voltage power supply and switch: U13, Q1 and Q2, circuit elements for controlling the selection of read or write: U9A and B, U11A and B and U12A, B, C and F; 8 pair of differential playback preamplifiers: AR1A and AR4D; 2 multiplexers: U7 and U8, with a bank of isolation resistors: U15 and U16, a bank of read mode load resistors: U5 and U6, a bank of read mode isolation diodes: CR1 through CR16, single track amplifier decoder circuitry: AR5A through AR5D and U10, and the data playback output circuitry: U9C and U9D.

(1) Write Mode

When the recorder is in the write mode of operation, data is written to tape on any one of eight tracks (channels) as selected by the write circuit electronics in response to track select signals transmitted by the Transport Interface card, A3, to this card, A2, under the direction of the CPU. Figure 18 is a simplified diagram of the read and write circuitry and is typical of all 8 channels. At this card, the write data-to-tape operation begins when bi-phase data and a -WRITE ENABLE signal are received. The LOW write enable signal is inverted to a HIGH by a hex inverter, U12B, and applied to three places:

- An input terminal of each of 2 two-input terminal NAND gates, U9A and U9B
- Q1 of the write current reference voltage power supply.

Bi-phase input data is applied directly to the other input terminal of U9A and through a HEX inverter, U12A, to the other NAND gate, U9B. Thus, the bi-phase data applied to these two NAND gates is always out of phase with each other. Consequently, during a write operation, the output signal at U9A is always the complement of the signal at U9B. These logic level output signals from U9A and U9B (the bi-phase data) are applied to the enable input terminals of the write drivers (U1 and U2) the outputs of which are selected by the application of track select signals. The three track select signal bits A, B and C select the track or channel for recording in accordance with the address decodes given in Figure 19.

31-30-32

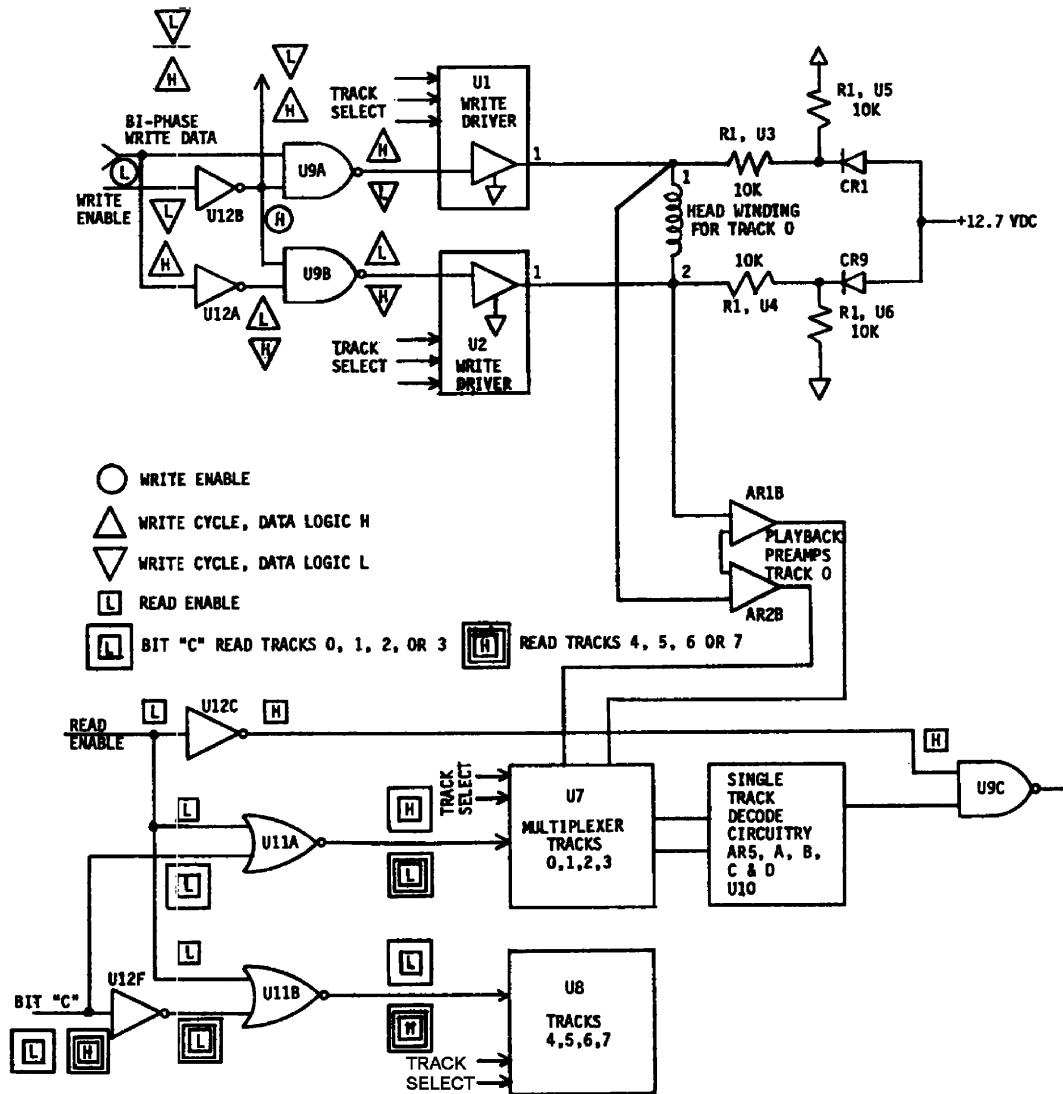
Page 30
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Read/Write Circuit Simplified Diagram
Figure 18

ID-180551

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

TRACK SELECT BITS			TRACK NO.
A	B	C	
0	0	0	0
1	0	0	1
0	1	0	2
1	1	0	3
0	0	1	4
1	0	1	5
0	1	1	6
1	1	1	7

Logic Signals for Track Select
Figure 19

(2) Read (Playback) Mode

During a tape read operation, playback data is simultaneously available from the eight pair of playback preamplifiers and the single track playback amplifier-decoder circuit. Each of the eight pair of differential playback preamplifiers is permanently wired across each read/write head winding. Consequently, to play back all eight tracks simultaneously in parallel requires only that the write mode be disabled and that the tape be moved across the heads in either direction. Parallel output signals from the playback preamplifiers are applied through the isolation resistors of U15 and U16 to pins in the front panel ground equipment connector labeled PB-A and PB-B (0 through 7). The resistors of U15 and U16 function to protect the multiplexers, U7 and U8, and the eight pair of preamplifiers, AR1A through AR4D.

Serial playback of a selected individual track is provided by single track playback circuitry. This playback circuitry is designed to convert the playback preamplifier signal into TTL compatible bi-phase signals. The single track playback circuitry consists of four circuits:

- A control enable logic circuit
- An analog multiplexer (which is configured to function as a dual 8 to 1 data selector)
- The single track amplifier/decoder circuit
- Data playback output circuitry.

(a) Control Enable Logic

The control enable logic circuit consists of U11A, U11B and U12F. U11A and U11B are NOR gates which function as inverted logic NAND gates. Each must produce a HIGH to enable the multiplexer to which each is connected: U11A to U7 and U11B to U8. U11A produces a HIGH when the read enable signal and track select bit C are both LOW. Track select bit C is LOW for tracks 0 through 3 and HIGH for tracks 4 through 7. When track select bit C goes HIGH, it is inverted to a LOW by U12F. This HIGH enables multiplexer, U8.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

(b) Analog Multiplexer

The analog multiplexers, U7 and U8, are dual 4 to 1 CMOS analog switches. When U7 is enabled by a HIGH from U11A, track select signal bits A and B, which are applied to pins 1 and 16, select output signals from among the playback preamplifiers for tracks 0, 1, 2 or 3. When U8 is enabled by a high from U11B, track signal bits A and B select output signals from among the playback preamplifiers for tracks 4, 5, 6 and 7. Thus, the two multiplexers together function as a dual 1 of 8 data selector and apply the selected output signals to PB-A and PB-B and the next stage: the single track amplifier decode playback circuitry.

(c) Single Track Amplifier-Decode Playback Circuitry

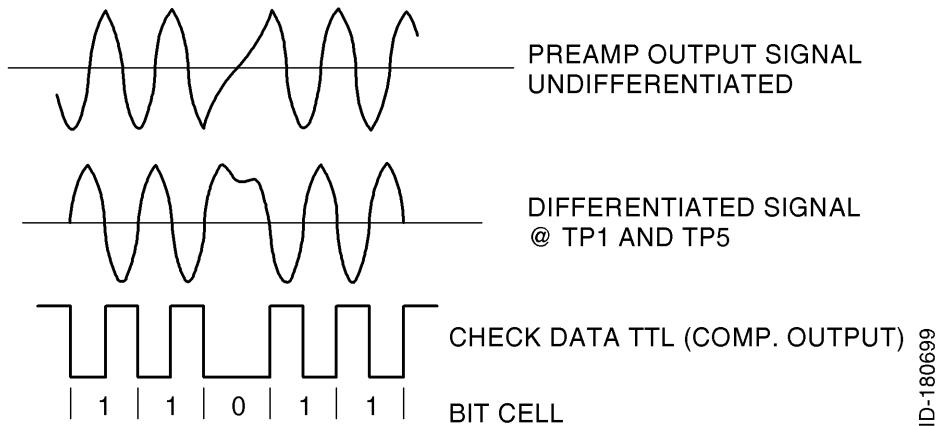
During a tape read operation, bi-polar signals are induced into each head winding as tape is moved across the heads. A playback signal voltage of approximately 2 millivolts peak-to-peak is developed across head load resistors for each head winding. The resistors are located in U4, U6, U3 and U5 and provide 20 kilohms to ground on both sides of each head winding. Isolation diodes, CR1 through CR16, prevent the playback signals from being coupled from one track to another through the write current reference voltage power supply which has its output voltage set to 0.0 VDC during a tape read operation. The bi-polar signals from each head winding are applied through a pair of differential playback preamplifiers, for example, AR1B and AR2B or other pairs, as appropriate, and amplified by a factor of approximately 40. The bandwidth of each pair of preamplifiers is designed to provide phase compensation between 5.7 kilohertz (data 0) and 11.4 kilohertz (data 1) to minimize effects of **bit shift**.

From the selected playback preamplifier, bi-polar signals are applied to single track amplifier-decoder circuitry. In this circuitry, the bi-polar signals are differentiated, amplified and converted to TTL logic level compatible bi-phase data signals. The single track playback amplifier-decoder circuitry consists of AR5A, AR5B, AR5C, AR5D and U10. AR5A and AR5B are designed to function together as an amplifier and signal shaper whose purpose is to transform peaks of the preamplifier signals into corresponding zero crossing signals. To do this, AR5A and AR5B are designed to function as a differential differentiator. See Figure 20 for typical voltage waveforms. Cutoff frequency of the differential differentiator circuit is set to approximately 48 kilohertz by values of C25 and R34. The AC gain of the stage is also set by these values as well as the values of R35 and R36. Upper cutoff frequency of the amplifier is determined by values of C26 and C27. Converting bi-polar peak signals to zero crossing signals facilitates the transition to TTL logic level signals. AR5C functions as a high common mode rejection analog subtractor unity gain amplifier. Its output signal is a single ended signal which is proportional to the differential amplitudes of the out-of-phase signals produced by AR5A and AR5B. AR5C, then, is a differential to single ended converter. AR5D is a wide band non-inverting amplifier with a gain of approximately 4.5. Its output is AC coupled to the next stage, the zero crossing comparator which is U10 and its associated circuitry.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Typical Input/Output Waveforms
Figure 20

U10 is designed to convert zero crossings of the differentiated playback signal into TTL compatible logic level bi-phase data signals and to provide the single track channel with a high degree of noise immunity. Improved noise immunity in the circuit is facilitated by superimposing the output signal from AR5D on to a 2.5 VDC reference voltage provided by zener diode, U14, and applying it to the inverting terminal of U10 through R44. To further ensure noise and jitter free logic transitions, the input reference voltage is applied to the non-inverting terminal of U10. This reference feedback voltage is established by a voltage divider network consisting of R45 and R46. When the output signal at pin 7 of U10 is a logic HIGH, the reference feedback voltage is set to +2.7 VDC. When the output signal is a logic LOW, the reference feedback voltage is set to +2.3 VDC. This changing of the input reference voltage with changes of output logic state, functions as an output to input hysteresis. Figure 21 is a waveform analysis of the operation of the comparator, U10, and is interpreted as follows: When the instantaneous signal voltage input at pin 3 of U10 is less than the reference voltage set at pin 2 of U10, the output of the comparator is a logic HIGH. Conversely, when the instantaneous signal voltage at pin 2 of U10 is greater than the reference voltage at pin 2, the comparator output is a logic LOW.

(d) Data Playback Output Circuitry

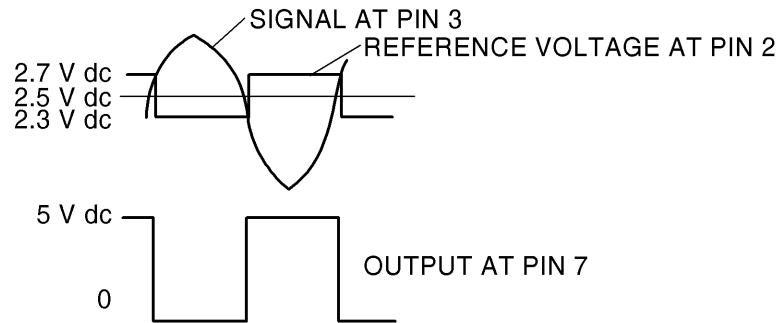
During a tape read operation, TTL logic level bi-phase data signals from the comparator, U10, are transmitted via two NAND gates connected in series, U9C and U9D, to two places. These two places are: Pin K of J2 via the A3 card, the front panel of the ground equipment connector, and the STC circuit in the Recording System Controller, A4. However, before these TTL level bi-phase signals can be transmitted to these two places, U9C and U9D must be enabled. To enable each NAND gate requires a logic level HIGH signal. The logic level HIGH signal which enables U9C is applied during a tape read operation only and is derived from the -PLAYBACK ENABLE signal which is sent to the read/write card by OPORT-0 on the Transport Interface card, A3. On the read/write card, the

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

PLAYBACK ENABLE signal is inverted to a logic HIGH by hex inverter, U12C, and applied to U9C. U9D is enabled by a logic HIGH transmitted by U9A. U9A continuously produces a logic HIGH when the write data to tape mode is disabled. Data transmitted to the STC during a tape read operation is the data provided for the CHEKRD function. During a write data to tape operation, write data from U9A is transmitted via U9D to the STC. This data is provided for the CHEKWR function.



Comparator U10 Waveforms
Figure 21

Since the playback preamplifiers and the playback amplifier-decoder circuitry are operational during both a write and read mode of operation, there is a possibility that write current transients could be transmitted to the playback data comparator, U10, during a write operation resulting in the comparator output changing state. Therefore, to lessen the possibility of that occurring, the reference signal voltage applied to the comparator is made unbalanced during a write-to-tape operation. Since during a write-to-tape operation, the signal produced by U12C is continuously a logic LOW, the LOW is applied through CR18 and R60 to the junction of R45 and R46. When the cathode of CR18 is held at a logic LOW, the resulting current flows through it. R60 and the feedback voltage divider network is sufficient to pull the reference feedback voltage to less than 1.8 VDC. This change in reference voltage causes the hysteresis of U10 to become unbalanced, which causes the output of U10 to tend to remain in a LOW state whenever the read-from-tape function is not enabled. Subsequently, when the read-from-tape function is first enabled (i.e., the -PLAYBACK ENABLE goes LOW), the hysteresis of U10 returns to a balanced condition after a delay period determined by the RC time constant of R60 and C52.

(3) Write Current Reference Voltage Power Supply

This power supply consists of a three terminal voltage regulator, U13; an NPN transistor, Q1; and a PNP transistor, Q2. Q1 and Q2 function as switches and switch the write current reference voltage ON during a write-to-tape operation. When a -WRITE ENABLE signal is received by this card, U12B, a hex inverter, applies a logic HIGH to the junction of R27 and R28 in the base terminal circuit of Q1. This HIGH causes a current to be applied to the base of Q1 to forward bias it ON. When Q1 turns ON, it conducts to saturation and changes the bias on Q2 to allow it to switch ON. When Q2 switches ON,

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

it also is made to conduct to saturation and 12.7 VDC is applied from U13 through Q2 to the isolation diodes, for example CR1 and CR9, and provides approximately 12.0 VDC through the write current resistors of U3 and U4 to the windings of the heads for the eight tracks.

G. Power Supply Assembly, A1 (see Figure 125)

(1) Power Supplies

The recorder power supply is designed to convert the applied 115 VAC, 400 Hz, received from the aircraft power source into the DC power required at various voltage levels for the operation of the recorder. The power supply design employs a toroidal power transformer for its qualities of low leakage, inductance and size. The transformer is of the multiple winding design with one primary and three secondary windings. The 115 VAC, 400 Hz, is applied to the primary and develops across each secondary winding 10, 38 and 20 VAC RMS, respectively. Connected across each secondary is a full wave bridge rectifier and a filter capacitor. The output from the bridge rectifier connected across the 10 VAC RMS winding is applied to two voltage regulators, AR1 and AR2. Each regulator is a three terminal, positive 5 VDC regulator capable of handling output currents to 3 amperes. Each regulator also has superior load regulation and temperature coefficients. The +5 VDC from AR1 supplies the +5 VDC requirements of PWB's A2, A3 and A5. The +5 VDC from AR2 supplies the +5 VDC requirements of PWB's A4, A6, A7 and A8. The A5 and A6 or A5, A6 and A7 assemblies are installed in the FDR version recorders only.

The output from the bridge rectifiers connected across the 38 VAC RMS winding is applied to regulators AR3, AR5, AR6 and AR7. These regulators supply the +15, -15, +12, -12 and -5 VDC, respectively. The AR3 regulator which produces the plus and minus 15 VDC is a dual tracking regulator. It is factory set to provide a balanced plus and minus 15 VDC output. The principal of a tracking regulator is to regulate one branch and make the other branch follow it. In this application, the negative branch is regulated while the positive branch is made to follow. Negative regulation is accomplished by providing a constant voltage reference for the negative error amplifier contained within the chip. Current limit sensing is provided by internal transistors which are normally held OFF by the connection of externally connected resistors between the base and emitter of each. These resistors are: R1 for the positive branch and R2 for the negative branch.

When load current passing through these resistors develops enough voltage drop across them, internal transistors are turned on and current is diverted. Current limit sensing may operate in either branch independently of the other. Current handling capability of the regulator circuit may be extended through the use of externally connected power transistors. In this application, Q1 and Q2 perform such a function. A path for regulator standby current is provided by resistors R3 for the positive branch and R4 for the negative branch. Each of the other regulators; AR5, AR6 and AR7 is a three terminal positive or negative fixed voltage regulator.

The output from the bridge rectifiers connected across the 20 VAC RMS winding is applied to voltage regulator, AR4. AR4 is an adjustable three terminal positive voltage regulator which has been factory set to provide the +16 VDC required by the motor windings. The three terminals of the regulator are the input, output and adjustment.

Connected between the adjustment terminal, the output terminal and ground return (motor return) is a resistive divider network consisting of R5 and R6. These resistors function to develop 1.25 VDC across R5 which acts to regulate the output of AR4 to +16 VDC. Also, connected between R5 and the ground return for the regulator is another resistor, R7, with a switching transistor, Q3. These components function to decrease the regulator output voltage when Q3 conducts. When Q3 is not turned on, R6 only is in the circuit. When Q3 is switched on by the application of the motor hold signal, R7 is effectively placed in parallel with R6. The total set resistance is thus reduced and causes the output voltage of the regulator to be reduced to a nominal +2.5 VDC during the time a motor hold signal is applied to Q3.

(2) Line Voltage Monitor

U3 is a line voltage monitor circuit (supply voltage level detector) which is designed to cause a system -RESTART signal to be generated whenever the output voltage of the bridge rectifier (CR1A) goes out of range. Its input is connected to the unregulated output of bridge rectifier, CR1A, which is connected across the power transformer 10 VAC RMS winding. The output of U3 is an open collector wire OR'ed TTL compatible signal which pulls the -RESTART line to a logic LOW whenever the monitored voltage drops to less than 7.8 VDC. It then holds the -RESTART line LOW until power up, when the voltage rises to above 8.3 VDC. At that point the output of the device (U3) goes HIGH. This allows the output of the reset circuit to the clock generator (A4U2) to go HIGH and clear the reset condition throughout the system.

(3) Stepper Motor Drive Circuitry

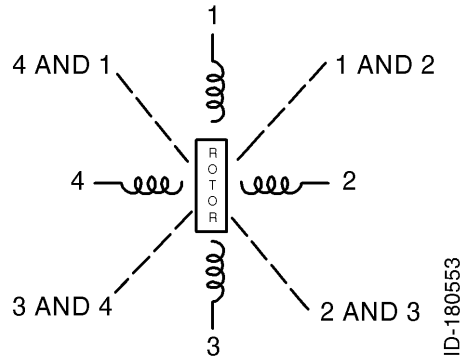
U1 and U2 are high voltage, high current Darlington transistor arrays which are used to drive the transport stepper motor. The stepper motor consists of a permanent magnet rotor and a stator with four wire wound poles. These four motor windings are designated motor windings 1, 2, 3 and 4. When the motor windings are energized by logic signals from the Darlington transistor arrays, north and south poles are generated. Motor rotation occurs as a result of interaction between the rotor magnet and magnetomotive force produced by the stator windings. When motor windings are energized in an organized pattern, a stable equilibrium point (null) is generated by the resulting north and south poles. The rotor, on interacting with the magnetomotive force, moves to the nearest null point. Thus, to step the motor in either direction, the motor windings are energized in a specific pattern. Figure 22 is a simplified schematic diagram of a typical stepper motor. It shows the eight null points that are produced when the pattern of pulses shown in Figure 23 are applied to this stepper motor configuration.

The stepper motor operates at 700 steps per second and provides tape speed of 5 inches per second (ips) to the UFDR transport. Adequate starting torque is provided to the stepper motor by stepping it on start up from 0 to 200 steps per second and then ramping it to a steady state of 700 steps per second. Total ramp-up and ramp-down activity is accomplished in 32 motor steps. Control of stepper motor operation is through the winding enable signals from the A3 card which are generated by the software.

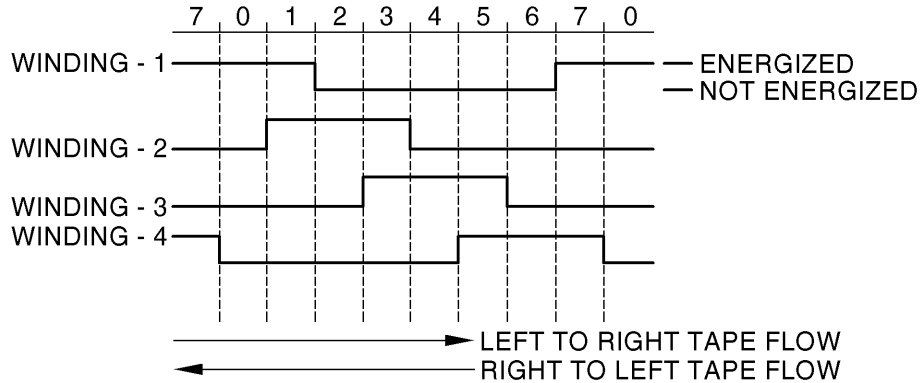
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Typical Stepper Motor
Figure 22



Pattern of Stepper Motor Signal Phases
Figure 23

TESTING AND TROUBLESHOOTING

1. General

This section contains information for the functional testing, calibration and troubleshooting of the UFDR. It also contains **schematic diagrams** of the equipment. The test procedure given is the minimum required to show compliance with specifications and to prove airworthiness of UFDR Recorders in a field service environment. The functional test procedure is the minimum level of testing required to be performed before flight operations may be resumed after a UFDR has been subjected to any corrective maintenance action.

A. Test Equipment Required

Test equipment is listed in Figure 901. Equivalent test equipment may be substituted provided that the substituted equipment has an accuracy equal to or greater than that of the specified equipment. Prior to performing any portion of the functional test or calibration procedure, verify that the calibration date for each item of test equipment used is current.

B. Power Requirements

A power source of 115 ± 1.0 VAC, 400 ± 10 Hz, single phase, 2 amperes minimum, is required.

C. Test Conditions

Unless otherwise specified, perform the functional test and calibration procedure under the following conditions:

Temperature	Room Ambient, $25 \pm 5^{\circ}\text{C}$
Altitude	Normal ground
Vibration	None
Humidity	Room ambient to 90% relative humidity

D. Flight Data Transcription Requirements

When a UFDR is removed from aircraft service for testing and/or troubleshooting, determine if a copy or transcription of recorded flight data is required. If a copy or transcription of flight data is required, **DO NOT** perform any testing or troubleshooting procedure beyond that specified below until a successful copy or transcription has been made. If a copy or transcription is required, but it is not known whether or not the UFDR is operational, the UFDR may be connected to the UFDR Functional Tester to perform the tests specified in paragraphs 2B and 2C. If the UFDR fails the requirements of these paragraphs, testing and/or troubleshooting is required before a copy or transcription can be made from this unit. Alternate methods can be used to make the required copy or transcription. The tape and/or tape transport assembly can be installed in another UFDR, known to be operational, or the tape can be transcribed onto a copy recorder.

E. Test Procedure Requirements

When a UFDR is removed from aircraft service for maintenance actions and a transcription is not required or has already been made, perform the Functional Test Procedure only to the extent required to determine that a fault condition exists. When a fault condition is determined

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

to exist, use the Troubleshooting portion of this section to isolate the fault. After corrective maintenance action(s) are completed, perform the entire Functional Test Procedure.

F. ATE Users

For ATE users, an ATLAS document has been prepared for UFDR part number 980-4100-D/B. It is available from Honeywell by ordering document number 050-0893-000. Also available, is a Stimuli Response Annotation document. It may be obtained by ordering document number 980-4100-704. This document correlates manual test procedures with ATE procedures.

G. Readout and Test Unit (RTU) Users

A method of **Return-to-Service** testing of the UFDR is available. This method utilizes the Readout and Test Unit (RTU), Part Nos. 980-4106-00X or 960-0306-00X. Refer to the RTU Component Maintenance Manual, Doc. No. 012-0507-001, for capabilities and test procedures. This PC-based RTU method is an option for those operators with RTU equipment and is intended to show compliance with UFDR acceptance test specifications prior to returning equipment to service. Refer to paragraph 7, UFDR RTU Test Procedure.

H. Acceptance Test/Transcription Unit (ATU) Users

Return-to-Service testing of the UFDR, using the ATU, Part No. 964-0434-002 or -005, is available. Refer to the ATEasy User's Manual, Copyright 1991 or later. This PC-based ATU method is an option for those operators with an ATU, and intended to show compliance with UFDR acceptance test specifications prior to returning equipment to service. Refer to paragraph 9, ATU Test Procedure.

I. Underwater Locating Device (ULD) Test

If it is necessary to remove the ULD, refer to DISASSEMBLY.

(1) ULD Battery Test

Use a voltmeter with an impedance of 10 megohms to perform this test:

- (a) Clean ULD case and water switch with mild detergent and a soft cloth.
- (b) Put negative (–) lead of voltmeter on water switch pin and positive (+) lead on ULD case, or the mounting kit, if installed.
- (c) Read the voltmeter. Check the battery CODE on the ULD label. Minimum acceptable voltage for the DK100 is 7.10V with no code, 3.55V with code A, 3.00V with code B, and 2.97V with codes C or D. Minimum acceptable voltage for the ELP362D is 6.00V. If the battery voltage is below the minimum acceptable voltage, remove the ULD from service. Refer to the Technical Manual for the DK100, DK120, or the User's Manual for the ELP362D battery replacement information.

(2) ULD Test Procedure (Using Ultrasonic Test Set)

Test the ULD with an Ultrasonic Test Set (42, Figure 901) per these steps:

- (a) Set the Test Set INPUT SELECTOR switch to INT.
- (b) Activate the Test Set and check for operation by jingling a set of keys, coins, or other available ultrasonic noise-maker near the test set microphone. Set TUNING control between 35 and 40kHz.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (c) Establish a temporary circuit across the device water switch using a piece of metal braid or other flexible conductor. Tune Test Set for best signal.
- (d) Determine pulse repetition rate by counting pulses for 10 seconds and dividing by 10.
- (e) Determine the pulse duration by displaying the output of the Test Set on a calibrated oscilloscope (2, Figure 901).
- (f) Determine the approximate operating frequency by tuning the Test Set for zero beat with the acoustic signal and reading the frequency from the dial setting.

NOTE: *Results of the preceding tests with the device operating in air will usually be in only approximate agreement with the values shown on the individual Dukane test sheet, because the test sheet values were obtained with the device operating in water under conditions approaching free field operation. In air, the pulse duration is longer and the pulse repetition rate slower than in water. However, an accelerated or increasing pulse repetition rate indicates that the battery is weak. If normal operation does not occur after installation of a new battery, return the device to Dukane for service.*

(3) ULD Test Procedure (Using Pinglite Tester PL-1)

Test any ULD with a Pinglite Tester PL-1 (43, Figure 901).

- (a) Establish a temporary short circuit between the device water switch center pin and the case, using a piece of metal braid or other flexible conductor.
- (b) Press end of Pinglite Tester to beacon body, about 1 inch from switch end of ULD.
- (c) Press and hold the OPERATION pushbutton switch on Pinglite Tester. A light flash indicates that the Tester is operating.
- (d) The BEACON indicator lamp on the Pinglite Tester flashes with each output pulse from the ULD, to indicate that the ULD is operational.
- (e) No flashing on the Tester indicates a dead battery or defective ULD. An accelerated or increasing pulse repetition rate indicates a weak battery. If normal operation does not occur after installation of a new battery, return device to manufacturer for service.

(4) ULD Test Procedure (Using Datasonics Test set)

Test any ULD with a Datasonics Acoustic Test Set, Model ATS-260 (44, Figure 901).

Test any ULD with the Datasonics Acoustic Test Set, Model ATS-260, per the manufacturer's User's Manual, ELP-362D, Emergency Locator Beacon.

(5) ULD Test Procedure (Using Pinglite Tester PL-3)

Test any ULD with a Pinglite Tester PL-3 (43, Figure 901).

- (a) Press tester to beacon water switch.
- (b) The ULD operation is verified by the LED on the Pinglite tester flashing with each output pulse or by listening for the Ping.

31-30-32

Page 103
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (c) No flashing on the Tester indicates a dead battery or defective ULD. An accelerated or increasing pulse repetition rate indicates a weak battery. If normal operation does not occur after installation of a new battery, return device to manufacturer for service.

2. **Functional Test Procedure**

A. **Insulation Resistance Check**

NOTE: *This step is optional, at the operators discretion. It is recommended that the insulation resistance check be made after any 115VAC and/or 28VDC input circuitry repairs have been made.*

Using a High Potential Tester (HPT) (32, Figure 901) with voltage set at 100 VDC maximum, measure resistance between J1A-1 and J1A-6 (CASE GROUND) and between J1A-2 and J1A-6. Reverse the HPT leads and repeat measurements. In each case, resistance shall be >20 megohms.

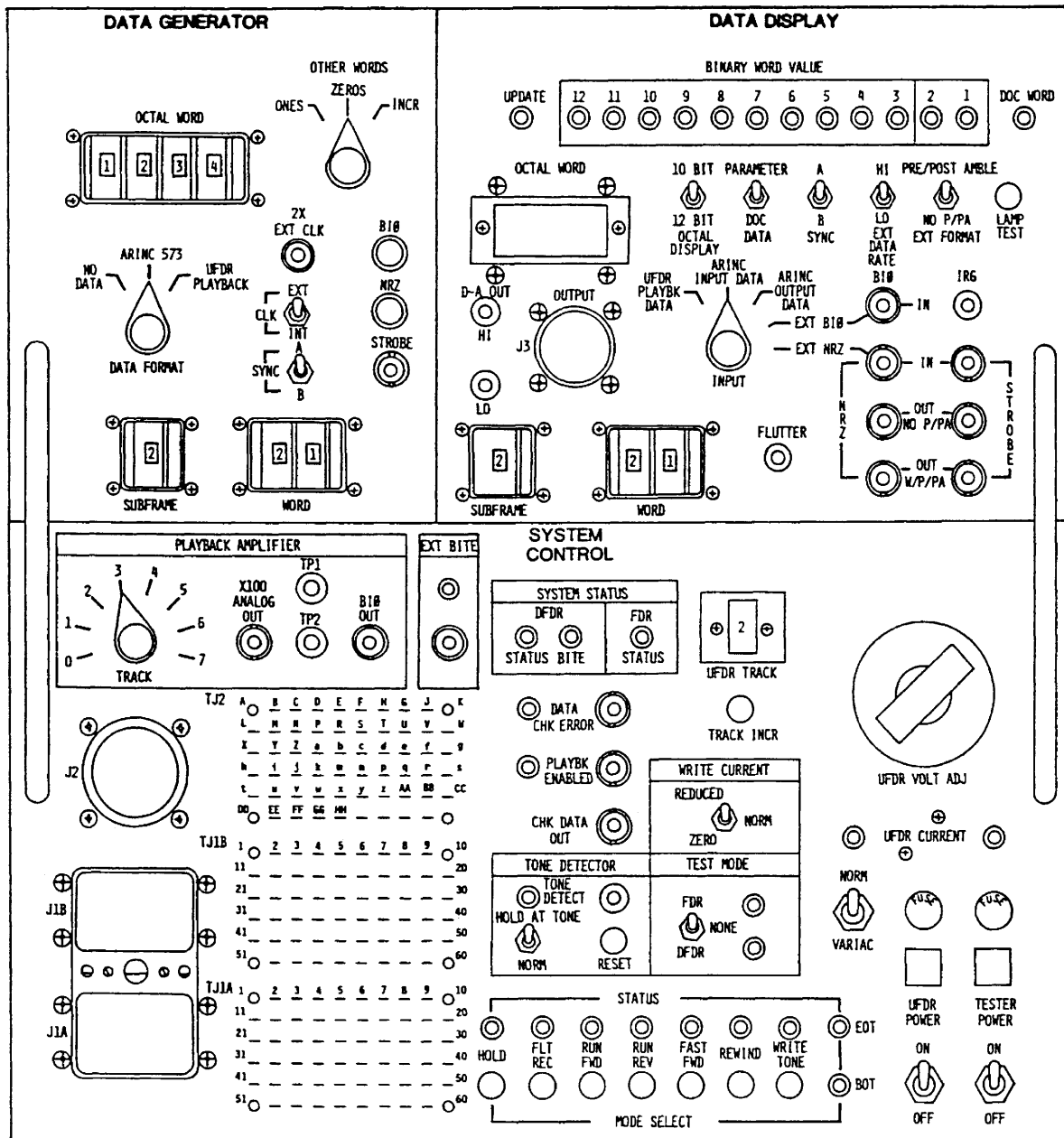
B. **Initial Setup**

- (1) Verify that the UFDR POWER and TESTER POWER switches on the front panel of the UFDR Functional Tester (hereinafter called the Tester) are in the OFF position.
- (2) Using the Tester UFDR Aircraft Interface cable, connect J1A/J1B on the front panel of the Tester to the UFDR aircraft connector, J1A/J1B, at the rear of the UFDR. Refer to Figure 101.
- (3) Using the Tester External Equipment Cable, connect J2 on the front panel of the Tester to J2 on the front panel of the UFDR, the unit under test (UUT).
- (4) Connect the Tester power cable (at the rear, lower right corner of the Tester) to a 115 VAC, single phase, 400 Hz power source.
- (5) Position the TESTER POWER switch to ON and verify that the TESTER POWER lamp goes on.
- (6) Position the VARIAC/NORM switch to NORM.
- (7) Position the WRITE CURRENT switch to NORM.
- (8) Position the TONE DETECTOR switch to NORM.
- (9) Press the HOLD MODE SELECT switch and verify that HOLD LED is on.
- (10) Position the DATA GENERATOR DATA FORMAT switch to NO DATA.
- (11) Position the UFDR POWER switch to ON and verify that the UFDR POWER lamp goes on.
- (12) Position the TEST MODE switch to DFDR, and then to NONE.

Honeywell

COMPONENT MAINTENANCE MANUAL

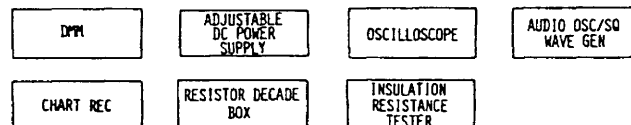
Universal Flight Data Recorder / Part No. 980-4100/980-4101



TO TESTER J1
704-1044-001

RECORDER

TO TESTER J2
704-1045-001 (6 FOOT)
OR
704-1045-002 (25 FOOT)



Functional Tester Control Panel
Figure 101

31-30-32

Page 105
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

ID-180555

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

(13) Verify the following Tester conditions:

SYSTEM STATUS DFDR BITE LED	off
SYSTEM STATUS DFDR STATUS LED	on
HOLD LED	on
TEST MODE DFDR LED	on
OCTAL WORD indicator indicates a random number	
BINARY WORD VALUE LED's indicate same random number in binary	
UFDR TRACK indicator indicates a track number (as-received tape track position)	

- (14) Verify that the BITE LED on the UUT is off.
- (15) Momentarily position UFDR POWER switch to OFF and then to ON.
- (16) Verify that DFDR mode LED stays on with reapplication of power.
- (17) Position the TEST MODE switch to DFDR.
- (18) Position the DATA GENERATOR DATA FORMAT switch to ARINC 573.
- (19) Verify that the UFDR and Tester BITE LED's are OFF and the HOLD LED is ON.
- (20) Observe the Tester UFDR track indication display and press the TRACK INCR switch through the complete track sequence. Verify that the tracks sequence in order from 0 through 7, starting from the as-received track position.

C. Nonflight Recording Modes Operating Tests

These tests are designed to check the control logic and tape drive signals for all nonflight recording modes of the UUT.

- (1) With the tape at its as-received position in the UUT and the initial setup procedure completed, press Tester TONE DETECTOR RESET switch and release it.
- (2) Press the Tester WRITE TONE MODE SELECT switch and release it. Verify that the WRITE TONE and RUN FWD LEDs go on.
- (3) Wait a few seconds for the write tone mode to operate and then press the HOLD MODE SELECT switch and release it when the HOLD LED goes on.
- (4) Press the REWIND MODE SELECT switch and release it. Verify that the REWIND LED goes on.
- (5) Press the RUN REV MODE SELECT switch and verify that the HOLD LED goes on. Release it and verify that RUN REV LED goes on. Wait approximately 1 minute before continuing.
- (6) Position the TONE DETECTOR switch from NORM to HOLD AT TONE; press and release the RESET switch.
- (7) Press the FAST FWD MODE SELECT switch and verify that HOLD LED goes on. Release it and verify that FAST FWD LED goes on.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (8) Press the RUN FWD MODE SELECT switch and verify that the HOLD LED goes on. Release it and verify that RUN FWD LED goes on.
- (9) After a few moments of operation, the tape in the UUT should return to the as-received position as marked by the write tone signal in step (3). Verify that the TONE DETECT LED and the HOLD LED go on to indicate this condition.
- (10) Press HOLD MODE SELECT switch. Press TONE DETECTOR RESET switch and release it when TONE DETECT LED goes off.
- (11) Position the TONE DETECTOR switch to NORM.

D. Flight Recording Mode Operating Test

This test is designed to check the flight record mode of the UUT by writing data to the tape with normal write current. The UUT BITE system is checked when the write current signal is reduced to zero.

- (1) On the Tester, position DATA GENERATOR switches as indicated:

CLK	INT
OCTAL WORD	1234
SUBFRAME	0
WORD	02
DATA FORMAT	ARINC 573
SYNC	B
OTHER WORDS	INCR

- (2) On the Tester, position DATA DISPLAY switches as indicated:

INPUT	ARINC OUTPUT DATA
SUBFRAME	0
WORD	01
OCTAL DISPLAY	12 bit
DATA	PARAMETER
SYNC	B
EXT DATA RATE	LO
EXT FORMAT	NO P/PA

- (3) Press FLT REC MODE SELECT switch and verify that HOLD LED remains on. Release switch and verify that FLT REC LED goes on.
- (4) Verify the condition given for each of the following LED's.

SYSTEM STATUS DFDR STATUS	OFF
SYSTEM STATUS DFDR BITE	OFF
PLAYBK ENABLED	Flashing at a 1 second rate
DATA CHECK ERROR	Not flashing

31-30-32

Page 107
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

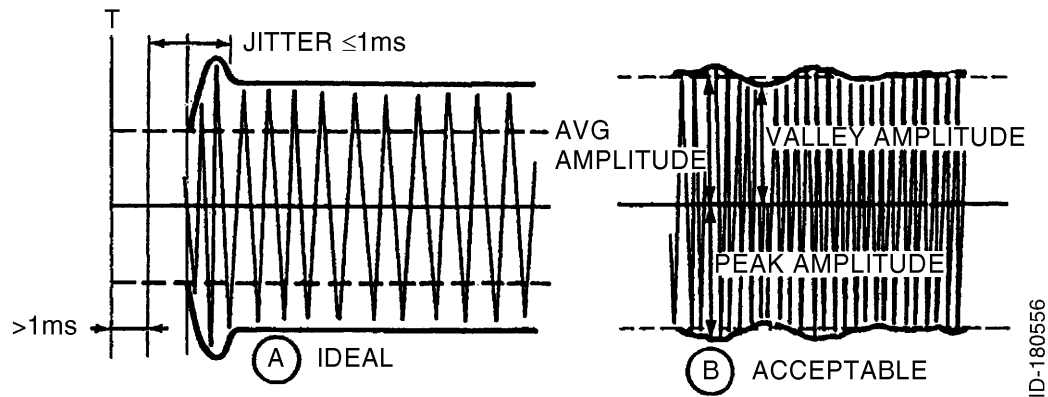
Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (5) Verify that the OCTAL WORD indicator displays the proper incrementing of the following sync words: 1107, 2670, 5107 and 6670.
- (6) Position the WRITE CURRENT switch from NORM to ZERO.
- (7) Count the flashes of the DATA CHECK ERROR LED.
- (8) After 8 consecutive flashes, verify that the Tester EXT BITE LED, SYSTEM STATUS DFDR STATUS LED, SYSTEM STATUS DFDR BITE LED and the UUT front panel BITE LED go on.
- (9) Return the WRITE CURRENT switch to NORM and verify that all LED's in steps (7) and (8) are cleared.
- (10) Press the HOLD MODE SELECT switch, release it and verify that the HOLD LED goes on.

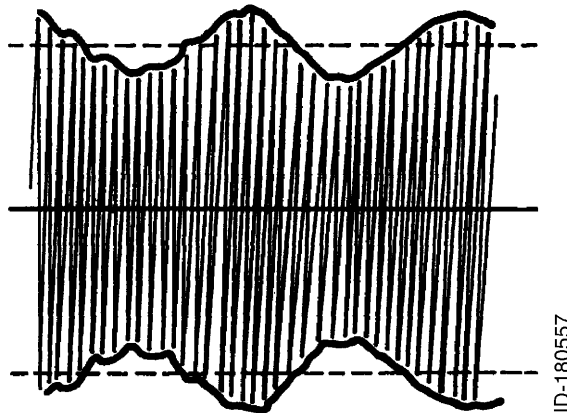
E. CHECKSTROKE Stability Test

This test verifies that the time displacements and ringing in the flight record mode caused by transport belt elasticity are within required limits.

- (1) Set DATA GENERATOR OCTAL WORD switches to 0000 and OCTAL WORDS to ZEROS. Connect the oscilloscope (Item 2, Figure 901) to the Tester PLAYBACK AMPLIFIER X100 ANALOG OUT BNC jack.
- (2) Trigger scope with a signal from the PLAYBK ENABLED BNC jack.
- (3) For the Forward Direction Test:
 - (a) Verify that the HOLD mode is still selected.
 - (b) Position the PLAYBACK AMPLIFIER TRACK select switch to 0.
 - (c) Press the TRACK INCR switch and verify that the UFDR TRACK indicator indicates track 0 selected.
 - (d) Press the FLT REC MODE SELECT switch, release it and verify that the FLT REC LED goes on.
 - (e) Adjust the scope to trigger on the negative going edge of the signal from the PLAYBK ENABLED jack.
 - (f) Measure the minimum and maximum times from the trigger point to the start of the playback signal. Make the measurement over a period of about 30 seconds.
 - (g) Verify that the time from the trigger point to the start of the playback signal is >1 millisecond, and that the jitter time is ≤ 1 millisecond.
 - (h) Adjust the scope to display the entire playback CHECKSTROKE waveform.
 - (i) Examine the envelope of the playback signal. The waveform should be similar to A or B of Figure 102. If it is similar to Figure 103, the transport assembly is faulty



Acceptable Waveforms for Playback Signals Modulation Envelope
Figure 102



Unacceptable Waveforms for Playback Signals Modulation Envelope
Figure 103

- (j) If the waveform is similar to B of Figure 102, determine if it is acceptable by measuring the amplitude of the peak, valley and average dimensions of the waveform and calculate the percent of modulation using the following formula:

$$\%MOD = \frac{\text{Peak Amplitude Minus Valley Amplitude}}{\text{Average Amplitude}} \times 100$$

- (k) If the percent of modulation is 35% or greater, the transport assembly is faulty.

NOTE: If the transport assembly is a new or reworked assembly, the percent of modulation must not exceed 25%.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (4) For the Reverse Direction Test:
 - (a) Press HOLD MODE SELECT switch and verify HOLD LED goes on.
 - (b) Position the PLAYBACK AMPLIFIER TRACK select switch to 1.
 - (c) Press the TRACK INCR switch and verify that the UFDR TRACK indicator indicates track 1 selected.
 - (d) Press the FLT REC MODE SELECT switch, release it and verify that the FLT REC LED goes on.
 - (e) Repeat steps (3)(d) through (3)(k).
 - (f) Press HOLD MODE SELECT switch and verify HOLD LED goes on.

F. Write-to-Tape Test Procedure

Test records data on tape for subsequent playback tests. Data consists of five minutes of different words on each of eight tracks. Data on each track is preceded and succeeded by a tone marker.

- (1) Press the REWIND MODE SELECT switch and allow the tape to run for about four minutes or until BOT is encountered.
- (2) At the end of four minutes or at BOT, press the FAST FWD MODE SELECT switch and allow the tape to run for about two minutes.
- (3) At the end of two minutes, press the HOLD MODE SELECT switch.
- (4) On the Tester, position DATA GENERATOR switches as indicated:

CLK	INT
OCTAL WORD	0000
SUBFRAME	0
WORD	02
DATA FORMAT	ARINC 573
SYNC	B
OTHER WORDS	INCR

- (5) On the Tester, position DATA DISPLAY switches as indicated:

INPUT	ARINC OUTPUT DATA
SUBFRAME	0
WORD	02
OCTAL DISPLAY	12 bit
DATA	PARAMETER
SYNC	B
EXT DATA RATE	LO
EXT FORMAT	NO P/PA

31-30-32

Page 110
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (6) Verify that the Tester UFDR TRACK indicator indicates track 0. If it does not, press the TRACK INCR switch until it does.
- (7) Press the WRITE TONE switch, release it and wait five seconds.
- (8) Press the FLT REC MODE SELECT switch, and release it.
- (9) Allow the UUT to operate in the flight record mode for about five minutes and check the following while waiting:
 - (a) The Tester OCTAL WORD display indicates the octal word selected by the positioning of the DATA GENERATOR OCTAL WORD thumbwheel switches.
 - (b) The UPDATE LED flashes at a one second rate.
- (10) After about five minutes of flight record mode of operation, press the WRITE TONE MODE SELECT switch, release it when the HOLD LED goes on and wait five seconds.
- (11) After five seconds, press the HOLD MODE SELECT switch, and release it.
- (12) Write data to each of the remaining tracks using the following procedure:
 - (a) Press the TRACK INCR switch on the Tester to select the next track and verify that the UFDR TRACK indicator on the Tester indicates the proper track number.
 - (b) Set the Tester DATA GENERATOR OCTAL WORD thumbwheel switches to the octal word assigned to that track as follows:

TRACK	OCTAL WORD
0	0000
1	1000
2	2000
3	3000
4	4000
5	5000
6	6000
7	7000

- (13) Repeat steps (7) through (11).

G. Playback Tests

These tests use the data recorded on the tape in the write-to-tape tests described in paragraph 2F. These tests consist of a check of data decoding and track integrity, an IRG noise level check, a playback signal amplitude check, an IRG stability check and a forward and reverse direction flutter test.

- (1) Initial Setup
 - (a) Position the DATA GENERATOR DATA FORMAT switch to NO DATA.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (b) On the Tester, position DATA DISPLAY switches as indicated:

INPUT	UFDR PLAYBK
SUBFRAME	0
WORD	02
OCTAL DISPLAY	12
DATA	PARAMETER
SYNC	B
EXT DATA RATE	HI
EXT FORMAT	NO P/PA

(2) Data Decoding and Track Integrity Tests

These tests are performed on each track in sequence. Select each succeeding track after the playback test has been completed on the selected track. Select each track by pressing the TRACK INCR switch and observing the proper track number displayed by the UFDR TRACK indicator.

- (a) Press the TRACK INCR switch and select track 0. Verify that the UFDR TRACK indicator displays the proper track number.
- (b) Press the RUN REV MODE SELECT switch, release it and allow the tape to run for about 30 seconds.
- (c) After 30 seconds elapses, press the HOLD MODE SELECT switch and release it.
- (d) Position the TONE DETECTOR switch to HOLD AT TONE.

NOTE: *It may be necessary to press and hold the TONE DETECTOR RESET switch if the TONE DETECT LED is on when the RUN FWD MODE SELECT switch is pressed. Release it when the TONE DETECT LED goes off.*

- (e) Press the RUN FWD MODE SELECT switch and release it.
- (f) After a short period of time, verify that the HOLD LED and the TONE DETECT LED go on. (This identifies the tone at the beginning of the data on track 0.)
- (g) Press the TONE DETECTOR RESET switch and hold it pressed until the TONE DETECT LED goes off.
- (h) Verify that the OCTAL WORD display indicates the correct code for the selected track, i.e., 0 0 0 0 for track 0, etc.
- (i) Verify that the UUT tape automatically stops at the tone, marking the end of the recording (second tone marker). This is indicated by the TONE DETECT and the HOLD LED's going on.
- (j) Press the TRACK INCR switch and select the next track (track 1). Verify that the UFDR TRACK indicator displays the proper track number.
- (k) Press the RUN FWD MODE SELECT switch and release it.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (l) Verify that the OCTAL WORD display indicates the correct code for the selected track, i.e., 1 0 0 0 for track 1, etc.

NOTE: See paragraph 2F(12)(b) for the correct octal code assigned to each track.

- (m) Verify that the UUT tape automatically stops at the tone, marking the end of the recording (second tone marker). This is indicated by the TONE DETECT and the HOLD LED's going on.
- (n) Repeat steps (j) through (m) for tracks 2 through 7.

(3) IRG Noise Level Check

This test measures the peak-to-peak noise level in the inter-record-gap for each track.

- (a) Connect the oscilloscope (Item 2, Figure 901) channel 2 to PLAYBACK AMPLIFIER X100 ANALOG OUT jack and channel 1 to TP2 jack.
- (b) Trigger the scope with a signal from the IRG test point on the Tester.
- (c) Position the PLAYBACK AMPLIFIER TRACK select switch to 0.
- (d) Press the TRACK INCR switch and select track 0. Verify that the UFDR TRACK indicator displays the proper track number.
- (e) Press and release the RUN FWD MODE SELECT switch.
- (f) Press and hold the TONE DETECTOR RESET switch until the TONE DETECT LED goes off.
- (g) Verify that the scope indicates a maximum noise level of 320 millivolts peak-to-peak.
- (h) Also, verify that the UUT tape automatically stops at the tone, marking the end of the recording. This is indicated by the TONE DETECT and the HOLD LED's going on.
- (i) Press the TRACK INCR switch and select the next track (track 1, 2, etc.). Verify that the UFDR TRACK indicator indicates the proper track number. Position the PLAYBACK AMPLIFIER TRACK select switch to the next track selected.
- (j) Press the RUN FWD MODE SELECT switch and release it.

NOTE: It may be necessary to press and hold the TONE DETECTOR RESET switch if the TONE DETECT LED is on when the RUN FWD MODE SELECT switch is pressed. Release it when the TONE DETECT LED goes off.

- (k) Verify that the scope indicates a noise level of not more than 320 millivolts peak-to-peak.
- (l) Verify that the UUT tape automatically stops at the tone, marking the end of the recording. This is indicated by the TONE DETECT and the HOLD LED's going on.
- (m) Repeat steps (i) through (l) for tracks 2 through 7.

Honeywell

COMPONENT MAINTENANCE MANUAL

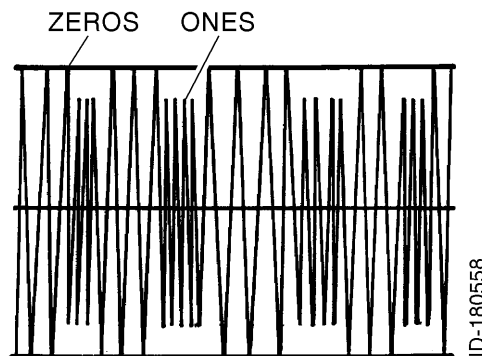
Universal Flight Data Recorder / Part No. 980-4100/980-4101

(4) Playback Signal Amplitude

This test is designed to measure the peak-to-peak amplitude of the 1's and 0's for each track.

- Set scope to internal trigger and adjust time base to display mixed 1's and 0's.
- Press the TRACK INCR switch and verify that the UFDR TRACK indicator indicates track 0 selected. Position the PLAYBACK AMPLIFIER TRACK switch to 0.
- Press and release the RUN FWD MODE SELECT switch.
- Press and hold the TONE DETECTOR RESET switch until the TONE DETECT LED goes off.
- Verify that the scope indicates (Ref. Figure 104) a peak-to-peak level for 1's (high frequency) and 0's (low frequency) as follows:
- Verify that the UUT tape automatically stops at the tone, marking the end of the recording. This is indicated by the TONE DETECT and the HOLD LED's going on.

PLAYBACK SIGNAL	1 LEVEL	0 LEVEL
Mod 0 thru 4		
Heads Replaced	5.0 Vp-p min.	±3 db relative to 1
Original Heads	3.5 Vp-p min.	±6 db relative to 1
Mod 5 and on		
Heads Replaced	3.7 Vp-p min.	±3 db relative to 1
Original Heads	3.0 Vp-p min.	±6 db relative to 1



Scope Indication of 1's and 0's
Figure 104

31-30-32

Page 114
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

(g) Press the TRACK INCR switch and select the next track (track 1, 2, etc.). Verify that the UFDR TRACK indicator indicates the proper track number. Position the PLAYBACK AMPLIFIER TRACK select switch to the next track selected.

(h) Press the RUN FWD MODE SELECT switch and release it.

NOTE: *Press and hold TONE DETECTOR RESET switch if TONE DETECT LED is on when RUN FWD MODE SELECT switch is pressed. Release it when the TONE DETECT LED goes off.*

(i) Repeat steps (d) and (e) above.

(j) Repeat steps (f), (g) and (h) for tracks 2 through 7.

(5) IRG Stability Check

This test is designed to measure the time stability of the IRG in the forward and reverse directions (tracks 0 and 1).

NOTE: *The IRG Stability Check and the Flutter Test may be performed on any one pair of odd and even tracks, i.e., tracks 0 and 1 or tracks 2 and 3, etc.*

(a) Press the TRACK INCR switch and verify that the UFDR TRACK indicator indicates track 0. Position the PLAYBACK AMPLIFIER TRACK select switch to 0.

(b) Move scope from the X100 ANALOG jack to the IRG test point.

(c) Press and release the RUN FWD MODE SELECT switch.

(d) Press and hold the TONE DETECTOR RESET switch until the TONE DETECT LED goes off.

(e) Verify that the minimum and maximum duration of IRG's over the period of the recording is 9 to 13 milliseconds.

(f) Verify that the UUT tape automatically stops at the tone, marking the end of the recording. This is indicated by the TONE DETECT and HOLD LED's going on.

(g) Press the TRACK INCR switch and select the next track (track 1, 2, etc.). Verify that the UFDR TRACK indicator indicates the proper track number. Position the PLAYBACK AMPLIFIER TRACK select switch to the next track selected.

(h) Press the RUN FWD MODE SELECT switch and release it.

NOTE: *Press and hold TONE DETECTOR RESET switch if TONE DETECT LED is on when RUN FWD MODE SELECT switch is pressed. Release it when the TONE DETECT LED goes off.*

(i) Repeat steps (e) and (f).

(6) Flutter Test

This test is designed to measure the peak-to-peak level of TTL level signals in forward and reverse directions (track 0 and 1).

NOTE: *This test may be performed on any pair of odd and even tracks.*

(a) Move the scope connection from IRG to FLUTTER test point.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (b) Connect the EXTERNAL SYNC jack of the scope to IRG test point.
- (c) Press the TRACK INCR switch and verify that the UFDR TRACK indicator indicates the proper track number.

***NOTE:** As a result of having completed the previous test, the tape is stopped at the second tone marker on track 1; therefore, if it is desirable, either track 2 or track 0 may be selected for the even track flutter test.*
- (d) Press and release the RUN FWD MODE SELECT switch.
- (e) Press and hold the TONE DETECTOR RESET switch until the TONE DETECT LED goes off.
- (f) Use the trailing edge of the IRG signal to sync the scope display.
- (g) Observe the scope indication throughout the duration of the test recording and verify that the highest peak-to-peak value is not greater than 1 volt.
- (h) Verify that the UUT tape automatically stops at the tone, marking the end of the recording. This is indicated by the TONE DETECT and the HOLD LED's going on.
- (i) Press the TRACK INCR switch and select the next track (an odd track). Verify that the UFDR TRACK indicator indicates the proper track number.
- (j) Repeat steps (d) through (h).

When all of the tests of paragraph 2G have been completed on all of the applicable tracks, position the TONE DETECTOR switch from HOLD AT TONE to NORM.

H. Gain Margin Test

For this test, a section of each track is recorded at reduced write current and the UUT is checked for the absence of data check errors at the reduced write current.

- (1) Position the DATA GENERATOR DATA FORMAT switch to ARINC 573.
- (2) Position the DATA DISPLAY INPUT switch to ARINC OUTPUT DATA. Connect the oscilloscope to the PLAYBACK AMPLIFIER X100 ANALOG OUT test jack, and the oscilloscope trigger to the PLAYBK ENABLED BNC jack.
- (3) Press the TRACK INCR switch to select the first track (track 0) and verify that the UFDR TRACK indicator displays the proper track number.
- (4) Press and release the FLT REC MODE SELECT switch.
- (5) Verify that the PLAYBK ENABLED LED flashes.
- (6) Verify that the DATA CHK ERROR LED remains off.

***NOTE:** The DATA CHK ERROR LED may flash initially a maximum of 7 times, but must remain off after the initial flashing.*

- (7) Position the WRITE CURRENT switch to REDUCED.
- (8) For observation period of at least 1 minute, verify that CHECKSTROKE envelope stays essentially the same as for normal write current except for reduction in overall level (i.e., no repetitive dropouts or dropdowns attributable to reduced current).

31-30-32

Page 116
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

- (9) Verify that the DATA CHK ERROR LED shows no repetitive flashing throughout the observation period.
- (10) Position the WRITE CURRENT switch to NORM.
- (11) Press and release the HOLD MODE SELECT switch.
- (12) Repeat steps (3) through (11) for each of the remaining tracks.

I. System Status Indicator

The Tester is designed to provide system status indication for two system conditions: DFDR STATUS and DFDR BITE. The DFDR STATUS LED goes on when any of these conditions exist: Nonflight recording mode selected, loss of AC power, loss of input data or a BITE fault condition. The BITE LED goes on when the UFDR (the UUT) BITE system detects a BITE fault condition. A BITE fault condition can exist if the input data fails to meet the BITE system data integrity test.

- (1) With UUT in HOLD mode, verify that DFDR STATUS LED is on.
- (2) Select each of the following nonflight recording modes and verify that the DFDR STATUS LED remains on: RUN FWD, RUN REV, FAST FWD, WRITE TONE and REWIND.
- (3) Press and release the FLT REC MODE SELECT switch.
- (4) Verify that the DFDR STATUS LED goes off.
- (5) Position the UFDR POWER switch to OFF.
- (6) Verify that the DFDR STATUS LED goes on.
- (7) Position the UFDR POWER switch to ON.
- (8) Verify that the DFDR STATUS LED goes off.
- (9) Position the DATA GENERATOR DATA FORMAT switch to NO DATA.
- (10) Verify that the DFDR STATUS LED goes on.
- (11) Position the DATA FORMAT switch to ARINC 573 and verify that the DFDR STATUS LED goes off.
- (12) Data Integrity Test
 - (a) Connect a square wave generator (Item 3, Figure 901) to the DATA GENERATOR BIPHASE BNC jack.
 - (b) Adjust square wave generator for a 0 to 5V amplitude square wave at a frequency of 384 ± 2 Hz. Verify that the DFDR STATUS and BITE LED's go on.
 - (c) Position the DATA FORMAT switch to NO DATA.
 - (d) Disconnect the square wave generator. Verify that the DFDR STATUS LED remains on and the BITE LED's go off.
 - (e) Adjust the square wave generator to 768 ± 2 Hz and reconnect it to the BIPHASE jack. Verify that BITE LED's go on.
 - (f) Disconnect the square wave generator and position the DATA GENERATOR DATA FORMAT switch back to ARINC 573.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (g) Verify that the DFDR STATUS and DFDR BITE LED's go off.
- (h) Press and release the HOLD MODE SELECT switch.

J. Abnormal Power Condition Tests

These tests are designed to verify that the UUT can operate without errors under the following extremes of primary line voltage: minimum, 95 VAC RMS: maximum, 134 VAC RMS.

- (1) Position the UFDR POWER switch to OFF.
- (2) Position the NORM/VARIAC switch to VARIAC and set the VARIAC to midrange.
- (3) Adjust the DMM (Item 4, Figure 901) to indicate AC volts and connect its test leads to TJ1A-1(HI) and TJ1A-2(LO).
- (4) Position the UFDR POWER switch to ON.
- (5) Adjust the UFDR VOLT ADJ (the variac) to produce a DMM indication of 115 ± 1.0 VAC RMS.
- (6) Observe that operation of the UUT is normal and proper.
- (7) Adjust UFDR VOLT ADJ for DMM indication of $95 -0, +1$ VAC RMS.
- (8) Verify that error free flight recording operations continue.
- (9) Momentarily position the UFDR POWER switch to OFF.
- (10) When power is reapplied, verify that error free operation continues and that no track change has taken place.
- (11) Press the HOLD MODE SELECT switch.
- (12) With the UFDR VOLT ADJ, slowly reduce applied primary line voltage toward zero and stop at point where the BITE LED goes on.
- (13) Record the voltage at which the DFDR BITE LED goes on.
- (14) With the UFDR VOLT ADJ, slowly increase the applied primary line voltage toward 115 VAC RMS, and stop at the point where the BITE LED goes off.
- (15) Record the voltage at which the DFDR BITE LED goes off.
- (16) Verify that no track changes occurred.
- (17) Subtract the voltage recorded in step (13) from that recorded in step (15).
- (18) Verify that resulting value of hysteresis is 2.0 to 6.0 V.
- (19) With the UFDR VOLT ADJ, slowly increase the applied primary line voltage to $134 +1, -0$ VAC RMS.
- (20) Press and release the FLT REC MODE SELECT switch.
- (21) Verify that error free flight recording operations continue.
- (22) Momentarily position the UFDR POWER switch to OFF.
- (23) When the power is reapplied, verify that error free operation continues and that no track change has taken place.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (24) With the UFDR VOLT ADJ, reset the primary line voltage to 115 ± 1.0 VAC RMS.
- (25) Position the NORM/VARIAC switch to NORM.
- (26) Ensure that the tape end is away from EOT or BOT by pressing and releasing the REWIND switch and allowing the tape to run for about 20 seconds.
- (27) At the end of 20 seconds, press and release the FAST FWD MODE SELECT switch and allow the tape to run for about 5 seconds.
- (28) At the end of five seconds, press and release the HOLD MODE SELECT switch.
- (29) If the UFDR TRACK indicator does not indicate an even track, press the TRACK INCR switch and select an even track.
- (30) Press and release the FLT REC MODE SELECT switch.
- (31) Position the UFDR POWER switch to the OFF and ON position several times including several times per second.
- (32) When the UFDR POWER switch is returned to ON continuously, verify that no track change has occurred.
- (33) Press and release the HOLD MODE SELECT switch.

K. Data Output Signal Level Test

This test verifies that the data output signal levels meet the following specifications:

Rise Time	10 to 30 microseconds
Fall Time	10 to 30 microseconds
Overshoot	2.0 volts, maximum
Amplitude (after overshoot)	10 to 15 volts p-p
Droop	3 volts maximum

- (1) Connect the oscilloscope differentially to TJ1A-9 and TJ1A-10 with scope ground at TJ1A-6.
- (2) Position the DATA GENERATOR OTHER WORDS switch to ZEROS.
- (3) Verify that the scope indication is a waveform similar to that shown in Figure 105 and that its parameters are within the specifications given above.

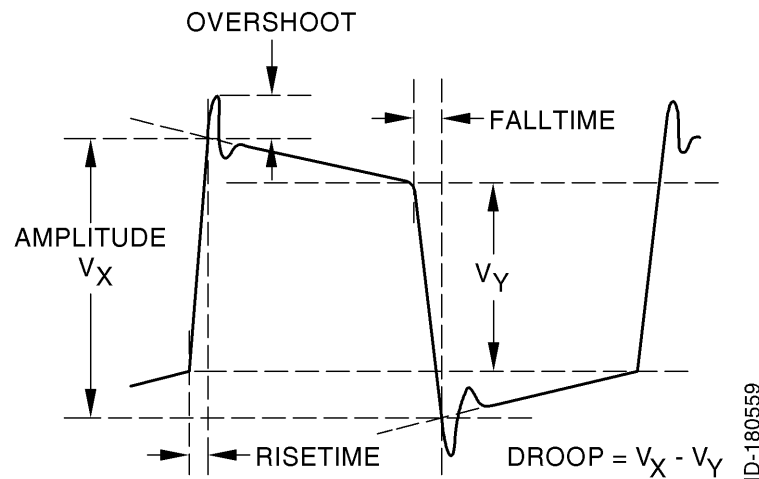
L. End of Functional Test

This concludes the Functional Test Procedure. Position the UFDR and TESTER POWER switches to OFF and disconnect the UUT from the test equipment. Record the individual track count.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Data Output Signal Waveform
Figure 105

3. **Power Supply, A1, Calibration Procedure**

WARNING: DURING CALIBRATION, WHEN POWER SUPPLY IS CONNECTED TO EXTENDER BOARD, BE EXTREMELY CAREFUL AS THERE IS 115 VAC COMING DIRECTLY INTO A1 POWER SUPPLY. TURN OFF TESTER POWER SWITCH WHEN HANDLING POWER SUPPLY, I.E., REMOVAL FROM CHASSIS, INSTALLING INTO CHASSIS, OR INSTALLING SELECT RESISTOR ONTO BOARD.

- Verify that UFDR POWER and TESTER POWER switches are positioned to OFF. Remove two screws (10, IPL Figure 1), lock washers (15), flat washers (20), and dust cover (5, 5A or 5B) from chassis assembly.
- Remove two screws (75, 90) and retainer (70, 70A). Remove and set aside A2, A3, A4 and A8 printed wiring boards.
- Remove eight screws (100 and 102) and four lock nuts (103), and pull power supply (95, 95A, 95B) from chassis or remove six screws (102A) and pull power supply (95C, 95D) from chassis.
- Install power supply extender board (Item 6, Figure 901) into chassis connector where power supply board was removed.
- Remove five screws (10, IPL Figure 2 or 15), flat washers (15), and EMI shield (5, 5A) from power supply PWB assembly (25, 25A, 25B).
- Remove two screws (30) and two washers (35) to loosen power supply assembly (25). There is a two inch service loop in the wiring between power supply assembly and power supply cover when these two assemblies are opened 180 degrees from each other.

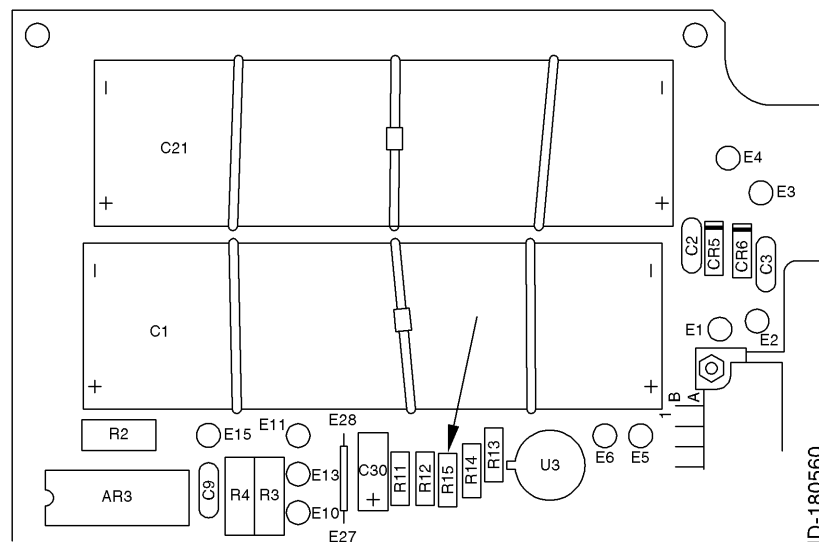
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- G. Install power supply assembly connector P1 into extender board and position rest of power supply so that wiring is not damaged or stressed and so that components on board are accessible.
- H. Apply electrical power to the test setup in accordance with the procedure given in paragraph 2B.
- I. Perform the following checks to determine if calibration is necessary or possible without prior repair.

NOTE: The A1A1 board contains a connector having right angle pins soldered to the board. These right angle pins are free of circuit board protective coating and provide good points to probe in the following steps. The top row of pins is the B row as shown in Figure 106.



Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (4) Slowly increase the UFDR VOLT ADJ control while monitoring the oscilloscope for a logic state change from 0 to 1. The DMM shall indicate 8.30 ± 0.08 VDC where logic state changes from 0 to 1. Note the actual value.
- (5) Slowly decrease the UFDR VOLT ADJ control while monitoring the oscilloscope for a logic state change from 1 to 0. The DMM shall indicate 0.50 to 0.70 VDC less than the voltage noted in step (4).

NOTE: For A1A1 boards with part number 722-2440-003 (i.e., boards with R16 installed), the off point tolerance is 0.70 to 0.95 VDC below 8.30 ± 0.08 VDC.

J. If steps I(4) and I(5) are out of tolerance, recalibrate in accordance with steps K and on.

K. Position UFDR POWER switch to OFF.

L. Unsolder and remove resistor A1A1R15 (see Figure 106 for location). Attach a decade resistor box, set at 200K, in place of A1A1R15 and perform the following procedure.

- (1) Position UFDR POWER switch to ON.
- (2) Decrease the UFDR VOLT ADJ control for a logic 0 state and then slowly increase control until the logic state changes from 0 to 1. Verify that the DMM indicates from +6 to +10 VDC. If not, terminate calibration and repair board.

NOTE: The calibration sequence that follows shall consist of monitoring the logic level on the oscilloscope and adjusting the decade resistance box such that the 0 to 1 transition occurs when the DMM indicates 8.30 ± 0.02 VDC.

- (3) Set UFDR VOLT ADJ control for a DMM indication of 8.30 ± 0.02 VDC. Adjust the decade resistance box until the logic level just changes state from 0 to 1 as indicated on the scope.
- (4) With the resistance decade box set at the value obtained in step L(3), slowly decrease UFDR VOLT ADJ control until the logic level changes state from 1 to 0. Slowly, to minimize overshoot, increase UFDR VOLT ADJ control until the logic level changes state from 0 to 1. Note the DMM indication.
- (5) If the DMM indication noted in step (4) is above 8.3 VDC, increase the decade resistance value and repeat step (4). Conversely, if the DMM indication is below 8.3 VDC, decrease the decade resistance value and repeat step (4).
- (6) Repeat steps (4) and (5) as necessary to find the proper resistance value for the logic trip point (0 to 1) to occur at a DMM indication of 8.30 ± 0.02 VDC.

NOTE: The circuit being calibrated in steps (4) through (6) has hysteresis. Once the logic state has changed from 0 to 1 at the particular DMM indication, the logic state will not revert back to the 0 level until the DMM is dropped a certain value below the value for a logic state 1 level. The device is properly calibrated when logic state transition from 0 to 1 occurs at 8.30 ± 0.02 VDC and 1 to 0 transition occurs at 0.70 to 0.95 VDC or 0.50 to 0.70 VDC, as applicable, below 8.30 ± 0.02 VDC.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- M. Position UFDR POWER switch to OFF. Remove decade resistance box and note the resistance value. Select closest valued resistor (165, IPL Figure 7) and install for A1A1R15. It is permissible to install resistors in parallel to obtain the required calibration. Solder resistor(s) in place.
- N. Position UFDR POWER switch to ON and repeat steps I(2) through I(5) to verify calibration. If calibration is satisfactory, remove 4.7K resistor, oscilloscope, and DMM.
- O. Position UFDR POWER and TEST POWER switches to OFF. Reassemble unit as follows.
 - (1) Remove power supply assembly from extender board and extender board from chassis. Apply protective coating to repaired areas in accordance with the **ASSEMBLY** Section, Step 2C.
 - (2) Reinstall power supply PWB assembly (25, IPL Figure 2 or 15) onto cover and secure with two screws (30) and two washers (35). Prior to installing screws, apply primer (Item 14, Figure 701) and staking compound (Item 15, Figure 701) to threads.
 - (3) Reinstall EMI shield (5) onto power supply assembly and secure with five screws (10) and five flat washers (15). Prior to installing screws, apply primer (Item 14, Figure 701) and staking compound (Item 15, Figure 701) to threads.
 - (4) Install power supply (95, 95A, 95B, IPL Figure 1) into chassis and secure with eight screws (100 and 102) and four lock nuts (103) or install power supply (95C, 95D) into chassis and secure with six screws (102A). Verify that connectors are fully inserted.
 - (5) Reinstall A2, A3, A4 and A8 printed wiring boards. Secure with retainer (70) and two screws (75, 90).
 - (6) Install dust cover (5 or 5A) and secure with two screws (10), lock washers (15) and flat washers (20).
- P. Perform Functional Test Procedure given in paragraphs 2C through 2L.

4. **Read/Write Interface. A2. Calibration Procedure**

- A. Verify that UFDR POWER and TESTER POWER switches are set to OFF. Remove two screws (10, IPL Figure 1), lock washers (15), flat washers (20), and dust cover (5, 5A or 5B) from chassis assembly.
- B. Remove two screws (75, 90) and retainer (70, 70A).
- C. Remove A2 assembly from chassis guides and install printed wiring extender board (Item 7, Figure 901) in A2 slot.
- D. Install A2 assembly into extender board.
- E. Apply electrical power to the test setup in accordance with the procedure given in paragraph 2B.
- F. Connect a jumper between P1A29 and P1A1.
- G. Connect the DMM, set to DC, (-) lead to A2 header GND pin and (+) lead to A2 header TP4. Verify that DMM indicates 12.80 ± 0.13 VDC. If not, calibrate as follows.
 - (1) Position UFDR POWER switch to OFF. Locate, unsolder, and remove resistor R54. See Figure 107 for location.

31-30-32

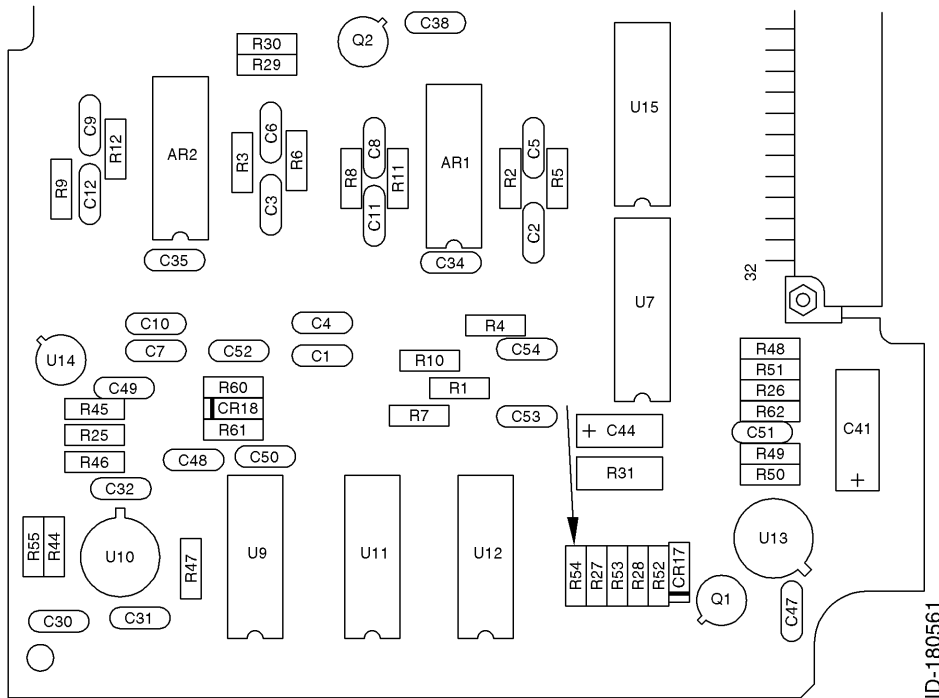
Page 123
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Read/Write Interface, A2., Resistor Location
Figure 107

- (2) In place of R54, attach a decade resistance box set to 200K.
- (3) Position UFDR POWER switch to ON.
- (4) Adjust the decade resistance box for a DMM indication of 12.80 ± 0.02 VDC.
- (5) Position UFDR POWER switch to OFF.
- (6) Remove decade resistance box and note the resistance value. Select closest valued resistor (150, IPL Figure 3) and install for R54. It is permissible to install resistors in parallel to obtain the required calibration. Solder resistor(s) in place.
***NOTE:** Higher valued resistors increase the voltage to TP4 and lower valued resistors decrease the voltage.*
- (7) Position UFDR POWER switch to ON and repeat step G.
- (8) If calibration is satisfactory, position UFDR POWER and TEST POWER switches to OFF. Disconnect DMM.

- H. Remove A2 assembly from extender board and extender board from chassis. Apply protective coating to repaired areas in accordance with **ASSEMBLY** section, Step 2C.
- I. Reinstall A2 assembly into chassis and verify that connectors are fully inserted.
 - (1) Reinstall retainer (70) and two screws (75, 90).

31-30-32

Page 124
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (2) Install dust cover (5, 5A or 5B) and secure with two screws (10), lock washers (15) and flat washers (20).

J. Perform Functional Test Procedure given in paragraphs 2C through 2L.

5. Recording System Controller. A4. Calibration Procedure

This procedure contains information for functional test and/or calibration of the under voltage monitor circuit, A4U17, located on Assembly A4.

A. Setup Procedure

- (1) Verify that UFDR POWER and TESTER POWER switches are set to OFF. Remove two screws (10, IPL Figure 1), lock washers (15), flat washers (20) and dust cover (5, 5A, or 5B) from chassis assembly.
- (2) Remove two screws (75, 90) and retainer (70).
- (3) Remove A4 assembly from chassis guides.
- (4) Install A4 assembly into extender board (Item 7, Figure 901).

NOTE: *The extender board, with the A4 assembly installed should not be reinstalled for the following test.*

- (5) Apply electrical power to the DC Power Supply (Item 8, Figure 901) and adjust it to produce $+5 \pm 0.15$ VDC. Position the DC Power Supply switch to OFF.
- (6) Connect the positive lead from the power supply to pin 32 on the extender board. Connect the common (ground) lead to pin B1.
- (7) Connect the scope input lead to pin B29 of the extender board. Connect the scope common to pin B2.

B. Check Procedure

NOTE: *The Check Procedure has a wider tolerance than the calibration procedure.*

- (1) On the DC power supply, slowly decrease the +5 VDC and observe the scope for a logic state change from HIGH to LOW.
- (2) When the logic state changes from HIGH to LOW, verify that power supply is at a point of 0.15 to 0.25 VDC below 4.70 ± 0.5 VDC.
- (3) On the DC Power Supply, slowly increase the +5 VDC and observe the scope for a logic state change from LOW to HIGH.

NOTE: *For A4 assemblies, part number 722-2438-00X (i.e., assemblies with R30 installed), the off point tolerance is 0.30 to 0.42 VDC below 4.70 ± 0.05 VDC.*

- (4) When the logic state changes from LOW to HIGH, verify that the power supply is at a point of 4.70 ± 0.05 VDC.

C. Calibration Procedure

If the voltage values in the check procedure are out of tolerance on the high side, the resistance value of A4R23 is too high. If out of tolerance on the low side, the resistance value of A4R23 is too low. Calibrate the circuit using the following procedure:

31-30-32

Page 125
22 Dec 2006

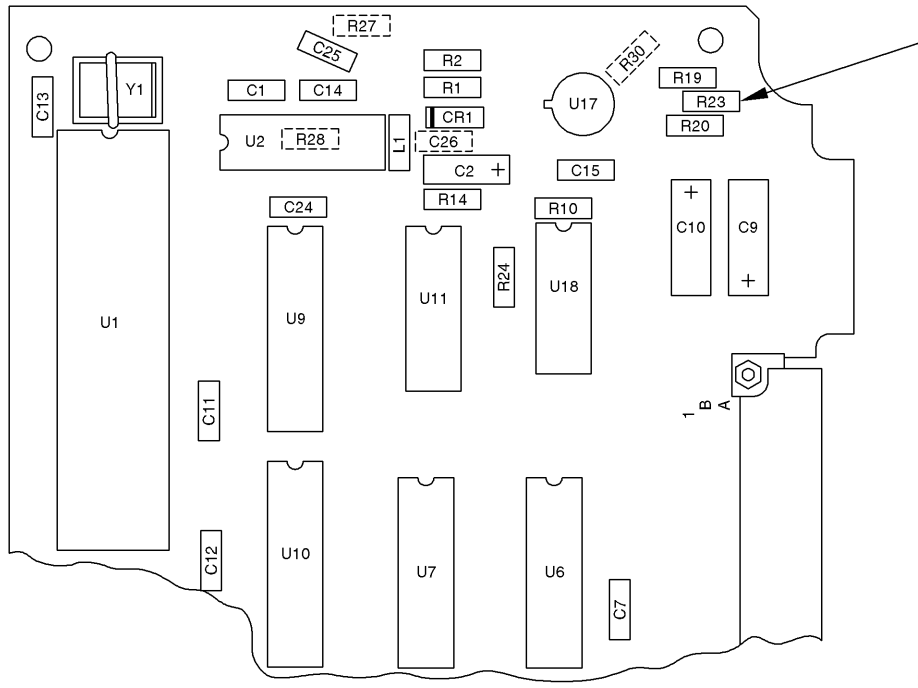
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (1) Position the DC Power Supply switch to off. Locate A4R23 (see Figure 108) and cut it out of the circuit, leaving the leads to serve as connection points.



Recorder System Controller, A4, Resistor Location
Figure 108

- (2) Adjust a decade resistance box (Item 13, Figure 901) to the approximate value of the removed A4R23 and connect the box to the leads left in step (1).
- (3) Position the DC Power Supply switch to on.
- (4) Increase or decrease the resistance of the decade resistance box as indicated by the results of the check procedure in paragraph 5B, above.
- (5) Repeat the check procedure (paragraph 5B) but use the tighter tolerance of 4.70 ± 0.02 VDC instead of 4.70 ± 0.05 VDC.
- (6) Readjust the decade resistance box and repeat the check procedure as required.
- (7) When the check procedure indicates satisfactory results, set the DC Power Supply to off, remove the decade resistance box, and install a select resistor (180, IPL Figure 5) closest in value to the final setting of the decade resistance box. Do not solder in place at this time.
- (8) Repeat check procedure (paragraph 5B) using the wider tolerance.

31-30-32

Page 126
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (9) If the check procedure indicates satisfactory results, position the DC Power Supply switch to off and solder A4R23 in place. Disconnect test equipment and reinstall the A4 assembly.
- (10) If the check procedure indicates unsatisfactory results, repeat steps 1 through 7 above.

6. Troubleshooting (UFDR Tester Users)

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES ARE SUBJECT TO DAMAGE BY EXCESSIVE LEVELS OF VOLTAGE. REFER TO DOD-HDBK-263 FOR PROPER HANDLING PROCEDURES.

A. General

NOTE: *Paragraph 8 contains troubleshooting information for Readout and Test Unit (RTU) users.*

This portion of the Test and Troubleshooting section contains information which is designed to aid in the location of malfunctions which may occur in the equipment. Since the UFDR Tester (Item 1, Figure 901) is designed to test the UFDR and indicate both SYSTEM STATUS and BITE STATUS conditions, suspected malfunctioning UFDR's should be connected to the UFDR Tester and the Tester set up in accordance with the procedure given in paragraph 2B, steps (1) through (10). Once the Tester and suspected malfunctioning UFDR have been connected together, the next step is to apply power and begin a preliminary diagnostic procedure which will tend to isolate the cause of the malfunction to one of the major subassemblies. These are the Transport Assembly; the Power Supply Assembly, A1; the Read/Write Interface Assembly, A2; the Transport Interface Assembly, A3; the Recording System Controller Assembly, A4; and the Aircraft Wiring Interface Assembly, A8.

B. Built-In Test Equipment (BITE)

Since the UFDR has a built-in test system which is capable of detecting 95% of all fault conditions, it will be used to help isolate the source of malfunction(s). There are actually two fault indicating systems in the UFDR: the BITE indicating system and the Recording System STATUS indicating system. In the aircraft, both provide fault indications in the cockpit. The STATUS indicator is intended to indicate faults which are likely to be external to the UFDR. There are four conditions which could cause the STATUS indicator to illuminate. These are:

- Loss of input AC Power.
- Loss of input data.
- Signal at IPORT-2 indicating that UFDR is NOT in flight recording mode (normal for nonflight recording modes).
- A BITE fault condition is signaled.

There are three general conditions which can cause a BITE fault indication to be signaled. These are:

- Software detected fault condition (OPORT-2, Bit 6 (A3U14-16 set HIGH)).
- Lock-up circuit detected fault condition (A3U19A set LOW)
- Track select fault condition (A3U13A set LOW).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

BITE signals are transmitted by an OR gate circuit consisting of A3U20B, A3U11F, A3U12A and A3U6D to the A8 assembly to activate relay A8K1. This relay is normally deactivated until a BITE fault signal is sent to the A8 assembly. When A8K1 is activated, the BITE indicators on the front panel of the UFDR and the Tester are illuminated and the fault indicator in the aircraft cockpit is extinguished.

C. Causes of STATUS Indicator ON

(1) Software Detected Causes

Software detected system STATUS fault signals are transmitted by OPORT-2 (A3U14), bit 4, to deactivate relay A8K2. The relay is normally activated; however, when a STATUS fault signal is transmitted, it deactivates and the Tester STATUS indicator illuminates. Some software detected faults that can illuminate the STATUS indicator are:

- (a) Data receiver circuit faulty (A4): Signals lost or latched.
- (b) Input signals are 1's or all 0's - See paragraph 6F. (Check for fault indication at A3TP5.)
- (c) Demodulation 1-shot faulty (A4U4).
- (d) Data bus error (data bus is shared by transmitter and receiver): bus latched to single state.
- (e) No interrupt signal to service incoming data.
- (f) IPORT-2 (A3U15) faulty (causes software to detect: not in the flight record mode).
- (g) OPORT-2 (A3U14) faulty (causes software to detect: not in the flight record mode or biphasic data loss).
- (h) A BITE fault condition occurs.

(2) Hardware Detected Causes

Hardware detected STATUS fault conditions are generally faults which remove power from the relay, A8K2. Some hardware faults which can cause the STATUS indicator to illuminate are:

- (a) EMI filter faulty: Causes AC input loss.
- (b) Thermal switch open.
- (c) Open connection in STATUS indicator system: Relay contacts, logic gates latched, loss of +5 VDC or +24 VDC.

D. Causes of Software Detected Fault Condition

When OPORT-2 (A3U14) bit 6 is set HIGH, it indicates that the software has detected a fault condition in one of the following:

- (1) EOT/BOT sensors.
- (2) Data Process Fault 1: CHEKIN \Not= CHEKWR: OPORT-1 (A3U3) bit 1 set LOW - can be checked at A3TP6.
- (3) Data Process Fault 2: RAM buffer contains all 1's or all 0's - can be checked at A3TP5.

- (4) Data Read/Write Fault: CHEKRD \Not= CHEKWR; OPORT-2 (A3U14) bit 5 set LOW - can be checked at J2-L.
- (5) Track select fault.

E. Causes of Data Process Fault 1: CHEKIN \Not= CHEKWR

When data process fault 1 is detected by the software, a logic LOW is transmitted via OPORT-1 (A3U3) to test point A3TP6.

- (1) Faulty Recording System Controller Assembly, A4
 - (a) RAM (A4U6/U7) data buffer A/B faulty (data bit error or latched to a single state).
 - (b) Master Clock Generator (A4U2) faulty (02 TTL clock latched - provides clock pulses to the data receiver/transmitter).
 - (c) Data Transmitter (A4U5) faulty (data lost, serial output latched).
 - (d) 4XTXCLK (A4U4) faulty (data bit missing).
 - (e) CHEKDATA counter (A4U4) faulty (check data error).
 - (f) No interrupt signal to service data transmission (causes system lock-up, also).
- (2) Faulty Transport Interface Assembly, A3
 - (a) OPORT-0 (A3U2-5) write enable signal, bit 1, latched HIGH.
 - (b) OPORT-0 (A3U2-5) write enable signal, bit 1, latched LOW. (Also causes a data read/write fault - OPORT-2, bit 5 set LOW - can be checked at J2-L).
 - (c) OPORT-0 (A3U2-6) TXEMPT signal, bit 2, latched HIGH or LOW.
 - (d) OPORT-0 (A3U2-2) -PLAYBACK DATA ENABLE signal, bit 0 latched LOW (also causes a data read/write fault).
 - (e) OPORT-0 (A3U2-11) write control signal latched (also causes data read/write fault and track select fault).
 - (f) NRZ to biphase modulator circuitry faulty (see paragraph 3D(8) **DESCRIPTION AND OPERATION** section for circuitry description).
- (3) Fault Read/Write Interface Assembly, A2
 - (a) Write enable circuit (A2U12A/B and/or U9A/B/D faulty).
 - (b) Multiplexer circuit (A2U1/U2) faulty.

F. Causes of Data Process Fault 2: RAM buffer contains all 1's or 0's

When a data process fault 2 is detected by the software, a logic LOW is transmitted via OPORT-1 (A3U3), bit 0, to A3TP5.

- (1) Fault Recording System Controller Assembly, A4
 - (a) Data Receiver (A4U13A/B, A4U15B, A4U5) faulty.
 - (b) Master Clock (A4U2) faulty: 02 TTL clock latched (clock pulses for data transmitter). (Also causes data process fault 1 indication - logic LOW at A3TP6).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (2) Faulty Airframe Wiring Interface Assembly, A8
Data Receiver (A8Q1/Q2/U1/U5B) faulty.

G. Causes of Data Read/Write Fault Condition: CHEKRD \Not= CHEKWR

When CHEKRD does not equal CHEKWR, OPORT-2 (A3U14), bit 5, is set to a logic LOW. Eight consecutive read/write faults are required to cause OPORT-2, bit 6, to signal a BITE condition. The signal from OPORT-2, bit 5, can be observed on J2-L.

- (1) Tape Movement Faulty (Tape Transport Assembly)
 - (a) Tape dirty, pitted, worn or jammed.
 - (b) Tape motion erratic due to faulty tape transport assembly.
 - (c) Belt(s) broken.
 - (d) Tape broken and/or off reel(s).
 - (e) Motor erratic or stopped: motor bearings seized.
 - (f) Capstan and/or reel bearing(s) seized.
 - (g) Wrong sensor ON, i.e., EOT when tape is at BOT.
 - (h) Sensor not on and data is written to tape clear leader.
 - (i) Read/Write Head(s) dirty, worn or damaged.
 - (j) Erase head dirty, worn or damaged.
- (2) Power Supply Faulty
 - (a) Motor power supply open or unregulated.
 - (b) ± 15 V power supply output open; unregulated, excess ripple.
- (3) Faulty Read/Write Interface Assembly, A2
 - (a) Write current circuit (A2U13/Q1/Q2) faulty. (Check at A2TP4.)
 - (b) Playback multiplexers (A2U7/U8) faulty.
 - (c) Playback preamps (A2AR1A through AR4D) faulty.
 - (d) Head channel multiplexers (A2U1/U2) faulty.
 - (e) Playback data check circuitry (A2AR5A through AR5D, U10 and U9C) faulty. (Can be checked at A2TP1, TP2, TP3 and TP5.)
- (4) Faulty Transport Interface Assembly, A3
 - (a) Motor phase selection in error or not selected.
 - (b) OPORT-2 (A3U2) faulty: signals latched. Output signals from port are: read enable, write enable, TX EMPTY, track memory save, track select fault.
 - (c) OPORT-1 (A3U3) faulty: signals latched. Output signals from port are: C3OUT, INT enable, motor hold, motor phases 1, 2, 3 and 4.

- (d) OPORT-2 (A3U14) faulty: signals latched. Output signals from port are: erase enable, software detect fail, data check error, system STATUS, external control A, B and C.
- (5) Faulty Recorder System Controller Assembly, A4
 - (a) CPU faulty: address word error - bit(s) latched or lost.
 - (b) CPU faulty: data word error - bit(s) latched or lost.

H. Causes of Lock-Up Circuit Detected Fault

A system lock-up state occurs whenever the lock-up detector, A3U19A, produces a logic LOW at its Q output terminal. The lock-up detect circuit is designed to produce a logic LOW any time the CPU does NOT read IPORT-2 for more than six seconds. Therefore, anything that prevents or delays the CPU from reading IPORT-2 causes a lock-up signal to be produced. When the lock-up circuit signals a fault condition, a logic HIGH is transmitted by A3U19A-4 to A3TP3.

- (1) Faulty Power Supply Assembly, A1
 - (a) -5V open or unregulated (provides power to CPU).
 - (b) -12V output open (affects -5 VDC).
 - (c) +12V output open, unregulated or too much ripple.
 - (d) +20V output open (no power to write drive circuits).
 - (e) +5V output too low (CPU will be in reset state).
 - (f) +5V has too much ripple or is unregulated (induces logic error throughout system) (software monitored faults will probably also result).
 - (g) Output of restart circuit, A1U3, stays at a logic LOW at all times (output normally goes to a logic LOW to reset CPU when AC input goes below tolerance).
- (2) VCMOS Supply (A3 Assembly) to RAM
 - (a) No output (protects RAM from data loss due to momentary power loss).
- (3) Faulty Recording System Controller Assembly, A4
 - (a) CPU address word error (A0 - A7): bit latched or lost.
 - (b) CPU data word error (D0 - D7): bit latched or lost.
 - (c) CPU read/write control error: bit latched or lost.
 - (d) PROM (A4U8) data bit error or address word error.
 - (e) RAM (A4U6/U7) latched to a single state or data bit error.
 - (f) Interrupt control (A4U14) output latched HIGH or LOW or loss of any interrupt signal.
 - (g) Master clock (A4U2) ready signal latched LOW or HIGH, reset signal latched LOW or HIGH, strobe signal latched LOW or HIGH, clock pulses lost or latched.
 - (h) +5 VDC monitor (A4U17) output latched LOW.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

I. Causes of Track Select Fault Condition: A3U13-13 Set LOW and/or OPORT-2, Bit 6, set HIGH

(1) Faulty Power Supply, A1: -12 VDC low or unregulated (provides power to A3U13).

(2) Transport Interface Assembly, A3, Faulty

(a) IPORT-0 (A3U1) faulty (CPU track select readback signals).

NOTE: If signals latched, EOT/BOT also software detected.

(b) OPORT-0 (A3U2) faulty (Track select signals to A3U13).

NOTE: If signals in this port are latched, a data read/write fault (logic LOW at J2-L) and a data process fault 1 (logic LOW at A3TP6) also are indicated.

(c) Track select data latch (A3U13) faulty.

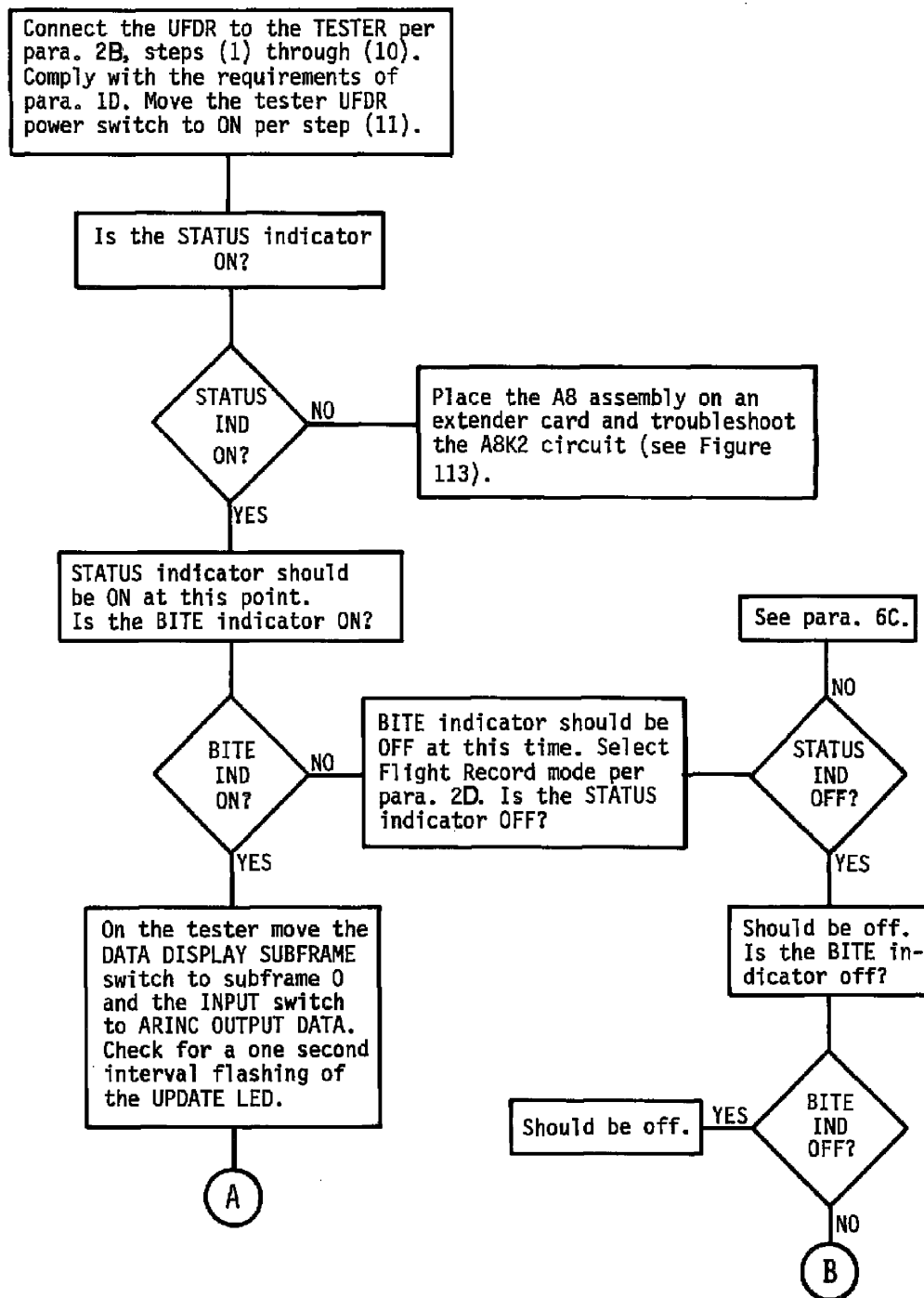
J. Fault Logic Diagrams

NOTE: Paragraph 8 contains fault logic diagram information for Readout and Test Unit (RTU) users.

The fault logic diagrams given in Figures 109 through 115 provide guidance for isolating faults explicitly indicated by the BITE and/or STATUS indicator system(s) of the UFDR. Malfunctions resulting from degraded recorder operation require the performance of the Functional Test Procedure given in paragraphs 2B through 2L of this section.

The fault logic diagrams given in Figure 109 provide guidance for isolating faults only to the card (assembly) level. The fault logic diagrams given in Figures 110 through 115 provide guidance for isolating faults to the component and/or circuit level on the affected card or subassembly. Although an attempt has been made to cover a wide spectrum of possible faults, no attempt has been made to provide an exhaustive analysis of all conceivable faults that could occur. Rather, the aim is to provide guidance in the solving of some kinds of malfunctions with the view that they will suggest other avenues of effort if a particularly difficult problem arises.

Before attempting to diagnose a suspected malfunctioning UFDR, first become thoroughly familiar with the information given in paragraphs 6A through 6I, then perform the tests given in Figure 109. When the apparent cause of the malfunction has been isolated to a particular PWB Assembly or to the Tape Recorder Assembly, perform the tests called out in the fault logic diagrams for that assembly.



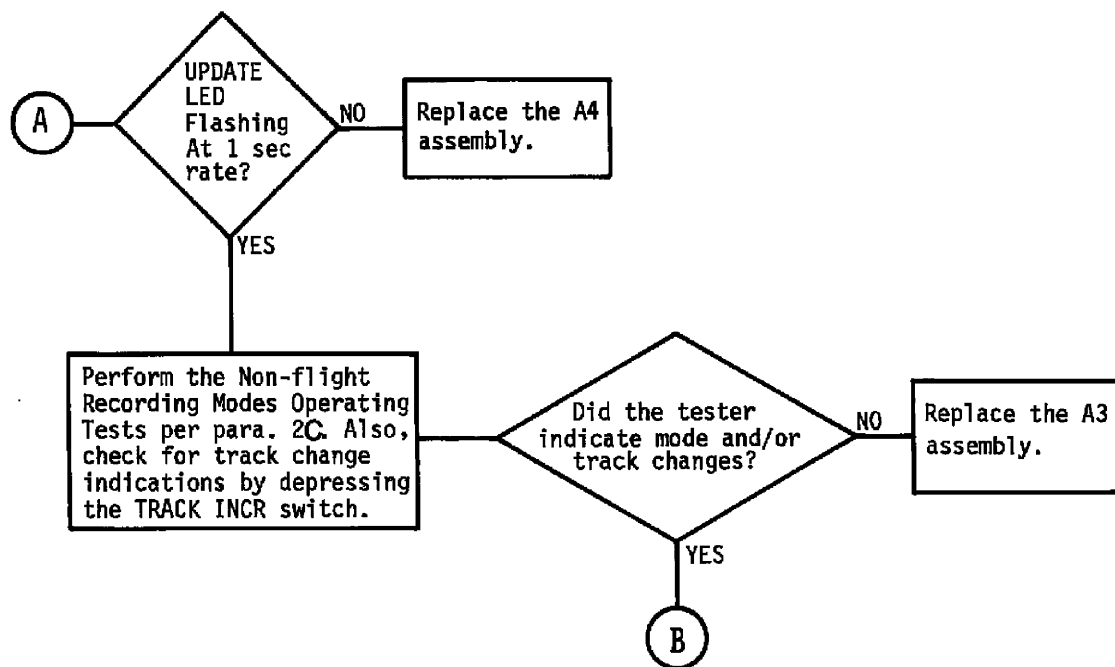
ID-180563

Assembly Level Fault Logic Diagram
Figure 109 (Sheet 1 of 6)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180564

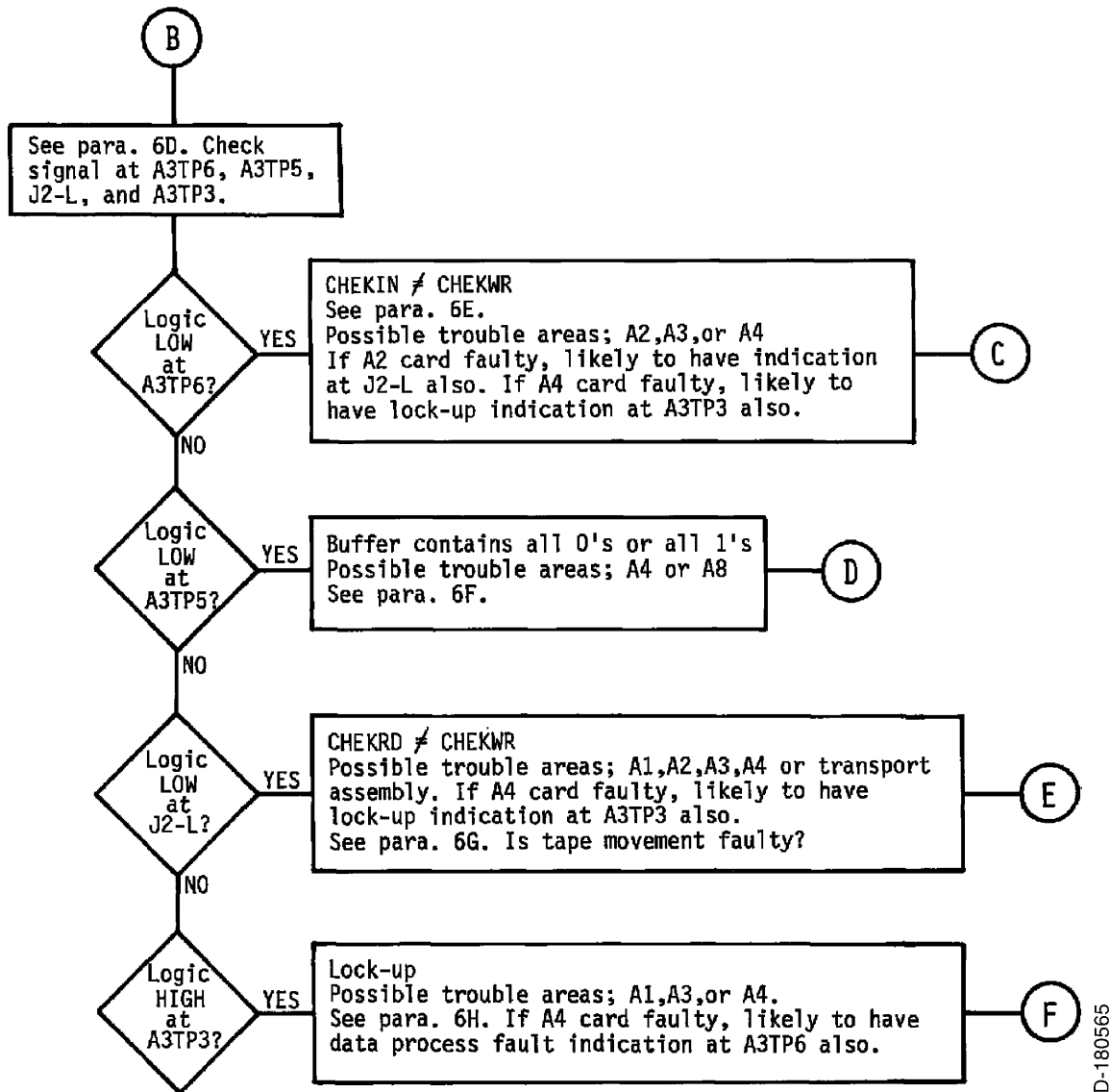
Assembly Level Fault Logic Diagram
Figure 109 (Sheet 2 of 6)

31-30-32

Page 134
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

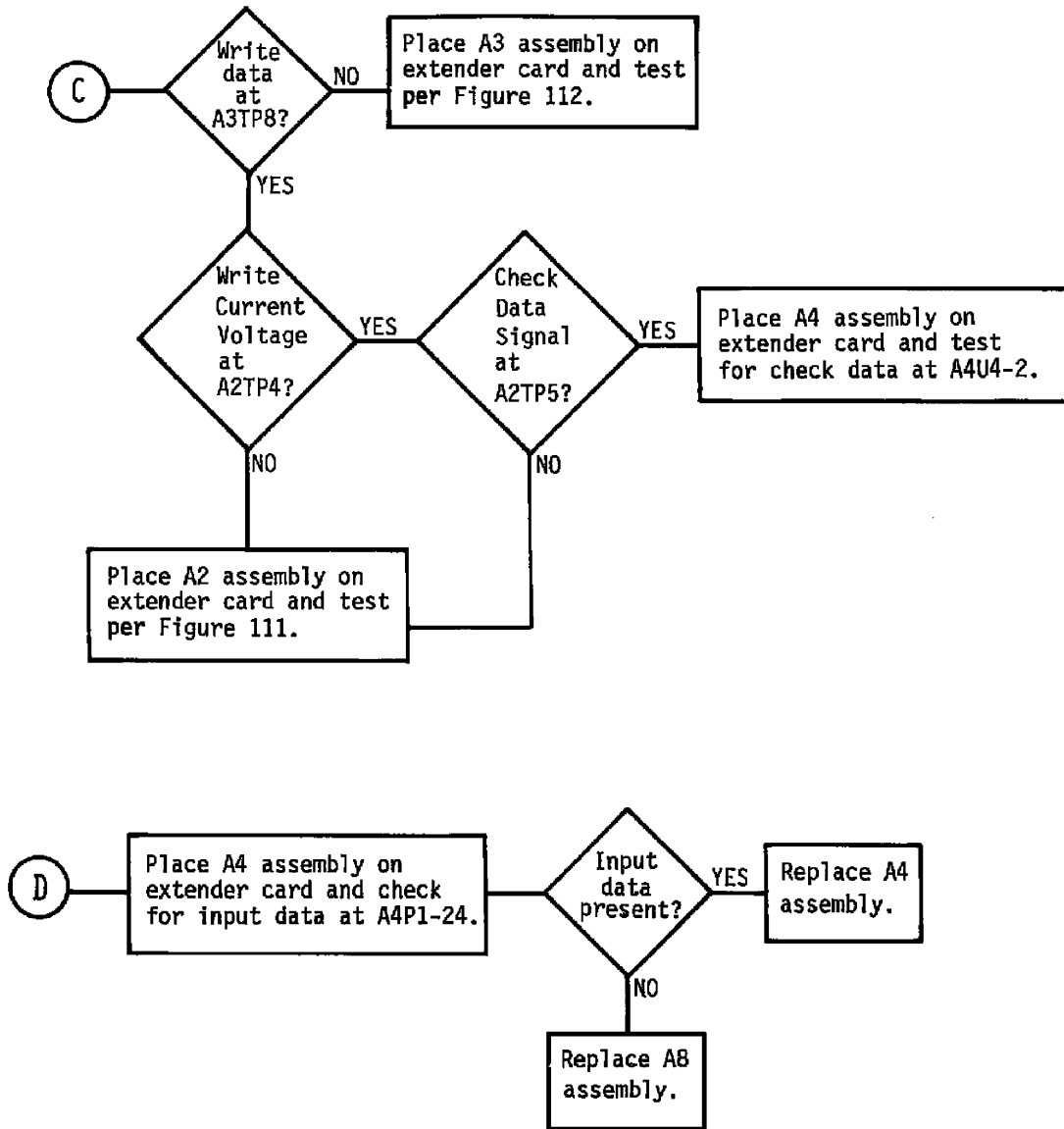


Assembly Level Fault Logic Diagram
Figure 109 (Sheet 3 of 6)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180566

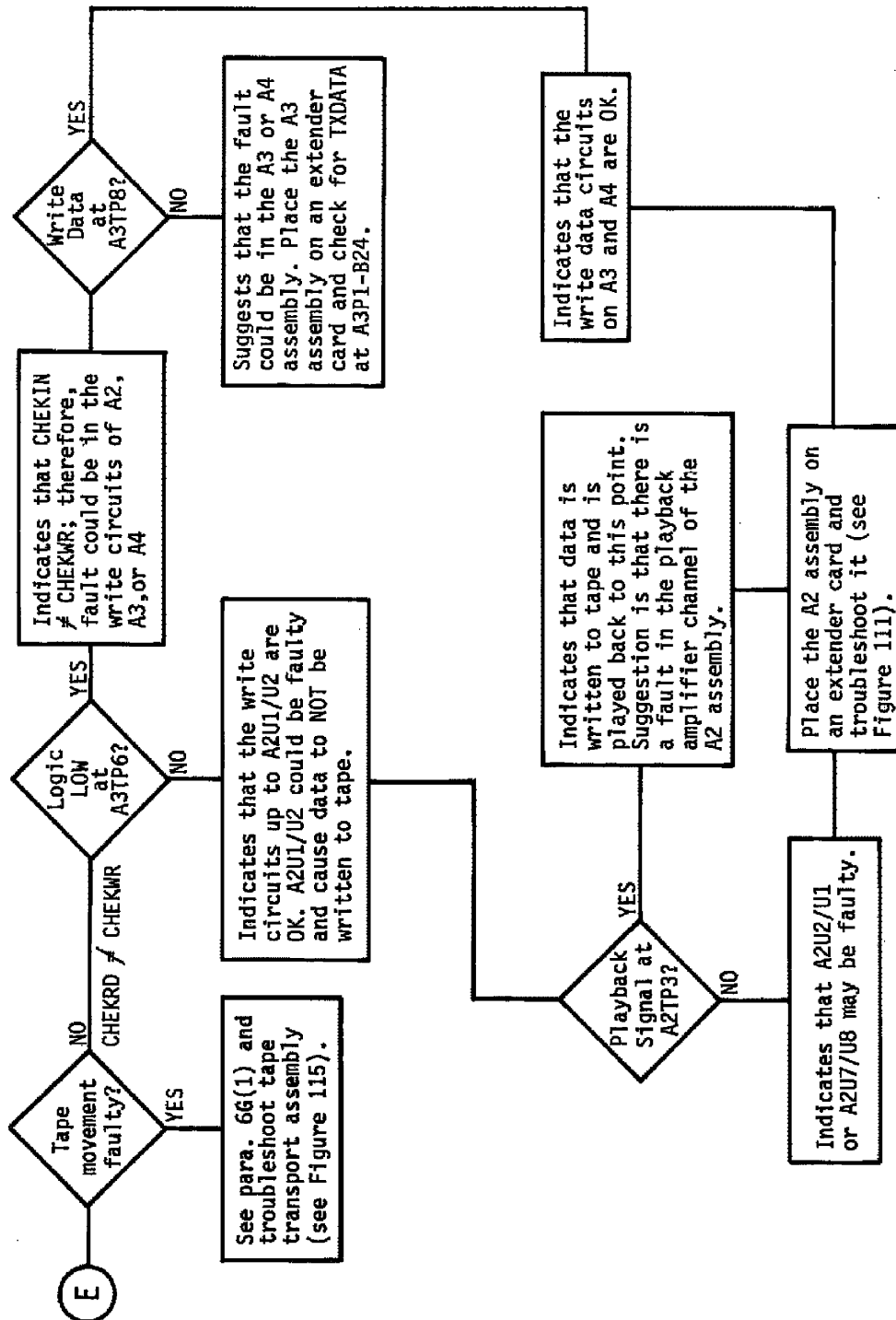
Assembly Level Fault Logic Diagram
Figure 109 (Sheet 4 of 6)

31-30-32

Page 136
22 Dec 2006

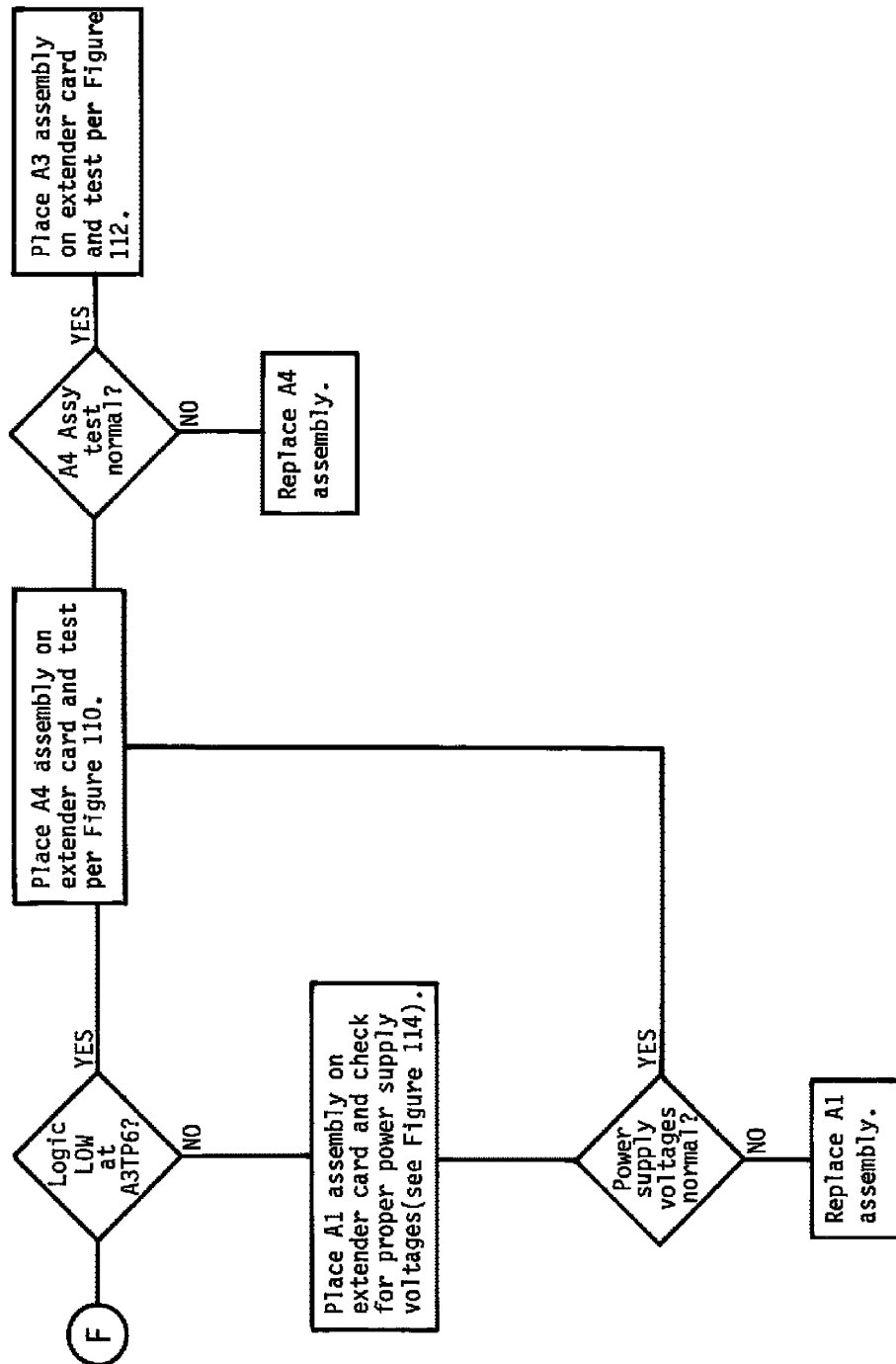
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA



ID-180567

Assembly Level Fault Logic Diagram
Figure 109 (Sheet 5 of 6)



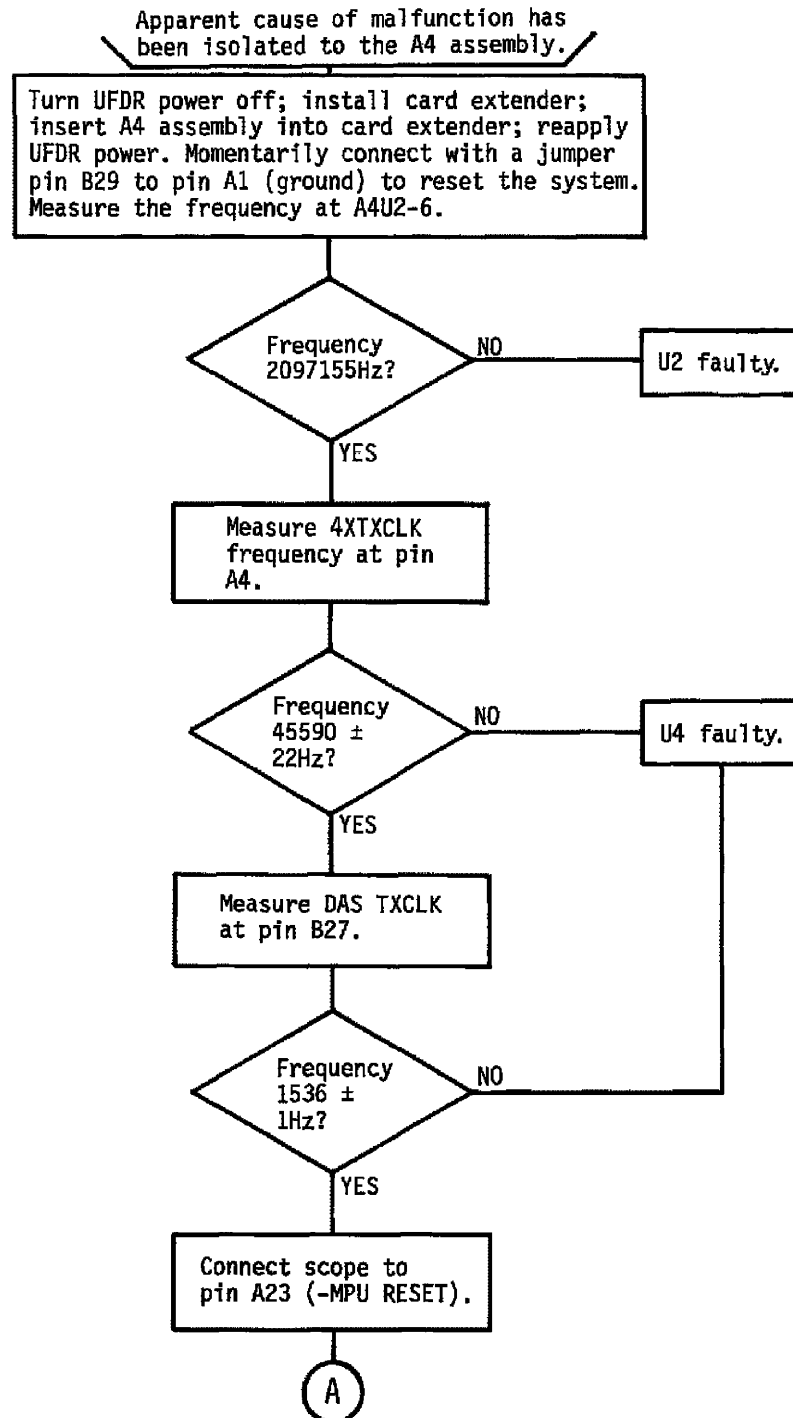
ID-180568

Assembly Level Fault Logic Diagram
Figure 109 (Sheet 6 of 6)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Recording System Controller Assembly, A4, Fault Logic Diagram
Figure 110 (Sheet 1 of 5)

31-30-32

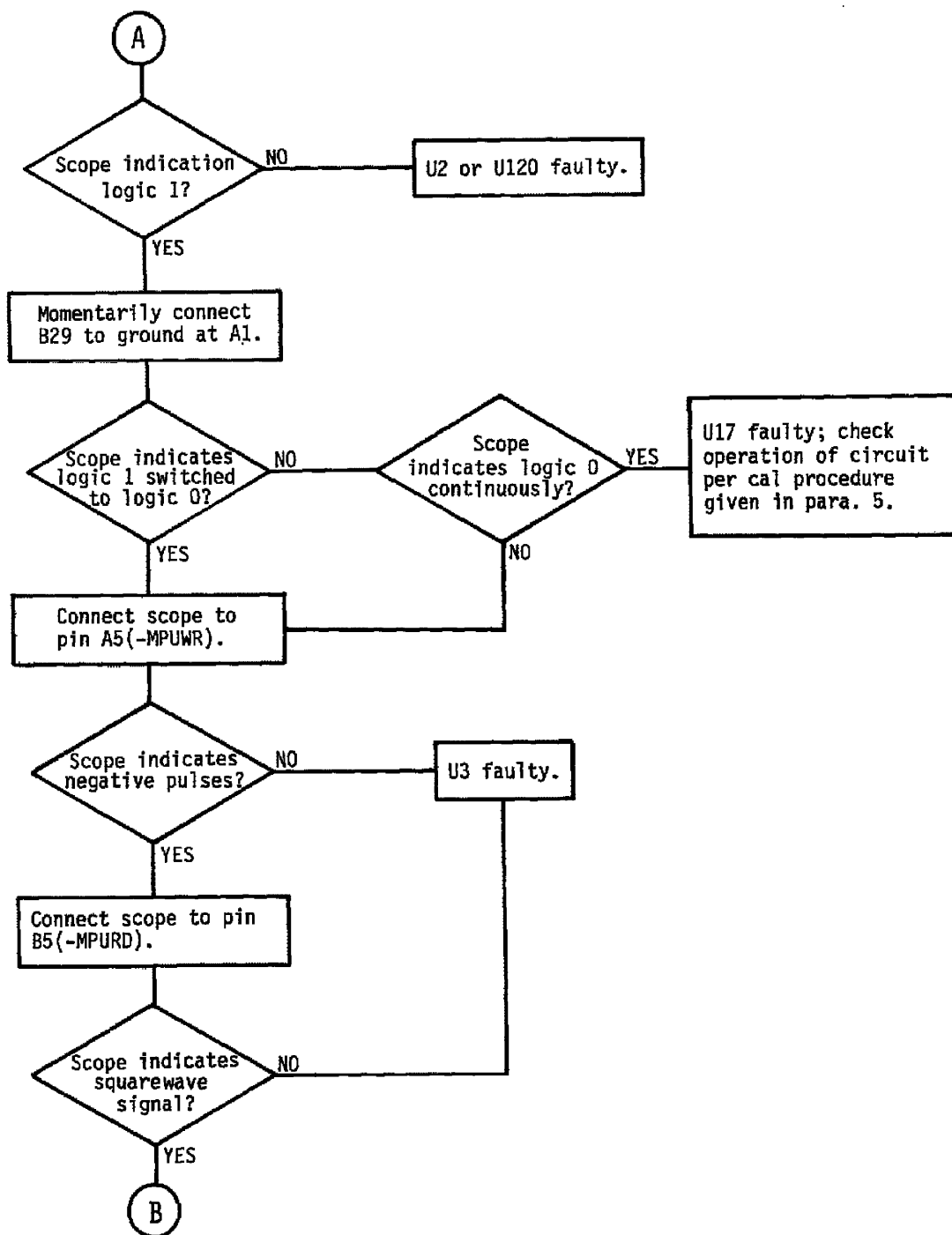
Page 139
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180570

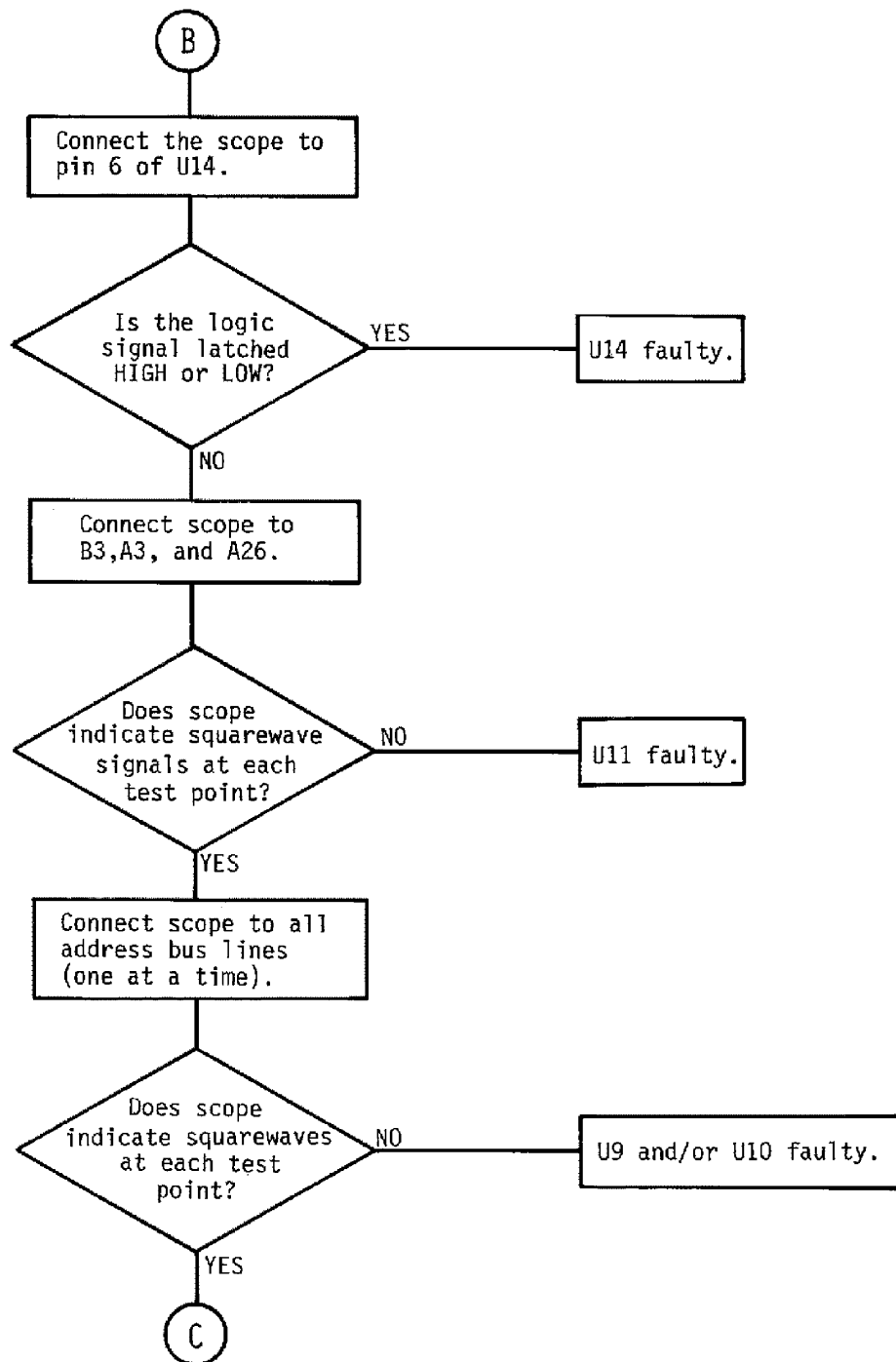
Recording System Controller Assembly, A4, Fault Logic Diagram
Figure 110 (Sheet 2 of 5)

31-30-32

Page 140
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTY y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA



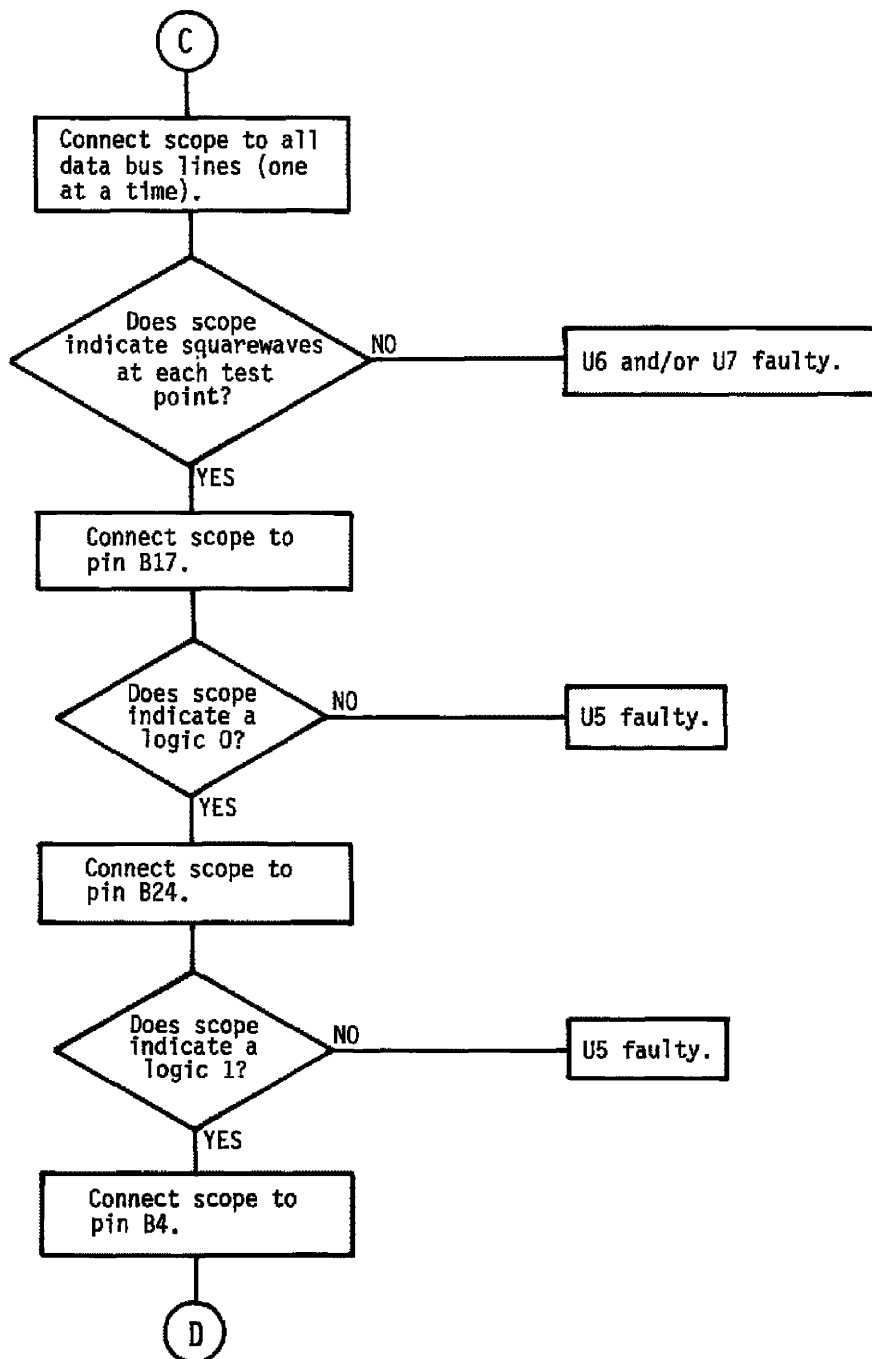
ID-180571

Recording System Controller Assembly, A4, Fault Logic Diagram
Figure 110 (Sheet 3 of 5)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180572

Recording System Controller Assembly, A4, Fault Logic Diagram
Figure 110 (Sheet 4 of 5)

31-30-32

Page 142
22 Dec 2006

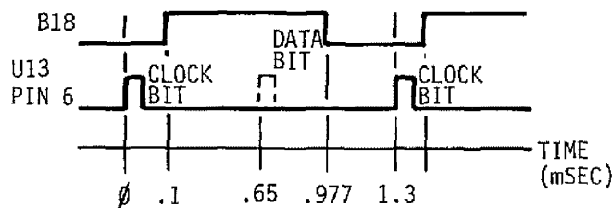
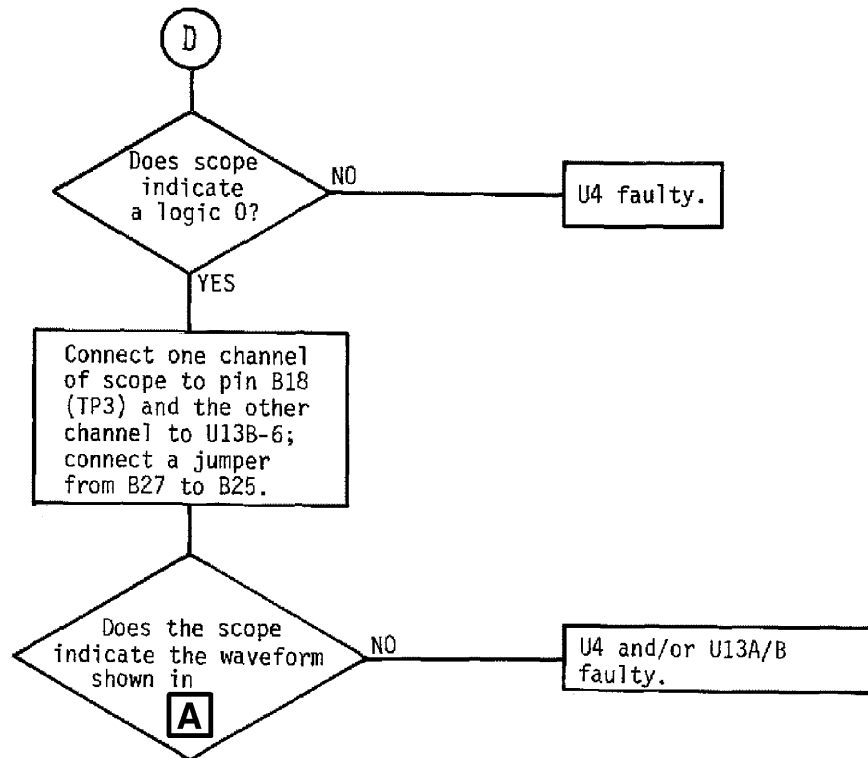
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



A

ID-180573

Recording System Controller Assembly, A4, Fault Logic Diagram
Figure 110 (Sheet 5 of 5)

31-30-32

Page 143
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Apparent cause of malfunction has been isolated to the A2 assembly. CHEKIN $\neq$ CHEKWR and/or CHEKRD $\neq$ CHERWR. See para. 6E(3) and 6G(3).

Turn UFDR power off; install card extender; insert A2 assembly into extender; reapply power. Connect DMM to A2TP4. Connect jumper between A2P1-A29 and A2P1-A1.

DMM indicates 12.8 ± 0.13 VDC?

NO

If DMM indication is 0.0 VDC, U13,Q1,Q2 or circuit component(s) faulty. If DMM is out of tolerance, perform calibration procedure given in para. 4.

YES

Set-up flight record mode per para. 2D. Connect scope to A2TP2/TP3.

Playback data present?

NO

Connect scope to A2U1-12.

YES

Connect scope to A2TP5.

B

Write data present?

NO

A2U9A/B faulty.

YES

A

Playback data present?

YES

A2U9C/D faulty.

NO

A2AR5A,B,C or D faulty.

ID-180574

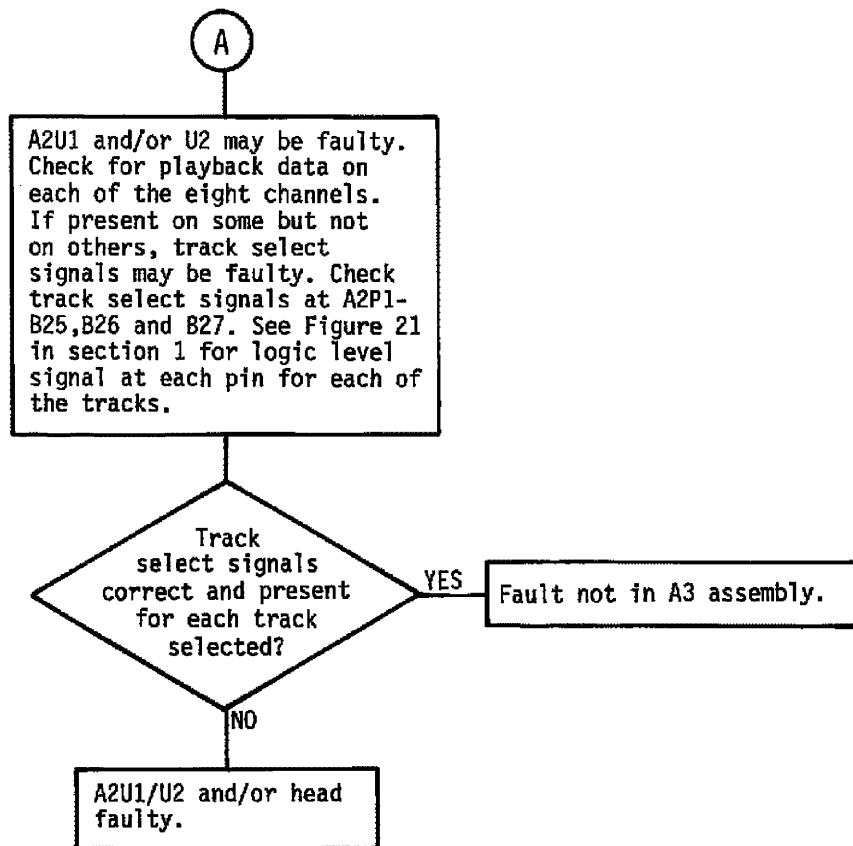
Read/Write Interface Assembly, A2, Fault Logic Diagram
Figure 111 (Sheet 1 of 3)

31-30-32

Page 144
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTv y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA



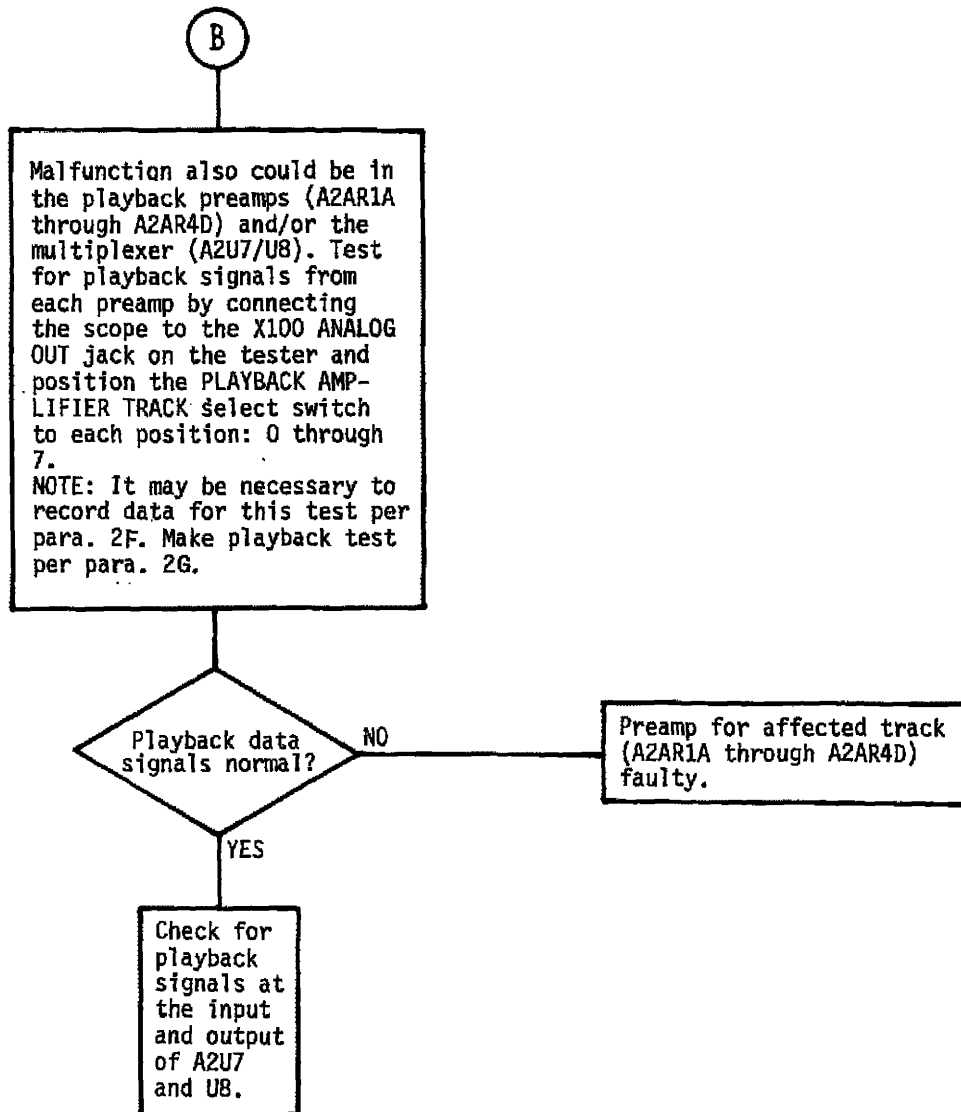
ID-180575

Read/Write Interface Assembly, A2, Fault Logic Diagram
Figure 111 (Sheet 2 of 3)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180576

Read/Write Interface Assembly, A2, Fault Logic Diagram
Figure 111 (Sheet 3 of 3)

31-30-32

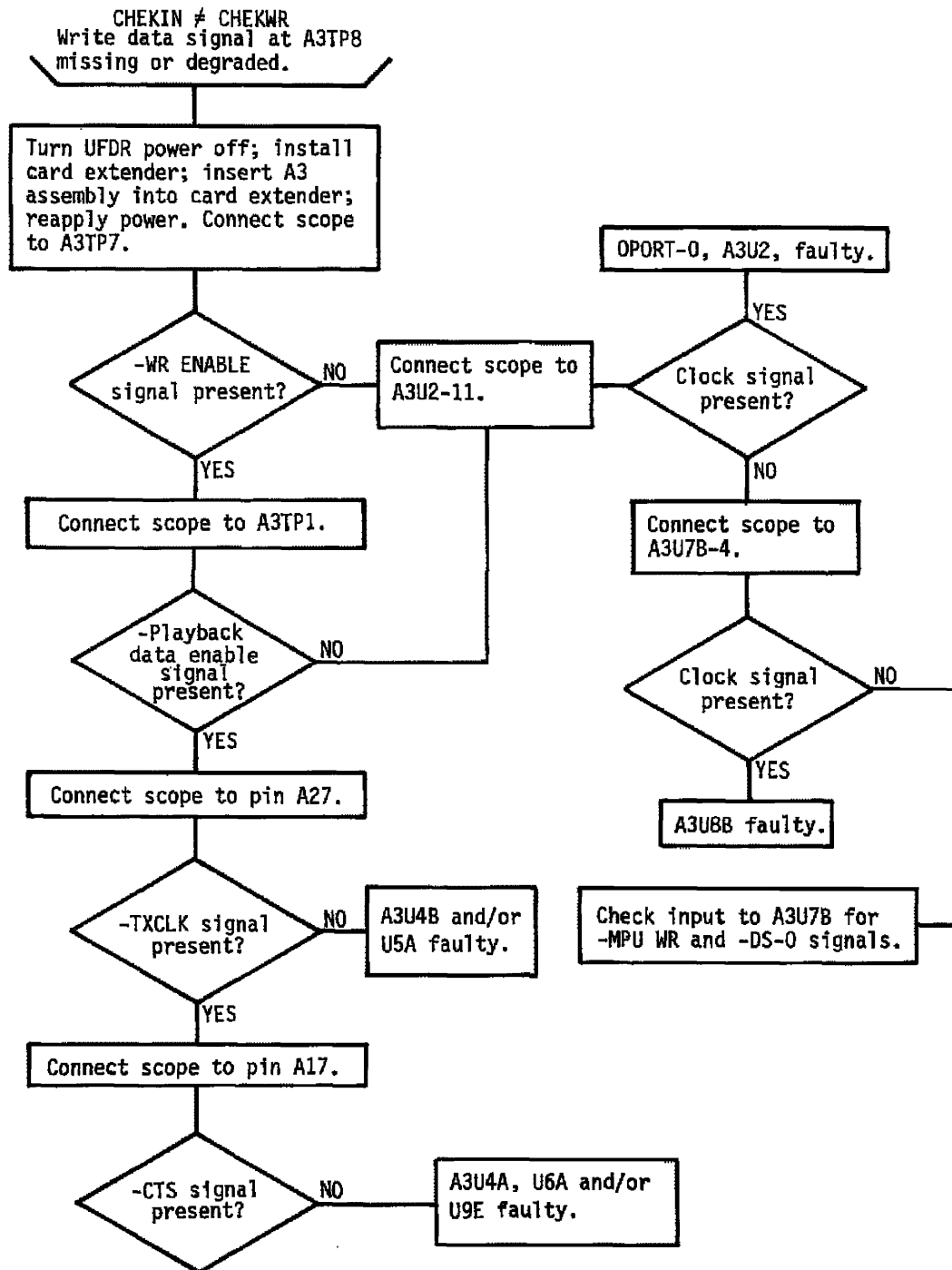
Page 146
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Transport Interface Assembly, A3, Fault Logic Diagram
Figure 112 (Sheet 1 of 3)

31-30-32

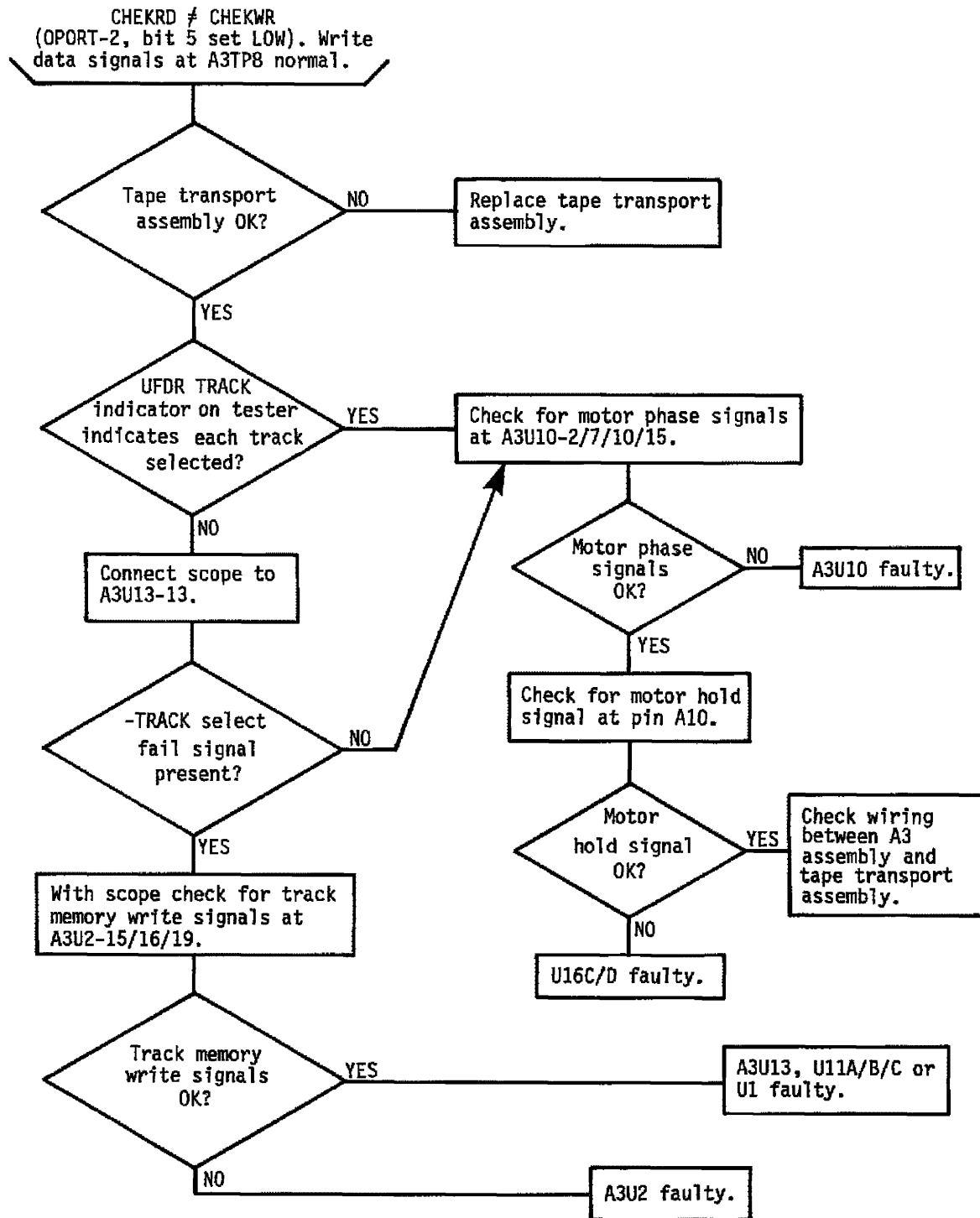
Page 147
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180578

Transport Interface Assembly, A3, Fault Logic Diagram
Figure 112 (Sheet 2 of 3)

31-30-32

Page 148
22 Dec 2006

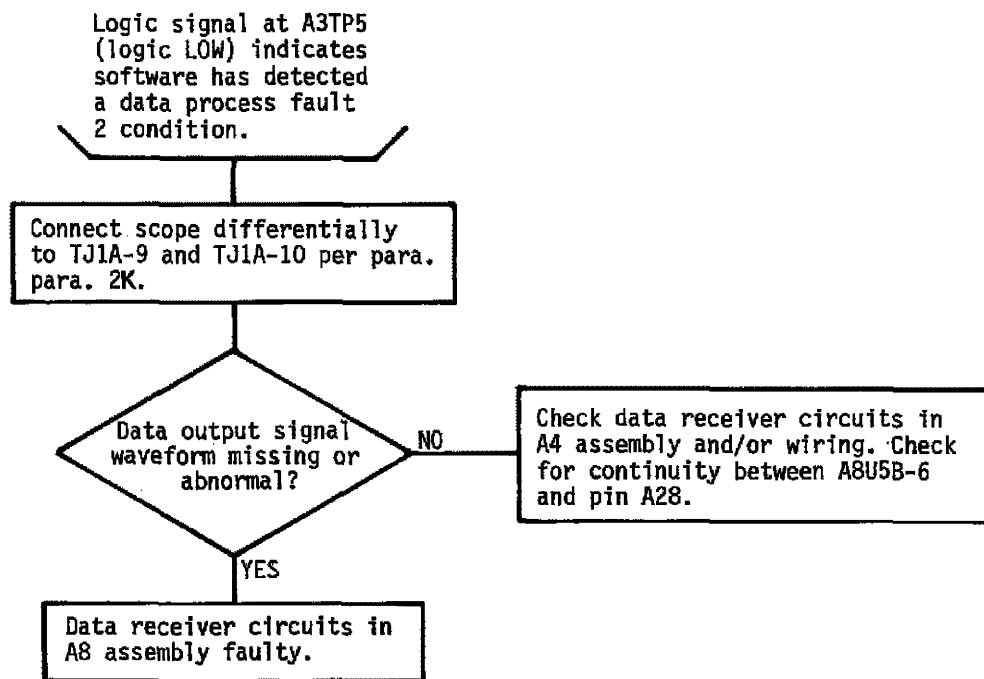
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTv y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Transport Interface Assembly, A3, Fault Logic Diagram
Figure 112 (Sheet 3 of 3)

31-30-32

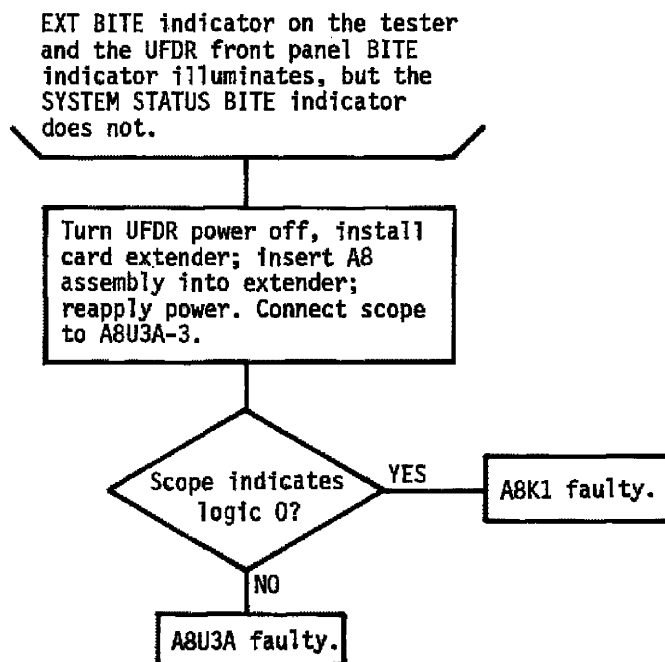
Page 149
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180580

Aircraft Wiring Interface Assembly, A8, Fault Logic Diagram
Figure 113

31-30-32

Page 150
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

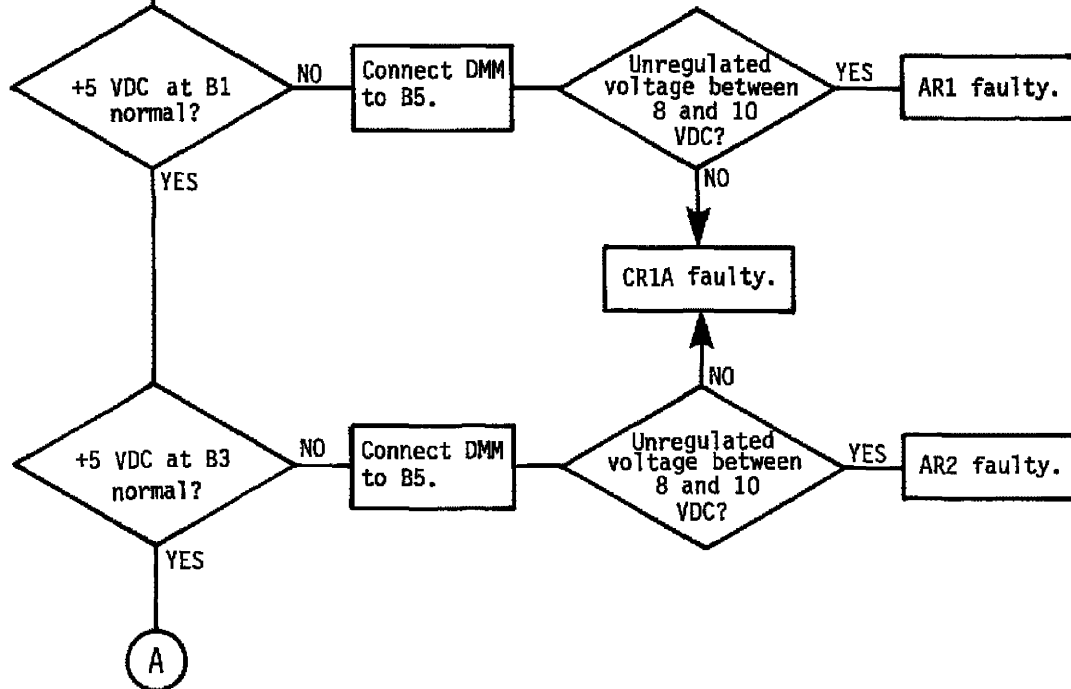
Lock-up signalled by logic signal at A3TP3; signal at A3TP6 is a logic HIGH. Suspect power supply is faulty.

Turn UFDR power off; install power supply extender; insert power supply assembly into extender; reapply power.

Connect DMM to each of the test points and check for voltage indication.

TEST POINTS

Signal pin	GND pin	Nom. Voltage	Min DC	Max DC
B1	B31	+5V.-A	4.9	5.2
B3	B31	+5V.-B	4.9	5.2
B26	B31	+12V.	11.65	12.5
B24	B31	-12V.	-11.65	-12.5
B30	B27	+15V.	14.85	15.2
B29	B27	-15V.	-14.85	-15.2
B19	B14	+16V.	14.9	16.2
B25	B31	-5V.	-4.85	-5.2
B6	B27	+20 UNREG		



Power Supply Assembly, A1, Fault Logic Diagram
Figure 114 (Sheet 1 of 2)

ID-180581

31-30-32

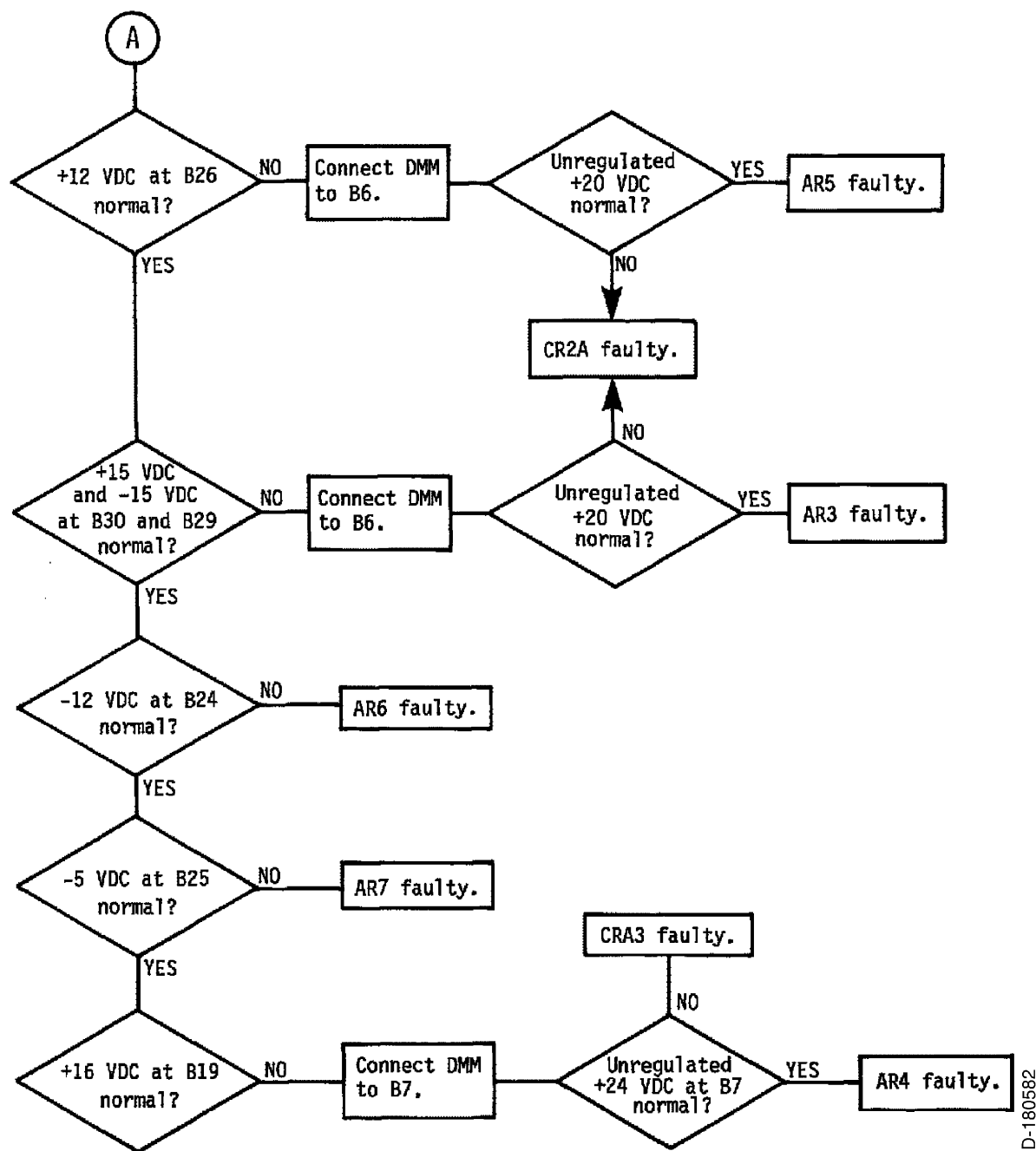
Page 151
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

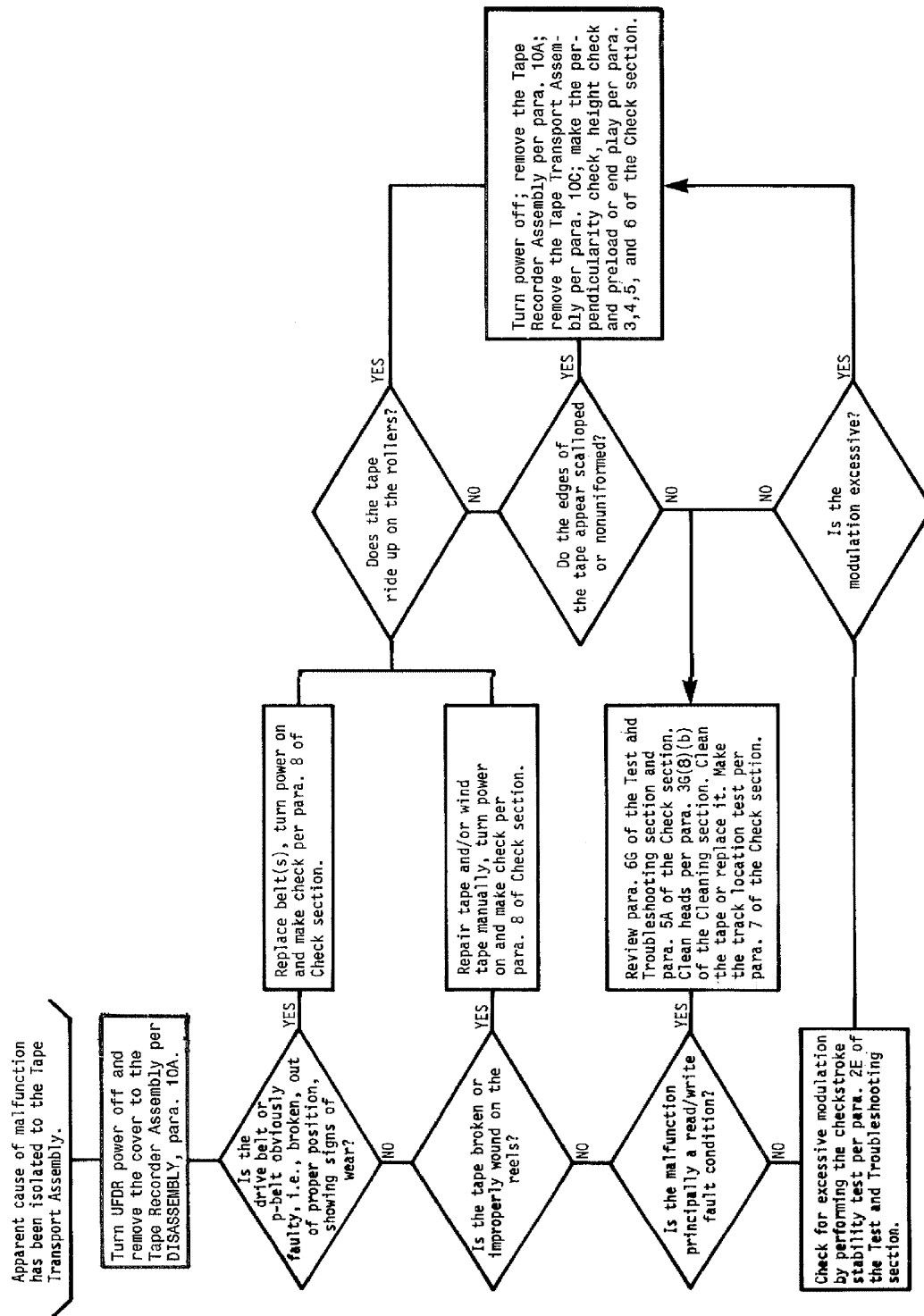


Power Supply Assembly, A1, Fault Logic Diagram
Figure 114 (Sheet 2 of 2)

31-30-32

Page 152
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.



ID-180583

Tape Reproducer Assembly, Fault Logic Diagram
Figure 115

31-30-32

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

7. UFDR RTU Test Procedure

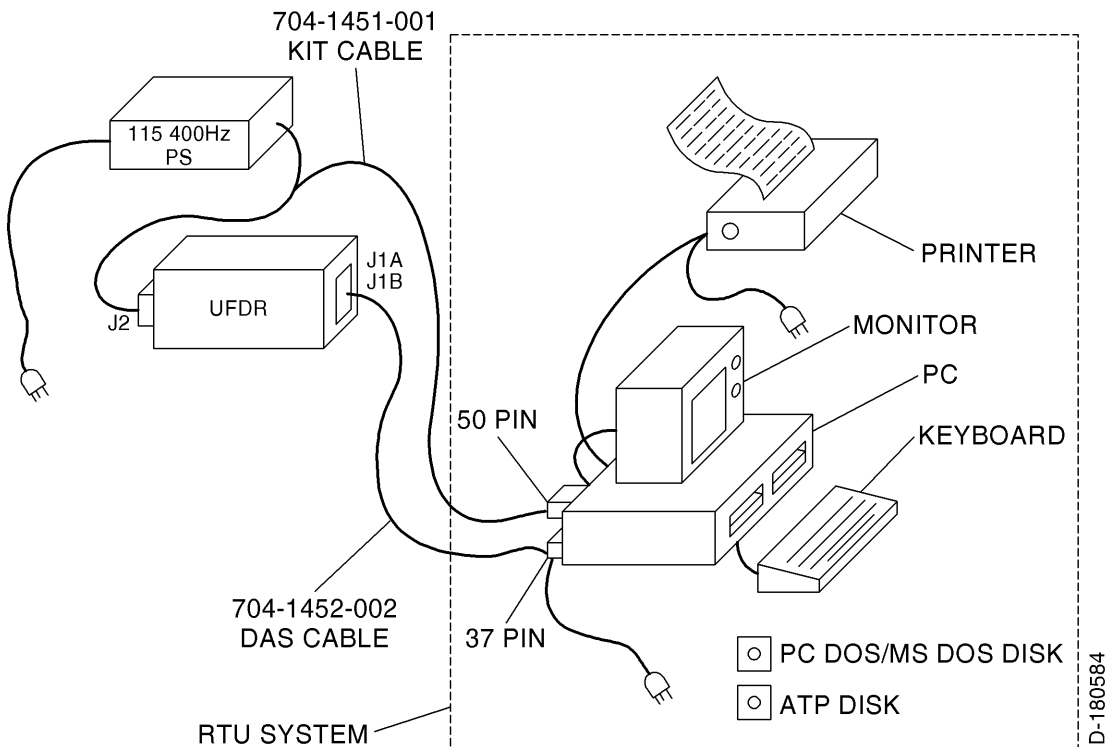
A. Test Equipment Required

- (1) Honeywell Readout and Test Unit (RTU), Part No. 980-4106-004, or -007 (includes cables), installed in a host PC (IBM, IBM XT, Compaq +, or Compaq) with 320 KB (640 KB recommended) memory and dual disk drive, or 1 disk drive and 1 hard disk, or equivalent or Part No. 960-0306-00X, which includes the host PC and appropriate parts. Instructions for performing software installation and readout of UFDR's are included in the RTU CMM, Part No. 980-4106-00X/960-0306-00X, Doc. No. 012-0507-001.
- (2) TTEST software diskette, Part No. 760-0284-10X.
- (3) 115V 400 Hz power supply, Model Invertron, manufactured by California Instruments Div. (V16067), 9025 Balboa Ave., San Diego, CA 92123-1509 or equivalent.

B. Equipment Setup

Set up the equipment for the on-line tests per these steps:

- (1) Verify that the 400 Hz power supply is turned off. Connect the RTT cable, Part No. 704-1451-001, to the front of the UFDR and to the appropriate connector on the host PC as shown in Figure 116.
- (2) Connect the power supply leads on the RTT cable to the external 400 Hz power supply.



UFDR Test Setup
Figure 116

31-30-32

Page 154
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAUTION: USE CAUTION WHEN CONNECTING THE GREEN SAFETY GROUND. FAILURE TO DO SO MAY DAMAGE BOTH THE UFDR AND THE RTU/PC. USE A FAST-BLOW FUSE, 2 AMPERES MAXIMUM, IN THE 400HZ POWER SUPPLY.

- (3) Connect the DAS cable, Part No. 704-1452-002 if using PN 980-4106-007 or 704-1452-003 if using PN 960-0306-XXX, to the rear of the UFDR and the appropriate connector on the host PC.
- (4) Connect printer, PC, and display as shown in Figure 116.

CAUTION: DO NOT TURN ON THE 400 HZ POWER SUPPLY UNTIL THE TEST SETUP IS PROPERLY COMPLETED AND PC POWER IS ON. IF THIS ORDER IS NOT FOLLOWED, THE UFDR MAY ENTER RECORD MODE AND OVERWRITE DATA CONTAINED ON THE TAPE.

- (a) Turn on the PC, display, and printer power.
- (b) Boot the PC DOS system per the manufacturer's procedures for booting the PC.

C. Return to Service Information

This procedure covers the testing required to show compliance with UFDR acceptance test specifications. It is required to verify UFDR operation prior to affixing the SERVICEABLE PART tag to the UFDR and returning it to service.

- (1) Initial dispensation of UFDR prior to Return to Service Test.
 - (a) On receipt of a UFDR from aircraft service, an initial determination shall be made as to whether or not a copy or transcription of the recorded data is required.
 - (b) In cases where a copy or transcription is required, this Return to Service Test shall not be conducted until a successful copy or transcription has been made.
 - (c) In cases where a copy or transcription is not required (or has successfully been made), and the unit is considered to be in need of service, it is unnecessary to run this entire test. If the nature of the problem requiring service is unknown, this test need only be run to the point at which the first failure occurs. The operator may then switch to the fault diagnostic routines to troubleshoot the failures.
- (2) After repair activity, or at any other time that a serviceable part status is to be established, the entire test shall be performed, except as noted in C(3).
- (3) The 25-hour tape verification test is not required.
- (4) Pass/Fail Criteria

If the unit under test fails to pass any of the automatic tests, the UFDR shall be subjected to the manual tests and diagnostic tests as noted on the menus in order to determine the proper maintenance action.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

(5) Inspection of Unit Under Test

The following data about the UFDR is requested by the program:

- Part Number
- Unit Serial Number
- Manufacturing (Mfg.) date
- Tape Transport Serial Number (if replaced).

D. Transport Test

This procedure describes the purpose and gives the measured parameters for each test. These tests are performed as part of the automatic test procedure. Review paragraph 7C(1) to ensure that UFDR data is not lost.

- (1) Insert the TTEST705 diskette (write-protect tab removed), a working copy of Part No. 760-0284-10X, or if the host PC is equipped with a hard disk drive and if the software has been installed, select the hard disk subdirectory by typing **CD/UFDR** and pressing **<RTN>**. The display will show **A:>** or **C:/UFDR>** (if a hard disk is used).
- (2) Type **TTEST705** and press **<RTN>**.
- (3) The program will be loaded and will begin by running a mini-selftest on the RTU circuit cards before displaying the top level menu (procedure takes approximately 2 minutes).
- (4) Wait for the top level menu to appear on the monitor display. The display will appear as shown in Figure 117.

Honeywell RTU(c) 1985
UFDR ATP-705 Transport Test 760-0284-10X

F1 - HELP	F2 - Automatic ATP-705
F3 - Return to Service Test	F4 - Manual Mode Submenu
F5 -	F6 - Diagnostic Mode
F7 -	F8 -
F9 - RTU Selftest	F10 - Exit to DOS

-----**WARNING**-----
This Program may overwrite
the data recorded in the UFDR.

Top Level Menu
Figure 117

- (5) Press the F2 key for the automatic ATP-705. The screen display shown in Figure 118 will result.

31-30-32

Page 156
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Honeywell RTU(c) 1985

UFDR ATP-705 Transport Test 760-0284-10X

-----WARNING-----

This Program may overwrite
the data recorded in the UFDR.

F1:Continue

F10:Abort

Transport Test Warning Screen
Figure 118

- (6) Press the F1 key for the UFDR ATP-705 header display. The screen display shown in Figure 119 will result.

NOTE: Paragraph references on the screen refer to Honeywell documents and are not applicable in the field

Mfg part no. 980-4100-____ Serial no. ____ Mfg date ____
Mod dots xx xx xx xx xx xx xx xx xx xx
Is this a production test? (Y/N) ____
4.1.1 Workmanship ____ 4.1.2 ULB battery date ____
4.1.3 Insul. Resistance 20.0 ____ 4.1.4 Ground Fault Isolator (Y/N) ____
4.1.5 Burn-in Results (if known) ____
Max d. ck err's 16 ____ Max sequential d. ck err's 4 ____
Total d. ck err's 64 ____ Min records/ track 11250 ____
Service Activity (if applicable) ____
Transport serial no. xxxx (if replaced) ____
Replacements: (Y/N) Belt ____ Tape ____ Heads ____
Other: ____
DAS information (if applicable) ____
FAA or CAA tolerances? (FAA/CAA) ____
Abs. Pneu. xducer P/N 300-0424-____ S/N ____
Dif. Pneu. xducer P/N 300-0425-____ S/N ____
Op. Name: ____
Comments: ____
Automatic Printout? (Y/N) ____
Is data correct? (Y/N) ____

ID-180585

UFDR Transport Test Header Screen
Figure 119

31-30-32

Page 157
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (7) All of the data requested should now be filled in. The procedure for entering data on this screen is as follows: The operation should look at the box where the cursor is positioned and enter the correct data, followed by **<RTN>**, or press **<RTN>** if the data is correct. The Off-Line and Burn-In Test data is irrelevant for field maintenance and may be left blank. The Operator should enter data in the box named **Service Activity**, such as, **changed belts** or **replaced XA4**. Proceed in this manner until all the data has been entered, and the display again asks **is data correct?**. Verify that the data is now correct. If the data is correct, type **Y** and press **<RTN>**. If an error was made, type **N** and press **<RTN>**. The cursor will again move from box to box to allow the operator to change the data. For each case, if the data is correct, the operator can press **<RTN>**. The data will not change if only **<RTN>** is pressed. After the last question has been answered with a yes, the display will be as shown in Figure 120.
- (8) Turn on the 400 Hz power supply, adjust to 117 VAC, and press the F5 key (continue).
- NOTE:** If the UFDR and 400 Hz power supply are adjusted prior to this step, an answer of **Y <RTN>** to the last question in Figure 120 will start the automatic testing.
- (9) The RTU now starts the tests. The first few tests require the operator to make adjustments while the RTU performs the test. In each case the operator instructions will be shown at the bottom of the display.
- (10) This completes the test. Remove the data sheets from the printer and sign them. Turn off the power supplies and disconnect the cables.

NOTE: If data sheets from the RTU Automatic Test show that the **Abnormal Power Conditions Test Voltage Hysteresis Test** failed perform the procedure in the **REPAIR** section, Para. 20, prior to removing and replacing parts.

Mfg part no. 980-4100-DXUS				Serial no. 0000		Mfg date 8200		
Mod dots xx xx xx xx xx xx xx xx xx xx								
Is this a production test? (Y/N) Y								
4.1.1 Workmanship Y		4.1.2 ULB battery date 01/01/81						
4.1.3 Insul. Resistance 20.0		4.1.4 Ground Fault Isolator (Y/N) Y						
4.1.5 Burn-in Results (if known)								
Max d. ck err's 16		Max sequential d. ck err's 4						
Total d. ck err's 64		Min records/ track 11250						
Service Activity (if applicable)								
Transport serial no. xxxx		(if replaced)						
Replacements: (Y/N)		Belt N		Tape N		Heads N		
Other:								
DAS information (if applicable)								
FAA or CAA tolerances? (FAA/CAA) FAA								
Abs. Pneu. xducer P/N 300-0424-		S/N						
Dif. Pneu. xducer P/N 300-0425-		S/N						
Op. Name:								
Comments:								
Automatic Printout? (Y/N) Y								
Is data correct? (Y/N) Y								
F1:Retry		UFDR is not online			F5:Continue			F10:Abort

ID-180586

Transport Test Header Screen with Data
Figure 120

31-30-32

Page 158
22 Dec 2006

E. Manual Acceptance Test

The **Manual Mode Submenu** shown in Figure 117 allows the operator to run (or rerun) any one or more of the individual tape transport tests on the UFDR, or to position the tape, to write or seek tones, etc. Review paragraph 7C(1) to ensure that UFDR data is not lost.

F. Diagnostic Mode

The Diagnostic Mode shown in Figure 117 allows the operator to exercise the UFDR through submenus which provide lower level testing than either the automatic or manual modes. It is intended for use in fault isolating when the automatic or manual tests display a fault. Review paragraph 7C(1) to ensure that UFDR data is not lost.

8. Troubleshooting (RTU Users)

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES ARE SUBJECT TO DAMAGE BY EXCESSIVE LEVELS OF VOLTAGE. REFER TO DOD-HDBK-263 FOR PROPER HANDLING PROCEDURES.

A. General

- (1) This section contains information designed to assist Readout and Test Unit (RTU) users in locating faults that may occur in the UFDR. It is intended to be used after the RTU Test Procedure of paragraph 7 is performed.

***NOTE:** Successful use of the RTU for troubleshooting may require extensive knowledge of both UFDR and RTU operation.*

- (2) TTEST705 faults may occur separately or in groups depending upon the type of fault. Types of faults are:
 - (a) BITE failures.
 - (b) Status failures.
 - (c) Power Test failures.
 - (d) Nonflight Recording Modes failures.
 - (e) Flight Recording failures.
 - (f) Playback failures.
 - (g) Tape Sensor (BOT/EOT) failures.
- (3) A BITE fault may occur during TTEST705. Examine the Detailed Test Results and determine if the test failed or if a BITE failure occurred during the test. If so, the tested circuit may be OK and the BITE failure must be isolated. Comments on failures are listed at the end of each test.
- (4) The RTU can be used in either of 2 ways to isolate faults of the ATP-705 Transport Test: Manual Mode Submenu (**F4**) or Diagnostic Mode (**F6**).
 - (a) Manual Mode Submenu (**F4**) provides selected testing of any UFDR tests performed during the Automatic ATP-705 Test. Brief explanations (**F1**) of the UFDR tests are included.
 - (b) Diagnostic Mode (**F6**) provides the ability to directly control the inputs to the UFDR.

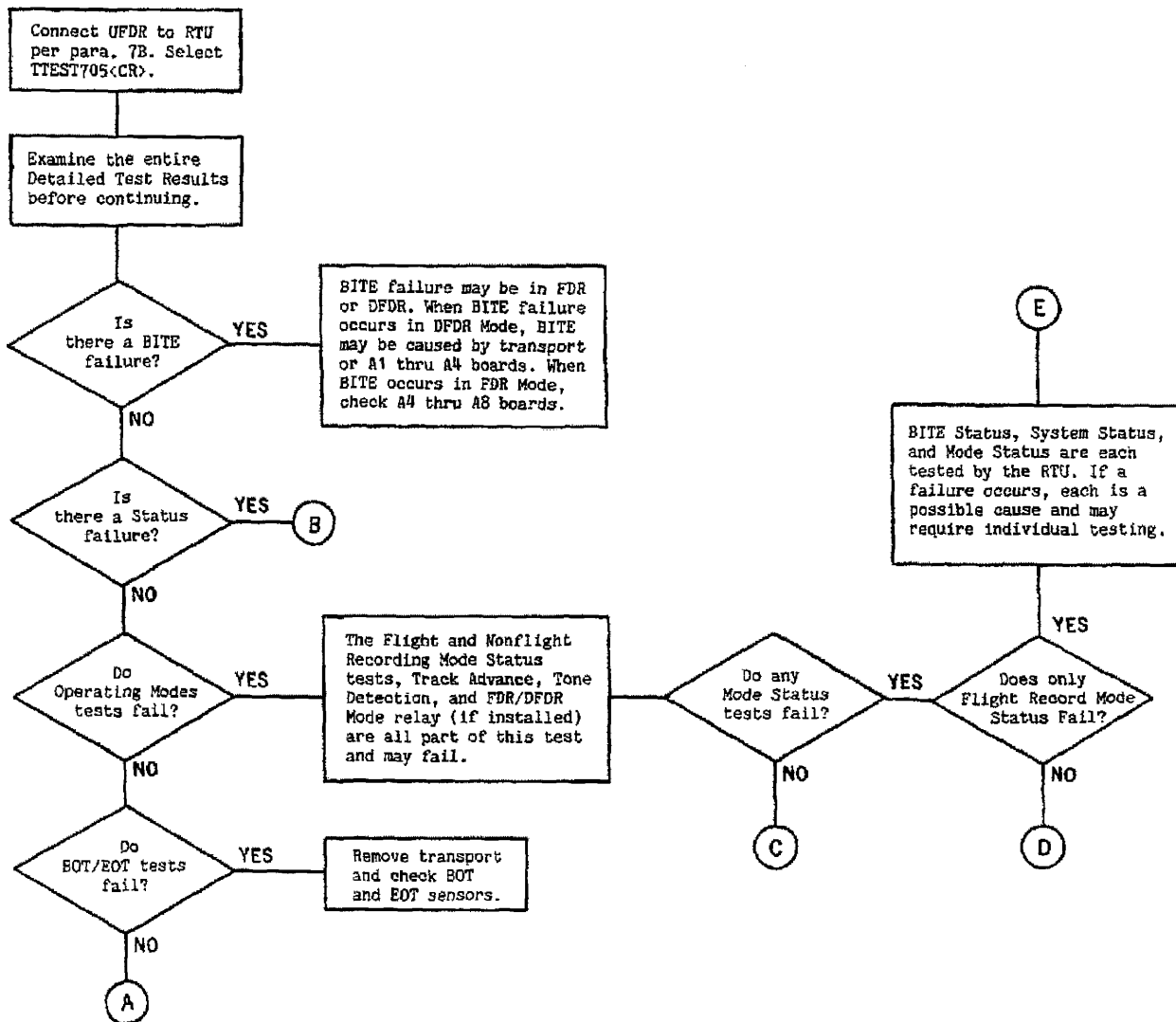
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

B. Fault Logic Diagram

The fault logic diagram of Figure 121 provides guidance for isolating most faults to the card (assembly) level. In some cases, it may be possible to isolate faults to the component or circuit level.



ID-180587

UFDR Fault Logic Diagram
Figure 121 (Sheet 1 of 7)

31-30-32

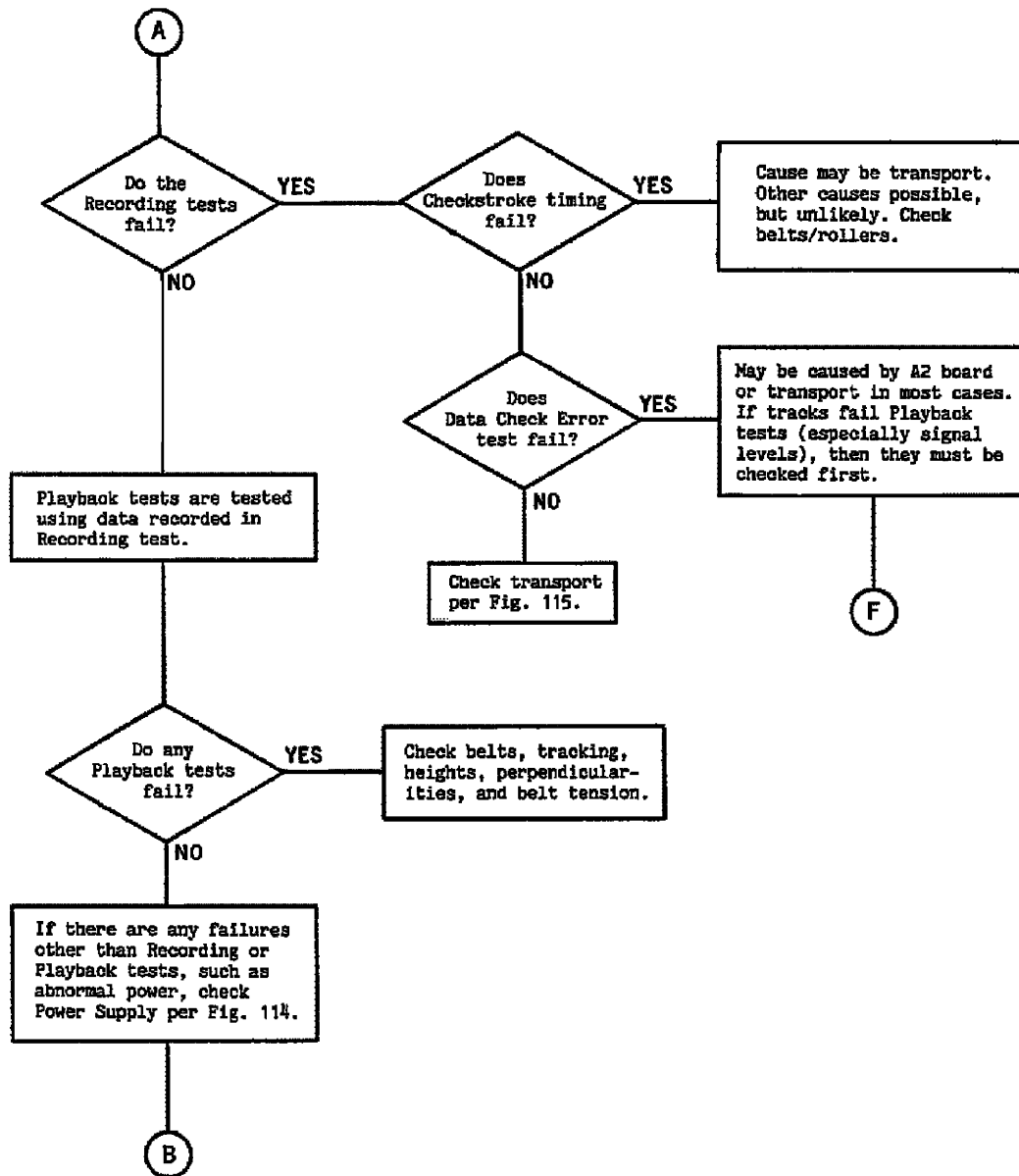
Page 160
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180588

UFDR Fault Logic Diagram
Figure 121 (Sheet 2 of 7)

31-30-32

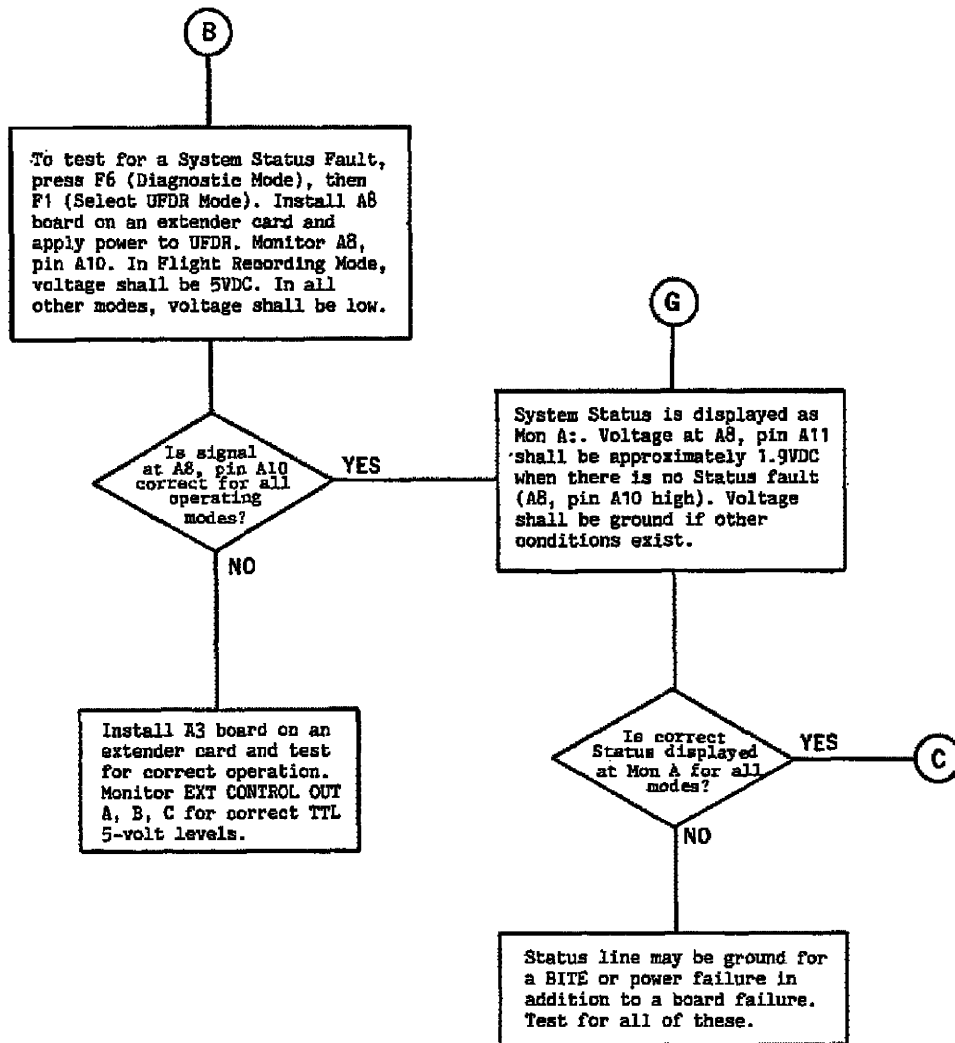
Page 161
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



UFDR Fault Logic Diagram
Figure 121 (Sheet 3 of 7)

31-30-32

Page 162
22 Dec 2006

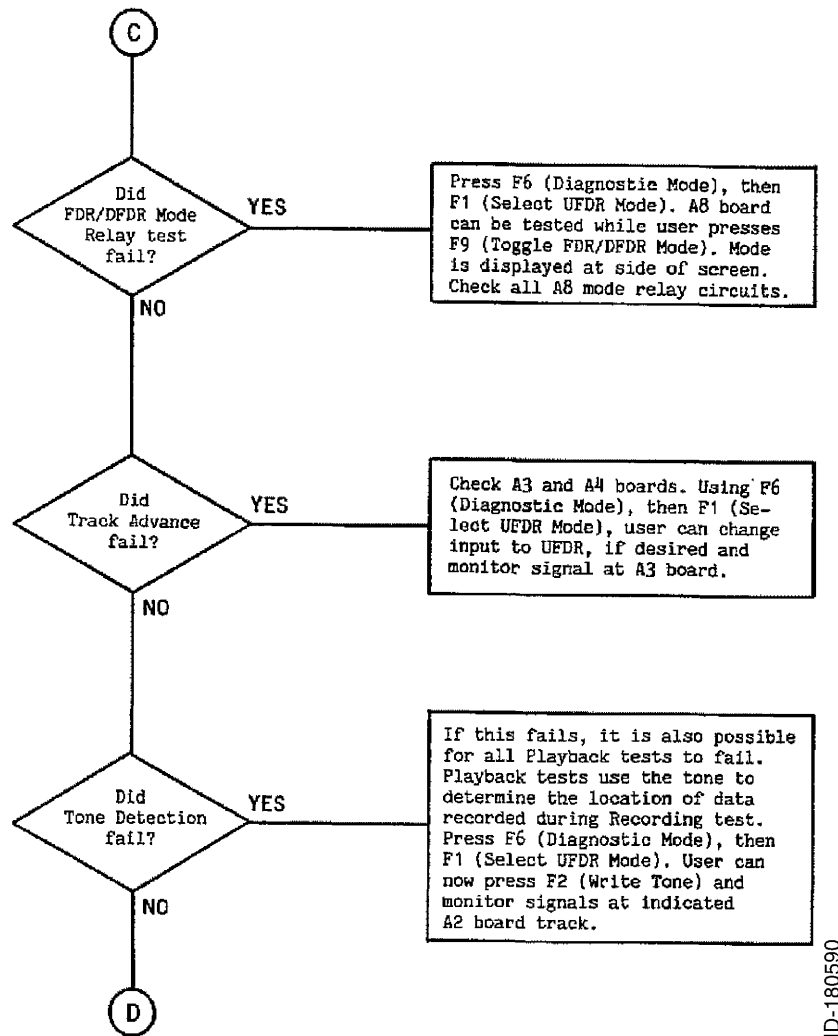
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTv y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180590

UFDR Fault Logic Diagram
Figure 121 (Sheet 4 of 7)

31-30-32

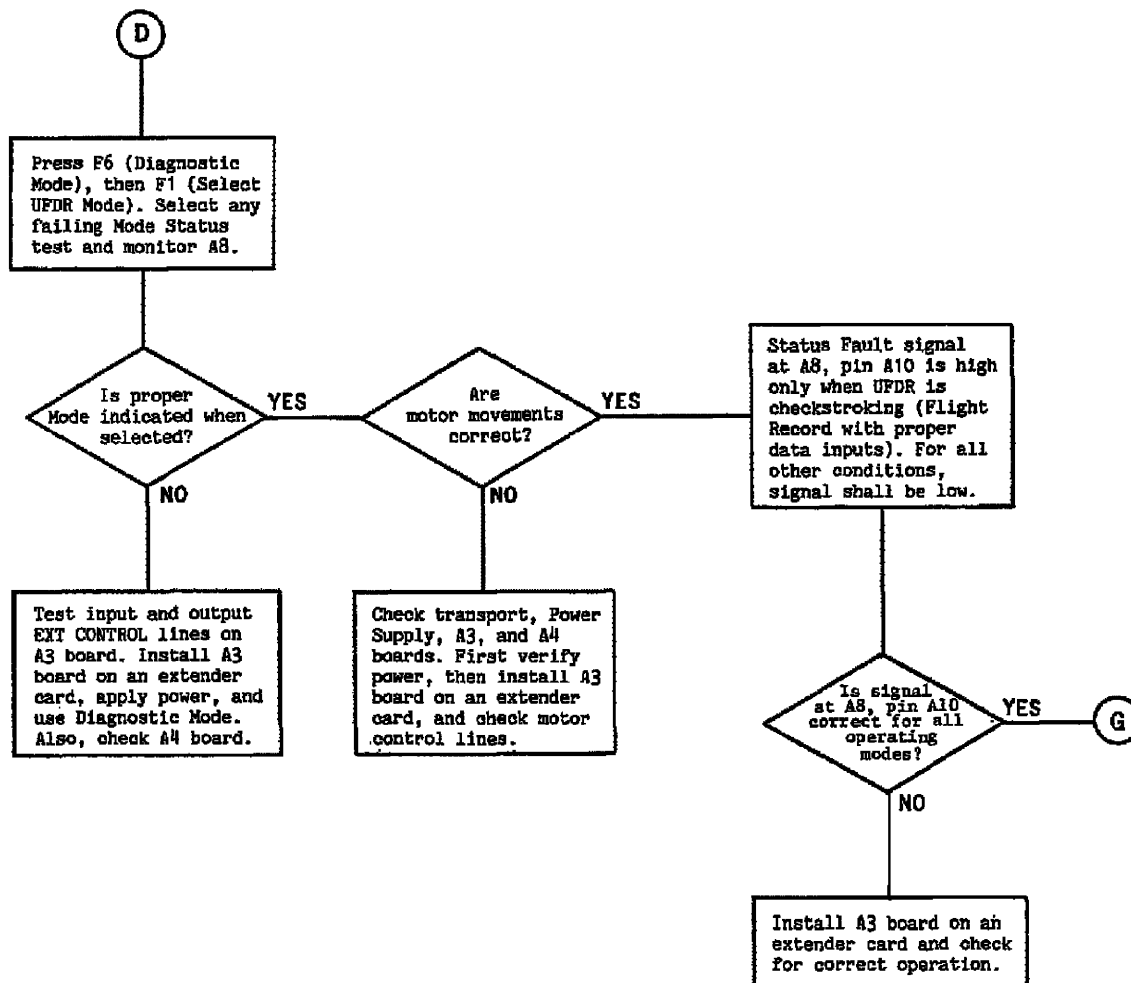
Page 163
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180591

UFDR Fault Logic Diagram
Figure 121 (Sheet 5 of 7)

31-30-32

Page 164
22 Dec 2006

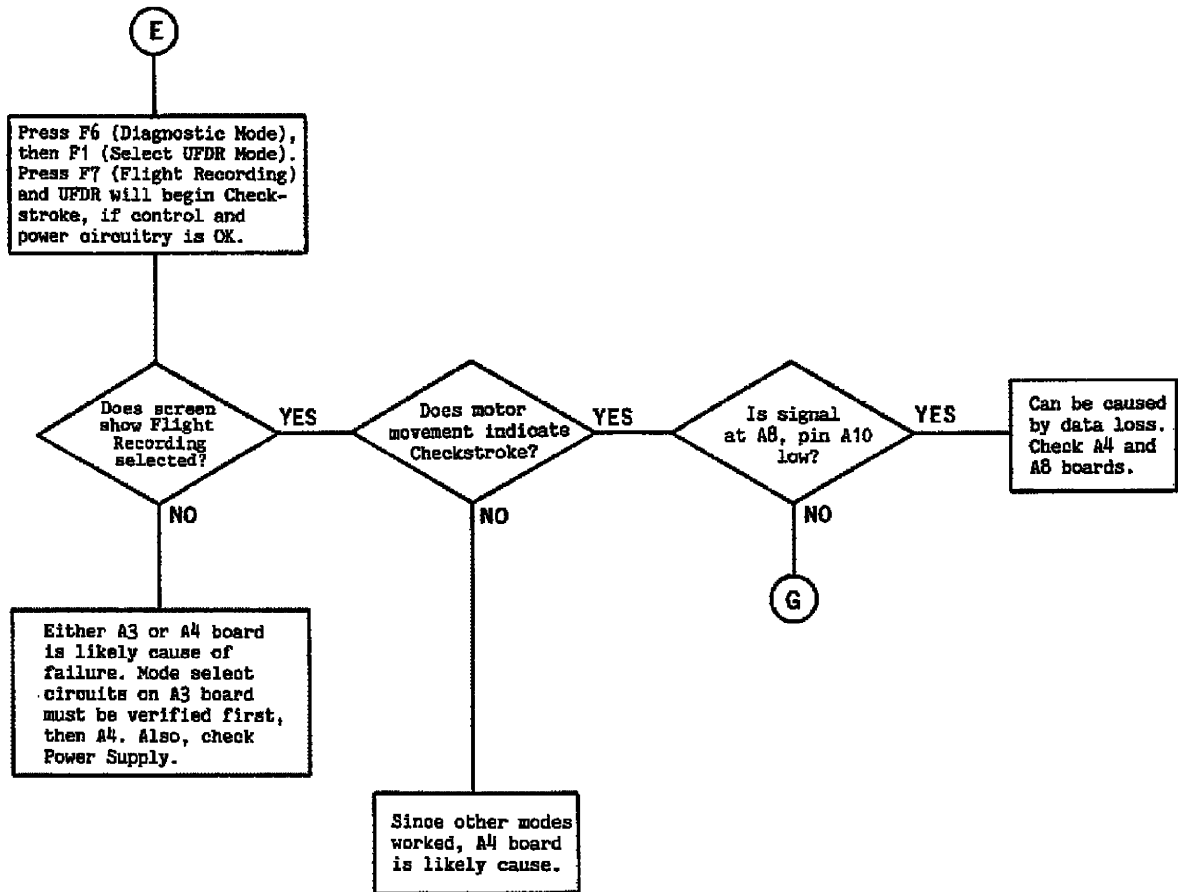
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTY y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180592

UFDR Fault Logic Diagram
Figure 121 (Sheet 6 of 7)

31-30-32

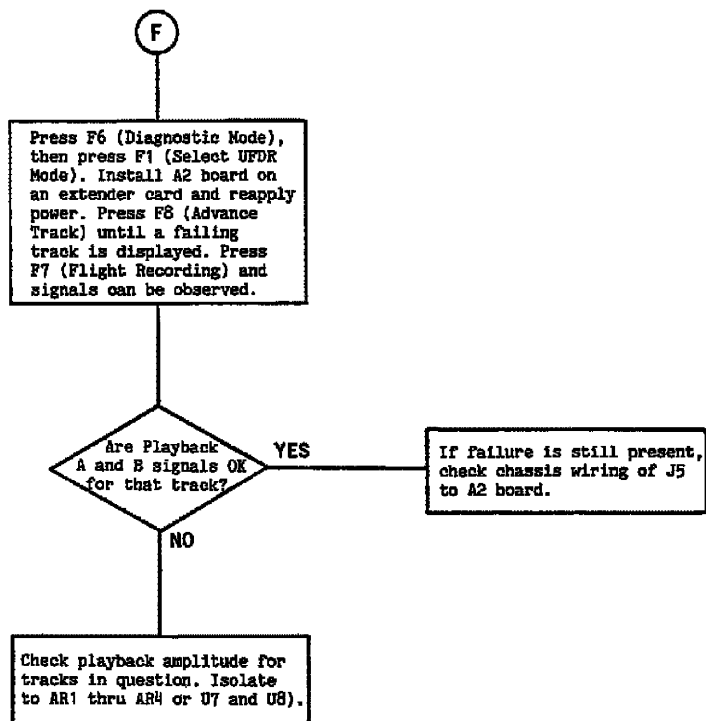
Page 165
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



UFDR Fault Logic Diagram
Figure 121 (Sheet 7 of 7)

31-30-32

Page 166
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity
	Updated all sheets to latest revision. Sheet 2: Replaced schematic Sheet 3: Replaced schematic and added ADCN A-1 and ADCN A-2 to 617-4640-306 Sheet 4: Combined sheets 4 and 5	

Chassis and Tape Recorder Assembly Schematic
Figure 122 (Sheet 1 of 4)

31-30-32

Page 167
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

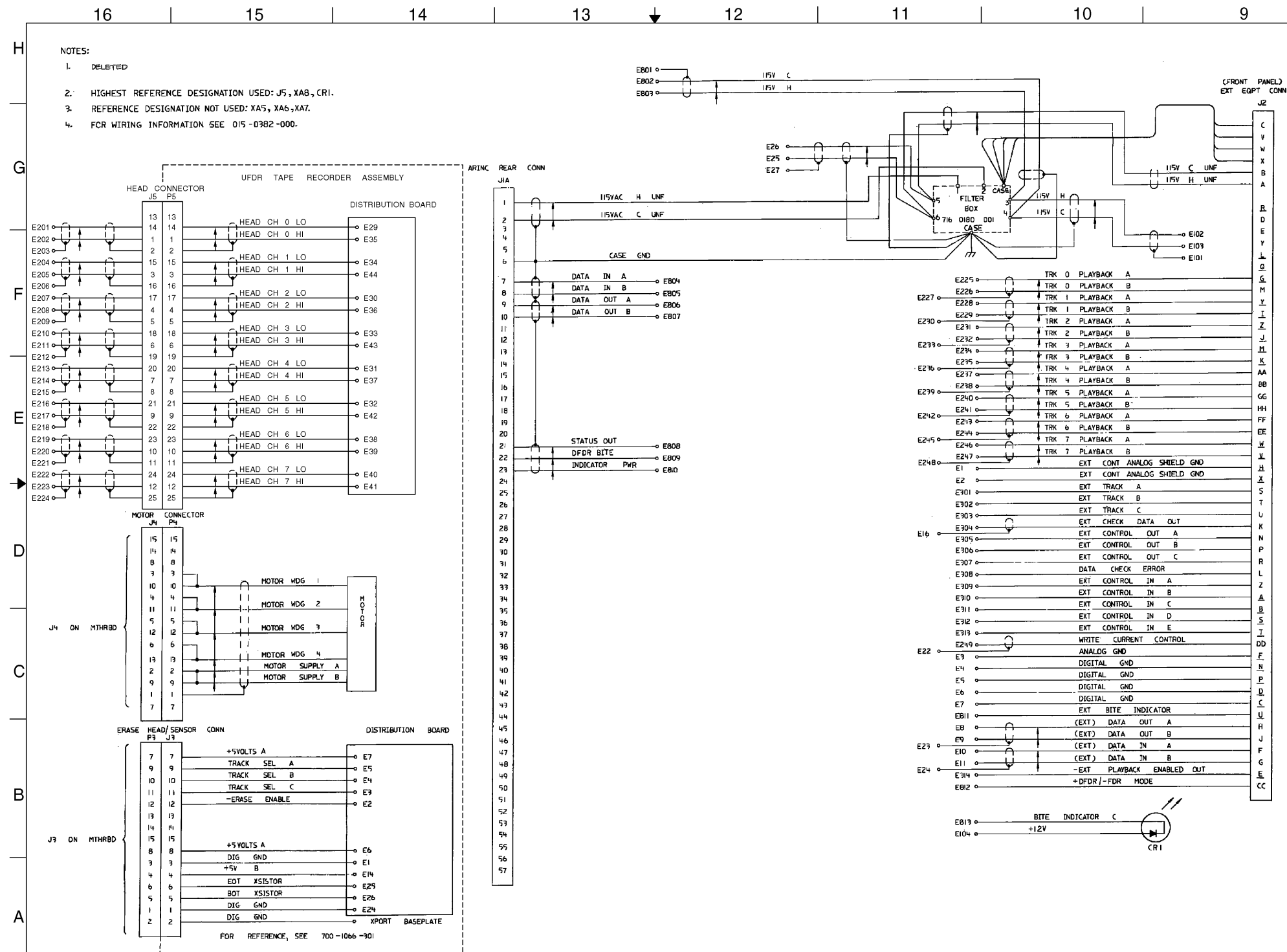
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101

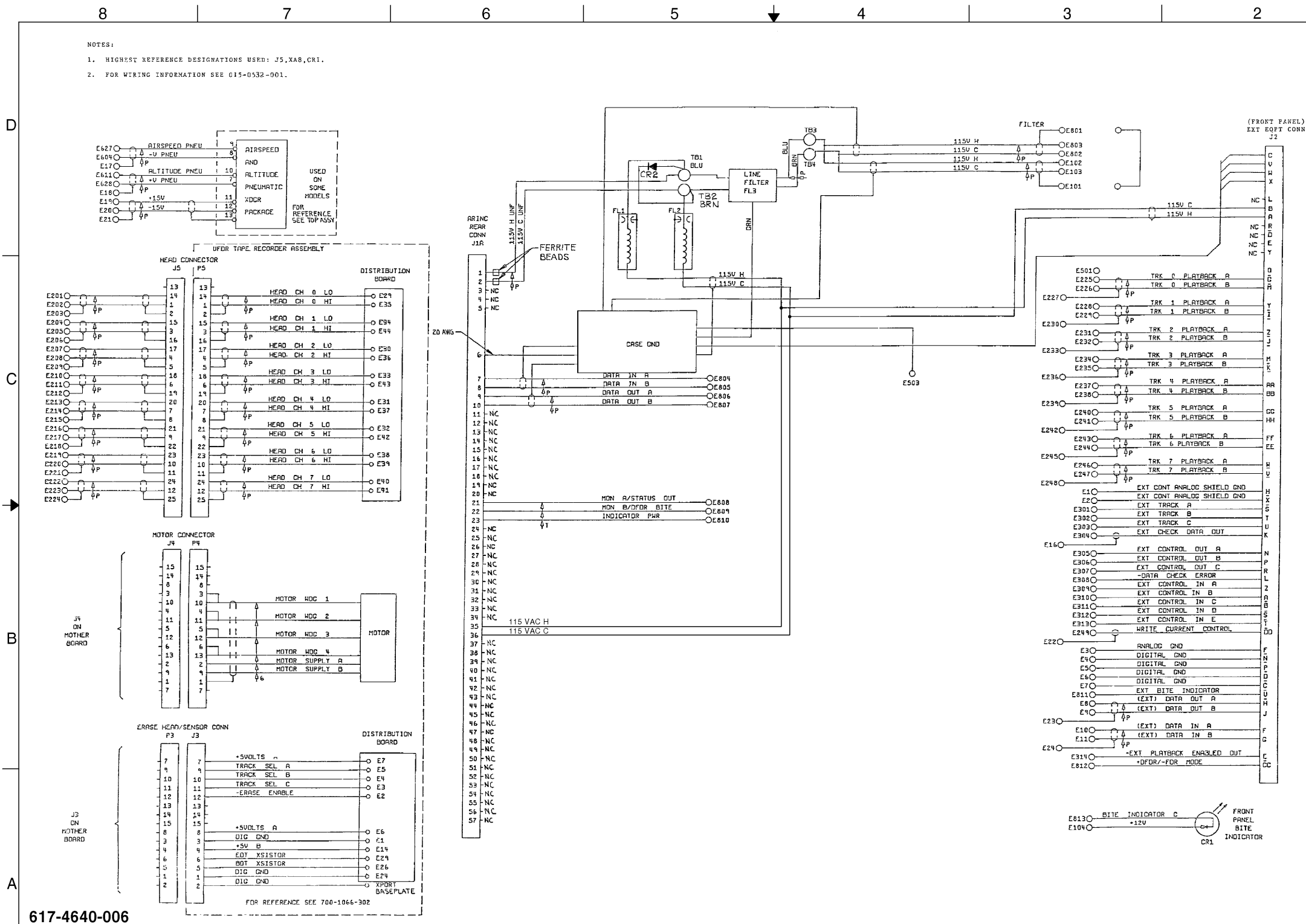


613-3716-001, -002, -005, -006

Chassis and Tape Recorder Assembly Schematic
Figure 122 (Sheet 2 of 4)

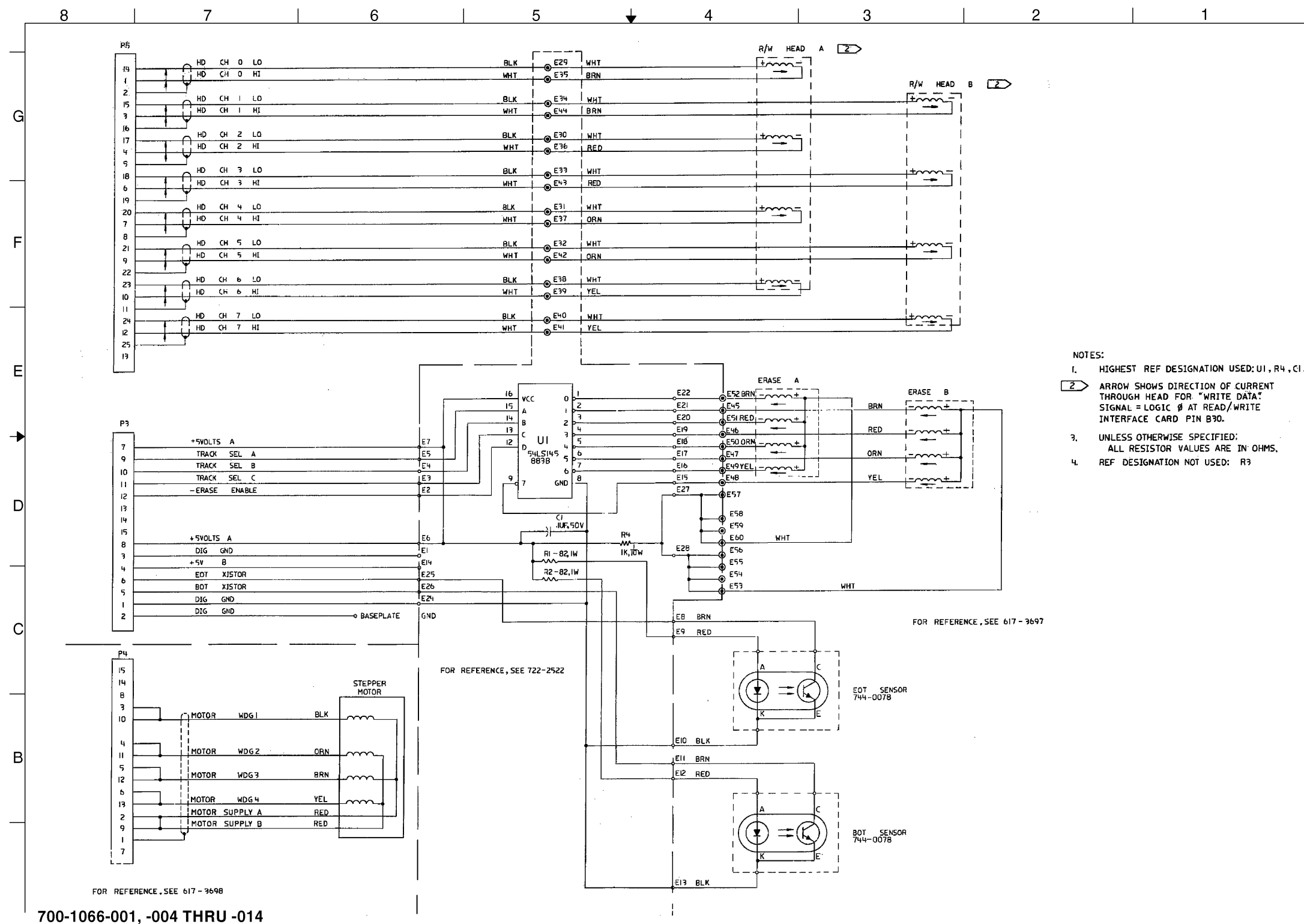
ID-180594 E617-3716-301-1-B E617-3716-301-2-B

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



Chassis and Tape Recorder Assembly Schematic
Figure 122 (Sheet 3 of 4)

ID-180595 E617-4640-306-1-A ADCN-1 ADCN-2



700-1066-001, -004 THRU -014

Chassis and Tape Recorder Assembly Schematic
Figure 122 (Sheet 4 of 4)

31-30-32

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity

Filter Schematic
Figure 123 (Sheet 1 of 2)

31-30-32

Page 175
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

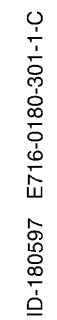
31-30-32

Page 176
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



31-30-32

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity
	Updated all sheets to latest revision. Sheet 2: Combined sheets 2 and 3. Sheet 3, 4: Added schematic 722-3142-302 and added ADCN -1.	

Motherboard Schematic
Figure 124 (Sheet 1 of 4)

31-30-32

Page 179
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

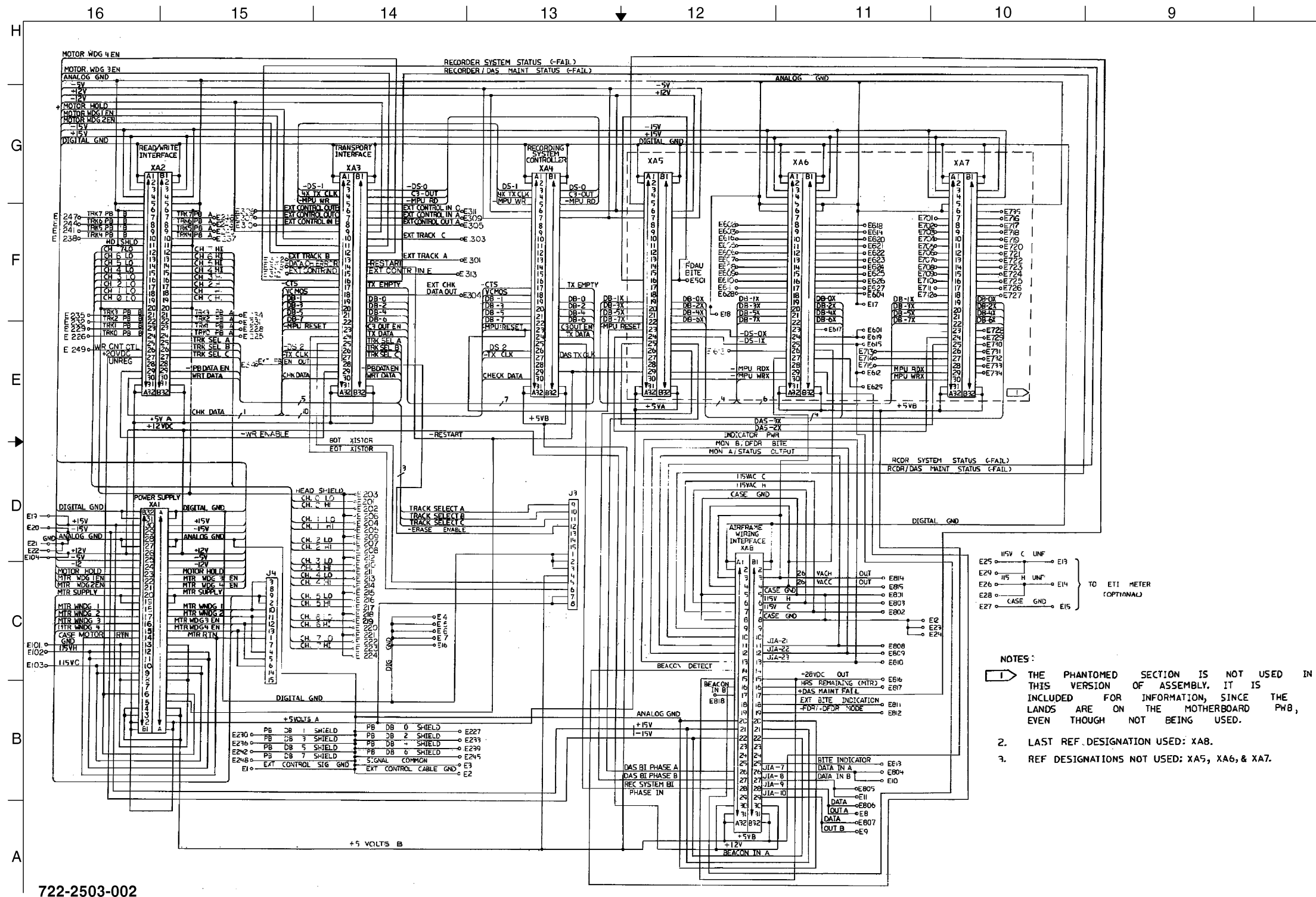
Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

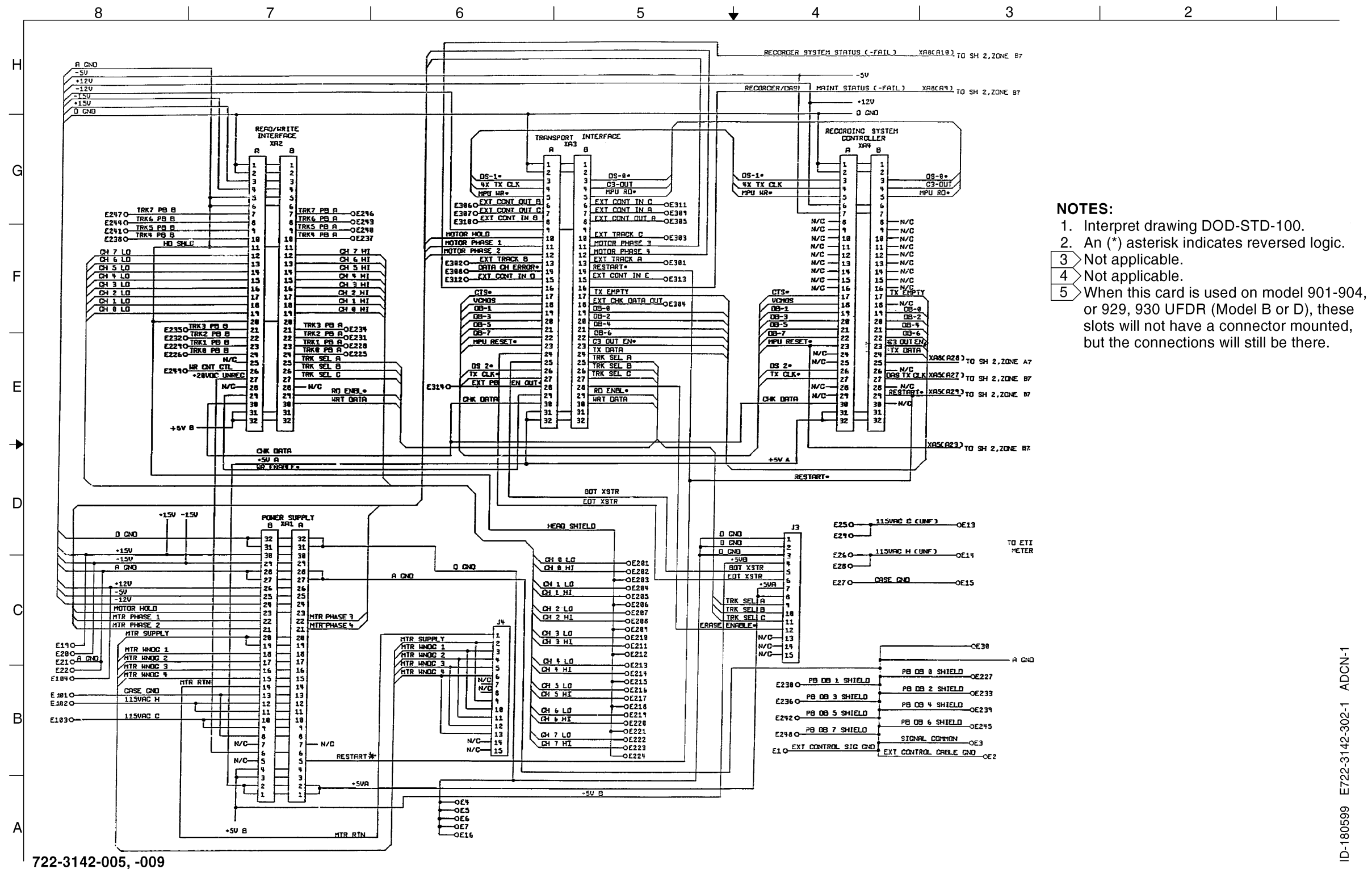
THIS PAGE INTENTIONALLY LEFT BLANK



722-2503-002

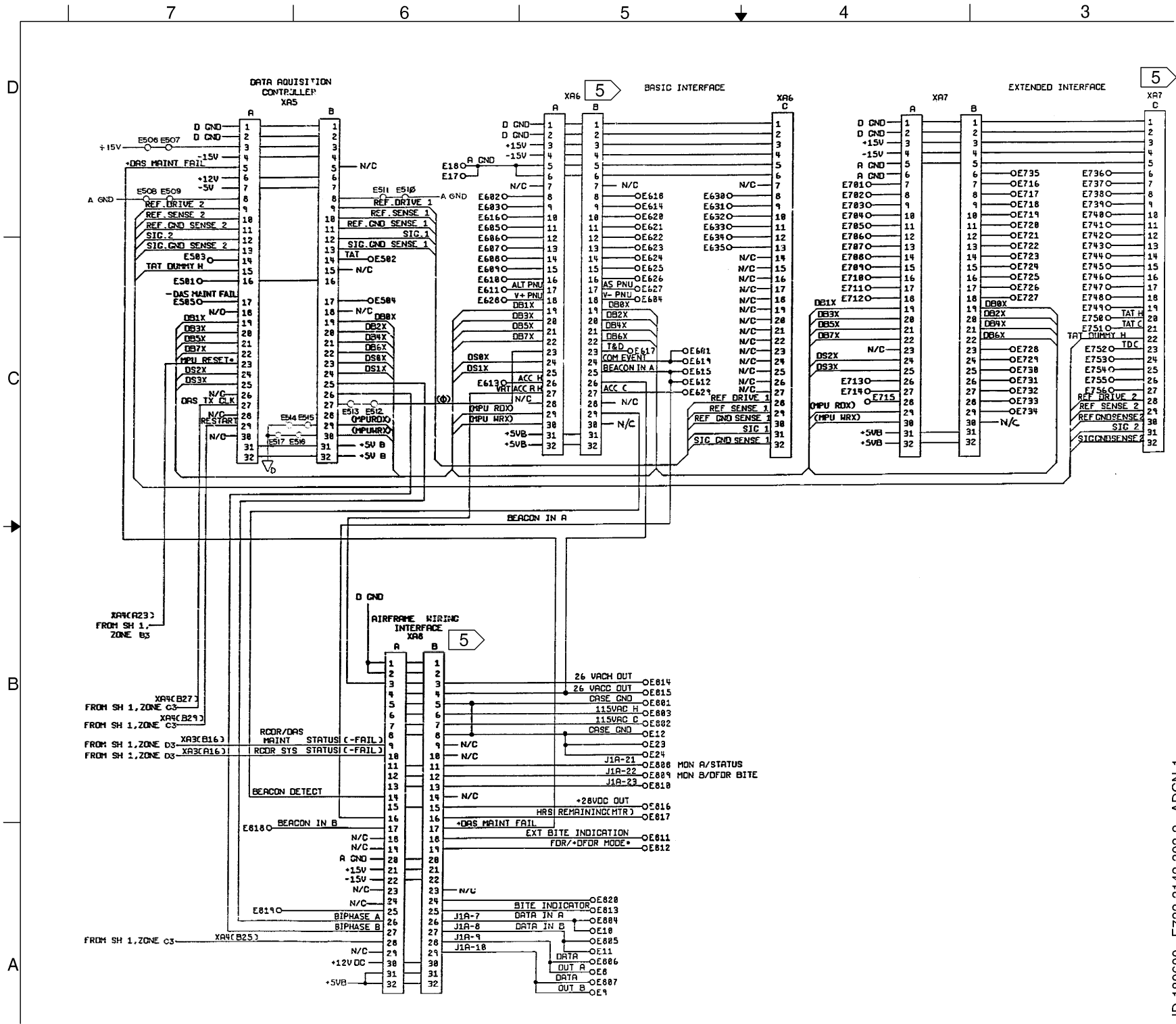
Motherboard Schematic
Figure 124 (Sheet 2 of 4)

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



- NOTES:**
1. Interpret drawing DOD-STD-100.
 2. An (*) asterisk indicates reversed logic.
 3. Not applicable.
 4. Not applicable.
 5. When this card is used on model 901-904, or 929, 930 UFDR (Model B or D), these slots will not have a connector mounted, but the connections will still be there.

Motherboard Schematic
Figure 124 (Sheet 3 of 4)



Motherboard Schematic
Figure 124 (Sheet 4 of 4)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity
	Updated all sheets to latest revision. Sheet 2: Combined sheets 2 and 3 for schematic 720-0123-301. Sheet 3, 4: Added schematics 720-0123-302 and 720-0123-303 respectively.	

Power Supply, A1, Schematic
Figure 125 (Sheet 1 of 4)

31-30-32

Page 187
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

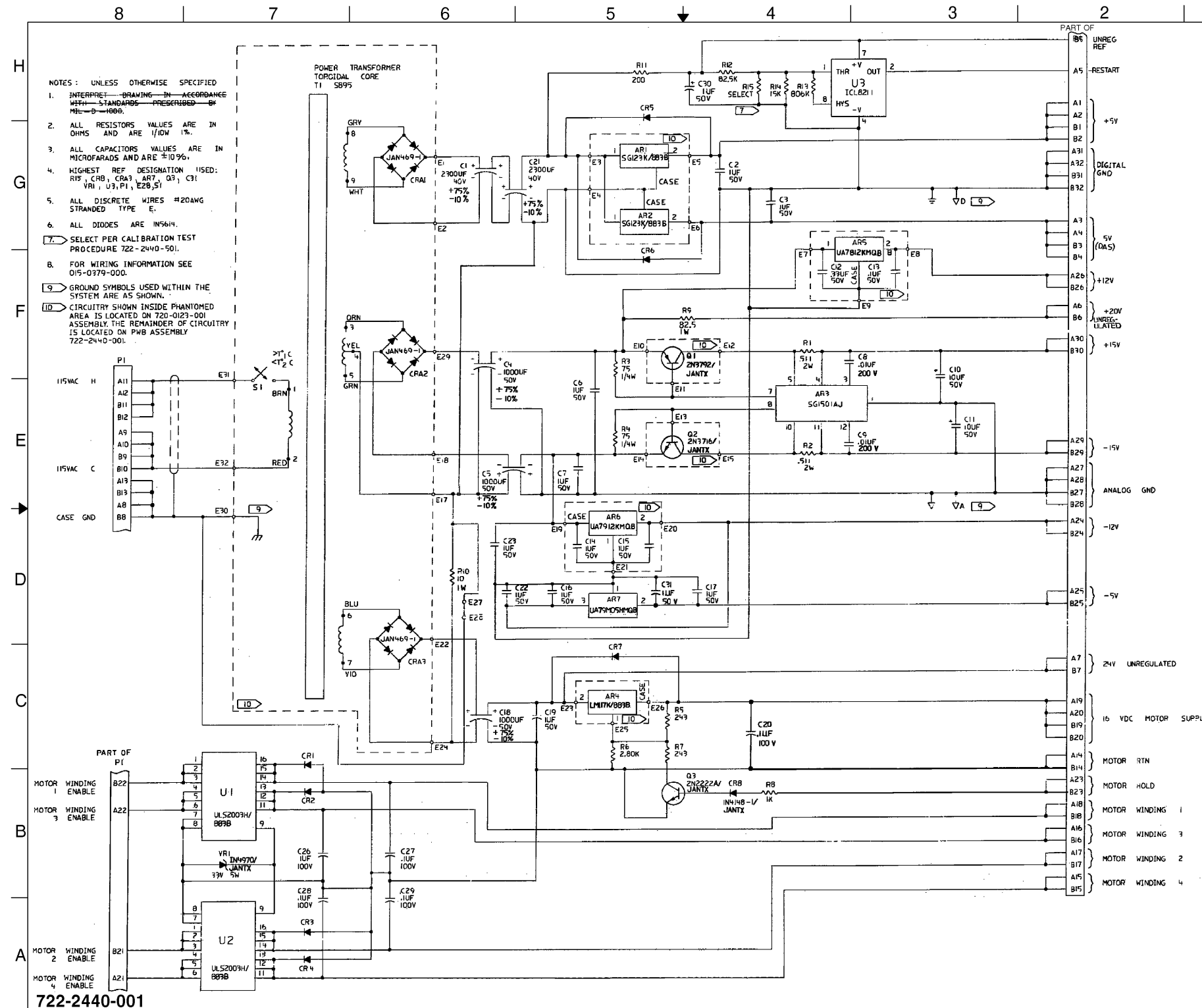
Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

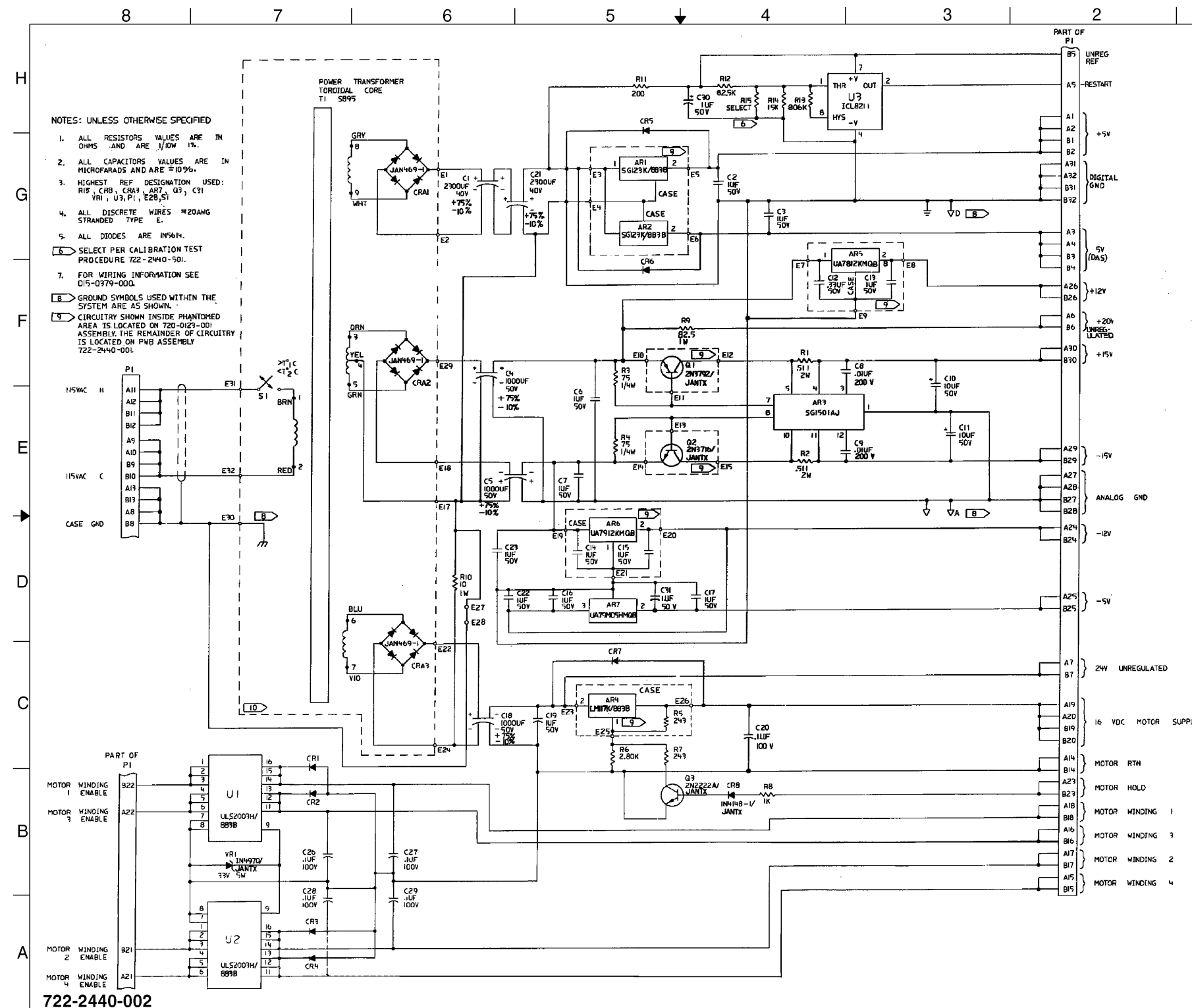
Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK



ID-180601 E720-0123-301-1-C

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101

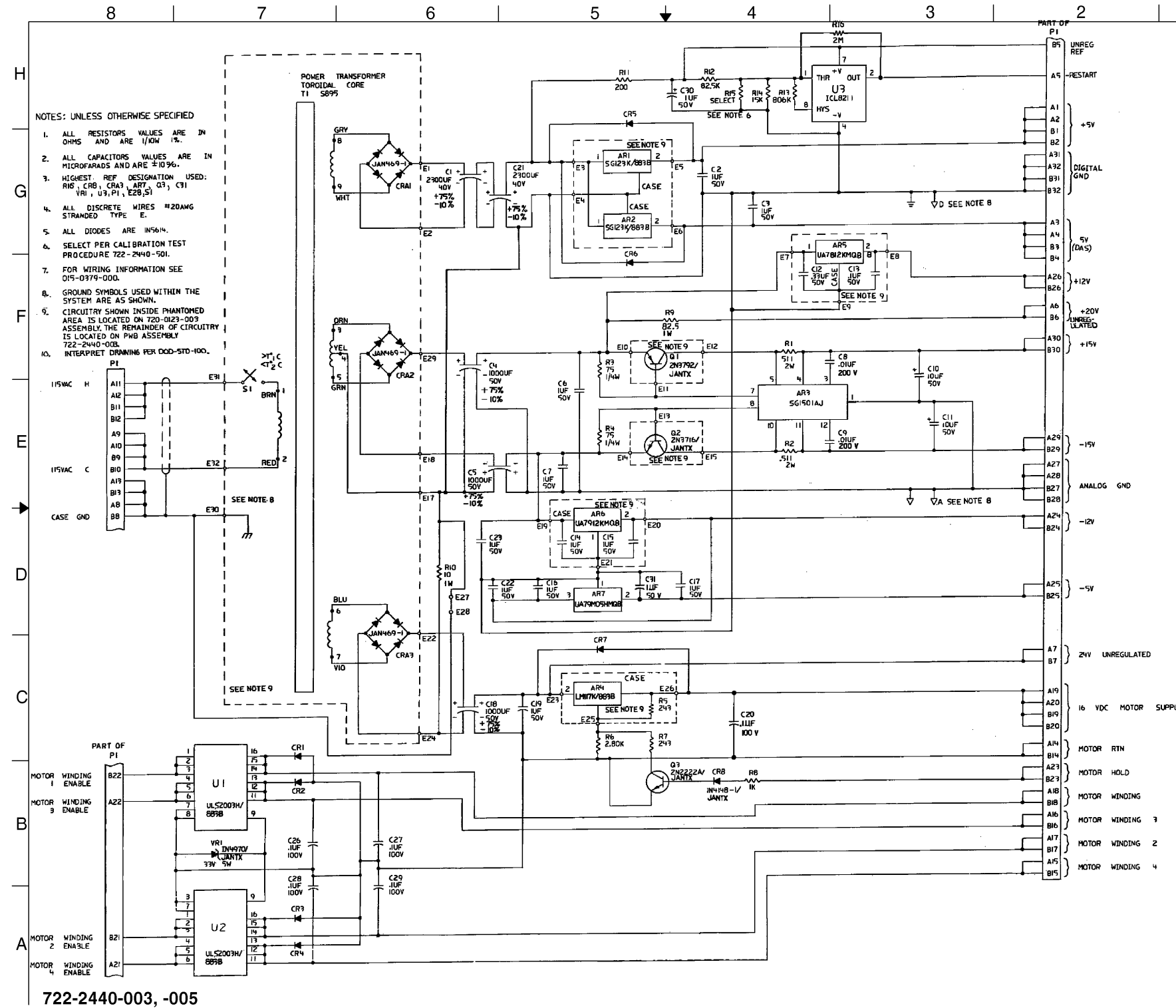


Power Supply, A1, Schematic
Figure 125 (Sheet 3 of 4)

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Power Supply, A1, Schematic
Figure 125 (Sheet 4 of 4)

ID-180603 E720-0123-303-1

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity
	Updated all sheets to latest revision. Sheet 2: Combined sheets 2 and 3 for schematic 722-2442-301. Sheet 3: Added for schematic 722-2442-302.	

Read/Write Interface, A2, Schematic
Figure 126 (Sheet 1 of 3)

31-30-32

Page 195
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

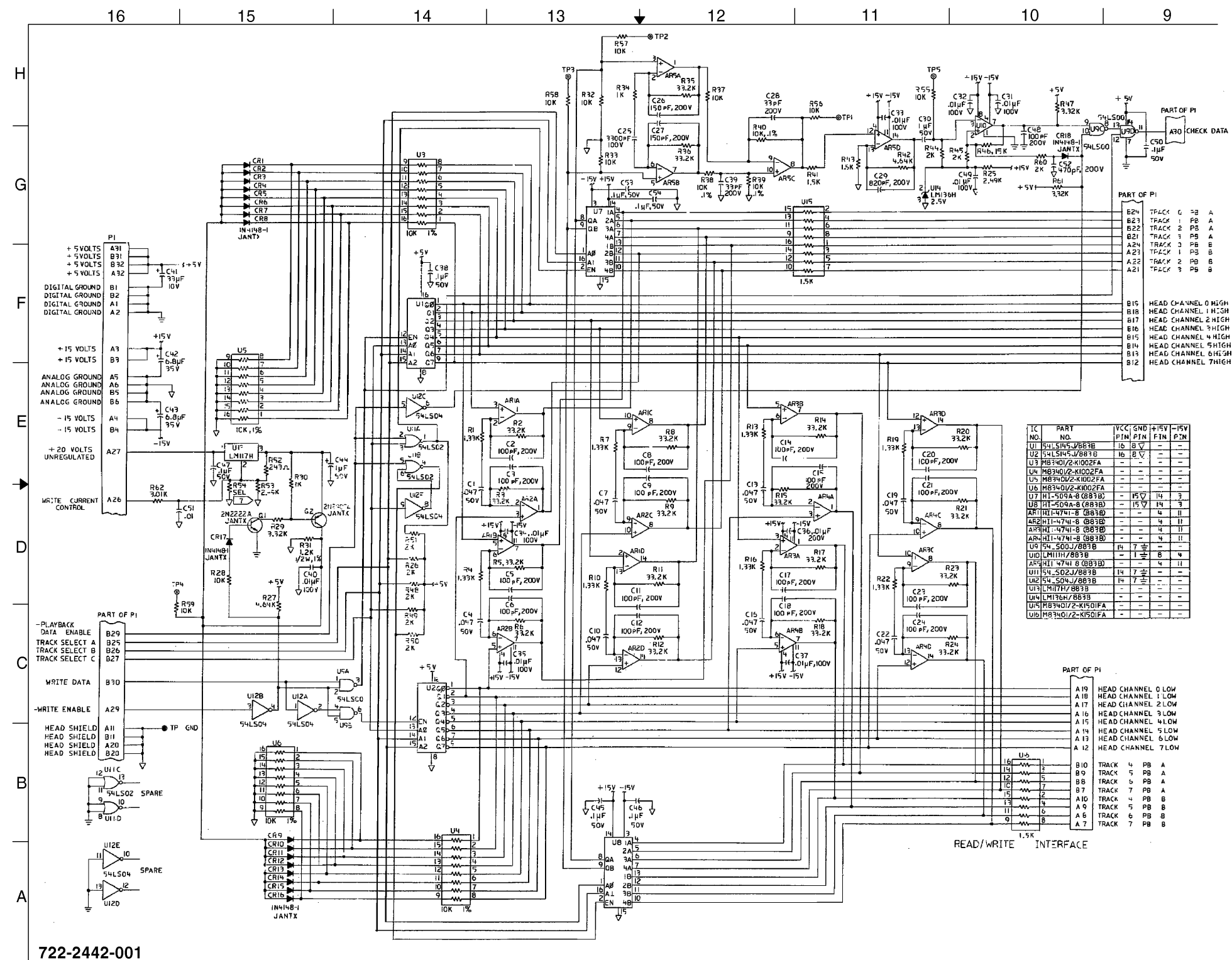
31-30-32

Page 196
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



NOTES: UNLESS OTHERWISE SPECIFIED

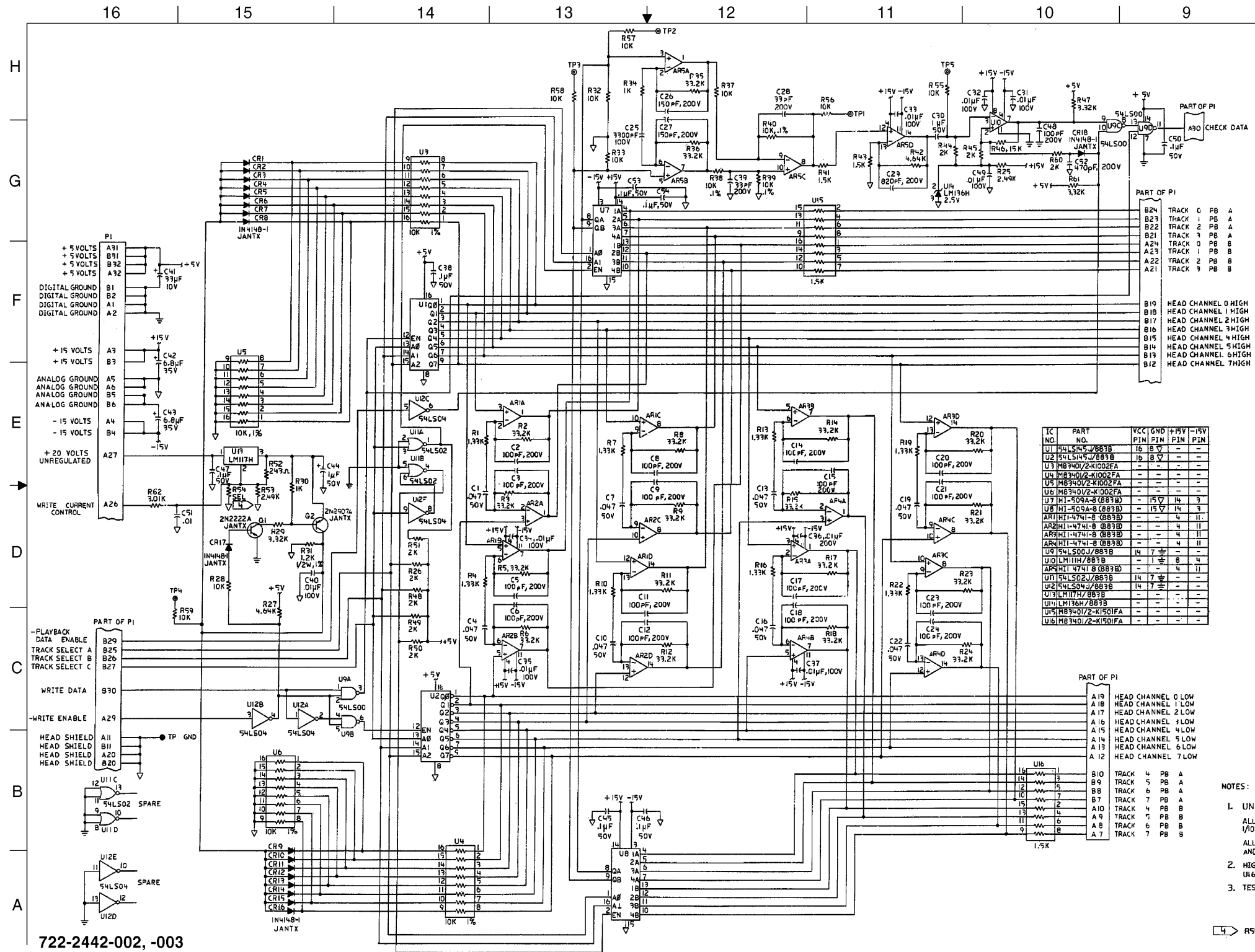
1. ~~INTERPRET DRAWING IN ACCORDANCE WITH STANDARDS PRESENTED BY MIL-D-100.~~
2. ALL RESISTOR VALUES ARE IN OHMS AND ARE $\pm 10\%$
3. ALL CAPACITORS VALUES ARE IN MICROFARADS AND $\pm 10\%$
4. HIGHEST REFERENCE DESIGNATION USED:
U16; CR18, RB2, C54, A195, Q2.
5. TEST POINTS SHALL BE ORIENTED AS FOLLOWS:
SPARE TP 1-5 KEY GND
0 0 0 0 0 0 0 0 0 0

7. R34 SELECTED FOR V1P1 = 12.8M.

IC NO.	PART	NCC	GND	+5V
		PIN	PIN	PIN
U1	54LS145 J8818	16	8 ▽	-
U2	54LS145 J8818	16	8 ▽	-
U3	M8301/2-K1002FA	-	-	-
U4	M8301/2-K1002FA	-	-	-
U5	M8301/2-K1002FA	-	-	-
U6	M8301/2-K1002FA	-	-	-
U7	14-504A-8 (8818)	-	15 ▽	14 ▽
U8	14-504A-8 (8818)	-	15 ▽	14 ▽
U9	ARH11-4741-8 (8818)	-	-	4 ▽
U10	ARH211-4741-8 (8818)	-	-	4 ▽
U11	ARH31-4741-8 (8818)	-	-	4 ▽
U12	ARH41-4741-8 (8818)	14	7 ▽	-
U13	U11M11-8818	-	1 ▽	8 ▽
U14	ARH41-4741-8 (8818)	-	1 ▽	4 ▽
U15	541 502J/8818	14	7 ▽	-
U16	541 504V/8818	14	7 ▽	-
U17	LM17H/8818	-	-	-
U18	LM136H/8818	-	-	-
U19	54LS145 J8818	-	-	-
U20	54LS145 J8818	-	-	-
U21	54LS145 J8818	-	-	-
U22	54LS145 J8818	-	-	-
U23	54LS145 J8818	-	-	-
U24	54LS145 J8818	-	-	-
U25	54LS145 J8818	-	-	-
U26	54LS145 J8818	-	-	-
U27	54LS145 J8818	-	-	-
U28	54LS145 J8818	-	-	-
U29	54LS145 J8818	-	-	-
U30	54LS145 J8818	-	-	-
U31	54LS145 J8818	-	-	-
U32	54LS145 J8818	-	-	-
U33	54LS145 J8818	-	-	-
U34	54LS145 J8818	-	-	-
U35	54LS145 J8818	-	-	-
U36	54LS145 J8818	-	-	-
U37	54LS145 J8818	-	-	-
U38	54LS145 J8818	-	-	-
U39	54LS145 J8818	-	-	-
U40	54LS145 J8818	-	-	-
U41	54LS145 J8818	-	-	-
U42	54LS145 J8818	-	-	-
U43	54LS145 J8818	-	-	-
U44	54LS145 J8818	-	-	-
U45	54LS145 J8818	-	-	-
U46	54LS145 J8818	-	-	-
U47	54LS145 J8818	-	-	-
U48	54LS145 J8818	-	-	-
U49	54LS145 J8818	-	-	-
U50	54LS145 J8818	-	-	-
U51	54LS145 J8818	-	-	-
U52	54LS145 J8818	-	-	-
U53	54LS145 J8818	-	-	-
U54	54LS145 J8818	-	-	-
U55	54LS145 J8818	-	-	-
U56	54LS145 J8818	-	-	-
U57	54LS145 J8818	-	-	-
U58	54LS145 J8818	-	-	-
U59	54LS145 J8818	-	-	-
U60	54LS145 J8818	-	-	-
U61	54LS145 J8818	-	-	-
U62	54LS145 J8818	-	-	-
U63	54LS145 J8818	-	-	-
U64	54LS145 J8818	-	-	-
U65	54LS145 J8818	-	-	-
U66	54LS145 J8818	-	-	-
U67	54LS145 J8818	-	-	-
U68	54LS145 J8818	-	-	-
U69	54LS145 J8818	-	-	-
U70	54LS145 J8818	-	-	-
U71	54LS145 J8818	-	-	-
U72	54LS145 J8818	-	-	-
U73	54LS145 J8818	-	-	-
U74	54LS145 J8818	-	-	-
U75	54LS145 J8818	-	-	-
U76	54LS145 J8818	-	-	-
U77	54LS145 J8818	-	-	-
U78	54LS145 J8818	-	-	-
U79	54LS145 J8818	-	-	-
U80	54LS145 J8818	-	-	-
U81	54LS145 J8818	-	-	-
U82	54LS145 J8818	-	-	-
U83	54LS145 J8818	-	-	-
U84	54LS145 J8818	-	-	-
U85	54LS145 J8818	-	-	-
U86	54LS145 J8818	-	-	-
U87	54LS145 J8818	-	-	-

PART OF PI	
A 19	HEAD CHANNEL 0 LOW
A 18	HEAD CHANNEL 1 LOW
A 17	HEAD CHANNEL 2 LOW
A 16	HEAD CHANNEL 3 LOW
A 15	HEAD CHANNEL 4 LOW
A 14	HEAD CHANNEL 5 LOW
A 13	HEAD CHANNEL 6 LOW
A 12	HEAD CHANNEL 7 LOW
U-5	
2	8 0 TRACK 4 PB A
2	8 0 TRACK 5 PB A
5	8 B TRACK 6 PB A
7	8 7 TRACK 7 PB A
2	7 0 TRACK 8 PB B
4	6 9 TRACK 9 PB B
6	6 5 TRACK 7 PB B
8	4 7 TRACK 7 PB B

Read/Write Interface, A2, Schematic
Figure 126 (Sheet 2 of 3)



Read/Write Interface, A2, Schematic
Figure 126 (Sheet 3 of 3)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity
	Updated all sheets to latest revision. Sheet 2: Combined sheets 2 and 3 for schematic 722-2441-301. Sheet 3: Added for schematic 722-2441-302. Sheet 4: Added for schematic 722-2441-303. Sheet 5: Added for schematic 722-2441-306. Sheet 6: Added for schematic 722-2441-307.	

Transport Interface Assembly, A3, Schematic
Figure 127 (Sheet 1 of 6)

31-30-32

Page 198.3
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

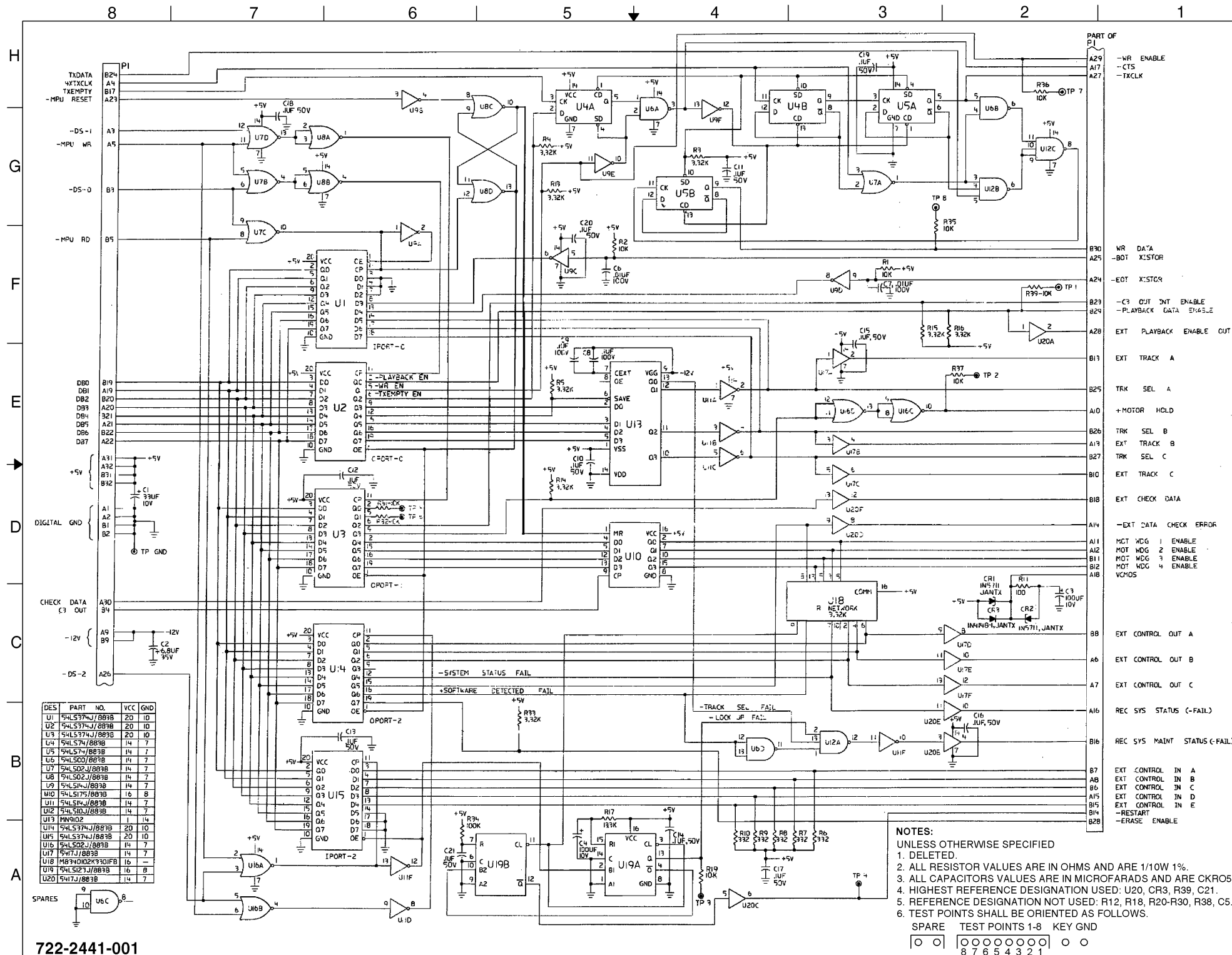
Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

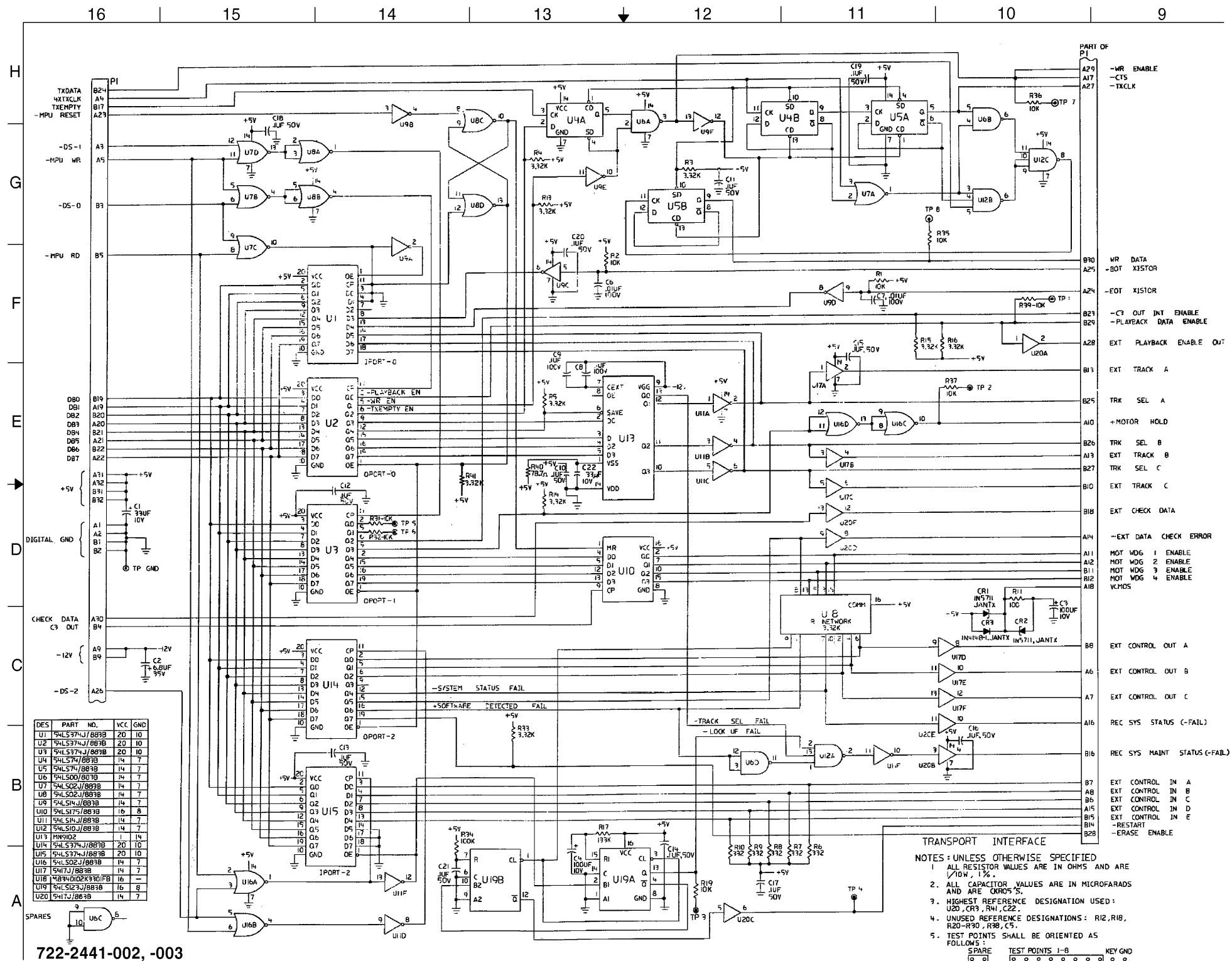
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

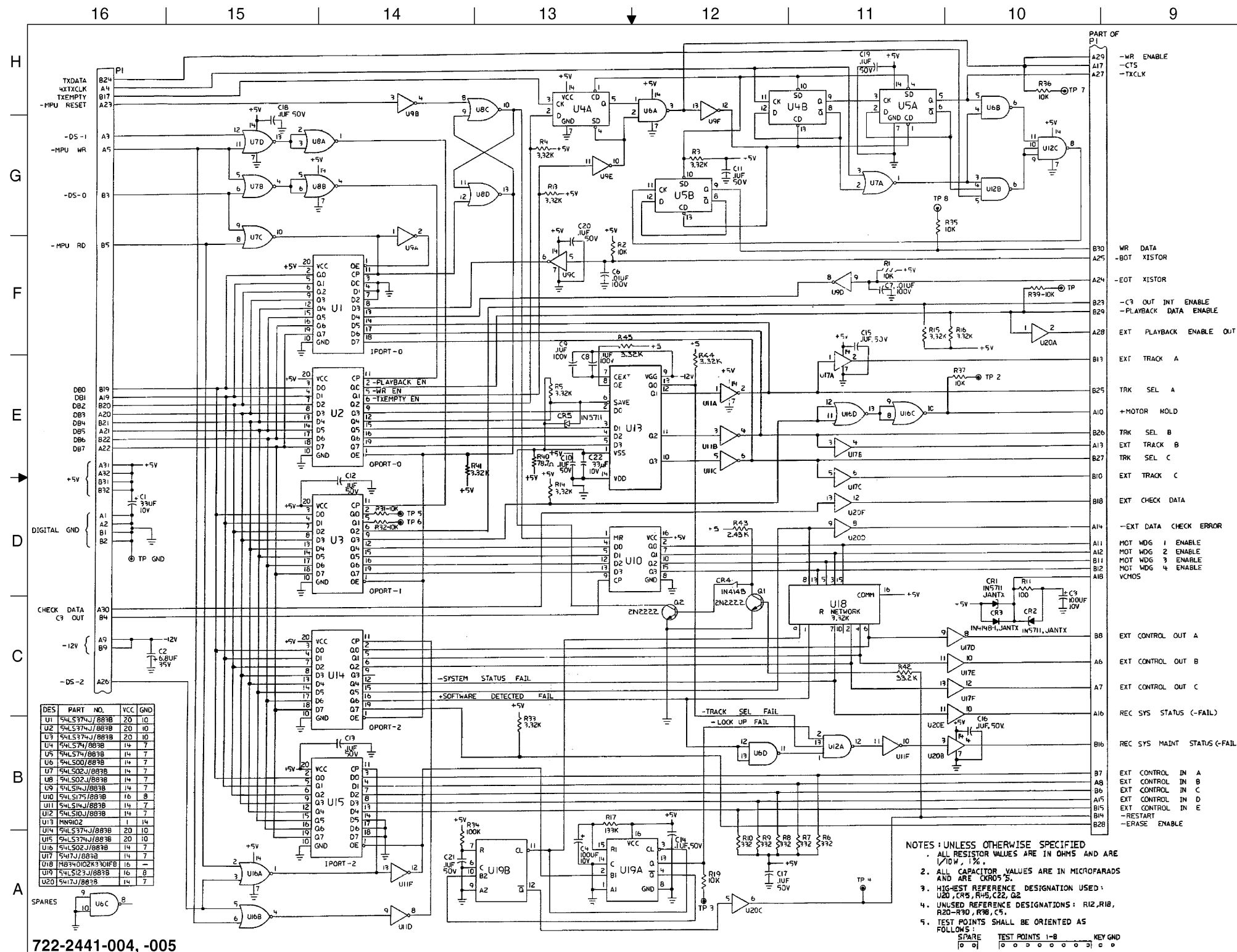


Transport Interface Assembly, A3, Schematic
Figure 127 (Sheet 2 of 6)

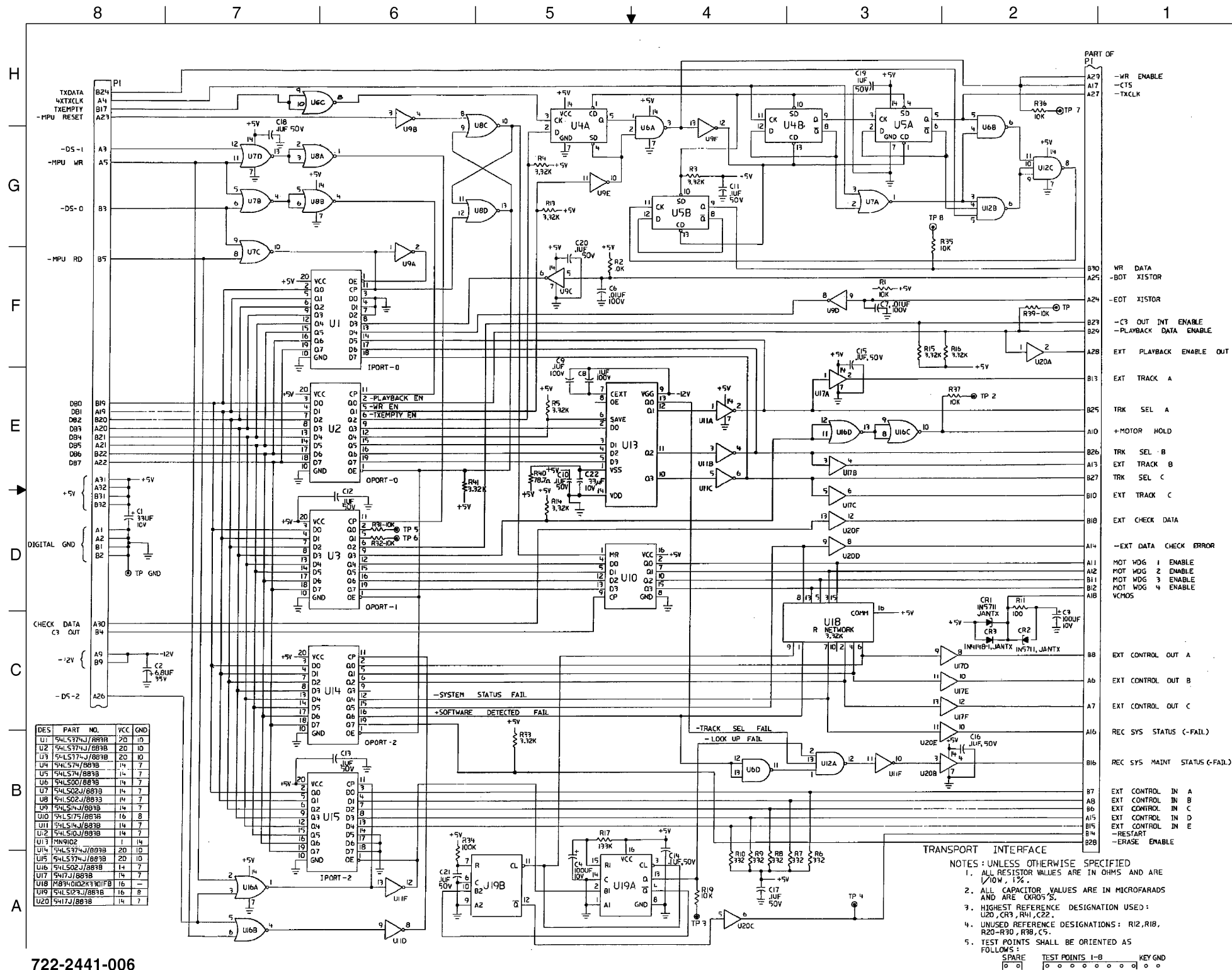


722-2441-002, -003

Transport Interface Assembly, A3, Schematic
Figure 127 (Sheet 3 of 6)

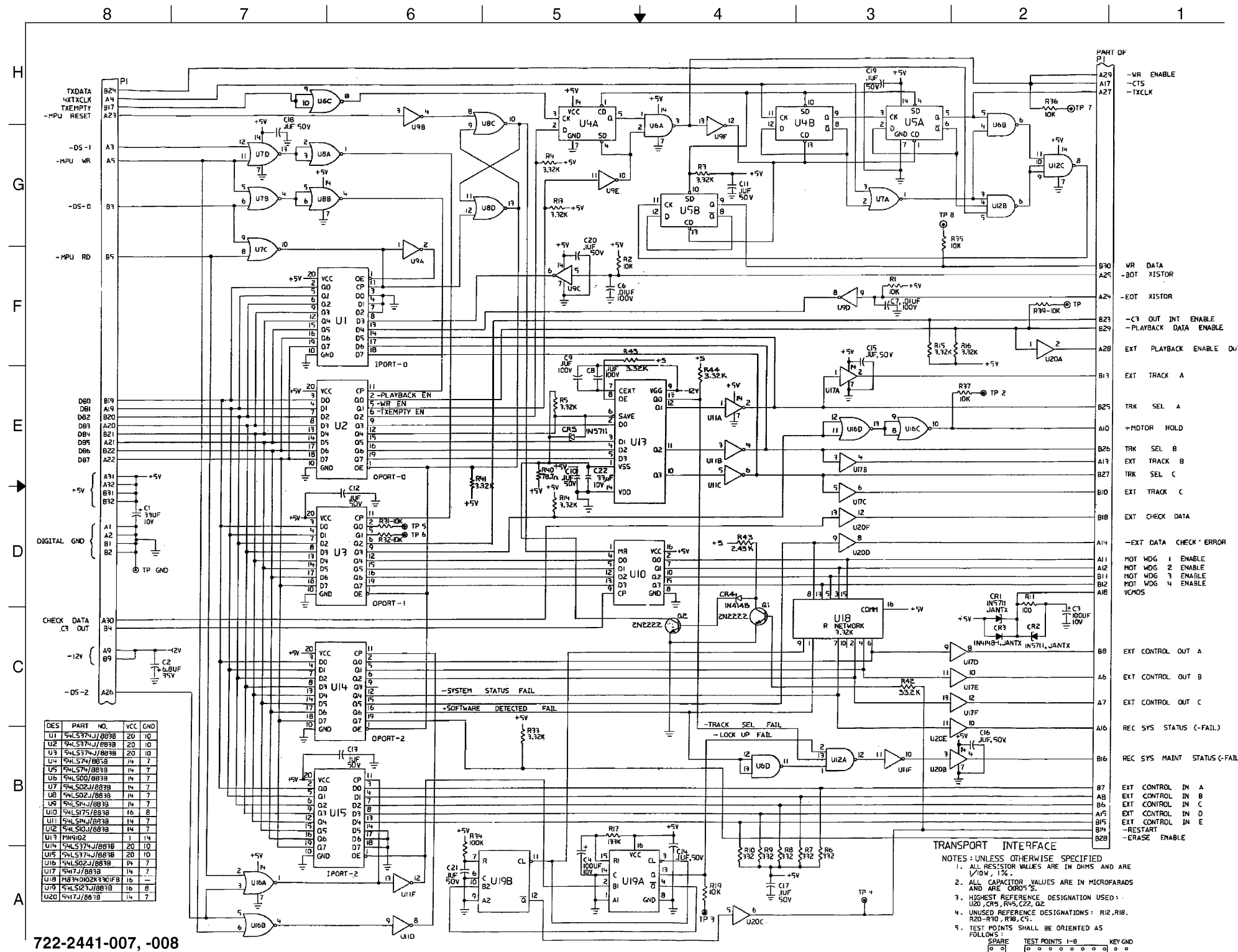


ID-180608 E722-2441-303-1 E722-2441-303-2



722-2441-006

Transport Interface Assembly, A3, Schematic
Figure 127 (Sheet 5 of 6)



722-2441-007, -008

Transport Interface Assembly, A3, Schematic
Figure 127 (Sheet 6 of 6)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity
	Updated all sheets to latest revision. Sheet 2: Combined sheets 2 and 3 for schematic 722-2438-301. Sheet 3: Added for schematic 722-2438-302. Sheet 4: Added for schematic 722-2438-303. Sheet 5: Added for schematic 722-2438-304 and added ADCN -1.	

Recording System Controller, A4, Schematic
Figure 128 (Sheet 1 of 5)

31-30-32

Page 198.15
22 Dec 2006

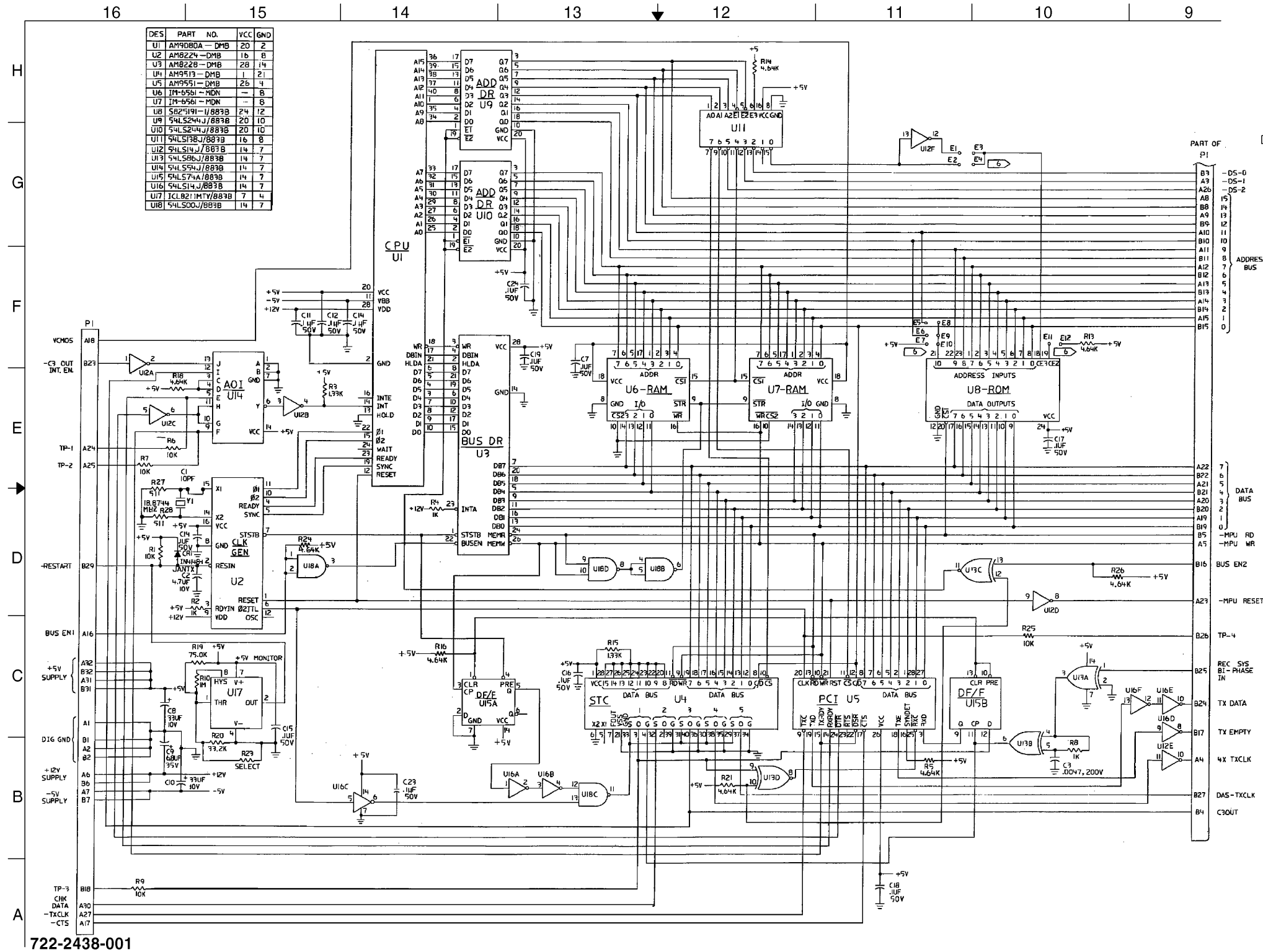
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

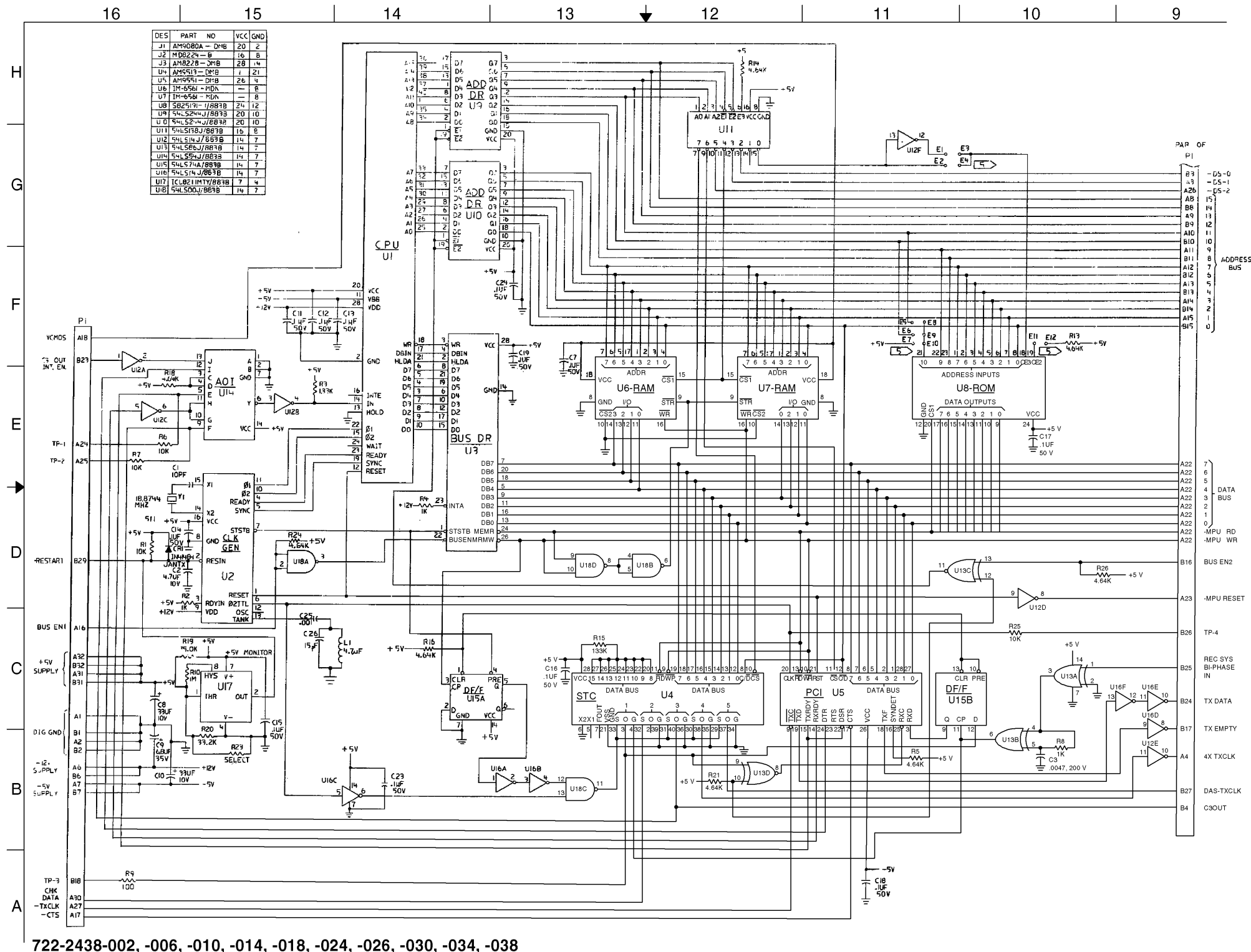
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK



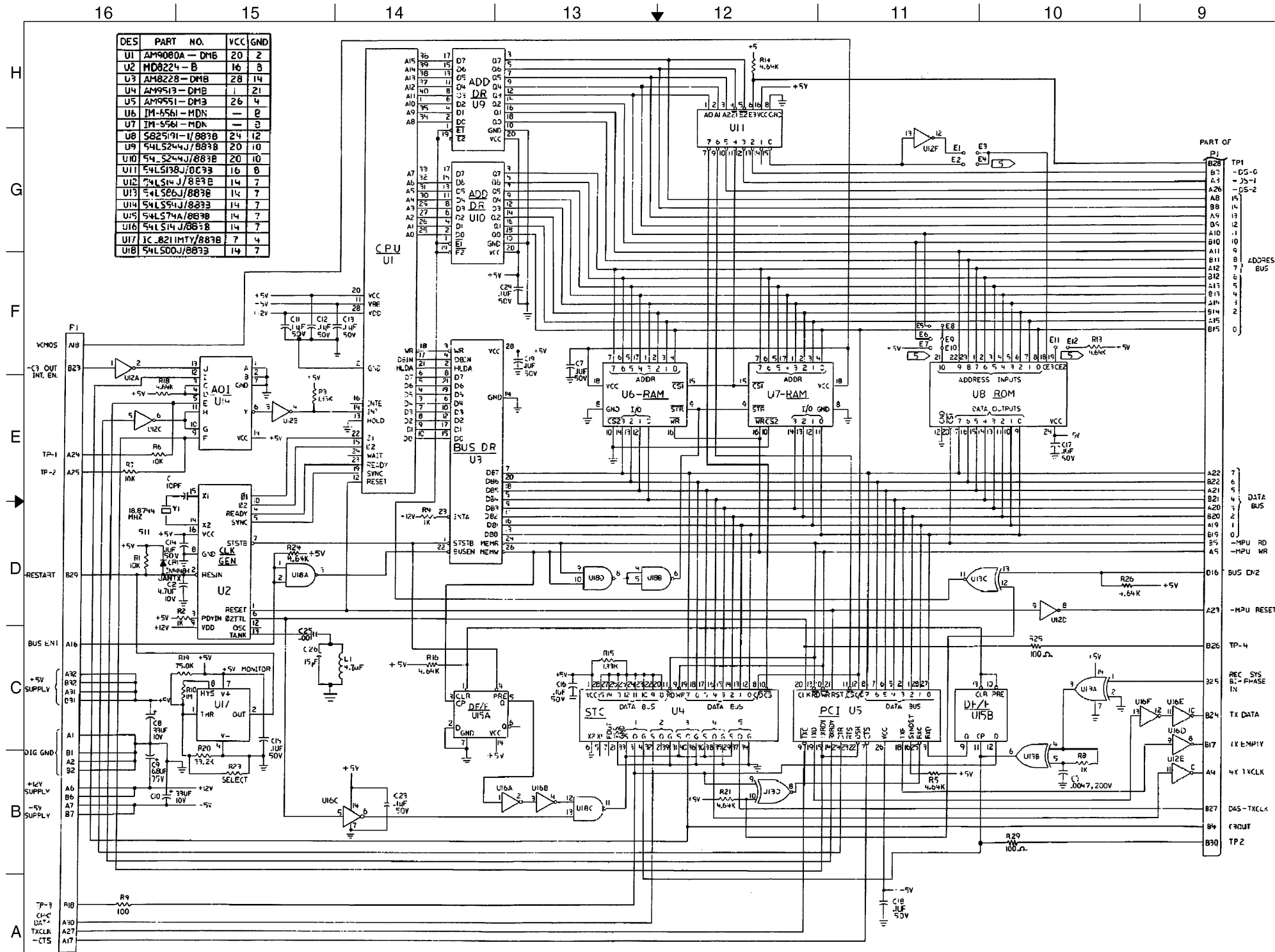
Recording System Controller, A4, Schematic
Figure 128 (Sheet 2 of 5)



- NOTES: UNLESS OTHERWISE SPECIFIED
1. ALL RESISTOR VALUES ARE IN OHMS & ARE 1/10W 1%. V1, R26, C20, P1, E12, U1.
 2. ALL CAPACITOR VALUES ARE IN MICROFARADS & ARE CK409'S.
 3. HIGHEST REFERENCE DESIGNATION USED: U18, CRI, V1, R26, C20, P1, E12, U1.
 4. REFERENCE DESIGNATIONS NOT USED: R11, R17, C4, C5, C6, C20-C22, R22, R12.
5. "E" NUMBER CONNECTION GUIDE:
- | ROM | CONNECT |
|--------|------------------------|
| E25191 | E1-E3, E11, E12, E6-E9 |
| 2716 | E2-E4, E6-E11, E7-E10 |
| 2732 | E2-E4, E6-E11, E5-E8 |
6. THIS REVISION LEVEL APPLIES TO ASSEMBLY 722-2438-002.

722-2438-002, -006, -010, -014, -018, -024, -026, -030, -034, -038

Functional Tester Control Panel
Figure 128 (Sheet 3 of 5)



NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL RESISTOR VALUES ARE IN OHMS & ARE 1/10W, 1%. 100K

2. ALL CAPACITOR VALUES ARE IN MICROFARADS & ARE CKR05'S.

3. HIGHEST REFERENCE DESIGNATION USED: U18, C11, V1, R29, C26, P1, E12, U1, TP2.

4. REFERENCE DESIGNATION NOT USED: R11, R17, C4, C5, C6, C20-C22, R22, R12, R9, R27, R28

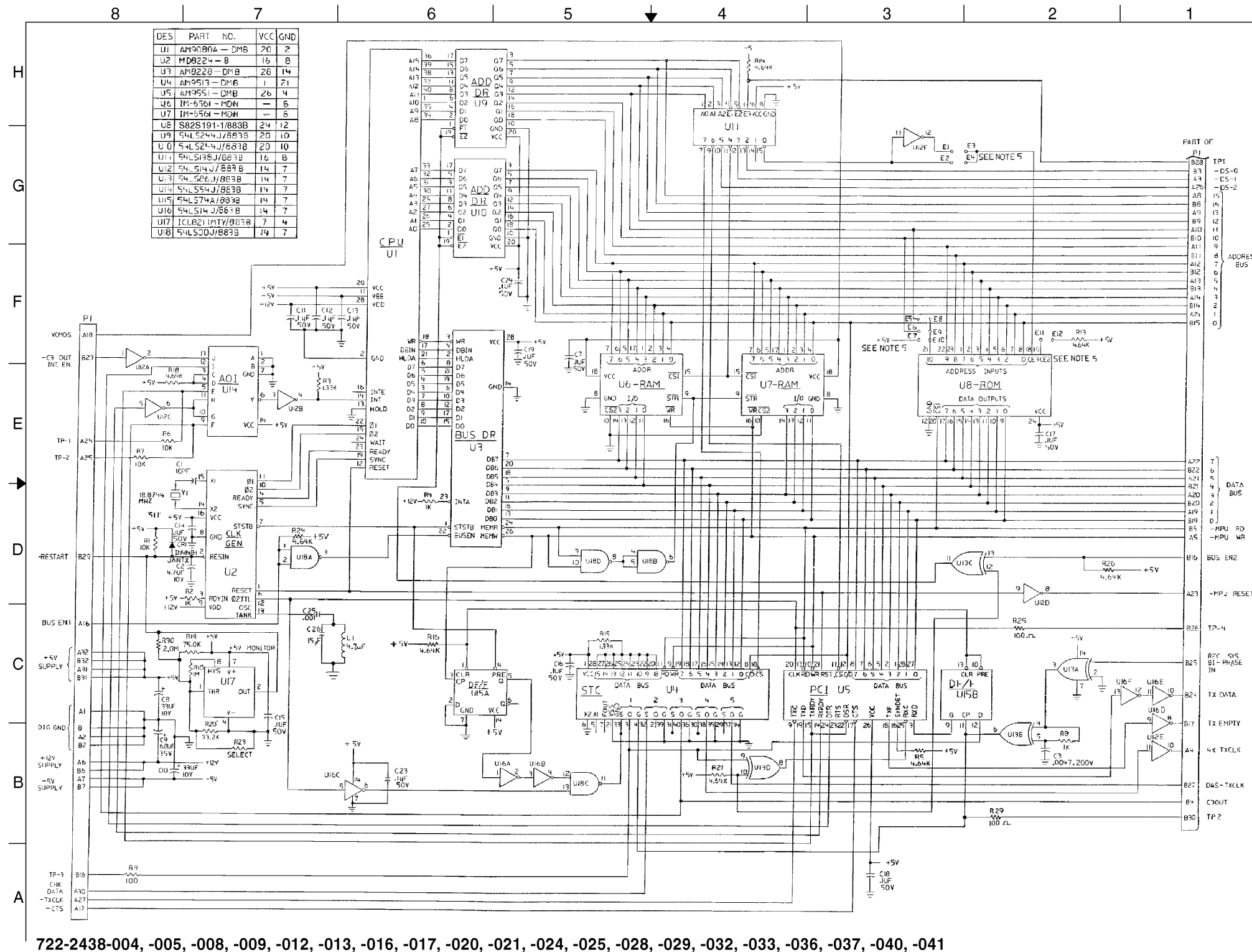
5. "E" NUMBER CONNECTION GUIDE:

ROM	CONNECT
B25191	E1-E3, E11-E12, E6-E9
2716	E2-E4, E6-E11, E7-E10
2732	E2-E4, E6-E11, E5-E8

6. THIS REVISION LEVEL APPLIES TO ASSEMBLY 722-2438-003.

722-2438-003, -007, -011, -015, -019, -023, -027, -031, -035, -039

Recording System Controller, A4, Schematic
Figure 128 (Sheet 4 of 5)



- NOTES: UNLESS OTHERWISE SPECIFIED
1. ALL RESISTOR VALUES ARE IN OHMS & ARE 1/10W, 1%.
 2. ALL CAPACITOR VALUES ARE IN MICROFARADS & ARE .001'S.
 3. HIGHEST REFERENCE DESIGNATION USED: U18, C11, Y1, R30, C26, P1, E12, L1, TP2.
 4. REFERENCE DESIGNATION NOT USED: R11, R17, C4, C5, C6, C20-C22, R22, R12, R9, R27, R28.
 5. "E" NUMBER CONNECTION GUIDE:
- | ROM | CONNECT |
|-------|-----------------------|
| 825P1 | E1-E3, E11-E12, E6-E9 |
| 2716 | E2-E4, E6-E11, E7-E10 |
| 2732 | E2-E4, E6-E11, E5-E8 |
6. THIS REVISION LEVEL APPLIES TO ASSEMBLY 722-2438-004.
 7. INTERPRET DRAWING PER DOD-STD-100.

722-2438-004, -005, -008, -009, -012, -013, -016, -017, -020, -021, -024, -025, -028, -029, -032, -033, -036, -037, -040, -041

Recording System Controller, A4, Schematic
Figure 128 (Sheet 5 of 5)

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Revision Reference Sheet

Revision No.	Revision Description	Serial No. Effectivity
	Updated all sheets to latest revision. Sheet 2: Combined sheets 2 and 3 for schematic 722-2501-301 and added ADCN A-1. Sheet 3: Added for schematic 722-2501-302.	

Airframe Wiring Interface, A8, Schematic
Figure 129 (Sheet 1 of 3)

31-30-32

Page 198.25
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

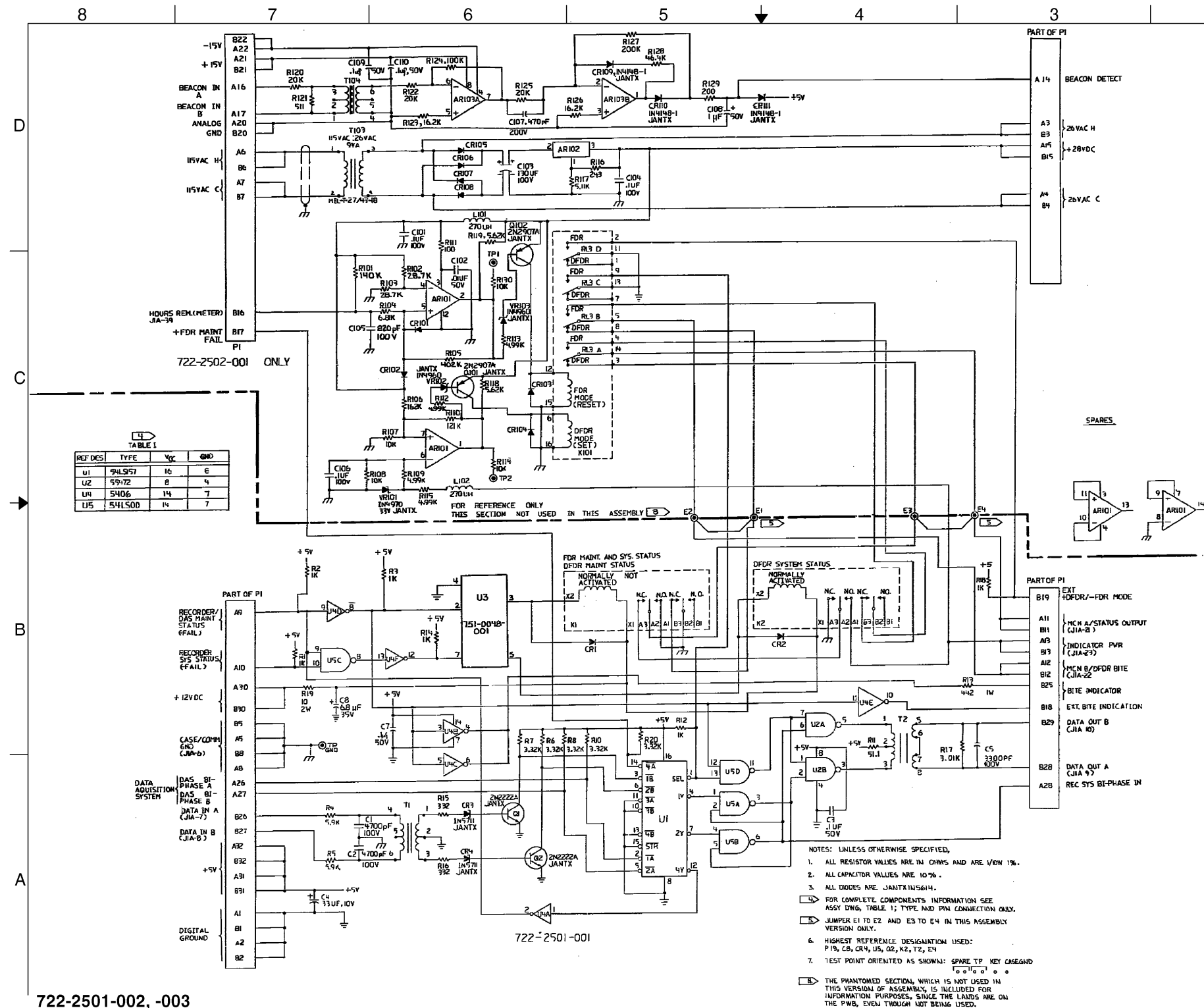
Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK



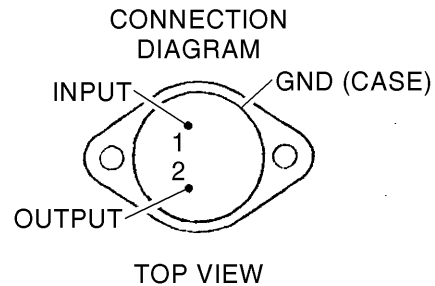
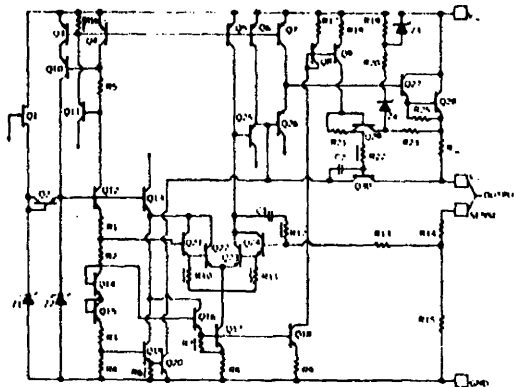
722-2501-002, -003
Airframe Wiring Interface, A8, Schematic
Figure 129 (Sheet 3 of 3)

Honeywell

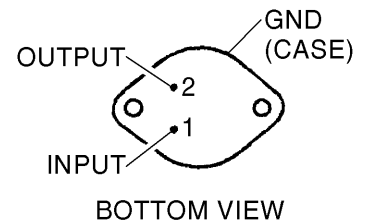
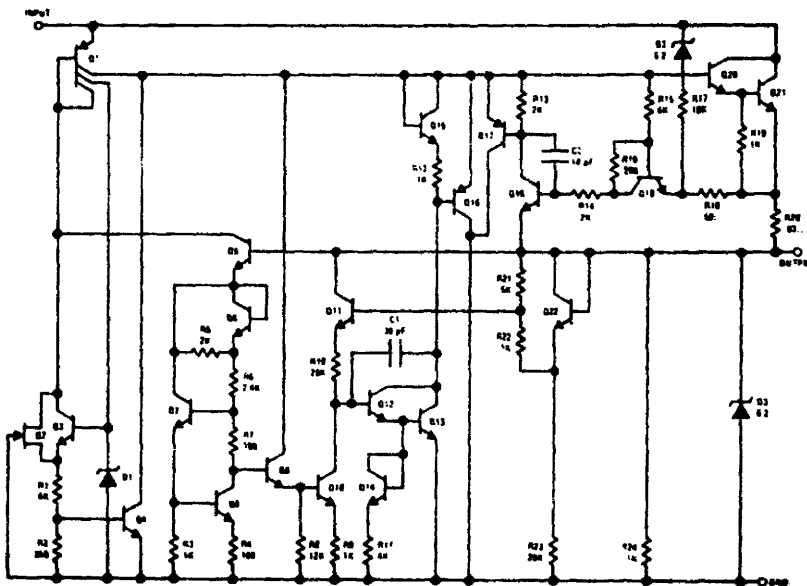
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

A1 (AR1, 2)
SG123K883B



A1 (AR1, 2)
LM123K883B



Integrated Circuit Diagram
Figure 130 (Sheet 1 of 14)

31-30-32

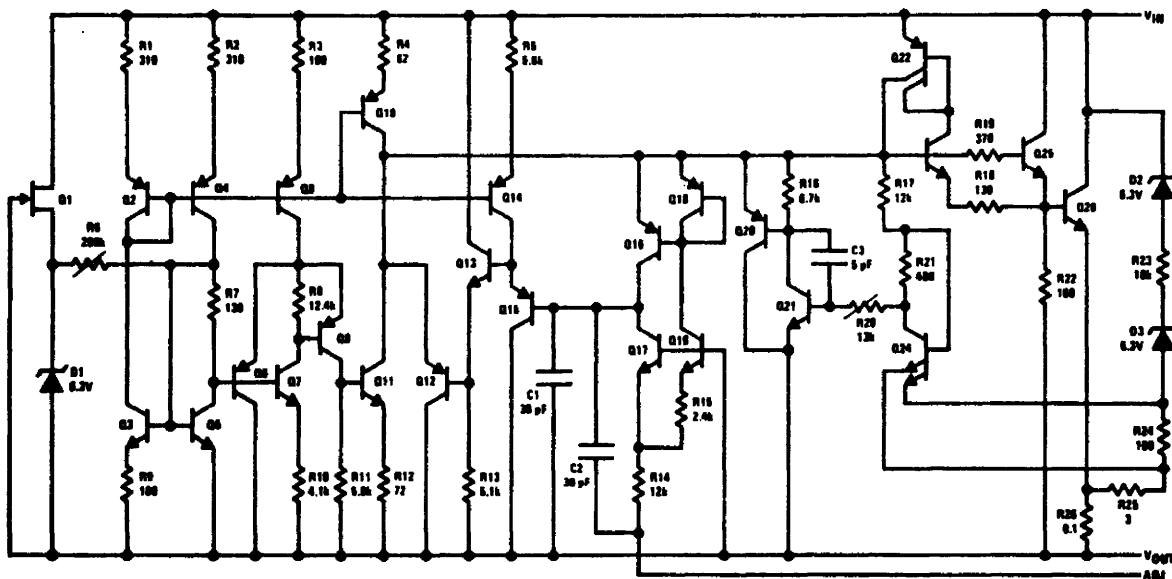
Page 198.31
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

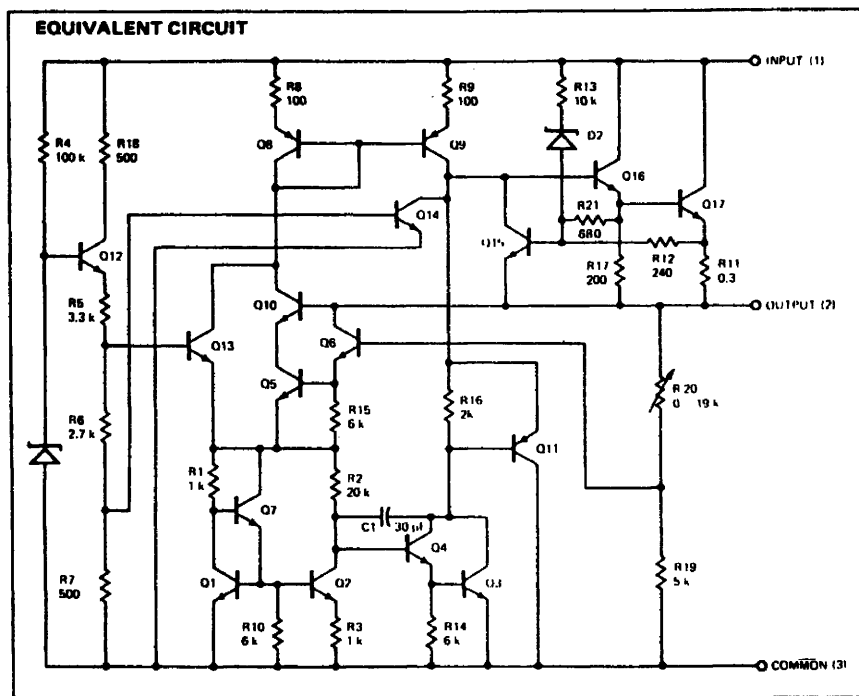
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

A1AR4
A2U13
LM117K883B
LM117H883B



A1AR5
UA7812KMQB



TOP VIEW

A diagram of a 3-pin connector. It consists of a central circle with two smaller circles inside it, one above and one below the center. The top circle is labeled '2' and the bottom circle is labeled '1'. The outer ring of the connector is labeled 'COMMON (3)' at the top, 'OUTPUT' at the top right, and 'INPUT' at the bottom right. The entire diagram is enclosed in a rectangular box.

ID-180611

Integrated Circuit Diagram
Figure 130 (Sheet 2 of 14)

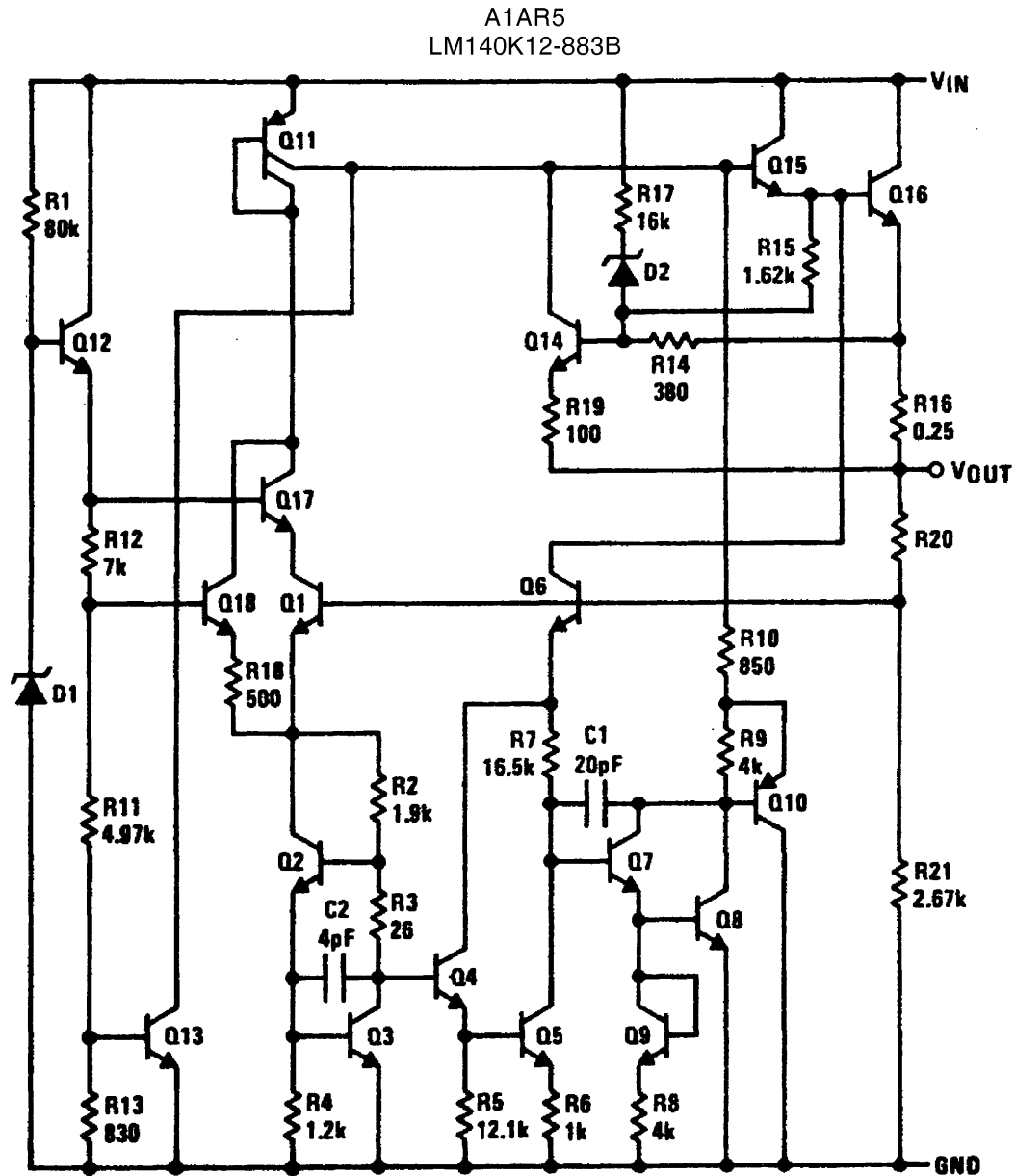
31-30-32

Page 198.32
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Integrated Circuit Diagram
Figure 130 (Sheet 3 of 14)

31-30-32

Page 198.33
22 Dec 2006

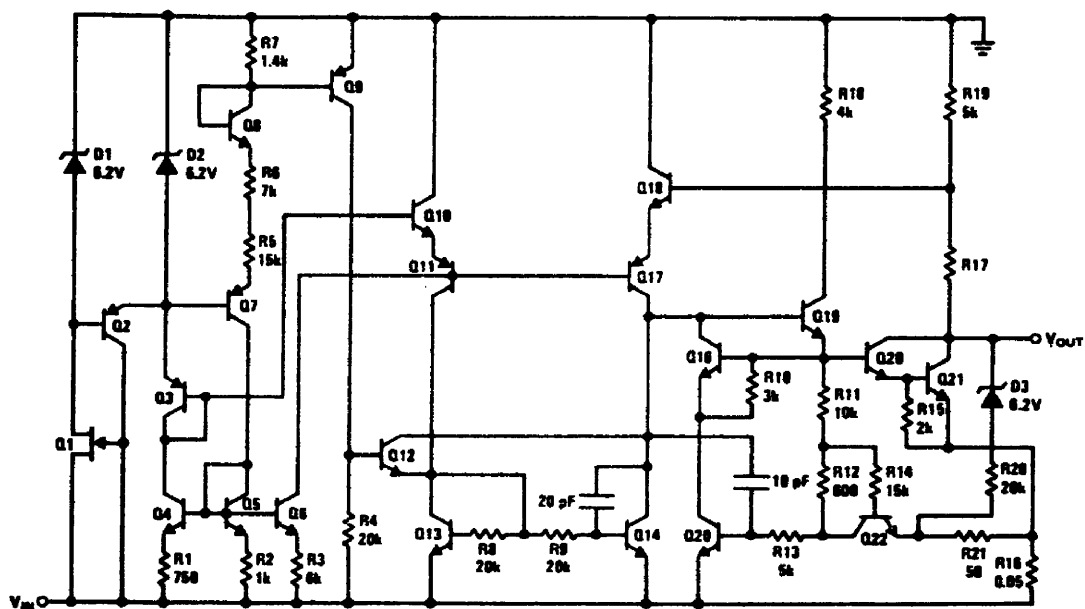
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIVALENT CIRCUIT

The diagram illustrates a complex equivalent circuit for a power MOSFET. It features a multi-stage gate driver and a power output stage. The gate driver includes transistors Q1, Q2, Q3, and Q4, which are configured to drive the gate of the main power MOSFET (Q5). The power stage consists of a common-emitter amplifier (Q6) followed by a push-pull output stage (Q7, Q8). The circuit includes various resistors (R1 through R100) for biasing and load, capacitors (C1 through C100) for coupling and bypassing, and diodes (D1 through D100) for protection and rectification. The input is labeled 'IN' and the output is labeled 'OUT'. The circuit is powered by a 15V supply and a 0V supply.

A1AR6
LM120K12-883B



Integrated Circuit Diagram
Figure 130 (Sheet 4 of 14)

Page 198.34
22 Dec 2006

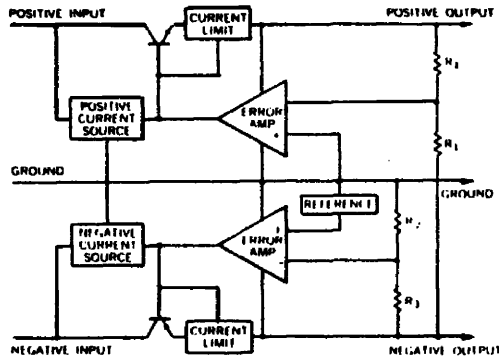
ID-180613

Honeywell

COMPONENT MAINTENANCE MANUAL

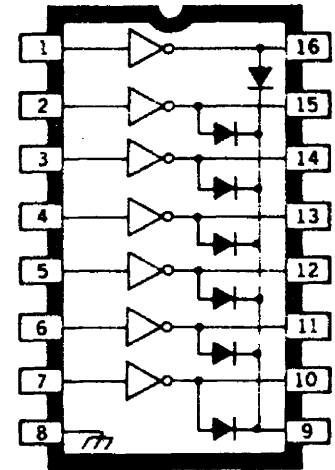
Universal Flight Data Recorder / Part No. 980-4100/980-4101

A1A1AR3
SG1501AJ883B

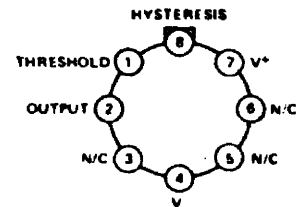
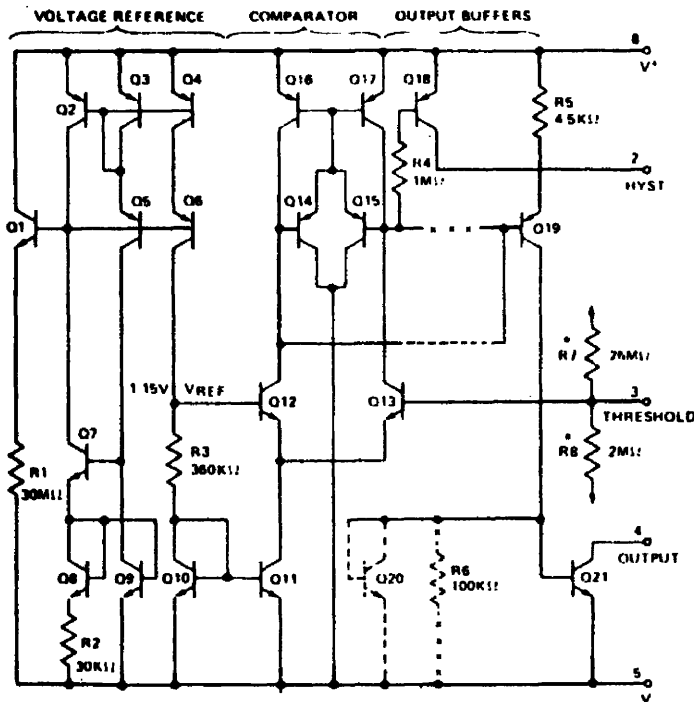


Block Diagram

A1A1 (U1, 2)
ULS2003H883B



A1A1U3
A4U17
ICL8211MTY



Integrated Circuit Diagram
Figure 130 (Sheet 5 of 14)

31-30-32

Page 198.35
22 Dec 2006

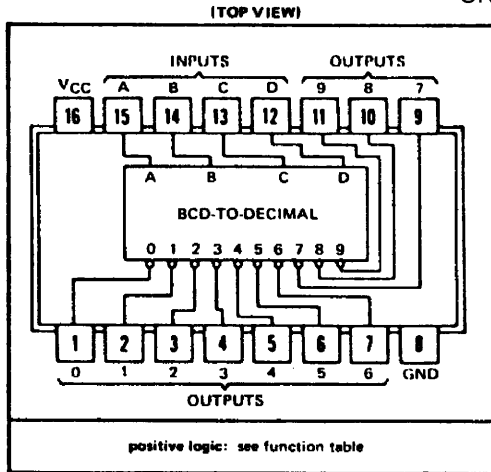
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

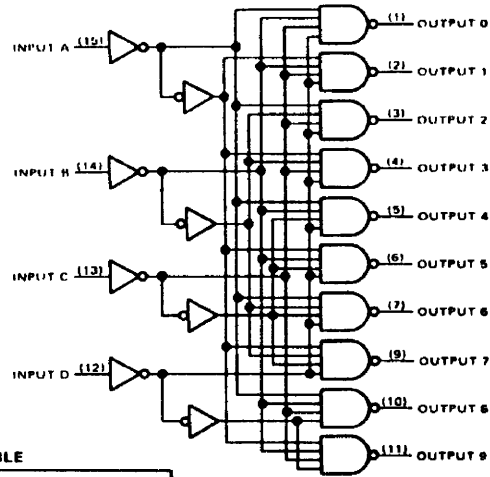
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

A2 (U1, 2)
U1
SNJ54LS145J00



functional block diagram

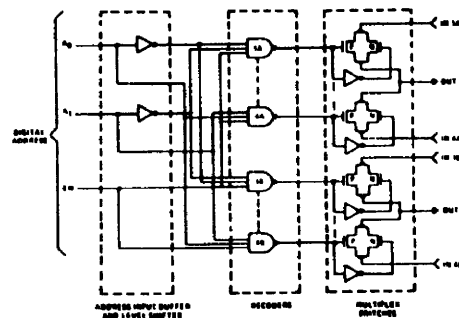
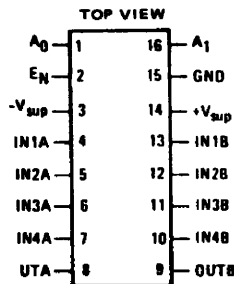


FUNCTION TABLE

NO.	INPUTS				OUTPUTS									
	D	C	B	A	0	1	2	3	4	5	6	7	8	9
0	L	L	L	L	L	H	H	H	H	H	H	H	H	H
1	L	L	L	H	H	L	H	H	H	H	H	H	H	H
2	L	L	H	L	H	H	L	H	H	H	H	H	H	H
3	L	L	H	H	H	H	L	L	H	H	H	H	H	H
4	L	H	L	L	H	H	H	L	L	H	H	H	H	H
5	L	H	L	H	H	H	H	L	L	H	H	H	H	H
6	L	H	H	L	H	H	H	H	L	L	H	H	H	H
7	L	H	H	H	H	H	H	H	H	L	L	H	H	H
8	H	L	L	L	H	H	H	H	H	H	H	L	L	H
9	H	L	L	H	H	H	H	H	H	H	H	H	L	L
INVALID	H	L	H	L	H	H	H	H	H	H	H	H	H	H
	H	L	H	H	H	H	H	H	H	H	H	H	H	H
	H	H	L	L	H	H	H	H	H	H	H	H	H	H
	H	H	L	H	H	H	H	H	H	H	H	H	H	H
	H	H	H	L	H	H	H	H	H	H	H	H	H	H

H = high level (off), L = low level (on)

A2 (U7, 8)
HI1-0509A8



Integrated Circuit Diagram
Figure 130 (Sheet 6 of 14)

31-30-32

Page 198.36
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

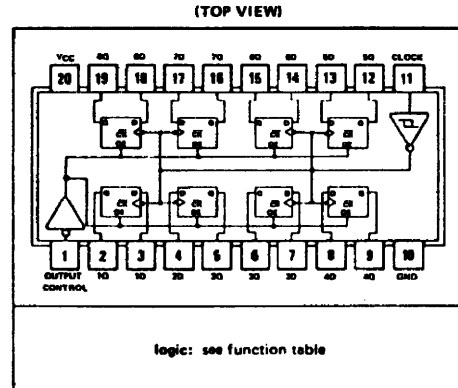
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

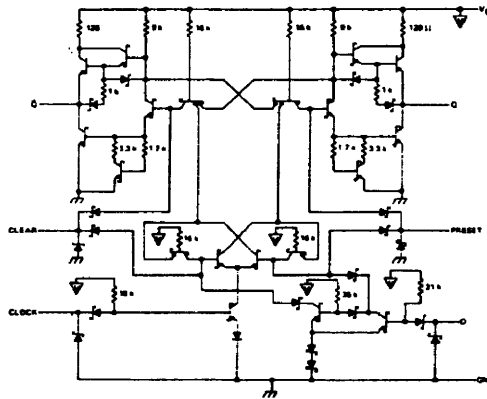
A3 (U1, 2, 3, 14, 15)
SNJ54LS374J00

FUNCTION TABLE

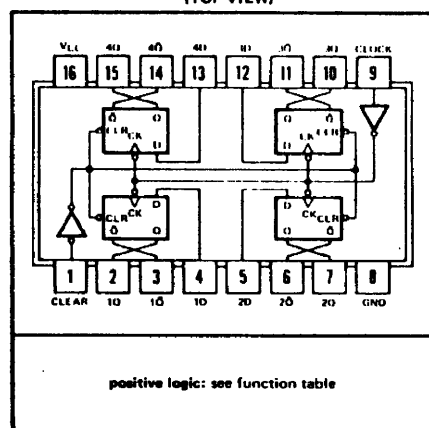
OUTPUT CONTROL	CLOCK	D	OUTPUT
L	↑	H	H
L	↑	L	L
L	L	X	Q_0
H	X	X	Z



A3 (U4, 5)
SNJ54LS74J00



A3U10
A4U15
SNJ54LS175J00



**FUNCTION TABLE
(EACH FLIP-FLOP)**

INPUTS			OUTPUTS	
CLEAR	CLOCK	D	Q	$\bar{Q}$
L	X	X	L	H
H	↑	H	H	L
H	↑	L	L	H
H	L	X	Q_0	$\bar{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
 Q_0 = the level of Q before the indicated steady state input conditions were established
↑ = '175, 'LS175, and 'S175 only

Integrated Circuit Diagram
Figure 130 (Sheet 7 of 14)

31-30-32

Page 198.37
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

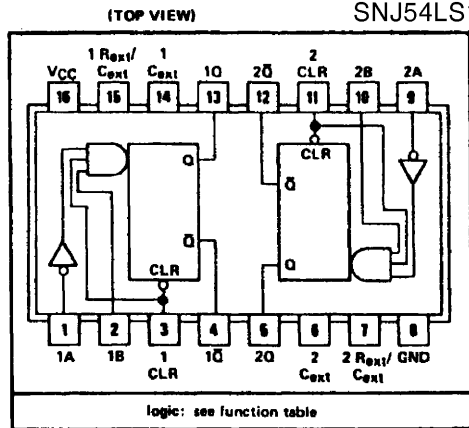
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

A3U19

SNJ54LS123J00

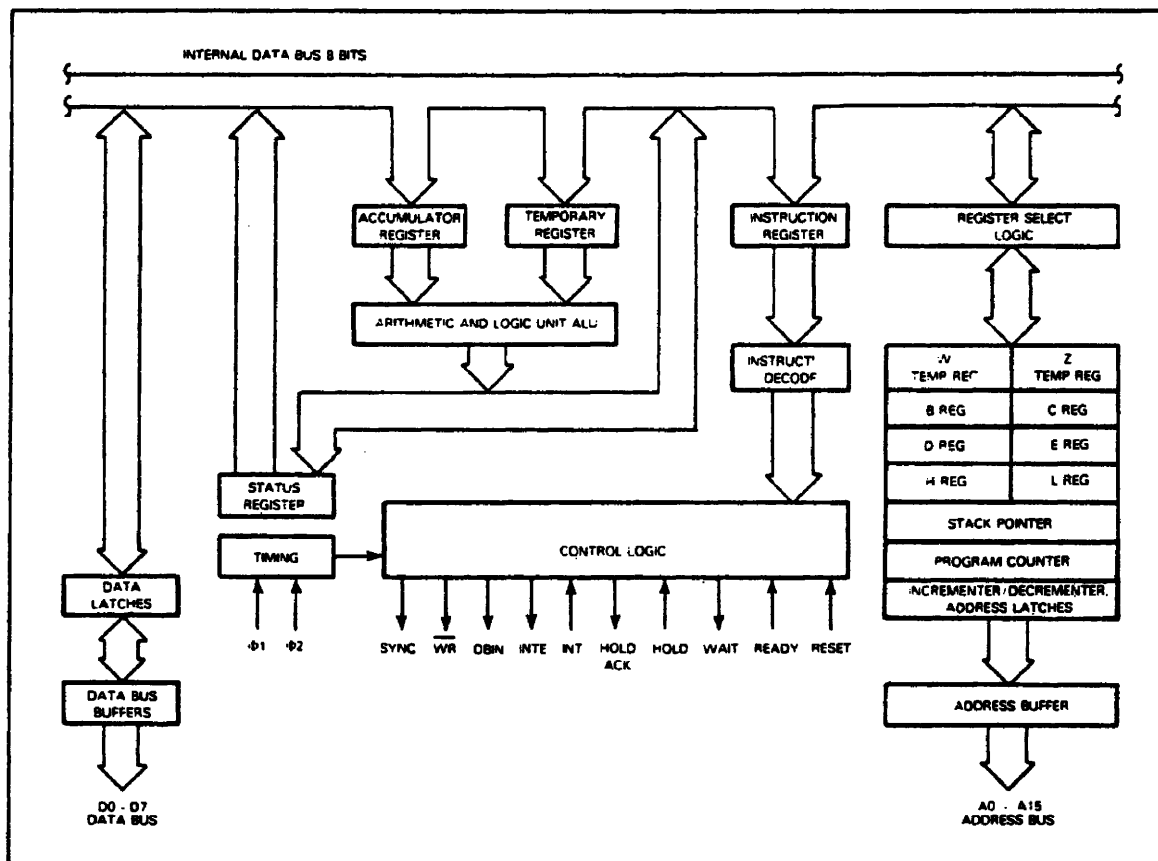


FUNCTION TABLE

INPUTS		OUTPUTS	
CLEAR	A B	Q	Q̄
L	X X	L	H
X	H X	L	H
X	X L	L	H
H	L 1	L	U
H	1 H	L	U
1	L H	L	U

A4U1

AM9080ADMB883B



Integrated Circuit Diagram
Figure 130 (Sheet 8 of 14)

31-30-32

Page 198.38
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

ID-180617

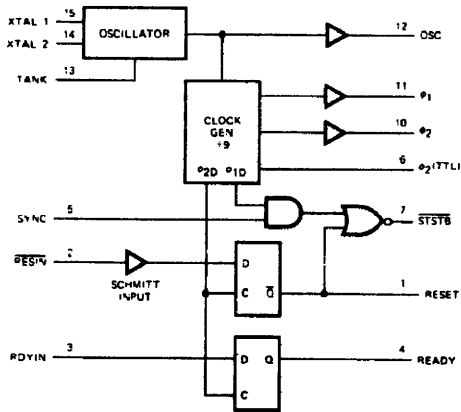
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

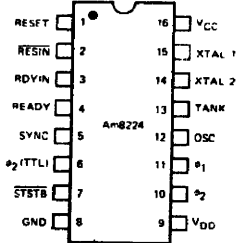
A4U2 MD8224B

LOGIC DIAGRAM



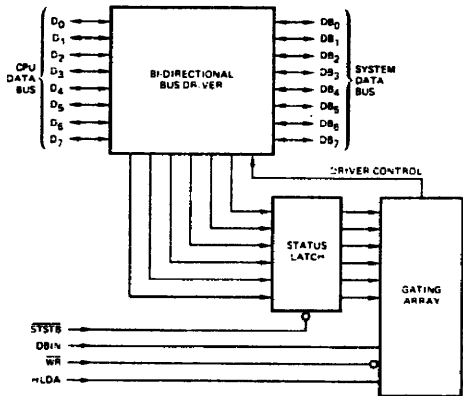
XTAL 1	CONNECTIONS FOR CRYSTAL
XTAL 2	
TANK	USED WITH OVERTONE XTAL
OSC	OSCILLATOR OUTPUT
ϕ_2 (TTL)	ϕ_2 CLK (TTL LEVEL)
VCC	+5.0V
VDD	+12V
GND	0V
RESIN	RESET INPUT
RESET	RESET OUTPUT
RDYIN	READY INPUT
READY	READY OUTPUT
SYNC	SYNC INPUT
STSTB	STATUS STB (ACTIVE LOW)
ϕ_1	Am9080A/8080A CLOCKS
ϕ_2	

Top View

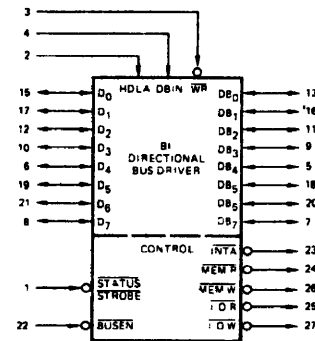


A4U3 AM8228DMB883B

LOGIC DIAGRAM

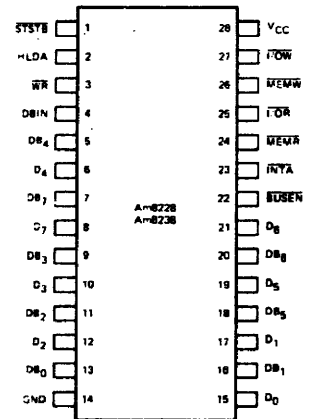


LOGIC SYMBOL



VCC = Pin 28
GND = Pin 14

Top View



Integrated Circuit Diagram
Figure 130 (Sheet 9 of 14)

31-30-32

Page 198.39
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

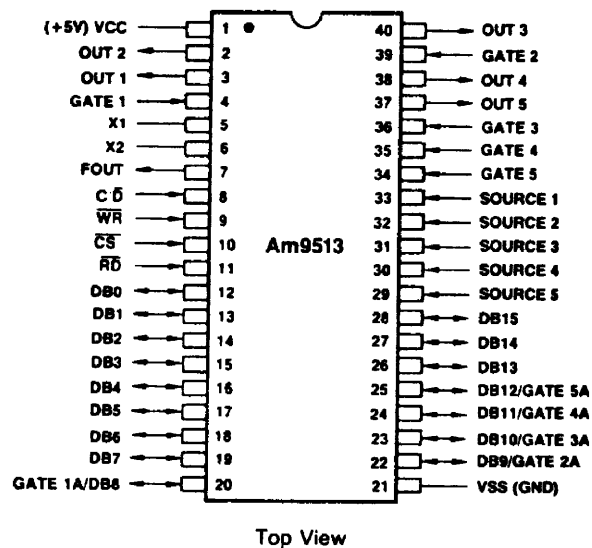
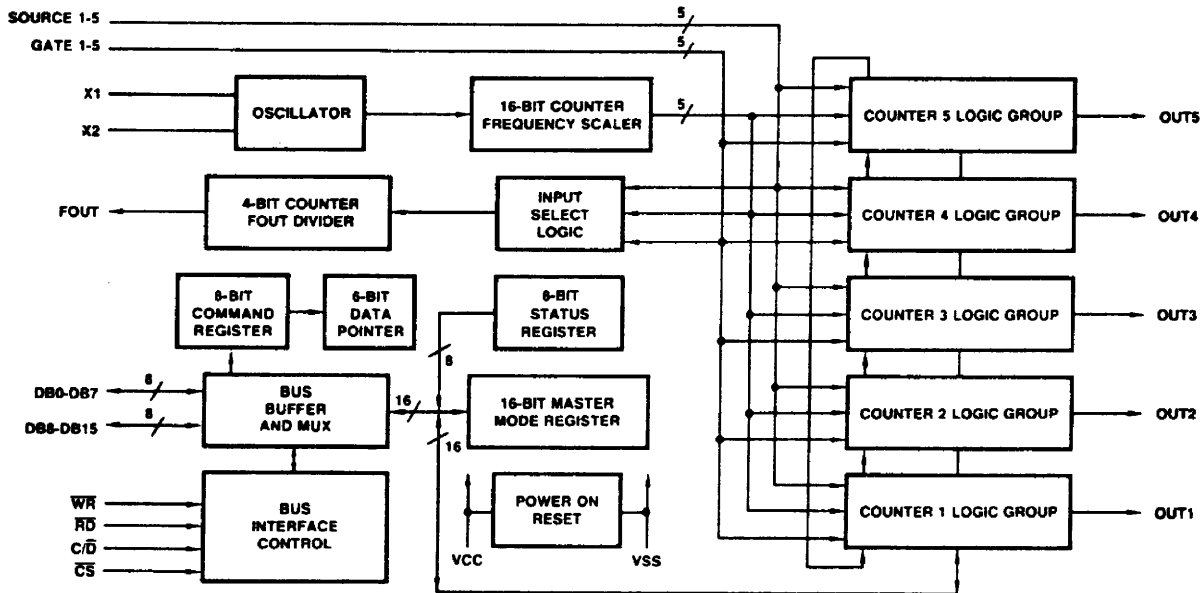
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

A4U4
AM9513DMB

GENERAL BLOCK DIAGRAM



Integrated Circuit Diagram
Figure 130 (Sheet 10 of 14)

31-30-32

Page 198.40
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

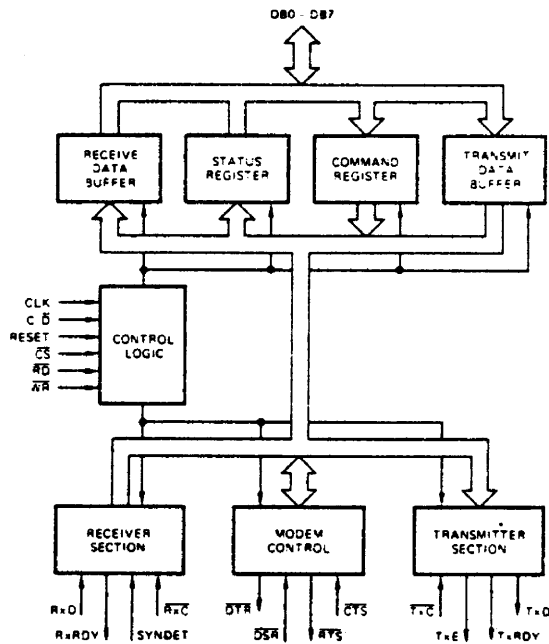
ID-180619

Honeywell

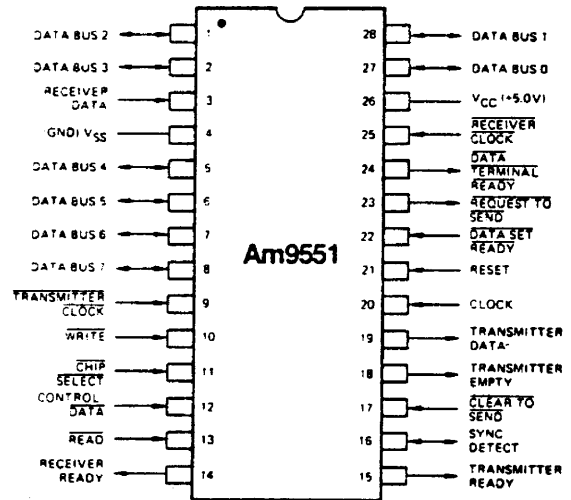
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

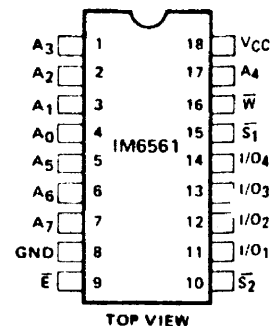
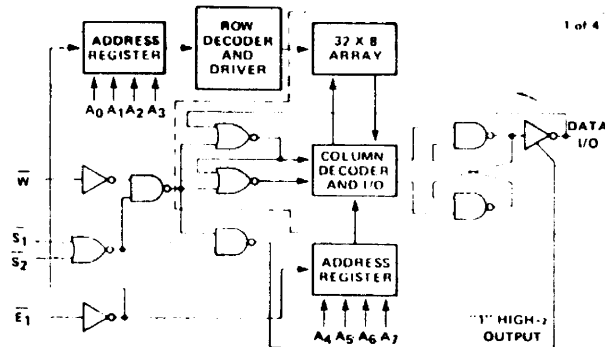
A4U5
AM9551DMB



Top View



A4 (U6, 7)
1M6561MDN883B



Integrated Circuit Diagram
Figure 130 (Sheet 11 of 14)

31-30-32

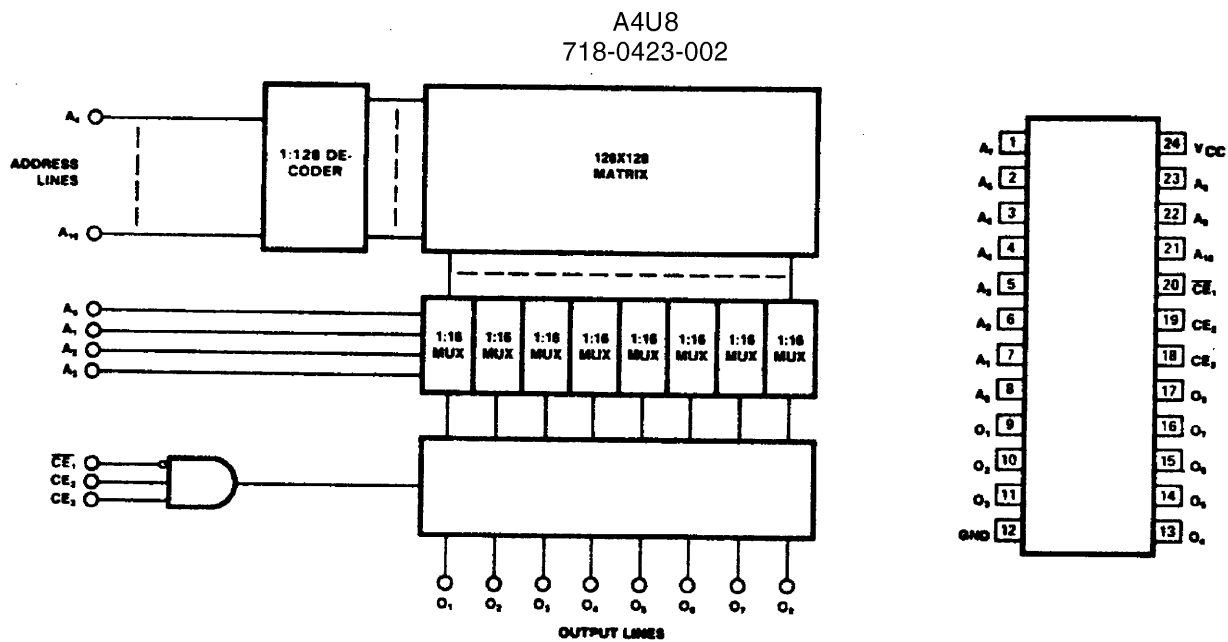
Page 198.41
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

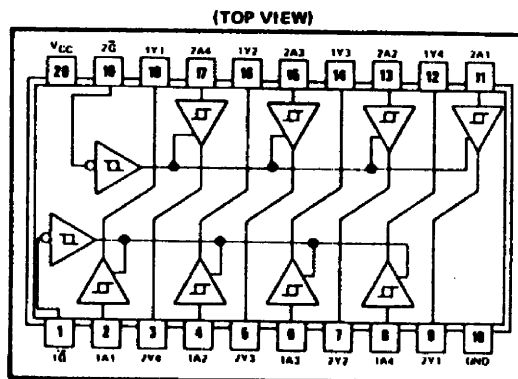
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



A4 (U9, 10)
SNJ54LS244J00



ID-180621

Integrated Circuit Diagram
Figure 130 (Sheet 12 of 14)

31-30-32

Page 198.42
22 Dec 2006

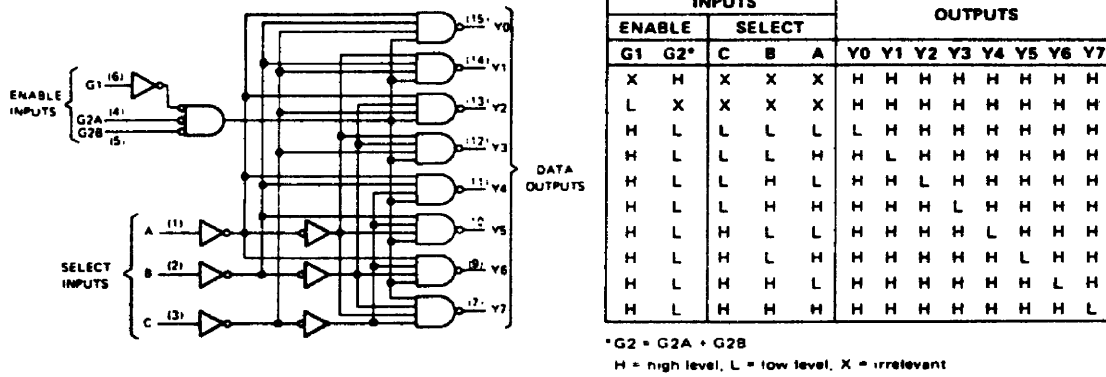
Honeywell

COMPONENT MAINTENANCE MANUAL

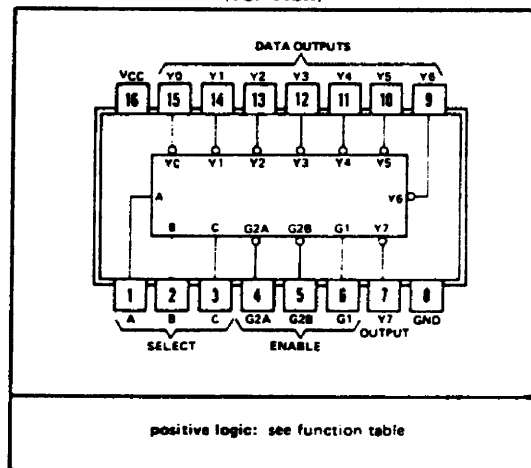
Universal Flight Data Recorder / Part No. 980-4100/980-4101

A4U11
SNJ54LS138J00

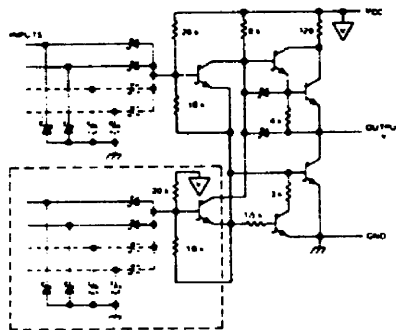
FUNCTION TABLE



(TOP VIEW)



A4U14
SNJ54LS54J00



Integrated Circuit Diagram
Figure 130 (Sheet 13 of 14)

31-30-32

Page 198.43
22 Dec 2006

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

(TOP VIEW)

INPUTS OUTPUT INPUTS OUTPUT

VCC STROBE 4A 4B 4Y 3A 3B 3Y

16 15 14 13 12 11 10 9

G 4A 4B 4Y 3A 3B

S 3Y

1A 1B 1Y 2A 2B 2Y

1 2 3 4 5 6 7 8

SELECT 1A 1B 1Y 2A 2B 2Y GND

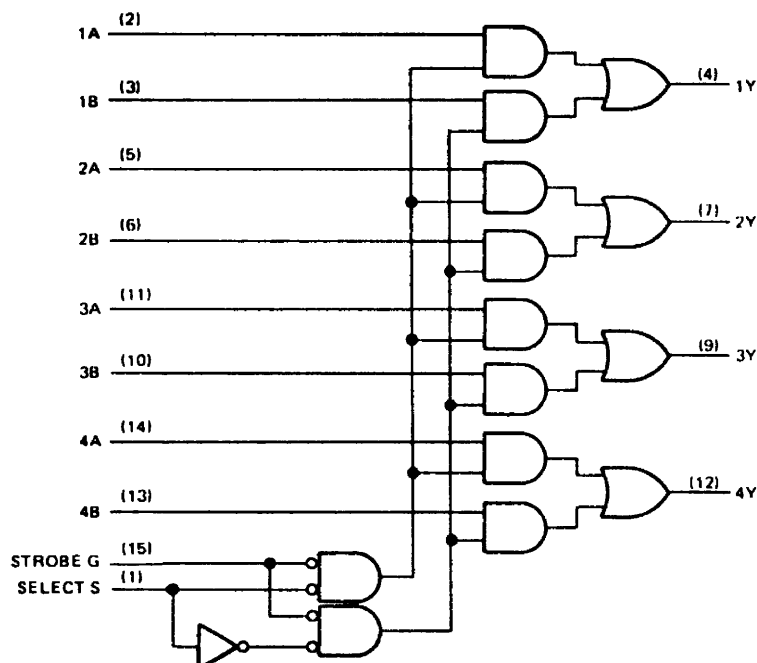
INPUTS OUTPUT INPUTS OUTPUT

positive logic:

Low level at S selects A inputs
High level at S selects B inputs

INPUTS				OUTPUT Y	
STROBE	SELECT	A	B	'157, '15157	'LS158 'S158
H	X	X	X	L	H
L	L	L	X	L	H
L	L	H	X	H	L
L	H	X	L	L	H
L	H	X	H	H	L

H = high level, L = low level, X = irrelevant



31-30-32

Page 198.44
22 Dec 2006

ID-180623

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

9. ATU Test Procedure

A. Inspection of UFDR

NOTE: Record all test results on the Acceptance Test Report data sheet and printout.

NOTE: Make an entry on the data sheet for steps (1) through (4).

- (1) Enter Accessory Code, Manufacturing Dash No., and Unit Serial No.
- (2) If the Tape Transport Assembly has been replaced, transfer the Tape Transport Assembly Serial No. from prior documentation to the data sheet.
- (3) Examine inspection records pertaining to any repair activity of the Tape Transport Assembly for completeness and acceptability.
- (4) Ensure that the accompanying repair tag contains both the diagnostic and repair details.

B. Base Software Installation

NOTE: This installation applies only on a one-time basis.

- (1) Install Microsoft Windows version 3.1 or above (set up to run in the 386 Enhanced mode) per the Microsoft Windows documentation.
- (2) Install ATEasy Version 1.10 or above per the ATEasy installation instructions (refer to the ATEasy User's Manual). A hard drive with a minimum of 3MB is required. Use the default directory name **ATEasy**.

C. ATP Software Installation

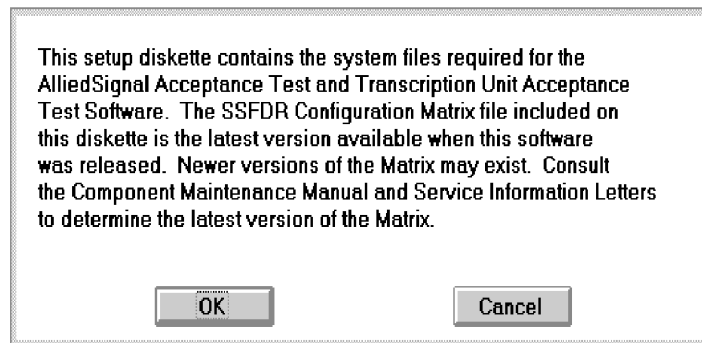
It is recommended that this section be performed at the beginning of a test session (the period during a given day that testing of the specific UUT is being done) but not necessarily before testing each UUT. If these steps have not been taken during the test session, the test program may not access the latest and correct files, causing the program to fail tests.

- (1) Start the ATU computer and start Windows by typing **WIN** at the command prompt. Insert Disk 1 of the ATU Setup Disk, 998-2439-XXX (Figure 137) set into the **A:** drive of the computer.
- (2) From the Program Manager, choose **File Run**, and enter **A:\SETUP** into the command field. Click **OK** or press **Enter**.
- (3) After a few moments, the dialog of instructions as shown in Figure 131 will appear. Note that the SSFDR Configuration Matrix mentioned in this dialog is PN 998-2273-5XX (see Figure 137 for latest version). Choose **OK**.
- (4) A dialog that provides a means to identify which version of the ATU software to be installed will appear as shown in Figure 132 below. Choose the button that matches the configuration of ATU being setup and click **OK**.
- (5) The next dialog box provides a means for selecting which software to install as shown in Figure 133 below. Unless there is a reason to do otherwise, select **Install All Software** and click **OK**.

Honeywell

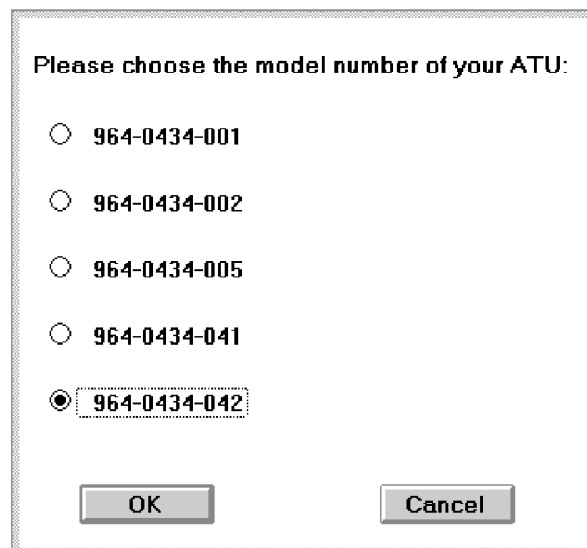
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180624

Initial Instructions Dialog
Figure 131



ID-180625

ATU Configuration Select Dialog
Figure 132

31-30-32

Page 198.46
22 Dec 2006

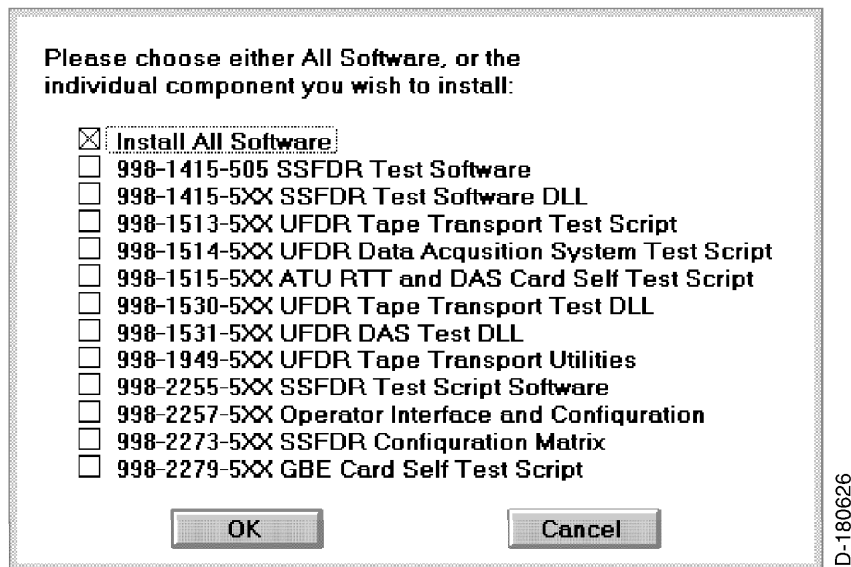
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Software Choice Dialog
Figure 133

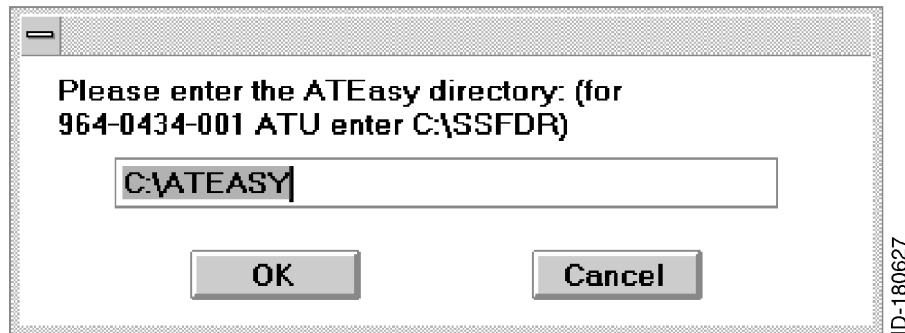
NOTE: This dialog provides check boxes for all software part numbers so that if it is known that one of the software versions has been damaged or needs replacing, the setup program provides a means to do so. Managing the configurations is made easier, however, by selecting **Install All Software**. If this is always done, it is assured that a complete set of the latest configurations of software will always be available. This is the recommended method for use of the setup disk.

- (6) The next dialog box, Figure 134, provides a place to enter an alternate directory for the ATEasy root location. Normally this will be left as **C:\ATEASY**, but may be changed if there is more than one instance when ATEasy is required. If the ATU is a **-001 ATU**, enter **C:\SSFDR** since there is no ATEasy on this version of ATU (the setup program also installs MS-DOS executables for the **-001 ATU**. Use this dialog to specify the path for the software). Click **OK**.
- (7) The software copy process begins. After all the files on Disk 1 are copied, a dialog box requesting Disk 2 be put in the floppy disk drive will appear as shown in Figure 135. Follow the instructions and insert the second disk.
- (8) When the software installation process is completed, the setup program automatically sets up a Program Group, which contains icons for each of the software items selected to be installed. See Figure 136, below.

Honeywell

COMPONENT MAINTENANCE MANUAL

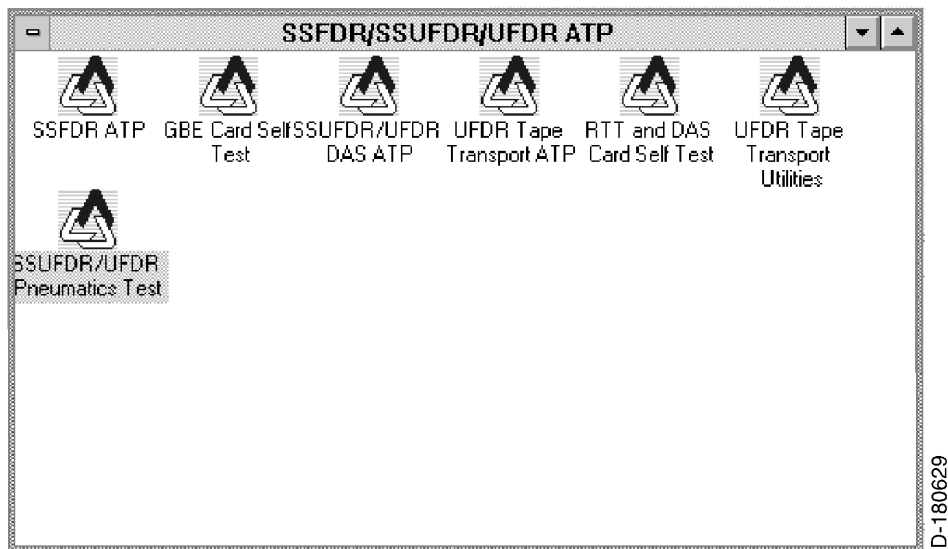
Universal Flight Data Recorder / Part No. 980-4100/980-4101



ATEasy Directory Dialog
Figure 134



Change Disk Dialog
Figure 135



SSFDR/SSUFDR/UFDR ATP Program Group
Figure 136

31-30-32

Page 198.48
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (9) Each of these icons automatically starts ATEasy and automatically runs the associated script. If it is required that the script is loaded, but not automatically executed, each icon must be selected from the **File** menu by choosing **Properties** and change the command string. For example, SETUP creates the icon for the UFDR Tape Transport Test by setting the command string to include the string **/rrtt.prg**. This would have to be changed to **/lr tt.prg**. The first letter "r" after the slash (/) tells ATEasy to load and run the script. Change this to letter "l" (lowercase L) to tell ATEasy to only load the script. Each icon that was required to load only would have to be changed by looking for the "r" and the script name which ends with ".prg".
- (10) Note, that the icons shown in Figure 137, above, include all icons possible for -042 ATU configuration. There may be fewer icons depending on which ATU part number is being set up.

NOTE: *Note that although the software being installed comes with a disk labelled 998-2439-5XX, in fact many other part numbers are being installed. These other part numbers will be seen in the test log when the ATP is performed. This is by design. Use Figure 137 to determine which software is installed on the ATU when installing from the 998-2439-5XX ATU Setup Disk.*

NOTE: *Note that a copy of all the software part numbers need not be kept as the ATU Setup Disk contains the exact same software. Thus, resources and energy can be saved by using only the ATU Setup Disk for maintaining this software.*

D. Install the ADRAS for Windows Software

This software comes with a setup utility that performs all the software setup for you automatically. This software must be run in Windows.

- (1) Start the Windows 3.1 operating system. Click on **File** from the Windows **Program Manager**. At the drop down menu, click on **Run**. The Run dialog appears as shown in Figure 138 below.
- (2) Insert the part number 998-1520-50X ADRAS for Windows Engine/Setup in the 3 1/2 inch floppy drive. In the **Command Line** type **drive:SETUP** per the example shown above and press **Enter**, where **drive** is the 3 1/2 inch floppy drive of the PC. When the setup utility starts to execute, the setup initiation screen appears as shown in Figure 139 below.
- (3) Click on the box by **AFW Engine** to toggle it to an x. Click on the box by any other options you plan to install at this time. Note that diskettes must be available for each option you want to install.
- (4) A dialog will ask for the path to install to. Select the path by clicking **OK** or pressing **Enter**, or change it by typing in a different path.
- (5) Then a message will tell you to install diskettes, always starting with ADRAS for Windows Engine/Setup diskette show below in Figure 140. Insert diskettes when requested.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

UFDR = HII P/N 980-4100-XXX; SSFDR = HII P/N 980-4700-XXX; SSUFDR = HII P/N 980-4120-XXX I = Disk Included with ATU S = Part of the ATU Setup Disk B = Part of AFW ATU Bundle									
Part Number	Rev	Description	LRU	964-0434 ATU Model					
				-001	-002	-003	-005	-041	-042
998-1415-505	-	SSFDR Test Software	SSFDR, SSUFDR	S	S			S*	S*
998-1415-509	-	SSFDR Test S/W DLL	SSFDR, SSUFDR		S		S	S	S
998-1469-501	-	GBE Test Software	Tests GBE Card in ATU	S	S				
998-1513-513	-	RTT Test Script	UFDR		S		S		S
998-1514-510	-	DAS Test Script	UFDR, SSUFDR		S		S		S
998-1515-507	-	ATU Self Test Script	Tests RTT and DAS Card in ATU		S		S	S	S
998-1520-507	-	ADRAS for Windows Engine	UFDR, SSFDR, SSUFDR	B	B	B	B	B	B
998-1520-567	-	ADRAS for Windows ATU Bundle		I	I	I	I	I	I
998-1527-504	-	CSV Output Option	UFDR, SSFDR, SSUFDR	B	B	B	B	B	B
998-1530-502	-	RTT DLL	UFDR		S		S	S	S
998-1531-503	-	DAS DLL	UFDR, SSUFDR		S		S	S	S
998-1539-505	-	UFDR Input Option	UFDR	B	B	B	B	B	B
998-1556-503	-	SSFDR, SSUFDR Readout	SSFDR, SSUFDR	B	B	B	B	B	B
998-1913-502	-	Cardtalk Version 3.15	UFDR, SSFDR, SSUFDR	I	I	I	I	I	I
998-1949-501	-	UFDR Tape Transport Utilities	UFDR		S		S		S
998-2255-516	-	SSFDR ATEasy Test Script	SSFDR, SSUFDR					S	S
998-2257-509	-	Operator Interface and Configuration	SSFDR, SSUFDR		S		S	S	S
998-2273-506	-	SSFDR Configuration Matrix	SSFDR, SSUFDR					S	S
998-2279-504	-	GBE Card Self Test Script	Tests GBE Card in ATU					S	S
998-2439-518	-	SSFDR/SSUFDR/UFDR ATP Setup Disk	Provides all test scripts in one Windows Setup disk	I	I		I	I	I
998-2697-501	-	GBE Card Config. Matrix						S	S
* Use for Download only on -041 and -042 ATU.									

ID-180630

ATU Model Software List
Figure 137

31-30-32

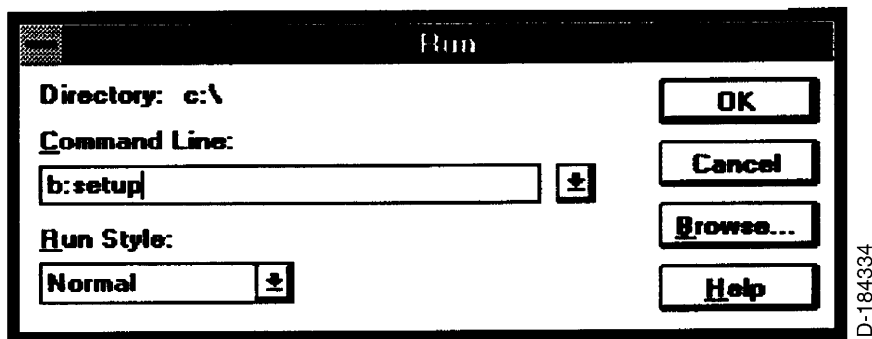
Page 198.50
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

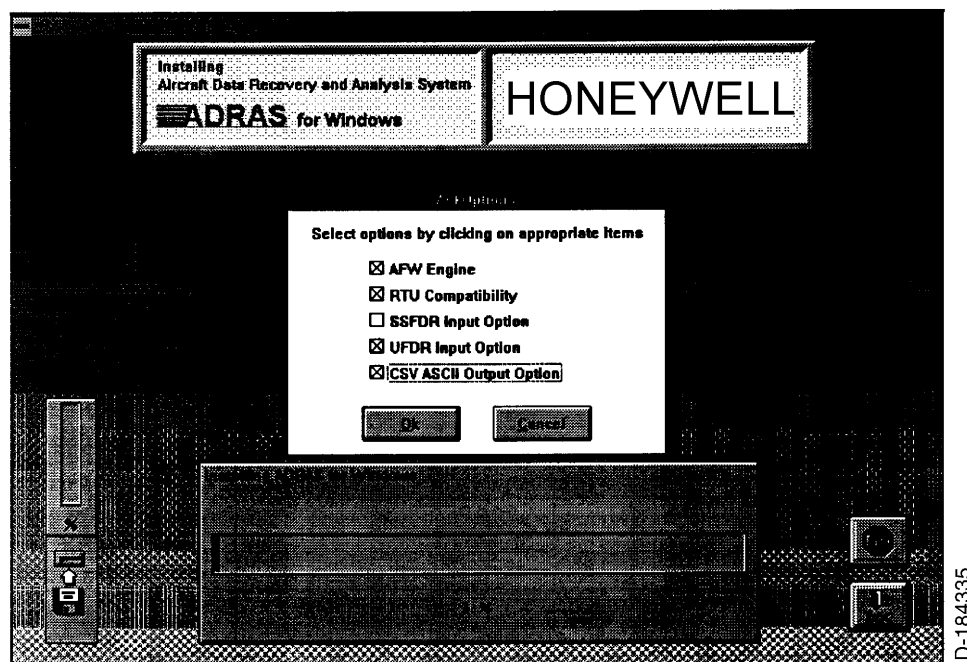
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Run Dialog
Figure 138



Setup Initiation
Figure 139

31-30-32

Page 198.51
22 Dec 2006

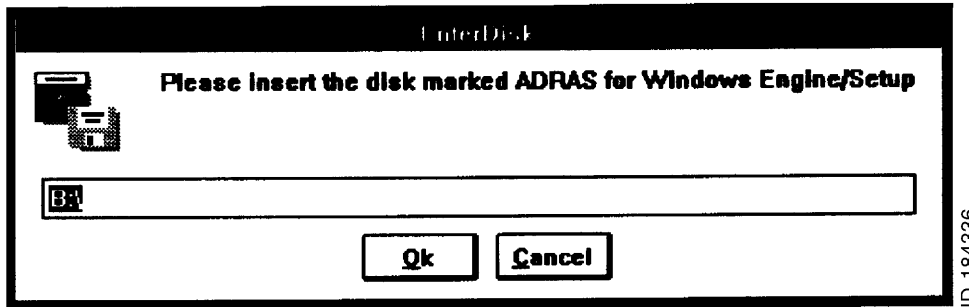
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Enter Disk Window
Figure 140

- (6) When setup has finished, the proprietary notice will be on the setup screen for a while. Wait for the screen to pass.
- (7) After ADRAS for windows setup closes, a new AFW Program Group with the ADRAS for Windows icon will be in your Program Manager. Note, that you can drag the icon to another group, or panel, if you wish.
- (8) Store the software diskettes in a safe place. As a precaution, you should always make a backup copy of the source diskette.
- (9) It is suggested that subdirectories named DATABASE and DATA (C:\AFW \DATABASE and C:\AFW\DATA) to avoid mixing them with the ADRAS for Windows program files. Copy your airline specific parameter conversion database to the DATABASE directory.

E. Configuring ATEasy for the ATU

NOTE: This procedure should be performed by the designated system administrator.

- (1) Turn on the PC and start Windows by typing **win**. Double click on the **ATEasy** icon.
- (2) When the Password dialog box appears, type the password **ATEZ**.

NOTE: This is a level 3 password, which gives the operator complete control of the environment, including defaults and passwords. Therefore, this password should only be used by the designated system administrator. (The system administrator should consider changing the level 3 password from the published password after this one-time ATEasy configuration has been performed). The system administrator should create a level 1 password for all personnel to use. It should be the password primarily used for formal UFDR ATP. Consult the ATEasy User's Manual for more information on the use of passwords.

- (3) Activate the **Utilities** menu and choose **Defaults**. The Defaults dialog box appears. Under **Initial Files - Configuration**, change the file name to **ATU.CFG** (refer to Figure 141). Note that in **Defaults**, **Print upon run completion** can be selected to print the Test Log. When finished, click on **OK**.
- (4) Activate the **File** menu and choose **Exit ATEasy**. (ATEasy must be restarted in order to recognize the default changes.)

31-30-32

Page 198.52
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

The screenshot shows a 'Defaults' dialog window with several sections:

- Directories:** Program(PRG) : PRG, Driver(INS) : INS, Log (Lxx) : LOG
- Test Log:** Company : AlliedSignal Avo, ☐ Prompt Serial #, ☒ Clear on start, ☐ Print upon run completion
- Initial Files:** Configuration : ATU.CFG, Program : DAS.PR6
- Menu return:** ☒ 1st, ☐ 2nd, ☐ 3rd
- Test type:** ☒ MIN/MAX, ☐ REF-2, ☐ REF-X, ☐ OTHER
- Misc.:** ☐ Auto Dolt!, ☐ Real-Time Watch update

Buttons at the bottom include 'OK', 'Cancel', and 'Save defaults to file'. A vertical ID number 'ID-180631' is on the right side of the dialog.

Defaults Dialog Window
Figure 141

F. Checking UFDR Insulation Resistance

NOTE: This step is optional, at the operator's discretion. It is recommended that the insulation resistance check be made only after any 115VAC and/or 28VAC input circuitry repairs have been made.

- (1) Use a high potential tester (32, Figure 901) with the voltage set to 100VDC maximum to measure the UFDR insulation resistance between J1A-1 and J1A-6 (case ground), and between J1A-2 and J1A-6. Resistance shall be greater than 20 megohms in each case.
- (2) Reverse the high potential tester leads, and repeat measurements. Resistance shall be greater than 20 megohms in each case.

G. UFDR ATU Test Setup

- (1) Turn off all test equipment power.
- (2) Connect the RTT Cable, Part No. 704-1451-001, to the RTT card on the ATU. Connect Part No. 704-1452-004 DAS cable to the DAS card on the ATU (this cable has a 50 pin D type connector on the PC end). Refer to Figure 142.
- (3) Connect the PC, Printer and Variable Power Supply (41, Figure 901) to a 115VAC, 50/60Hz power source. Connect the power cable (part of the RTT cable) to the Variable Power Supply. Connect the DAS and RTT cables to the UFDR. Refer to Figure 142.
- (4) Connect the Printer to the PC. Verify that the Printer is set up for use. Refer to the Windows User's Guide.

31-30-32

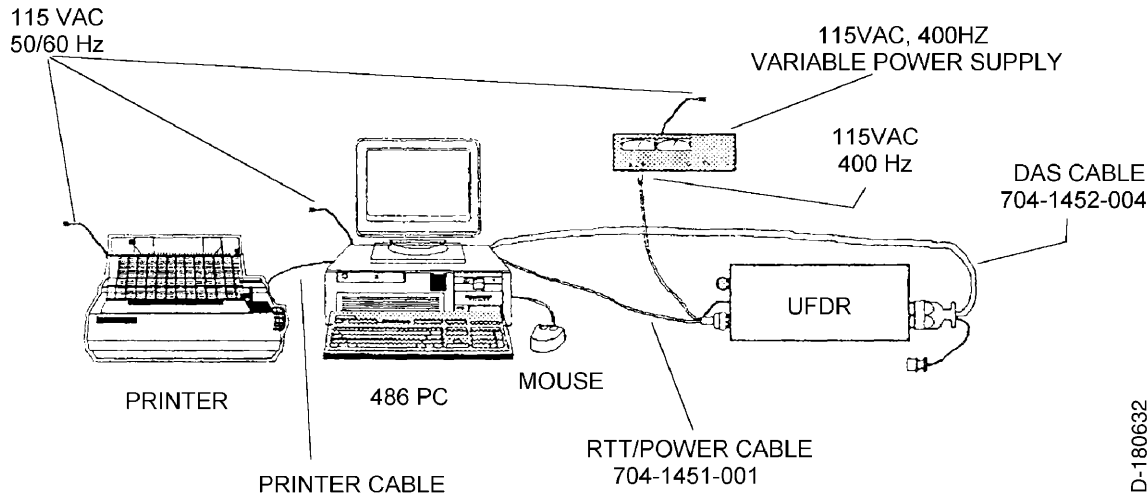
Page 198.53
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Equipment Setup
Figure 142

ID-180632

H. Tape Transport Test

NOTE: This portion of the procedure automatically produces a test log. Upon successful completion of this portion of the testing, print the test log and attach it to the ATP data sheets.

- (1) Connect a DMM (4, Figure 901) to the Variable Power Supply output. Turn on the PC, Printer, and Variable Power Supply. Adjust the Variable Power Supply output to 115 ± 1 VAC.
- (2) Click on **File**, then click on **Open [PRG] Directory**. In the dialog box for selecting the .PRG file, select **RTT.PRG**, select **Open**, then click on **START**, or press the **F2** key.
- (3) In the Select Profile dialog (refer to Figure 143), click on the arrow. If the UFDR has had Tape Transport rework of any kind, select **Acceptance Test**; if no rework has been performed on the UFDR, select **Return To Service Test** as shown in Figure 144. Click on **Select**. The Return To Service Test selection tests the Tape Transport only at the Beginning of Tape.
- (4) The Part Number dialog box appears. Enter the UFDR Part Number. When the Part Number is correct, click on **OK**.
- (5) In the dialog box, enter the UFDR Serial Number. When the Serial Number is correct, click on **OK**. (The Serial Number is needed for test reports and as part of the default file name for the report disk file that will be saved in a later step.)
- (6) A dialog box presents a default Log Output file name **FDR + <Serial Number>.LOG**. By default, ATEasy writes the Log Output file to the default file name in the **LOG** subdirectory of the **ATEasy** directory. Each time a test is run on the same UFDR, the previous log file is overwritten.
- (7) A dialog box for comment data appears. Any pertinent text may be entered here.

31-30-32

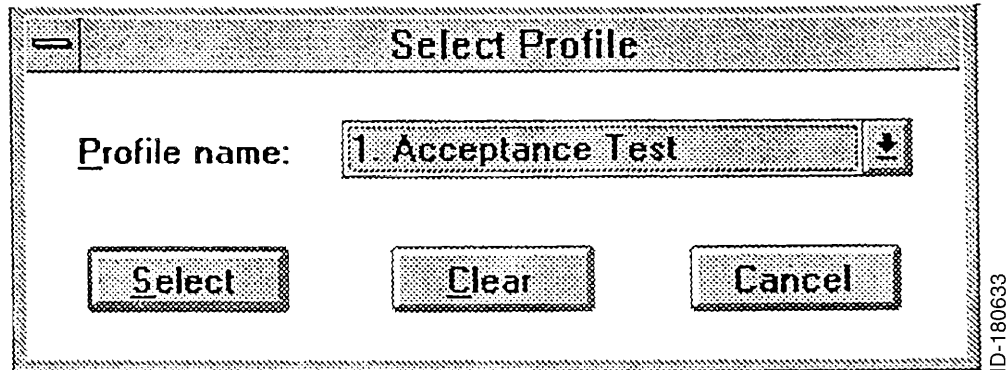
Page 198.54
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

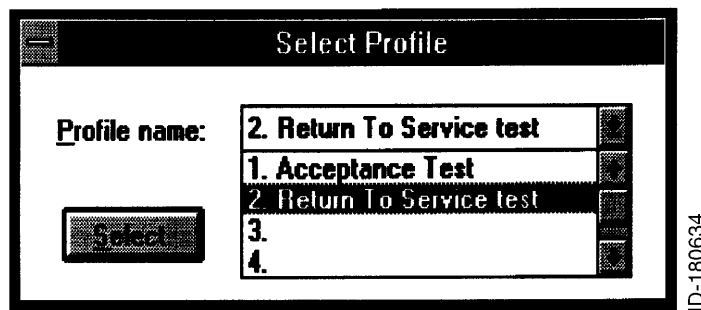
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Select Profile Dialog
Figure 143



Return to Service Test Selection
Figure 144

- (8) Additional manual actions and visual verifications are requested at times throughout the test program via dialog boxes; respond as appropriate. Some tests take much time with no change of screen display. Screen messages always display the status of the test cases. If test difficulties prevent the software from conducting the entire test, ATEasy aborts the profile, and does not create a log output file.
- (9) Test report data remains in memory, ready to print. To print the report, activate the **LOG** menu, and select **Print**.

NOTE: To save a copy of the file that won't get overwritten, print to file (**Save As**). The default file name seen now is **xxxxxxx.L0X**; the filename can be changed. The extension automatically increments the next time the test is run (**L00, L01, etc.**).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

ATLAS TEST SPECIFICATION

1. Introduction

This section lists the ATLAS Test Specifications for testing the following:

- PN 980-4100-BXUN
- PN 980-4100-BXUS

2. General

The purpose of the ATLAS Test Specifications is to provide the ATLAS Test requirements for the LRUs listed above.

A. ATLAS Test Specification Listings are as follows:

- 050-0893-000

B. The ATLAS Test Specifications is made available on a Personal Computer 3 1/2 inch disk meeting the following specifications:

- Compatible with ANSI Standard X3.33-1973 or ANSI Standard X3.39-1973.
- ASCII format.

C. Order PC 3 1/2 inch disks from the following address:

Honeywell

15001 NE 36th Street

PO Box 97001

Redmond, WA 98073-9701

Attention: Order Administration

Telephone: 425-885-8719

Fax: 425-885-8988

31-30-32

Page 201
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

DISASSEMBLY

1. General

Disassemble the unit only to the extent necessary for repair or replacement of components. Note that disassembly instructions are in the general or recommended sequence necessary to provide access to the next lower assembly or item.

NOTE: Refer to **TESTING AND TROUBLESHOOTING** section to establish condition of unit or most probable cause of its malfunction.

CAUTION: TO PREVENT DAMAGE TO UNIT, TURN OFF EXTERNAL POWER TO UNIT AND REMOVE IT FROM INSTALLATION RACK BEFORE DISASSEMBLING IT.

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES ARE SUBJECT TO DAMAGE BY EXCESSIVE LEVELS OF VOLTAGE. REFER TO DOD-HDBK-263 FOR PROPER HANDLING PROCEDURES.

2. General Techniques

A. Label assemblies, subassemblies, parts, and electrical leads to be removed. Note placement angle and polarity before removal.

CAUTION: USE MINIMUM HEAT AND A HEAT SINK WHEN DESOLDERING COMPONENTS TO PREVENT DAMAGE TO COMPONENTS AND PRINTED WIRING BOARD.

WARNING: CUTTING AND SCRAPING CONFORMAL COATING MAY CAUSE AIRBORNE PARTICLES TO ENTER THE EYES. WEAR GOGGLES TO PROTECT EYES.

B. Locate and label assemblies or items to be removed. Check all parts, subassemblies, and assemblies for polarity, placement angle, position, insulating requirements, adjacent heat-sensitive components or substances, and other conditions affecting removal.

Rearrange or remove any assembly that will facilitate disassembly using the instructions in this section. If components are covered with conformal coating, remove coating as necessary from around defective components with a chisel-tip soldering iron. Remove defective components which have bent leads by clipping leads to 1/16 inch from board surface on component side. Remove component and unsolder remaining lead wire from circuit side of board. Remove defective components having leads going straight into board by unsoldering on circuit side and pulling leads through from component side.

C. Screws that have been installed with adhesive can be loosened by applying heat from soldering iron to screwhead.

3. Recorder. Universal Flight Data (IPL Figure 1)

Before starting to disassembly the UFDR, connect it to the UFDR Tester and rewind the tape onto the loaded reel. This facilitates replacing the tape should it be found necessary.

A. Remove 2 screws (10), washers (15), washers (20), and dust cover (5, 5A, 5B).

B. If plug (25) is damaged, replace it.

NOTE: There is no reason to remove or replace screws (27, 28) on some front panels. These screws are used strictly for filling empty holes.

31-30-32

Page 301
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- C. If underwater locating device (ULD) (30) or underwater acoustic beacon (30A) is installed and its removal is required, remove screws (35, 35A), mounts (40, 40A), and ULD.

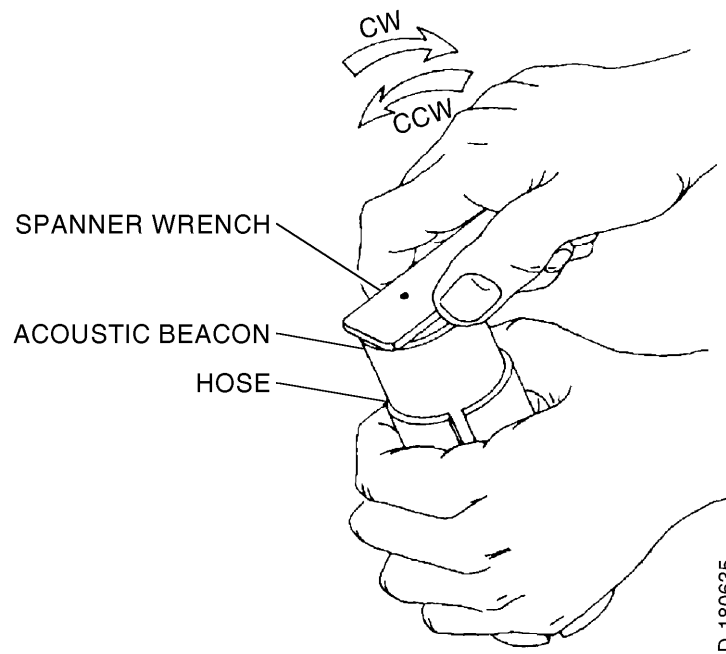
CAUTION: DO NOT DISASSEMBLE ULD (30A). DO NOT DISASSEMBLE ULD (30B) IF SERIAL NUMBER BEGINS WITH AN S BECAUSE DISASSEMBLY COULD RESULT IN RELEASE OF HAZARDOUS CHEMICALS.

WARNING: NO BATTERY MAINTENANCE IS REQUIRED FOR THE DK100 ULD (30A) (4 YEAR WARRANTY PER THE TECHNICAL MANUAL), OR FOR THE ELP362D (30B) (6 YEAR WARRANTY PER THE USER'S MANUAL) IF THE SERIAL NUMBER BEGINS WITH AN S. DO NOT REMOVE BATTERY AT ANY TIME. AVOID ANY SITUATION THAT COULD POSSIBLY CRUSH OR PENETRATE ULD CASE. DO NOT DISPOSE OF ULD. AT OR NEAR EXPIRATION DATE PRINTED ON CASE, RETURN ULD TO MANUFACTURER FOR SERVICE. FAILURE TO OBSERVE THESE PRECAUTIONS COULD RESULT IN RELEASE OF HAZARDOUS CHEMICALS.

- D. Disassembly of ULD (30) as follows:

CAUTION: DO NOT CLAMP ULD IN A VISE OR SIMILAR TOOL AS CASE DAMAGE MAY RESULT.

- (1) Grasp the body of the ULD using a 4-inch length of 1.25-inch split rubber hose or equivalent (see Figure 301).



ULD Disassembly
Figure 301

- (2) Using a spanner wrench (9, Figure 901), remove end cap (45), marked BATTERY ACCESS, by unscrewing counterclockwise.

31-30-32

Page 302
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (3) Remove and discard (or save for reuse, if not damaged) o-ring (50). Remove cushions (55, 60) and battery (65) from beacon housing (P/O 30).
 - (4) For information, see Technical Manual, Underwater Acoustic Beacons, N15F210B and N15F210B, Mod 1, or Technical Manual, Underwater Acoustic Beacon, Model DK100, Dukane Corp. (V94970), Seacom Div., St. Charles, IL 60174-3348.
- E. PWB Assembly removal as follows:
- (1) Remove screws (75, 90) and retainer (70).
 - (2) Replace silicon rubber bumper (91) and/or anchor nut (92) if damaged. Silicon rubber bumper may be removed with a thin-bladed tool. Anchor nut may be removed by drilling or clipping rivets (93). Do not enlarge 0.177-inch diameter rivet holes by more than 0.003 inch of original diameter.
 - (3) Pull Read/Write Interface, A2 (105), Transport Interface Assembly, A3 (110), Recording System Controller Assembly, A4 (115), or Airframe Wiring Interface, A8 (120) out of card guides with card puller (15, Figure 901).
- F. Power Supply Assembly, A1, removal as follows:
- (1) Remove A2, A3, and A4 per paragraph 3E.
 - (2) For (95, 95A, 95B), remove 8 screws (100, 102), 4 nuts (103), and Power Supply. For (95C, 95D), remove 6 screws (102A) and Power Supply. Pull the A1 connector from its receptacle. The A1 connector is mounted on Power Supply PWB Assembly, A1A1 (25, IPL Figure 2) or (25, IPL Figure 15).
- G. Replace plug (125) if damaged.
- H. Remove screws (135) and identification plate (130). Before discarding, transfer all information (i.e., Mfg Part No., Serial No., etc.), including Mod status, from the old plate to the new plate.
- NOTE:** *When maintenance actions are the result of a modification document, enter data relative to the modification.*
- I. Modification plate (127) and patent label (147) may be removed with a thin-bladed tool or new label may be installed over old label.
- J. Tape Recorder Assembly (150) removal as follows:
- For certain maintenance actions, such as changing tape, cleaning heads, adjusting rollers, replacing the peripheral belt, replacing the tape guide rollers, replacing the capstan or capstan bearing, and/or checking tape tracking, the Tape Recorder Assembly need not be removed. If maintenance action does not require removal of the Tape Recorder Assembly, remove only the cover per paragraph 10A.
- (1) Remove UFDR dust cover per paragraph 3A.
 - (2) Remove PWB assemblies per paragraph 3E.
 - (3) Remove Power Supply Assembly per paragraph 3F.
 - (4) Remove 4 screws (155), washers (160), and washers (165).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (5) Remove From Chassis, PN 617-3716-XXX (IPL Figure 9) as follows:
 - (a) Remove 2 screws (95) or 2 screws (95A) and 1 fiber washer (97) which attach the motherboard (90, 90A) to the bracket (190, 190A).
 - (b) Remove 4 of 6 screws (195); the 2 upper screws on each side of bracket (190, 190A).
 - (c) Loosen the 2 bottom bracket screws (195) nearest the motherboard and pivot the bracket toward the motherboard.
- (6) Removal from Chassis, PN 617-4640-006 (IPL Figure 16) as follows:
 - (a) Remove 2 screws (175, IPL Figure 16) which attach the motherboard (170) to bracket (255).
 - (b) Remove 4 of 6 screws (260); the 2 upper screws on each side of bracket (255).
 - (c) Loosen the 2 bottom bracket screws (260) nearest the motherboard and pivot the bracket toward the motherboard.
- (7) Separate connectors P5 (395, IPL Figures 10 and 14), P4 (65, IPL Figure 11) or (90, IPL Figure 13), and P3 (405, IPL Figures 10 and 14) from their receptacles J5, J4, and J3, respectively.
- (8) Remove Tape Recorder Assembly.

4. **Power Supply A1** (IPL Figure 2)

- A. Remove 5 screws (10), washers (15), and EMI shield (5) from Power Supply PWB Assembly (25).

CAUTION: DO NOT DEBURR HOLES AFTER DRILLING OUT STANDOFFS OR SPACERS. DEBURR ONLY THE HOLE EDGES USING HAND TOOLS. DO NOT USE POWER TOOLS TO REMOVE BURRS OR SHARP EDGES.

- B. Remove standoffs (20) or spacers (20A) by drilling or clipping. Use care not to enlarge 0.136-inch diameter holes in (5) or 0.166-inch diameter holes in (5A) by more than 0.003 inch of original diameter.
- C. Replace aluminum washer (21) as necessary.
- D. Sponge rubber (23) or foam tape may be pulled off, or removed with a thin-bladed tool.
- E. Remove 2 screws (30), washers (35), and Power Supply PWB Assembly, A1A1. When opened 180°, there shall be a service loop of 2.0 ± 0.5 inches, so that troubleshooting and repair may be performed without unwiring the assembly. If the assembly has to be replaced, identify each wire before unsoldering to make reassembly more convenient.
- F. Unsolder and remove capacitors (40A, 45A, 50A) and resistor (53).
- G. Remove 2 screws (85A), nuts (95), internal star washers (90), washers (100), shoulder washers (105B), lug (110A), insulator (115), voltage regulators (55, 60, 65, 80), and/or transistors (70, 75).
- H. Remove screw (125), washer (130), 2 nuts (135), washers (140), insulator (145), and transformer (120).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- I. Remove 1 nut (155) and thermal switch (150).
- J. Remove 3 screws (165), nuts (170), washers (175), terminal (178B), and rectifiers (160).
- K. Remove grommets (180).
- L. Remove anchor nuts (190) by drilling or clipping rivets (195A). Use care not to enlarge 0.098-inch diameter rivet holes by more than 0.005 inch or original diameter.

5. Read/Write Interface Assembly. A2 (IPL Figure 3)

Replace components (5 through 255) as necessary to clear faults. Remove components per Para. 2B.

6. Transport Interface Assembly. A3 (IPL Figure 4)

Replace components (5 through 148, 155, 160) as necessary to clear faults. Remove components per Para. 2B.

7. Recording System Controller Assembly. A4 (IPL Figure 5)

Replace components (5 through 235) as necessary to clear faults. Remove components per Para. 2B.

8. Airframe Wiring Interface Assembly. A8 (IPL Figure 6)

Replace components (5 through 135, 145 through 195) as necessary to clear faults. Remove components per Para. 2B.

9. Power Supply Assembly. A1A1 (IPL Figure 7)

Replace components (5 through 220) as necessary to clear faults. Remove components per Para. 2B.

10. Tape Recorder Assembly (IPL Figure 8)

It is recommended that the disassembly of the Tape Recorder Assembly be accomplished in a clean room atmosphere. Use a Laminar Flow bench or equivalent. The assembly contains bearings, pulleys, rollers, belts, magnetic heads, and magnetic tape, which must be kept clean and free from dirt and dust to provide trouble-free operation.

- A. To remove top cover (5, 5A), remove 4 screws (10) and 4 screws (15); 2 that attach Motor Drive Assembly (45, 45A) to cover and 2 that attach harness shield (35) to cover.
- B. Remove gasket (20) and set it aside to avoid damaging it while performing maintenance actions. To remove the gasket, remove 2 screws (40) and the harness shield from bottom cover (5B, 5C).
- C. Tape Transport Assembly removal as follows:
 - (1) Remove top cover per paragraph 10A and top plate per paragraph 16A(1).
 - (2) If necessary or desirable, remove loaded reel (55A, IPL Figure 10) per paragraph 12A(4).
 - (3) If necessary or desirable, remove loaded reel (45, IPL Figure 14) per paragraph 16A(4).

NOTE: Located on the underside of the Tape Transport Assembly are the motor drive pulley, the motor drive belt, 2 motor drive rollers, and the distribution board assembly. If it is necessary to replace the drive belt, perform other

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

maintenance actions, and/or inspect these components, the Tape Transport Assembly must be removed.

CAUTION: USE CARE WHEN REMOVING THE TAPE TRANSPORT ASSEMBLY. DO NOT REMOVE UNLESS NECESSARY. DRIVE BELT MUST BE PROPERLY RETENTIONED.

(4) Remove Tape Transport Assembly (25) by removing 3, 2, or 4 flat head screws (30, 30A, 30B, 30C) and also removing the Motor Assembly (90, 90B, IPL Figure 11) from the Motor Drive Assembly (45, 45A, 45B, IPL Figure 8).

(5) Remove 2 screws (50, IPL Figure 11), washers (55), D washer (57), and Motor Assembly.

NOTE: *To observe drive belt while removing Motor Assembly, remove 4 screws (10, IPL Figure 11), washers (11), and cover (5).*

(6) Tilt the Motor Assembly toward the Transport Assembly to release tension on the drive belt and remove the Motor Assembly from the Motor Drive Assembly.

NOTE: *Removal of the Motor Drive Assembly, the motor mount plate (55, 55A), and the drive belt feedthru (60) are usually unnecessary unless the environmental enclosure cover(s) (5, 5A, 5B, 5C) need servicing. When necessary removal of 2 screws (50) releases all 3 items.*

(7) Remove Tape Transport Assembly from bottom cover.

NOTE: *Do not remove spacers (70, IPL Figure 8) unless the cover liner has become separated from the cover. If necessary, rebond spacers to the cover liner with adhesive (16, Figure 701).*

(8) Inspect covers (5, 5A, 5B, 5C) per the **CHECK** section. Repair covers per the **REPAIR** section.

11. **Chassis Assembly** (IPL Figure 9)

Generally, complete disassembly of the chassis assembly should not be necessary; however, from time to time, it may be necessary to remove some of the chassis components. This procedure is designed to meet that need.

A. Filter Removal

- (1) Remove 2 screws (25) and filter (20).
- (2) Mark wires for identification before removing from filter.
- (3) Remove solder lugs (28) or terminals (28A).

B. Elapsed Time Indicator (ETI) Removal

- (1) Remove screws (35), nuts (40), and ETI (30).
- (2) Mark wires for identification before removing from ETI.

C. Connector J1 Removal

Remove 8 screws (10), nuts (15), and J1 (5). Remove 4 screws (165) and plate (160, 160A, 160B, 160C).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

D. Motherboard Removal

- (1) Remove filter per paragraph 11A(1); do not unsolder leads.
- (2) Remove ETI per paragraph 11B(1); do not unsolder leads.
- (3) Remove J1 per paragraph 11C.
- (4) Remove 12 screws (95) or 12 screws (95A), and 1 washer (97).
- (5) Remove 4 of 6 screws (195) that attach bracket (190, 190A). Remove 2 screws from each side. Loosen the 2 remaining screws (1 on each side) and pivot the bracket away from the motherboard.
- (6) Remove motherboard (90, 90A).

NOTE: *Filter (20), ETI (30), and connector (5) will remain tethered to the motherboard by their wires, which are part of the wire harness tied to the motherboard. The motherboard also remains tethered to the chassis by the harness. Cut harness ties (17) and straps (60, 60A) as necessary to move the motherboard to a working position.*

- (7) If it is necessary to remove connectors (100), remove the shield (170, IPL Figure 1) by removing 4 screws (175), washers (180), and washers (185). Identify and unsolder the connector pins.
- (8) The sponge rubber (187), or foam tape, on the shield may be removed with a thin-bladed tool.
- (9) To remove each connector (100, IPL Figure 9), remove 2 screws (105), washers (115B), washers (117), and nuts (110). Lock washers (117) have been deleted from new production assemblies and need not be replaced if damaged.

NOTE: *The sealant at the base of each connector should peel off as the connector is lifted.*

- (10) If it is necessary to remove either J3 or J4 (120), remove the 2 jackposts (125). Identify and unsolder the connector pins and lift the connector from the motherboard.

NOTE: *The sealant at the base of each connector should peel off as the connector is lifted.*

- (11) If damaged, replace standoffs (150). Standoffs may be removed by drilling or clipping. Do not enlarge 0.138-inch diameter holes by more than 0.003 inch of original diameter.
- (12) If damaged, replace terminals (145). Do not attempt to straighten bent terminals. If any terminal is bent more than 20° from perpendicular to the mounting surface, replace it. Terminals may be removed by drilling or clipping. Do not enlarge 0.113-inch diameter holes by more than 0.003 inch of original diameter.
- (13) Replace standoffs (140) if damaged or loose. Standoffs may be removed by drilling or clipping. Do not enlarge 0.138-inch diameter holes by more than 0.003 inch of original diameter.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

E. Shockmount Removal

- (1) Remove 2 screws (175) and shockmount (170). Retain ground strap (180) for reinstallation.
- (2) To remove upper shockmount (245), remove ULD assembly per paragraph 3C, then remove 2 screws (250), and nuts (255).
- (3) To remove lower shockmount (245), remove the identification plate (130, IPL Figure 1) by removing 6 screws (135). Then remove 2 screws (250, IPL Figure 9), and nuts (255).

F. Connector J2 Removal

To remove J2 (80), remove nut, which is part of connector assembly.

G. LED Indicator Removal

To remove LED indicator (85) or lamp assembly (85A), remove 1 screw (89), retainer (88), and spacer (87).

H. Connector J5 Removal

To remove J5 (45), remove potting shell (55) from the connector bracket (57) by removing the pair of jackposts (50). Remove 2 screws (58) and connector bracket.

I. Tywrap Clamp Removal

Remove screws (70), nuts (75), and tywrap clamps (65, 65A).

12. Tape Transport Assembly (IPL Figure 10)

A. Tape Reel and Tape Reel Pin Removal

- (1) Remove 4 screws (10) and top plate (5). If any rubber bumper (15) requires replacement, remove screw (20), washer (25), washer (30), and rubber bumper.

CAUTION: PERIPHERAL BELT (P-BELT) SHALL NOT BE STRETCHED BEYOND 18.0 INCHES MAXIMUM INTERNAL CIRCUMFERENCE.

CAUTION: FOR RELIABLE TAPE RECORDER OPERATION, REPLACE TAPE REELS AND PERIPHERAL BELT STRICTLY ACCORDING TO THE PROCEDURES SPECIFIED IN THE ASSEMBLY SECTION.

- (2) Remove P-belt (50B) and examine it per the **CHECK** section. Discard belt if it does not meet the given criteria. If belt is satisfactory for reinstallation, clean it per the **CLEANING** section. Store cleaned belt on a storage support form (normally) supplied with new belts).
- (3) If the tape is to be replaced or removed for other maintenance actions, rewind it onto the loaded reel (55A).

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING TOOL.

- (4) Remove E-ring (60), shims (65, 70, 75), loaded reel (55A), ball bearings (80), and shim (85). Note locations of shims and save for reinstallation.
- (5) Remove E-ring (95), shims (100, 105, 110), tape reel (90A), ball bearings (115), and shim (120). Note locations of shims and save for reinstallation.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (6) Remove screws (280), tape reel pins (275), and shims (285, 290, 295). Note locations of shims and save for reinstallations.

B. Motor Drive (Flat) Belt Removal

- (1) Remove Tape Transport Assembly per paragraph 10C.
- (2) Remove 4 screws (40) and bottom plate (35).
- (3) Hand hold the Tape Transport Assembly and hand clamp the tape pulley (200) and the motor drive pulley (215) to prevent movement. Remove screw (220) from shaft (235).
- (4) Remove drive pulley (215) and motor drive belt (45A, 420), and replace screw (220) in shaft (235).
- (5) Examine motor drive belt per the **CHECK** section and discard if it does not meet the given criteria.

C. Drive Pulley Bearing Removal

- (1) Remove peripheral belt per paragraph 12A(2).
- (2) Remove loaded reel per paragraph 12A(4).
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Remove motor drive belt (45A, 420) and drive pulley (215) per paragraph 12B.
- (5) Remove screw (205) and tape pulley (200).
- (6) Remove ball bearing (230), wavy washer (225), and shaft (235) from baseplate by pushing on shaft.
- (7) Remove ball bearing (210) from baseplate.

D. Tape Guide Roller and Roller Pin Removal

- (1) Remove top cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 12.

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- (3) Remove E-rings (155), shims (160, 165, 170), tape guide rollers (150), ball bearings (175), and shims (180). Note locations of shims and save for reinstallation.
- (4) To remove any of the roller pins (300), first remove Tape Transport Assembly per paragraph 10C. Then remove screw (305), roller pin (300), and shims (310, 315, 320). Note locations of shims and save for reinstallation.

E. Drive Belt Roller and Roller Pin Removal

- (1) Remove Tape Transport Assembly per paragraph 10C.

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- (2) Remove E-rings (335) and drive belt rollers (330, 425).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (3) Remove roller pins (340, 340A).

NOTE: *Roller pins (340) are press fit into the tape deck; roller pins (340A) are screwed into the tape deck.*

F. Magnetic Head Removal

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 12.
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Before removing a read/write or erase magnetic head, mark wires for identification and unsolder at the distribution board (250).
- (5) Remove 2 screws (195) and head.

NOTE: *For units that contain matched sets of magnetic heads (185, 187, 190) or (185A, 187A, 190A), individual heads within a set may be replaced provided that head sets are NOT mixed. See NOTE of paragraph 11S of the **ASSEMBLY** section for further information.*

- (6) If the unit contains magnetic head set (185, 187, 190), identify the shim(s) (197) used with each head for use in reinstallation.

NOTE: *If the magnetic head set is being replaced by the alternate set (185A, 187A, 190A), no shims are required.*

- (7) Inspect all magnetic head tape surfaces per the **CHECK** section. Either replace worn or damaged heads, or rework head surfaces per the **REPAIR** section.

G. EOT/BOT Sensors Removal

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 12.
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Before removing sensors (125, 130), mark wires for identification and unsolder at the distribution board (250).
- (5) Separate wires from bundle.
- (6) Remove 2 screws (135) and sensor.

H. Distribution Board Removal

The distribution board (250) is hardwired and should not require complete removal for replacement of parts.

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 12.
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Release the distribution board (250) and shield (265) by removing 3 screws (255), washers (260), solder lug (270), and phenolic washers (262).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (5) Remove 2 screws (350), washers (355), washers (360), and harness cover (345).
- (6) Remove 2 screws (370), washers (375), washers (380), and harness bracket (365).
- (7) Carefully move the distribution board away from the deck.

I. Peripheral Belt Roller and Roller Pin Removal

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 12.

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- (3) Remove E-rings (245) and belt rollers (240).
- (4) Inspect rollers per the **CHECK** section.
- (5) Inspect roller pins per the **CHECK** section.
- (6) Remove screws (305), shims (325), and roller pins (300). Note locations of shims and save for reinstallation.

NOTE: To remove roller pins, remove the Tape Transport Assembly per paragraph 10C.

13. Motor Drive Assembly (IPL Figure 11)

- A. Remove 4 screws (10), washers (11), and cover (5). Gasket (12) is attached to cover with adhesive.
- B. Remove 2 screws (50), washers (55), D washer (57), cable clamp (60), and Motor Assembly (90, 90A, 90B).
- C. Tilt the Motor Assembly toward the Transport Assembly to release tension on the drive belt and remove the Motor Assembly.
- D. Remove 2 screws (50, IPL Figure 8) and motor mount (45).

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- E. Remove E-rings (20) and rollers (15).

NOTE: Roller pins (35) are permanently attached to tension plate.

- F. Replace tension plate (25) if roller pins (35) are bent or damaged. Remove 2 screws (30), washers (32), washers (33), and tension plate.
- G. Replace connector (65), screwlocks (70), and clamp (75), and both ferrules (80, 85), if damaged.
- H. If pin (100) is loose, replace it.

NOTE: The shaft of stepper motor (105, 105A) and drive pulley (95B, 95C) are drilled and pinned as a matched assembly.

NOTE: It is recommended that the Motor Assembly (90, 90A, 90B) not be disassembled to any lower level.

31-30-32

Page 311
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

14. Distribution Board (IPL Figure 12)

Replace components (5 through 20) on printed wiring board (30 or 30A) as necessary to clear faults. Remove components per the **REPAIR** section.

If damaged, replace terminals (35 or 37). Do not attempt to straighten bent terminals. If any terminal is bent more than 20° from perpendicular to the mounting surface, replace it. Terminals may be removed by drilling or clipping. Do not enlarge 0.044-inch diameter holes by more than 0.003 inch of original diameter.

15. Motor Drive Assembly (IPL Figure 13)

- A. Remove screw (76), washer (77, 78), and cable clamp (65).
- B. Remove 2 screws (55), washers (60), and Stepper Motor Assembly (105).
- C. Tilt the Stepper Motor Assembly toward the Transport Assembly to release tension on the drive belt and remove Stepper Motor Assembly.
- D. Remove 2 screws (50, IPL Figure 8) and motor mount (55).

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- E. Remove E-rings (10), shims (15, 20, 25), rollers (35), and ball bearings (30 or 31). Note locations of shims and save for reinstallation.
- F. Replace pins (40, 40A) if bent or damaged. Remove screws (45) and pins.
- G. Replace connector (90), clamp (95), clamp screws, and both ferrules (80, 85), if damaged. Replace clamp screws with new screws (102).
- H. Remove screws (125) and drive pulley (120).
- I. Refer to paragraph 18 of **REPAIR** section for motor bearing replacement.

16. Tape Transport Assembly (IPL Figure 14)

A. Tape Reel Removal

- (1) Remove 4 screws (10), washers (12, 13, 14), and top plate (4). If any rubber bumper (15) requires replacement, remove screw (20), washer (25), washer (30), and rubber bumper.

CAUTION: PERIPHERAL BELT (P-BELT) SHALL NOT BE STRETCHED BEYOND 18.0 INCHES MAXIMUM INTERNAL CIRCUMFERENCE.

CAUTION: FOR RELIABLE TAPE RECORDER OPERATION, REPLACE TAPE REELS AND PERIPHERAL BELT STRICTLY ACCORDING TO THE PROCEDURES SPECIFIED IN THE ASSEMBLY SECTION.

- (2) Remove P-belt (40) and examine it per the **CHECK** section. Discard belt if it does not meet the given criteria. If belt is satisfactory for reinstallation, clean it per the **CLEANING** section. Store cleaned belt on a storage support form (normally supplied with new belts).
- (3) If the tape is to be replaced or removed for other maintenance actions, rewind it onto the loaded reel (45, 45A, 45B, 45C).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- (4) Remove 3 screws (55), split washers (65), washers (60), and loaded reel (45, 45A, 45B, 45C) or tape reel (50, 50A).

B. Disassembly of Reel Hub Assembly

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

Remove E-ring (85), shims (90, 91, 92, 92A, 93, 94), wavy washer (95), and ball bearings (77) or (100). Note locations of shims and save for reinstallation.

NOTE: *Ball bearings (77) are press fit into reel hub bearing assembly (75B, 75C).*

C. Tape Reel Pin Removal

Remove screws (120), washers (122) or screws (120B), tape reel pins (80), shims (105, 110, 115, 116), and tape reel pins. Note locations of shims and save for reinstallation.

NOTE: *To remove tape reel pins, remove Tape Transport Assembly per paragraph 10C.*

D. Flat Belt and Tape Transport Pulley Removal

- (1) Remove Tape Transport Assembly per paragraph 10C.
- (2) Hand hold the Tape Transport Assembly and hand clamp the tape drive pulley (205) and the tape transport pulley (220) to prevent movement. Remove screw (225, 225A) from shaft (240).

NOTE: *Screw (225A) shall not be reused. Replace with new screws (225A).*

- (3) Remove tape transport pulley (220) and belt (35), and replace screw (225, 225A) in shaft (240).

NOTE: *Screw (225A) shall not be reused. Replace with new screws (225A).*

- (4) Examine belt per the **CHECK** section and discard if it goes not meet the given criteria.

E. Tape Transport Pulley Bearing Removal

- (1) Remove peripheral belt per paragraph 16A(2).
- (2) Remove loaded reel per paragraph 16A(4).
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Remove belt and tape transport pulley per paragraph 16D.
- (5) Remove screw (210, 210A) and tape drive pulley (205).

NOTE: *Screw (210A) shall not be reused. Replace with new screws (210A).*

- (6) Remove ball bearing (235), wavy washer (230), and shaft (240) from baseplate by pushing on shaft.
- (7) Remove ball bearing (215) from baseplate.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

F. Tape Guide Roller and Pin Removal

- (1) Remove top cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 16A.

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- (3) Remove E-ring (145, 445, 525), shims (150, 155, 160, 162, 450, 455, 460, 465, 530, 535, 540, 545) wavy washer (165), rollers (175, 475, 475B, 555, 555A), ball bearings (170, 470, 550), and shims (150, 155, 160, 162, 450, 455, 460, 465, 530, 535, 540 545). Note locations of shims and save for reinstallations.
- (4) To remove any of the roller pins (180, 480, 555), first remove Tape Transport Assembly per paragraph 10C. Then remove screw (200, 200A, 201, 490, 495, 500), roller pin (180, 480, 555), and shims (185, 190, 195, 196, 505, 510, 515, 520). Note locations of shims and save for reinstallations.

G. Disassembly of Drive Belt Roller Guide Assembly

- (1) Remove Tape Transport Assembly per paragraph 10C.

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- (2) Remove E-ring (305), shims (310, 315, 320), roller (330), and ball bearings (325). Note locations of shims and save for reinstallation.
- (3) Remove 2 head screws (340), shims (341), if applicable, and pins (335, 335A).

H. Magnetic Head Removal

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 16A.
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Before removing a read/write or erase magnetic head, mark wires for identification and unsolder at the distribution board (270).
- (5) Remove 2 screws (435), head (420, 425, 430), and shims (421, 422, 423, 426, 427, 428, 431, 432, 433).
- (6) Inspect all magnetic head tape surfaces per the **CHECK** section. Either replace worn or damaged heads, or rework head surfaces per the **REPAIR** section.

I. EOT/BOT Sensors Removal

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 16A.
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Before removing sensors (125, 130), mark wires for identification and unsolder at the distribution board (270).
- (5) Separate wires from bundle.

31-30-32

Page 314
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (6) Remove 2 screws (135) and sensor.

J. Distribution Board Removal

The distribution board (270) is hardwired and should not require complete removal for replacement of parts.

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, loaded reel, and P-belt per paragraph 16A.
- (3) Remove Tape Transport Assembly per paragraph 10C.
- (4) Release the distribution board (270) and shield (275) by removing 3 screws (280), washers (285), lug (295), and washers (282).
- (5) Remove 2 screws (350), washers (355), washers (360), and harness cover (345).
- (6) Remove 2 screws (370), washers (375), washers (380), and harness bracket (365).
- (7) Carefully move the distribution board away from the deck.

K. Disassembly of P-Belt Pulley Assembly

- (1) Remove cover per paragraph 10A.
- (2) Remove top plate, tape reel, and P-belt per paragraph 16A.

CAUTION: TO PREVENT LOSS OF E-RING TENSION DUE TO OVERSPREADING, USE PROPER E-RING REMOVAL TOOL.

- (3) Remove E-rings (250), shims (261, 261A, 261B, 261C), and belt idler pulley (255, 255A, 255B, 255C).
- (4) Inspect rollers per the **CHECK** section.
- (5) Inspect roller pins per the **CHECK** section.

NOTE: To replace roller pins, remove the Tape Transport Assembly per paragraph 10C.

- (6) Remove screws (265, 265A), shims (262, 262A, 262B, 262C), and roller pins (260, 260A). Note locations of shims and save for reinstallation.

17. Power Supply. A1 (IPL Figure 15)

- A. Remove 5 screws (10), washers (15), and EMI shield (5) from Power Supply PWB Assembly, A1A1 (25).
- B. Remove standoffs (20) or spacers (20A) by drilling or clipping. Use care not to enlarge 0.136-inch diameter holes in (5) or 0.166-inch diameter holes in (5A) by more than 0.003 inch of original diameter.
- C. Replace aluminum washer (21) as necessary.
- D. Sponge rubber (23) or foam tape may be pulled off, or removed with a thin-bladed tool.
- E. Remove 2 screws (30), washers (35), and Power Supply PWB Assembly, A1A1. When opened 180°, there shall be a service loop of 0.75 inch, maximum, between board edge and support assembly (195) edge so that troubleshooting and repair may be performed without

31-30-32

Page 315
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

unwiring the assembly. If the A1A1 assembly has to be replaced, identify each wire before unsoldering to make reassembly more convenient.

- F. Unsolder and remove capacitors (40, 45, 50) and resistor (55).
- G. Remove 2 screws (90), nuts (100), internal star washers (95), washers (105), shoulder washers (110), lug (115), insulator (120), voltage regulators (60, 65, 70, 85), and/or transistors (75, 80).
- H. Remove screws (130), washer (135), 2 nuts (140), washers (145), insulator (150), and transformer (125).
- I. Remove nut (160) and thermal switch (155).
- J. Remove 3 screws (170), nuts (175), washers (180), terminal (185A), and rectifiers (165).
- K. Remove grommets (190).
- L. Remove clinch nuts (200) by drilling or clipping. Use care not to enlarge 0.269 - 0.271 inch diameter holes.

18. **Chassis Assembly** (IPL Figure 16)

Generally, complete disassembly of the chassis assembly should not be necessary; however, from time to time, it may be necessary to remove some of the chassis components. This procedure is designed to meet that need.

A. EMI Filter Removal

- (1) Mark wires for identification before removing from filter.
- (2) Remove 2 nuts (50), terminals (45), screws (40), EMI filter (35), and bracket (30).

B. EMI Filter and Diode Removal

- (1) Unsolder and remove diode (60).
- (2) Mark wires for identification before removing from filter.
- (3) Remove nuts (part of EMI filters) and EMI filters (55).

C. Elapsed Time Indicator (ETI) Removal

- (1) Remove 2 nuts (80), screws (75), and ETI (70).
- (2) Mark wires for identification before removing from ETI.

D. Connector J1 Removal

Remove 8 screws (15), nuts (20), and J1 (5). Remove 4 screws (250) and connector plate (245).

E. Motherboard Removal

- (1) Remove filter per paragraph 18A; do not unsolder leads.
- (2) Remove ETI per paragraph 18C; do not unsolder leads.
- (3) Remove J1 per paragraph 18D.
- (4) Remove 12 screws (175).

31-30-32

Page 316
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (5) Remove 4 of 6 screws (260) that attach bracket (255). Remove 2 screws from each side. Loosen the 2 remaining screws (1 on each side) and pivot bracket away from motherboard.

- (6) Remove motherboard (170).

NOTE: EMI filter (35), ETI (70), and connector (5) will remain tethered to the motherboard by their wires, which are part of the wire harness tied to the motherboard. The motherboard also remains tethered to the chassis by the harness. Cut the cable ties (25) and straps (110) as necessary to move the motherboard to a working position.

- (7) If it is necessary to remove connectors (180), remove the shield (170, IPL Figure 1) by removing 4 screws (175), washers (180) and washers (185). Identify and unsolder the connector pins.

- (8) The sponge rubber (187), or foam tape, on the shield may be removed with a thin-bladed tool.

- (9) To remove each connector (180), remove 2 screws (185), nylon washers (190), and nuts (195).

NOTE: The sealant at the base of each connector should peel off as the connector is lifted.

- (10) If it is necessary to remove either J3 or J4 (200), remove the 2 jackposts (205). Identify and unsolder the connector pins and lift the connector from the motherboard.

NOTE: The sealant at the base of each connector should peel off as the connector is lifted.

- (11) If damaged, replace standoffs (225). Standoffs may be removed by drilling or clipping. Do not enlarge 0.138-inch diameter holes by more than 0.003 inch of original diameter.

- (12) If damaged, replace terminals (220). Do not attempt to straighten bent terminals. If any terminal is bent more than 20° from perpendicular to the mounting surface, replace it. Terminals may be removed by drilling or clipping. Do not enlarge 0.113-inch diameter holes by more than 0.003 inch of original diameter.

- (13) Replace standoffs (215) if damaged or loose. Standoffs may be removed by drilling or clipping. Do not enlarge 0.138-inch diameter holes by more than 0.003 inch of original diameter.

- (14) Nutplates (230) may be removed by drilling or clipping rivets (235). Do not enlarge 0.098-inch diameter rivet holes by more than 0.003 inch of original diameter.

F. Shockmount Removal

- (1) Remove screws (275) and shockmounts (265). Retain ground strap (270) for reinstallation.
- (2) To remove upper shockmount (325), remove the ULD assembly per paragraph 3C, then remove 2 screws (330), and nuts (335).
- (3) To remove lower shockmount (325), remove the identification plate (130, IPL Figure 1) by removing 6 screws (135), then remove 2 screws (330) and nuts (335).

31-30-32

Page 317
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

G. Connector J2 Removal

To remove J2 (140), remove nut, which is part of connector assembly.

H. LED Indicator Assembly Removal

To remove LED indicator (145), remove 1 screw (160), nut (165), retainer (155), and spacer (150).

I. Connector J5 Removal

To remove J5 (85), remove potting shell (95) from the connector bracket (100) by removing the pair of jackposts (90). Remove 2 screws (105A) and connector bracket.

J. Tywrap Base Removal

To remove the tywrap bases (120), remove screws (125) and nuts (130).

K. Card Guide Removal

- (1) Card guides (360, 370, 395, 405) may be removed by drilling or clipping rivets (365, 375, 400, 410). Do not enlarge 0.067-inch diameter rivet holes by more than 0.003 inch of original diameter.
- (2) Remove screws (285) and power supply guides (280).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CLEANING

1. **General**

This section presents instructions for cleaning the components, subassemblies, and parts of the Universal Flight Data Recorder (UFDR).

2. **Cleaning Materials**

Required materials are listed in Figure 401.

NOTE: *Equivalent substitutes may be used for all cleaning materials.*

MATERIALS	SPECIFICATION/SOURCE
Cloth, Cotton, Lint-free	Materials are obtainable from any commercial source.
Applicators, cotton tipped	
Brush, Soft-Bristle, Nonmetallic	
Detergent, Low-Suds	
Water, Soft	
Alcohol, Isopropyl	
Acetone or Methyl Ethyl Ketone (MEK), Technical Grade	Competition Chemicals Inc. Iowa Falls, IA 51026
Simichrome Polish	

Cleaning Materials
Figure 401

3. **Cleaning Procedures**

The following paragraphs contain instructions and procedures for cleaning the various parts of the dismantled and disassembled assemblies and parts preparatory to inspection.

WARNING: OBSERVE ALL FIRE PRECAUTIONS WHEN USING FLAMMABLE MATERIALS FOR CLEANING PURPOSES. THESE MATERIALS SHOULD ONLY BE USED OUTSIDE OR IN A VENTILATED BOOTH PROVIDED WITH EXPLOSION-PROOF ELECTRICAL EQUIPMENT AND AN EXHAUST FAN HAVING SPARK-PROOF BLADES.

WARNING: PERFORM OPERATIONS INVOLVING CLEANING SOLVENT UNDER A VENTILATED HOOD. AVOID BREATHING SOLVENT VAPOR; WEAR A SUITABLE MASK WHEN NECESSARY. USE GOGGLES, GLOVES, AND APRON TO PREVENT IRRITATION FROM PROLONGED CONTACT. CHANGE CLOTHING THAT HAS BECOME SATURATED WITH SOLVENT.

WARNING: WEAR GOGGLES WHEN USING AIR JET TO BLOW DUST AND DIRT FROM PARTS. WARN PERSONS IN THE IMMEDIATE VICINITY.

31-30-32

Page 401
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAUTION: USE CARE NOT TO CONTAMINATE TAPE. DO NOT ALLOW ALCOHOL OR SOLVENT TO COME IN CONTACT WITH TAPE.

NOTE: *Reference to air jet signifies a hand-operated air nozzle supplied with clean, dry, compressed air at a maximum pressure of 28 pounds per square inch.*

A. Printed Wiring Board Assemblies

CAUTION: AVOID AIR-BLASTING SMALL COILS, LEADS, AND OTHER DELICATE PARTS BY TOO CLOSE AN APPROACH WITH AIR JET NOZZLE.

CAUTION: DO NOT BRUSH OR SPRAY CLEAN CIRCUIT BOARD.

- (1) Dust and foreign matter may be removed with an air jet provided that board circuits are grounded and ionized air is directed at the board.

NOTE: *If wiring and cables are disturbed, restore them to their proper positions or dress after cleaning.*

- (2) After replacing components and jumpers, clean all solder joints with isopropyl alcohol (Ref. Figure 401).

B. Connectors

- (1) Wipe dust and dirt from bodies, shells, and cable clamps with an alcohol moistened cloth. Dry with a clean, dry, lintless cloth.

- (2) Remove dust from inserts with a small, soft-bristle brush and an air jet.

CAUTION: DO NOT PERMIT SOLVENT TO RUN INTO SLEEVES OR CONDUIT COVERING ANY WIRES OR CABLES CONNECTED TO CONTACT TERMINALS OF THE INSERT, AS SWELLING AND SOFTENING WILL RESULT.

- (3) Wash dirt and any traces of lubricant from inserts, insulation, and contacts with alcohol, applied sparingly with a small, soft-bristle brush.

- (4) Dry insert with an air jet.

C. Machined Metal Parts or Assemblies

- (1) Blow dust from surfaces, holes, and recesses with an air jet.

- (2) Remove bulk of surface grease and dirt with rags.

- (3) Immerse machined metal parts in solvent and scrub until clean, working over all surfaces and into all holes and recesses with a suitable nonmetallic brush. Flat, wood-backed brushes with soft fibre bristles are recommended for surfaces; round brushes, like those used for washing bottles and test tubes are recommended for holes and recesses.

- (4) Remove parts from solvent and permit solvent to drain.

- (5) Immerse parts in clean solvent and drain. Do not permit solvent to remain trapped in holes or recesses during drain. Use an air jet to remove any trapped solvent.

- (6) Dry in dust-free, dry area or suitable enclosure. Radiant heat used in a ventilated enclosure is recommended for drying.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

D. Miscellaneous Metal Parts

Metal parts, such as mounting plates, mounting clamps, brackets, heatsinks, and attaching hardware should be cleaned in a suitable cleaning machine or washed in solvent to remove all accumulated dirt and dust. Dry with a clean, dry, lintless cloth and an air jet.

NOTE: *An ultrasonic cleaner can be used to clean small metal parts. Refer to the manufacturer's instructions for exact uses and procedures. Use the drying methods normally used in cleaning a part. Lubricate as required.*

E. Nonmetallic Parts

Nonmetallic parts, excluding static sensitive components.

(1) Clean all surfaces with a soft-bristle brush and an air jet.

CAUTION: DO NOT SATURATE MYLAR, NYLON, EPOXY, FIBERGLASS, OR TEFLON PARTS, AS SWELLING AND SOFTENING MAY RESULT. THESE PARTS MAY BE CLEANED WITH A SOLUTION OF 2 OUNCES OF DETERGENT POWDER TO 1 GALLON OF WATER AND SUITABLE BRUSHES TO REMOVE SURFACE DIRT OR FOREIGN MATTER.

(2) Wipe clean with a solvent-moistened, lintless cloth.

(3) Dry with a clean, dry, lintless cloth and an air jet.

F. Bearings

CAUTION: DO NOT SPIN BALL BEARINGS WITH AIR JET OR ALLOW ALCOHOL OR SOLVENT TO COME IN CONTACT WITH BALL BEARINGS.

Bearings should be cleaned and/or reworked only by factory personnel or by a factory approved shop.

NOTE: *Bearings are prelubricated before usage. Refer to Figure 603 for bearing type and lubrication.*

G. Tape Transport Cleaning Procedure

NOTE: *Honeywell recommends that the following procedure be accomplished whenever the unit is being serviced at a shop maintenance level.*

NOTE: *This procedure should be accomplished in a dust-free area such as a clean room, or within the confines of a laminar-flow clean bench.*

- (1) Remove screws (10, 15, IPL Figure 8) and top cover (5 or 5A). Place tape transport in a fixture or otherwise immobilize it so that parts are not damaged during the cleaning process.
- (2) Remove flat belt (45A, 420, IPL Figure 10) or (35, IPL Figure 14) and examine as follows:
 - (a) Examine belt for scratches, nicks, cuts, cracks, folds, bubbles, tears, or obvious wear. If any of these faults are apparent, or doubt exists regarding serviceability, replace with new belt.
 - (b) If examined belt is satisfactory, wipe clean with a lintless cloth moistened in a solution of 1 part detergent powder to 120 parts water. Allow to air dry.

31-30-32

Page 403
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (c) Check belt per **CHECK** section.
- (d) Reinstall per **ASSEMBLY** section.

CAUTION: PERIPHERAL BELTS SHALL NOT BE STRETCHED BEYOND 18.0 INCHES MAXIMUM INTERNAL CIRCUMFERENCE.

- (3) Remove peripheral belt (50B, IPL Figure 10; 40, IPL Figure 14) per **DISASSEMBLY** section.
 - (a) Examine belt for scratches, nicks, cuts, bubbles, burrs, tears, or obvious wear. Belt shall contain no evidence of material contamination, uneven or incomplete cure, or flaws. If any of these faults are apparent, or doubt exists regarding serviceability, replace with new belt.
 - (b) If the examined belt is satisfactory, wipe clean with a lintless cloth moistened in a solution of 1 part detergent powder to 120 parts water. Allow to air dry. Discoloration of belt surface is allowable.
 - (c) Check belt per **CHECK** section.
 - (d) Store cleaned belt on a 'storage hat' support form normally supplied with new belts.

CAUTION: WHEN CLEANING SURFACES IN THE FOLLOWING STEPS, DO NOT ALLOW CLEANING AGENT TO GET INTO BEARINGS. DAMPEN, DO NOT SATURATE, A COTTON SWAB OR LINTLESS CLOTH.

- (4) Clean rollers (15, IPL Figure 11) or (35, IPL Figure 13), drive pulley (95B, 95C, IPL Figure 11) or (120 or 145, IPL Figure 13), and drive belt roller guides (330, 425, IPL Figure 10) or (330, IPL Figure 14), where belt contacts pulleys, with a lintless cloth or cotton swab dampened in isopropyl alcohol (Ref. Figure 401). Allow to air dry.

WARNING: ACETONE AND METHYL ETHYL KETONE ARE TOXIC AND FLAMMABLE. DO NOT BREATHE VAPORS. USE IN A WELL-VENTILATED AREA FREE FROM SPARKS, FLAME, AND HOT SURFACES. WEAR SPLASH GOGGLES, SOLVENT-RESISTANT GLOVES, AND OTHER PROTECTIVE GEAR. IN CASE OF EYE CONTACT, FLUSH WITH WATER AND SEEK MEDICAL ATTENTION. IN CASE OF SKIN CONTACT, WASH WITH SOAP AND WATER.

- (5) Clean tape drive pulley (capstan) (200, IPL Figure 10) or (205, 205A, 205B, 205C, IPL Figure 14), tape transport pulley (flywheel) (215, IPL Figure 10) or (220, IPL Figure 14), 2 idler pulleys (240, IPL Figure 10) or (255, 255A, 255B, 255C, IPL Figure 14), where belt contacts pulleys, and roller pins (300, IPL Figure 10) or (260, 260A, IPL Figure 14) with a lintless cloth or cotton swab dampened in acetone, MEK, or isopropyl alcohol (Ref. Figure 401). Allow to air dry.

NOTE: *If black debris is embedded in roller pin(s) (260, 260A), and is not removable with acetone, MEK, or isopropyl alcohol, replace pin(s).*

- (6) Vacuum dust from both control plates (5B, 5C, 5D, IPL Figure 14).

CAUTION: REPLACE PERIPHERAL BELT PER ASSEMBLY SECTION. DAMAGE TO BELT AND/OR TAPE CAN RESULT FROM IMPROPER HANDLING.

- (7) Reinstall peripheral belt per **ASSEMBLY** section.

31-30-32

Page 404
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAUTION: WHEN HANDLING TAPE, EXERCISE EXTREME CAUTION TO PREVENT TAPE DAMAGE. ANY TAPE OVERSTRESS CAN CAUSE SIGNAL DROPOUTS OR TAPE BREAKAGE.

CAUTION: WHEN WORKING AROUND AND CLEANING HEADS, USE EXTREME CAUTION TO AVOID DAMAGING HEADS. NO HARSH DETERGENTS OR SOLVENTS SHALL BE USED.

(8) Turn loaded reel just enough to provide slack necessary to remove magnetic tape from tape contacting surfaces while performing the following cleaning procedure. It is recommended that tweezers with mylar sleeved tips be used to gently pull tape away from parts mentioned in steps (a) and (b).

(a) Clean tape contacting surfaces of the 5 tape guide rollers (175, 475, 475A, 475B, 555, 555A, IPL Figure 14) with a lintless cloth or cotton swab moistened in isopropyl alcohol. Allow to air dry.

(b) Clean surfaces of magnetic heads with a cotton swab and Simichrome polish. Remove polish residue with acetone, MEK, or isopropyl alcohol (Ref. Figure 401).

NOTE: *If excess wear is noted after cleaning heads, rework head surface per **REPAIR** section.*

(c) After parts in steps (a) and (b) are dry, carefully replace tape and wind slack back onto loaded reel.

(d) Clean both surfaces of magnetic tape with ribbed watch paper (33, Figure 701) per **ASSEMBLY** section.

(9) Blow dust from inside of top and bottom covers (5, 5A, 5B, 5C, IPL Figure 8) with an air jet, and clean with a lintless cloth moistened in isopropyl alcohol. Allow to air dry.

(10) Examine inside of top and bottom covers (5, 5A, 5B, 5C, IPL Figure 8) for delamination or separation of the liner from the cover. For corrective action, refer to **REPAIR** section.

H. Underwater Locating Device (ULD) (30, IPL Figure 1)

(1) Clean threads and o-ring groove in body of ULD and threads of end cap with a lintless cloth moistened in isopropyl alcohol. Allow to air dry.

(2) Verify that ULD case and water switch are free of grease film. Wipe clean with lintless cloth moistened in isopropyl alcohol. Allow to air dry.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

I. Underwater Acoustic Beacon (UAB) (30A, IPL Figure 1)

CAUTION: DO NOT DISASSEMBLE UNDERWATER ACOUSTIC BEACON (UAB)(30A).

WARNING: NO BATTERY MAINTENANCE IS REQUIRED FOR THE DK100 UAB. DO NOT REMOVE BATTERY AT ANY TIME. AVOID ANY SITUATION THAT COULD POSSIBLE CRUSH OR PENETRATE UAB CASE. DO NOT DISPOSE OF UAB. AT OR NEAR EXPIRATION DATE PRINTED ON CASE, RETURN UAB TO MANUFACTURER FOR SERVICE. FAILURE TO OBSERVE THESE PRECAUTIONS COULD RESULT IN RELEASE OF HAZARDOUS CHEMICALS.

Clean switch end of UAB with a clean, soft cloth and mild detergent. Thoroughly dry with a clean cloth. Frequently clean switch end insulator to prevent the occurrence of leakage currents across the switch, which will affect battery life.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CHECK

1. **General**

The check procedures detailed in the following paragraphs are required to ensure that the repaired equipment will meet new equipment standards of operation. Check includes a thorough check for excessive wear or deterioration, as well as damage or other obvious defects. Do not retain a part for assembly if it is found to be worn in excess of the allowable service limits. If any doubt exists regarding the serviceability of a part, replace the part. Fits and clearances and other dimensional limits are given in the **FITS AND CLEARANCES** section and referenced in the **ASSEMBLY** section.

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES ARE SUBJECT TO DAMAGE BY EXCESSIVE LEVELS OF VOLTAGE. REFER TO DOD-HDBK-263 FOR PROPER HANDLING PROCEDURES.

NOTE: *Excessive wear may be defined as any obvious deterioration or deformation of parts that may contribute to the malfunction of the unit, or that may render the unit inoperative. Any part exceeding the tolerance limits may also be considered excessively worn or unacceptable.*

A. **Component Physical Examination**

- (1) Using adequate light, and under moderate magnification, visually check all parts for scratches, corrosion, nicks, cracks, foreign matter, breaks, or apparent obvious wear.
- (2) Check all springs for cracks, breaks, permanent set, fatigue, distortion, or corrosion. Check springs for proper tension and position.
- (3) Check all threaded surfaces for cross thread, stripping, chips, or foreign matter.
- (4) Check all cast parts for cracks. Carefully check areas around bosses, flanges, and other critical areas.
- (5) Check nameplate and decals for legibility.

B. **Component Electrical Examination**

- (1) Wiring: loose or poor connections.
- (2) Insulation: worn, burned, or broken areas.
- (3) Connectors: loose, corroded, or bent pins and loose conductive material between pins.
- (4) Printed wiring boards: overheating, physical damage, and board deterioration.
- (5) Printed wiring on board: small cracks or shorts.
- (6) Detailed parts: displacement, damaged sleeving, and shorted leads.

2. **Specific Checks (See Figure 10 or Figure 14)**

- A. Check wavy washers and retainers for breaks, permanent set, fatigue, distortion, or corrosion. During assembly, verify that E-ring retainers are installed with the smooth (shiny) side toward hardware being retained.

31-30-32

Page 501
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- B. Check all bearings for signs of wear, ease of rotation (binding), chattering, and erratic operation. Any bearing that does not appear to be perfect in operation should be replaced. Bearings should be cleaned and/or reworked only by factory personnel or by a factory approved shop.
- C. Examine the flat (drive) belt as follows:
- (1) Examine belt for scratches, nicks, cuts, cracks, folds, bubbles, or tears.
 - (2) Place the belt under one pound of tension. The internal circumference shall be 13.40 to 13.52 inches (for reference only, inside diameter is 4.29 inches). At the same tension, the belt edges shall be straight equal to or less than 0.0007 inch per inch of belt.

NOTE: Step (2) may be performed at the discretion of operator.

- D. Examine the peripheral belt (50, IPL Figure 10 or 40, IPL Figure 14) as follows. Step (2) may be performed at the operator's discretion.

- (1) Examine the peripheral belt for scratches, nicks, cuts, bubbles, burrs, or tears. Belt shall contain no evidence of material contamination, uneven or incomplete cure, or flaws.

CAUTION: BELTS STRETCHED IN STEP (2) SHALL NOT BE STRETCHED BEYOND 18.0 INCHES MAXIMUM INTERNAL CIRCUMFERENCE.

- (2) Stretch belt twice uniformly to 17.5 inches internal circumference and note tension and belt width each time. The tension of both stretches combined shall be 67 to 75 ounces for the -004 and 67 to 80 for the -005. The belt width, each time, shall be 0.22 to 0.235 inches. The inner and outer surfaces shall be smooth and without embossed patterns of any type. Belt may be shiny.

- E. Check vespel peripheral belt rollers and roller pins, and replace if damaged in any of the following ways:

NOTE: Disassemble peripheral (P-belt) roller assemblies according to the **DISASSEMBLY** section.

- (1) Perpendicularity of roller pins with respect to the baseplate and each other exceeds 0.001 inch.
- (2) Black debris is embedded in roller pin, and is not removable using isopropyl alcohol.
- (3) Roller hole diameter taper, measured with commercially available test pins, is more than 0.0005 inch, end to end.
- (4) Roller hole diameter, measured with a commercially available test pin, is worn to more than 0.0955 inch.
- (5) Roller pin diameter is worn to less than 0.074 inch.

- F. Check shockmount and card guides for proper position.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

G. Check magnetic heads as follows:

CAUTION: ANY TIME A DC OHMMETER IS USED, THE HEADS MUST BE DEGAUSSED AFTER EXAMINATION OR CHECKING. REFER TO REPAIR SECTION FOR DEGAUSSING PROCEDURE.

- (1) As viewed through adequate magnification, no scratches wider than 0.001 inch may cross the gap, and no scratches wider than 0.001 inch can appear on laminations, casting areas, or on the face of the head vertically in line with the window areas.
- (2) Coil phasing must be consistent to produce same pole piece polarity track to track.
- (3) Check for cracks, nicks, or loose wiring.
- (4) Check heads for proper positioning, wear, corrosion, or other obvious defects. A procedure for inspecting and refinishing the pole tip surface area of head assemblies that are worn as a result of normal tape-to-head contact is provided in the **REPAIR** section.
- (5) Check heads at tape-to-head interface for contamination residue. A procedure for cleaning the head assemblies is provided in the **CLEANING** section.

H. Check magnetic tape for contamination residue. A procedure for cleaning tape is provided in the **ASSEMBLY** section.

I. Check tape recorder environmental enclosure covers for delamination of the liner from the insulation inside the cover, a damaged liner, or liner separation from the cover edge. Repair the environmental enclosure covers according to the REPAIR section.

3. **Perpendicularity Check**

A. General

A perpendicularity check can be made with a balanced dial indicator and a height test set, or with an electronic height gage (35, Figure 901). A balanced dial indicator is one that has figures in both directions from zero in the middle of the scale. Thus, the dial indicator can give an indication in either direction above or below the plane of reference. The height set includes a surface plate (38, Figure 901) for establishing a plane of flatness and a height locator gage for transmitting position information to the dial indicator. Any kind of precision height locator gaging blocks may be used. For example, Brown & Sharpe makes a column of gaging blocks (39, Figure 901), which can be raised or lowered 1 inch by means of a micrometer head with an accuracy of 0.0001 inch. Any height finder gage may be used with the balanced dial indicator, provided it is compatible with the dial indicator used. If an electronic height gage is used, the height locator gaging blocks are not needed.

B. Procedure

- (1) Mount the baseplate (415, 415A, 415B, IPL Figure 14) in fixture CF 617-4422-001 (16, Figure 901).
- (2) Set the fixture on the surface plate and orient it so that the pin(s) of interest are in the horizontal plane.
- (3) Adjust the height finder gage to establish point contact between the probe of the height finder gage and a point at the base of the pin (see Figure 501).
- (4) With the contact point satisfactorily established, zero the dial indicator.

31-30-32

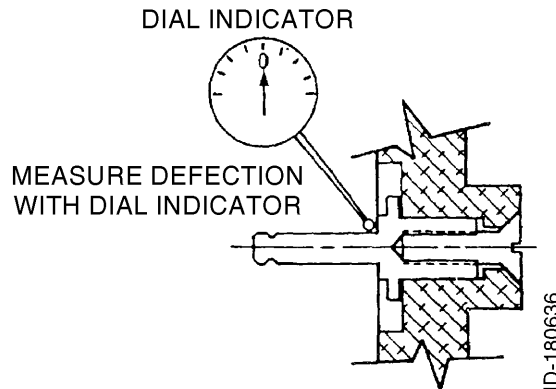
Page 503
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Perpendicularity Check
Figure 501

- (5) Move the probe of the height finder from the base of the pin to the end of the pin.
- (6) Read the dial indicator and verify that the dial indication is within tolerance (see Figure 502).

PINS	TOLERANCES
Tape Guide Pins (14-180A)	0.0005 inch
Drive Belt Roller Pins (14-335, 335A)	0.001 inch
Tape Reel Pins (14-80, 80A)	0.001 inch
Peripheral Belt Pins (14-260, 260A)	0.001 inch

Perpendicularity Check Requirements
Figure 502

- (7) If the dial indication is not within tolerance, loosen the pin fastener, rotate the pin in 90 degree increments, tightening the pin fastener, and repeating steps (3) through (6) for each increment. If the dial indication is still not within tolerance, replace the pin.
- (8) If the dial indication is within tolerance, turn the fixture with the tape deck mounted in it 90 degrees to establish a second plane for the perpendicularity check.
- (9) Repeat steps (3) through (7).
- (10) If dial indication is within tolerance in both pin planes of reference, repeat procedure for other pins as required.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

4. Height Check

A. General

To ensure proper tape tracking, the height of the 5 tape guide roller assemblies (140, 440, 485, IPL Figure 14) and 2 reel hub assemblies (70, IPL Figure 14) must be properly adjusted. To adjust these components, a height check is made.

NOTE: *If using height locator gaging blocks (39, Figure 901), perform steps (1) through (10); if using electronic height gage (35, Figure 901), perform steps (1) through (4) and (9) through (10).*

B. Procedure

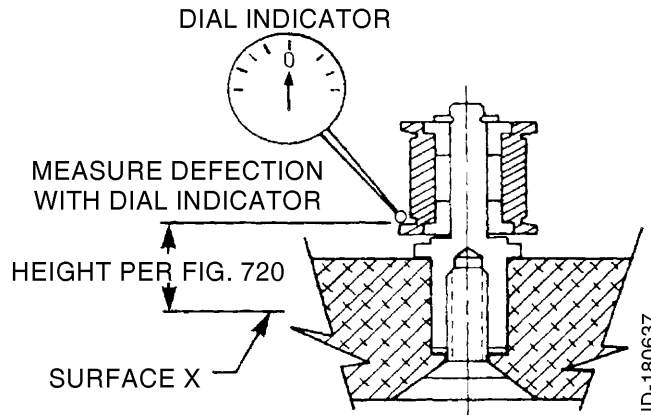
- (1) Mount the baseplate (415, 415A, 415B, IPL Figure 14) in fixture CF 617-4422-001 (16, Figure 901).
- (2) Set the fixture on the surface plate and orient it so the tape deck is right side up.
- (3) Adjust the height finder gage to establish point contact between the probe of the height finder gage and surface X (lowest head mounting surface) in front of the even erase head. The even erase head is nearest to the stepper motor.
- (4) Zero the dial indicator.
- (5) Using the height locator gaging blocks, determine the height of surface X above the surface plate; for example, 5.2541 inches.
- (6) Add appropriate height from Figure 711 or Figure 720 to height found in step (5):
 - (a) For tape guide roller assemblies: height from Figure 720, e.g., $5.2541 + 0.2520 = 5.5061$ inches.
 - (b) For reel hub assemblies: height from Figure 711, e.g., $5.2541 + 0.3560 = 5.6101$ inches.
 - (c) Using height locator gaging blocks, set gaging block(s) to sum of the appropriate height found in step (6).
- (7) Adjust the height finder gage to one of these heights as appropriate and zero the dial indicator.
- (8) Bring the probe of the height finder gage into contact with one of the following surfaces as appropriate:
 - (a) Lower flange of the tape guide rollers (140, 440, 485, IPL Figure 14) (see Figure 503).
 - (b) Top of flange of hub assembly (70, IPL Figure 14) (see Figure 504).
- (9) Observe the dial indicator for one of the following indications:
 - (a) Tape guide rollers: height differential among all rollers of the same height requirement of less than 0.001 inch.
 - (b) Hub Assemblies: ± 0.002 inch.

Honeywell

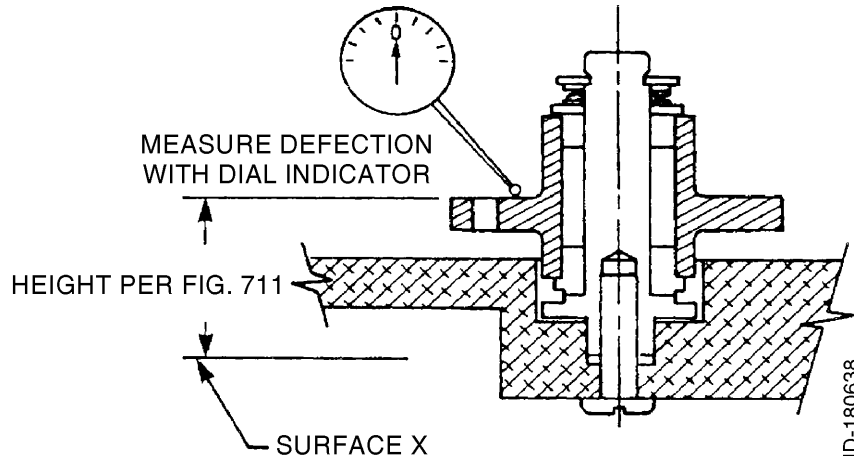
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (10) If the dial indications are out of tolerance, add or subtract the following shims as required:
- (a) Tape guide roller; 150A, 155, 160A, 162, IPL Figure 14.
 - (b) Hub assembly; 105, 110, 115, 116, IPL Figure 14.



Tape Guide Roller Height Check
Figure 503



Hub Assembly Height Check
Figure 504

5. Preload Check

A. General

It is important that bearing preload specifications be maintained in the tape deck. Preloads are specified to ensure minimum axial and radial play, to reduce the effects of vibration, and to realize optimum bearing life under all environmental conditions. Proper bearing preloads are critical to the tape recording quality. Maintaining the proper bearing preload assures optimum

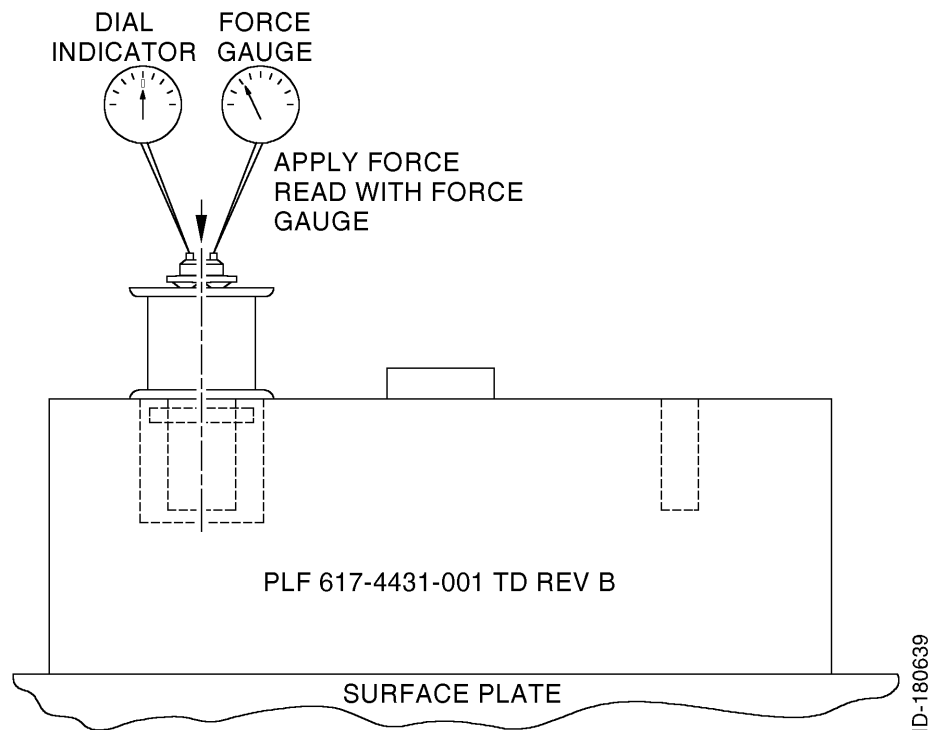
31-30-32

Page 506
22 Dec 2006

intervals between repairs. Insufficient preload causes bearings to be loose in operation and produce 'jitter' and mistracking in the tape recorder. It thereby reduces recording quality and shortens bearing life. Bearings assembled with too much preload wear out prematurely.

B. Procedure

- (1) If data jitter or tape mistracking is observed, remove peripheral belt and/or magnetic tape from tape transport according to the **DISASSEMBLY** section.
- (2) Using finger pressure on the tape guide and belt drive assemblies, check for end play and axial freedom.
- (3) Remove and check the preload of any assemblies suspected of being faulty. Removal procedures are provided in the **DISASSEMBLY** section.
- (4) Mount assembly to be checked in preload fixture PLF 617-4431-001 (TD REV B) (31, Figure 901) as shown in Figure 505. Mount preload fixture in fixture CF 617-4422-001 (16, Figure 901).



Preload Check
Figure 505

- (5) Adjust height finder gage to establish point contact between probe of the height finder gage and an appropriate point on the pin of the roller assembly which is to be preload checked (see Figure 505).
- (6) Zero the dial indicator.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (7) Using a force gage, axially deflect the pin with an amount of force required to deflect the pin and observe the dial indicator (see Figure 506 for bearing preload requirements).

BEARING ASSEMBLY	AXIAL DEFLECTION	REQUIRED FORCE
Flywheel (14-220)	0.0002 inch	20 - 40 oz.
Reel Hubs (14-70,70A)	0.0002 inch	5 - 9 oz.
Reel Hubs (14-70B,70C)	0.0005 inch	24 - 40 oz.
Tape Guide Roller (14-140,A,B,C)	0.0005 inch	2 - 4 oz.

Preload Requirements
Figure 506

- (8) If the force required is too high, shims must be removed from the assembly; if the force required is too low, shims must be added to the assembly. Remove or replace shims at the top of the assemblies only so that roller height is not affected.

6. End Play Check

A. General

An end play or axial freedom check is similar to a preload check and is performed for essentially the same reasons. Figure 507 lists assemblies which require an end play check.

PINS	TOLERANCE
Peripheral Belt Pulley (14-245B,C,D,E)	0.001 to 0.002 inch
Drive Belt Roller Guide (14-300A,C)	0.0005 to 0.002 inch
Roller Guide (13-5,A,B,C)	0.0005 to 0.002 inch
Tape Guide Roller (14-440,A,B,C; 485,A,B,C,D,E)	0.0005 to 0.002 inch

End Play Requirements
Figure 507

B. Procedure

- (1) Mount the baseplate (415, 415A, 415B, IPL Figure 14) in fixture CF 617-4422-001 (16, Figure 901).
- (2) Place the fixture on the surface plate and orient it so the tape deck is right side up.
- (3) Adjust the height finder gage to establish point contact between the probe of the height finder gage and an appropriate point on the assembly which is to be end play checked.
- (4) Zero the dial indicator.
- (5) Axially deflect the assembly and observe the dial indicator.

31-30-32

Page 508
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (6) If the dial indicator indicates too much end play, shims will have to be added to the assembly; if the dial indicator indicates too little end play, shims will have to be removed.
- (7) Repeat the procedure until the end play tolerance is met.

7. Track Location Check

A. General

The purpose of this check is to verify that the magnetic heads are properly wired, that all four tracks of each head function, and that the spacing between tracks is satisfactory. To perform this check requires that the tape recorder assembly be operated and that data be recorded on a section of tape in two predetermined patterns. After the patterns of data are recorded, the section of tape with the data is cut out from the rest of the tape and developed in a solution for visual inspection.

CAUTION: WHEN REPLACING THE SCRATCH PERIPHERAL BELT AND TAPE REEL WITH KNOWN GOOD ITEMS, ALWAYS USE THE PROCEDURES IN PARAGRAPH 14 OF THE ASSEMBLY SECTION TO AVOID DAMAGING THE BELT OR TAPE.

B. Procedure

- (1) Install a scratch reel loaded with old tape in good condition, and a scratch peripheral belt on the tape transport.

NOTE: *A full reel is not required for this check.*

- (2) Install the Tape Recorder Assembly (150A, 150B, IPL Figure 1) into the UFDR chassis assembly and connect plugs P3, P4, and P5.

NOTE: *The top cover (5, IPL Figure 8) need not be installed at this time.*

NOTE: *Instead of installing the Tape Recorder Assembly (150A, 150B, IPL Figure 1) into the chassis assembly, it may be connected into the chassis assembly with jumper cables, if available.*

- (3) Connect the UFDR to the UFDR Tester (1, Figure 901) according to the instructions given in paragraph 2B of the **TESTING AND TROUBLESHOOTING** section.
- (4) Set up the Tester according to paragraph 2D, steps (1) and (2), of the **TESTING AND TROUBLESHOOTING** section.
- (5) Using the REWIND and FAST FWD switches on the tester, cycle the tape back and forth for one minute to stabilize tension and tracking. Select HOLD.
- (6) Optimize the head signal level outputs per these steps:

NOTE: *Step [7B(6)] is applicable only if the heads are replaced or repositioned for any reason.*

- (a) Using the TRACK INCR switch on the tester, select TRACK 0.
- (b) Connect oscilloscope (2, Figure 901) to tester X100 ANALOG OUT jack and EXTERNAL TRIGGER to tester PLAYBK ENABLED jack. Set VERTICAL to 2 volts/div. and HORIZONTAL to 10 ms/div. Set tester WRITE CURRENT switch to NORM.

31-30-32

Page 509
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (c) Actuate FLT REC switch and note the amplitude of the playback signal.
- (d) Loosen the screws that attach the even track read/write head (420, IPL Figure 14) 1/2 turn.
- (e) Monitor the amplitude of the playback signal while rotating the head back and forth over the limits of its adjustment range. Also, monitor the playback signal while moving the head away from and toward the tape over the limits of its adjustment range.
- (f) Select the optimum head position for achieving the maximum playback signal amplitude. Tighten the head screws, ensuring that the signal amplitude does not drop.
- (g) Select HOLD and track 7. Actuate the tester FLT REC switch. Note the amplitude of the playback signal.
- (h) Loosen the screws that attach the odd track read/write head (425, IPL Figure 14) 1/2 turn. Repeat steps (e) and (f).
- (i) Select HOLD.
- (j) Select REWIND, rewind the tape to near the beginning of tape and select HOLD.
- (7) Record a pattern of data on the tape using this procedure:
 - (a) Using the TRACK INCR switch on the tester, select each track in this sequence: 0, 2, 4, 6, 1, 3, 5, 7.
 - (b) After each track is selected, actuate FLT REC switch and observe that four or five data records are recorded on each track selected.

NOTE: *To determine that a number of data records is recorded, observe the PLAYBK ENABLED LED on the tester blink on and off (count the number) or observe the tape transport assembly and count the number of times the tape reel reverses (checkstrokes).*
 - (c) When track 7 (last track) has been recorded (at the count of four, for example), select HOLD.
 - (d) Using a Phillips head screwdriver, manually turn the capstan and advance tape onto the take-up reel a few inches.
- (8) Record a second pattern of data using this procedure:

NOTE: *This pattern of data is for checking tape track spacing.*

 - (a) Using the TRACK INCR switch on the tester, select the tracks in this sequence: 0, 1, 2, 3, 4, 5, 6, 7.
 - (b) After each track is selected, actuate FLT REC switch and observe that 10 data records are recorded on each track.
 - (c) When track 7 (last track) has been recorded, select HOLD.
- (9) Using the Phillips head screwdriver, manually turn the capstan and advance the tape a few inches onto the take-up reel.

31-30-32

Page 510
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

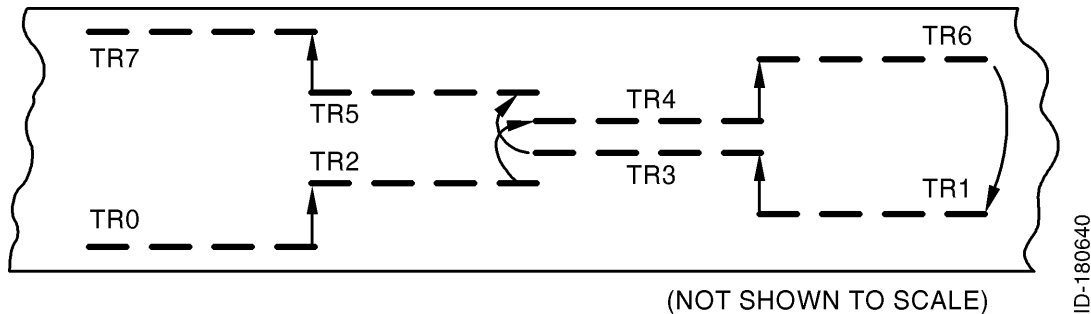
Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (10) Cut the tape near the loaded reel and remove the section of tape with the data recorded on it.
- (11) Immerse the cut length of tape into the Magna-See solution (30, Figure 701) and swish it around for a few seconds or dust the sample tape with iron powder (43, Figure 701) or immerse the sample tape into a solution of iron powder and a volatile liquid such as acetone or fluorinert.
- (12) Withdraw the cut length of tape from the Magna-See solution and examine it for the two patterns shown in Figures 508 and 509.

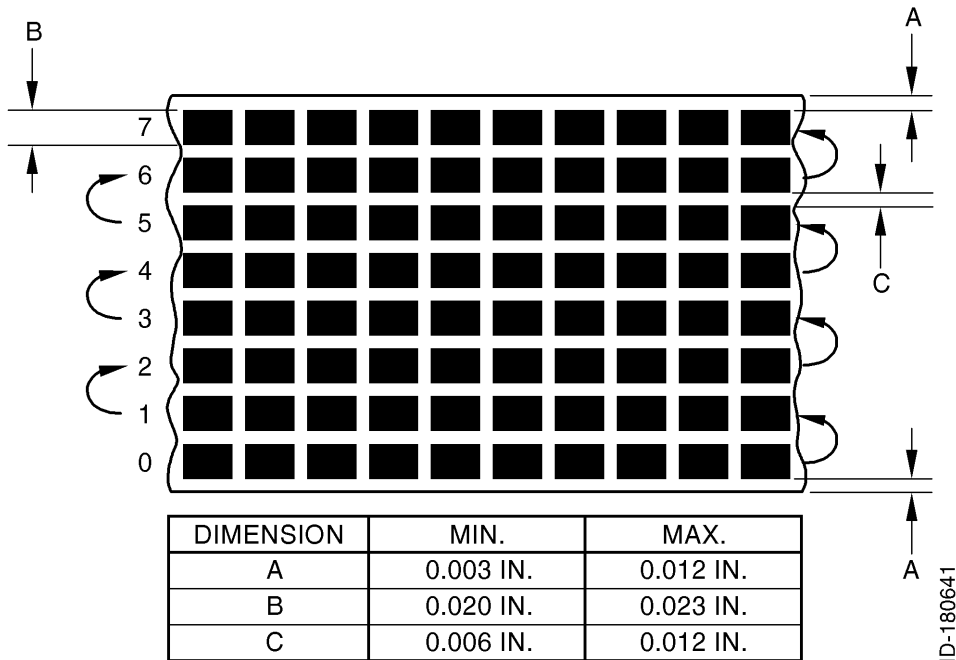
NOTE: When examining tape for the pattern shown in Figure 508, look for the presence of each track in proper sequence.

NOTE: When examining the tape for the pattern shown in Figure 509, look for adequate spacing between each track.

- (13) Add or remove shims (421, 422, 423, 426, 427, 428, 431, 432, 433; IPL Figure 14) as required to achieve the required track location dimensions. Shims must be equally installed under pairs read/write heads and erase heads. Repeat steps (7) through (12).
- (14) Remove the head attaching screws 1 at a time, and apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw threads.



Track Wiring Check Pattern
Figure 508



Track Spacing Pattern
Figure 509

8. Tape Stacking, P-Belt Running Location, and Tape Reversing Time Checks

A. General

These three checks provide an indication of whether the peripheral belt drag roller system is properly aligned and whether the peripheral belt develops proper tension.

B. Procedure

- (1) Actuate the REWIND switch on the tester and allow the tape to move to BOT.
- (2) When the tape reaches BOT, actuate the FAST FWD switch and simultaneously start the stop watch.
- (3) As the tape is spooling from BOT to EOT, observe the following:
 - (a) The peripheral belt runs in the center of the tape pack and does not overhang the top or bottom.
 - (b) The tape does not ride up on the rollers and the edges of the tape do not appear scalloped or nonuniform when viewed from directly above.
 - (c) At midreel, reel-to-reel spacing is 0.020 inch, minimum.
 - (d) When EOT is reached, stop the stopwatch and verify that the tape traversing time is 900 seconds, minimum.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

9. Tape Tracking Check

A. General

This check verifies tape guide assembly (140, 440, 485, IPL Figure 14) and reel hub assembly (70, 70A, 70B, IPL Figure 14) alignment and proper tracking of the tape through the tape guide system.

B. Procedure

- (1) Mount Tape Transport Assembly in run-in fixture (36, Figure 901).
- (2) Connect Tape Transport Assembly to motor mover (37, Figure 901). Note that the UFDR Tester (1, Figure 901), in conjunction with a UFDR, may be used to run the Tape Transport Assembly. However, this setup requires shop-made cables to connect the UFDR to the Tape Transport Assembly.
- (3) Install a 'scratch' loaded reel (45, 45A, 45B, IPL Figure 14), an empty reel (50, 50A, IPL Figure 14), and a 'scratch' peripheral belt (P-belt) (40, IPL Figure 14) on Tape Transport Assembly as shown in Figure 727.
- (4) Run the tape in 1 direction for approximately 3 minutes, then in the opposite direction for approximately 3 minutes. Verify that the tape does not ride up on the roller flanges and that the edges of the tape are not curled or nonuniform.
- (5) If tape tracking is not satisfactory, loosen screw, rotate tape guide pin in 15\Deg increments, and tighten screw. Repeat step (4) for each increment. Or, remove E-ring, shims, and roller, invert roller, and reinstall roller, shims, and E-ring. Ensure that shims are reinstalled in their proper positions. If requirements cannot be met, replace tape guide assembly.
- (6) When all requirements have been met, apply adhesive to screw threads per the Reel Hub Assembly and Tape Guide Roller Assembly Installation paragraphs in the **ASSEMBLY** section.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

REPAIR

1. **General**

Repair consists of repairing minor surface damage to the Universal Flight Data Recorder (UFDR) and repairing or replacing damaged electrical components on printed wiring boards. Ref. Figure 601 for materials required for repair. Replace all components not meeting the requirements outlined in the **CHECK** or **TESTING AND TROUBLESHOOTING** sections. Remove loose conductive material according to the **CLEANING** section.

NOTE: *Equivalent substitutes may be used for all repair materials.*

ITEM	MANUFACTURER AND SPECIFICATION	USE
1. Solder	Per ANSI/J-STD-006.	Leads and wires.
2. Wire	Electrical, insulated, high temperature AWG 22, 26; per MIL-W-16878/4.	Jumper wire.
3. Wire	Solid electrical; AWG 22, 26; per QQ-W-343, Type S, tinned.	Bus wire.
4. Sleeving	Teflon per AMS3655.	Component leads, jumper wires
5. Fine grit sandpaper or crocus cloth	Commercially available.	Finishing metal surfaces.
6. Paint	International orange enamel polyurethane base per FED-STD-595, color 12197.	Paint for cover and front panel.
7. Conformal coating	PC18 or PC29M manufactured by Hysol Div., Dexter Corp. (V04347), 211 Franklin St., Olean NY 14760 or Conathane 1155 per MIL-I-46058, Type UR.	Printed wiring board coating.
8. Isopropyl alcohol	Commercially available.	Clean mounting surface for reflective tape.
9. Reflective Tape	Part No. 3290, white, one inch wide, manufactured by Minnesota Mining and Mfg. Co., Decorative Products Div., 3M Center, St. Paul, MN 55101 (V34360).	Cover tape.
10. Lubricant	Windsor Lube L245X per MIL-L-6085A.	Bearing Lube.
11. Lacing cord	No. 4 lacing cord per MIL-T-43435.	Cable ties.
12. Solvent (MEK)	Commercially available.	Clean interior, enclosure
13. Liner	600-3961-001, -002, -003, manufactured by Honeywell, Redmond, WA 98052 (V97896).	Enclosure liner.

Repair Materials
Figure 601 (Sheet 1 of 2)

31-30-32

Page 601
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

ITEM	MANUFACTURER AND SPECIFICATION	USE
14. Insulation	600-3962-001, manufactured by Honeywell, Redmond, WA 98052 (V97896).	Enclosure insulation.
15. Primer	SS4004, manufactured by GE Silicones, Hudson River Road, Waterford, NY 12188 (V01139).	Prime environmental enclosure cover interior, liner and insulation.
16. Case resin adhesive	RTV60 with catalyst T-12, manufactured by GE Silicones, Hudson River Road, Waterford, NY 12188 (V01139).	Inject between enclosure liner and insulation.
17. Cast resin adhesive	RTV88 with catalyst T-12, manufactured by GE Silicones, Hudson River Road, Waterford, NY 12188 (V01139).	Bond enclosure insulator and liner to cover.
18. Adhesive	RTV162, manufactured by GE Silicones, Hudson River Road, Waterford, NY 12188 (V01139).	Bond liner to enclosure cover.
19. Insert	MS21209C0420.	Tape recorder mounting holes.
20. Lapping material	Ultralap®, green, 0.5 micron, 2 mil thick, 100 sheet packages, 8.5 by 11 inch sheets, manufactured by Moyco Ind. 2054W Clearfield St., Philadelphia, PA 19132 (V41946)	Head assembly field rework.
21. Lapping material	Ultralap®, gray, 3 micron, 2 mil thick, 100 sheet packages, 8.5 by 11 inch sheets, manufacturer same as 20.	Head assembly and field rework.
22. Distilled water	Commercially available.	Head lapping procedure.
23. Paper Towels	Commercially available; clean and absorbent.	Head lapping procedure.
24. Silicone rubber	RTV60W/12 Cat., manufactured by GE Silicones, Hudson River Road, Waterford, NY 12188 (V01139).	Pot filter assembly.
25. Insulator	Honeywell Part Number 614-0645-001.	Filter assembly.
26. Adhesive	0151, Epoxy, clear, manufacturer same as item 7.	Secure inserts.
27. Adhesive	RTV118, manufacturer same as item 24.	Secure motor label.
28. Lubricant	Texaco Starfak, Low Temp Grease EP (Code 2656), manufactured by Texaco Lubricants Co. (V0TTJ3), Affiliate of Texaco, Inc., 158 Private Dr 8323, Proctorville, OH 45669.	Bearing lubricant.

Repair Materials
Figure 601 (Sheet 2 of 2)

31-30-32

Page 602
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

2. PWB Repair Procedure

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES ARE SUBJECT TO DAMAGE BY EXCESSIVE LEVELS OF VOLTAGE. REFER TO DOD-HDBK-263 FOR PROPER HANDLING PROCEDURES.

- A. Remove and replace components only as necessary to clear faulty operation disclosed by **TESTING AND TROUBLESHOOTING** section procedures.
- B. Prior to unsoldering or disconnecting a component, identify the leads, pins, or other electrical connecting mechanism to ensure correct installation of the replacement component.
- C. If the components on the board are coated with conformal coating, remove the coating per paragraph 3.
- D. Soldered components are secured to board by leads protruding through and soldering to the circuit side. To remove components mounted with bent leads, clip the leads 1/16 inch from the board surface on the component side of the board and unsolder the remaining portions from the circuit side. Push the remaining portion through the circuit side. To remove components with leads going straight through the board, unsolder on circuit side and pull the leads through from the component side.

CAUTION: PREVENT DAMAGE TO THE PWB BY APPLYING HEAT CAREFULLY. USE A HEATSINK IF NECESSARY.

- E. Unsolder the faulty component.

3. Conformal Coating Removal

- A. When removing defective electronic components from a PWB, remove conformal coating only to the extent required to remove component.

WARNING: SAFETY GLASSES MUST BE WORN WHEN CHIPPING THE COATING OFF PWB ASSEMBLIES.

- B. The coating may be removed by careful cutting and scraping or by any of the following procedures:

CAUTION: PREVENT DAMAGE TO THE PWB BY APPLYING HEAT CAREFULLY. USE A HEATSINK IF NECESSARY.

- (1) Soften the coating by oven heating the PWB to 160° F, maximum.
- (2) Remove the coating with a chisel-tip soldering iron.
- (3) Remove the coating in conjunction with the part removed by using a vacuum-solder removal device, unless the surrounding encapsulated area exhibits any evidence of scorching or burning.

4. Installation of Replacement Components

When installing replacement components, observe the following:

- A. Remove old solder as required.
- B. Sleeve (4, Figure 601) component leads and jumpers as required to prevent shorting.
- C. Observe fits and clearances of Figure 801 and paragraph 5.

31-30-32

Page 603
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

D. Make connections mechanically secure before resoldering.

CAUTION: USE A HEATSINK WHEN SOLDERING A SOLID-STATE DEVICE AND MINIMUM HEAT TO PREVENT DAMAGE TO PWB.

E. Solder per ANSI/J-STD-001A and IPC-A-610B.

CAUTION: CLEAN SOLDER JOINTS ONLY WHEN PWB CIRCUITS ARE GROUNDED.

F. Clean all new solder joints with isopropyl alcohol (8, Figure 601).

G. If required, apply conformal coating (7, Figure 601) to all replaced components, new solder joints, and repaired areas per paragraph 6.

5. **Fits and Clearance Requirements**

Observe the following general requirements and the requirements given in the **FITS AND CLEARANCES** section.

A. Components with axial type leads (capacitors, diodes, resistors) that are installed parallel to the board surface shall be mounted 0.032 inch maximum above the board surface.

B. Small semiconductors (integrated circuits, transistor) with bent leads (stress relief) and components dissipating one (1) watt or more shall be mounted 0.100 to 0.200 inch from the board surface.

C. Small semiconductors (integrated circuits, transistors) with straight leads, capacitors with part numbers M39014-01 and M39014-02, trimpots, and metal cased relays shall be mounted 0.040 to 0.080 inch above board surface.

6. **Procedure for Applying Conformal Coating**

CAUTION: APPLY CONFORMAL COATING ONLY WHEN PWB CIRCUITS ARE GROUNDED.

WARNING: AVOID SKIN CONTACT WITH CONFORMAL COATING MATERIAL. IF SKIN CONTACT OCCURS, CLEAN WITH DENATURED ALCOHOL OR WATERLESS CLEANER. WASH THOROUGHLY WITH MILD SOAP AND WATER.

Apply conformal coating (7, Figure 601) to all replaced components, repaired areas, and new solder joints per these steps:

A. Ensure that PWB assembly is clean and dry.

B. Coating may be sprayed or applied with a brush, or assembly may be dipped.

C. Cure coating per manufacturer's instructions.

7. **Reflective Tape Installation**

A. If reflective tape (9, Figure 601) requires replacement, remove old tape with a thin-bladed tool.

B. Clean tape mounting surface with isopropyl alcohol and wipe dry with a clean cotton cloth.

C. Install reflective tape as shown in Figure 602. Remove backing and press firmly into place.

8. **Degaussing Procedure**

A. Remove the tape transport from the bottom cover, remove the tape transport top cover, and place the tape transport on a test bench during degaussing.

31-30-32

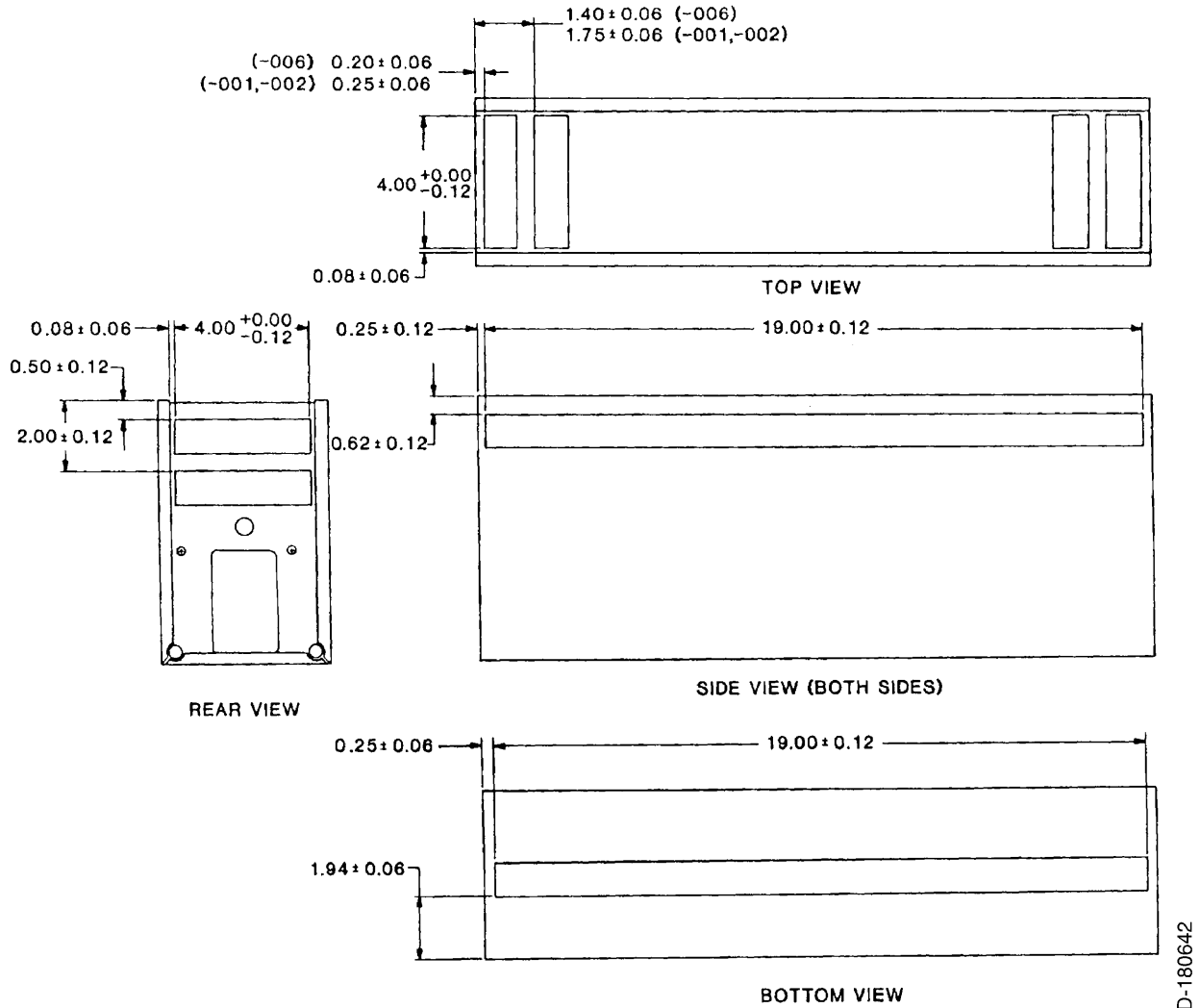
Page 604
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- B. Remove the loaded reel from the baseplate as use of the degausser can cause partial erasure of the tape.
- C. Prior to degaussing, thoroughly clean heads per **CLEANING** section.
- D. Install a short length of plastic tubing or shrink sleeving on the end of the degausser head (12, Figure 901) to prevent scratching the heads during degaussing.



Reflective Tape Locations
Figure 602

CAUTION: NEVER APPLY OR REMOVE CURRENT TO THE DEGAUSSER WITH THE DEGAUSSER NEAR THE MAGNETIC HEADS.

- E. Plug the degausser into an AC outlet.

31-30-32

Page 605
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAUTION: DO NOT USE RAPID MOTION OVER THE HEAD AS THIS WILL CAUSE THE HEAD TO BECOME MAGNETIZED.

- F. Pass the degausser slowly over each head gap area several times to ensure complete degaussing.
- G. After degaussing is complete, slowly remove the degausser from the head area before disconnecting the degausser from the AC outlet.

9. Bearing Information

- A. Bearings that are to be periodically checked and replaced are listed in Figure 603.

NOTE: *It is recommended that improperly operating bearings be replaced with new bearings. If any doubt exists regarding the serviceability of a bearing, replace it.*

NOTE: *Any bearing cleaning or relubricating operation must be performed by qualified personnel in a suitable shop environment.*

BEARING	PART NUMBER	IPL FIGURE-ITEM	LUBRICANT
Tape Reel, Loaded Reel	400-0056-040 (SFR1563PPK58)	10-80, 115	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601).
Drive Belt Roller	400-0056-041 (SFR133PPK58) ³	10-175, 13-30A, 14-325A, 470	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601).
Motor Drive Rollers	400-0056-042 (SFR1333PPK58) ³	13-30, 14-325, 14-470A	
Tape Guide Rollers	400-0056-045 (SFR144PPC317K58) ³	14-170A, 550	
	400-0056-052 (SFR133SSW4) ¹ (S3332FCHH5P35) ² (SSRIF3332ZZHA3P35) ³	13-30B, 14-325B, 14-470B	
	400-0056-053 (SFR133SSW3) ¹ (S3332FCHH5P25) ² (SSRIF3332ZZHA3P25) ³	13-30C, 14-325C, 14-470C	
	400-0172-001 (SFR133SSW0113) ¹ (S3332FCHH7P25L01) ² (SFR133PPK25L01) ³	13-30D, 14-325D, 14-470D	

Bearing Lubrication Information
Figure 603 (Sheet 1 of 2)

31-30-32

Page 606
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

BEARING	PART NUMBER	IPL FIGURE-ITEM	LUBRICANT
Tape Drive Pulley	400-0057-027 (SR1665PPK25) ³	10-210, 14-215	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601).
Tape Transport Pulley	400-0057-028 (SR43PPK15) ³	10-230, 14-235	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601).
Stepper Motor	SR1883PPK15-7 ³	13-112	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601).
Reel Hub Assembly	400-0058-003 (SFRW156SS25) ¹ (SFR1565PPEEK25) ³	14-100	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601).
Tape Guide Rollers; Drive Belt Rollers	400-0056-060, -061, -062, -063, -064, -065	14-326,471,326A, 471A,170B,550A, 326B,471B,326C, 471C,326D,471D, 13-31,A,B,C,D	Texaco Starfak, grease plated, no fill (28, Figure 601).
Tape Drive Pulley	400-0057-030,-031	14-215A,235A	Texaco Starfak, 1/3 fill (28, Figure 601).
Stepper Motor	400-0057-032,-033	13-143,118,165	Texaco Starfak, 1/3 fill (28, Figure 601).
Reel Hub Assembly	400-0058-005 (SSRIF5632ZZEEHA5C0XP58LG54GPLU) ³	14-77A	Texaco Starfak, grease plated, no fill (28, Figure 601).
Reel Hub Bearing Assembly	400-0058-004 (SFRW156SS58C0X10-11) ¹ (S5632FCHHE5P58L1Z01) ² (SSRIF5632ZZEEHA5C0X1P58L01) ³	14-77	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601)
Reel Hub Assembly	400-0058-003 (SFRW156SS25) ¹ (SFR1565PPEEK25) ³	14-100	Windsorlube L245X, 1 drop with #27 needle (10, Figure 601)

1 Barden Corp. (V70854), Precision Bearing Div., 200 Park Ave., Danbury, CT 06810-7505.

2 MPB Corp. (V40920), Miniature Precision Bearings Div., Precision Pk., P.O. Box 547, Keene, NH 03431.

3 New Hampshire Ball Bearings (V83086), Rt. 202, Peterborough, NH 03458.

Bearing Lubrication Information

Figure 603 (Sheet 2 of 2)

31-30-32

Page 607
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- B. Bearings and bearing lubrication in UFDR applications are considered critical to reliable Tape Transport Assembly operation. Bearing part numbers provided in the IPL ensure the correct lubrication and fill requirements for these applications.
- C. Manufacturer's part numbers and lubrication information are given in Figure 603 for those operators with the facilities and personnel for bearing repair. When ordering bearings from the manufacturer, be certain to specify the lubricant type and fill as well as the part number. If this is not done, the bearings will be lubricated at the vendor's option with a lubricant type and fill not suitable for use in UFDR applications.

10. Bearing Preload Information

- A. Bearing preloads specified for the Tape Transport Assembly must be maintained to ensure minimum axial and radial play and vibration effects, and to realize optimum bearing life under all environmental conditions. Maintaining proper bearing preloads also ensures optimum intervals between repairs.
- B. Proper bearing preloads are critical to tape recording quality.
 - (1) Insufficient preload causes bearings to be loose in operation and produce excess jitter and mistracking in a tape recording, thereby reducing recording quality and shortening bearing life.
 - (2) Excess preload causes bearings to wear out prematurely.

11. Bearing Handling Information

- A. If any doubt exists regarding the serviceability of a bearing, replace it with a new one. Use only bearings specified in the IPL and ensure that bearings are lubricated as specified in Figure 603.
- B. To ensure that bearings achieve their design life, and that they perform without objectionable noise, temperature rise, or shaft excursions, the following precautions are recommended:
 - (1) Handle bearings with care, keeping them in the sealed original container until ready for use.
 - (2) Work with clean dry hands and tools in clean surroundings.
 - (3) Do not wipe or wash bearings prior to or after installation.
 - (4) Do not use wooden mallets, brittle or chipped tools, or dirty fixtures and tools when mounting bearings.
 - (5) Do not spin any bearing with an air blast.
 - (6) Use care not to scratch or nick bearings.
 - (7) Do not strike or press on race flanges.
 - (8) Install bearings with tools that provide uniform steady pressure, rather than hammering on a drift or sleeve.
 - (9) Ensure that bearings are evenly started on shafts or in bores.
 - (10) Check shafts and bores before installing bearings to ensure that proper fits will be maintained.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (11) When installed bearings on shafts, never strike the outer race or press on it to force the inner race. Apply pressure on inner race only.
- (12) When installing bearings in bores, never strike the inner race or press on it to force the outer race. Apply pressure on outer race only.

NOTE: Follow the precautions given in steps (11) and (12) when removing bearings.

- (13) Do not press, strike, or otherwise force the seal or shield on factory-sealed bearings.

12. **Finished Metallic Surfaces**

- A. Polish out minor scoring or scratches with fine grit sandpaper or crocus cloth (5, Figure 601).
- B. Clean all polished areas with clean cloth dipped in isopropyl alcohol (8, Figure 601). Wipe dry with a clean lint-free cloth.
- C. Finish with paint (6, Figure 601).

13. **Thermal Enclosure Repair**

A. Determining Extent of Delamination

Two types of delamination can occur. The extent of delamination must be determined to select the correct repair procedure.

- (1) If delamination is in the top cover, remove the gasket from the top cover assembly. If delamination is in the bottom cover, remove tape transport, drive motor, and harness shield from the bottom cover per the **DISASSEMBLY** section.
- (2) Examine the delaminated area to determine repair procedure.
 - (a) If liner is separated from the edge of the cover and the insulation, use the repair procedure of paragraph B.
 - (b) If liner is damaged in any way, chipped, cracked, or broken, use the repair procedure of paragraph B.
 - (c) If liner is still secure around the cover edge, but the liner is separated from the insulation, use the repair procedure of paragraph C.

B. Repair Procedure, Liner and Insulator Replacement

WARNING: THE INSULATION MATERIAL CONTAINS A VERY TOXIC CHEMICAL THAT CAN CAUSE BLISTERS IF ALLOWED TO CONTACT SKIN. TAKE EXTREME PRECAUTIONS TO CONTAIN THE DUST WHEN REMOVING THE INSULATION FROM THE COVERS. SOAKING THE INSULATION IN WATER FOR APPROXIMATELY TWO HOURS BEFORE REMOVAL AND HOLDING UNDER RUNNING WATER DURING THE PROCEDURE WILL PREVENT THE POWDER MATERIAL FROM BLOWING AROUND THE ROOM.

WARNING: PERFORM OPERATIONS INVOLVING INSULATION REMOVAL IN A WELL VENTILATED AREA TO AVOID BREATHING INSULATION DUST LINGERING IN THE AIR: WEAR A SUITABLE FACE MASK WHEN NECESSARY. USE GOGGLES, GLOVES, AND APRON TO PREVENT IRRITATION FROM PROLONGED CONTACT WITH INSULATION DUST WHILE CHISELING, SAND BLASTING, OR SANDING THE INSULATION AND ADHESIVES. CHANGE CLOTHING THAT HAS BECOME SOILED WITH

31-30-32

Page 609
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

DUST. CAREFULLY COLLECT ALL RESIDUE AND PROPERLY DISPOSE OF IT TO PROTECT OTHERS WHO MAY HAVE TO WORK NEARBY OR IN THE SAME AREA AFTER THE PROCEDURE HAS BEEN COMPLETED.

- (1) Use a chisel (19, Figure 901) to remove liner and insulator.
- (2) Use sand blast and/or 60 grit crocus cloth (5, Figure 601) to remove adhesive.

WARNING: METHYL ETHYL KETONE AND TOLUENE ARE TOXIC AND FLAMMABLE. DO NOT BREATHE VAPORS. USE IN A WELL VENTILATED AREA FREE FROM SPARKS, FLAME, OR HOT SURFACES. WEAR SPLASH GOGGLES, SOLVENT-RESISTANT GLOVES, AND OTHER PROTECTIVE GEAR. IN CASE OF EYE CONTACT, FLUSH WITH WATER AND PROMPTLY SEEK MEDICAL ATTENTION. IN CASE OF SKIN CONTACT, WASH WITH SOAP AND WATER.

- (3) Thoroughly clean interior of cover with solvent (12, Figure 601).

WARNING: DEGREASER SOLVENT TEMPERATURE IS SUFFICIENT TO ACTIVATE INTUMESCENT PAINT ON EXTERIOR SURFACE OF COVER. DO NOT SUBMERGE COVER IN DEGREASER SOLVENT OR SPRAY EXTERIOR SURFACE.

- (4) Spray cover interior with degreaser wand to remove MEK residue.

CAUTION: THE INSULATION MATERIAL IS VERY BRITTLE AND FRAGILE. USE EXTREME CAUTION WHEN HANDLING THE INSULATOR TO AVOID BREAKING AND CHIPPING THE CORNERS. IF CHIPPED OR BROKEN, THE NEW INSULATOR MUST BE REPLACED; A COSTLY PROPOSITION.

WARNING: PERFORM OPERATIONS INVOLVING PRIMER UNDER A VENTILATED HOOD. AVOID BREATHING DUST THAT MAY RESULT FROM HANDLING THE INSULATOR DURING THE PRIMING PROCESS: WEAR A SUITABLE FACE MASK. USE GOGGLES, GLOVES, AND APRON TO PREVENT IRRITATION FROM PROLONGED CONTACT. CHANGE CLOTHING THAT HAS BECOME SATURATED WITH PRIMER.

- (5) Prime (15, Figure 601) all bonding surfaces of the cover, insulator, and liner.
- (6) Cure the primer at room temperature for at least one hour.
- (7) Apply adhesive (17, Figure 601) to convex surfaces of the insulator and to the primed inside surface of the cover within 48 hours after applying the primer.
- (8) Position insulator firmly into cover to eliminate air gaps.
- (9) Cure the adhesive for two hours at 200° F.
- (10) Allow the cover to cool to room temperature before proceeding.
- (11) Apply adhesive (17, Figure 601) to primed surfaces of the insulator and liner within 48 hours after applying the primer.

NOTE: *Apply a sufficient quantity of adhesive to allow the adhesive to bleed through the four vent holes drilled in the liner, when cover is mounted in the locating jig.*

Honeywell

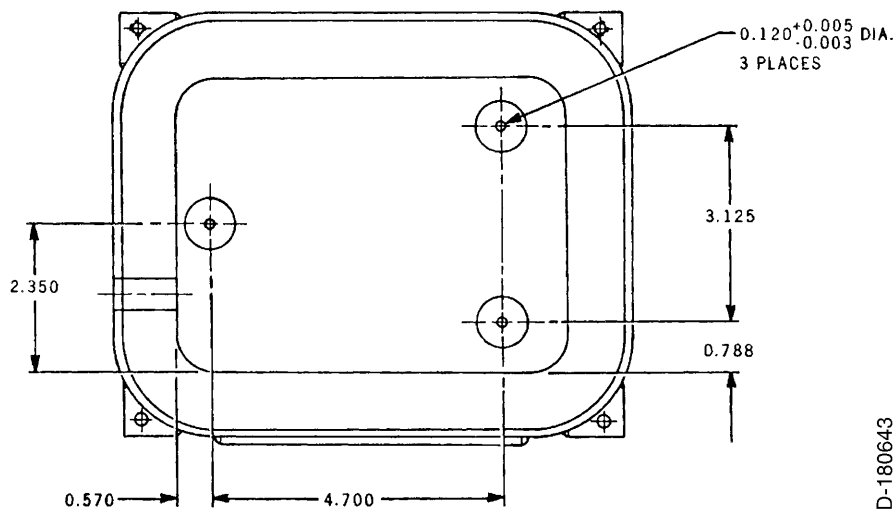
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (12) Position the liner on the cover assembly and seat the assembly in the locating jig (20, Figure 901). The locating jig will apply sufficient pressure to properly locate the liner in the cover.
- (13) Examine vent holes to verify that adhesive has bled through all four holes. This assures bonding of three pads to insulator.
- (14) Cure the cover assembly for two hours at 200° F.
- (15) Allow the cover assembly to cool to room temperature before removing it from the fixture.
- (16) Apply adhesive (18, Figure 601) between the liner and the edge of the cover around the entire circumference of the liner.
- (17) Cure the adhesive for 48 hours at room temperature.

CAUTION: DO NOT DRILL THROUGH THE FIBERGLASS LINER UNDER THE PHENOLIC PLUGS.

- (18) On the bottom cover only, drill (22, Figure 901) holes, 0.325 inch deep, in liner phenolic plugs, located as shown in Figure 604.
- (19) Countersink (23, Figure 901) the above holes 120° by 0.130 to 0.160 inch diameter.
- (20) Tap (24, Figure 901) the above holes 0.250 inch deep. I
- (21) Install screwlock insert (19, Figure 601) into each of the above holes flush or below surface of phenolic plug. Secure insert with adhesive (26, Figure 601).



Tape Transport Mounting Hole Pattern
Figure 604

C. Repair Procedure, Liner Separation From Insulator

- (1) Carefully drill (21, Figure 901) holes through the liner in four corners as shown in Figure 605.

31-30-32

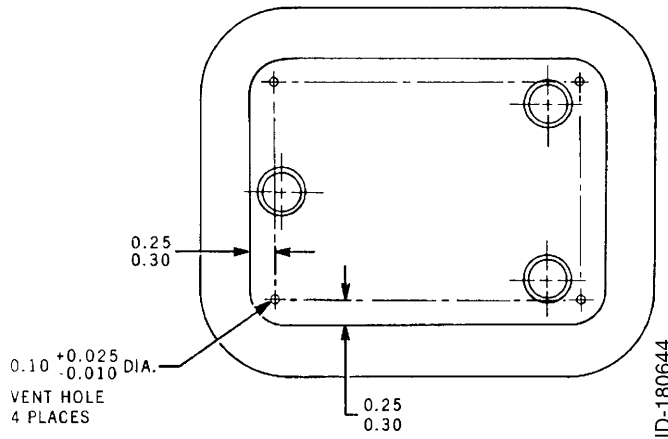
Page 611
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Liner Hole Pattern
Figure 605

- (2) Depending upon the extent of delamination, drill (21, Figure 901) one or more holes through the liner, located over the delaminated area.

WARNING: PERFORM OPERATIONS INVOLVING PRIMER UNDER A VENTILATED HOOD. AVOID BREATHING PRIMER VAPOR: WEAR A SUITABLE FACE MASK WHEN NECESSARY. USE GOGGLES, GLOVES AND APRON TO PREVENT IRRITATION FROM PROLONGED CONTACT. CHANGE CLOTHING THAT HAS BECOME SATURATED WITH PRIMER.

- (3) Using a syringe or air powered nozzle, inject primer (15, Figure 601) into each hole, including the four corner holes.
- (4) Using a syringe or air powered nozzle, inject adhesive (16, Figure 601) into each hole drilled over the delaminated area. Inject sufficient adhesive to cause adhesive to appear at one or more of the four corner holes.
- (5) Clamp the cover and liner in the locating jig (20, Figure 901) so liner is in correct position.
- (6) Cure adhesive for two hours at 200° F.
- (7) Allow cover to cool to room temperature before removing from locating jig.
- (8) Remove excess adhesive after cooling to room temperature.

14. Field Rework of UFDR Head Assemblies

This section describes the procedure for refinishing the pole tip surface of UFDR magnetic head assemblies that are worn as a result of normal tape-to-head contact. UFDR head assemblies covered by this procedure are listed below:

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

NOMENCLATURE	PART NUMBER
Read/Write Head	754-0011-00X
Read/Write Head	317-1044-00X
Erase Head	317-1039-00X
Erase Head	317-1045-00X

NOTE: *UFDR head assembly field rework shall be performed only by qualified personnel in a certified repair depot.*

A. Inspection Procedure

- (1) Using a DMM (4, Figure 901), check the coils for opens. If any coils are found open, the head shall be discarded.

NOTE: *Any time a DC ohmmeter is used, the heads must be degaussed per paragraph 8.*

- (2) Optically check the pole tip area of the magnetic head with a 30-power microscope (30, Figure 901) for serious cracks or nicks.
- (3) Measure the starting pole tip depth of the head assembly while it is installed on its mounting surface, using 30-power magnification. Record the value. If value is less than 0.005 inch, replace the head. Refer to Figure 606, Section A.

NOTE: *It may be necessary to polish the end of the head with lapping material (21, Figure 601) to get a clear view of the pole tip depth area.*

- (4) Wear Depth Measurement

This measurement is to determine if there is sufficient tip depth left to be able to relap the head to a usable condition.

- (a) Loosen the head attaching screws and move the head from its mounting position. Install head holding fixture (29, Figure 901) at the head position and secure to baseplate. Install head, face up, on the fixture. Refer to Figure 607.
- (b) Position the microscope (30, Figure 901) directly over the head gap line and establish reference readings by focusing on the unworn surfaces adjacent to each side of the tape path outside the machined relief cuts (where applicable). Zero the microscope at the lowest to the two readings. Refer to Figure 606, Section B.
- (c) Adjust the microscope to measure wear depth in the tape wear area. Take several readings in order to find the worst case, generally found just inside the edges of the tape wear area. Add 0.001 inch to the worst case reading and record the results.
- (d) Subtract the worst case reading (+0.001 inch) of step (c) from the starting pole depth measurement of paragraph (3). The resulting figure represents the new tip depth if a rework/relap were to be accomplished, and must be equal to or greater than 0.0040 inch (0.1016 mm). If the figure is less than 0.0040 inch, the magnetic head is worn beyond acceptable rework limits and shall be discarded.

31-30-32

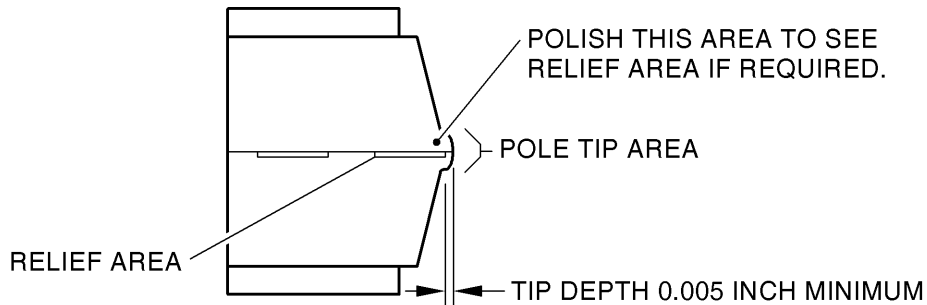
Page 613
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

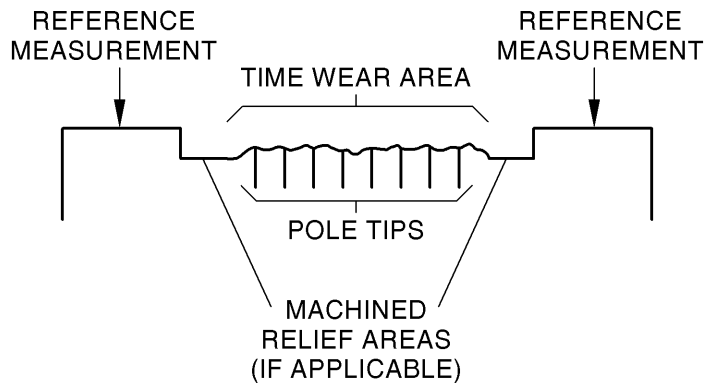
Honeywell

COMPONENT MAINTENANCE MANUAL

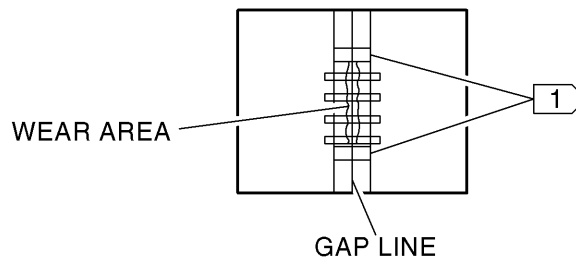
Universal Flight Data Recorder / Part No. 980-4100/980-4101



A. MEASURE STARTING POLE DIP DEPTH 0.005 INCH MINIMUM.



- B. 1. ESTABLISH REFERENCE AT LOWEST SIDE.
2. MEASURE WORST CASE READING IN TAPE WEAR AREA. ADD 0.001 TO THIS READING.
3. SUBTRACT WORST CASE (+0.001) READING FROM A STARTING POLE TIP DEPTH. MUST BE 0.0040 INCH OR GREATER FOR RELAP.



NOTES:

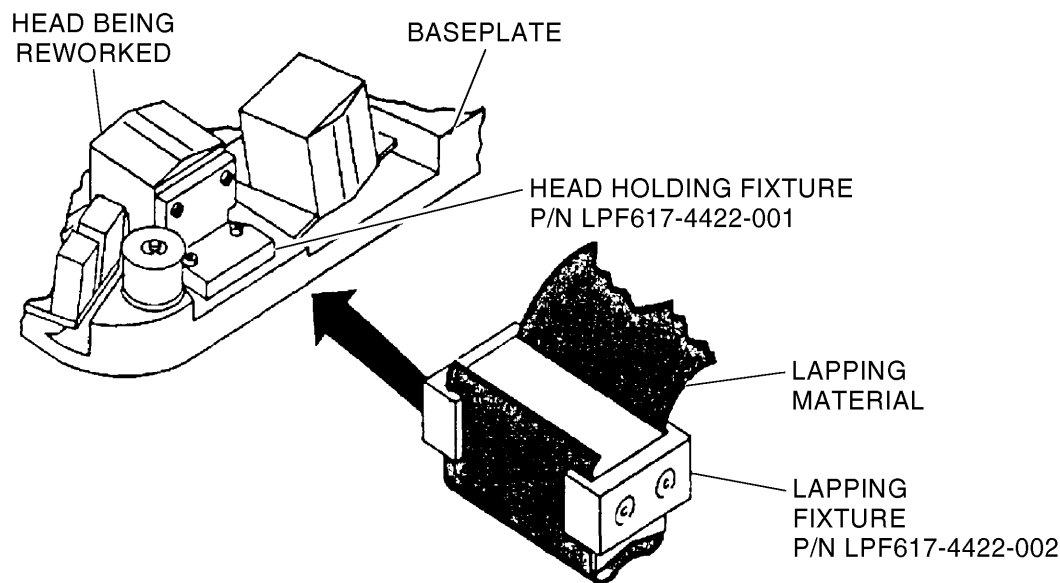
- 1 2 relief cuts are shown. However, the location and quantity of cut(s) will vary (from 2 to none), depending on the head part number.

ID-180645

Head Assembly Measurement
Figure 606

31-30-32

Page 614
22 Dec 2006



Magnetic Head Pole Tip Lapping Setup
Figure 607

B. Magnetic Head Assembly Relap Procedure

- (1) Clean all dust or residue from the lapping fixture (28, Figure 901). Feed both ends of the 3-micron lapping material through slots on each side of the lapping tool. Apply isopropyl alcohol (8, Figure 601) to the abrasive side of the lapping material surface where the lapping will be done.

CAUTION: PROTECT THE BASEPLATE ASSEMBLY FROM MOISTURE AND CONTAMINATION DURING THE LAPPING PROCEDURE BY CUTTING A HOLE IN A CLOTH OR PAPER TOWEL AND PLACING IT OVER THE BASEPLATE, EXPOSING ONLY THE MAGNETIC HEAD BEING LAPPED.

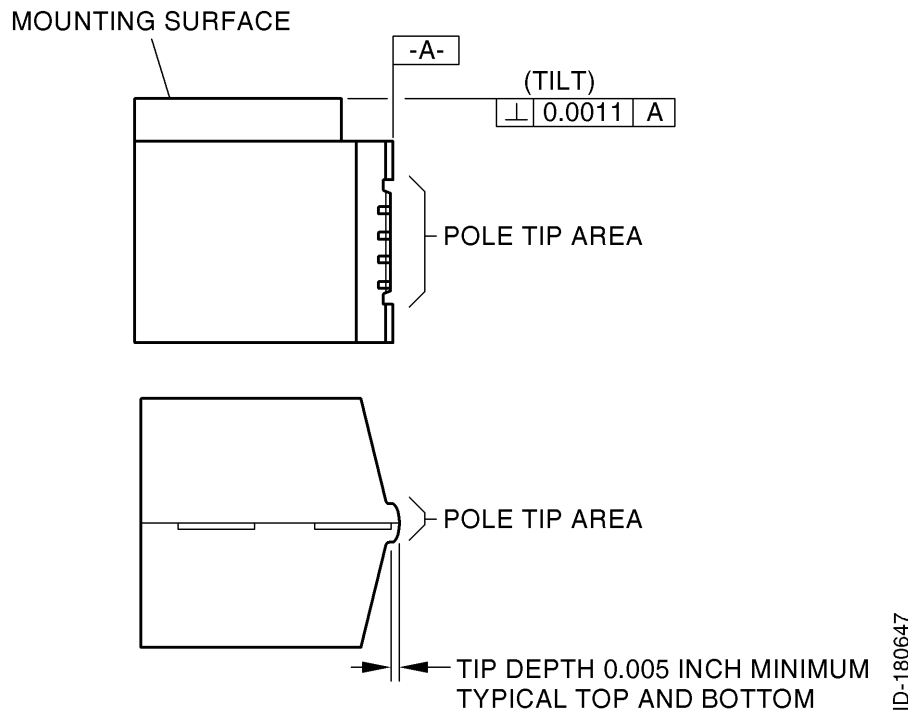
- (2) Grip the lapping tool assembly with one hand, while holding the lapping material firmly in the tool. With a constant pressure (approximately 3 pounds, 1.5 KG), push and pull the lapping tool across the magnetic head assembly, perpendicularly to the direction of tape motion (see Figure 607). The polishing passes must be made equal in both directions and should not exceed 10 passes in any one area of the lapping material
- (3) After approximately 20 passes, the head assembly shall be inspected under the 30-power microscope to determine when all the tape wear defects have been removed. Also, measure tip depth so that corrections can be made to maintain uniform tip depth throughout the relapping operation. This procedure will help maintain head tilt within specification. Correct tip depth by applying less pressure to the side with the least tip depth during the lapping operation. When the major surface wear areas have been removed, proceed to paragraph (4).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (4) Repeat paragraphs (1) through (3), except use the 0.5 micron lapping material (20, Figure 601) to apply the final finish of less than 4 microinches (1 micron).
- (5) Clean the head using a cotton swab and Simichrome polish (Ref. Figure 401). Remove the polish residue with isopropyl alcohol (8, Figure 601).
- (6) Check the magnetic head for good gap definition under the 30-power microscope. The gap shall appear straight and clean.
- (7) Check the head's final tip depth and tilt. All measurements shall be within the limits specified in Figure 608.



Tip Depth and Tilt Final Measurements
Figure 608

C. Supplemental Lapping Procedure

CAUTION: THIS PROCEDURE REMOVES MATERIAL FROM THE POLE TIP SURFACE MORE RAPIDLY THAN THE PROCEDURE OF PARAGRAPH 14B. IT IS RECOMMENDED THAT THIS PROCEDURE BE PERFORMED ONLY BY AN EXPERIENCED TECHNICIAN.

- (1) Perform paragraphs 14A(1) through (4)(d).
- (2) Wrap 3-micron lapping material (21, Figure 601) around a flat, smooth metal block (approximate dimensions of 1 x 1 X 2 inches). Tape the lapping material in place. Observe caution of paragraph 14B(1).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (3) Firmly grip the block and place it against the head with the 2-inch length of the block parallel with the pole tips (perpendicular to the tape path). With a constant pressure (approximately 3 pounds), push and pull the block across the magnetic head parallel to the direction of tape motion. The polishing passes must be made equal in both directions. The polishing passes must be made equal in both directions. Replace the lapping material as required to provide equal cutting action across the entire surface of the head (approximately every 10 passes).
- (4) Perform paragraphs 14B(3) through (7).

15. Filter Assembly Repair Procedure

A. Test Equipment Required

The following list of test equipment is required to test the repaired filter assembly:

- (1) High Potential Tester (32, Figure 901).
- (2) Function Generator (3, Figure 901).
- (3) Oscilloscope (2, Figure 901).
- (4) DMM (4, Figure 901).
- (5) Stopwatch (33, Figure 901).
- (6) Variac (34, Figure 901).
- (7) Test Fixture (Figure 609).

B. Repair Procedure

- (1) Remove filter assembly from UFDR per **DISASSEMBLY** section.
- (2) If required to isolate the faulty component(s), perform the test procedure given in paragraph 15C.
- (3) Remove the faulty component(s).
 - (a) Unsolder the faulty component(s) per paragraph 4.
 - (b) Remove potting compound as required.

NOTE: *Do not disturb the potting compound unless it is necessary to remove a faulty component. If it is necessary to remove any of the potting compound, remove all of it and report the filter assembly upon completion of reinstalling replacement component(s).*

- (c) Install replacement component(s) using the applicable techniques given in paragraph 2 of the **ASSEMBLY** section.
- (d) If potting material was removed, install new insulator (25, Figure 601) and repot to a depth of 0.6 ± 0.06 inch using silicone rubber (24, Figure 601). Repot per the procedure specified by the manufacturer of the potting material.

31-30-32

Page 617
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

C. Test Procedure

(1) Verify the following resistances before performing other tests:

CAUTION: HIGH VOLTAGE IS PRESENT DURING THIS TEST. OBSERVE ALL SAFETY PRECAUTIONS TO ENSURE OPERATOR AND EQUIPMENT SAFETY. DO NOT TOUCH THE FILTER BOX OR TEST FIXTURE SPRING CONTACTS WHEN S1 IS IN THE ON POSITION.

CAUTION: IF LAMP, DS1, ON THE TEST FIXTURE GOES ON AT ANY TIME DURING THE TEST, IMMEDIATELY MOVE S1 TO OFF AND DISCONNECT THE FILTER ASSEMBLY FROM THE TEST FIXTURE. THEN, CHECK FOR 51 OHMS $\pm$ 1% BETWEEN FL3 AND GND ON THE TEST FIXTURE. ALSO, CHECK AND/OR REPLACE TEST FIXTURE FUSE, F1.

FROM	TO	RESISTANCE
FL1 Terminal	Case	>20 Megohms
FL2 Terminal	Case	>20 Megohms
FL1 Terminal	FL2	>20 Megohms
FL1 Terminal	FL3	<1.0 ohm
FL2 Terminal	FL4	<1.0 ohm
FL1 Terminal	FL5	<0.5 ohm
FL2 Terminal	FL6	<0.5 ohm

(2) Initial setup as follows:

- Connect the filter assembly to be tested to the test fixture as shown in Figure 609.
- On the test fixture, set the switches to the INITIAL SETTING position given in Figure 610.
- Connect the (+) terminal of the function generator to GEN HI on the test fixture and the (-) terminal to GND.
- Using the DMM set to indicate ohms, verify that the resistance between FL4 and GND on test fixture is 51 ohms $\pm$ 1%.

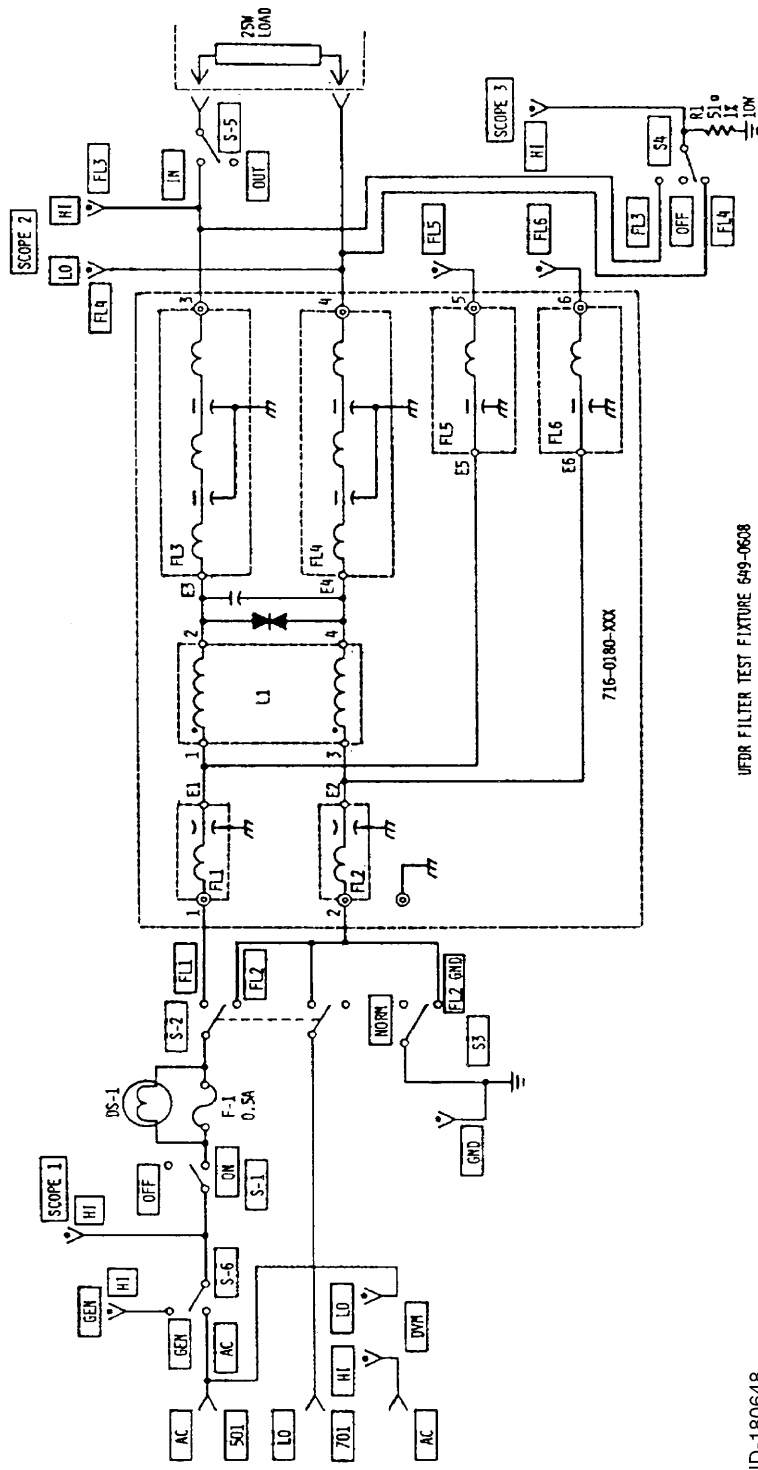
(3) Test of Filter Attenuation at 1MHz as follows:

- Set the function generator to produce a sine wave at a frequency of 1MHz $\pm$ 1%.
- Connect the oscilloscope channel 1 (+) test lead to SCOPE 1 HI and the (-) test lead to GND.
- Adjust the function generator for a signal amplitude of 0.5 Vp-p indication on scope.
- Connect the oscilloscope channel 2 (+) test lead to SCOPE 2 FL2 and the (-) test lead to GND.
- Check for a channel 2 scope indication of a peak-to-peak signal $\leq$ 0.1 of the channel 1 signal.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Filter Assembly Test Setup
Figure 609

31-30-32

Page 619
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

SWITCH	INITIAL SETTING	1MHZ TEST 1	1MHZ TEST 2	1MHZ TEST 3	AC CURRENT	FILTER PHASE TEST
S1	OFF	ON	ON	OFF	OFF	OFF
S2	FL1 OR X	FL1	FL2	FL2	FL1	FL1
S3	NORM	NORM	NORM	NORM	GND	FL2
S4	FL4	FL3	FL4	OFF	OFF	OFF
S5	OUT	OUT	OUT	OUT	OUT	IN
S6	GEN OR X	GEN	GEN	GEN	AC	AC

X = DON'T CARE (ANY POSITION)

Test Fixture Test Switch Positions
Figure 610

- (f) Set the test fixture switches to the position shown in Figure 610 for the 1MHz TEST 2.
- (g) Move scope channel 2 (+) test lead to SCOPE 2 LO FL4.
- (h) Check for a channel 2 scope indication of a peak-to-peak signal ≤ 0.1 of the channel 1 signal.
- (i) Set the test fixture switches to the position shown in Figure 610 for the 1MHz TEST 3.
- (j) Disconnect the filter assembly and the function generator from the test fixture.
- (4) Insulation Resistance Test as follows:
 - (a) Connect the (+) test lead from the High Potential Tester (HPT) to the FL1 terminal of the filter assembly and the (-) test lead to the case of the filter assembly.
 - (b) Do the following simultaneously:
 - Apply 300VDC from the HPT.
 - Measure an elapsed time of 45 to 60 seconds.
 - Check for a minimum DC resistance of 100 megohms.
 - Check for a leakage current of less than 3 microamps.
 - (c) Discharge FL1 by pressing the DISCHARGE switch on the HPT.
 - (d) Reverse the test leads of the HPT; i.e., connect (+) lead to case and (-) lead to FL1.
 - (e) Repeat step (4)(b).
 - (f) Discharge FL1 by pressing the DISCHARGE switch on the HPT.
 - (g) Connect the (+) test lead from the HPT to the FL2 terminal of the filter assembly and the (-) test lead to the case of the filter assembly.

31-30-32

Page 620
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (h) Do the following simultaneously:
 - Apply 300VDC from the HPT.
 - Measure an elapsed time of 45 to 60 seconds.
 - Check for a minimum DC resistance of 100 megohms.
 - Check for a leakage current of less than 3 microamps.
- (i) Discharge FL2 by pressing the DISCHARGE switch on the HPT.
- (j) Reverse the test leads of the HPT; i.e., connect (+) lead to case and (-) lead to FL2.
- (k) Repeat step (4)(h).
- (l) Discharge FL2 by pressing the DISCHARGE switch on the HPT.
- (5) Overvoltage Test as follows:
 - (a) Connect the filter assembly to the test fixture.
 - (b) Connect the (+) output test lead of the variac to AC and the (-) output test lead to LO.
 - (c) Connect the (+) and (-) input test leads of the variac to a 400 Hz, 115 VAC power source.
 - (d) On the test fixture, set the switches to the AC CURRENT positions given in Figure 610.
 - (e) Set the variac to produce an output voltage of 115 VAC.
 - (f) Set the DMM to indicate 400 milliamps, then connect the DMM (+) lead to DMM HI on the test fixture and the (-) lead to DMM LO.
 - (g) Do the following simultaneously:
 - Set S1 to ON.
 - Measure an elapsed time of 60 ± 5 seconds.
 - Check that the current indicated by the DMM does not exceed 400 milliamps.
 - (h) Set S1 to OFF.
 - (i) Adjust the variac to produce 150 ± 1.0 VAC.
 - (j) Do the following simultaneously:
 - Set S1 to ON.
 - Measure an elapsed time of 10 ± 1.0 seconds.
 - (k) Set S1 to OFF.
 - (l) Perform the resistance test of paragraph 15C(1).
- (6) Filter Phase Test as follows:
 - (a) Connect a 25 watt load (500-ohms, 50 watt resistor or 25 watt light bulb) to the load terminals of the test fixture.
 - (b) Set the test fixture switches to the position shown in Figure 610 for FILTER PHASE TEST 2.

31-30-32

Page 621
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (c) Connect oscilloscope channel 1 (+) lead to FL3 HI, ground lead to FL4, and channel 2 (+) lead to SCOPE HI.
- (d) Set S1 to ON.
- (e) Check that the output waveform at channel 2 is in phase with the channel 1 input waveform and within 5.0 VACp-p of the channel 1 waveform.
- (f) Set S1 to OFF.
- (7) End of test.

Disconnect the filter assembly from the test fixture.

16. Tape Reel Loading Procedure

A. Equipment Requirements

- (1) Tape Winder, FAC-826 or FAC-1207, manufactured by Honeywell.

NOTE: Drawings for local fabrication are available upon request from Honeywell.

- (1) Ionizing Air Blower.

B. Material Requirements

- (1) Solvent, Acetone or Methyl Ethyl Ketone (MEK), Technical Grade, commercially available.
- (2) Isopropyl Alcohol, commercially available.
- (3) Finger Cots, commercially available.
- (4) Cotton Swabs, commercially available.
- (5) Lintless Cleaning Wipes, commercially available.
- (6) Permanent Black Marking Material, commercially available.
- (7) Mylar Splicing Tape, Part No. P-256 1/2" (516-6003-002), manufactured by Minnesota Mining and Manufacturing Co. (V26066), Industrial Tape Div., 3M Center, St. Paul, MN 55101-1428.
- (8) Adhesive (27, Figure 701).
- (9) Tape Reel, Part No. 600-4507-001 or 600-5128-001.
- (10) Magnetic Tape (Mylar), Part No. 8206-1/4-3600r183 (516-6074-001), Magnetic Tape (Kapton), Part No. 465A (516-6040-001), 600-5509-002 (without windows), and 600-5509-006 (with windows) are no longer available. When existing stock is depleted, buy Part No. 820617RGY (516-6074-003), manufactured by Quantegy, Inc. (V1S165), 2330 Marvyn Parkway, P.O. Box 190, Opelika, AL 36801-0190. This tape is available from Honeywell on a loaded reel, Part No. 617-4262-006.

Magnetic Tape (Mylar), Part No. 797-15DW11 (516-6055-001), used in loaded reel, Part No. 617-4262-004 is optional to Part No. 617-4262-006.

Magnetic Tape (Mylar), Part No. 705-173111 (516-6074-002) used in loaded reel, Part No. 617-4262-005, is no longer allowed for installation in the UFDR.

Honeywell

COMPONENT MAINTENANCE MANUAL

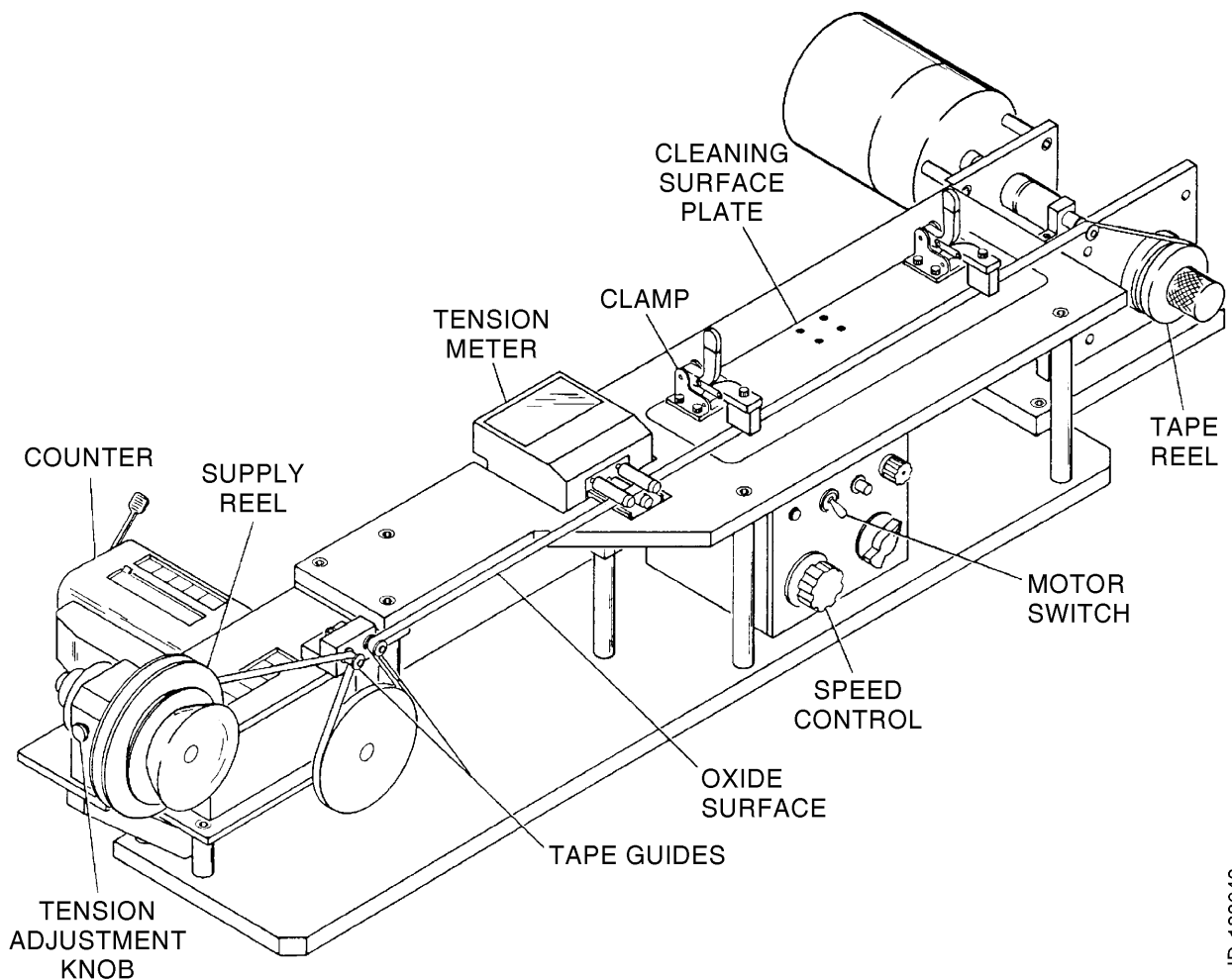
Universal Flight Data Recorder / Part No. 980-4100/980-4101

C. Tape Winder Preparation

- (1) Clean work area and tape winder of all dust and contamination using isopropyl alcohol and lintless cleaning wipes.
- (2) Use clean finger cots. Never touch tape with bare hands or contaminated finger cots.

D. Tape Loading Procedure

- (1) Set the counter at the left rear of the tape winder to 388 by opening the cover (push it left) and turning the thumbwheels.
- (2) Load the supply reel onto the dereeler drag shaft at the left side of the tape winder. Ensure that the tape unwinds from the top of the reel with the oxide surface (shiny side) facing down as shown in Figure 611.



Tape Winder
Figure 611

ID-180649

31-30-32

Page 623
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

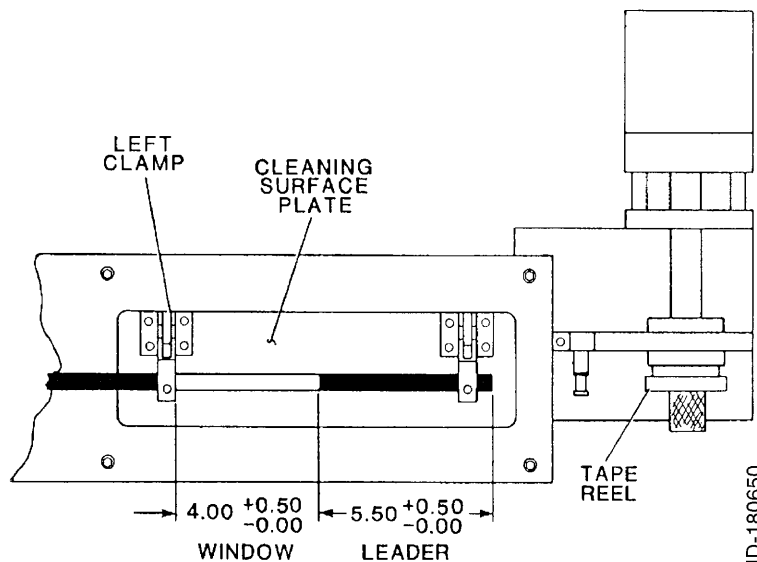
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (3) Thread the tape through the footage counter, tape guides, and tension meter rollers. Ensure that the supply reel is in alignment with the tape guides.
- (4) Turn the tension adjustment knob (at left of supply reel) to obtain a dynamic tension reading on the meter of 4 ± 1 ounces when pulling the tape through by hand.
- (5) Clamp the tape at each side of the cleaning surface plate. For mylar tape, extend the tape $9.50 +1.00, -0.00$ inches past the right edge of the left clamp as shown in Figure 612. For kapton tape, extend the tape 19.00 ± 0.25 inches past the left edge of the right clamp as shown in Figure 613.

WARNING: ACETONE AND METHYL ETHYL KETONE ARE TOXIC AND FLAMMABLE. DO NOT BREATHE VAPORS. USE IN A WELL-VENTILATED AREA FREE FROM SPARKS, FLAME, AND HOT SURFACES. WEAR SPLASH GOGGLES, SOLVENT-RESISTANT GLOVES, AND OTHER PROTECTIVE GEAR. IN CASE OF EYE CONTACT, FLUSH WITH WATER AND SEEK MEDICAL ATTENTION. IN CASE OF SKIN CONTACT, WASH WITH SOAP AND WATER.

- (6) For mylar tape, remove the backing with acetone or MEK for a distance of $4.00 +0.50, -0.00$ inches (from right edge of left clamp as shown in Figure 612. Alcohol may be used instead of MEK or acetone. Use of alcohol eliminates risk of solvent residue interfering with tape recording. For kapton tape, remove the backing for a distance of 8.00 ± 0.25 inches (from left edge of right clamp to right edge of left clamp) as shown in Figure 613. Be careful not to splash solvent beyond the defined window area. Carefully remove all backing within the window area without damage to the tape base, leaving a clearly-defined line between the window and the recording area. Damage of any kind to the tape and wrinkles are unacceptable.



BOT/EOT Windows, Mylar Tape
Figure 612

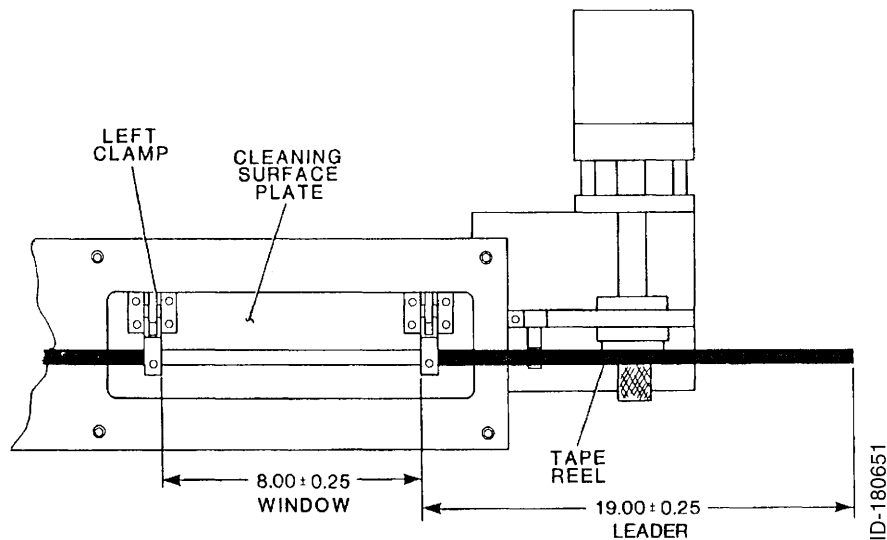
31-30-32

Page 624
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



BOT/EOT Windows, Kapton Tape

Figure 613

- (7) Clean away all loosened backing from the tape and work area with isopropyl alcohol and lintless swabs. Thoroughly dry the tape and work area with a stream of ionized air.
- (8) Carefully unclamp the tape and turn it over, being careful not to excessively twist the tape. Realign the tape and clamp it to the cleaning surface plate.
- (9) Remove the oxide from the tape base to match the backing window as in steps (6) and (7).
- (10) Unclamp the tape and verify that the window area is free of oxide and backing with the clearly-defined window $4.00 +0.50, -0.00$ inches long for mylar tape or 8.00 ± 0.25 inches for kapton. Verify the tape is completely dry and free of solvent or solvent residue. Buff window and adjacent tape area with lintless cloth to remove all visible residue.
- (11) If tape reel, Part No. 600-5128-001, is used, install a shim (supplied with tape winder) and the tape reel on the takeup shaft with the flat side facing out. If tape reel, Part No. 600-4507-001, is used, install the tape reel on the takeup shaft with the flat side facing out.
- (12) Verify that the tape is correctly threaded through the tape winder. Smoothly spread a very small amount of adhesive (27, Figure 701) on the tape reel and attach the end of the tape to the reel with the oxide (shiny) surface facing the reel. Verify that the tape is straight and winds evenly on the reel.
- (13) If the footage counter is not set to 388, push the lever, at the rear of the counter, down. Set the MOTOR switch to on and slowly increase the SPEED control to winding speed (20 to 30 on dial). Monitor the tape tension meter and maintain a steady reading of 3 to 5 ounces by using the speed control and the tension adjustment knob at the left of the supply reel shaft.

31-30-32

Page 625
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (14) As the counter approaches 0 feet, reduce the winding speed. The tape winder automatically stops at the count of 0. Length of coated tape (between windows) shall be 388 ± 1 feet.
- (15) Repeat steps (6) through (10) to prepare the EOT window.
- (16) Cut the tape at the left end of the window. Secure the end of the tape to the tape pack using a short length of mylar splicing tape. Ensure that the tape does not extend beyond the edge of the tape pack.
- (17) Remove the loaded tape reel and, if it is to be stored, place it in a clean petri dish with the part marking facing up. Rubber stamp the tape manufacturer's lot number on the reel using permanent black marking material. Put the cover on the petri dish.

17. Power Supply Cable Clamp Installation

- A. Damage to the wire bundle of the 720-0123-001 Power Supply may occur if the wire bundle is not flush with the cover surface. This is caused by interference between the wire bundle and the BOT/EOT connector when the Power Supply is installed in the chassis.
- B. Material Requirements as follows:
 - (1) Black vinyl tape, Part No. P-29 (516-6006-001), 0.5 inch wide, manufactured by Permacel Div. of Johnson and Johnson (V99742), US Highway 1, P.O. Box 671, New Brunswick, NJ 08903.
 - (2) Cable clamp, Part No. F4NY312-0 (400-5003-003), manufactured by Lynx, Inc. (V65664), 45 Francis St., P.O. Box 67, Leominster, MA 01453-4911.
 - (3) Washer, Part No. MS15795-805 (434-0030-004).
- C. Cable Clamp installation as follows:
 - (1) Remove the cable tie that is closest to the red grommet that is closest to transformer (120, IPL Figure 2).
 - (2) Pull lightly on the wire bundle to expose as much of the wire bundle as possible.
 - (3) Starting at strain relief, spiral wrap an 8-inch length of black vinyl tape (P-29) around wire bundle. Tightly cover approximately 3 inches of exposed wire bundle. Refer to Figure 614.
 - (4) Remove screws (10), washers (15), and EMI shield (5A). Remove screws (30), washers (35), and Power Supply PWB (25A).
 - (5) Remove nut (170), washer (175), and screw (165) that attach bridge rectifier (160) that is closest to the main connector.
 - (6) Install cable clamp (F4NY312-0) on wrapped part of wire bundle.
 - (7) Install washer (MS15795-805) on screw (165). Insert screw through cable clamp and bridge rectifier and secure with washer and nut removed in step (5).
 - (8) Install Power Supply PWB Assembly and EMI shield and secure with screws and washers removed in step (4).
 - (9) Use finger pressure to flatten the wrapped wire bundle against cover (185).

31-30-32

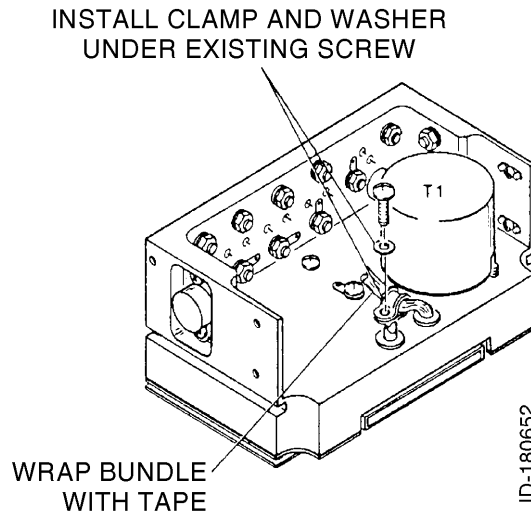
Page 626
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Cable Clamp Installation
Figure 614

18. Stepper Motor PN 360-0082-002/004 Bearing Replacement

This procedure applies to Stepper Motor, Part No. 360-0082-002 and -004 (135, 135A, IPL Figure 13).

NOTE: *Honeywell does not stock piece parts for the motor. These parts are available from the vendor: (V48240) Hansen Mfg. Co., Inc., 1934 Virgil Blvd., P.O. Box 23, Princeton, IN 47670.*

A. Materials Required

- (1) Motor Label, Part No. 315-7-1 (V48240) (Honeywell PN 402-6063-001).
- (2) Bearing, Part No. SSRI614ZZHA3P25LG54 (V48240) (Honeywell PN 400-0057-032) (137, IPL Figure 13).

B. Disassembly

- (1) Remove label from motor housing with a thin-bladed tool. Discard label.
- (2) Remove socket head screw (150, IPL Figure 13) and drive pulley (145). Retain all hardware for reassembly.
- (3) Remove snap ring and shim.

NOTE: *Front of motor is end from which shaft protrudes.*

- (4) Push squarely on front end of shaft and force rotor out of rear of housing.
- (5) Remove bearing, shims, and wavy washers from front of housing.
- (6) Remove bearing from housing or rotor. Leave any shims on rotor in place.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

C. Clean and Check

- (1) Clean all parts with a clean, lint-free cloth and compressed air.
- (2) Check all parts for signs of excessive wear or damage. Motors with cracks, large rotor chips, rotor scraping, dents, signs of bearing slippage, or similar damage are considered nonserviceable and should be discarded. Rotors with fretting corrosion (red rust) that is not easily removable should be discarded. Do not sand rotor shafts.

D. Assembly

- (1) Using only light finger pressure on outer race of bearing, install a new bearing (137, IPL Figure 13) in rear of motor housing.
- (2) Install shims and wavy washers removed in paragraph 18B(5) as shown in Figure 615.
- (3) Using only light finger pressure on outer race of bearing, install a new bearing (137, IPL Figure 13) in front of motor housing.
- (4) Install rotor in rear of housing while holding front bearing in place.
- (5) Install shim and snap ring removed in paragraph 18B(3). Orient snap ring such that flat side is facing up. Snap ring should be easy to install. If not, check assembly.

NOTE: *If motor is assembled per Figure 615, it is not necessary to verify preload.*

- (6) Install drive pulley and socket head screw removed in paragraph 18B(2). Adjust pulley to dimension shown in Figure 722, and tighten screw to 5.75 ± 0.25 in-lb.
- (7) Mark the date code (in the form of year-month-day) and the word REWORKED on label (Ref. paragraph 18A) with permanent black marking material (an indelible marker or ball point pen may be used):

REWORKED:
y-m-d

- (8) Install label on rear of motor housing. Use finger pressure only around outer edge of label to secure it to housing, but not to rotor. If label becomes attached to rotor, replace label.

19. Preventive Maintenance Information

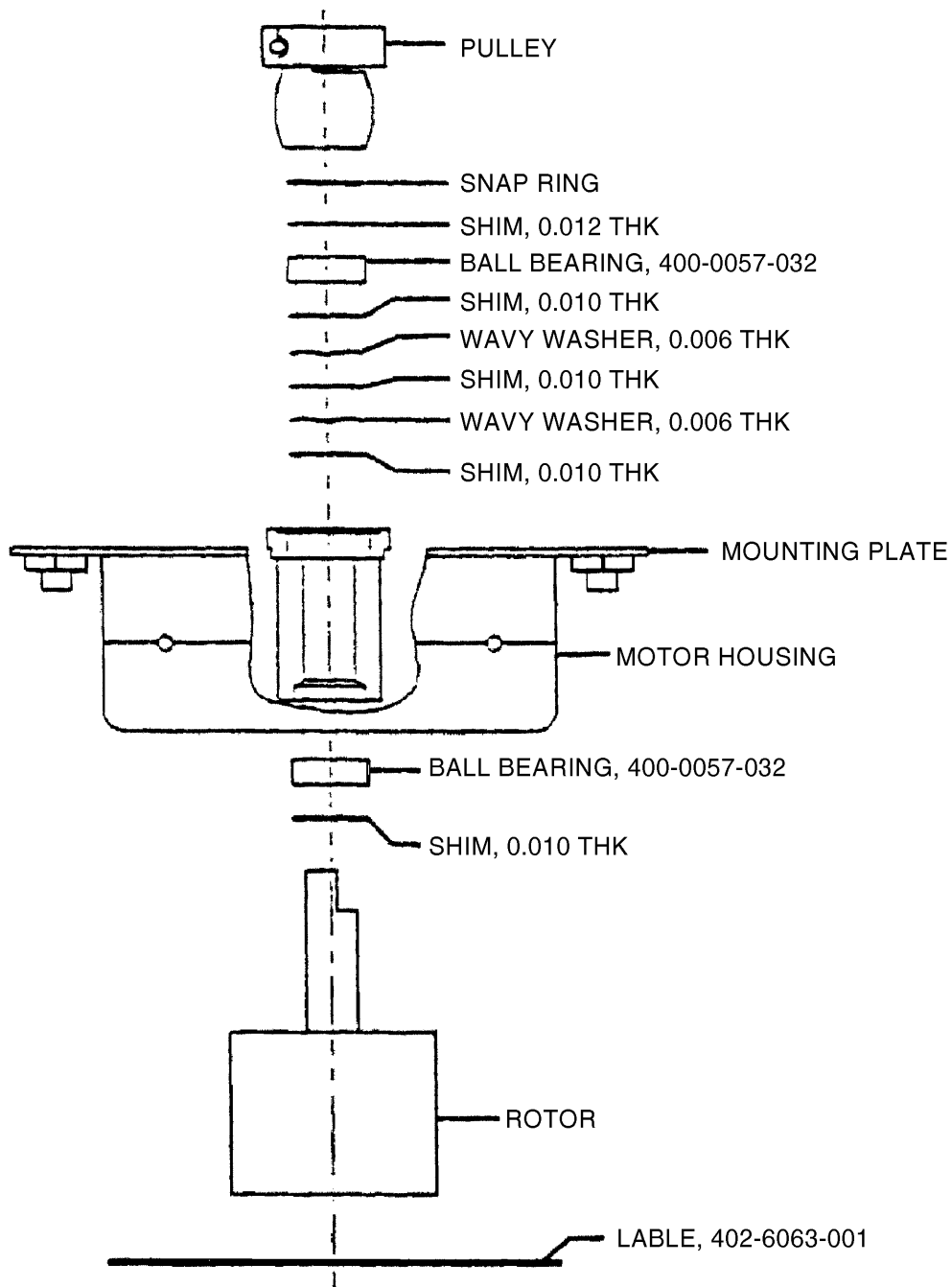
A review of operator records indicates that the UFDR annual operating time averages 2,500 to 3,000 hours. The service life of the UFDR is defined as 60,000 hours of operation or 20 years, whichever occurs first.

In order to maintain minimum performance standards during the useful service life of the UFDR, and to extend the time between removals and repairs, it is recommended that a preventive maintenance program for the UFDR be implemented.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180653

Stepper Motor, PN 360-0082-002/004

Figure 615

31-30-32

Page 629
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Components to be periodically checked and cleaned, or repaired, are listed in Figure 616. The applicable **DISASSEMBLY, CHECK, CLEANING, and ASSEMBLY** paragraphs are referenced for each maintenance item. Refer to paragraphs 9, 10, and 11 for bearing information.

If the tape recorder is in the shop and is opened for any reason, it is recommended that the loaded reel and the motor drive belt be replaced, and that the tape transport be cleaned per the **CLEANING** section, paragraph 3G, before being returned to service. These actions will provide UFDR performance of greatest reliability.

NOTE: For additional information and considerations for establishing a maintenance schedule, refer to Honeywell Document No. 032-0002, Recommended Maintenance of Honeywell Universal Flight Data Recorder.

MAINTENANCE ITEM	IPL FIG-ITEM	DISASSEMBLY PARA.	CLEANING PARA.	CHECK PARA.	ASSEMBLY PARA.
Flat Belt	10-45A,420 14-35	12B 16D	3G(2)(b)	2C	11V, 6M(3), 7N(5)
Peripheral Belt	10-50B,14-40	12A(2), 16A(2)	3G(3)(b)	2D	11U,6P,14D(7)
Tape Guide Rollers	10-150,14-140 -140C,440-440C,	12D,16F	3G(8)(a)	3B,4B, 5B,6B	11L,6Q,7V
Magnetic Heads	10-185,185A,187, 187A,190,190A 14-420,425,430	12F,16H	3G(8)(b)	2G	11S,6J,7K
Tape Drive Pulley (Capstan)	10-200 14-205 - 205C	12C, 16E	3G(5)	- - -	11P,6M,7N
Tape Transport Pulley (Flywheel)	10-215 14-220	12B,16D	3G(5)	5B	11O, 6M,7N
Drive Belt Roller Guide	10-330,425 14-330	12E,16G	3G(4)	3B,4B, 5B,6B	11I,6N,7S
Motor Drive Rollers	11-15 13-35	13E,15E	3G(4)	5B	10G, 8A(2)
Drive Pulley	11-95B,95C 13-120	13H,15H	3G(4)	- - -	10A(1)
Belt Idler Pulley	10-240 14-255 - 255C	12I(3) 16K(3)	3G(5)	2E	11K,6O 7T,7U
Control Plates	14-5D	- - -	3G(6)	- - -	7O

Preventive Maintenance Items
Figure 616

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

20. Power Supply (A1A1) -002 or less to -003 Modification

If the unit fails the RTU Test **Abnormal Power Conditions Test Voltage Hysteresis Test** (as noted on the data sheets) perform the following procedure and retest.

- A. Check the power supply A1A1 board part number. If it is PN 722-2440-002 or lower proceed to step B. If it is dash -003 or higher troubleshoot and repair fault.
- B. Remove power supply A1 from unit per **DISASSEMBLY** section, Para. 3F.
- C. Separate power supply A1A1 from heatsink per **DISASSEMBLY** section Para. 4E.
- D. Locate U3 and remove conformal coating per **DISASSEMBLY** section from around U3 pins 1 and 2.
- E. Install a new resistor R16 (56, IPL Figure 7) on the circuit side at the approximate location shown in IPL Figure 7.
- F. Sleeve (4, Figure 701) R16 leads.
- G. Wrap leads around A1A1U3 (see schematic Figure 125) pins 1 and 2 0.5 to 1 turn and solder.
- H. Mark reference designation R16 close to the new part with permanent white marking material.
- I. Locate PN 722-2440-002 on the board and change the -002 to -003.
- J. Clean all new solder joints with Isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per **ASSEMBLY** section para. 2C.
- K. Reassemble power supply per **ASSEMBLY** section Para. 5J. Reinstall power supply into chassis and retest.

21. Stepper Motor PN 360-0082-001/003 Bearing Replacement

This procedure applies to Stepper Motor, Part No. 360-0082-001 and -003 (110, 110A, IPL Figure 13).

NOTE: *Honeywell does not stock piece parts for the motor. These parts are available from the vendor: (V82227) Airpax Co., Cheshire Div., PO Box 768, Cheshire, CT 06410.*

A. Materials Required

Bearing, Part No. SSRI814ZZHA3P15LG54 (V48240) (Honeywell PN 400-0057-033) (112A, IPL Figure 13).

B. Disassembly

- (1) Insert a knife blade between the aluminum label and the motor housing. Use caution not to bend the label for it must be reinstalled. Carefully work the blade around the circumference of the motor to loosen the glue holding the label. Gently lift off the label. Set aside label, printed side down, in a clean location and protect adhesive backing from contamination.
- (2) Remove socket head screw (125, IPL Figure 13) and drive pulley (120). Retain all hardware for reassembly.
- (3) Remove snap ring and shim.

NOTE: *Front of motor is end from which shaft protrudes.*

31-30-32

Page 631
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (4) Push squarely on front end of shaft and force rotor out of rear of housing.
- (5) Remove bearing, shims, and wavy washers from front of housing.
- (6) Remove bearing from housing or rotor. Leave any shims on rotor in place.

C. Clean and Check

- (1) Clean all parts with a clean, lint-free cloth and compressed air.
- (2) Check all parts for signs of excessive wear or damage. Motors with cracks, large rotor chips, rotor scraping, dents, signs of bearing slippage, or similar damage are considered nonserviceable and should be discarded. Rotors with fretting corrosion (red rust) that is not easily removable should be discarded. Do not sand rotor shafts.

D. Assembly

- (1) Using only light finger pressure on outer race of bearing, install a new bearing (112A, IPL Figure 13) in rear of motor housing.
- (2) Install shims and wavy washers removed in paragraph 20B(5) as shown in Figure 617.
- (3) Using only light finger pressure on outer race of bearing, install a new bearing (112A, IPL Figure 13) in front of motor housing.
- (4) Install rotor in rear of housing while holding front bearing in place.
- (5) Install shim and snap ring removed in paragraph 20B(3). Orient snap ring such that flat side is facing up. Snap ring should be easy to install. If not, check assembly.

NOTE: *If motor is assembled per Figure 617, it is not necessary to verify preload.*

- (6) Install drive pulley and socket head screw removed in paragraph 20B(2). Adjust pulley to dimension shown in Figure 722, and tighten screw to 5.75 ± 0.25 in-lb.

NOTE: *The label may be cleaned with isopropyl alcohol to remove adhesive. Apply adhesive (27, Figure 601) to label and reinstall on motor.*

- (7) Reinstall label to back of motor being careful to ensure proper alignment with motor housing. Use finger pressure around the outer edge of the label to secure it firmly to the back of the motor. If label becomes attached to rotor, remove and replace label.
- (8) Mark the date code (in the form of year-month-day) and the word REWORKED on label with permanent black marking material (an indelible marker or ball point pen may be used):

REWORKED:
y-m-d

- (9) If the motor has the customer part number 360-0082-001 on label, cross out the -001 and remark with -003.

31-30-32

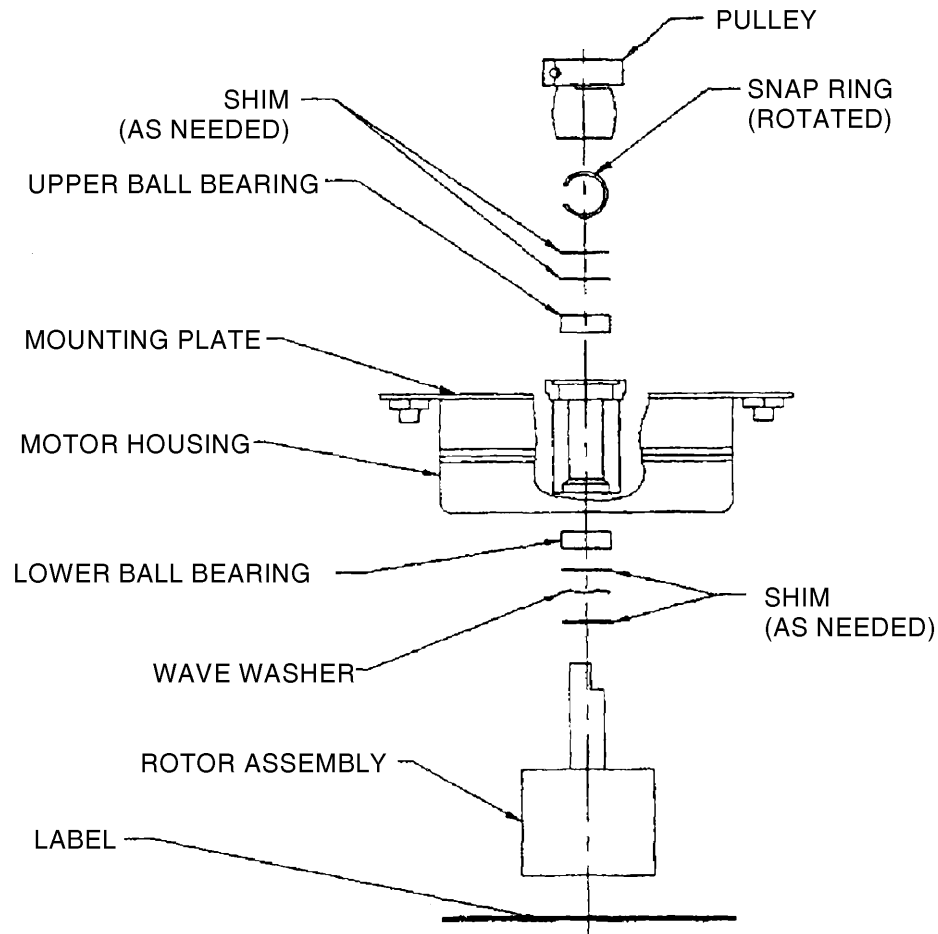
Page 632
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180654

Stepper Motor, PN 360-0082-001/003

Figure 617

31-30-32

Page 633
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

ASSEMBLY (INCLUDING STORAGE)

1. General

- A. Assemble the equipment covered by this manual according to the following procedures. Observe all identification marks when soldering electrical leads to components.
- B. Refer to Figure 701 for materials required during assembly, to **FITS AND CLEARANCES** section for tolerances, clearances, etc., and to **SPECIAL TOOLS, FIXTURES, AND EQUIPMENT** section for special equipment required during assembly.

2. General Techniques

CAUTION: USE HEATSINK WHEN SOLDERING A SOLID-STATE DEVICE. USE MINIMUM HEAT WHEN SOLDERING COMPONENTS ON A PRINTED WIRING BOARD.

CAUTION: USE CARE WHEN HANDLING THE DUST COVER TO AVOID SCRATCHING PAINTED SURFACES.

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES ARE SUBJECT TO DAMAGE BY EXCESSIVE LEVELS OF VOLTAGE. REFER TO DOD-HDBK-263 FOR PROPER HANDLING PROCEDURES.

A. Remove old solder when necessary and make connections mechanically secure before resoldering.

B. Solder per ANSI/J-STD-001A and IPC-A-610B.

CAUTION: CLEAN SOLDER JOINTS AND APPLY CONFORMAL COATING ONLY WHEN PWB CIRCUITS ARE GROUNDED AND IONIZED AIR IS DIRECTED AT PWB. REFER TO REPAIR SECTION.

C. Apply conformal coating (2, Figure 701) to all replaced components, repaired areas, and to new solder joints per these steps:

- (1) Ensure that printed wiring assemblies are clean and dry. Clean all new solder joints with isopropyl alcohol (3, Figure 701).
- (2) Coating may be sprayed or applied with a brush or assembly may be dipped.
- (3) Cure coating per manufacturer's instructions.

D. Sleeve (4, Figure 701) component leads and jumpers as required to prevent shorting.

E. Observe the following techniques when installing retainer rings.

- (1) Always use the proper retainer ring tools when installing rings to prevent loss of tension due to overspreading.
- (2) Ensure that retainer rings are installed with the smooth (shiny) side toward hardware being retained to maintain the proper retaining qualities and preload values.

3. Assembly Materials

Materials required for assembly are listed in Figure 701.

NOTE: *Equivalent substitutes may be used for all assembly materials.*

31-30-32

Page 701
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

ITEM	MANUFACTURER AND SPECIFICATION	USE
1. Solder	Per ANSI/J-STD-006.	Component leads and wires.
2. Conformal coating	PC18 or PC29M manufactured by Hysol Div., Dexter Corp. (V04347), 211 Franklin St., Olean NY 14760 or Conathane 1155 per MIL-I-46058, Type UR.	Printed wiring board coating.
3. Isopropyl alcohol	Commercially available.	Clean new solder joints.
4. Sleeving	Teflon per ASTM D3295, Type I, Class B.	Sleeve leads and jumpers.
5. Wire	High temperature electrical wire per MIL-W-16878/4, AWG 20, 22, 24, 26, 30	Electrical wiring.
6. Bus wire	Electrical wire per QQ-W-343, Type S, tinned, AWG 20, 22, 26, 28	Electrical wiring.
7. Setscrew	MS51051-10, 4-40, 0.187 long.	Motor/pulley drilling.
8. Primer	SS4004, manufactured by GE Silicones, Hudson River Road, Waterford, NY 12188 (V01139).	Secure gasket to cover.
9. Adhesive	RTV112, same source as item 8.	Secure gasket to cover.
10. Braided shield	8663, manufactured by Cooper Belden Electronic Wire and Cable (V16428), Sub. of Cooper Industries, Inc., NW N St., Richmond, IN 47374	Shield for motor leads.
11. Sleeving	M23053/5-2XX-C, heat shrinkable; (XX = 01 through 10)	Motor leads; wire bundle.
12. Encapsulant	Stycast 2651, manufactured by Grace, WR, and Co. (V04552), Emerson and Cuming, Dewey and Almy Chemical Div., 869 Washington ST., Canton, MA 02021-2513.	Pot P5 shell; J3/J4 sockets.
13. Cable	GWF5143, manufactured by Gore, WL, and Associates, Inc. (V17217), 555 Paper Mill Road, P.O. Box 9329, Newark, DE 19714; 13207, manufactured by Teledyne Thermatics a Teledyne, Inc., Co. (V12515), Hwy 301 S, P.O. Box 909, Elm City, NC 27822; or MTSE840212, manufactured by Condtek Electronics, Inc. (V57765), 2615 Oak St., Santa Ana, CA 92707.	Read/write heads to P5.

Assembly Materials
Figure 701 (Sheet 1 of 3)

31-30-32

Page 702
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

ITEM	MANUFACTURER AND SPECIFICATION	USE
14. Primer	747-56, Grade T, Form R per MIL-S-22473, manufactured by Loctite Corp. (V05972), 705 N. Mountain Rd., Newington, CT 06111	Primer for item 15.
15. Adhesive	Grade C per MIL-S-22473, same source as item 14.	Retain screws.
16. Adhesive	46551, same source as item 14.	Bond spacers.
17. Lacing Cord	No. 4 per MIL-T-43435.	Wire bundle ties.
18. Sealant	PR1201Q, Type II, Class 2 per MIL-S-8516.	Fillet around connectors.
19. Ink	73XNM, black, manufactured by Independent Ink, Inc. (V12744), 14705 S. Avalon Blvd., Gardena, CA 90248.	Permanent black marking material.
20. Adhesive	1C, Patch Kit, manufactured by Hysol Div. of the Dexter Corp. (V04347), 211 Franklin St., Olean, NY 14760-1211.	Secure components to boards.
21. Tape, cloth	Type GFT, 3/4 inch per MIL-I-15126.	Build up motor cable.
22. Primer	Chemlok 607, manufactured by Lord Corp. (V30676), Chemical Products Group, 2000 W Grandview Blvd., P.O. Box 10038, Erie, PA 16514.	Bond bumper to board retainer.
23. Lubricant	810-346, manufactured by Dukane Corp. (v94970), Seacom Div., 2900 Dukane Dr., St. Charles, IL 60174-3348	Lubricate ULD o-ring and threads.
24. Foam Tape	Part No. 4508, 1/2 inch, manufactured by Minnesota Mining and Manufacturing Co. (V26066), Industrial Tape Division, 3M Center, St. Paul, MN 55101-1428.	Cushion between EMI shields and boards.
25. Teflon Tape	P422, gray, 3/8 inch wide, manufactured by Permacel Div. of Johnson and Johnson (V99742), US Highway 1, US Highway 1, P.O. Box 671, New Brunswick, NJ 08903	Insulate main bundle out of baseplate.
26. Solvent	Catalog No. 753-53, same source as item 14.	Remove excess Loctite 404 or dissolve bond.
27. Adhesive	U26, UHU Glue Stick, manufactured by Lingner-Fischer GMBH, 7 Hermann Str., D-7580 Buehl (Baden), West Germany. Distributed by Boise Cascade Corp. (V86004), Paper Group, 1600 SW 4th Ave., P.O. Box 1414, Portland, OR 97207.	Adhere tape to reel.

Assembly Materials
Figure 701 (Sheet 2 of 3)

31-30-32

Page 703
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

ITEM	MANUFACTURER AND SPECIFICATION	USE
28. Lubricant	Molykote G paste, manufactured by Dow Corning Corp. (V71984), 390 S Saginaw Rd., P.O. Box 997, Midland, MI 48640	Start tape transport cover screws.
29. Adhesive	85-21, Loctite B, same source as item 14.	Bond bearings.
30. Sprague Mag	SM-2, manufactured by Sprague Magnetics, Inc. (V0AX86), 15720 Stagg St., Van Nuys, CA 91406	Track location check.
31. Cable tie	SST1.5M, manufactured by Panduit Corp. (V06383), 17301 Ridgeland, Tinley Park, IL 60477-3048	Wire bundle ties.
32. Sleeving	RSK201 (blue) and RSK101 (red), manufactured by Thomas & Betts Corp. (V56501), 920 Route 202, Raritan, NJ 08869-1420.	Shield termination, harness
33. Watch paper	Commercially available.	Check tape cleanliness.
34. Adhesive	Loctite 601 or 609, same source as item 14.	Bond bearings.
35. Sealant	RTV-630, same source as item 8.	Pot J5.
36. Resin defoamer	No. 1, manufactured by National Aerosol Products Co. (V59986), Sub. of Grow Group, Inc., 2193 E 14th, St, Los Angeles, CA 90021-2843	Pot J5
37. Sealant	RTV116, same source as item 8.	Pot J5; secure A4L1 to board.
38. Sealant	RTV103, same source as item 8.	Pot J5.
39. Activator	Catalyst 9, same source as item 12.	Activator for item 12.
40. Adhesive	Loctite 290, same source as item 14.	Bond jackposts.
41. Adhesive	Tak Pak Kit per MIL-A-46050, Type II, Class 3. Same source as item 14.	Secure wire to board.
42. Adhesive	H20E, 1 lb. kit, conductive epoxy, manufactured by Epoxy Technology, Inc. (V33246), 14 Fortune Dr., P.O. Box 567, Billerica, MA 01821-3922.	Bond spacers to EMI shield.
43. Iron Powder	2228-01, manufactured by Baker JT Chemical Co., 995 Zephyr Ave., Hayward, CA 94544-7917 (V5B970) or 5320-500, manufactured by Mallinckrdt Chemical, 223 West Side Ave., Jersey City, NJ 07303 (V2A915).	Track location check.

Assembly Materials
Figure 701 (Sheet 3 of 3)

31-30-32

Page 704
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

4. **Chassis Assembly** (IPL Figure 16)

Refer to Figure 122 for wire replacement information. Secure wire bundles with lacing cord (17, Figure 701).

A. Assembly of Chassis

Install chassis parts per the following steps. Apply primer (14, Figure 701) and adhesive (15, Figure 701) to all screws.

- (1) Install bracket (255) and secure to plates (345, 380) with screws (260).
- (2) Install connector plate (245) and secure with screws (250).
- (3) Install front panel (300) and secure with screws (305).
- (4) Install hooks (290) and secure with screws (295).
- (5) Install handle (310) and secure with screws (315) and washers (320).
- (6) Install cable bracket (415) and secure with screws (420) and nuts (425).

B. Card Guide Installation

- (1) Install card guides (360, 370, 395, 405) and secure with rivets (365, 375, 400, 410).
- (2) Install guides (280) and secure with screws (285). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.

C. Shockmount Installation

- (1) Install shockmounts (325) and secure with screws (330) and nuts (335). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.
- (2) Install shockmounts (265) and secure with screws (275). Install ground strap (270) on lower shockmount. Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.

D. Tywrap Base Installation

- (1) Install tywrap bases (120) and secure with screws (125) and nuts (130).
- (2) Secure wire bundle to tywrap bases with straps (110). Install cable tie (25) over wire bundle.

E. Connector J5 Installation

- (1) Install connector bracket (100) and secure with screws (105A).
- (2) If replacement of J5 (85) is required, replace either with connector (87), strain relief backshell (96), and socket contacts (97), or with connector (85) and potting backshell (95). If (85) and (95) are used, pot per these steps:
 - (a) Preassemble connector (85), potting backshell (95), and jackposts (90).
 - (b) Pot wires with sealant (35, Figure 701) and air cure for 2 hours, using defoamer (36, Figure 701) as required to eliminate surface bubbles.
 - (c) Preseal with red sealant (37, Figure 701) and pot with black sealant (38, Figure 701).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

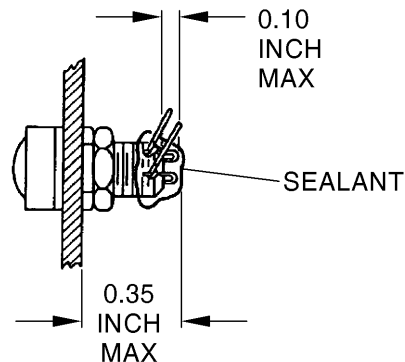
- (3) Stamp J5 on both sides of connector with black ink (19, Figure 701), using a minimum character height of 0.25 inch.
- (4) Install J5 (85) and secure with jackposts (90).

F. Connector J2 Installation

- (1) Install J2 (140, 140A) in front panel and secure with nut and washer (supplied with connector). Apply adhesive (15, Figure 701) to connector threads. Note that washer is attached to chain on connector cap.
- (2) Install connector cap (135).

G. LED Indicator Assembly Installation

- (1) Preassemble LED indicator assembly (145), spacer (150), retainer (155), and nut (supplied with LED).
- (2) Bend leads at rear of LED to allow a minimum clearance of 0.060 inch between leads and threaded portion of LED. Maximum protrusion at rear of LED is 0.10 inch. Refer to Figure 702.



Installation of LED Indicator Assembly
Figure 702

- (3) Connect wires to leads as shown in Figure 122 and solder. Clean solder joints with isopropyl alcohol (3, Figure 701). Cover leads, solder connections, and end of LED with sealant (38, Figure 701).
- (4) Install LED indicator assembly in front panel. Secure with screw (160) and nut (165). Maximum protrusion of LED behind front panel shall be 0.35 inch as shown in Figure 702.

H. Elapsed Time Indicator (ETI) Installation

- (1) Install ETI and secure with screws (80) and nuts (75).
- (2) Wire per Figure 122.

31-30-32

Page 706
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

I. EMI Filter Installation

- (1) Install EMI filters (55) and CR2 (60) on bracket (30). Sleeve (4, Figure 701) leads of CR2 and solder to terminals marked 1B and 2B.
- (2) Install bracket (30) and EMI filter (35) with screws (40), terminals (45), and nuts (50). Install terminals under nuts. EMI filter (35) is potted and is not repairable. Ref. Figure 123 for schematic.

J. Connector J1 Installation

- (1) Install J1 (5) and secure with screws (15) and nuts (20).
- (2) Route wire harness from J4 (200) side of board. Keep wires away from areas where printed wiring boards will be installed. No wiring shall touch installed boards. Ensure that wire bundles are routed such that they do not come within 0.150 inch of outside of Tape Recorder Assembly.

K. Assembly of Motherboard PWB

Ref. Figure 124 for Motherboard PWB schematic and to Figure 122 for Tape Recorder Assembly schematic.

- (1) Install standoffs (215), terminals (220), and standoffs (225). Ensure that they fit tightly after swaging.
- (2) Install nutplates (230) and secure with rivets (235). Ensure that rivets fit tightly.
- (3) Install connectors J3 and J4 (200) and secure with jackposts (205). Jackposts are supplied with lock washers and nuts. Discard lock washers and nuts. Apply adhesive (40, Figure 701) to jackpost threads. After connectors are installed, fill sockets J3-14 and 15 and J4-7 and 8 with encapsulant (12, Figure 701).
- (4) Install connectors XA1, XA2, XA3, XA4, XA8 (180) and secure with screws (185), nylon washers (190A), and nuts (195). If nuts (195A) are used, apply adhesive (40, Figure 701) to nuts after assembly. Apply a fillet of sealant (18, Figure 701) around connectors. Sealant shall not bridge adjacent components, extend under mating area between connectors and board, nor cover any wire termination holes. The sealant prevents the conformal coating from wicking up through the contact holes in the connector shell, which prevents plugging in mating connector at next higher assembly.
- (5) Replace 20 AWG wire (6, Figure 701) from E101 to E801 as required. Sleeve (4, Figure 701) wire and secure to board with adhesive (41, Figure 701) as necessary.
- (6) Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.

L. Motherboard PWB Installation

Install motherboard (170) and secure with screws (175). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.

Honeywell

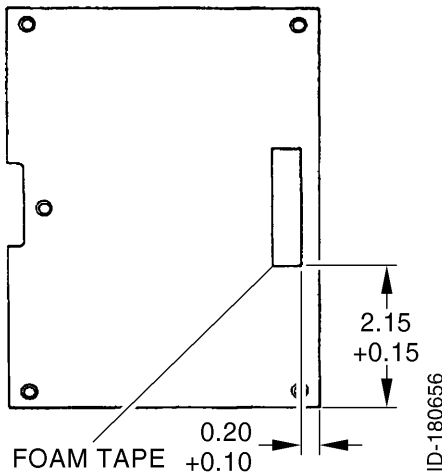
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

5. Power Supply A1 (IPL Figure 15)

Replace components (40 through 85, 125, 155, 165) and 20 AWG wire (5, Figure 701) as necessary to repair faults. Ref. Figure 125 for schematic and wiring information.

- A. Secure wire bundle with lacing cord (17, Figure 701) or cable tie (31, Figure 701).
- B. Install clinch nuts (200) on support assembly (195). Ensure that they fit tightly after swaging.
- C. Replace grommets (190) as necessary. These grommets prevent wire damage from metal support assembly hole.
- D. Install CR1, CR2, and CR3 (165) and secure with screws (170), washers (180), lug (185), and nuts (175). Install lug under screw that attaches CR2.
- E. Install S1 (155, 155A) and secure with nut (160).
- F. Install insulator (150) and T1 (125). Secure with screw (130), washer (135), washers (145), and nuts (140). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to all screw threads.
- G. Install insulator (120), AR1, AR2 (60), AR4 (85), AR5 (65), AR6 (70), Q1 (80), Q2 (75), and secure with screws (90), washers (95), shoulder washers (110), lug (115), washers (105), and nuts (100). Torque screws (90) to 5 ± 1 in-lb.
- H. Install C12 (40), C13 (45), C14, C15 (50), and R5 (55) as shown in views B and C of IPL Figure 15. Maximum length of capacitor and resistor leads shall be 0.50 inch. Sleeve (4, Figure 701) leads and solder in place.
- I. If A1A1 Power Supply PWB (25) is being replaced, wire per Figure 125. Ensure that maximum length of wire bundle service loop is 0.75 inch from edge of support assembly to edge of PWB when assembly is opened 180°. Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.
- J. Install A1A1 Power Supply PWB and secure with screws (30) and washers (35). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.
- K. To install standoffs (20) on EMI shield (5), apply a band of adhesive (42, Figure 701) around center of shank, place shank through hole, place aluminum washer (21) over shank, and swage over washer. Ensure that spacer fits tightly. Clean excess adhesive from around standoffs and cure at $175 \pm 10^\circ$ for 90 minutes minimum. DC resistance between any 2 standoffs shall be ≤ 5 ohms.
- L. To install spacers (20A) on EMI shield (5A), place shank through hole and swage. Ensure that spacer fits tightly. Resistance check not required (knurled shank).
- M. Install a 3.00 ± 0.15 -inch length of foam tape (24, Figure 701) in location shown in Figure 703. Do not use sponge rubber (23). Clean mounting surface with isopropyl alcohol (3, Figure 701).
- N. Install EMI shield (5) on Power Supply PWB (25) and secure with washers (15) and screws (10). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.



Foam Tape Installation on EMI Shield
Figure 703

6. Tape Transport Assembly (1, 1A, IPL Figure 14)

- A. The assembly procedure given in the following paragraphs applies only to modified Tape Transport Assemblies (617-4422) which have been modified in accordance with Service Bulletin 2. Assemble the Tape Transport Assembly in the sequence given unless otherwise indicated. If the Tape Transport Assembly has not been totally disassembled, omit those steps which do not apply.

CAUTION: DURING ASSEMBLY OF TAPE DECK, EXERCISE CARE DURING HANDLING TO PROTECT TAPE DECK COMPONENTS (BELTS, SHAFTS, TAPE HEADS, ETC.) FROM DAMAGE. PROVIDE SUPPORT FOR BASEPLATE SO THAT COMPONENTS DO NOT CONTACT WORK BENCH SURFACE.

- B. If damaged, replace feedthru (410) on P3 - P5 wire bundle as shown in Figure 705. Feedthru's may be slit in order to be installed over wire bundle.
- C. If wires from distribution board (270) to connectors (395 or 405) are damaged, replace with AWG 26 wire (5, Figure 701). See schematic diagram, Figure 122.
- D. Replace, if damaged, connector P3 (405) and clamp (400). If clamp (400) is replaced, remove screws supplied with clamp and replace with screws (402). Install heatshrink sleeving (16, Figure 701) under clamp as shown in Figure 705. See schematic diagram Figure 122 for wire identification. Cut connector P3-14 and 15 contacts off to 0.060 inch maximum after installation.
- E. If shell (385) or clamp (400) is replaced, stamp P3 or P5 on both sides in 0.25 inch minimum letters with permanent black ink (19, Figure 701) in accordance with Figure 705.
- F. Replace, if damaged, connector P5 (395) and shell (385). Fill potting shell (385) with encapsulant (12, Figure 701) above potting shell 0.070 ± 0.010 inch. All exposed wiring shall be covered by potting material. After potting, cut end tabs of shell flush with shell body. Sleeve (16, Figure 701) wire bundle from next to connector P5 down wire bundle

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

3.00 ± 0.30 inches as shown in Figure 705. If teflon tape (25, Figure 701) is damaged, replace as necessary in accordance with the dimensions shown in Figure 705.

- G. Install harness bracket (365) onto baseplate (415/415A) and secure with two lock washers (380), flat washers (375) and screws (370).
- H. Install harness cover (345) onto baseplate and secure with two lock washers (360), flat washers (355) and screws (350).
- I. Temporarily install distribution board (270) onto baseplate using screws (280).
- J. If magnetic head (420, 425, 430) replacement is required, thoroughly clean mating surfaces of baseplate and heads as shown in Figure 727. Clean with isopropyl alcohol (3, Figure 701).
 - (1) Inspect magnetic heads in accordance with the field rework procedure in the **REPAIR** section. Replace or rework any heads that fail to meet the specified requirements.
 - (2) Degauss magnetic heads in accordance with the procedure in the **REPAIR** section. After thoroughly degaussing, mount heads onto baseplate and secure using screws (435). Do not apply primer or adhesive to threads at this time.
- K. If EOT (125) or BOT (130) sensor replacement is required, mount sensor onto baseplate and secure with screws (135). Do not apply primer or adhesive to threads at this time.
- L. Install distribution as follows:
 - (1) Verify that U1, located on circuit side of board, is down in rectangular hole through baseplate. Install lug (295) and secure board (270), shield (275/275A) and lug with phenolic washers (290), flat washers (285), and screws (280). Install phenolic washers (290) between distribution board and shield. Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.
 - (2) Verify that wire bundles are routed approximately as shown in Figure 727. If bundle ties are damaged, replace with lacing cord (17, Figure 701). For schematic diagram, see Figure 122. Ensure that wires are 0.06 inch or more from all moving parts.
- M. Install capstan shaft (240) through baseplate (415/415A). Verify that shaft flange is tight against baseplate.

CAUTION: IN STEPS (2) AND (3), BELOW, EXERCISE EXTREME CARE NOT TO GET PRIMER OR STAKING COMPOUND INTO BEARINGS.

- (1) Install wavy washer (230) and bearing (235) over shaft and into slot in baseplate. Loosely secure to shaft with screw (225). Final tightening is accomplished in Step (3) after belt (35) is installed.
- (2) From the tape side of baseplate, install bearing (215) over shaft and into baseplate slot. Install tape pulley (205) onto shaft (240) and secure with screw (210). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw threads prior to installation.
- (3) Insert motor drive belt (35) into capstan cavity of baseplate (415/415A) with the smooth side of belt facing inward. Install pulley (220) on capstan shaft (240) and secure with screws (225A). Use care to avoid pinching or creasing motor drive belt. Check for free rotation and push axially on pulley (220) to ensure wavy washer deflects and recoils upon release, and there is no binding of bearings when assembly is axially deflected.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Check for 20 - 40 ounce preload when assembly is axially deflected 0.0002 inch. (See **CHECK** section, paragraph 5.)

- N. If drive belt roller guides (300/300A) are replaced, secure pins (335, 335A) into baseplate (415/415A) with screw (340) and shim (341). Check that perpendicularity requirements of **CHECK** section, paragraph 3, are met. Apply primer (14, Figure 701) and staking compound (15, Figure 701) to threads and screw into baseplate (415/415A). Before adhesive sets, recheck the perpendicularity requirement. Install drive belt roller(s) (330), ball bearings (325) and the shims (310, 315, 320), removed during disassembly, onto roller pin(s) (335/335A) and secure with E-ring (305). Observe paragraph 2E procedure when installing E-ring retainers. Check the rollers for free rotation.
- O. Install peripheral belt roller pin(s) (260/260A) and shims (262), removed during disassembly, onto baseplate. These roller pins are used to hold the belt rollers (255/255A). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw (265/265A) threads and secure pins to baseplate. Install belt rollers (255/255A) onto roller pins (260/260A) and secure with E-ring (250). Observe paragraph 2E procedure when installing E-ring retainers. Check for free rotation and a minimum of 0.001 inch axial freedom.
- P. Peripheral Belt (Vespel) roller assemblies (245/245A, IPL Figure 14).

NOTE: *Retrofitted Tape Transport Assemblies, Post SB 5, (617-4422-003) are configured with eccentric roller pins (260A, IPL Figure 14) for the peripheral belt (vespel) roller assemblies. Therefore, when the eccentric roller pins are replaced because of a maintenance action, the pins must be aligned to provide proper peripheral belt tracking. In the following procedure, steps (3), (4), (7), (8), (9) and (10) apply only to tape transport assemblies configured with the eccentric roller pins (260A).*

NOTE: *Check vespel peripheral belt rollers and roller pins in accordance with the **CHECK** section, paragraph 2E. Replace if damaged in any of the specified ways. Clean vespel peripheral belt rollers and roller pins in accordance with the **CLEANING** section.*

- (1) Check for and remove all burrs and other protrusions and/or contaminations from the two holes and the area near where the assemblies are to be installed.
- (2) Insert each roller pin (260 or 260A, as appropriate) with two shims each (262) into the pin hole.
- (3) Rotate eccentric pins with offsets toward center of tape transport (see Figure 715).
- (4) Fasten each pin in place using screws (265/265A). Do not apply adhesive at this time.
- (5) Install pulley assembly (245A) and E-ring (250). Observe paragraph 2E procedure when installing E-ring retainer.
- (6) Spin each P-belt roller by hand. If each roller does not spin freely, check for a minimum of 0.001 inch axial freedom. To check for proper axial freedom, perform the end play procedure given in the **CHECK** section, paragraph 6.
- (7) If the reel hub assemblies and the capstan assembly are already installed, proceed to the next step. If they are not installed, install before continuing.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (8) Install 'scratch' P-belt (40) over capstan and the two corner P-belt rollers (the rollers on the eccentric pins). Position the P-belt inboard of the two hub assemblies.
- (9) Position the eccentric pins (260A, IPL Figure 14) to achieve a 0.030 inch maximum parallelism between transport edge and belt along length of belt. If not, loosen either or both pin screws (265A) and move position of pin(s) until P-belt is inboard of the edge of the tape transport.
- (10) Remove the P-belt.
- (11) Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw (265A) threads.

Q. Assemble tape guide rollers (175, 475) in accordance with Figure 725 and the following procedure:

- (1) Install one shim (150A, 155, 160A, or 450, 455, 460), one bearing (170A, 470), one tape guide roller (175, 475) and second bearing (170A, 470) onto roller pin (180A, 480). If shim (150/150A, 155, 160/160A, 465 or 450, 455, 460, 465) stack-up was noted during disassembly, use same stack-up as a starting point.
- (2) Mount shims (150A, 155, 160A, 465 or 450, 455, 460, 465), wavy washer (165, 165A) and E-ring (145A, 445) onto roller pin (180A, 480). Shim as required to obtain an end play of 0.0005 to 0.0020 inch between roller (175, 475) and pin (180A, 480). Observe paragraph 2E procedure when installing E-ring retainer. Check 2 - 4 ounce preload when assembly is axially deflected 0.0005 inch. (See **CHECK** section, paragraph 5.)

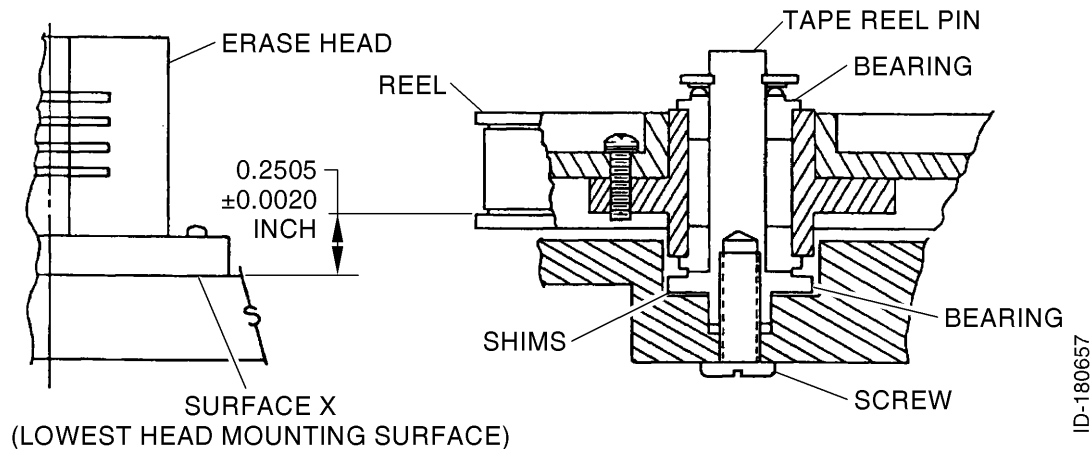
NOTE: *If roller pins (180A, 480) were not removed from baseplate during disassembly, verify dimensions of Step (3). Reshim only if necessary.*

CAUTION: IN STEP (3), EXERCISE EXTREME CARE NOT TO GET PRIMER OR ADHESIVE INTO BEARINGS.

- (3) Install five roller pins (180A, 480) and assembled parts, shims (185, 190, 195, 196) and screw (200 and 201, or 200A) onto baseplate in accordance with Figure 725. Shim as required between roller flanges and baseplate to obtain dimensions 0.2605 ± 0.0010 inch between baseplate surface X and flanges of tape guide rollers as shown. Surface X is the erase head mounting surface as shown by the cross hatched area in Figure 727. Check that the height differential between rollers is less than 0.0010 inch. Repeat steps as necessary to assure proper dimension. Adjust EOT and BOT sensors to clear tape guide flanges. Apply primer (14, Figure 701) and adhesive (15, Figure 701) to sensor screw threads and secure sensors to baseplate. Check the rollers for free rotation. Check that perpendicularity requirements of **CHECK** section, paragraph 3, are met. After shimming has been completed and checks have been met, apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw (200 and 201, or 200A) threads and secure roller pins to baseplate. Due to the criticality of the tape guide rollers, recheck the guide heights and perpendicularities to assure that the dimensional requirements have been met.

R. Assemble reel hubs in accordance with Figure 704 and the following procedure:

- (1) Install one bearing (100), the reel hub (75) and the second bearing (100) onto tape reel pin (80, 80A).



Reel Assembly
Figure 704

- (2) Install one shim (90/90A, 91, 92), a wavy washer (95, 95A), and another shim (90/90A, 91, 92) on the tape reel pin (80, 80A) and secure with E-ring (85). Observe paragraph 2E procedure when installing E-ring retainers.

NOTE: If tape reel pins (80, 80A) were not removed from baseplate during disassembly, verify dimension of step (3). Reshim only if necessary.

CAUTION: IN STEP (3), EXERCISE EXTREME CARE NOT TO GET PRIMER OR STAKING COMPOUND INTO BEARINGS.

- (3) Install the tape reel pins (80, 80A) and assembled parts, shims (105, 110, 115, 116), and screws (120/120B) onto baseplate in accordance with Figure 704. Check the rollers for free rotation. Shim as required between tape reel pin flanges and baseplate to obtain dimension 0.3600 ± 0.0020 inch between baseplate surface X and hub flanges as shown in Figure 704. Surface X is the erase head mounting surface as shown by the cross hatched area in Figure 727. Check that the height differential between hubs is less than 0.0005 inch. Check that perpendicularity requirements are met by repeating the height check at 90 degree increments around the circumference of the hub table surface. The height differential between points on the hub shall be less than 0.0005 inch. Repeat steps as necessary to assure proper dimension. After shimming has been completed, apply primer (14, Figure 701) and adhesive (15, Figure 701) to each screw (120/120B) and secure each tape reel pin to baseplate. Check the rollers for free rotation.
- S. If bumpers (15) are damaged, replace as necessary. Secure bumpers to top plate (5A) with lock washers (30), flat washers (25) and screws (20).
- T. Install top plate (5A) onto baseplate and secure with four screws (10). Screws and top plate will be removed and reinstalled in tape recorder assembly, paragraph 13K.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

7. Tape Transport Assembly (1C, 1E and on, IPL Figure 14)

- A. The assembly procedures given in the following paragraphs apply to Tape Transport Assemblies, Part Numbers 617-4422-005 and -007 and on, which have been modified per Service Bulletins 5, 8, 11, and 13. Assemble the Tape Transport Assembly in the sequence given unless otherwise indicated. If the Tape Transport Assembly has not been totally disassembled, omit those steps which do not apply.

CAUTION: DURING ASSEMBLY OF TAPE DECK, EXERCISE CARE DURING HANDLING TO PROTECT TAPE DECK COMPONENTS (BELTS, SHAFTS, TAPE HEADS, ETC.) FROM DAMAGE. PROVIDE SUPPORT FOR BASEPLATE SO THAT COMPONENTS DO NOT CONTACT WORK BENCH SURFACE.

- B. Install feedthru (410) on P3-P5 wire bundle as shown in Figure 705. Slit feedthru and install over wire bundle.
- C. Replace 26 AWG wires (5, Figure 701) and cables (13, Figure 701) from distribution board (270) to connectors (395 or 405) as necessary. Refer to schematic, Figure 122, for wire identification and to Figure 705 for dimensions and cable shield connections.
- D. Install clamp (400) over wires with heatshrink sleeving (11, Figure 701) under clamp as shown in Figure 705. If clamp is being replaced, remove screws supplied with clamp and replace with new screws (402). Install connector P3 (405) per schematic, Figure 122. Cut pins 14 and 15 of connector to 0.060 inch maximum.
- E. If clamp is new, stamp P3, 0.25 inch minimum height, on both sides with permanent black ink (19, Figure 701) as shown.
- F. Install nylon shell (385) over wires. Install connector P5 (395) per schematic, Figure 122. Fill nylon shell with encapsulant (12, Figure 701). Maximum overfill above nylon shell shall be 0.070 inch. All exposed wiring shall be covered by potting material. After potting, cut end tabs of shell flush with shell body. Sleeve (11, Figure 701) wire bundle from next to connector P5 down wire bundle 3.00 ± 0.30 inches as shown in Figure 705. Replace teflon tape (25, Figure 701) as necessary per dimensions shown.
- G. If nylon shell is new, stamp P5, 0.25 inch minimum height, on both sides with permanent black ink (19, Figure 701) as shown.
- H. Install harness bracket (365) and secure with screws (370), washers (380), and washers (375).
- I. Install harness cover (345) and secure with screws (350), washers (360), and washers (355).
- J. Temporarily install distribution board (270) onto baseplate with screws (280).
- K. Before installing magnetic heads (420, 425, 430), thoroughly clean mating surfaces of baseplate and heads with isopropyl alcohol (3, Figure 701).
- (1) Inspect magnetic heads per paragraph 14 of **REPAIR** section. Replace or rework heads that fail to meet specified requirements.
 - (2) Thoroughly degauss magnetic heads per paragraph 8 of the **REPAIR** section before installing them.

31-30-32

Page 714
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



31-30-32

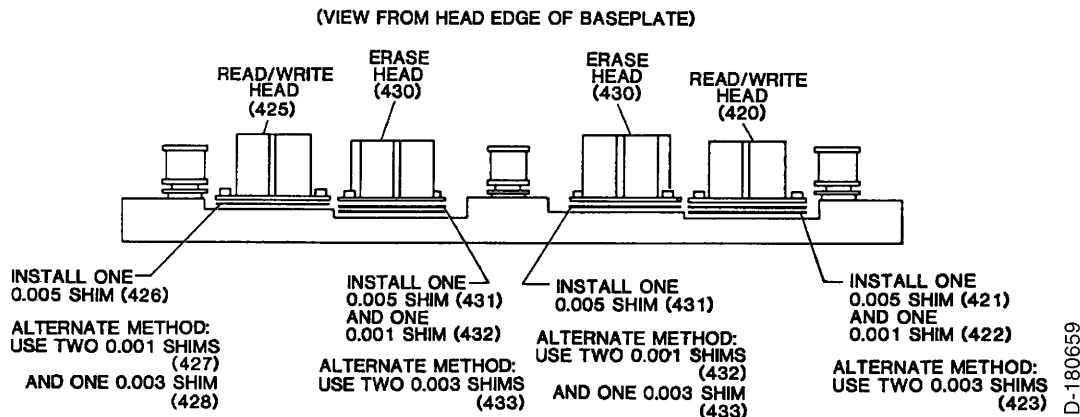
D-180658

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (3) For Tape Transport Assemblies (1J, 1K, 1Q), install magnetic heads with no shims. For Tape Transport Assemblies (1L, 1M, 1N, 1P, 1R, 1S, 1T, 1U), install shims (421, 422, 423, 426, 427, 428, 431, 432, 433) and heads. Ensure that the correct shims and quantities of shims are used with each head as shown in Figure 706. Secure heads with screws (435). Do not apply primer or adhesive to screws at this time; they will be applied after all head checks and/or alignments have been made. Refer to paragraph 14 of this section and to paragraph 7 of the **CHECK** section.



Head and Shim Installation
Figure 706

- L. Install EOT (125) and BOT (130) sensors and secure with screws (135). Do not apply primer or adhesive to screws at this time; they will be applied after tape guide roller assemblies are installed.
- M. Install distribution board (270) per these steps:
- (1) For Pre Mod 27 (SB 17) units, install shield (275A) and distribution board and secure with screws (280), washers (285, 290), and lug (295). For Post Mod 27 (SB 17) units, install lug (295) and secure board (270), shield (275/275A) and lug with phenolic washers (290), flat washers (285), two top screws (280A), spacers (282), and 1 bottom screws (280B). Install fibre washers (290) between shield and distribution board. Verify that U1, located on circuit side of board, is properly positioned through rectangular hole in shield and baseplate. Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.
 - (2) Verify that wire bundles are routed approximately as shown in Figure 727. Secure wires as required with lacing cord (17, Figure 701) and sleeving (11, Figure 701). Ref. Figure 122 for schematic. Ensure that wires are 0.06 inch or more from all moving parts.
- N. Tape drive pulley (capstan) installation as follows:
- CAUTION:** DO NOT APPLY ANY ADHESIVE TO SCREWS (210A, 225A).
- (1) Install ball bearing (215) in top (tape) side of baseplate (415, 415A, 415B).
 - (2) Install shaft (240) from lower side of baseplate.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (3) Install wavy washer (230) on shaft followed by ball bearing (235) from lower side of baseplate. Loosely secure to shaft with screw (225A). Tape transport pulley (flywheel) (220) is installed after flat belt (35) is installed.
- (4) Install tape drive pulley (205) on shaft and secure with screw (210A). Torque screw to 10 - 12 in lbs.

NOTE: Screw (210A) shall not be reused. If removed, replace with new screws.

- (5) Insert flat belt (35) into capstan cavity of baseplate with smooth side of belt facing inward. Carefully loop belt over tape transport pulley (220) and install pulley on shaft (240) with belt between guide rollers. Secure with screw (225A). Ensure that flat belt is not pinched or creased and is free to move with pulley rotation. Torque screw to 10 - 12 in-lbs.

NOTE: Screw (225A) shall not be reused. If removed, replace with new screws.

- (6) Check for free pulley rotation and push axially on tape transport pulley (220) to ensure that wavy washer deflects and recoils upon release, and that there is no binding of bearings when assembly is axially deflected. When assembly is axially deflected 0.0002 inch, preload shall be 20 to 40 ounces. Check preload per paragraph 5 of **CHECK** section.

O. Control plate installation as follows:

- (1) Ensure that old control plates are completely removed.
- (2) Clean surfaces of top plate assembly (4) and baseplate with isopropyl alcohol (3, Figure 701).

CAUTION: USE EXTREME CARE TO PREVENT DAMAGE TO NEW CONTROL PLATES.

- (3) Remove control plate (5D) from its package and peel off backing.
- (4) Orient control plate over top plate such that the larger perimeter cutout faces where the capstan would be with the top plate installed, and large holes are centered over top plate reel shaft holes. Press into place with finger pressure.
- (5) Remove remaining control plate (5D) from its package and peel off backing.
- (6) Orient control plate over baseplate such that the larger perimeter cutout faces the capstan and large holes are centered over relieved areas of baseplate. Press into place with finger pressure. Ensure that control plate adheres to baseplate inside relieved areas.

P. Assembly of reel hub assembly (70, 70A) as follows:

CAUTION: USE EXTREME CARE NOT TO GET PRIMER OR ADHESIVE IN BEARINGS.

- (1) Sparingly apply primer (14, Figure 701) to inner (bearing mating) surfaces of reel tape housing (75, 75A). Sparingly apply adhesive (34, Figure 701) to 4 equally-spaced places on surface of each ball bearing (100). Press ball bearings into housing and remove any excess adhesive with a cotton-tipped applicator. Refer to Figure 707.

NOTE: If available, an ultraviolet lamp and magnifier may be used to view the assembly. The adhesive fluoresces and becomes readily visible under ultraviolet light.

31-30-32

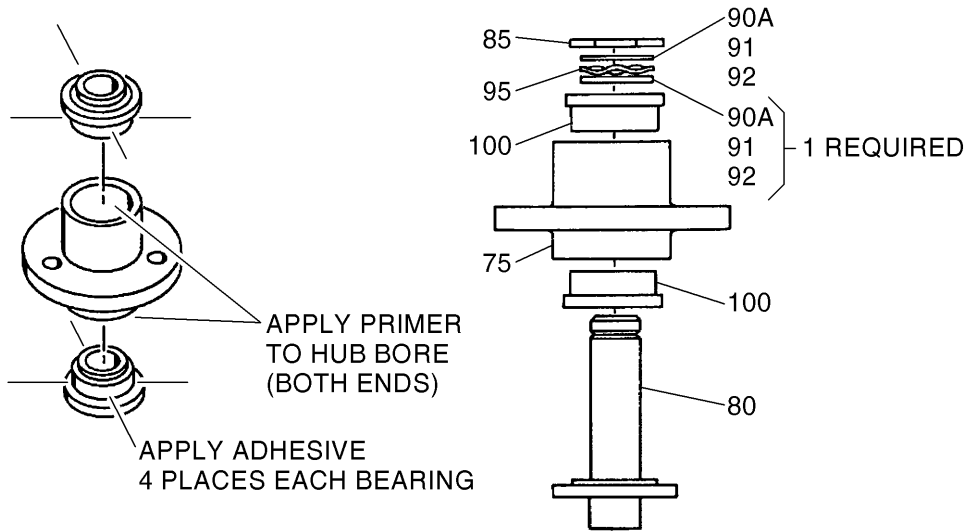
Page 717
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180660

Reel Hub Assembly (70, 70A)
Figure 707

- (2) Install reel housing/bearing assembly on tape reel pin (80).

CAUTION: AVOID EXCESSIVE DEFLECTION OF WAVY WASHER DURING ASSEMBLY.

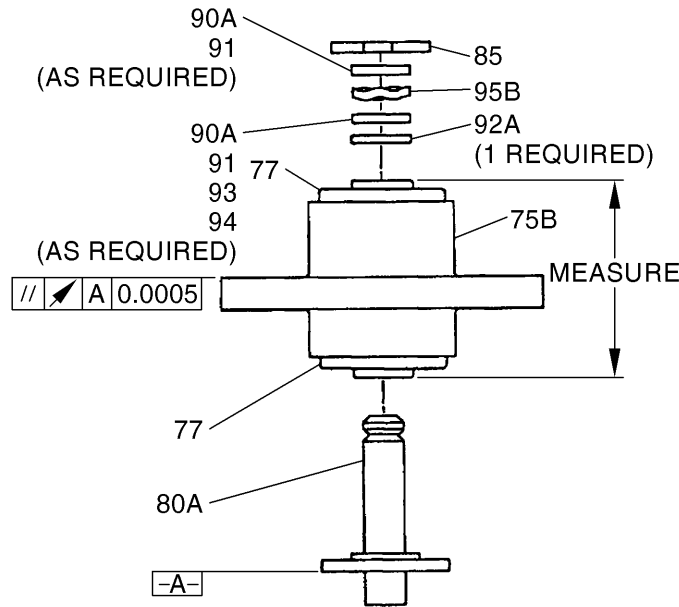
- (3) Install 1 shim (90A, 91, or 92), wavy washer (95), and shims (90A, 91, or 92) on reel tape pin (80) as shown in Figure 707.
- (4) Install E-ring retainer (85). Observe paragraph 2E procedure when installing E-ring retainer.
- (5) Check the reel hubs for free rotation. If any roughness is felt, replace bearings (100).
- (6) Check preload per paragraph 5 of **CHECK** section. Preload shall be 5 to 9 ounces when pin is deflected 0.0002 inch. Reshim as required to obtain specified preload.

Q. Assembly of reel hub assembly (70B) as follows:

- (1) Press ball bearings (77) into housing (76). Refer to Figure 708.
- (2) Measure the distance across the bearing inner race end surfaces of reel hub bearing assembly (75B,C) as shown in Figure 708. Note the dimension.
- (3) Install tape reel pin (80A) in reel hub bearing assembly.

CAUTION: AVOID EXCESSIVE DEFLECTION OF WAVY WASHER DURING ASSEMBLY.

- (4) Install 1 shim (92A). Install shims (90A, 91, 93, 94) and wavy washer (95B) in locations shown in Figure 708. Select the appropriate combination of shims from Figure 709.
- (5) Install E-ring retainer (85). Observe paragraph 2E procedure when installing E-ring retainer.



Reel Hub Assembly (70B, 70C)
Figure 708

DIMENSIONS ACROSS BEARING INNER RACE ENDS (INCH)					
0.5380	0.5400	0.5420	0.5440	0.5460	0.5480
TO	TO	TO	TO	TO	TO
0.5399	0.5419	0.5439	0.5459	0.5479	0.5499
TOTAL REQUIRED SHIM HEIGHT (INCH)					
0.026	0.024	0.022	0.020	0.018	0.016
SHIM COMBINATIONS (INCH)					
0.012	0.012	0.012	0.012	0.012	0.012
0.006	- - -	- - -	- - -	- - -	- - -
0.008	0.012	0.010	0.008	0.006	0.004

Shim Requirements, Reel Hub Assembly (70B, 70C)
Figure 709

- (6) Check the reel hubs for free rotation. If any roughness is felt, replace bearing (77).
- (7) Check preload per paragraph 5 of **CHECK** section. Preload shall be 2.0 ± 0.5 oz. when pin is deflected 0.0005 inch. If preload is less than 1.5 oz., select shim combinations as in Figure 709 to increase total required shim height in 0.002-inch increments to obtain specified preload. If preload ever exceeds 3.0 oz., remove and discard wavy washer. Install new wavy washer and repeat preload check.

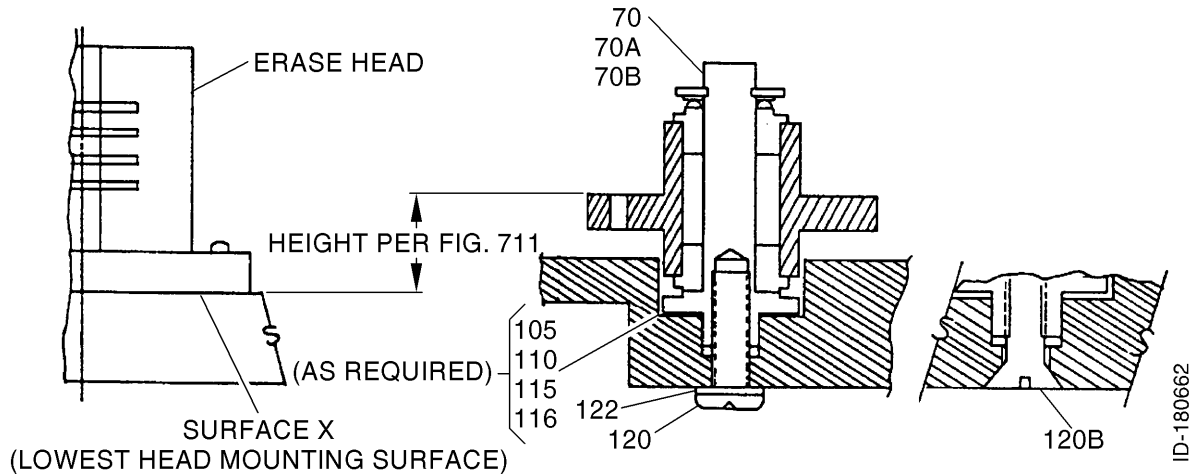
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

R. Reel hub assembly installation as follows:

- (1) Install shims (105, 110, 115, 116) in reel hub counter bore; install reel hub assembly (70, 70A, 70B) in counter bore through shims. Secure with screw (120) and washer (122) or screw (120B) as appropriate. Firmly tighten screw. Repeat for the other reel hub assembly. Do not apply primer or adhesive to screws at this time.
- (2) Perform height check for each reel hub assembly per Figure 710 and these steps:



Height Check, Reel Hub Assembly
Figure 710

- (a) Mount Tape Transport Assembly in checking fixture (16, Figure 901) and place on surface plate (38, Figure 901).
- (b) Place probe of height gage (35, Figure 901) on surface X and zero the meter. Surface X is the lowest erase head mounting surface as shown by the cross hatched area in Figure 727.
- (c) Move probe of height gage to reel hub flange top surface.
- (d) The reel hub height shall be as listed in Figure 711 for the various Tape Transport Assemblies.

HEIGHT (INCH)	TAPE TRANSPORT ASSEMBLY PART NUMBER
0.148 +0.004, -0.000	617-4422-005, -007
0.356 ± 0.001	617-4422-008, -009, -016, -017, -026, -027
0.361 ± 0.001	617-4422-010, -011, -014, -015, -024, -025
0.356 ± 0.002	617-4422-012, -013, -023

Reel Hub Heights
Figure 711

31-30-32

Page 720
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (3) Check perpendicularity of reel hub assemblies per these steps:
 - (a) Mark a point on top of each hub flange near outer edge.
 - (b) Place height gage probe on marked spot and zero gage. Rotate hub 90° (maintain height gage probe on reference mark) and check deviation from zero point.
 - (c) Repeat step (b) until the reference spot has been measure at 0°, 90°, 180°, and 270°. Maximum allowable deviation is 0.001 inch.
 - (d) If perpendicularity is not within specification, carefully remove discrepant assembly and rotate it 90°. Tighten screw and recheck perpendicularity. Assembly may be rotated in 90° increments until 1 full revolution is achieved. If proper perpendicularity still cannot be obtained, replace assembly.

NOTE: *If assembly is rotated, repeat height and perpendicularity checks.*

CAUTION: USE EXTREME CARE NOT TO GET PRIMER OR ADHESIVE IN BEARINGS.

- (4) For Tape Transport Assemblies that use screw (120B) (after shimming has been completed), remove screw, sparingly apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads, and secure each tape reel pin to baseplate. For Tape Transport Assemblies that use screw (120), do not use primer and adhesive.

NOTE: *Do not apply adhesive to screws if a tape tracking check is to be performed. Refer to Tape Tracking Check in **CHECKING** section.*

S. Drive belt roller guide assembly installation as follows:

- (1) Install pins (335, 335A) and secure each with screw (340) and shims (341, 342, or 343) as required to ensure that pins are fully tight against baseplate; however, total thickness of added shims must be kept to a minimum. Check perpendicularity requirement per paragraph 3 of the **CHECK** section. When this requirement is met, remove screw, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads, and reinstall. Before adhesive sets, recheck the perpendicularity requirement.

CAUTION: USE EXTREME CARE NOT TO GET PRIMER OR ADHESIVE IN BEARINGS.

- (2) Sparingly apply primer (14, Figure 701) to inner (bearing mating) surface of roller (330). Sparingly apply adhesive (29, Figure 701) to outer race of ball bearings (325). Press ball bearings into rollers and remove any excess adhesive with a cotton-tipped applicator. Refer to Figure 712.

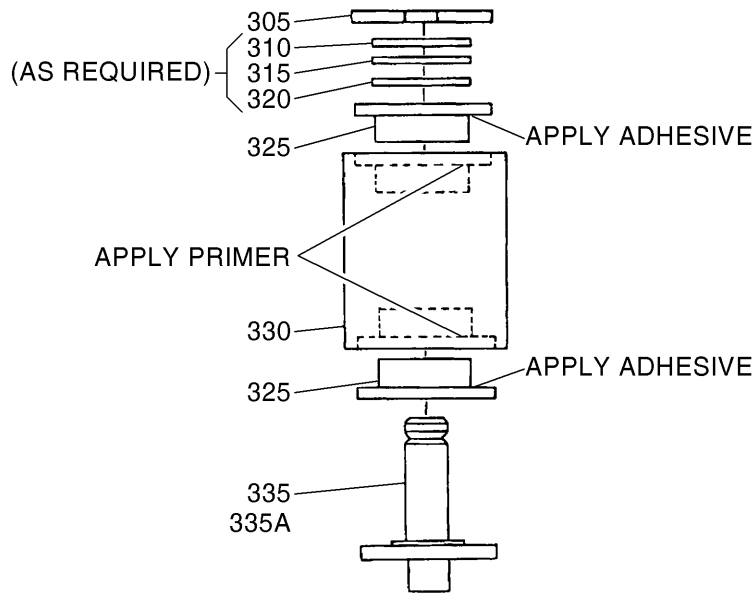
NOTE: *If available, an ultraviolet lamp and magnifier may be used to view the assembly. The adhesive fluoresces and becomes readily visible under ultraviolet light.*

- (3) Install roller and ball bearing assembly on pin (335, 335A). Install shims (310, 315, 320) as required to achieve an end play of 0.0005 to 0.0020 inch and secure with E-ring retainer (305). Refer to Figure 712. Observe paragraph 2E procedure when installing E-ring retainer. Check end play per paragraph 6 of **CHECK** section.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Drive Belt Roller Guide Assembly
Figure 712

- (4) Manually check each roller for free rotation. If any roughness is felt, replace ball bearings.

NOTE: There are 2 types of peripheral belt (P-belt) pulley assemblies used on Tape Transport Assemblies: straight-pin and eccentric-pin. Perform step T if the straight-pin type is used; step U if the eccentric-pin type is used.

- T. Straight-pin peripheral belt assembly installation as follows:

NOTE: Clean peripheral belt pulleys and roller pins per paragraph 3G(5) of the **CLEANING** section and check per paragraph 2E of the **CHECK** section. Replace any part that does not meet the specified criteria.

- (1) Ensure that the 2 holes and the areas near where the pins are to be installed are free of contamination.
- (2) Install 2 shims (262), 1 shim (262B), and roller pin (260) into mounting recess of baseplate.
- (3) For Tape Transport Assemblies, 617-4422-012, -013, and -023, thickness of shims is fixed at 0.013 inch.
- (4) For Tape Transport Assemblies, 617-4422-005, -007 through -011, -014 through -017, -024 through -027, temporarily secure with screw (265) and measure the dimension between surface X (lowest magnetic head mounting surface) and shoulder surface of each roller pin as shown in Figure 713. Reshim as required with shims (262, 262A, 262B, 262C) to achieve the dimension of 0.232 ± 0.001 inch. Check perpendicularity of pins per paragraph 3 of the **CHECK** section.

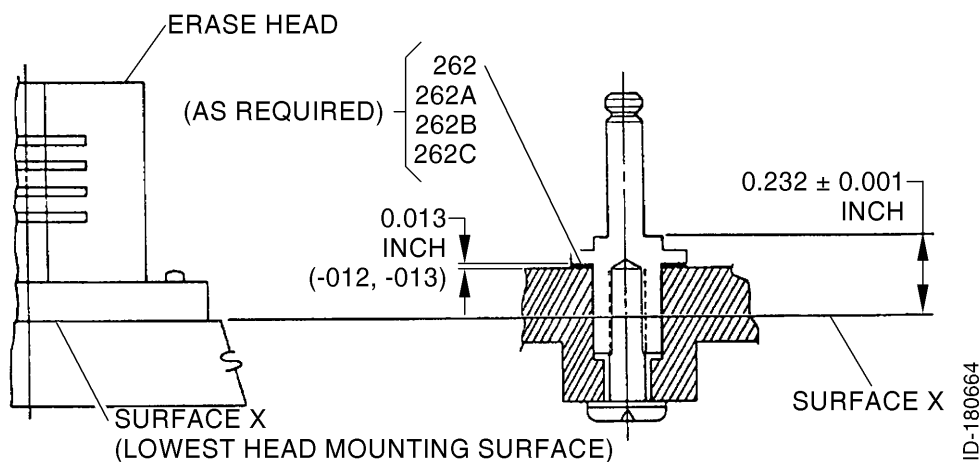
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

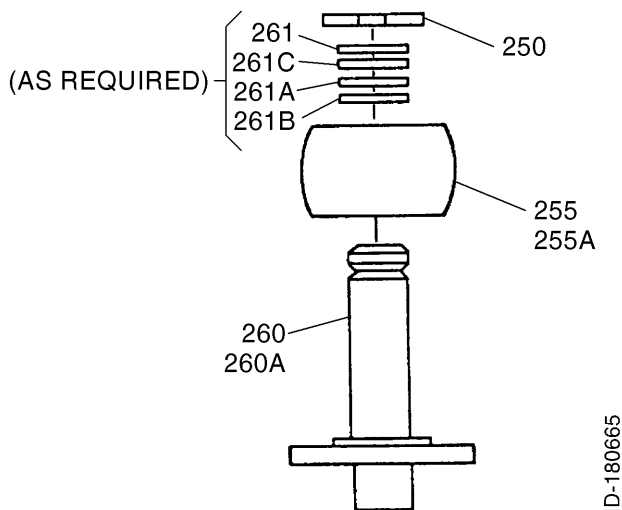
- (5) Install belt idler pulley (255, 255B, 255C) and 2 shims (261A) on roller pin (260). Secure with E-ring retainer (250). Observe paragraph 2E procedure when installing E-ring retainer.

NOTE: *Peripheral belt pulleys must be installed on pins in the same direction as they were before disassembly.*

- (6) Check end play per paragraph 6 of the **CHECK** section. Reshim as required with shims (261, 261A, 261B, 261C) to establish an end play of 0.001 to 0.002 inch. Refer to Figure 714.



Height Check, Peripheral Belt Roller Pin Figure 713



Peripheral Belt Pulley Assembly Figure 714

Honeywell

COMPONENT MAINTENANCE MANUAL

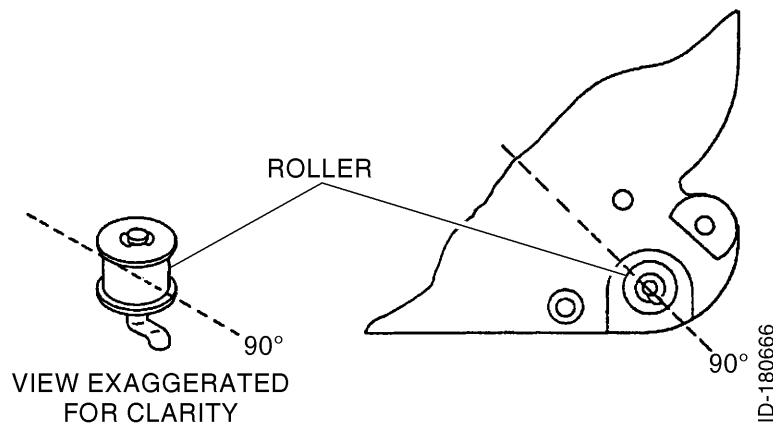
Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (7) Manually check each pulley for free rotation. If any roughness is felt, replace the pulley.
- (8) Remove screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw (265, 265A) threads, and reinstall screws. Verify the dimension of Figure 713.

U. Eccentric-pin peripheral belt assembly installation as follows:

NOTE: Clean peripheral belt pulleys and roller pins per paragraph 3G(5) of the **CLEANING** section and check per paragraph 2E of the **CHECK** section. Replace any part that does not meet the specified criteria.

- (1) Ensure that the 2 holes and the areas near where the pins are to be installed are free of contamination.
- (2) Install 2 shims (262), 1 shim (262B), and eccentric roller pin (260A) into mounting recess of baseplate. Temporarily secure with screw (265A) and measure the dimension between surface X (lowest magnetic head mounting surface) and shoulder surface of each eccentric roller pin as shown in Figure 713. Reshim as required with shims (262, 262A, 262B, 262C) to achieve the dimension of 0.232 ± 0.001 inch.
- (3) Loosen screws (265A), rotate eccentric roller pins so that offsets are toward center of tape transport as shown in Figure 715 and tighten screws.



Preliminary Eccentric Roller Pin Alignment
Figure 715

- (4) Check perpendicularity of pins per paragraph 3 of **CHECK** section.
- (5) Install belt idler pulley (255A, 255B, 255C) and 2 shims (261A) on roller pin (260A). Secure with E-ring retainer (250). Observe paragraph 2E procedure when installing E-ring retainer.

NOTE: Peripheral belt pulleys must be installed on pins in the same direction as they were before disassembly.

- (6) Check end play per paragraph 6 of the **CHECK** section. Reshim as required with shims (261, 261A, 261B, 261C) to establish an end play of 0.001 to 0.002 inch. Refer to Figure 714.

31-30-32

Page 724
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (7) Manually check each pulley for free rotation. If any roughness is felt, replace the pulley.
- (8) If the tape drive pulley (capstan) and reel hub assemblies are already installed, proceed to the next step. If not, install before continuing.
- (9) Install 'scratch' peripheral belt (P-belt) (40) over capstan and P-belt rollers. Position P-belt inboard of reel hub assemblies.
- (10) Rotate eccentric pins (260A) such that offset is oriented toward inside of Tape Transport Assembly and belt is parallel to edge of Tape Transport Assembly within a maximum of 0.030 inch.
- (11) Remove P-belt.
- (12) Remove screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads, and reinstall screws.

V. Assembly of tape guide roller assemblies as follows:

(1) Tape Guide Roller Assembly (140)

CAUTION: USE EXTREME CARE NOT TO GET PRIMER OR ADHESIVE IN BEARINGS.

- (a) Sparingly apply primer (14, Figure 701) to inner (bearing mating) surfaces of each roller (175). Sparingly apply adhesive (29, Figure 701) to outer race of ball bearings (170A). Press ball bearings into rollers and remove any excess adhesive with a cotton-tipped applicator. Maintain a dimension of 0.250 to 0.251 inch between bearing flange surfaces with adhesive applied. Refer to Figure 716.

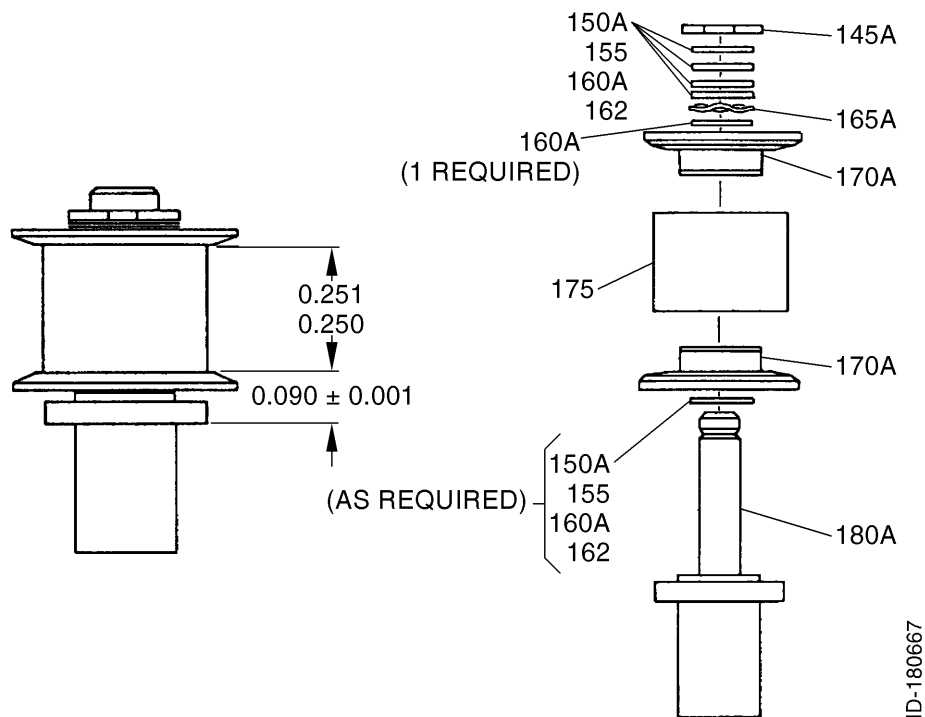
NOTE: *If available, an ultraviolet lamp and magnifier may be used to view the assembly. The adhesive fluoresces and becomes readily visible under ultraviolet light.*

- (b) Install shims (150A, 155, 160A, 162) and bearing/roller assembly on pin (180A). Adjust shims as required to obtain a dimension of 0.090 ± 0.001 inch between lower surface of pin flange and upper surface of lower bearing flange as shown in Figure 716.
- (c) Install 1 shim (160A), wavy washer (165A), shims (150A, 155, 160A, 162), and E-ring retainer (145A) on pin (180A). Observe paragraph 2E procedure when installing E-ring.

NOTE: *Do not use a 0.001-inch thick shim (160A) or a 0.0015-inch thick shim (162) directly under E-ring retainer.*

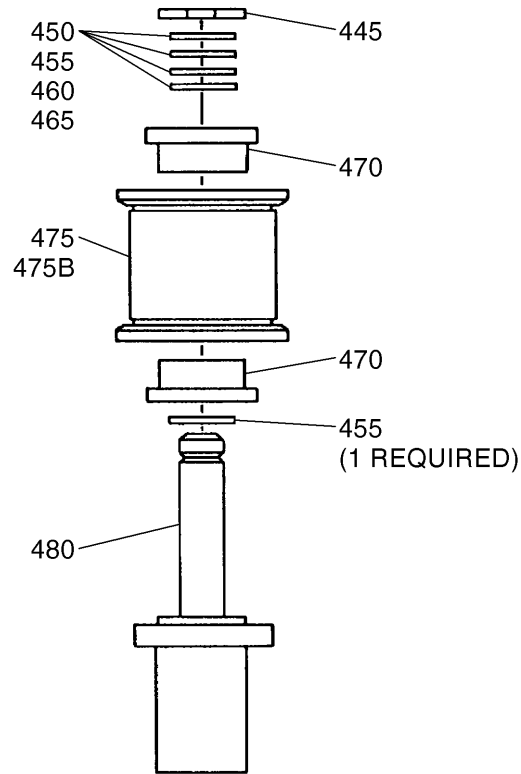
- (d) Check preload per paragraph 5 of **CHECK** section. Preload shall be 3.0 ± 1.0 ounces when pin is axially deflected 0.0005 inch. Reshim as required to obtain specified preload.

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



Tape Guide Roller Assembly (140)
Figure 716

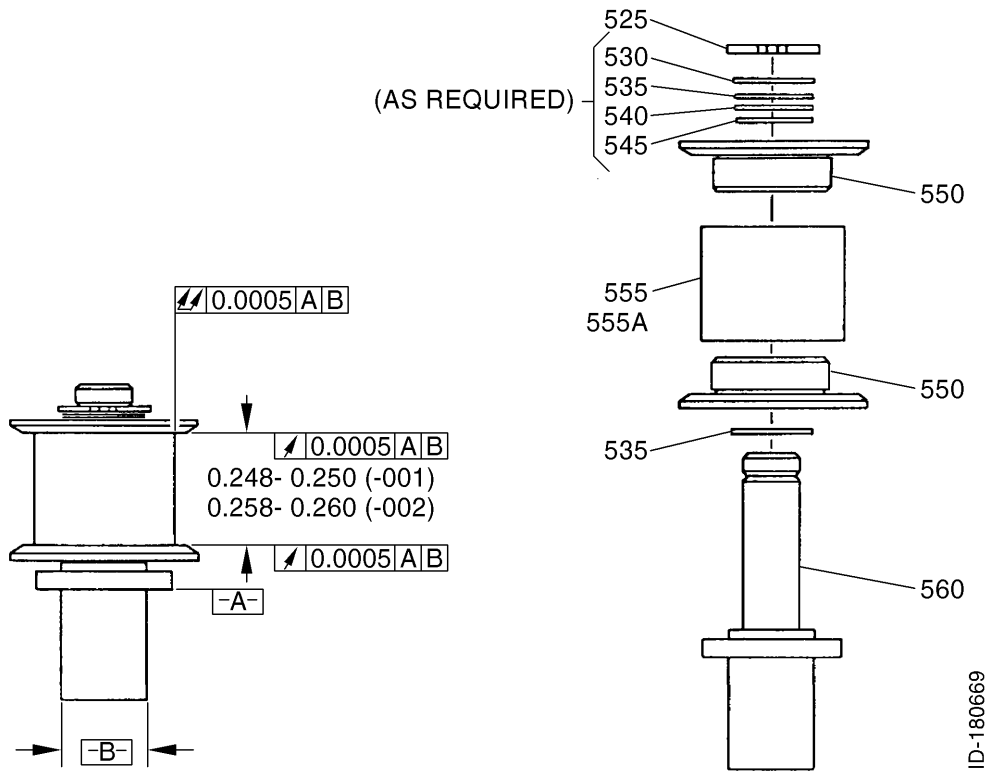
- (2) Tape Guide Roller Assembly (440)
 - (a) Install 1 shim (455) on pin (480). Refer to Figure 717.
 - (b) Install ball bearings (470) into ends of roller (475, 475B) and install assembly on roller pin.
 - (c) Install shims (450, 455, 460, 465), and E-ring retainer (445) on roller pin. Observe paragraph 2E procedure when installing E-ring retainer. Reshim as required to obtain an end play of 0.0005 to 0.0020 inch.



ID-180668

Tape Guide Roller Assembly (440, 440A)
Figure 717

- (3) Tape Guide Roller Assembly (485, 485B)
- (a) Press ball bearings (550) into rollers (555, 555A). Maintain a dimension between bearing flange surfaces of 0.248 to 0.250 inch for (485) and 0.258 to 0.260 inch for (485B). Refer to Figure 718.
 - (b) Install 1 shim (535) and bearing/roller assembly on roller pin (560).
 - (c) Install shims (530, 535, 540, 545) and E-ring retainer (525) on roller pin. Observe paragraph 2E procedure when installing E-ring retainer. Reshim as required to obtain an end play of 0.0005 to 0.0020 inch.

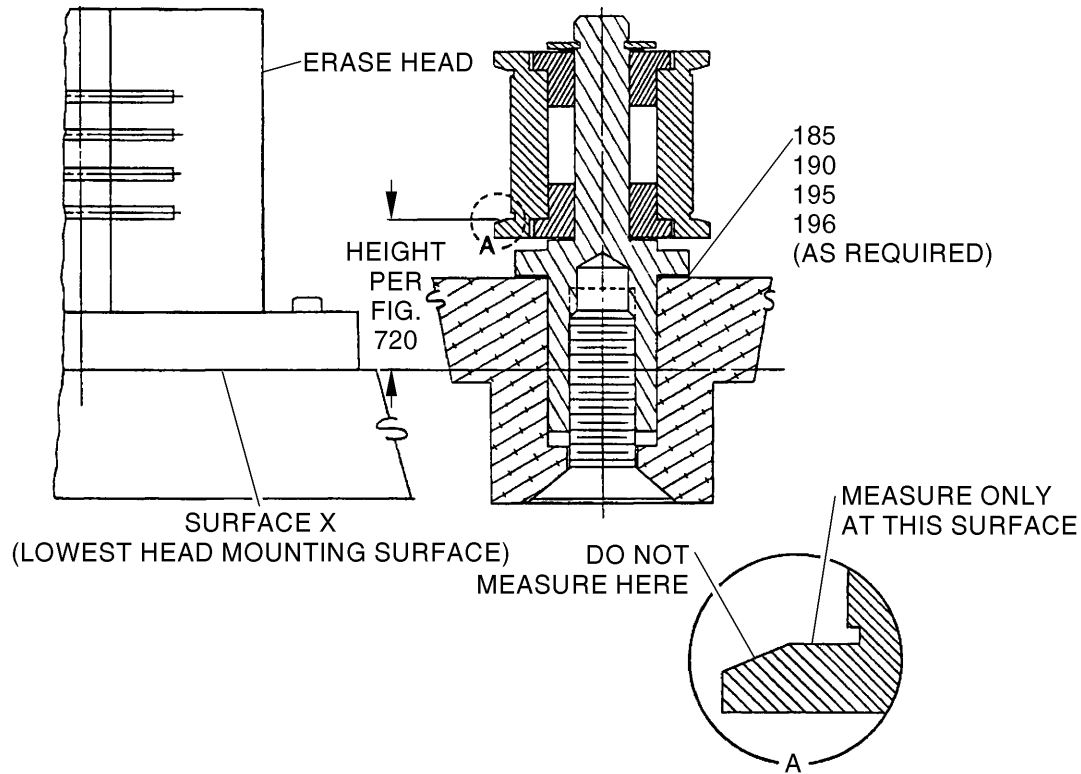


Tape Guide Roller Assembly (485, 485B)

Figure 718

W. Tape guide roller assembly installation as follows:

- (1) Check all tape guide rollers for free rotation. If any roughness is felt, replace ball bearings.
- (2) Install shims (185, 190, 195, 196) and tape guide roller assemblies (140, 440, 485, 485B). Secure assemblies (140, 440) with screws (200), washers (202), and screw (201), or screws (200A and 201). Secure assemblies (485, 485B) with screws (490), washers (502), and screw (500), or screws (495 and 500). Do not apply primer or adhesive to threads at this time.
- (3) Perform height check for each tape guide roller assembly per Figure 719 and these steps:
 - (a) Mount Tape Transport Assembly in checking fixture (16, Figure 901) and place on surface plate (38, Figure 901).
 - (b) Place probe of height gage (35, Figure 901) on surface X and zero the meter. Surface X is the lowest erase head mounting surface as shown in Figure 727.
 - (c) Move probe of height gage to upper surface of lower roller (or bearing) flange.
 - (d) The tape guide heights shall be as listed in Figure 720 for the various Tape Transport Assemblies.



ID-180670

Height Check, Tape Guide Roller Assemblies
Figure 719

HEIGHT (INCH)	TAPE TRANSPORT ASSEMBLY PART NUMBER
0.2555 ± 0.0010	617-4422-005, -007
0.257 ± 0.001	617-4422-008, -009, -016, -017, -026, -027
0.2620 ± 0.0010	617-4422-010, -011, -014, -015, -024, -025
0.2570 ± 0.0010	617-4422-012, -013, -023 (CENTER AND CORNERS)
0.2520 ± 0.0010	617-4422-012, -013, -023 (SIDES)

Tape Guide Heights
Figure 720

- (e) The difference in height between all tape guide rollers of the same height requirement shall be less than 0.001 inch. Reshim as required to obtain required dimensions.
- (4) Check perpendicularity of each tape guide roller assembly per paragraph 3 of **CHECK** section. If any tape guide roller assembly does not meet the perpendicularity requirements, using the described techniques, replace the assembly.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAUTION: USE EXTREME CARE NOT TO GET PRIMER OR ADHESIVE IN BEARINGS.

- (5) After all height and perpendicularity requirements have been met and shimming is completed, sparingly apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws (200A and 201 or 495 and 500) and secure roller pins to baseplate. Do not apply primer or adhesive to screws (200 or 490).

NOTE: Do not apply adhesive to screws if a tape tracking check is to be performed. Refer to Tape Tracking Check in **CHECK** section.

- (6) Recheck the guide heights and perpendicularities to ensure that the dimensional requirements have been met.
- (7) Remove BOT and EOT sensor screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads and reinstall screws. Adjust sensors to clear tape guide flanges and tighten screws. If proper adjustment cannot be made, replace sensor(s).

X. Secure bumpers (15) to top plate (5A) with screws (20), washers (25), and washers (30).

8. **Motor Drive Assembly (IPL Figure 13)**

A. Assembly of roller guide assembly as follows:

CAUTION: USE EXTREME CARE NOT TO GET PRIMER OR ADHESIVE IN BEARINGS.

- (1) Apply primer (14, Figure 701) and adhesive (29, Figure 701) to ball bearings (30). Install ball bearings in rollers (35). Refer to Figure 721.
- (2) Install rollers on pins (40, 40A). Install shims (15, 20, 25) and E-ring retainers (10). Shim to obtain end play of 0.0005 to 0.0020 inch. Observe paragraph 2E procedure when installing E-ring retainers.

B. Install roller guide assemblies (5) on motor mount (50) and secure with screws (45).

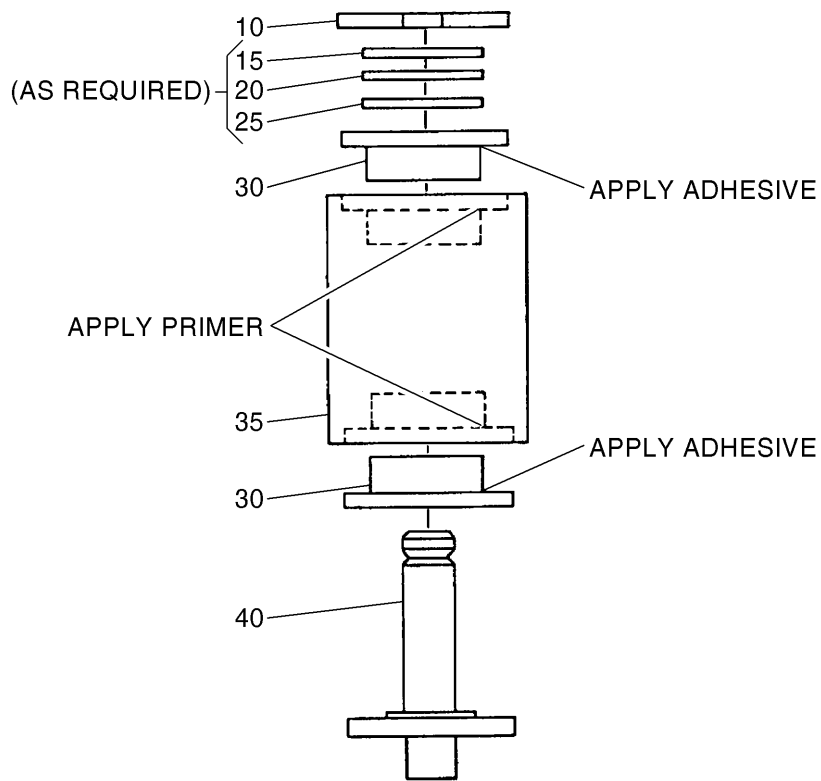
C. If stepper motor (110) is being replaced, dress cable and install drive pulley per these steps:

- (1) Install drive pulley (120) on shaft of stepper motor (110). Insert screw (125) in pulley, adjust pulley to dimension shown in Figure 722, and tighten screws to 5.75 ± 0.25 in-lb.
- (2) Install press nuts (115) in motor flange. Ensure that nuts fit tightly.
- (3) Install inner ferrule (85) on motor wires and slide ferrule to motor case. Refer to Figure 723.
- (4) Install braided shield (10, Figure 701) on motor wires over inner ferrule.
- (5) Install outer ferrule (80) on motor wires over shield. Position ferrules one over the other with end of shield just protruding. Outer ferrule shall be 0.50 ± 0.25 inch from motor case. Crimp outer ferrule.
- (6) Install sleeving (11, Figure 701) over shield. Cut sleeving to approximate length shown in Figure 723, but do not heatshrink.
- (7) Install clamp (95) over motor wires.
- (8) Install connector (90) per wiring diagram of Figure 723. Ensure that cable length will be 6.00 ± 0.25 inches between extreme end of connector and motor case. Cut wires to required length and solder to connector contacts.

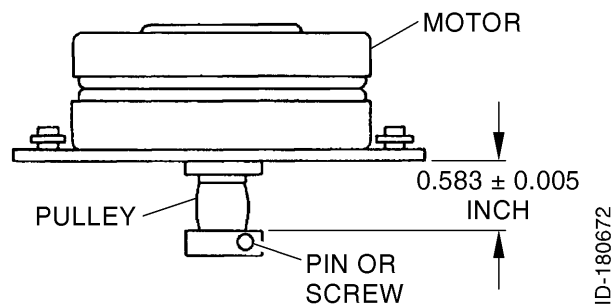
Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Roller Guide Assembly
Figure 721

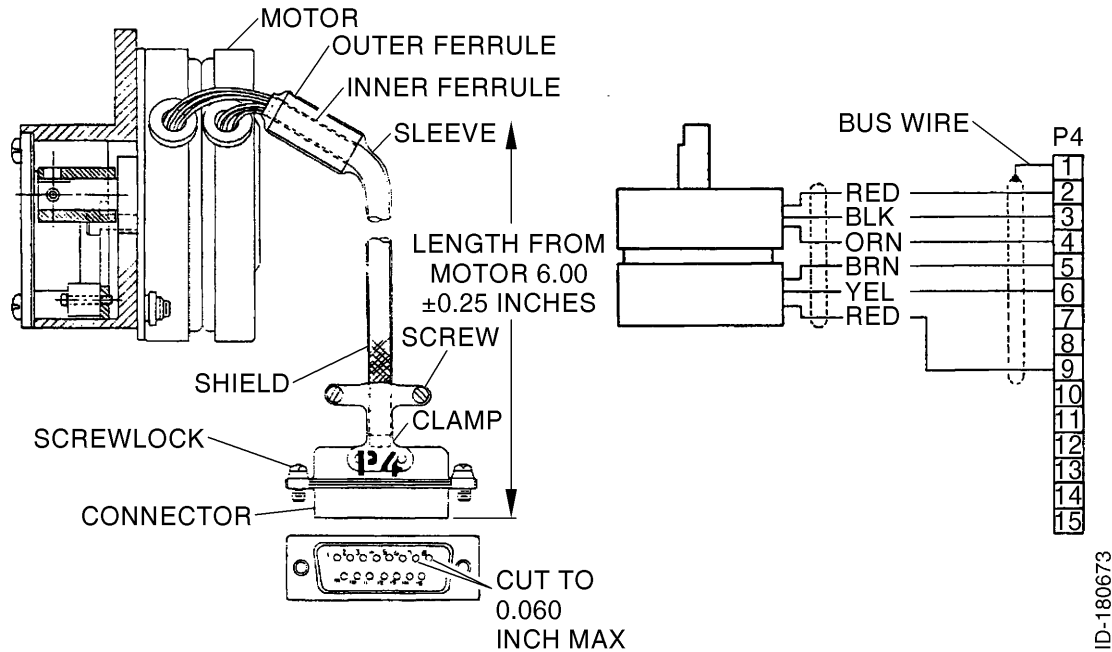


Stepper Motor Assembly
Figure 722

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Motor Cable Assembly
Figure 723

- (9) Solder a 22 AWG bus wire (6, Figure 701) to connector shield and to P4-1. Cut pins 7 and 8 of connector to 0.060 inch maximum.
- (10) Position sleeving as close as possible to motor and heatshrink. Remove and discard screws supplied with clamp (95) and install new screws (102). Slide clamp down to connector and tighten screws. Build up cable diameter as required with cloth tape (21, Figure 701).
- (11) If connector is new, stamp P4, 0.25 inch minimum height, on both sides with permanent black ink (19, Figure 701) as shown.

- D. Install motor mount (50) on stepper motor (110) and secure with screws (55) and washers (60).
- E. Install clamp (65) and secure with screw (76) and washer (77).

9. Distribution Board (IPL Figure 12)

- A. Replace components (5 through 20) on printed wiring board (30, 30A) as necessary to repair faults. Replace wires as necessary per Figure 724. Refer to Figure 122 for schematic.
- B. Maximum height of all components on component side shall be 0.156 inch above board surface. On circuit side, maximum protrusion of component leads and wires shall be 0.03 inch.
- C. Bend leads of C1 (5) so that body of C1 lays flat against board.
- D. Install U1 (10) on circuit side of board. Clip or clinch leads 0.04 inch maximum from board surface.

31-30-32

Page 732
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

FROM	TO	SIZE (AWG)
E15	E48	26
E16	E49	26
E17	E47	26
E18	E50	26
E19	E46	26
E20	E51	26
E21	E45	26
E22	E52	26
E27	E57-60	26
E28	E53-56	26

Wire Chart, Distribution Board
Figure 724

- E. Install terminals (35, 37) on board. Ensure that terminals fit tightly after swaging.
- F. Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.

10. **Motor Drive Assembly (IPL Figure 11)**

- A. If motor assembly (90, 90A, 90B) is purchased and replaced as an assembly, it need only be installed to use. If the stepper motor (105/105A) and drive pulley (95B, 95C) are purchased separately, assemble in accordance with Figure 722 and the following steps:
 - (1) Install drive pulley (95B, 95C) onto stepper motor (105/105A) shaft. Insert a setscrew (7, Figure 701) through pulley and into motor shaft flat.
 - (2) Adjust pulley to the dimension shown in Figure 722 and tighten setscrew to a torque of 1.75 ± 0.25 inch-pounds.
 - (3) Drill a 0.108 to 0.112 inch diameter hole through center of pulley and motor shaft 90 ± 2 degrees from setscrew hole. Deburr pulley and shaft.
 - (4) Install spring pin (100A) through pulley and motor shaft. Remove and discard setscrew.
- B. If tension plate (25) roller pins (35) are damaged, replace as follows:
 - (1) Roller pins (35) shall be perpendicular to surface of tension plate (40) (surface opposite pins) within 0.005 inch. If not, replace pins.
 - (2) Pins are welded into tension plate.
 - (3) Press new pins until flange is flush with tension plate and weld on opposite side. Use a heatsink when welding to prevent annealing of pins. Verify criteria of step (1).
- C. If gasket (12) is damaged, pull or scrape adhesive off of cover (5). Clean gasket mounting surface with isopropyl alcohol (3, Figure 701) and allow to dry. Prime (8, Figure 701) gasket

31-30-32

Page 733
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

and cover mating surface. Apply adhesive (9, Figure 701) to gasket and cover. The gasket and cover mismatch on all edges shall not exceed 0.015 inch, maximum.

- D. If motor (105/105A) was replaced in accordance with paragraph 10A, dress cable and install connector and inner and outer ferrule as shown in Figure 723 and the following:
- (1) After cable is dressed, it shall be 6.00 ± 0.25 inches long. Length is from where cable comes out of motor (105/105A) to extreme end of connector (65).
 - (2) Install inner ferrule (85) and outer ferrule (80) onto motor leads and run both ferrules up to the motor case.
 - (3) Install braided shield (10, Figure 701) onto motor leads and run shield to within 0.50 ± 0.25 inch from motor case as shown.
 - (4) Insert inner ferrule (85) between motor leads and shield and outer ferrule (80) over shield. Position ferrules one over the other, with end of shield just protruding, within 0.50 ± 0.25 inch of motor case, and crimp outer ferrule.
 - (5) Install sleeving (11, Figure 701) over shield. Cut sleeving at approximate length shown in Figure 723 but do not heatshrink.
 - (6) Install clamp (75) onto motor cable and position cable to clear adjacent components.
 - (7) Install connector (65) onto motor leads. Prior to soldering leads to connector contacts in accordance with wiring diagram, Figure 723, verify that cable length will be 6.00 ± 0.25 inch between extreme end of connector and motor case. Cut wires to required length and solder to connector contacts along with AWG 26 solid wires (5, Figure 701) as shown in Figure 723.
 - (8) Solder an AWG 22 bus wire (6, Figure 701) between connector shield and connector P4-7 as shown in Figure 723. Cut off connector P4-7 and 8 contacts to within 0.060 inch maximum from connector as shown.
 - (9) Slide clamp (75) down to connector (65) and secure with screwlocks (70).
 - (10) Heatshrink sleeving onto braided shield and ferrules as close to motor as possible as shown. Tighten two clamp screws onto cable.
 - (11) If connector (65) is new, stamp P4, 0.25 inch minimum height on both sides with permanent black ink (19, Figure 701) as shown in Figure 723.
- E. Install motor mount (45 or 45A) onto motor (105/105A). Place clamp (60) over cable and loosely secure with one D washer (57), two flat washers (55) and screws (50). Mounting hardware will have to be removed to install belt in step 11V.
- F. Install tension plate (25) into motor mount and loosely secure with screws (30), flat washers (32) and lock washers (33). These screws are tightened in a later step to adjust belt.
- G. Install two rollers (15) onto roller pins (35) and secure with E-ring (20). Observe paragraph 2E procedure when installing retainer.
- H. If motor drive assembly is not ready to install in unit, place cover (5) onto motor mount and finger tighten four screws (10). Install four lock washers (11) under screws. Cover will have to be removed to install belt in Step 11V.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

11. Tape Transport Assembly (IPL Figure 10)

- CAUTION:** DURING ASSEMBLY OF TAPE DECK, EXERCISE CARE DURING HANDLING TO PROTECT TAPE DECK COMPONENTS (BELTS, SHAFTS, TAPE HEADS, ETC.) FROM DAMAGE. PROVIDE SUPPORT FOR BASEPLATE SO THAT COMPONENTS DO NOT CONTACT WORK BENCH SURFACE. DEPENDING ON WHAT IS BEING REPAIRED, THE PLATES (5 OR 35) MAY BE USED.
- A. If damaged, replace feedthru (410) on P3 - P5 wire bundle as shown in Figure 705. Feedthru's may be slit in order to be installed over wire bundle.
 - B. If wires from distribution board (250) to connectors (395 or 405) are damaged, replace with AWG 26 wire (5, Figure 701). See schematic diagram, Figure 122.
 - C. Replace, if damaged, connector P3 (405), clamp (400) and screwlocks (390). Install heatshrink sleeving (11, Figure 701) under clamp as shown in Figure 705. See schematic diagram Figure 122 for wire identification. Cut connector P3-14 and 15 contacts off to 0.060 inch maximum after installation.
 - D. If shell (385) or clamp (400) is replaced, stamp P3 or P5 on both sides in 0.25-inch minimum letters with permanent black ink (19, Figure 701) in accordance with Figure 705.
 - E. Replace, if damaged, connector P5 (395), shell (385) and screwlocks (390). Fill potting shell (385) with encapsulant (12, Figure 701) above potting shell 0.070 ± 0.010 inch. All exposed wiring shall be covered by potting material. After potting, cut end tabs of shell flush with shell body. Sleeve (11, Figure 701) wire bundle as shown in Figure 705, from next to connector P5 down wire bundle 3.00 ± 0.30 inches. If teflon tape (25, Figure 701) is damaged, replace as necessary in accordance with the dimensions shown in Figure 705.
 - F. Install harness bracket (365) onto baseplate (415, 415A, or 415B) and secure with two lock washers (380), flat washers (375) and screws (370).
 - G. Install harness cover (345) onto baseplate and secure with two lock washers (360), flat washers (355) and screws (350).
 - H. If roller pins (340 or 340A) are replaced, press pins into baseplate (415, 415A or 415B) until pin flange is tight against baseplate. If roller pins (340A) are replaced, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads and screw into baseplate.
 - I. Install motor drive rollers (330/425) onto roller pins and secure with E-ring (335). Observe paragraph 2E procedure when installing E-ring retainer.
 - J. Install two roller pins (300) and four shims (325), two shims between each roller pin flange and baseplate, onto baseplate. These two roller pins are used to hold the belt rollers (240). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw (305) threads and secure pins to baseplate.
 - K. Install belt rollers (240) onto roller pins (300) and secure with E-ring (245). Observe paragraph 2E procedure when installing E-ring retainer.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

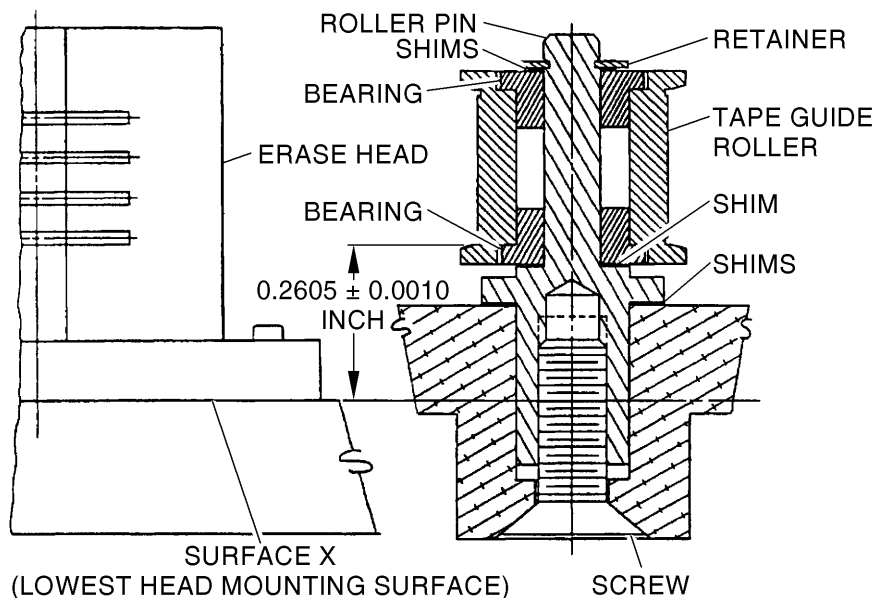
L. Assemble five tape guide rollers (150) in accordance with Figure 725 and the following procedure.

- (1) Install one shim (180), one bearing (175), tape guide roller (150) and a second bearing (175) onto roller pin (300). If shim (160, 165, 170) stack-up was noted during disassembly, use same stack-up as a starting point. Review bearing preload information in **CHECK** section, paragraph 5.
- (2) Mount shims (160, 165, 170) and E-ring retainer (155) onto roller pin (300). Shim as required to obtain an end play of 0.0005 to 0.0020 inch between roller (150) and pin (300). Observe paragraph 2E procedure when installing E-ring retainer.

NOTE: If roller pins (300) were not removed from baseplate during disassembly, verify dimension of step (3). Reshim only if necessary.

CAUTION: IN STEP (3), EXERCISE EXTREME CARE NOT TO GET PRIMER OR STAKING COMPOUND INTO BEARINGS

- (3) Install the roller pin (300) and assembled parts, shims (310, 315, 320) and screw (305) onto baseplate in accordance with Figure 725. Shim as required between roller pin flange and baseplate to obtain dimension of 0.2605 ± 0.0010 inch between baseplate surface X and flange of tape guide roller as shown. Surface X is the right hand erase head with baseplate facing operator as shown by the cross hatched area in Figure 727. After shimming has been completed, apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw (305) threads and secure roller pin to baseplate. Repeat steps as necessary to assure proper dimension.



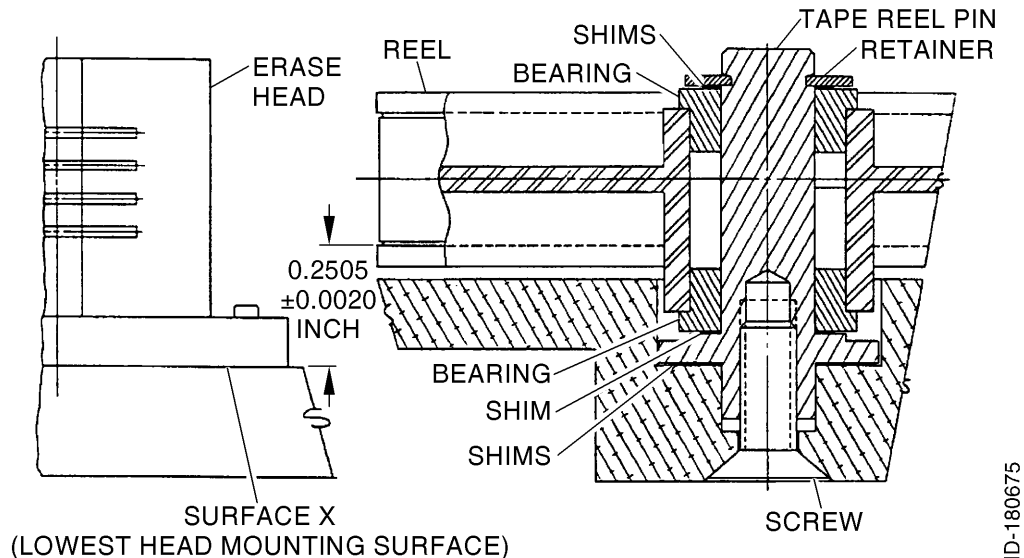
Tape Guide Roller Assembly
Figure 725

31-30-32

Page 736
22 Dec 2006

- M. Assemble loaded reel (55A) and reel (90A) in accordance with Figure 726 and the following procedure.

NOTE: Tape is preloaded on reel with a dynamic tension of 4 ± 1 ounce. Tape is secured to reel with a thin coating of UHU gluestick adhesive (27, Figure 701). Tape is wound on reel with oxide side of tape facing reel. See step U for cleaning adhesive and **WARNING** if tape is being replaced at this step.



Reel Assembly
Figure 726

- (1) Install one shim (85 or 120), one bearing (80 or 115), loaded reel or empty reel (55A or 90A), and second bearing (80 or 115) onto tape reel pin (275). If loaded reel shim (65, 70, 75) or empty reel shim (100, 105, 110) stack-up was noted during disassembly, use same stack-up as a starting point. Review bearing preload information in **CHECK** section, paragraph 5.
- (2) Mount shims (65, 70, 75 or 100, 105, 110) and E-ring (60 or 95) onto tape reel pin (275). Shim as required to obtain an end play of 0.0005 to 0.0020 inch between pins (275) and reels (55A or 90A). Observe paragraph 2E procedure when installing E-ring retainer.

NOTE: If tape reel pins (275) were not removed from baseplate during disassembly, verify dimension of step (3). Reshim only if necessary.

CAUTION: IN STEP (3), EXERCISE EXTREME CARE NOT TO GET PRIMER OR STAKING COMPOUND INTO BEARINGS.

- (3) Install the tape reel pins (275) and assembled parts, shims (285, 290, 295) and screws (280) into baseplate in accordance with Figure 726. Shim as required between tape reel pin flange and baseplate to obtain dimension 0.2505 ± 0.0020 inch between baseplate surface X and flange of reel as shown. Surface X is the right hand erase head

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

with baseplate facing operator as shown by the cross hatched area in Figure 727. After shimming has been completed, apply primer (14, Figure 701) and staking compound (15, Figure 701) to screw (280) and secure tape reel pin to baseplate. Repeat steps as necessary to assure proper dimension.

N. Install shaft (235) through baseplate. Verify that shaft flange is tight against baseplate.

CAUTION: IN STEPS O AND P, EXERCISE EXTREME CARE NOT TO GET PRIMER OR STAKING COMPOUND INTO BEARINGS.

O. Install bearing (230) over shaft and into slot in baseplate. Install wavy washer (225), pulley (215), and loosely secure to shaft with screw (220). Final tightening will be accomplished in step V after belt (45A or 420) is installed.

P. From the tape side of baseplate, install bearing (210) over shaft and into baseplate slot. Install tape pulley (200) onto shaft (235) and secure with screw (205). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw threads prior to installation.

Q. Install distribution board (250) onto baseplate. Verify that U1, located on circuit side of board, is down in rectangular hole through baseplate. Install distribution board shield (265) and lug (270), and secure board and terminal with phenolic washers (262), flat washers (260) and screws (255). Install phenolic washers (262) as required to maintain 0.035 to 0.060 inch spacing from baseplate to board laminate. Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.

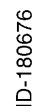
R. Verify that wire bundles are routed approximately as shown in Figure 727 and that cable bundle does not contact bottom plate (35). If bundle ties are damaged, replace with lacing cord (17, Figure 701). For schematic diagram, see Figure 122. Ensure that wires are 0.06 inch or more from all moving parts.

S. Magnetic head replacement as follows:

NOTE: Steps (4) and (5) provide instructions for installing matched sets of magnetic heads. Heads (185, 187, 190) are a matched set and heads (185A, 187A, 190A) are a matched set. These head sets cannot be mixed within a unit. Within each head set, individual heads can be replaced. In other words, heads in step (4) can be replaced individually and heads in step (5) can be replaced individually, but heads in step (5) cannot individually replace heads in step (4) and vice versa.

- (1) Inspect magnetic heads in accordance with the field rework procedure in the **REPAIR** section, paragraph 14. Replace or rework any heads that fail to meet the specified requirements.
- (2) Clean the surface of the magnetic heads per the **CLEANING** section, paragraph 3G(8)(b).
- (3) Degauss magnetic heads in accordance with the procedure in the **REPAIR** section.

Universal Flight Data Recorder / Part No. 980-4100/980-4101

Page 739
22 Dec 2006

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (4) If magnetic head (185, 187, 190) replacement is required, thoroughly clean mating surfaces of baseplate as shown in Figure 727. Clean with isopropyl alcohol (3, Figure 701). Check for and remove any burrs or other protrusions on these mating surfaces. Once preparation of mounting surfaces is complete, mount heads onto baseplate and secure using screws (195). Install one shim (197) under head (185) and one shim (197) under each erase head (190). Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.
- (5) If magnetic heads (185A, 187A, 190A) replacement is required, thoroughly clean mating surfaces of baseplate as shown in Figure 727. Clean with isopropyl alcohol (3, Figure 701). Check for and remove any burrs or other protrusions on these mating surfaces. Once preparation of mounting surfaces is complete, mount heads onto baseplate and secure using screws (195). Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.
- T. Install EOT (125) or BOT (130) sensor onto baseplate and secure with screws (135).
- U. Thread tape from loaded reel (55A) around five tape guide rollers (150), through BOT sensor (130) slot, across heads (185, 187, 190) or (185A, 187A, 190A), through EOT sensor (125) slot and onto empty reel as shown in Figure 727. Attach tape to reel with thin film of UHU gluestick adhesive (27, Figure 701) applied to tape. Remove excess adhesive outside of joint with a lintless cloth moistened in isopropyl alcohol (3, Figure 701). If tape replacement is being performed, apply alcohol (3, Figure 701) liberally to bond line. The alcohol will penetrate and dissolve adhesive. Roll enough turns of tape onto empty reel to assure that 4.00 ± 0.05 inch tape window is on the empty reel side of EOT sensor. Install peripheral belt (50B) as shown in Figure 727 to hold tape in place and to move tape.
- V. In installed in accordance with step O, remove screw (220), pulley (215) and wavy washer (225). Install flat belt (45A/420) onto drive pulley (215). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw threads and reassemble wavy washer, pulley and screw. Tighten screw and place belt between motor drive rollers (330/425) as shown in Figure 727.
- W. Install bottom plate (35) onto baseplate and secure with four screws (40). These screws and plate will be removed and installed in the tape recorder assembly procedure.
- X. If bumpers (15) are damaged, replace as necessary. Secure bumpers to top plate (5) with lock washers (30), flat washers (25) and screws (20).
- Y. Install top plate (5) onto baseplate and secure with four screws (10). Screws and top plate will be removed and reinstalled in tape recorder assembly procedure.

12. Chassis Assembly (IPL Figure 9)

- A. If card guides (295, 305) located on plate (290 through 290C) are damaged, replace as necessary, using rivets (300A, 310A).
- B. If card guides (270, 280) located on plate (265 through 265C) are damaged, replace as necessary, using rivets (275A, 285A).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- C. If parts replacement is required on front panel (-225), replace in accordance with the following:
- (1) Install shockmounts (245) onto panel (260) and secure with nuts (255) and screws (250A). Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.
 - (2) Install handle (235) onto panel and secure with screws (240) and lock washers (242).
- D. Replace damaged parts on chassis (-155/155A/155B/155C) in accordance with the following procedure. Apply primer (14, Figure 701) and adhesive (15, Figure 701) to all screws.
- (1) Install front panel (-225) onto plates (265 through 265C or 290 through 290C) and secure with screws (230 and 233A).
 - (2) Mount hooks (210) onto front panel and secure hooks, front panel and plates with screws (215 and 217A).
 - (3) Install guides (200) onto plates and secure with screws (205).
 - (4) Mount bracket (190, 190A) onto plates and secure with six screws (195).
 - (5) Install shockmounts (170) onto bracket (190, 190A) and secure with screws (175). Install ground strap (180) on lower shockmount.
 - (6) Mount cable bracket (315) onto plates and secure with two screws (320) and lock nuts (325).
 - (7) Install connector plate (160, 160A, 160B, 160C) onto plates (265 through 265C or 290 through 290C) and secure with screws (165).
- E. Replace damaged or faulty parts on motherboard (-90, -90A) in accordance with the following procedure. See Figure 124 for schematic diagram of motherboard and Figure 122 for tape recorder assembly schematic.
- (1) If terminals (145) require replacing, remove by drilling or clipping. Do not enlarge hole by more than 0.005 inch of the original diameter (0.113 +0.003, -0.001 inch). Make certain terminals fit tightly.
 - (2) If anchor nuts (130) require replacing, remove by drilling or clipping rivets (132, 132A). Do not enlarge hole by more than 0.005 inch of the original diameter (0.098, +0.010, -0.000 inch). Install new anchor nuts (130) using two rivets (132, 132A). Make certain rivets fit tightly.
 - (3) If standoffs (140, 150) are replaced, verify that they fit tightly after swaging.
 - (4) Install connectors J3 and J4 (120) onto board (135/135A) and secure with jackposts (125). Jackposts are supplied with washers, and nuts. Discard nuts as they are not required. Install washers between jackposts and connectors. After connectors are installed, fill sockets J3-14 and 15 and J4-7 and 8 with encapsulant (12, Figure 701).
 - (5) If replacement is necessary, install new connectors (100) onto board and secure with screws (105), nylon washers (115, 115A) and nuts (110). Lock washers (117) are no longer required and may be omitted, if damaged. Apply a fillet of sealant (18, Figure 701) around connectors XA1, XA2, XA3, and XA8. Sealant shall not bridge adjacent components, extend under mating area between connectors and board, not cover any wire termination holes. The sealant prevents the conformal coating from wicking up

31-30-32

Page 741
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

through the contact holes in the connector shell, which prevents plugging in mating connector at next higher assembly.

- (6) Clean all new solder joints with isopropyl alcohol (3, Figure 701). Use moistened, clean lintless cloth or Q-tip, depending on area being cleaned. Do not saturate board area or remaining conformal coating.
- (7) Apply conformal coating (2, Figure 701) to all repaired areas. Ref. paragraph 2C for specific instructions.

F. Assemble chassis assembly as follows. See Figure 122, schematic and wire diagram, for wire replacement information. If wire ties are damaged, replace with lacing cord (17, Figure 701).

- (1) If removed, install motherboard (-90, -90A) onto chassis (-155, -155A, 155B, 115C), and secure with twelve screws (95 or 95A). Install one fibre washer (97) as shown in IPL Figure 9. Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw threads.
- (2) If LED indicator (85) is replaced, preassemble spacer (87) onto LED, next to flange, retainer (88) next to spacer, and nut (supplied with LED) next to retainer. Bend leads at rear of LED to allow 0.060 inch minimum clearance between leads and threaded portion of LED. Maximum lead protrusion at rear of LED is 0.060 inch. Solder connections to LED leads. Cover leads, solder connections, and end of LED with encapsulant (12, Figure 701). Mount assembly to front panel and secure with screw (89). Maximum protrusion of LED behind front panel is 0.35 inch maximum as shown in Figure 702.
- (3) Mount connector (80 or 80B) onto front panel and secure with nut supplied with connector. Install cap (77) onto connector to protect pins.
- (4) If removed, mount tywrap clamps (65, 65A) onto chassis and secure with nuts (75) and screws (70).
- (5) Replace straps (60, 60A) on tywrap clamps if damaged. These straps hold the wire bundle to the chassis. If damaged or gone, replace cable tie (17) over wire bundle.
- (6) If replacement was required, install potting shell (55) onto connector (45). Fill potting shell with encapsulant (12, Figure 701) above potting shell 0.070 ± 0.010 inch. All exposed wiring shall be covered by potting material. If connector is new, stamp J5 on both sides, using 0.25 inch minimum letter height, with permanent black ink (19, Figure 701). Cut tabs off sides of potting shell flush with shell body after potting.
- (7) Mount connector bracket (57) onto bracket (190, 190A) and secure with screws (58). Install connector (45) onto connector bracket (57) and secure with jackposts (50).
- (8) Mount elapsed time indicator (30) onto chassis and secure with four nuts (40) and screws (35).
- (9) If solder lugs (28) are damaged, remove nuts supplied with filter (20) and replace. The filter housing is filled with potting. Refer to paragraph 15 of the **REPAIR** section for repair procedure. See Figure 123 for filter schematic.
- (10) Install filter (20) onto chassis and secure with two screws (25). Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to screw threads.
- (11) Install connector J1 (5) onto chassis and secure with eight nuts (15) and screws (10).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (12) Route wire harness from connector J4 (120, 120A) side of board being careful to keep wires away from areas where printed wiring boards will be installed. No wiring shall touch installed boards. Ensure wire bundles are routed such that they do not come within 0.150 inch of the outside of tape recorder assembly when it is installed.

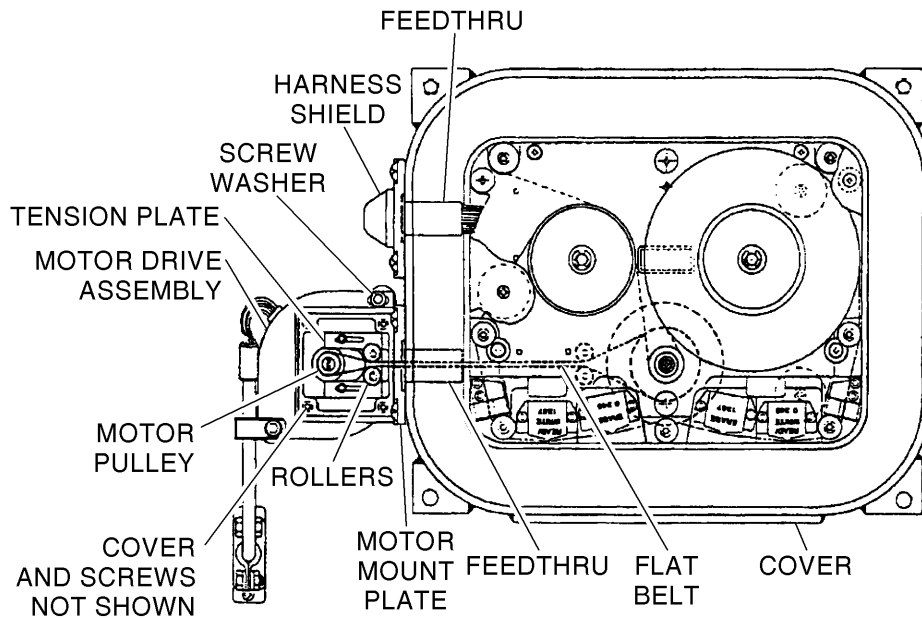
13. Tape Recorder Assembly (1, 1A, IPL Figure 8)

NOTE: Check the tape recorder environmental enclosure covers for delamination, damage and liner separation from the cover, in accordance with the **CHECK** section, paragraph 2I. Repair environmental enclosure covers, if required, in accordance with the **REPAIR** section, paragraph 13.

- A. Remove top plate (5, IPL Figure 10) and bottom plate (35, IPL Figure 10) from transport assembly (25A, 25B, 25C). Install bottom plate (35, IPL Figure 10) into cover (5 or 5A) and secure with screws (30, 30A, 30B or 30C).

CAUTION: IN STEPS B THROUGH I, HANDLE BELT CAREFULLY SO THAT IT IS NOT CRIMPED OR DAMAGED.

- B. Carefully insert flat belt (45A/420, IPL Figure 10) through feedthru (60) and install transport assembly onto bottom plate and feedthru (60) into cover (5, 5A) slot. Install feedthru with flat side up. Secure with four screws (40, IPL Figure 10). Prior to inserting screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads. See Figure 728.



Belt Installation (1, 1A)
Figure 728

- C. Install harness shield (35) over transport assembly wire bundle and onto cover (5A). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads of two screws (40) and install. Insert flange of feedthru (410, IPL Figure 10) into slot of shield and tighten screws (40) with torque of 20 ± 2 pound-inches. Flat side of feedthru will be on cover edge. See Figure 728.

31-30-32

Page 743
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- D. If not already accomplished, remove motor drive assembly (45A), screws (10, IPL Figure 11) and cover (5, IPL Figure 11). See Figure 728.
- E. Carefully insert belt through slot in motor mount plate (55, 55A) and into motor drive assembly (45A) as shown in Figure 728. Install motor mount plate and motor drive assembly onto cover (5, 5A) and secure with screws (50). Prior to installing screw, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads and torque screws to 20 ± 2 pound-inches.
- F. Loosen screws (30, IPL Figure 11) and slide tension plate (25, IPL Figure 11) as far as possible from motor pulley (95B, IPL Figure 11). See Figure 728.
- G. Remove two screws (50, IPL Figure 11), washers (55, IPL Figure 11) and motor assembly (90, 90A, IPL Figure 11). Reinstall motor assembly while sliding pulley through belt loop. Loosely install motor assembly cable clamp, two screws and washers. Place belt between the two rollers (15, IPL Figure 11) as shown in Figure 728.
- H. Adjust tension plate to mid-position and tighten screws (30, IPL Figure 11).
- I. Slide motor outward until a sliding force of 40 ± 10 ounces is obtained and tighten screws (50, IPL Figure 11). If motor sliding force is too high or too low within travel provided (0.15 inch), readjust tension plate and repeat this step.
- J. Install motor drive assembly (45, 45A) cover (5, IPL Figure 11) and secure with four screws (10, IPL Figure 11) and lock washers (11).
- K. Install top cover (5, IPL Figure 10 or 5A, IPL Figure 14) onto tape transport assembly (25A, 25B, 25C) and secure with four screws (10, IPL Figure 10 or 10, IPL Figure 14). Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.
- L. Mount gasket (20) onto bottom cover (5B, 5C) and install top cover (5, 5A) over feedthru and tape transport. Apply lubricant (28, Figure 701) sparingly to 3 or 4 starting threads of screws (10) and secure cover (5, 5A) to cover (5B, 5C) with four screws (10). Torque screws to 71 ± 2 pound-inches.
- M. Apply primer (14, Figure 701) and adhesive (15, Figure 701) to four screws (15) and secure motor drive assembly (45, 45A) and harness shield (35) to cover (5, 5A). Torque screws to 20 ± 2 pound-inches.

14. Tape Recorder Assembly (1B and on, IPL Figure 8)

CAUTION: HANDLE FLAT BELT CAREFULLY TO PREVENT CRIMPING OR DAMAGING IT.

NOTE: Check the tape recorder environmental enclosure covers for delamination, damage, and liner separation from the cover per paragraph 21 of the **CHECK** section. Refer to paragraph 13 of the **REPAIR** section for repair procedures.

A. Tape Transport Assembly Installation

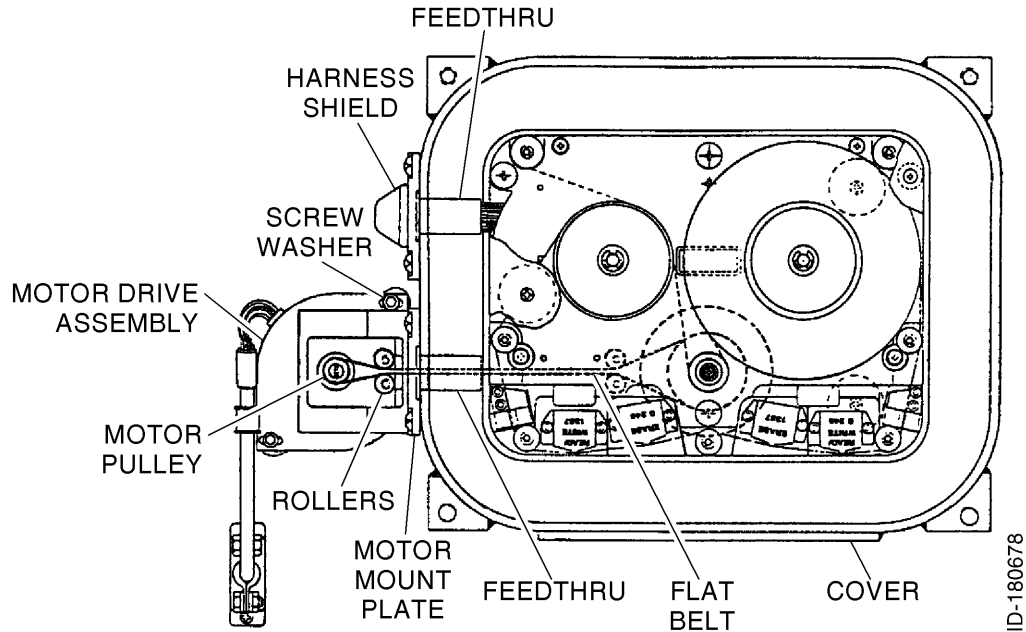
- (1) Bond spacers (70, IPL Figure 8) to bottom cover (5B, 5C) with adhesive (16, Figure 701). Refer to Figure 606 of the **REPAIR** section for mounting locations. Note that these are the Tape Transport Assembly attaching screw locations.
- (2) Install drive belt feedthru (60) in slot of bottom cover with flat side up. Insert flat belt (35, IPL Figure 14) through drive belt feedthru and carefully lower Tape Transport Assembly into bottom cover. Ensure that flat belt remains between roller guides on bottom of

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

transport. Secure transport to cover with screws (30A, 30C). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws. Refer to Figure 729.



Flat Belt Installation (1B and on)
Figure 729

- (3) Insert flat belt through motor mount plate (55A), with counterbore facing bottom cover, and into motor drive assembly (45A).
- (4) Install motor mount plate and motor drive assembly on bottom cover and secure with screws (50). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws. Torque screws to 20 ± 2 in-lb.
- (5) Loosen screws (55, IPL Figure 13) and slide motor toward cover. Place flat belt between rollers guides (5, IPL Figure 13) and over drive pulley as shown in Figure 729. Slide motor outward just far enough to remove slack in flat belt and tighten screws.
- (6) Install gasket (20, IPL Figure 8) on bottom cover.
- (7) Install harness shield (35) over wire bundle and secure to bottom cover with screws (40). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws. Insert flange of feedthru (410, IPL Figure 14) into slot of shield with flat side against gasket. Torque screws to 20 ± 2 in-lb. Refer to Figure 729.

B. Flat Belt Slip Torque Check

- (1) Check flat belt slip torque by holding motor shaft seized, while rotating a torque watch inserted into capstan shaft mounting screw. Slip torque in both directions shall be 38 ± 2 in-oz. Adjust belt tension as necessary to obtain proper slip torque.

31-30-32

Page 745
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

C. Motor Drive Belt Tracking Check

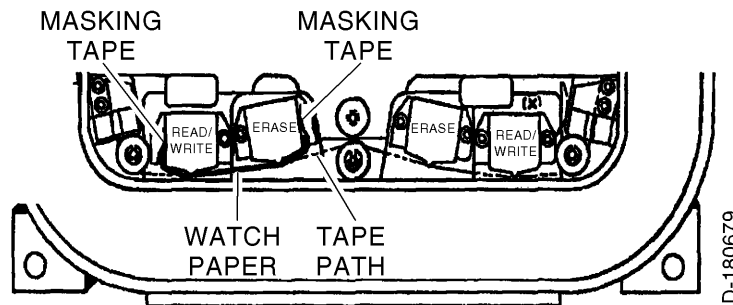
- (1) With power off, connect Tester (1, Figure 901) cables J3, J4, and J5 to Tape Transport Assembly. Connect function generator (3, Figure 901) to ARINC DATA IN jack on Tester.
- (2) Set Tester POWER switch to ON and select Flight Recording mode. Allow belt tracking to stabilize. Observe any tendency to 'binary track'. Loosen motor drive assembly mounting screws (50) and optimize belt tracking by changing the physical position of the motor drive assembly, i.e., by translation and/or rotation, to minimize the binary tracking displacement. When belt tracking is optimally adjusted for minimum binary tracking, tighten motor drive assembly mounting screws. Set Tester POWER switch to OFF.
- (3) Recheck slip torque per step B.

D. Magnetic Tape and P-belt Installation

- (1) Install a 'scratch' loaded reel (45, 45A, 45B, IPL Figure 14), an empty reel (50, 50A, IPL Figure 14), and a 'scratch' peripheral belt (P-belt) (40, IPL Figure 14) on Tape Transport Assembly as shown in Figure 727. Perform Track Location Check per paragraph 7 of the **CHECK** section. When these requirements are met, remove the 'scratch' P-belt.

NOTE: Step D(1), above, is optional and need only be performed if head replacement or tape guide work has been performed.

- (2) With the P-belt removed, use a force gage to measure wobble (displacement) on each reel by applying 8.0 ounces of force at upper side, outer edge of reel and measuring deflection, which shall be less than 0.001 inch. Repeat measurement for the opposite side of each reel. Repeat as necessary to assure proper dimension.
- (3) Remove 'scratch' loaded and empty reels.
- (4) Place a piece of watch paper (33, Figure 701) so that a single layer covers the width of the heads and is long enough to cover 2 adjacent heads, i.e., read/write and its associated erase head as shown in Figure 730. Attach the folded watch paper to the pair of heads with masking tape such that, when the new reel of tape is installed on the transport, the tape will contact the watch paper and not the masking tape or either of the heads. Repeat this procedure for the opposite pair of heads.



Watch Paper Installation (Tape Cleaning)
Figure 730

Honeywell

COMPONENT MAINTENANCE MANUAL

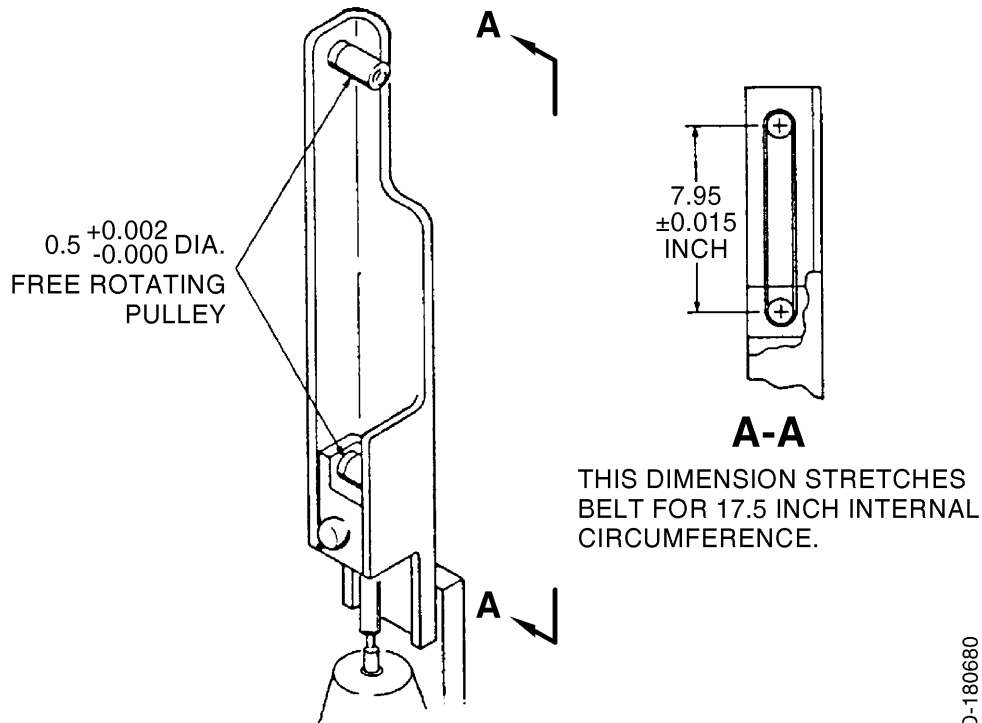
Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (5) Install an empty tape reel (50, IPL Figure 14) on the reel hub farthest from the motor and secure with screws (55), lock washers (65), and flat washers (60). Temporarily install an empty tape reel on the other reel hub.

CAUTION: HANDLING AND INSTALLATION OF PERIPHERAL BELTS MUST BE ACCOMPLISHED WITH EXTREME CARE TO AVOID OVERSTRETCHING. OVERSTRETCHING, LOCALLY OVER SHORT LENGTHS OF THE BELT OR OVER THE ENTIRE LENGTH, CAN IMPART NONVISIBLE DAMAGE WHICH MAY ADVERSELY AFFECT THE DATA PERFORMANCE OF THE RECORDER.

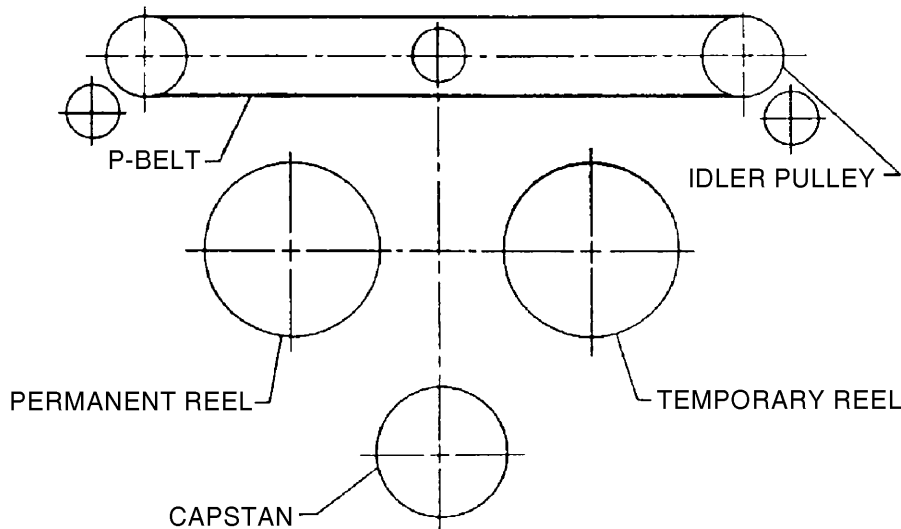
- (6) Check peripheral belt to be installed per these steps:
- (a) If installing a new belt, very carefully, pushing a small amount each time, push belt from storage fixture while working around its perimeter, to prevent stretching belt.
 - (b) Examine a used belt (without stretching) for dirt, tears, nicks, or wear. Discard belt if any discrepancies are present. If belt is satisfactory, wipe clean using a lintless cloth moistened in isopropyl alcohol and allow to air dry. Discoloration of belt surface is allowable.
 - (c) For either new or used belts, use a Belt Test Fixture (25, Figure 901) to measure the belt force at elongation. The force, measured when belt is stretched to 17.5 inches internal circumference as shown in Figure 731, shall be 67 to 75 ounces for -004 (40, IPL Figure 14) and 67 to 80 ounces for -005 (40A, IPL Figure 14).
- (7) Immediately install the peripheral belt by carefully and slowly stretching it:
- (a) Around the 2 idler pulleys as shown in Figure 732.
 - (b) Around the capstan pulley and the empty reels as shown in Figure 733.
 - (c) Carefully and slowly stretch the belt using 3 fingers as shown in Figure 734. Remove the temporary empty reel and install a loaded reel (45, 45A, 45B, IPL Figure 14). Carefully release the belt. Rotate the capstan pulley several revolutions to let the belt stretch uniformly.
 - (d) Inspect the peripheral belt for dirt, tears, nicks, and holes by slowly rotating the capstan shaft. Replace belt if any discrepancies are present.
- (8) Install tape loading fixture (27, Figure 901) on tape transport at the 3 top plate mounting bosses as shown in Figure 735, and tighten fixture mounting screws.

NOTE: *Tape loading fixture is optional.*



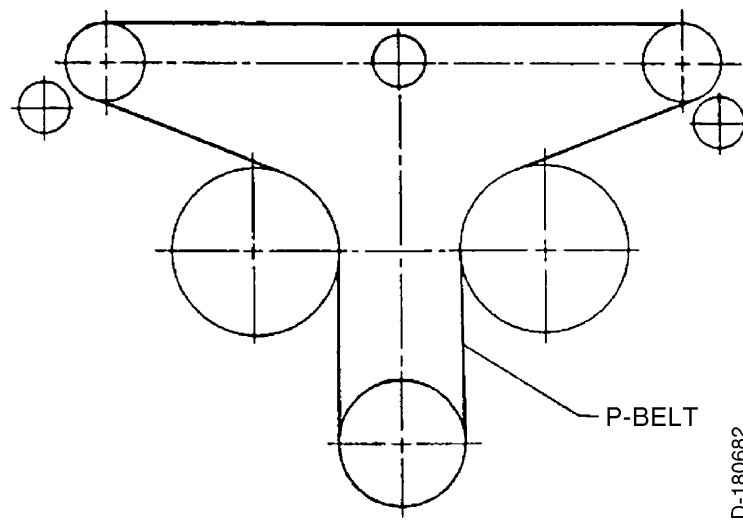
ID-180680

Peripheral Belt Force at Elongation Measurement Limits
Figure 731

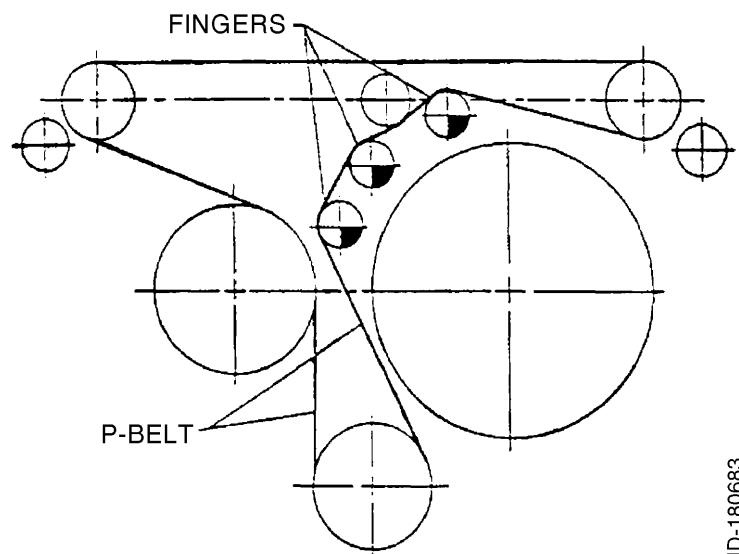


ID-180681

Peripheral Belt Installation on Idler Pulleys
Figure 732



Peripheral Belt Installation on Capstan Pulley
Figure 733

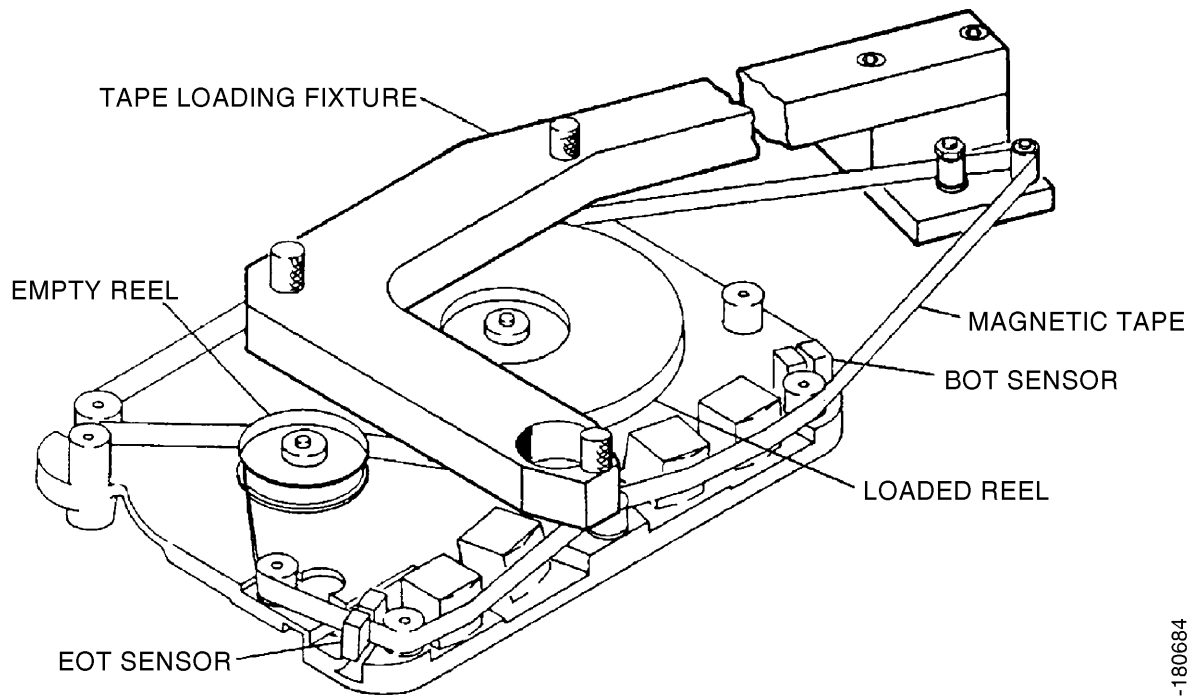


Peripheral Belt Installation on Loaded Reel
Figure 734

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180684

Tape Loading Fixture Mounting Location
Figure 735

- (9) Route the end of the tape from the loaded reel over the guides on the tape loading fixture and back through the transport center tape guide, heads, tape guides, and BOT sensor. (Feed tape off the pack by rotating the capstan shaft.) Sparingly apply adhesive (27, Figure 701) to the end of the tape 0.1 to 0.3 inch along the length of the tape. Attach the tape to the reel taking care to visually center the tape on the edge of the tape reel. Remove excess adhesive outside of joint with a lintless cloth moistened in isopropyl alcohol (3, Figure 701). If tape is being replaced, liberally apply isopropyl alcohol to bond line. The alcohol will penetrate and dissolve adhesive.

NOTE: Tape is preloaded on a reel with a dynamic tension of 4 ± 1 ounces. Refer to Tape Reel Loading Procedure in paragraph 15 of the **REPAIR** section.

- (10) Take the slack out of the tape path from reel-to-reel by sliding the empty reel against the peripheral belt until the tape is taut and properly routed over each tape guide.
- (11) Manually advance the tape onto the reel by rotating the capstan until the clear tape window is wound onto the pack. Observe that the tape is stacking uniformly.
- (12) Set the Tester (1, Figure 901) POWER switch to ON. Select 'REWIND' mode, and monitor progress of the tape stacking while allowing all the tape to transfer from reel-to-reel. At EOT, if optional tape loading fixture is used, remove the tape from around the loading fixture roller and remove loading fixture. Route the end of tape through the sensors and around heads and rollers. Sparingly apply adhesive (27, Figure 701) to the

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

end of the tape 0.1 to 0.3 inch along the length of the tape. Attach the tape to the reel taking care to visually center the tape on the edge of the tape reel. Remove excess adhesive outside of joint with a lintless cloth moistened in isopropyl alcohol (3, Figure 701).

- (13) Cycle the tape end-to-end (EOT-to-BOT) in FAST FWD and REWIND modes, a minimum of 2 full cycles, to establish proper pack tension. Monitor the tape to ensure that the pack stacks evenly, producing a smooth surface on the top of the pack, free from single or multiple turn mis-stacking, or otherwise out of flat by more than 0.010 inch. A tape pack that is poorly stacked after 2 passes requires that the tape be replaced and the procedures repeated. Perform Tape Stacking, Peripheral Belt Running Location, and Tape Traversing Checks per Paragraph 8 of the **CHECK** section.
- (14) While the tape is cycling, check that the tape passes uniformly over each tape guide, and that there is no lapping of the tape against the guides or heads. Any tape that mistracks all guides, or that appears scalloped or nonuniform while passing over guides and heads, should be replaced and the procedures repeated. If the tape appears to mistrack a single roller, recheck its perpendicularity per paragraph 3 of the **CHECK** section.
- (15) Remove and inspect the watch paper. If still soiled, attach fresh watch paper per Para. D(4) and repeat the cycling procedure. Inspect and replace watch paper after each cycle as necessary. If after six cycles the watch paper remains soiled, replace the tape and repeat the process. Set Tester POWER switch to OFF.

E. Drive Torque Test

Perform a Drive Torque Test to establish the proper unit torque required to drive the tape mechanism. The torque required to drive the mechanism is dissipated in the peripheral drive belt operation, continuous tensioning of the magnetic tape, overcoming head-to-tape friction, and ball and sleeve bearing friction.

Place a torque watch (17, Figure 901) with a Phillips tip in the screw located on top of the capstan and rotate the torque watch. The torque required to move the capstan in both directions shall be ≤ 10 ounces.

F. Bulk Tape Erasure

Bulk erase the tape, taking care not to leave any imprint on the tape, by moving the degausser (12, Figure 901) in circles around the full tape pack (without touching), and then slowly drawing the degausser away from the tape pack before releasing the erase button.

G. Install Top Plate

Install top plate (4, IPL Figure 14) on Tape Transport Assembly and install either 8 washers (12, IPL Figure 14), or 4 washers (13), or 4 washers (14) between the top plate and Tape Transport Assembly. Secure with 4 screws (10, IPL Figure 14). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws.

H. Install Top Cover

Install top cover (5, 5A). Sparingly apply lubricant (28, Figure 701) to 3 or 4 starting threads of screws (10) and secure top cover to bottom cover. Torque screws to 71 ± 2 in-lb.

Honeywell

COMPONENT MAINTENANCE MANUAL

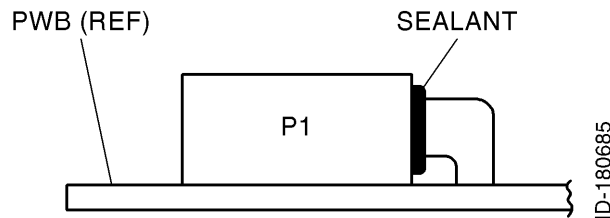
Universal Flight Data Recorder / Part No. 980-4100/980-4101

I. Secure Motor Drive Assembly

Apply primer (14, Figure 701) and adhesive (15, Figure 701) to 4 screws (15, IPL Figure 8) and secure motor drive assembly (45A) and harness shield (35) to top cover. Torque screws to 20 ± 2 in-lb.

15. Power Supply Assembly, A1A1 (IPL Figure 7)

- A. Replace components (5 through 220) on printed wiring board (225, 225A) as necessary to repair faults. Ref. Figure 125 for schematic.
- B. Install connector (105) and secure with screws (110), washers (120A), and nuts (115). Apply a fillet of sealant (18, Figure 701) around back of connector where right-angle pins enter body of connector. Refer to Figure 736. Sealant shall not bridge any adjacent components, cover wire termination holes, or extend under mating area between connector and board. Sealant prevents conformal coating from wicking up right-angle pins and into body of connector.



Sealing Connector
Figure 736

- C. Sleeve (4, Figure 701) leads of R16 (56), wrap leads around U3-1 and U3-2 0.5 to 1 turn on circuit side, and solder in place. See **REPAIR** section, Para. 20, for details.
- D. Install C1, C4, C5, C18, C21 (5, 130) and secure with adhesive (20, Figure 701). Apply a bead of adhesive 0.188, +0.063, -0.000 inch in diameter along capacitor mounting path. Adhesive may extend beyond capacitor, but shall not cover areas to be soldered. Sleeve (4, Figure 701) leads of capacitors. Leads and sleeving shall clear edge of board by 0.200 inch minimum.

NOTE: Strain relief on leads of C18 and C21 must be made toward interior of board.

- E. If R15 (165) requires replacement, select per calibration procedure in **TESTING AND TROUBLESHOOTING** section.
- F. On circuit side, maximum protrusion of leads shall be 0.05 inch.
- G. Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.
- H. If repairing a -002 board, it is recommended the following procedure be performed to make the -002 board into a -003 board.

NOTE: The addition of R16 will eliminate the possibility of oscillations of the output of U3. If U3 oscillations are suspected, then install R16 and reidentify as -003 per the following.

- (1) Install R16 resistor (56) between U3 pins 1 and 2.

31-30-32

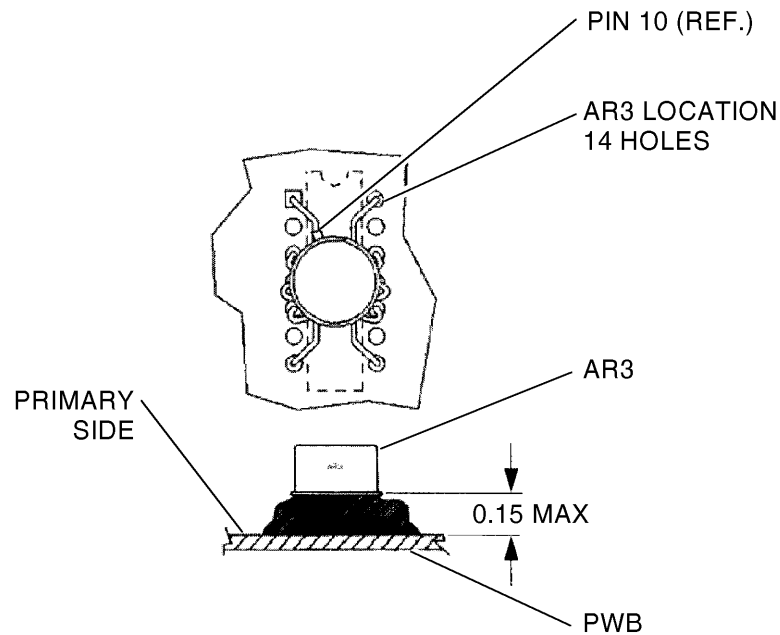
Page 752
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (2) Sleeve (4, Figure 701) R16 leads.
 - (3) Wrap R16 leads around U3 pins 1 and 2 0.5 to 1.0 turns and solder.
 - (4) Mark in the R16 reference designation as close to the component as possible with white permanent marking material.
 - (5) Locate Part No. 722-2440-002 on the board and change it so that it indicates 722-2440-003. User permanent white marking material.
- I. If replacing AR3 with optional part (220B or 220C), install the 10-lead device in the AR3 location (14-pin holes) per Figure 737. Apply adhesive (26, Figure 601) between the base of AR3 and the PWB to prevent shorting.



AR3 LOCATION (14 HOLES)	AR3 (220A, 220B) (10 PINS)	AR3 LOCATION (14 HOLES)	AR3 (220A, 220B) (10 PINS)
1	10	8	5
2	BLANK	9	BLANK
3	1	10	6
4	2	11	7
5	3	12	8
6	BLANK	13	BLANK
7	4	14	9

ID-180686

Installation of AR3 (220B, 220C, IPL Figure 7)
Figure 737

31-30-32

Page 753
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

16. Airframe Wiring Interface. A8 (IPL Figure 6)

- A. Replace components (5 through 135, 145, 185, 190, 195) on printed wiring board (140) and/or (150 through 175) on printed wiring board (180) as necessary to repair faults. Ref. Figure 129 for schematic.
- B. Install connector (45) and secure with screws (50), washers (60A), and nuts (55). Apply a fillet of sealant (18, Figure 701) around back of connector where right-angle pins enter body of connector. Refer to Figure 736. Sealant shall not bridge any adjacent components, cover wire termination holes, or extend under mating area between connector and board. Sealant prevents conformal coating from wicking up right-angle pins and into body of connector.
- C. Replace 22 AWG wires (6, Figure 701) between terminals E1 and E2 or E3 and E4 as necessary. Sleeve (4, Figure 701) wires. Maximum height of top of sleeving shall be 0.02 inch above board surface.
- D. When installing relay driver (145), PWB (180) shall be 0.02 inch minimum from board surface, circuit side of (180) shall not touch components of PWB (140), and overall height of (145) shall be 0.50 inch maximum from board surface.
- E. If terminal strips (175) on PWB (180) require replacement, break off a strip of 3 terminals and a strip of 4 terminals and install on circuit side. Largest diameter pin of each strip installs in PWB.
- F. When installing header assembly (190), break off a strip of 3 sockets (195) and install it in locations of pins 2, 3, and 4 of U3. Break off a strip of 4 sockets (195) and install it in locations of Pins 5 through 8 of U3. Install header assembly in sockets and secure with strap (185).
***NOTE:** Header assembly is encapsulated and is not repairable.*
- G. Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.

17. Recording System Controller. A4 (IPL Figure 5)

- A. Replace components (5 through 229) on printed wiring board (235, 235A) as necessary to repair faults. Ref. Figure 128 for schematic.
- B. Install connector (95) and secure with screws (100), washers (110A), and nuts (105). Apply a fillet of sealant (18, Figure 701) around back of connector where right-angle pins enter body of connector. Refer to Figure 736. Sealant shall not bridge any adjacent components, cover wire termination holes, or extend under mating area between connector and board. Sealant prevents conformal coating from wicking up right-angle pins and into body of connector.
- C. Replace 22 AWG wires (6, Figure 701) as necessary per Figure 738.
- D. If R23 (180) requires replacement, select per calibration procedure in **TESTING AND TROUBLESHOOTING** section.
- E. For assemblies using PWB (235A), install L1 (24) and secure to board with adhesive (37, Figure 701). Apply a small bead of adhesive to body of L1, extending halfway down lead length.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

FROM	TO	SIZE (AWG)	FROM	TO	SIZE (AWG)
<u>For -001 through -005</u>			<u>For -006 and on</u>		
E1	E3	22	E2	E4	22
E12	E11	22	E6	E11	22
E6	E9	22	E5	E8	22

Wire Chart, Recording System Controller, A4
Figure 738

- F. Sleeve (4, Figure 701) body of R30 (56), wrap leads around U17-1 and U17-2 0.5 to 1 turn on circuit side, and solder in place.
- G. Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.

18. Transport Interface Assembly, A3 (IPL Figure 4)

- A. Replace components (5 through 148, 155, 160) on printed wiring board (150, 150A, 150B) as necessary to repair faults. Ref. Figure 127 for schematic.
- B. Install connector (85) and secure with screws (90), washers (100A), and nuts (95). Apply a fillet of sealant (18, Figure 701) around back of connector where right-angle pins enter body of connector. Refer to Figure 736. Sealant shall not bridge any adjacent components, cover wire termination holes, or extend under mating area between connector and board. Sealant prevents conformal coating from wicking up right-angle pins and into body of connector.
- C. If header (130A) requires replacement, break off a 12-pin length and install in board. After installing, cut test point lead next to ground test point to a maximum length of 0.06 inch.
- D. Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.

19. Read/Write Interface, A2 (IPL Figure 3)

- A. Replace components (5 through 255) on printed wiring board (260, 260A) as necessary to repair faults. Ref. Figure 126 for schematic.
- B. Install connector (75) and secure with screws (80), washers (90A), and nuts (85). Apply a fillet of sealant (18, Figure 701) around back of connector where right-angle pins enter body of connector. Refer to Figure 736. Sealant shall not bridge any adjacent components, cover wire termination holes, or extend under mating area between connector and board. Sealant prevents conformal coating from wicking up right-angle pins and into body of connector.
- C. If R54 (150) requires replacement, select per calibration procedure in **TESTING AND TROUBLESHOOTING** section.
- D. If header (240) requires replacement, break off a 9-pin length and install in board. After installing, cut test point lead next to ground test point to a maximum length of 0.06 inch.
- E. If applicable, replace wire across R63 nodes, if damaged, with 22 AWG wire (5, Figure 701).
- F. If applicable, replace wire from U11-5 to U11-7, if damaged, with 30 AWG wire (5, Figure 701).

31-30-32

Page 755
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- G. Clean all new solder joints with isopropyl alcohol (3, Figure 701) and apply conformal coating (2, Figure 701) to all repaired areas per paragraph 2C.

20. Power Supply, A1 (IPL Figure 2)

- A. If anchor nuts (190) are damaged, replace. Secure nuts to support assembly (185, 185A, 185B, 185C) with rivets (195).
- B. If grommets (180) are damaged, replace as necessary. These grommets prevent wire damage from metal support assembly hole.
- C. If wiring is damaged, replace with AWG 20 wire (5, Figure 701). See Figure 125 for schematic diagram and wiring information.
- D. Sleeve (4, Figure 701) component leads and jumpers as necessary to prevent shorts.
- E. If wire bundle ties are damaged, replace with lacing cord (17, Figure 701).
- F. Mount bridge rectifiers (160) onto support assembly (185, 185A, 185B, 185C) and secure with flat washers (175), nuts (170), lug (178A) and screws (165). The lug is used with CR2 rectifier only.
- G. Install thermal switch (150/150A) onto support assembly 185, 185A, 185B, 185C) and secure with nut (155).
- H. Mount insulator (145) and transformer (120) onto support assembly. Secure with one flat washer (130) and one screw (125). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to transformer screw threads and install two washers (140) and two nuts (135). See Figure 125 for transformer wiring information and colors.
- I. If regulators (55 or 55A, 60, 65 or 65A, 80 or 80A) or transistors (70, 75) are faulty, replace as follows:
- (1) Install insulator (115) and regulator and/or transistors onto support assembly. Secure using screws (85A), shoulder washers (105B), lug (110A), washers (100), lock washers (90), and nuts (95).
 - (2) Torque screws (85A) from 4 to 6 pound-inches.
 - (3) Install decoupling capacitors C12 (40A), C13 (45A) and C14, C15 (50A) on opposite side of support assembly on AR5 and AR6 leads. Cut capacitor leads to 0.50 inch maximum length, sleeve (4, Figure 701) and solder in place.
 - (4) Install and solder resistor (53) between AR4 (80/80A), lug (110A) and AR4 lead 1 as shown in IPL Figure 2.
- J. If power supply PWB assembly (25/25A/25B) and support assembly were unsoldered and replaced, verify that when put back together, the wire bundle service loop is 2.0 ± 0.5 inch from edge of support assembly to edge of board when opened 180 degrees.
- K. Install power supply PWB assembly (25/25A/25B) onto support assembly and secure with two washers (35) and two screws (30). Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.

CAUTION: DO NOT DEBURR THE 0.136 INCH DIAMETER HOLES IN THE FOLLOWING PARAGRAPH. DEBURR ONLY THE HOLE EDGES USING HAND TOOLS. DO NOT USE POWER TOOLS TO REMOVE BURRS OR SHARP EDGES.

31-30-32

Page 756
22 Dec 2006

- L. Examine standoffs (20) or spacers (20A) on EMI shield (5 of 5A). If damaged, remove standoffs or spacers by drilling. Do not enlarge hole by more than 0.005 inch of its original diameter (0.136 +0.010, -0.000 inch). Always replace standoffs (20) with spacers (20A). Install spacers by placing shank through hole, placing aluminum washer (21) over shank and then swaging over the washer. Make certain the spacer fits tightly.
- M. Check sponge rubber (23) or foam tape on shield. Sponge rubber and foam tape are to be replaced only with foam tape (24, Figure 701). Replace as follows:
 - (1) Carefully pull or scrape rubber or tape off of shield.
 - (2) Clean rubber or tape mounting area with lintless cloth moistened in isopropyl alcohol (3, Figure 701).
 - (3) If visible, replace rubber or foam tape with foam tape (24, Figure 701) in same location as old rubber or tape. Dimension of tape replacement is shown in IPL Figure 2.
- N. Install EMI shield (5/5A) onto power supply PWB assembly (25/25A/25B) and secure with five washers (15) and screws (10). Prior to installing screws, apply primer (14, Figure 701) and adhesive (15, Figure 701) to threads.

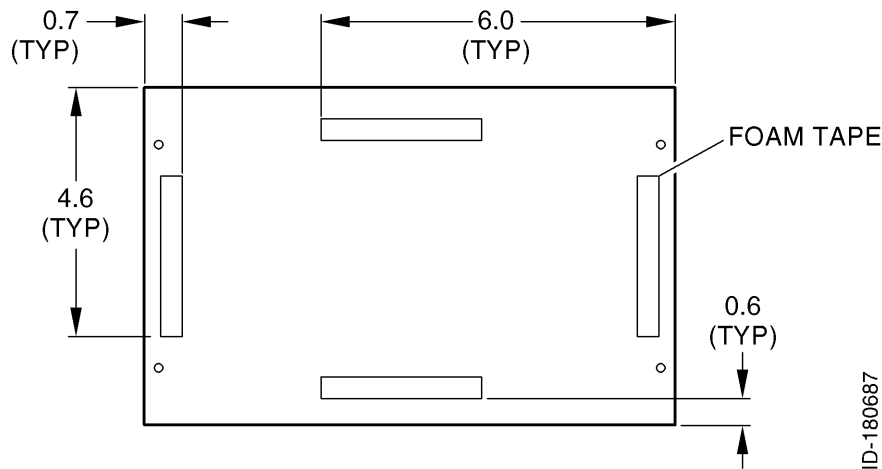
21. Recorder. Universal Flight Data (IPL Figure 1)

- A. Install shield (170) on motherboard standoffs and secure with screws (175), washers (180), and washers (185).
- B. Install foam tape (24, Figure 701) in locations shown in Figure 739. Do not use sponge rubber (187). Clean mounting surface with isopropyl alcohol (3, Figure 701).
- C. Mount Tape Recorder Assembly (150) on chassis shockmounts and secure with screws (155), washers (160), washers (165). Bend cable strap as required and secure under shockmount screw (155). Verify that wire bundles are at least 0.150 inch away from outside of Tape Recorder Assembly. Insert connectors P3, P4, and P5 into chassis connectors J3, J4, and J5, respectively. Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screwlocks before tightening.
- D. Install identification plate (130) on front panel and secure with screws (135). Apply primer (14, Figure 701) and adhesive (15, Figure 701) to screws. If a new plate is being installed, rubber stamp information on new plate that differs from that on old plate. Install patent label (147) only when identification plate (130D) is used to replace existing item.
- E. Install plug (125).
- F. Install A8 Airframe Wiring Interface (120), A4 Recording System Controller (115), A3 Transport Interface Assembly (110), and A2 Read/Write Interface (105) in chassis. Verify that boards are completely inserted so that connectors mate properly.
- G. Install A1 Power Supply (95) and secure with screws (100 and 102) and nuts (103), or install A1 Power Supply (95C) and secure with screws (102A).
- H. Replace silicon rubber bumper (91) on retainer (70) per these steps:
 - (1) Clean mounting surface with lintless cloth moistened with isopropyl alcohol (3, Figure 701).
 - (2) Apply primer (22, Figure 701) to mating surfaces of retainer and bumper.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Foam Tape Installation on Shield
Figure 739

(3) Apply adhesive (9, Figure 701) to mating surfaces of retainer and bumper and mount together. Avoid getting adhesive in slots of bumper. Mount bumper 1.44 ± 0.02 inch from straight end of retainer and on same side of retainer as anchor nut. Bumper and retainer width are the same.

(4) Install anchor nut (92) and secure with rivets (93).

I. Install retainer (70) and secure with screws (75, 90).

J. Install plug (25) into cover (5, 5A, 5B).

K. In Universal Flight Data Recorders (1, 1A, 2A, 3A, or 3B) install 5 screws (27) in unused screw holes if screws were removed or are missing. In Universal Flight Data Recorders (1B, 1C, 1D, or 1E), install 1 screw (28) in unused screw hole if screw was removed or is missing.

L. Install cover (5, 5A, 5B) and secure with screws (10), washers (15), and washers (20).

CAUTION: DO NOT DISASSEMBLE UNDERWATER LOCATING DEVICE (ULD)(30A).

WARNING: NO BATTERY MAINTENANCE IS REQUIRED FOR THE DK100 ULD (30A) (4 YEAR WARRANTY PER THE TECHNICAL MANUAL), OR FOR THE ELP362D (30B) (6 YEAR WARRANTY PER THE USER'S MANUAL) IF THE SERIAL NUMBER BEGINS WITH AN S. DO NOT REMOVE BATTERY AT ANY TIME. AVOID ANY SITUATION THAT COULD POSSIBLY CRUSH OR PENETRATE ULD CASE. DO NOT DISPOSE OF ULD. AT OR NEAR EXPIRATION DATE PRINTED ON CASE, RETURN ULD TO MANUFACTURER FOR SERVICE. FAILURE TO OBSERVE THESE PRECAUTIONS COULD RESULT IN RELEASE OF HAZARDOUS CHEMICALS.

M. Assemble underwater locating device (ULD) (30) per these steps:

(1) Install shock cushion (60) into ULD housing.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAUTION: BE SURE THAT BATTERY END MARKED 'INSERT THIS END' GOES IN FIRST, AS INDICATED BY ARROW. INCORRECT POLARITY WILL PERMANENTLY DAMAGE ULD.

(2) Install battery (65).

CAUTION: ANY SUBSTITUTION OF EITHER O-RING OR LUBRICANT MAY RESULT IN DECOMPOSITION OF O-RING AND LOSS OF SEAL.

(3) Thoroughly clean threads of ULD end cap (45) and threads of housing (P/O 30). Apply a thin coat of lubricant (23, Figure 701) to o-ring groove, housing threads, and new o-ring (50).

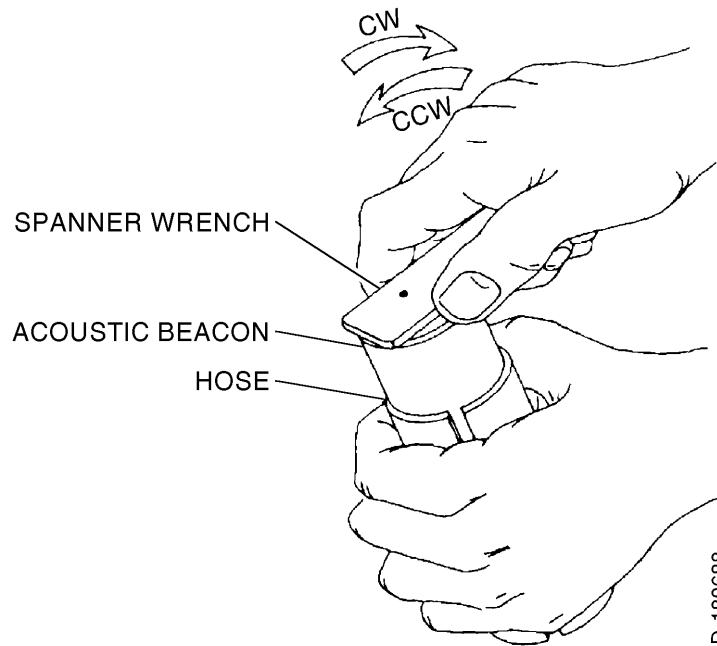
CAUTION: DO NOT STRETCH, NICK, OR CREASE O-RING.

(4) Carefully install o-ring in o-ring groove.

(5) Put shock cushion (55) on end cap and install cap in housing.

(6) Grasp body of ULD with a split rubber hose 4 inches long, 1.25 inches in diameter or equivalent. Refer to Figure 740.

CAUTION: DO NOT CLAMP ULD IN A VISE OR SIMILAR TOOL.



Underwater Locating Device (ULD)
Figure 740

(7) Using a spanner wrench (9, Figure 901), hand tighten cap clockwise until cap flanges contacts ULD body.

N. Install ULD between mounts (40, 40A) and secure to front panel with screws (35, 35A).

31-30-32

Page 759
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

22. Final Confidence Test

A final confidence test is required whenever a major maintenance action has been accomplished, and especially if the Tape Transport Assembly was the subject of the major maintenance action. To perform a final confidence test, perform the functional test procedure given in paragraphs 2B through 2L of the **TESTING AND TROUBLESHOOTING** section. Pay particular attention to the checkstroke stability test, the write-to-tape tests, and the gain margin test. In the checkstroke stability test, closely examine the modulation envelope and evaluate it for dropouts. Dropouts are signal amplitude losses where the signal momentarily disappears, causing a data check error. When making these examinations on all 8 tracks, monitor the waveform for at least 60 seconds on each track.

23. Storage After Assembly

It is recommended that units not be stored in an unpackaged condition for longer than 6 months. Units stored for 6 months or less can be functional tested and placed directly into service.

If it is anticipated that units will be stored for longer than 6 months, perform the Extended Storage procedures of paragraph B. When units are removed from extended storage, perform the procedures in Document Number 032-0002, Recommended Maintenance of Honeywell Universal Flight Data Recorder, 5000-hour column of Table A, prior to placing the unit into service. The recording tape need NOT be replaced on units removed from extended storage. Performance of the recommended scheduled preventive maintenance procedures on units stored longer than 6 months will reduce the possibility of placing a unit into service with degraded bearing and peripheral belt life expectancy.

A. Temporary Storage

- (1) Place equipment in a waterproof plastic bag.
- (2) Place a packet of silica gel or other desiccant in bag.
- (3) Seal open end of plastic bag.
- (4) Place equipment in corrugated cardboard carton (preferably the original shipping carton) with packing.
- (5) Store on rack away from vibration and shock.
- (6) Storage temperature range shall not exceed -55° C to +71° C (-67° F to +159.8° F).

B. Extended Storage

- (1) Place equipment in a waterproof plastic bag.
- (2) Seal open end of plastic bag.
- (3) Place equipment in corrugated cardboard carton (preferably the original shipping carton) with packing.
- (4) Store in clean, dry area, away from vibration and shock.
- (5) Storage temperature range shall not exceed -55° C to +71° C (-67° F to +159.8° F).
- (6) Humidity range shall be 10% to 95% noncondensing.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

C. Peripheral Belt Storage (40, Figure 14).

- (1) Belts are supplied on a plastic storage fixture, Part No. 600-5633-001.
- (2) Belts must be stored on and remain on storage fixture until just prior to installation in a unit.
- (3) Save storage fixture for reuse.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

FITS AND CLEARANCES

The fits and clearances that shall be observed and maintained for the unit to operate properly are listed in Figure 801.

PART/AREA	IPL FIG-ITEM	ADJUSTMENT AND TOLERANCE	ASSEMBLY PARAGRAPH
Chassis wire harness to tape recorder cover	1 - 150	Wire bundles are routed such that they do not come within 0.150 inch maximum of outside of tape recorder.	21C, 12F(12), 4J(2)
Standoffs or spacers	2 - 20 2 - 20A	If drilled out, enlarge hole no more than 0.005 inch.	20L
Wire bundle service loop	2 - 25 2 - 185 15 - 25 15 - 195A	If power supply assembly and cover are opened 180 degrees, the wire bundle service loop shall be 2.0 ± 0.5 inches from edge of support assembly to edge of board. Service loop shall be 0.75 inches	20J 5I
Capacitor, resistors	2 - 40A, 45A, 50A 15 - 40, 45, 50, 55	Leads shall not exceed 0.50 inch maximum length	20I(3) 5H
Screw	2 - 85A	Torque to 4 to 6 pound-inches.	20I(2)
Clinch nuts	13 - 115	If drilled out, enlarge hole no more than 0.005 inch.	8C(2)
Header	3 - 240A 4 - 130A	Clip test point on header next to ground test point to 0.06 inch long.	19D 18C
Relay driver	6 - 145	PWB (180) shall be 0.02 inch minimum from board surface; maximum overall height of (145) from board surface shall be 0.50 inch.	16D
Sleeving	6 -	Top of sleeving shall be 0.02 inch maximum above board surface.	16C
Wire leads on circuit side	6 - 12 - 12 - 10	Lead protrusion through board shall be 0.03 inch maximum; U1 leads shall be 0.04 inch maximum from board surface.	9B 9D
Screw	8 - 10	Torque to 71 ± 2 pound - inches.	13L, 14H
Screw	8 - 15	Torque to 20 ± 2 pound - inches.	13M, 14I
Screw	8 - 40	Torque to 20 ± 2 pound - inches.	13C, 14A(7)
Screw	8 - 50	Torque to 20 ± 2 pound - inches.	13E, 14A(4)

Fits and Clearances
Figure 801 (Sheet 1 of 4)

31-30-32

Page 801
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART/AREA	IPL FIG-ITEM	ADJUSTMENT AND TOLERANCE	ASSEMBLY PARAGRAPH
Potting Shell	9-55 10 - 385 14 - 385	Fill potting shell with sealant 0.070 ± 0.010 inch above shell.	12F(6) 11E 7F, 6F
LED indicator	9 - 85	Lead protrusion behind LED 0.060 inch maximum. LED to extend behind front panel no more than 0.35 inch.	12F(2) 4G(2), 4G(4)
Anchor nuts	9 - 130	If drilled out, enlarge hole no more than 0.005 inch.	12E(2)
Terminals	9 - 145 12 - 35, 37	If drilled out, enlarge hole no more than 0.005 inch.	12E(1) 9E
Tape transport wiring	10 - 14 -	Wires and bundles shall be a minimum of 0.06 inch from all moving parts.	11R 7M(2), 6L(2)
Tape reel	10 - 55, 90	Reel flange shall be 0.2505 ± 0.0020 inch above baseplate surface X.	11M(3)
Bearing	10 - 80 10 - 115	End play shall be 0.0005 to 0.0020 inch between pins and reels.	11M(2)
Tape guide roller	10 - 150 14 - 175	Height of tape guide roller flange shall be 0.2605 ± 0.0010 inch above baseplate surface X.	11L(3) 6Q(3)
Bearing	10 - 175 13 - 30 14 - 170A, 325, 470, 550	End play shall be 0.0005 to 0.0020 inch between roller and pin.	11L(2) 8A(2) 7S(3) 6Q(20) 7V(3)(c)
Distribution Board	10 - 250	Spacing between baseplate and board shall be 0.035 to 0.060 inch.	11Q
Connector pins	10 - 405 14 - 405	Cut connector pins P3-14 and 15 to 0.060 inch maximum from connector.	11C, 7D 6D
Wire bundle sleeve	10 - 14 -	Wire bundle sleeve to extend 2.70 to 3.30 inch down bundle from P5.	11E, 7F 6F
Cable assembly	10 - 14 -	Cable length from tape deck to P3 is 7.18 to 7.38 inches and to P5 is 7.63 to 7.88 inches.	7C
Gasket and cover	11 - 12 11 - 5	Gasket and cover mismatch shall not exceed 0.015 inch maximum on all edges.	10C
Roller pins and tension plate	11 - 35 11 - 40	Roller pins perpendicular to tension plate within 0.005 inch.	10B(1)

Fits and Clearances
Figure 801 (Sheet 2 of 4)

31-30-32

Page 802
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART/AREA	IPL FIG-ITEM	ADJUSTMENT AND TOLERANCE	ASSEMBLY PARAGRAPH
Connector pins	11 - 65 13 - 90	Cut off connector pins P4-7 and 8 to within 0.060 inch maximum from connector.	10D(8) 8C(9)
Motor Pulley	11 - 95 13 - 120	Sliding force required for proper belt tension is 40 ± 10 ounces.	13I
Shield to motor case	11 - 13 -	Cable shield and ferrules shall be within 0.50 ± 0.25 inch of motor case.	10D(3), 10D(4) 8C(5)
Cable length	11 - 13 -	Cable from motor case to extreme connector edge shall not exceed 6.00 ± 0.25 inch.	10D(1), 10D(7) 8C(8)
Pulley and motor shaft	11 - 95B,C 13 - 120	Adjust pulley to 0.583 ± 0.005 inch dimension.	8C(1) 10A(2)
Components	12 - various	Height above board surface shall not exceed 0.156 inch maximum.	9B
Screw	13 - 125	Torque to 5.75 ± 0.25 in.-lb.	8C(1)
P-Belt	14 - 40	Force shall be 70 to 85 ounces when belt is stretched to 17.5 inches internal circumference.	14D(6)(d)
Bearing	14 - 70, 70A	Preload shall be 5 to 9 ounces when pin is axially deflected 0.0002 inch.	7P(6)
Bearing	14 - 70B	Preload shall be 2 ± 0.5 pounds when pin is axially deflected 0.0005 inch.	7Q(6)
Reel hub	14 - 80, 80A	Hub flange shall be 0.3600 ± 0.0020 inch above baseplate surface X.	6R(3)
Tape guide roller	14 - 140, 440, 485, 485B	Height of tape guide roller flange shall be per Figures 719 and 720.	7W(3)
Tape guide roller	14 - 140	Dimension between bearing flange surfaces shall be 0.250 to 0.251 inch; dimension between pin flange lower surface and lower bearing flange upper surface shall be 0.090 ± 0.001 inch.	7V(1)(a) 7V(1)(b)
Bearing	14 - 140, 140A	Preload shall be 2 to 4 ounces when pin is axially deflected 0.0005 inch.	6Q(2)
Bearing	14 - 170A	Preload shall be 3.0 ± 1.0 ounces when pin is axially deflected 0.0005 inch.	7V(1)(d)
Bearing	14 - 205, 220	Preload shall be 20 to 40 ounces when pin is axially deflected 0.0002 inch.	6M(3) 7N(6)

Fits and Clearances
Figure 801 (Sheet 3 of 4)

31-30-32

Page 803
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART/AREA	IPL FIG-ITEM	ADJUSTMENT AND TOLERANCE	ASSEMBLY PARAGRAPH
Screw	14 - 210A, 225A	Torque to 12 in.-lbs.	7N(4) 7N(5)
Belt idler pulley	14 - 255, 255A, 255B, 255C	End play shall be 0.0010 to 0.0020 inch.	7T(6) 7U(6)
Straight pin	14 - 260	Shim thickness is fixed at 0.013 inch for 617-4422-012, -013, -023; dimension shall be 0.232 ± 0.001 inch from baseplate surface X to pin shoulder surface for -005, -007 thru -011, -014 and on.	7T(3) 7T(4)
Eccentric pin	14 - 260A	Dimension shall be 0.232 ± 0.001 inch from baseplate surface X to pin shoulder surface.	7U(2)
Eccentric pin	14 - 260A	Pins shall be rotated to achieve a 0.030 inch maximum parallelism between tape transport edge and peripheral belt, along belt length between corner rollers.	6P(9) 7U(10)
Tape guide roller	14 - 485, 485B	Dimension between bearing flanges shall be 0.248 to 0.250 inch for (485); 0.258 to 0.260 inch for (485B).	7V(3)(a)
Screw	15 - 90	Torque to 5 ± 1 in. - lb.	5G
LED	16 - 145	Minimum clearance between LED leads and LED threaded portion shall be 0.060 inch with maximum protrusion of 0.010 inch.	4G(2)

Fits and Clearances
Figure 801 (Sheet 4 of 4)

31-30-32

Page 804
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

SPECIAL TOOLS, FIXTURES, AND EQUIPMENT

The special tools, fixtures, and equipment required to repair, assemble, calibrate, test, and troubleshoot the unit are listed in Figure 901.

NOTE: *Equipment substitution is permissible as long as substituted equipment has equal or better performance characteristics.*

EQUIPMENT AND TYPE		SOURCE AND REMARKS
1.	Tester, Universal Flight Data Recorder, Part No. 980-8049-00X	Honeywell. The -002 tester tests all UFDR versions. The -003 tester tests all UFDR versions plus the Copy Recorder. The -004 tester tests only ARINC 573/717 UFDR versions plus the Copy Recorder.
2.	Oscilloscope, Model 5440	Tecktronix, Inc. (V80009), 14150 SW Karl Braun Dr., P.O. Box 500, Beaverton, OR 97077.
3.	Function Generator, Model 110	Wavetek Microwave, Inc. (V28821), 488 Tasman Dr., Sunnyvale, CA 94089-1705.
4.	Digital Multimeter (DMM), Model 8600A	John Fluke Mfg. Co. (V89536), 6920 Seaway Blvd., P.O. Box C9090, Everett, WA 98206.
5.	Chart Recorder, Gulton 4 Channel, Model TR 444	Deleted, no longer required.
6.	Extender Board, Power Supply, Part No. 617-4100-001	Same source as item 1. (Part of item 1.)
7.	Extender Board, Printed Wiring, Part No. 617-4101-001	Same source as item 1. (Part of item 1.)
8.	Variable DC Power Supply HP6291A or HP6206B	Hewlett Packard Co. (V28480), corporate Hq., 3000 Hanover St., Palo Alto, CA 94304.
9.	Spanner Wrench, Part No. 810-325	Dukane Corp. (V94970), Seacom Div., 2900 Dukane Dr., St. Charles, IL 60174-3395.
10.	Rubber Hose	Split hose, 4-inch length, 1.25 inch ID. Used to hold ULD when changing battery.
11.	Setscrew, Part No. MS51021-10	Required to align motor shaft and pulley if replaced during assembly.
12.	Degausser, Catalog No. 30-112	G C Electronics Co. (V72653), Div. of Hydrometals, Inc., 400 S Wyman St., Rockford, IL 61101-1314.
13.	Resistor Decade Box, Model DB62	Electro Scientific Industries, Inc. (V11837), 13900 NW Science Park Dr., Portland, OR 97229-5411.

Special Tools, Fixtures, and Equipment
Figure 901 (Sheet 1 of 4)

31-30-32

Page 901
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIPMENT AND TYPE		SOURCE AND REMARKS
14.	Resistor, Part No. RCR20G472JS	Resistor, 4.7K, 1/2W, 5%, commercially available.
15.	Card Puller, Part No. 657-0105-001	Same source as item 1. (Part of item 1.)
16.	Checking Fixture, Part No. CF617-4422-001 (900-0034-001)	Same source as item 1.
17.	Torque Watch (with chuck), Model 651C-3	Waters Mfg., Inc. (V94603), Sub of Minelco, Longfellow Center, Wayland, MA 01778.
18.	Test Fixture, Filter	Refer to Figure 609 for schematic of recommended test fixture.
19.	Chisel, 0.5 inch	Commercially available.
20.	Locating Jig. LJ607-1519-001 (900-0055-001)	Same source as item 1. See Figure 902.
21.	Drill, 0.100, +0.025, -0.010	Commercially available.
22.	Drill, 0.120, +0.005, -0.003	Commercially available.
23.	Countersink, 120 degrees	Commercially available.
24.	Tap, No. 23187-04	Commercially available.
25.	Belt Test Fixture, Part No. MIT406-1511-004 (900-0056-001)	Same source as item 1. (See Figure 903.)
26.	Readout and Test Unit (RTU), Part No. 980-4106-00X or 960-0306-00X	Same source as item 1. Refer to Component Maintenance Manual, Doc. No. 012-0507-001. The two RTU's differ in that PN 960-0306-00X includes a Personal computer (PC).
27.	Tape Loading Fixture, Part No. MIT700-1066-001 (900-0031-001)	Same source as item 1.
28.	Lapping Fixture, Part No. LPF617-4422-002 (900-0053-002)	Same source as item 1.
29.	Head Holding Fixture, Part No. LPF617-4422-001 (900-0052-001)	Same source as item 1.

Special Tools, Fixtures, and Equipment
Figure 901 (Sheet 2 of 4)

31-30-32

Page 902
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIPMENT AND TYPE		SOURCE AND REMARKS
30.	Universal Measuring Microscope, 30 power magnification, minimum	Unitron Instruments, Inc. (V06877), 175 Express Street, Plainview, NY 11803-2404.
31.	Bearing Preload Fixture, PLF617-4431-001 (TD Rev. B) (900-0028-001)	Same source as item 1.
32.	High Potential Tester, Model 1864	Genrad, Inc. (V24655), 300 Baker Ave., Concord, MA 01742-2113.
33.	Stopwatch	Commercially available.
34.	Variable Auto Transformer (Variac) Powerstat, Type 3PNB116B	Superior Electric Co. (V58474), 383 Middle St., Bristol, CT 06010-7438.
35.	Electronic Height Gage, Part No. 599-1015	Browne and Sharpe Mfg. Co. (V09058), Precision Park, P.O. Box 156, North Kingstown, RI 02852.
36.	Run-In Fixture, Part No. MIT600-4683-003	Same source as item 1. Use fixture with stepper motor and drive belt (from Tape Transport Assembly) and a motor mover (item 37), or shop-made jumper cables to connect UFDR to Tape Transport Assembly.
37.	Motor Mover	Refer to Figure 904 for schematic and 905 for parts list. Use with item 36.
38.	Surface Plate	DoAll Co. (V18056), 254 N Laurel Ave., Des Plaines, IL 60016-4321.
39.	Hite-Locater, Model 599-5831-18-1	Same source as item 35.

Special Tools, Fixtures, and Equipment
Figure 901 (Sheet 3 of 4)

31-30-32

Page 903
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIPMENT AND TYPE	SOURCE AND REMARKS
40. Automated Test/Transcription Unit Part No. 964-0434-002 or -005 or -042.	Honeywell Includes: PC, Part No. 300-1034-015; Monitor, Part No. 301-0021-001; ATEasy Software, Part No. 331-0017-001; RTT Cable, Part No. 704-1451-001; DAS Cable, Part No. 704-1452-004; RTT/DAS Interconnect Cable, Part No. 704-1477-002; ATU Wraparound Cable, Part No. 704-2562-001; ATU Test Cable Part No. 704-2464-001; RTT CCA, Part No. 722-3737-002; DAS CAA, Part No. 722-3738-002; Interface CCA, Part No. 722-4020-004; Mouse and Keyboard (included with PC); ATEasy Key (included with ATEasy software); Tape Transport Test Script Diskette, Part No. 998-1513-50X, DAS Test Script Diskette, Part No. 998-1514-50X; ATU Self-Test Script Diskette, Part No. 998-1515-50X; ADRAS for Windows Engine/Setup Diskette, Part No. 998-1520-50X; CSV Output Option Diskette, Part No. 998-1527-50X; RTT Card DLL Diskette, Part No. 998-1530-50X; DAS Card DLL Diskette, Part No. 998-1531-50X; UFDR Input Option Diskette, Part No. 998-1539-50X, SSFDR ATU DLL Diskette, Part No. 998-1415-50X. The -005 ATU is the same as the -002 ATU and the -042 ATU, except that it does not have ATU Test Cable, Part No. 704-2464-001 and Interface CCA, Part No. 722-4020-004 (needed for SSFDR). The -002 ATU and the -042 ATU are the same, except that the -042 ATU supports the 4X SSFDR. The ATU Test Cable Part No. for the -042 ATU is 704-2458-006, and the Interface CCA Part No. is 722-4020-006.
41. Variable Power Supply	Commercially available, 115 $\pm$ 5 VAC, 400 Hz $\pm$ 20 Hz, line to neutral, 3.0A.
42. Ultrasonic Test Set Model 42A12	Same source as item 9.
43. Pinglite ULD Tester, Model PL-1 or PL-3	Same source as item 9.
44. Acoustic Test Set, Model ATS-260	Datasonics, Inc. (V4U270), 1400 Route 28A, P.O. Box 8, Cataumet, MA 02534

Special Tools, Fixtures, and Equipment
Figure 901 (Sheet 4 of 4)

31-30-32

Page 904
22 Dec 2006

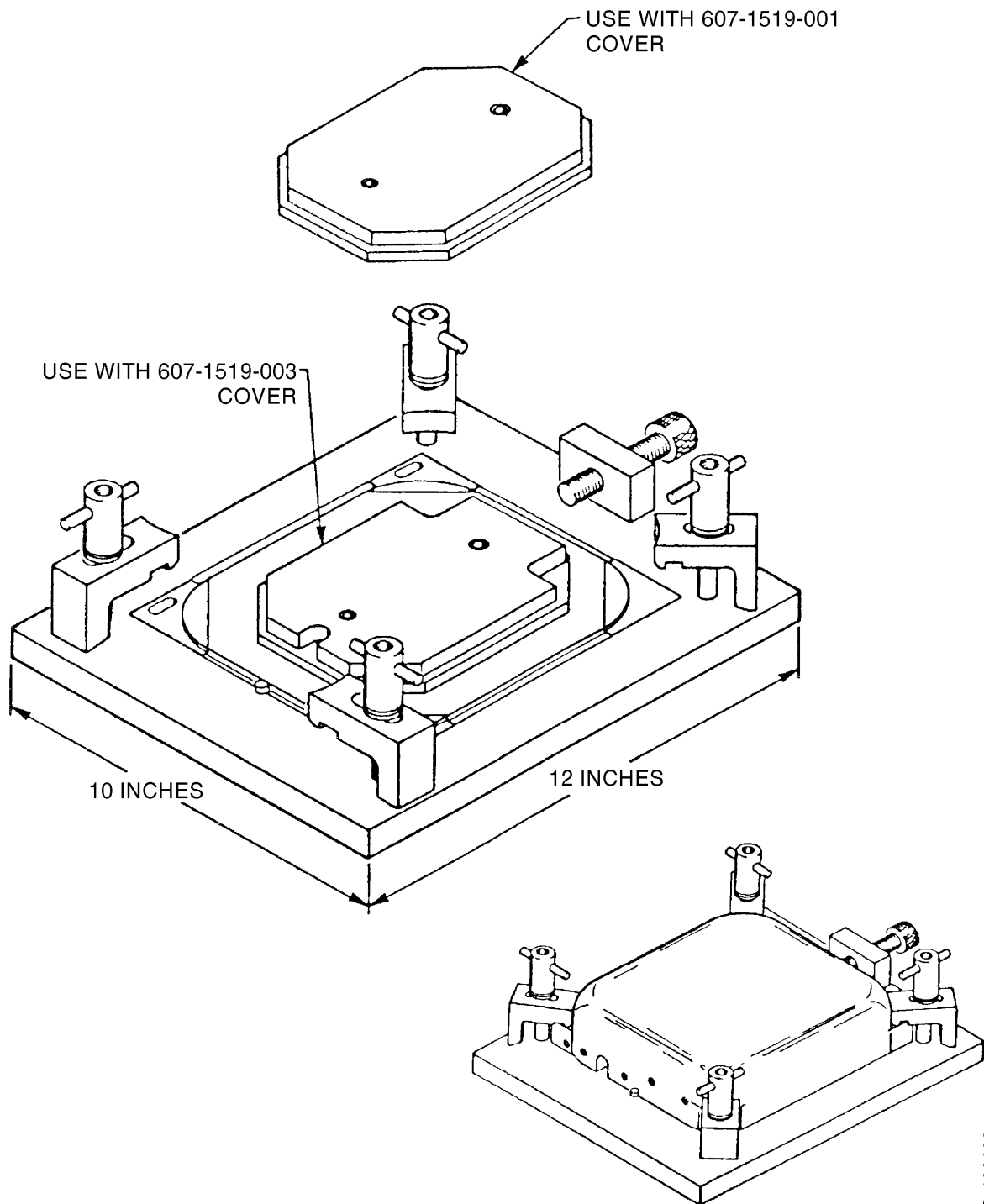
Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



Locating Jig, UFDR Environmental Enclosure Repair
Figure 902

31-30-32

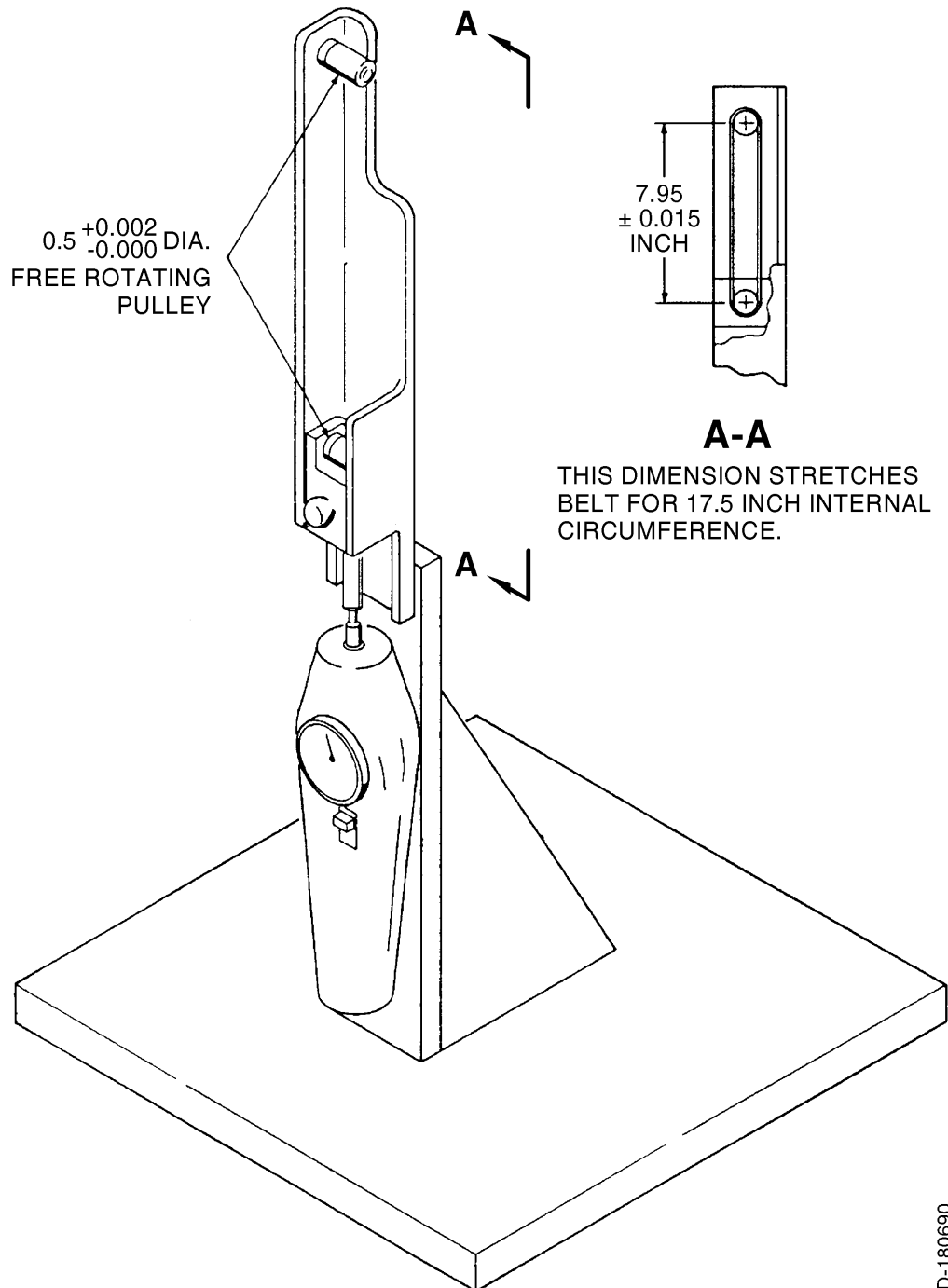
Page 905
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-180690

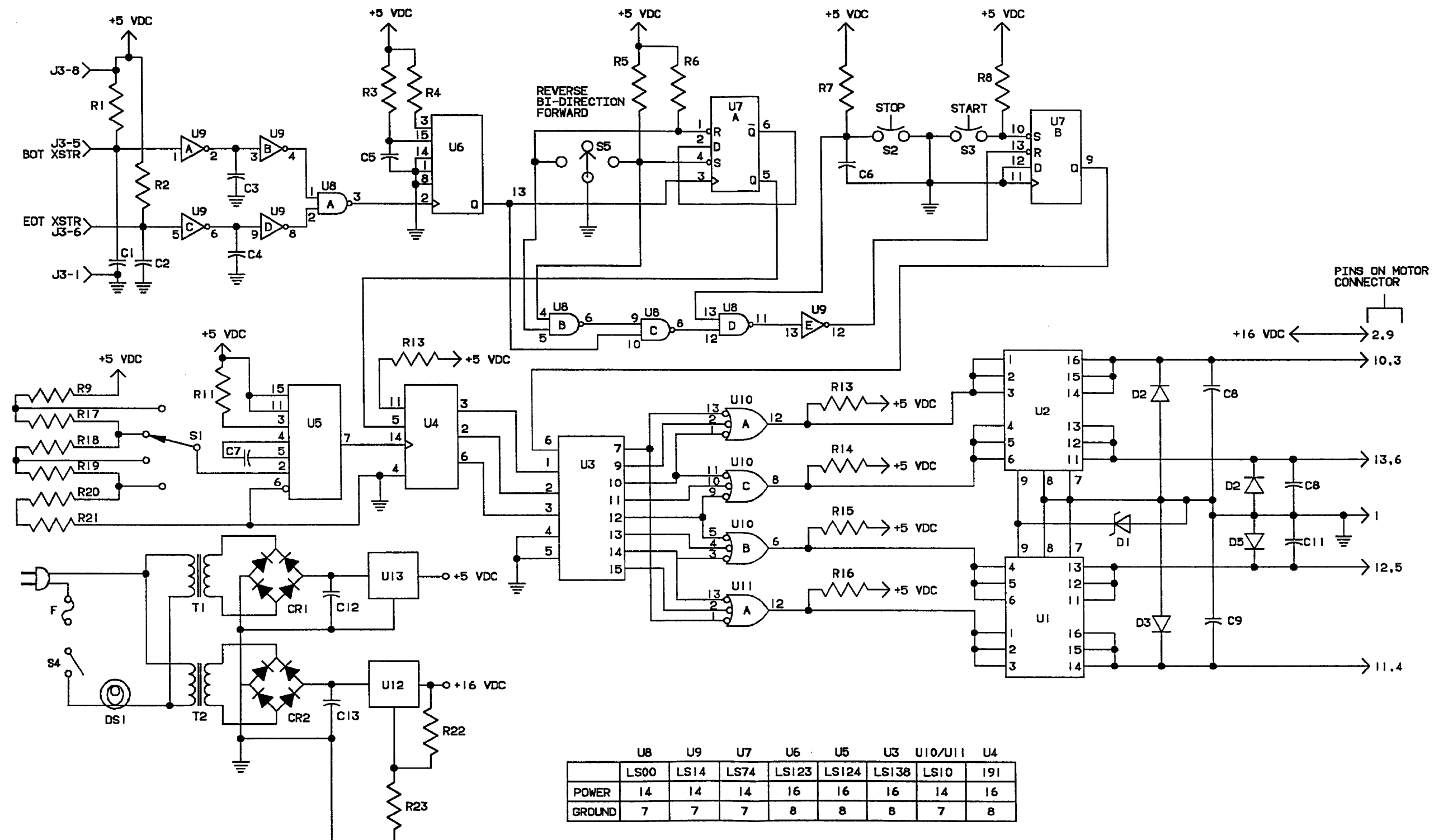
Belt Test Fixture
Figure 903

31-30-32

Page 906
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Esta hoja fue impresa desde la BTV y pertenece a la revisión del manual N° 19, de fecha 22/12/2006 - VERIFICAR SU VIGENCIA



UFR Motor Mover Schematic
Figure 904

31-30-32

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

REF. DES.	PART NUMBER	DESCRIPTION	VENDOR	QTY
U1, U2	SG2003/883B or ULS2003R883 (Honeywell PN 322-2000-002)	IC, Transistor Array	(V34333) (V56289)	2
U3	SNJ54LS138J or 9LS54LS138J883B (Honeywell PN 321-1902-002)	Decoder/Demux	(V01295) (V07933)	1
U4	SN74LS191N or SN74LS191N (Honeywell PN 322-0196-001)	IC, Counter	(V01295) (V04713)	1
U5	SN74LS124	IC, Vco	(V01295)	1
U6	SNJ54LS123J or 54LS123BEXJC (Honeywell PN 321-0182-013)	IC, Multivibrator	(V01295) (V04713)	1
U7	SNJ54LS74AJ or 54LS74ABCXJC (Honeywell PN 321-0182-005)	IC, D Flip-Flop	(V01295) (V04713)	1
U8	SNJ54LS00J or 54LS00BCXJC (Honeywell PN 321-0182-002))	IC, NAND Gate	(V01295) (V04713)	1
U9	SNJ54LS14J or 54LS14BCXJC (Honeywell PN 321-0182-012)	IC, Schmitt Trigger	(V01295) (V04713)	1
U10, U11	SNJ54LS10J or 54LS10BCXJJC (Honeywell PN 321-0182-003)	IC, NAND Gate	(V01295) (V04713)	2
U12	SG117K883B (Honeywell PN 322-0263-005)	IC, Regulator	(V34333)	1
U13	LM123K883C (Honeywell PN 322-0256-004)	IC, Regulator	(V27014)	1
R1, R2	RCR07G124JS (Honeywell PN 103-0000-113)	Resistor, 120K, 1/4W, 5%		2
R4, R11, R21	RCR07G102JS (Honeywell PN 103-0000-063)	Resistor, 1K, 1/4W, 5%		3
R3	RNC55H1333FS (Honeywell PN 104-0232-397)	Resistor, 133K, 1/10W, 1%		1
R5, R6, R8, R12	RCR07G472JS (Honeywell PN 103-0000-079)	Resistor, 4.7K, 1/4W, 5%		4
R7	RCR07G104JS (Honeywell PN 103-0000-111)	Resistor, 100K, 1/4W, 5%		1

UFDR Motor Mover Parts List
Figure 905 (Sheet 1 of 3)

31-30-32

Page 909
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

REF. DES.	PART NUMBER	DESCRIPTION	VENDOR	QTY
R9	RCR07G751JS (Honeywell PN 103-0000-060)	Resistor, 750 Ohms, 1/4W, 5%		1
R13, R14, R15, R16	RCR07G122JS (Honeywell PN 103-0000-065)	Resistor, 1.2K, 1/4W, 5%		4
R17, R20	RCR20G511JS (Honeywell PN 103-0001-056)	Resistor, 510 Ohms, 1/2W, 5%		2
R18, R19	RCR20G241JS (Honeywell PN 103-0001-048)	Resistor, 240 Ohms, 1/2W, 5%		2
R22	RNC55H2430FS (Honeywell PN 104-0232-134)	Resistor, 243 Ohms, 1/10W, 1%		1
R23	RNC55H2551FS (Honeywell PN 104-0232-232)	Resistor, 2.55K, 1/10W, 1%		1
C1 - C4	M39014-01-1573 (Honeywell PN 151-0247-055)	Capacitor, 0.01 UF, 100V, 10%		4
C5	107RSM025M (Honeywell PN 155-0245-004)	Capacitor, 100 UF, 25V, +30/-10%	(V74840)	1
C6	M39003-01-2286 (Honeywell PN 155-0102-242)	Capacitor, 10 UF, 20V, 10%		1
C7	M39014-02-1358 (Honeywell PN 151-0246-036)	Capacitor, 0.33 UF, 50V, 10%		1
C8 - C11	M39014-02-1350 (Honeywell PN 151-0246-030)	Capacitor, 0.1 UF, 100V, 10%		4
C12, C13	673D338H015GP5C (Honeywell PN 155-0257-014)	Capacitor, 3300 UF, 16V, -100/-10%	(V56289)	4
CR1, CR2	M19500/469-001 (Honeywell PN 340-0056-002)	Rectifier		2
D1	1N4970 (Honeywell PN 344-0365-013)	Diode, Zener		1
D2 - D5	1N5614 (Honeywell PN 342-0335-005)	Diode		4
S1	PA2002 (Honeywell PN 372-0016-025)	Switch, Rotary	(V71590)	1
S2, S3	8906K4173 (Honeywell PN 371-0107-001)	Switch, Toggle, 2PDT	(V81640)	2
S4	MSP105D/C12(0) (Honeywell PN 374-0013-008)	Switch, Pushbutton	(V95146)	1

UFDR Motor Mover Parts List
Figure 905 (Sheet 2 of 3)

31-30-32

Page 910
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

REF. DES.	PART NUMBER	DESCRIPTION	VENDOR	QTY
S5	JMT221 ((Honeywell PN 371-0033-002)	Switch, Toggle, 2PDT	(V55459)	1
T1	9A024 (Model TRC5J)	Transformer, 5V	(V62676)	1
T2	9A042 (Model TRC16M)	Transformer, 16V	(V62676)	1
Fuseholder	342014AL (Honeywell PN 401-4005-001)	Fuseholder	(V75915)	1
Heatsink	5423B (Honeywell PN 402-0063-001)	Heatsink, TO-3 (For U12/U13)	(V30161)	2
Button	C12-2 (Honeywell PN 404-9687-003)	Button (For S2, S3)	(V95146)	2
Knob	MS91528-1P2B (Honeywell PN 402-5002-004)	Knob (For S1)		1
DS1	32-2313T (Honeywell PN 312-0016-008)	Light Assembly, Amber	(V95263)	1
Power Cord	17237 (Honeywell PN 315-5015-002)	Cord, Power	(V16428)	1

UFDR Motor Mover Parts List
Figure 905 (Sheet 3 of 3)

31-30-32

Page 911
22 Dec 2006

Use or disclosure of information on this page is subject to the restrictions in the proprietary notice of this document.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

ILLUSTRATED PARTS LIST

1. Overview

A. Introduction

- (1) The ILLUSTRATED PARTS LIST (IPL) gives the parts that are used to make the product identified on the title page (T-1). It also supplies the necessary data to get replacement parts for repair and fault isolation.
- (2) The list of parts is given in general disassembly sequence. The illustrations show where you can find each part and are an important aid in disassembly and assembly procedures.

B. How to Find a Part

- (1) If the Part Number is known, refer to the Numerical Index to find the figure and item numbers in the Parts List.
- (2) If the equipment designator is known, refer to the Equipment Designator Index to find the figure and item numbers in the Parts List.
- (3) After you get the figure and item number, refer to the Parts List for more identification data.

2. Contents of the IPL

A. List of Suppliers

- (1) The names and addresses of companies that supply the items included in the Parts List are shown in the List of Suppliers. Refer to List of Suppliers, paragraph 3.
- (2) The supplier CAGE code is given in the Nomenclature column of the Parts List to identify the supplier.

B. Equipment Designator Index

- (1) The Equipment Designator Index is an alphanumeric list of the equipment designators given to the subassemblies and electrical/electronic parts. The designators are an aid to identify parts on the schematics and wiring diagrams. Refer to Equipment Designator Index, paragraph 4.
- (2) Each entry in the Equipment Designator Index identifies the figure and item numbers where the part is found in the Parts List. The Equipment Designator Index may also identify a geographical location of the equipment designator found on the graphic illustration. A hyphen between two designators shows a continuous sequence of designators (example: A1-A5 is the same as A1 thru A5).

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

C. Numerical Index

- (1) The Numerical Index is an alphanumeric list of all the Part Numbers shown in the Part Number column of the Parts List. Refer to Numerical Index, paragraph 5.
- (2) Also included in the index are the Honeywell Part Numbers that are equivalent to the manufacturer Part Number. Optional manufacturer Part Numbers are not included in the index.
- (3) The Fig. Item column gives all of the locations of a part. If a Part Number is in more than one Fig. Item location, the Part Number is shown only one time in the Part Number column.
- (4) The Total Req column shows the total number of parts that are used at each Fig. Item location.
- (5) The Airline Stock No. column has space for customers to use.

D. Parts List

- (1) An item number is given to each part in the Parts List and on the related figure. The item numbers show the general disassembly sequence. Refer to Parts List, paragraph 6.
- (2) Some of the parts are not shown on a figure, such as a fully assembled item, an unprogrammed microcircuit, and a selection of resistors. Items that are not illustrated have a dash before the item number in the Parts List.
- (3) An item number with a letter suffix (A thru Z except I and O) shows any one of these conditions has occurred:
 - A modification (by a service bulletin)
 - A change in configuration from the initial item
 - A change to a better part (no service bulletin)
 - Accomodation was made for a disassembly sequence.
- (4) The Part Number column shows the Part Number given by the primary source of the part, either Honeywell or a commercial supplier, or the Part Number of a U.S. standard part. When this column shows the Part Number of a commercial supplier or a U.S. standard part, the Nomenclature column may show the Honeywell Part Number. When this column shows the Honeywell Part Number for ordinary commercial hardware, the Nomenclature column gives a description of the part.
- (5) The Airline Stock No. column has space for customers to use.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (6) The Nomenclature column gives the applicable name for each assembly, subassembly, and part. The indentation of the Nomenclature data is used to show the relation of an item to the other items in the Parts List. Other data that may be shown in this column is as follows:
- Shows optional manufacturer Part Numbers (OPT MFG PN) with their supplier code when there is more than one supplier of the part. At the first location of a part with more than one source of supply, the applicable OPT MFG PN data is shown. Any subsequent locations of the same part refer to the first location for the OPT MFG PN.
 - Identifies parts that are electrostatic discharge sensitive with the identification (ESDS). Refer to the special precautions in the INTRODUCTION of this manual.
 - Identifies parts that can have dangerous effects with the warning *HAZARDOUS MATERIAL*. Refer to the special precautions in the INTRODUCTION of this manual.
 - Identifies parts for selection with the words "SEL FR PN MADE BY" These parts are commercially supplied, but Honeywell examines them and selects those that have the necessary quality. Parts for replacement must come from Honeywell.
 - Identifies parts that are airworthiness critical with the identification (AIRWORTHINESS CRITICAL ITEM). Refer to the special precautions in the INTRODUCTION of this manual.
 - Identifies parts that are moisture sensitive with the identification *MOISTURE SENSITIVE*. Refer to the special precautions in the INTRODUCTION of this manual.
 - Identifies parts that can be used when the recommended part is not available.
 - The Nomenclature column of the Parts List shows a supplier code with the letter V before it. This code (from *Cataloging Handbook H4/H8, Commercial and Government Entity (CAGE), Sections A and B*) identifies where to get the specified item.
 - No supplier code is shown in the Nomenclature column of the Parts List if one of these conditions is applicable:
 - The item is a U.S. standard part.
 - Honeywell is the supplier.

Honeywell

COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

- (7) The parts replacement words, in the NOMENCLATURE column, tell if one part replaces another as follows:
- OPTIONAL indicates a part is an optional alternative to other parts in the same item number variant group.
 - SUPSD BY shows the part in the PART NUMBER column is replaced by and is not interchangeable with the item number shown in the note.
 - SUPSD shows the part in the PART NUMBER column replaces and is not interchangeable with the item shown in the note.
 - REPL BY shows the part in the PART NUMBER column is replaced by and is interchangeable with the item number shown in the note.
 - REPL shows the part in the PART NUMBER column replaces and is interchangeable with the item number shown in the note.
 - MAY BE SUBST shows the part in the NOMENCLATURE column with its supplier code is an acceptable substitute for the part in the PART NUMBER column.
 - PRE SB XXXX indicates the part is used if service bulletin XXXX was not done.
 - POST SB XXXX indicates the is used if service bulletin XXXX was done.
 - NON-REPAIRABLE indicates the part cannot be repaired.
 - NON-PROCURABLE indicates the part cannot be purchased.
- (8) The EFF column identifies differences in specific configurations of the top assembly in each figure:
- If this column is empty, the part is used in all configurations of the top assembly for that figure.
 - If this column has a letter code in it, the part is only used in the configuration of the top assembly that has the same letter code.
 - If this column has more than one letter code, the part is used in each top assembly with any of the same letter codes.
- (9) The Units Per Assy column shows the quantity of each part that is used in the next higher assembly. The letters AR are an indication that the quantity changes as necessary. The letters RF identify a reference item.

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

3. List of Suppliers

CAGE	VENDOR NAME AND ADDRESS	CAGE	VENDOR NAME AND ADDRESS
00141	PRECISION INDUSTRIAL COMPONENTS LLC 86 BENSON ROAD P O BOX 1004 MIDDLEBURY, CT 06762-1004	07263	FAIRCHILD SEMICONDUCTOR CORP 333 WESTERN AVE ATTENTION: CORPORATE MARKETING SOUTH PORTLAND, ME 04106
00287	SPIROL INTERNATIONAL CORP. DANIELSON, CT 06239-1434	07933	FAIRCHILD SEMICONDUCTOR CORP SEMICONDUCTOR DIV HQ 3001 ORCHARD PKWY SAN JOSE, CA 95134-2017
01295	TEXAS INSTRUMENTS INC 6414 S HIGHWAY 75 STOP 837 SHERMAN, TX 75090-9493	08289	THE DELBERT BLINN COMPANY INC 4460-A BROOKS STREET MONTCLAIR, CA 91763-4135
01526	GENICOM CORPORATION WAYNESBORO, VA 22980-1999	0E3K6	LANSDALE SEMICONDUCTOR 2330 MICHIGAN AVE SANTA MONICA, CA 90404
02660	AMPHENOL CORPORATION SPECTRA-STRIP/ITD 40-60 DELAWARE AVENUE SIDNEY, NY 13838-1395	12954	MICROSEMI CORPORATION 8700 E THOMAS ROAD SCOTTSDALE, AZ 85251-5073
03038	LONG-LOK FASTENERS CORP 14119 CHADRON AVE HAWTHORNE, CA 90250-0820	13556	LABINAL COMPONENTS AND SYSTEMS INC CINCH MILITARY AEROSPACE DIV 8821 SCIENCE CENTER DR MINNEAPOLIS, MN 55428 - 3619
03694	PRECISION METAL PRODUCTS CO SAN DIEGO, CA 92100	14099	SEMTECH CORPORATION INC 200 FLYNN ROAD CAMARILLO, CA 93012-8790
04713	FREESCALE SEMICONDUCTOR INC 2100 E ELLIOT ROAD TEMPE, AZ 85284-1806	14283	AMPHENOL OPTIMIZE MFG CO 180 N FREEPORT DR #W-10 NOGALES, AZ 85621-2423
04729	UNICORP 291 CLEVELAND ST ORANGE, NJ 07050-2817	14519	DESIGNATRONICS INC 2101 JERICHO TPKE NEW HYDE PARK, NY 11040 - 4702
06383	PANDUIT CORP 17301 RIDGELAND AVENUE TINLEY PARK, IL 60477-0981	14936	GENERAL SEMICONDUCTOR INC 10 MELVILLE PARK ROAD MELVILLE, NY 11747-3113
06540	NEW HAVEN MFG CORP AMATOM ELECTRONIC HARDWARE DIV NEW HAVEN, CT 06515-1238	15653	ALCOA GLOBAL FASTENERS DE DBA ALCOA FASTENING SESTEMS DIV KAYNAR PRODUCTS 800 S COLLEGE BLVD FULLERTON, CA, 92831-3001
06776	ROBINSON NUGENT INC P O BOX 1208 NEW ALBANY, IN 47150-3264		

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAGE	VENDOR NAME AND ADDRESS	CAGE	VENDOR NAME AND ADDRESS
16258	SPACE LOK INC 13306 HALDALE AVE. GARDENA, CA 90249 - 2204	34335	ADVANCED MICRO DEVICES INC 901 THOMPSON PLACE SUNNYVALE, CA 94086-3453
17117	ELECTRONIC MOLDING CORPORATION WOONSOCKET, RI 02895	34355	LOCKHEED MARTIN SIPPICAN 7 BARNABAS RD MARION, MA 02738-1421
18324	PHILLIPS SEMICONDUCTORS INC 811 E ARQUES AVENUE SUNNYVALE, CA 94088-3409	34371	INTERSIL COMMUNICATIONS INC 675 TRADE ZONE BLVD MILPITAS, CA 95035-6803
18915	EG AND G BIRTCHEER INDUSTRIAL PRODUCTS 14100 DANIELSON STREET POWAY, CA 92064-6898	34649	INTEL CORPORATION 2200 MISSION COLLEGE BLVD SANTA CLARA, CA 95052-8119
19112	GARY ELECTRONICS LANGHORNE, PENNSYLVANIA	34899	FAIR-RITE PRODUCTS CORP 1 COMMERCIAL ROW WALLKILL, NY 12589-0288
22526	FCI USA INC 825 OLD TRAIL ROAD ETTERS, PA 17319-9392	46384	PENN ENGINEERING & MANUFACTURING CORP 5190 OLD EASTON ROAD DOYLESTOWN, PA 18901-9758
23880	STANDFORD APPLIED ENGINEERING COMPANY SANTA CLARA, CALIFORNIA	50434	AGILENT TECHNOLOGIES INC DBA SEMICONDUCTOR PRODUCTS GROUP DIV 350-370 WEST TRIMBLE RD SAN JOSE, CA 95131-1100
26858	BENTHOS INC 49 EDGERTON DRIVE NORTH FALMOUTH, MA 02556-6282	52072	CIRCUIT ASSEMBLY CORPORATION IRVINE, CALIFORNIA
27014	NATIONAL SEMICONDUCTOR CORP DIV HIGH RELIABILITY OPERATIONS 2900 SEMICONDUCTOR DR SANTA CLARA, CA 95052-8090	52648	PLESSEY TRADING CORP PLESSEY OPTOELECTRONICS AND MICROWAVE IRVINE, CA 92619
28520	HEYMAN MANUFACTURING COMPANY KENILWORTH, NEW JERSEY	54001	GLOBAL WULFSBERG SYSTEMS A DIV OF SUNDSTRAND DATA CONTROL INC 6400 WILKINSON DRIVE PRESCOTT, AZ 86301
32293	INTERSIL INCORPORATED SUB OF GENERAL ELECTRIC CO 2450 WALSH AVENUE SANTA CLARA, CA 95051	55104	TRI-STAR ELECTRONICS INTERNATIONAL INC 2201 ROSECRANS AVENUE EL SEGUNDO, CA 90245-4910
33095	SPECTRUM CONTROL INC 8031 AVONIA RD FAIRVIEW, PA 16415	55285	THE BERGQUIST CO 18930 W 78TH ST CHANHASSEN, MN 55317-9347
34333	MICROSEMI CORP INTEGRATED PRODUCTS 11861 WESTERN AVENUE GARDEN GROVE, CA 92841-2119		

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

CAGE	VENDOR NAME AND ADDRESS	CAGE	VENDOR NAME AND ADDRESS
55322	SAMTEC 520 PARK E BLVD NEW ALBANY, IN 47150-7251	71785	CINCH CONNECTORS INC 17000 FINLEY ROAD LOMBARD, IL 60148
55332	A M MACHINE CO INC 1300 CURTAIN AVE BALTIMORE, MD 21218-6287	71838	SPS TECHNOLOGIES AFG DIV 444 LEE RD CLEVELAND, OH 44128-2902
55566	RAF ELECTRONIC HARWARE INC 95 SILVERMINE ROAD SEYMOUR, CT 06483-3915	72962	HARVARD INDUSTRIES INC LEBANON, NJ 08833
56289	SPRAGUE ELECTRIC CO WORLD HEADQUARTERS 678 MAIN STREET SANFORD, ME 04073	73138	BI TECHNOLOGIES CORPORATION 4200 BONITA PLACE FULLERTON, CA 92835-1053
58998	FASTENER INNOVATION TECHNOLOGY INC 14601 S BROADWAY GARDENA, CA 90248-1811	75378	CTS CORPORATION 171 COVINGTON DRIVE BLOOMINGDALE, IL 60108-3107
59730	THOMAS AND BETTS CORP IOWA CITY, IA 52240	76385	MINOR RUBBER CO INC 49 ACKERMAN ST BLOOMFIELD, NJ 07003 - 4253
63265	UNITED CHEMI-CON INC 9801 W HIGGINS ROAD SUITE 430 ROSEMONT, IL 60018	77820	AMPHENOL AEROSPACE OPERATIONS 40-60 DELAWARE STREET SIDNEY, NY 13838-1304
64155	LINEAR TECHNOLOGY CORP 720 SYCAMORE DRIVE MILPITAS, CA 95035	77969	RUBBERCRAFT CORPORATION OF AMERICA TORRANCE, CALIFORNIA
65664	CATAMOUNT MANUFACTURING INCORPORATED 158 GOVERNOR DR P O BOX 720 ORANGE, MA 01364-0720	79136	WALDES KOHINOOR INCORPORATED LONG ISLAND, NEW YORK
71159	BRISTOL SOCKET SCREW CO WATERBURY, CT 06701	79963	ZIERICK MFG CO 131 RADIO CIR MOUNT KISCO, NY 10549
71218	BUD INDUSTRIES INC 4605 EAST 355TH ST WILLOUGHBY, OH 44094 - 4629	80205	NATIONAL AEROSPACE STANDARDS COMMITTEE AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA INC
71468	ITT INDUSTRIES INC 666 E DRYER ROAD SANTA ANA, CA 92705-5561	81095	TRIAD-ULTRAD DIV NATIONAL CITY, CA 91950
		81350	JOINT ARMY-NAVY SPECIFICATIONS PROMULGATED BY MILITARY DEPARTMENTS
		81860	BARRY WRIGHT CORPORATION DBA BARRY CONTROLS 82 SOUTH STREET HOPKINTON, MA 01748-2205

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

<u>CAGE</u>	<u>VENDOR NAME AND ADDRESS</u>	<u>CAGE</u>	<u>VENDOR NAME AND ADDRESS</u>
83086	NEW HAMPSHIRE BALL BEARING INCORPORATED PETERBOROUGH, NEW HAMPSHIRE	99378	ATLEE OF DELAWARE INC 10 BAYFIELD DRIVE NORTH ANDOVER, MA 01845-6014
83330	DIALIGHT CORP 1501 ROUTE 34 FARMINGDALE, NJ 07727-3932	OE3K6	
85480	BRADY W H CO CORP H Q INDUSTRIAL PRODUCTS DIV 6555 W GOOD HOPE ROAD MILWAUKEE, WI 85480		
86577	PRECISION METAL PRODUCTS CO P O BOX 6026 PEABODY, MA 01960		
86928	SEASTROM MANUFACTURING COMPANY 456 SEASTROM STREET TWIN FALLS, ID 83301-8526		
88044	AERONAUTICAL STANDARDS GROUP DEPT OF NAVY AND AIR FORCE		
88245	WINCHESTER ELECTRONICS LITTON SYSTEMS-USECO DIV VAN NUYS, CA REPLACED BY 81312		
92830	ASSOCIATED SPRING BARNES GROUP INC 18 MAIN ST BRISTOL, CT 06010-6547		
94970	DUKANE CORPORATION 2900 DUKANE DRIVE SAINT CHARLES, IL 60174-3395		
96906	MILITARY STANDARDS PROMULGATED BY MILITARY DEPARTMENTS UNDER AUTHORITY OF DEFENSE STANDARDIZATION MANUAL 4120 3-M		
97896	HONEYWELL INTERNATIONAL INC DBA HONEYWELL AEROSPACE 15001 NE 36 ST REDMOND, WA 98052-5317		
98159	RUBBERCRAFT CORP OF CALIFORNIA DBA RUBBER TECK DIVISION 15627 S BROADWAY ST GARDENA, CA 90248-2210		

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

4. Equipment Designator Index

EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER
A1	1 95	720-0123-001
	1 95A	720-0123-002
	1 95B	720-0123-003
	1 95C	720-0167-002
	1 95D	720-0167-004
A1A1	2 25	722-2440-001
	2 25A	722-2440-002
	2 25B	722-2440-003
A1AR1	2 55	SG123K883B
A1AR2	2 55	SG123K883B
A1AR4	2 80	LM117K883B
A1AR5	2 60	UA7812KMQB
A1AR6	2 65	UA7912KMQB
A1C12	2 40A	M39014-02-1358
A1C13	2 45A	M39014-02-1350
A1C14	2 50A	M39014-02-1419
A1C15	2 50A	M39014-02-1419
A1Q1	2 75	JANTX2N3792
A1Q2	2 70	JANTX2N3716
A1R5	2 53	RNC55H2430FS
A1S1	2 150	973-0031-561
	2 150A	974-0000-574
A1T1	2 120	310-0285-001
A2	1 105	722-2442-001
	1 105A	722-2442-002
	1 105B	722-2442-003
A2AR1	3 15	HA1-4741-8-883B
A2AR2	3 15	HA1-4741-8-883B
A2AR3	3 15	HA1-4741-8-883B
A2AR4	3 15	HA1-4741-8-883B
A2AR5	3 15	HA1-4741-8-883B
A2C1	3 50	M39014-01-1587
A2C10	3 50	M39014-01-1587
A2C11	3 30	M39014-01-1339
A2C12	3 30	M39014-01-1339
A2C13	3 50	M39014-01-1587
A2C14	3 30	M39014-01-1339
A2C15	3 30	M39014-01-1339
A2C16	3 50	M39014-01-1587
A2C17	3 30	M39014-01-1339
A2C18	3 30	M39014-01-1339
A2C19	3 50	M39014-01-1587
A2C2	3 30	M39014-01-1339
A2C20	3 30	M39014-01-1339
A2C21	3 30	M39014-01-1339
A2C22	3 50	M39014-01-1587
A2C23	3 30	M39014-01-1339
A2C24	3 30	M39014-01-1339
A2C25	3 20	CCR06CG332JP
A2C26	3 27	M39014-01-1342

EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER
A2C27	3 27	M39014-01-1342
A2C28	3 5	M39014-01-1330
A2C29	3 250	M39014-01-1356
A2C3	3 30	M39014-01-1339
A2C30	3 217	M39014-02-1419
A2C31	3 10	M39014-01-1575
A2C32	3 10	M39014-01-1575
A2C33	3 10	M39014-01-1575
A2C34	3 10	M39014-01-1575
A2C35	3 10	M39014-01-1575
A2C36	3 10	M39014-01-1575
A2C37	3 10	M39014-01-1575
A2C38	3 65	M39014-01-1593
A2C39	3 5	M39014-01-1330
A2C4	3 50	M39014-01-1587
A2C40	3 10	M39014-01-1575
A2C41	3 110	M39003-01-2017
A2C42	3 55	M39003-01-2064
A2C43	3 55	M39003-01-2064
A2C44	3 190	M39003-01-2116
A2C45	3 65	M39014-01-1593
A2C46	3 65	M39014-01-1593
A2C47	3 65	M39014-01-1593
A2C48	3 30	M39014-01-1339
A2C49	3 10	M39014-01-1575
A2C5	3 30	M39014-01-1339
A2C50	3 65	M39014-01-1593
A2C51	3 10	M39014-01-1575
A2C52	3 223	M39014-01-1351
A2C53	3 65	M39014-01-1593
A2C54	3 65	M39014-01-1593
A2C6	3 30	M39014-01-1339
A2C7	3 50	M39014-01-1587
A2C8	3 30	M39014-01-1339
A2C9	3 30	M39014-01-1339
A2CR1	3 130	JANTX1N4148-1
A2CR10	3 130	JANTX1N4148-1
A2CR11	3 130	JANTX1N4148-1
A2CR12	3 130	JANTX1N4148-1
A2CR13	3 130	JANTX1N4148-1
A2CR14	3 130	JANTX1N4148-1
A2CR15	3 130	JANTX1N4148-1
A2CR16	3 130	JANTX1N4148-1
A2CR17	3 130	JANTX1N4148-1
A2CR18	3 130	JANTX1N4148-1
A2CR2	3 130	JANTX1N4148-1
A2CR3	3 130	JANTX1N4148-1
A2CR4	3 130	JANTX1N4148-1
A2CR5	3 130	JANTX1N4148-1
A2CR6	3 130	JANTX1N4148-1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER	EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER
A2CR7	3 130	JANTX1N4148-1	A2R52	3 135	RNC55H2430FS
A2CR8	3 130	JANTX1N4148-1	A2R53	3 140	RNC55H2491FS
A2CR9	3 130	JANTX1N4148-1	A2R54	3 150	RNC55H1002FS
A2P1	3 75	G06D64P4BEBL	A2R55	3 40	RNC55H1002FS
A2Q1	3 125	JANTX2N2222A	A2R56	3 40	RNC55H1002FS
A2Q2	3 230	JANTX2N2907A	A2R57	3 40	RNC55H1002FS
A2R1	3 45	RNC55H1331FS	A2R58	3 40	RNC55H1002FS
A2R10	3 45	RNC55H1331FS	A2R59	3 40	RNC55H1002FS
A2R11	3 25	RNC55H3322FS	A2R6	3 25	RNC55H3322FS
A2R12	3 25	RNC55H3322FS	A2R60	3 100	RNC55H2001FS
A2R13	3 45	RNC55H1331FS	A2R61	3 210	RNC55H3321FS
A2R14	3 25	RNC55H3322FS	A2R62	3 105	RNC55H3011FS
A2R15	3 25	RNC55H3322FS	A2R7	3 45	RNC55H1331FS
A2R16	3 45	RNC55H1331FS	A2R8	3 25	RNC55H3322FS
A2R17	3 25	RNC55H3322FS	A2R9	3 25	RNC55H3322FS
A2R18	3 25	RNC55H3322FS	A2U1	3 95	SNJ54LS145J00
A2R19	3 45	RNC55H1331FS	A2U10	3 215	LM111H883B
A2R2	3 25	RNC55H3322FS	A2U11	3 200	SNJ54LS02J00
A2R20	3 25	RNC55H3322FS	A2U12	3 195	SNJ54LS04J00
A2R21	3 25	RNC55H3322FS	A2U13	3 115	LM117H883B
A2R22	3 45	RNC55H1331FS	A2U14	3 225	LM136H883B
A2R23	3 25	RNC55H3322FS	A2U15	3 70	M8340102K1501FA
A2R24	3 25	RNC55H3322FS	A2U16	3 70	M8340102K1501FA
A2R25	3 140	RNC55H2491FS	A2U2	3 95	SNJ54LS145J00
A2R26	3 100	RNC55H2001FS	A2U3	3 235	M8340102K1002FA
A2R27	3 145	RNC55H4641FS	A2U4	3 235	M8340102K1002FA
A2R28	3 40	RNC55H1002FS	A2U5	3 235	M8340102K1002FA
A2R29	3 210	RNC55H3321FS	A2U6	3 235	M8340102K1002FA
A2R3	3 25	RNC55H3322FS	A2U7	3 60	HI1-0509A8
A2R30	3 35	RNC55H1001FS	A2U8	3 60	HI1-0509A8
A2R31	3 120	RCR20G122JS	A2U9	3 205	SNJ54LS00J00
A2R32	3 40	RNC55H1002FS	A3	1 110	722-2441-001
A2R33	3 40	RNC55H1002FS		1 110B	722-2441-004
A2R34	3 35	RNC55H1001FS		1 110C	722-2441-003
A2R35	3 25	RNC55H3322FS		1 110D	722-2441-005
A2R36	3 25	RNC55H3322FS		1 110E	722-2441-006
A2R37	3 245	RNC55H1002BS		1 110F	722-2441-007
A2R38	3 245	RNC55H1002BS		1 110G	722-2441-008
A2R39	3 245	RNC55H1002BS	A3C1	4 50	M39003-01-2017
A2R4	3 45	RNC55H1331FS	A3C10	4 35	M39014-01-1593
A2R40	3 245	RNC55H1002BS	A3C11	4 35	M39014-01-1593
A2R41	3 255	RNC55H1501FS	A3C12	4 35	M39014-01-1593
A2R42	3 145	RNC55H4641FS	A3C13	4 35	M39014-01-1593
A2R43	3 255	RNC55H1501FS	A3C14	4 35	M39014-01-1593
A2R44	3 100	RNC55H2001FS	A3C15	4 35	M39014-01-1593
A2R45	3 100	RNC55H2001FS	A3C16	4 35	M39014-01-1593
A2R46	3 220	RNC55H1502FS	A3C17	4 35	M39014-01-1593
A2R47	3 210	RNC55H3321FS	A3C18	4 35	M39014-01-1593
A2R48	3 100	RNC55H2001FS	A3C19	4 35	M39014-01-1593
A2R49	3 100	RNC55H2001FS	A3C2	4 65	M39003-01-2064
A2R5	3 25	RNC55H3322FS	A3C20	4 35	M39014-01-1593
A2R50	3 100	RNC55H2001FS	A3C21	4 35	M39014-01-1593
A2R51	3 100	RNC55H2001FS	A3C22	4 50A	M39003-01-2017

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER
-------------------------	-------------	-------------

A3C3	4	15	M39003-01-2021
A3C4	4	15	M39003-01-2021
A3C6	4	105	M39014-01-1575
A3C7	4	105	M39014-01-1575
A3C8	4	140	M39014-02-1350
A3C9	4	140	M39014-02-1350
A3CR1	4	45	JANTX1N5711
A3CR2	4	45	JANTX1N5711
A3CR3	4	37	JANTX1N4148-1
A3CR4	4	38	JANTX1N4148-1
A3CR5	4	46	JANTX1N5711
A3P1	4	85	G06D64P4BEBL
A3Q1	4	148	JANTX2N2222A
A3Q2	4	148	JANTX2N2222A
A3R1	4	115	RNC55H1002FS
A3R10	4	145	RNC55H3320FS
A3R11	4	40	RNC55H1000FS
A3R13	4	80	RNC55H3321FS
A3R14	4	80	RNC55H3321FS
A3R15	4	80	RNC55H3321FS
A3R16	4	80	RNC55H3321FS
A3R17	4	20	RNC55H1333FS
A3R19	4	115	RNC55H1002FS
A3R2	4	115	RNC55H1002FS
A3R3	4	80	RNC55H3321FS
A3R31	4	115	RNC55H1002FS
A3R32	4	115	RNC55H1002FS
A3R33	4	80	RNC55H3321FS
A3R34	4	30	RNC55H1003FS
A3R35	4	115	RNC55H1002FS
A3R36	4	115	RNC55H1002FS
A3R37	4	115	RNC55H1002FS
A3R39	4	115	RNC55H1002FS
A3R4	4	80	RNC55H3321FS
A3R40	4	137	RNC55H78R7FS
A3R41	4	81	RNC55H3321FS
A3R42	4	160	RNC55H3322FS
A3R43	4	155	RNC55H2431FS
A3R44	4	126	RNC55H3321FS
A3R45	4	126	RNC55H3321FS
A3R5	4	80	RNC55H3321FS
A3R6	4	145	RNC55H3320FS
A3R7	4	145	RNC55H3320FS
A3R8	4	145	RNC55H3320FS
A3R9	4	145	RNC55H3320FS
A3U1	4	75	SNJ54LS374J00
A3U10	4	70	SNJ54LS175J00
A3U11	4	110	SNJ54LS14J00
A3U12	4	125	SNJ54LS10J00
A3U13	4	135	MN9102DG
A3U14	4	75	SNJ54LS374J00
A3U15	4	75	SNJ54LS374J00
A3U16	4	10	SNJ54LS02J00

EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER
-------------------------	-------------	-------------

A3U17	4	55	SNJ5417J00
A3U18	4	60	M8340102K3321FB
A3U19	4	25	SNJ54LS123J00
A3U2	4	75	SNJ54LS374J00
A3U20	4	55	SNJ5417J00
A3U3	4	75	SNJ54LS374J00
A3U4	4	120	SNJ54LS74J00
A3U5	4	120	SNJ54LS74J00
A3U6	4	5	SNJ54LS00J00
A3U7	4	10	SNJ54LS02J00
A3U8	4	10	SNJ54LS02J00
A3U9	4	110	SNJ54LS14J00
A4	1	115	722-2438-001
	1	115A	722-2438-002
	1	115B	722-2438-003
	1	115C	722-2438-004
	1	115D	722-2438-005
	1	115E	722-2438-006
	1	115F	722-2438-007
	1	115G	722-2438-008
	1	115H	722-2438-009
	1	115I	722-2438-010
	1	115J	722-2438-011
	1	115K	722-2438-012
	1	115L	722-2438-013
	1	115M	722-2438-014
	1	115N	722-2438-015
	1	115P	722-2438-016
	1	115Q	722-2438-017
	1	115R	722-2438-018
	1	115S	722-2438-019
	1	115T	722-2438-020
	1	115U	722-2438-021
	1	115V	722-2438-022
	1	115W	722-2438-023
	1	115X	722-2438-024
	1	115Y	722-2438-025
	1	115Z	722-2438-026
	1	116	722-2438-027
	1	116A	722-2438-028
	1	116B	722-2438-029
	1	116C	722-2438-030
	1	116D	722-2438-031
	1	116E	722-2438-032
	1	116F	722-2438-033
	1	116G	722-2438-034
	1	116H	722-2438-035
	1	116I	722-2438-036
	1	116J	722-2438-037
	1	116K	722-2438-038
	1	116L	722-2438-039
	1	116M	722-2438-040
	1	116N	722-2438-041

31-30-32

Page 1011
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER	EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER
A4C1	5 21	M39014-01-1321	A4U12	5 130	SNJ54LS14J00
A4C10	5 75	M39003-01-2017	A4U13	5 120	SNJ54LS86J00
A4C11	5 25	M39014-01-1593	A4U14	5 135	SNJ54LS54J00
A4C12	5 25	M39014-01-1593	A4U15	5 140	SNJ54LS74J00
A4C13	5 25	M39014-01-1593	A4U16	5 130	SNJ54LS14J00
A4C14	5 25	M39014-01-1593	A4U17	5 55	ICL8211MTY883B
A4C15	5 25	M39014-01-1593	A4U18	5 80	SNJ54LS00J00
A4C16	5 25	M39014-01-1593	A4U2	5 15	MD8224B
A4C17	5 25	M39014-01-1593	A4U3	5 165A	AM8228DMB
A4C18	5 25	M39014-01-1593	A4U4	5 160	AM9513DMB
A4C19	5 25	M39014-01-1593	A4U5	5 155	AM9551DMB
A4C2	5 50	M39003-01-2014		5 155C	MD8251A
A4C23	5 25	M39014-01-1593	A4U6	5 90	1M6561MDN883B
A4C24	5 25	M39014-01-1593	A4U7	5 90	1M6561MDN883B
A4C25	5 23	M39014-01-1357	A4U8	5 115	718-0423-002
A4C26	5 22	M39014-01-1324		5 115A	718-0423-003
A4C3	5 125	M39014-01-1569		5 115B	718-0423-004
A4C7	5 25	M39014-01-1593		5 115C	718-0423-005
A4C8	5 75	M39003-01-2017		5 115D	718-0423-007
A4C9	5 70	M39003-01-2064		5 115E	718-0423-008
A4CR1	5 45	JANTX1N4148-1		5 115F	718-0423-009
A4L1	5 24	M75101-3		5 115G	718-0423-010
	5 24A	MS75101-3		5 115H	718-0423-011
A4P1	5 95	G06D64P4BEBL		5 115I	718-0423-012
A4R1	5 35	RNC55H1002FS		5 115J	718-0423-013
A4R10	5 53	RNC55H1004FS		5 115K	718-0423-014
A4R13	5 30	RNC55H4641FS	A4U9	5 170	SNJ54LS244J00
A4R14	5 30	RNC55H4641FS	A4Y1	5 5	M3098-38CR60AU
A4R15	5 145	RNC55H1331FS	A8	1 120	722-2501-001
A4R16	5 30	RNC55H4641FS	A8C1	6 85	M39014-01-1569
A4R18	5 30	RNC55H4641FS	A8C2	6 85	M39014-01-1569
A4R19	5 60	RNC55H7502FS	A8C3	6 75	M39014-01-1593
A4R2	5 40	RNC55H1001FS		6 75A	M39014-01-1593
A4R20	5 65	RNC55H3322FS	A8C4	6 65	M39003-01-2017
A4R21	5 30	RNC55H4641FS	A8C5	6 90	M39014-01-1566
A4R23	5 180	RNC55H5901FS	A8C6	6 75	M39014-01-1593
A4R24	5 30	RNC55H4641FS	A8C7	6 75	M39014-01-1593
A4R25	5 36	RNC55H1000FS		6 75A	M39014-01-1593
	5 37	RNC55H1002FS	A8C8	6 15	M39003-01-2064
A4R26	5 30	RNC55H4641FS	A8CR1	6 35	JANTX1N5614
A4R29	5 38	RNC55H1000FS	A8CR2	6 35	JANTX1N5614
A4R3	5 145	RNC55H1331FS	A8CR3	6 105	JANTX1N5711
A4R30	5 56	RNC55H2004FS	A8CR4	6 105	JANTX1N5711
A4R4	5 40	RNC55H1001FS	A8K1	6 10	3SAV5004D1
A4R5	5 30	RNC55H4641FS	A8K2	6 10	3SAV5004D1
A4R6	5 35	RNC55H1002FS	A8P1	6 45	G06D64P4BEBL
A4R7	5 35	RNC55H1002FS	A8Q1	6 115	JANTX2N2222A
A4R8	5 40	RNC55H1001FS	A8Q2	6 115	JANTX2N2222A
A4R9	5 35A	RNC55H1002FS	A8R1	6 40	RNC55H1001FS
	5 38	RNC55H1000FS		6 40A	RNC55H1001FS
A4U1	5 175	AM9080ADMB883B	A8R10	6 70	RNC55H3321FS
A4U10	5 170	SNJ54LS244J00		6 70A	RNC55H3321FS
A4U11	5 85	SNJ54LS138J00	A8R11	6 5	RWR81S51R1FR

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER	EQUIPMENT DESIGNATOR	FIGURE/ITEM	PART NUMBER
A8R12	6 40	RNC55H1001FS	FL1	17 10	51-717-007
	6 40A	RNC55H1001FS	FL2	16 55	51-717-007
A8R13	6 25	RWR81S4420FR		17 10	51-717-007
A8R14	6 40A	RNC55H1001FS	FL3	16 35	GA3A1
	6 70	RNC55H3321FS		17 15	2360-006
A8R15	6 110	RNC55H3320FS	FL4	17 15	2360-006
A8R16	6 110	RNC55H3320FS	FL5	17 10	51-717-007
A8R17	6 95	RNC55H3011FS	J1	9 5	DPX2MA57P00P34-
A8R18	6 40	RNC55H1001FS		16 5	DPX2MA57P57P34-
	6 40A	RNC55H1001FS		16 5A	DPX2MA57P57P34-
A8R19	6 30	RWR80S10R0FR		16 5B	440-0005-009
A8R2	6 40	RNC55H1001FS		16 5C	440-0005-028
	6 40A	RNC55H1001FS	J2	9 80	KPT07A22-55S
A8R20	6 70	RNC55H3321FS		16 140	KPT07A22-55S
	6 70A	RNC55H3321FS	J3	9 120A	DACK15SBF
A8R3	6 40A	RNC55H1001FS		16 200	DACK15SBF
	6 70	RNC55H3321FS	J4	9 120A	DACK15SBF
A8R4	6 100	RNC55H5901FS		16 200	DACK15SBF
A8R5	6 100	RNC55H5901FS	J5	9 45	DBM25S
A8R6	6 70	RNC55H3321FS		16 85	DBM25S
	6 70A	RNC55H3321FS		16 87	M24308-2-283F
A8R7	6 70	RNC55H3321FS	L1	17 40	712-0084-001
	6 70A	RNC55H3321FS	P3	10 405	DAM15P
A8R8	6 70	RNC55H3321FS	P4	11 65	DAM15P
	6 70A	RNC55H3321FS	P5	10 395	DBM25P
A8R9	6 40	RNC55H1001FS	R1	12 20	RWR81S82R5FR
A8T1	6 125	SP21	R2	12 20	RWR81S82R5FR
A8T2	6 135	M27-165-10	R4	12 15	RNC55H1001FS
A8U1	6 80	SNJ54LS157J00	U1	12 10	SNJ54LS145J00
A8U2	6 20	SNJ55472JG00	U2	10 125	744-0078-002
	6 20A	SNJ55472JG883B	U3	10 130	744-0078-001
	6 22	SNJ55472JG	VR1	17 35	JANTX1N6072A
A8U3	6 20	SNJ55472JG00	XA1	9 100	G06D64P3BBBL
	6 20A	SNJ55472JG883B		16 180	G06D64P3BBBL090
	6 145	722-3523-001	XA2	9 100	G06D64P3BBBL
A8U3C1	6 170	M39003-01-2257		16 180	G06D64P3BBBL090
A8U3CR2	6 165	JANTX1N4148-1	XA3	9 100	G06D64P3BBBL
A8U3Q1	6 150	JANTX2N5582		16 180	G06D64P3BBBL090
A8U3Q2	6 150	JANTX2N5582	XA4	9 100	G06D64P3BBBL
A8U3R1	6 160	RNC55H2321FS		16 180	G06D64P3BBBL090
A8U3R2	6 160	RNC55H2321FS	XA8	9 100	G06D64P3BBBL
A8U3VR1	6 155	JAN1N4615		16 180	G06D64P3BBBL090
A8U4	6 120	SNJ5406J00			
A8U5	6 130	SNJ54LS00J00			
BT1	1 30	N15F210B			
C1	12 5	M39014-02-1350			
	12 5A	M39014-01-1593			
	17 30	M83421-01-5204P			
CR1	9 85	5082-4587			
	16 145	HLMP0454			
CR2	16 60	JANTX1N6072A			
FL	17 10	51-717-007			
FL1	16 55	51-717-007			

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

5. Numerical Index

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
AM8224DM		5 15D	1	DACK15SBF		9 120A	2
AM8224DMB		5 15C	1			16 200	2
AM8228DMB		5 165A	1	DAM15P		10 405	1
AM8228DMB883B		5 165	1			11 65	1
AM9080ADI		5 175D	1			13 90	1
AM9080ADMB		5 175B	1			13 90A	1
AM9080ADMB883B		5 175	1			13 127	1
AM9513-5DMB		5 160F	1			13 127A	1
AM9513ADI		5 160C	0			13 152	1
AM9513ADIB		5 160D	1			13 152A	1
AM9513ADMB		5 160B	1			14 405	1
AM9513DC		5 160E	1			14 405A	1
AM9513DMB		5 160	1	DB19678-2		16 96	1
AM9513SDMB		5 160A	0			16 96A	1
AM9551DI		5 155A	1	DB50906-1		9 55	1
AM9551DIB		5 155B	1			16 95	1
AM9551DMB		5 155	1	DBM25P		10 395	1
AN960-8		1 165A	4			14 395	1
AN960C4L		13 77	1	DBM25S		9 45	1
AN960C6		13 78	1			16 85	1
AN960C6L		14 12	8	DK100		1 30A	1
B362-06192-2		1 65C	1	DK120		1 30C	1
B6-1		14 92	AR	DM5406J883		6 120B	1
		14 92A	1	DM54LS00J883		3 205D	1
B6-2		14 91	AR			4 5D	1
B6-3		14 94	AR			5 80D	1
B6-4		14 93	AR			6 130D	1
B6-5		14 90A	AR	DM54LS02J883		3 200D	1
C30		4 147	3			4 10C	3
CAS01-18B37		5 112	2	DM54LS04J883		3 195D	1
CCR06CG332JP		3 20	1	DM54LS10J883		4 125D	1
CR60AU18M87440		5 5A	1	DM54LS123J883		4 25C	1
D110551		9 50	1	DM54LS138J883		5 85D	1
		9 125	2	DM54LS14J883		4 110E	2
		16 90	1			5 130D	2
		16 205	2	DM54LS175J883B		4 70D	1
D20419-21		1 195	3	DM54LS374J883B		4 75C	5
		10 390	4	DM54LS74AJ883		4 120D	2
		11 70	2			5 140E	1
		13 100	0	DM54LS86J883		5 120D	1
		14 390	0	DP8228MJMIL		5 165C	1
D6-140		11 57	1	DPX2MA57P00P34-		9 5	1
		13 70	0	DPX2MA57P57P34-		16 5	1
D6SE140GY		13 78A	1			16 5A	1
D8224		5 15A	1	DS55462J8-883		6 20E	2
DA15SU		9 120	2			6 22C	1
DA20961		10 400	1	ELP362D		1 30B	1
		11 75	1	ELP362DS		1 30D	1
		13 95	1	F12NC4284-2-82		15 200	6
		14 400	1	F6NY250-0		13 65A	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
FE832		9 267	3	JANTX1N4148-1		4 38	1
		9 292	3			5 45	1
		16 355	3			6 165	1
		16 390	3			7 80	1
FS2084-4-28		8 10A	4	JANTX1N4970		7 70	1
G06D64P3BBBL		9 100	5	JANTX1N5614		6 35	2
		16 180A	5			7 40	7
G06D64P3BBBL090		16 180	5	JANTX1N5711		4 45	2
G06D64P4BEBL		3 75	1			4 46	1
		4 85	1			6 105	2
		5 95	1	JANTX1N6072A		16 60	1
		6 45	1			17 35	1
		7 105	1	JANTX2N2222A		3 125	1
G2525-4		9 235	1			4 148	2
		16 310	1			6 115	2
GA3A1		16 35	1			7 100	1
GSC156		11 85	1	JANTX2N2907A		3 230	1
		13 85	1	JANTX2N3716		2 70	1
GSC219		11 80	1	JANTX2N3792		2 75	1
		13 80	1			15 80	1
HA1-4741-2		3 15A	5	JANTX2N5582		6 150	2
		3 15B	5	JANTX469-1		2 160	3
HA1-4741-8-883B		3 15	5	JANTXV2N3716		15 75B	1
HCPS9-3XXXW		16 115A	7	JNTX2N3716		15 75	1
HI1-0509A2		3 60A	2	JNTX469-1		15 165	3
HI1-0509A8		3 60	2	KPSE07A22-55S		9 80B	1
HLMP0454		9 85A	1			16 140A	1
		16 145	1	KPT07A22-55S		9 80	1
HM1-6561-2		5 90C	2			16 140	1
HM1-6561-8		5 90A	2	LKS632-1		13 140	0
HM1-6561-883		5 90D	2			13 170	2
HM1-6561B8		5 90B	2	LKS632-2		13 115	0
HRSCBH4		15 165B	3			13 119	2
HRSCBH6		15 165C	3			13 144	2
IC624SGT		5 117A	1	LL25U82J28		1 35A	4
ICA246STG30		5 117B	1	LM111H883B		3 215	1
ICL8211MTY		5 55A	1	LM111H883C		3 215A	1
		7 55A	1	LM117H883B		3 115	1
ICL8211MTY883B		5 55	1	LM117K		2 80C	1
		7 55	1			2 80D	1
ID8080A		5 175C	1			2 80E	1
IL8228BYA		5 165D	1			15 85C	1
IL8228Y1		5 165E	1			15 85D	1
JAN1N4615		6 155	1			15 85E	1
JAN1N5614		6 35A	2	LM117K883B		2 80	1
		7 40A	7			15 85	1
JAN1N5711		4 45A	2	LM117K883C		15 85B	1
		4 46A	1	LM120H5-0-883B		7 125A	1
		6 105A	2	LM120K12		2 65C	1
JAN2N3716		15 75A	1			15 70A	1
JAN2N5582		6 150A	2	LM120K12-883		15 70C	1
JANTX1N4148-1		3 130	18	LM120K12-883B		2 65A	1
		4 37	1	LM123K		2 55C	2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
LM123K		2 55D	2	M39014-01-1593		3 65	7
		15 60C	2			4 35	12
		15 60D	2			5 25	12
LM123K883B		2 55A	2			6 75	3
		15 60A	2			6 75A	2
LM123K883C		2 55B	2			12 5A	1
		15 60B	2	M39014-02-1338		7 155	2
LM136AH2-5		3 225D	1	M39014-02-1350		2 45A	1
LM136AH2-5883C		3 225C	1			4 140	2
LM136AH883B		3 225A	0			7 65	5
LM136H2-5		3 225E	1			12 5	1
LM136H883B		3 225	1			15 45	1
LM136H883C		3 225B	1	M39014-02-1350S		2 45	0
LM140K		2 60B	1	M39014-02-1358		2 40A	1
LM140K12		2 60C	1			15 40	1
		2 60D	1	M39014-02-1358S		2 40	0
		15 65A	1	M39014-02-1419		2 50A	2
		15 65B	1			3 217	1
M19500-469-01		2 160A	3			7 45	10
		15 165A	3			15 50	2
M24308-2-283F		16 87	1	M39014-02-1419S		2 50	0
M27-165-10		6 135	1	M39018-7-0284M		7 5	2
M3098-38CR60AU		5 5	1	M39018-7-0292M		7 130	3
M39003-01-2014		5 50	1	M75101-3		5 24	1
M39003-01-2017		3 110	1	M8340102K1002FA		3 235	4
		4 50	1	M8340102K1501FA		3 70	2
		4 50A	1	M8340102K3301GB		4 60A	1
		5 75	2	M8340102K3321FB		4 60	1
		6 65	1	M83421-01-5204P		17 30	1
M39003-01-2021		4 15	2	MC1568G		7 220C	1
M39003-01-2064		3 55	2	MC1568L		7 220B	1
		4 65	1	MD308G25P1		14 395B	1
		5 70	1	MD308G25S1		9 45B	1
		6 15	1			16 85B	1
M39003-01-2116		3 190	1	MD8080AB		5 175A	1
		7 20	1	MD8224		5 15B	1
M39003-01-2134		7 145	2	MD8224B		5 15	1
M39003-01-2257		6 170	1	MD8228B883C		5 165B	1
M39003-01-2304		6 15A	1	MD8251A		5 155C	1
M39014-01-1321		5 21	1	MET11		6 125A	1
M39014-01-1324		5 22	1	MF1001-06		9 130	12
M39014-01-1330		3 5	2			16 230	12
M39014-01-1339		3 30	17	MF1001-08		1 92	1
M39014-01-1342		3 27	2			2 190	4
M39014-01-1351		3 223	1	MILS19500-469		15 165D	3
M39014-01-1356		3 250	1			15 165E	3
M39014-01-1357		5 23	1	MN9102DG		4 135	1
M39014-01-1566		6 90	1	MS15795-802		1 145	0
M39014-01-1569		5 125	1	MS15795-803		1 185	4
		6 85	2			2 100	14
M39014-01-1575		3 10	10			10 25	3
		4 105	2			10 355	2
M39014-01-1587		3 50	8			10 375	2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
MS15795-803		11 32	2	MS21042-02		4 95	2
		14 25	3			5 105	2
		14 355	2			6 55	2
		14 375	2			7 115	2
		15 105	14			16 99	2
MS15795-805		1 20	2	MS21042-06		2 155	1
		1 85	0			15 160	1
		2 140	2	MS21044C04		2 95	14
		2 175	3			9 15	8
		15 145	2			9 40	0
		15 180	3			9 75	4
MS15795-807		1 165	4			15 100	14
MS16624-5012		14 145B	1			16 20	8
		14 445A	1			16 80	0
		14 525	1			16 130	6
MS16633-4009		13 10A	1			16 165	1
		14 250A	1	MS21044C06		2 135	2
		14 305A	1			2 170	3
MS16995-17		14 280A	2			15 140	2
MS16995-18		11 50	2			15 175	3
		13 55	2			16 50	2
MS16995-2		10 135	4	MS21044C08		1 80	0
		10 195	8			1 103	4
		13 125	1	MS24693C2		9 58	2
		13 150	1			14 201	1
		14 135	4			14 500	1
		14 435	8	MS24693C25		9 95A	12
MS17821-4-9		7 12	2			9 250	4
		7 137	3			9 320	2
		9 17	1			16 175	12
		16 25	1			16 420	2
		16 110A	6	MS24693C272		1 90	1
MS20341-2S		9 110	10	MS24693C3		8 30	3
		16 195	10			9 35	0
MS20426A3-4		2 195	8			9 70	4
MS20426A3-5		2 195A	8			10 10	4
MS20426AD2-3		9 275A	12			10 205	1
		9 285	2			10 220	1
		9 300A	12			10 280	2
		9 310	2			10 305	7
		16 365	12			14 10	4
		16 400	12			14 120B	2
MS20426AD2-4		9 275	12			14 200A	4
		9 285A	2			14 210	1
		9 300	12			14 225	1
		9 310A	2			14 265A	2
		16 375	2			14 495	4
		16 410	2			16 75	0
MS20426AD3-6		1 93	2			16 105A	2
MS20470A3-5		9 132A	24			16 125	6
		16 235	24	MS24693C32		16 40	2
MS20470A3-6		9 132	24	MS24693C34		14 120A	2
MS21042-02		3 85	2	MS24693C47		9 233	2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
MS24693C48		9 195	6	MS35649-222		16 195A	10
		9 230	2	MS35649-264		9 255	4
		9 233A	2			16 335	4
		16 260	6	MS51954-13		11 10	4
		16 305	4	MS51957-1		1 135	6
MS24693C5		8 30B	4	MS51957-12		10 350	2
		8 30C	1			14 350	2
		10 40	4	MS51957-13		1 175A	4
MS24693C50		1 100	4			10 20	3
		1 102B	6			10 370	2
MS24693C51		1 75	1			14 20	3
		1 102	4			14 370	2
		1 102A	6	MS51957-14		1 175	4
MS24693C8		8 30A	2			13 76	1
MS24693S4		9 89	1			13 102	2
		16 160	1			14 120	2
MS3181-22N		9 77	1			14 200	4
MS3181-22NA		16 135	1			14 265	2
MS3367-4-9		9 17A	1			14 402	2
		9 60	4			14 490	4
		9 60A	6	MS51957-15		9 10	8
		16 25A	1			9 25	2
		16 110	6			16 15	8
MS35214-14		2 30	2			16 105	2
		15 30	2	MS51957-16		2 85	14
MS35214-16		2 10	5	MS51957-17		2 85A	14
		15 10	5			15 90	14
MS35216-56		2 125	1	MS51957-2		14 55	6
		15 130	1	MS51957-26		9 95	12
MS35333-70		2 90	14			10 255	3
		14 122	2			14 280	3
		14 202	4			14 280B	1
		14 267	2	MS51957-28		9 175	4
		14 502	4			13 75	0
		15 95	14			16 275	4
MS35338-134		1 140	0	MS51957-30		1 10	2
		9 117	0	MS51957-31		2 165	3
		10 145	0			15 170	3
MS35338-135		1 180	4	MS51957-4		9 105	10
		10 30	3			16 98	2
		10 360	2			16 185	10
		10 380	2	MS51957-40		1 27	5
		11 11	4			1 28	1
		11 33	2	MS51957-43		8 15	4
		14 30	3			8 40	2
		14 360	2			9 165	4
		14 380	2			9 217	2
MS35338-136		1 15	2			16 250	4
MS35338-137		1 160	4	MS51957-45		9 215	2
		9 242	2			9 217A	2
		16 320	2			9 240	2
MS35338-39		14 65	6			16 295	4
MS35338-42		1 160A	4			16 315	2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
MS51957-50		1 155	4	RNC55H1000FS		4 40	1
MS51957-6		3 80	2			5 36	1
		4 90	2			5 38	2
		5 100	2	RNC55H1001FS		3 35	2
		6 50	2			5 40	3
		7 110	2			6 40	5
MS51959-3		9 205	6			6 40A	6
		16 285	6			7 85	1
MS75101-3		5 24A	1			12 15	1
N15F210B		1 30	1	RNC55H1002BS		3 245	4
N6550-4		9 170	2	RNC55H1002FS		3 40	8
		9 245	2			3 150	1
		16 265	2			4 115	9
		16 325	2			5 35	3
NAS1189E04P5L		14 210A	1			5 35A	1
		14 225A	1			5 37	1
NAS1189E06P5L		9 250A	4	RNC55H1003FS		4 30	1
		16 330	4	RNC55H1004FS		5 53	1
NAS1352C08-20		1 35B	4	RNC55H1052FS		5 188	AR
NAS43DD1-8		14 282	2			7 173	AR
NAS620-4		14 60	6	RNC55H1053FS		5 220	AR
NAS620B10L		2 130	1			7 205	AR
		15 135	1	RNC55H1072FS		3 151	AR
NAS620B4L		2 15	5	RNC55H1132FS		5 189	AR
		2 35	2			7 174	AR
		15 15	5	RNC55H1133FS		5 221	AR
		15 35	2			7 206	AR
NAS620C2		9 115	10	RNC55H1152FS		3 152	AR
		10 140	0	RNC55H1212FS		5 190	AR
NAS620C6		11 55	2			7 175	AR
		13 60	2	RNC55H1213FS		5 222	AR
NAS620C6L		10 260	3			7 207	AR
		14 285	3	RNC55H1242FS		3 153	AR
NAS623-2-12		1 155A	4	RNC55H1302FS		5 191	AR
NOTUSED		5 150	0			7 176	AR
		7 150	0	RNC55H1303FS		5 223	AR
		8 65	0			7 208	AR
		9 185	1	RNC55H1331FS		3 45	8
		9 220	0			5 145	2
		12 25	0	RNC55H1332FS		3 154	AR
NY04-090G		2 105E	0	RNC55H1333FS		4 20	1
		15 110A	14	RNC55H1402FS		5 192	AR
NY04-090N		2 105F	0			7 177	AR
		15 110B	14	RNC55H1403FS		5 224	AR
NY04-125G		2 105B	14			7 209	AR
		15 110	14	RNC55H1432FS		3 155	AR
NY04-125N		2 105A	0	RNC55H1501FS		3 255	2
NY04-125T		2 105C	0	RNC55H1502FS		3 220	1
NY04-90G		2 105D	0			5 193	AR
PHD600-828		1 35C	4			7 60	1
PT07A22-55S		9 80A	1			7 178	AR
		16 140B	1	RNC55H1503FS		5 225	AR
RRC20G122JS		3 120	1			7 210	AR

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
RNC55H1542FS		3 156	AR	RNC55H3162FS		3 166	AR
RNC55H1622FS		5 194	AR	RNC55H3320FS		4 145	5
		7 179	AR			6 110	2
RNC55H1623FS		5 226	AR	RNC55H3321FS		3 210	3
		7 211	AR			4 80	8
RNC55H1652FS		3 157	AR			4 81	1
RNC55H1742FS		5 195	AR			4 126	2
		7 180	AR			6 70	7
RNC55H1743FS		5 227	AR			6 70A	5
		7 212	AR	RNC55H3322FS		3 25	18
RNC55H1782FS		3 158	AR			4 160	1
RNC55H1872FS		5 196	AR			5 65	1
		7 181	AR			5 204	AR
RNC55H1873FS		5 228	AR			7 189	AR
		7 213	AR	RNC55H3402FS		3 167	AR
RNC55H1912FS		3 159	AR	RNC55H3572FS		5 205	AR
RNC55H2000FS		7 25	1			7 190	AR
RNC55H2001FS		3 100	8	RNC55H3652FS		3 168	AR
RNC55H2002FS		5 197	AR	RNC55H3832FS		5 206	AR
		7 182	AR			7 191	AR
RNC55H2003FS		5 229	AR	RNC55H3922FS		3 169	AR
		7 214	AR	RNC55H4122FS		5 207	AR
RNC55H2004FS		5 56	1			7 192	AR
		7 56	1	RNC55H4222FS		3 170	AR
RNC55H2052FS		3 160	AR	RNC55H4422FS		5 208	AR
RNC55H2152FS		5 198	AR			7 193	AR
		7 183	AR	RNC55H4532FS		3 171	AR
RNC55H2212FS		3 161	AR	RNC55H4641FS		3 145	2
RNC55H2321FS		6 160	2			5 30	8
RNC55H2322FS		5 199	AR	RNC55H4752FS		5 209	AR
		7 184	AR			7 194	AR
RNC55H2372FS		3 162	AR	RNC55H4872FS		3 172	AR
RNC55H2430FS		2 53	1	RNC55H5110FS		5 17	2
		3 135	1			5 20	0
		7 95	2	RNC55H5112FS		5 210	AR
		7 95A	1			7 195	AR
		15 55	1	RNC55H5232FS		3 173	AR
RNC55H2431FS		4 155	1	RNC55H5492FS		5 211	AR
RNC55H2491FS		3 140	2			7 196	AR
RNC55H2492FS		5 200	AR	RNC55H5622FS		3 174	AR
		7 185	AR	RNC55H5901FS		5 180	1
RNC55H2552FS		3 163	AR			6 100	2
RNC55H2672FS		5 201	AR			7 165	1
		7 186	AR	RNC55H5902FS		5 212	AR
RNC55H2742FS		3 164	AR			7 197	AR
RNC55H2801FS		7 90	1	RNC55H6042FS		3 175	AR
RNC55H2872FS		5 202	AR	RNC55H6341FS		5 181	AR
		7 187	AR			7 166	AR
RNC55H2942FS		3 165	AR	RNC55H6342FS		5 213	AR
RNC55H3011FS		3 105	1			7 198	AR
		6 95	1	RNC55H6492FS		3 176	AR
RNC55H3092FS		5 203	AR	RNC55H6811FS		5 182	AR
		7 188	AR			7 167	AR

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
RNC55H6812FS		5 214	AR	SG1501AJ883B		7 220	1
		7 199	AR	SG2003J		7 75C	2
RNC55H6982FS		3 177	AR	SG2003J883B		7 75B	2
RNC55H7321FS		5 183	AR	SN54LS145F883B		12 10B	1
		7 168	AR	SN54LS157J883		6 80E	1
RNC55H7322FS		5 215	AR	SN54LS244JB		5 170D	2
		7 200	AR	SN54LS374JB		4 75D	5
RNC55H7502FS		3 178	AR	SN55462JG		6 20D	2
		5 60	1			6 22B	1
RNC55H7871FS		5 184	AR	SNJ5406J		6 120A	1
		7 169	AR	SNJ5406J00		6 120	1
RNC55H7872FS		5 216	AR	SNJ5417J		4 55A	2
		7 201	AR	SNJ5417J00		4 55	2
RNC55H78R7FS		4 137	1	SNJ54LS00J		3 205A	1
RNC55H8062FS		3 179	AR			4 5A	1
RNC55H8063FS		7 50	1			5 80A	1
RNC55H8252FS		7 30	1			6 130A	1
RNC55H8451FS		5 185	AR	SNJ54LS00J00		3 205	1
		7 170	AR			4 5	1
RNC55H8452FS		5 217	AR			5 80	1
		7 202	AR			6 130	1
RNC55H8662FS		3 180	AR	SNJ54LS02J		3 200A	1
RNC55H9091FS		5 186	AR	SNJ54LS02J00		3 200	1
		7 171	AR			4 10	3
RNC55H9092FS		5 218	AR	SNJ54LS04J		3 195A	1
		7 203	AR	SNJ54LS04J00		3 195	1
RNC55H9312FS		3 181	AR	SNJ54LS10J		4 125A	1
RNC55H9761FS		5 187	AR	SNJ54LS10J00		4 125	1
		7 172	AR	SNJ54LS123J		4 25A	1
RNC55H9762FS		5 219	AR	SNJ54LS123J00		4 25	1
		7 204	AR	SNJ54LS138J		5 85A	1
RNC60H75R0FS		7 15	2	SNJ54LS138J00		5 85	1
RWR80S10R0FR		6 30	1	SNJ54LS145J		3 95A	2
RWR80SR511FS		7 160	2			12 10A	1
RWR81S10R0FR		7 140	1	SNJ54LS145J00		3 95	2
RWR81S4420FR		6 25	1			12 10	1
RWR81S51R1FR		6 5	1	SNJ54LS14J		4 110A	2
RWR81S82R5FR		7 35	1			5 130A	2
		12 20	2	SNJ54LS14J00		4 110	2
S0S632-8		9 287	4			5 130	2
		9 312	4	SNJ54LS157J		6 80A	1
S1027-2B		5 5B	1	SNJ54LS157J00		6 80	1
S54LS138F883B		5 85C	1	SNJ54LS175J		4 70A	1
S54LS145F883B		3 95B	2	SNJ54LS175J00		4 70	1
S54LS175FBEA		4 70C	1	SNJ54LS244J		5 170A	2
S54LS244F883B		5 170B	2	SNJ54LS244J00		5 170	2
SBE20STG30		6 195A	1	SNJ54LS374J		4 75A	5
SG117K883B		2 80A	1	SNJ54LS374J00		4 75	5
		15 85A	1			4 76	0
SG117K883C		2 80B	1	SNJ54LS54J		5 135A	1
SG123K883B		2 55	2	SNJ54LS54J00		5 135	1
		15 60	2	SNJ54LS74AJ		4 120A	2
SG1501AJ		7 220A	1			5 140A	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
SNJ54LS74J00		4 120	2	06/16/5610		6 60	2
		5 140	1			7 120	2
SNJ54LS86AJ		5 120A	1			9 115A	10
SNJ54LS86J00		5 120	1			16 190	10
SNJ55462JG		6 20C	2	062-0437MDP		11 100A	1
		6 22A	1	07/10/5710		14 342	AR
SNJ55472JG		6 20B	2	07/12/5710		14 343	AR
		6 22	1	1032NBSEXRR		9 266	3
SNJ55472JG00		6 20	2			9 291	3
SNJ55472JG883B		6 20A	2			16 350	3
SP21		6 125	1			16 385	3
SR1883PPK15-7		13 112	2	1032NBSEXRR091		9 266A	3
SS0120T2		6 195B	1			9 291A	3
SST1-5M		7 12A	2			16 350A	3
		7 137A	3			16 385A	3
TC814		9 65	4	10408-1		2 65D	1
		9 65A	6			15 70D	1
		16 120	6	10624-01-446		5 117	1
TL56		10 270	1	109-0437MDP		11 100	1
		14 295	1	110-2026-057		16 97A	24
TS120TA		6 175A	AR	117DBM25P		14 395A	1
UA7812KM		2 60A	1	117DBM25S		9 45A	1
		15 65C	1			16 85A	1
UA7812KMQB		2 60	1	119-2020-080		9 16A	57
		15 65	1			16 10A	57
UA7912KM		2 65B	1	1392C1		9 145	7
		15 70B	1			16 220	7
UA7912KMQB		2 65	1	1392C4-5		9 145A	7
		15 70	1	1410-14		9 28A	6
UA79M05HMQB		7 125	1			17 45	6
ULS2003H883B		7 75	2	14120-01-446		6 195	1
ULS2003R883C		7 75A	2	1412-4		2 110A	7
W0242-006S		14 95A	1			15 115	7
W0367-006S		14 165A	1	1416-6		16 45	4
WMC0LWTB500		16 115	7	1485-8		2 178B	1
Z427		2 180	2			15 185A	1
		15 190	2	1488-6		2 178A	1
Z673000801		16 65	2			14 295A	1
01/04/2006		11 60	1			15 185	1
		13 65	1	1-5KE220CA		16 60A	1
030-9081-000		9 16	57			16 60B	1
		16 10	57			16 60C	1
031-1007-004		16 97	24			16 60D	1
06/15/5610		3 90A	2	18091BB115-14		2 20A	5
		4 100A	2			15 20A	5
		5 110A	2	1M6561MDN883B		5 90	2
		6 60A	2	1N5711		4 45B	2
		7 120A	2			4 46B	1
		9 115B	10			6 105B	2
		16 190A	10	20000-96		4 147A	3
06/16/5610		3 90	2	2000B4-5		12 37	26
		4 100	2	2002-10TTE		6 175	AR
		5 110	2	2002B1		12 35	32

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
2002B4-5		12 35A	32	360-0082-002		13 135	1
2162		9 97	1	360-0082-003		13 110A	0
2204-07000CAP		11 30	2			13 117	1
2254PH141		10 262	AR	360-0082-004		13 135A	0
2254PH141-1		14 290	3			13 142	1
22NKM62		9 325	2	360-0082-006		13 160	1
		16 425	2	390		2 110	7
2360-006		17 15	2			2 178	1
24-0009-1400		17 20	6	3SAV5004D1		6 10	2
24-0010-0700		17 25	6	400-0056-040		10 80	2
24SX2-7-3L		9 280	1			10 115	2
		16 370	1	400-0056-041		10 175	10
24SX2-7-3R		9 305	1			13 30A	2
		16 405	1			14 325A	2
2673000801		16 65A	2			14 470	2
2829-75-2		5 10	1	400-0056-042		13 30	2
		6 185	1			14 325	2
		7 10	4			14 470A	2
		7 135	6	400-0056-043		14 170	2
29		9 28	6	400-0056-045		14 170A	2
2N5582		6 150B	2			14 550	2
310-0285-001		2 120	1	400-0056-052		13 30B	2
		15 125	1			14 325B	2
310-0501-001		6 135A	1			14 470B	2
316-6004-001		1 30I	1	400-0056-053		13 30C	2
316-6004-003		1 30E	1			14 325C	2
316-6004-004		1 30H	1			14 470C	2
316-6004-005		1 30G	1	400-0056-060		13 31A	2
316-6004-006		1 30F	1			14 326	2
317-1039-001		10 190	2			14 471	2
317-1044-001		14 425	1	400-0056-061		13 31	2
317-1044-002		10 185A	1			14 326A	2
		10 187A	1			14 471A	2
		14 420	1	400-0056-062		14 170B	2
317-1044-003		14 425A	1			14 550A	2
317-1044-004		14 420A	1	400-0056-063		13 31B	2
		14 430A	0			14 326B	2
317-1045-001		10 190A	2			14 471B	2
		14 430	2	400-0056-064		13 31C	2
317-1045-002		10 190B	1			14 326C	2
317-1045-003		10 190C	1			14 471C	2
317-1045-004		14 430B	2	400-0056-065		13 31D	2
321-0213-006		4 77	0			14 326D	2
3324BLACK		13 65B	1			14 471D	2
35-1CR2-7		9 270	6	400-0057-027		10 210	1
		9 295	6			14 215	1
		16 360	6	400-0057-028		10 230	1
		16 395	6			14 235	1
3531C4-5		9 145B	7	400-0057-030		14 215A	1
		16 220A	7	400-0057-031		14 235A	1
360-0082-001		11 105	1	400-0057-032		13 137	0
		13 110	1			13 143	2
360-0082-002		11 105A	1	400-0057-033		13 112A	0

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
400-0057-033		13 118	2	434-0201-007		10 180	5
		13 165	2			13 15	AR
400-0058-003		14 100	2			14 261	AR
400-0058-004		14 77	2			14 310	AR
400-0058-005		14 77A	2			14 460	AR
400-0172-001		13 30D	2	434-0201-008		10 165	AR
		14 325D	2			13 20	AR
		14 470D	2			14 261A	AR
400-7077-001		17 50	1			14 315	AR
401-8065-001		9 235A	1			14 455	AR
402-3036-001		2 145	1	434-0201-009		10 170	AR
		15 150	1			13 25	AR
403-6015-011		14 90	AR			14 261B	AR
406-1508-001		10 50	1			14 320	AR
406-1508-002		10 50A	1			14 341	AR
406-1508-003		10 50B	1			14 450	AR
406-1510-001		14 35	1	434-0201-010		10 310	AR
406-1510-005		10 420	1			14 185	AR
406-1511-004		14 40	1			14 262A	AR
406-1511-005		14 40A	1			14 505	AR
431-0343-013		14 340	2	434-0201-011		10 315	AR
431-0343-014		13 45	2			14 190	AR
431-0433-001		8 50	2			14 262B	AR
431-0433-002		8 50A	0			14 510	AR
431-0497-014		13 45A	2	434-0201-012		10 320	AR
434-0071-002		14 165	1			10 325	4
434-0086-001		14 150	AR			14 195	AR
434-0086-023		14 160A	AR			14 262	AR
		14 530	AR			14 515	AR
434-0086-024		14 155	AR	434-0201-013		14 116	AR
		14 535	AR	434-0201-014		14 196	AR
434-0086-025		14 150A	AR			14 262C	AR
		14 540	AR			14 520	AR
434-0086-026		14 14	4	434-0274-001		14 95B	1
434-0086-029		14 160	AR	440-0005-009		16 5B	1
434-0086-136		14 162	AR	440-0005-028		9 5A	1
		14 545	AR			16 5C	1
434-0171-003		14 95	1	440-0507-002		9 100A	5
434-0201-001		10 65	AR			16 180B	5
		10 85	1	46NSEX8		9 287A	4
		10 100	AR			9 312A	4
		10 120	1	5075B4-5		12 37A	26
434-0201-002		10 70	AR	5082-4587		9 85	1
		10 105	AR	5100-12C		14 145A	1
434-0201-003		10 75	AR	5133-18		10 60	1
		10 110	AR			10 95	1
434-0201-004		10 285	AR			14 85	1
		14 105	AR	5133-9H		10 155	5
434-0201-005		10 290	AR			10 245	2
		14 110	AR			10 335	2
434-0201-006		10 295	AR			11 20	2
		14 115	AR			13 10	1
434-0201-007		10 160	AR			14 145	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
5133-9H		14 250	1	600-3967-001		10 415	1
		14 305	1	600-3967-002		10 415A	1
		14 445	1	600-3967-003		10 415B	1
51-717-007		16 55	2	600-3967-004		14 415C	1
		17 10	4	600-3967-008		14 415A	1
51-717-021		16 55A	2	600-3969-001		10 90	1
		17 10A	4	600-3969-002		10 90A	1
54LS00BCA		3 205C	1	600-3970-001		10 410	1
		4 5C	1			14 410	1
		5 80C	1	600-3971-001		10 330	2
		6 130C	1			11 15	2
54LS00BCXJC		3 205B	1	600-3975-001		11 95	0
		4 5B	1	600-3975-002		11 95A	0
		5 80B	1	600-3975-003		11 95B	1
		6 130B	1	600-3975-004		11 95C	1
54LS02BCA		3 200C	1	600-3991-001		10 45	1
		4 10B	3	600-3991-002		10 45A	1
54LS02BCXJC		3 200B	1	600-4103-001		10 385	1
		4 10A	3			14 385	1
54LS04BCA		3 195C	1	600-4108-001		10 150	5
54LS04BCXJC		3 195B	1			14 475	1
54LS10BCA		4 125C	1	600-4108-002		14 475A	1
54LS10BCXJC		4 125B	1	600-4108-003		14 475B	1
54LS123BEXJC		4 25B	1	600-4143-001		1 187	4
54LS14BCA		4 110D	2			2 23	1
		5 130C	2			15 23	1
54LS14BCXJC		4 110B	2	600-4507-001		14 50	1
		5 130B	2	600-4674-001		10 425	2
54LS14DMBQ		4 110C	2			11 15A	2
54LS157BEXJC		6 80B	1	600-4679-001		13 120	1
54LS157DMQB		6 80C	1			13 145	1
54LS157J883B		6 80D	1	600-4681-001		13 35	1
54LS175BEXJC		4 70B	1			14 330	1
54LS244DMQB		5 170C	2	600-4683-002		14 415	1
54LS374BRAJC		4 75B	5	600-4683-003		14 415B	1
54LS54DMBQ		5 135B	1	600-5128-001		14 50A	1
54LS74ABCA		4 120C	2	602-2417-001		9 190	1
		5 140D	1	602-2417-002		9 190A	1
54LS74ABCXJC		4 120B	2			16 255	1
		5 140B	1	602-2461-001		9 57	1
54LS74DMQB		4 120E	2	602-2482-001		10 365	1
		5 140C	1			14 365	1
54LS86BCA		5 120C	1	602-2913-001		9 315	1
54LS86BCXJC		5 120B	1			16 415	1
5602-12-62		9 97A	1	602-2955-001		15 195	1
56040-4-28		8 10	4	602-2955-005		15 195A	1
5608-9		2 105	0	602-3139-002		16 30	1
5710-23-020		14 13	4	603-1001-001		10 200	1
58LNK060		9 267A	3			14 205	1
		9 292A	3	603-1001-003		14 205A	1
		16 355A	3	603-1001-004		14 205C	1
		16 390A	3	603-1001-005		14 205B	1
600-3964-001		8 60	1	603-1002-001		10 215	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
603-1002-001		14 220	1	617-3837-003		11 90B	1
603-1003-001		10 240	2	617-4205-001		14 5	1
		14 255	1	617-4205-002		14 5A	1
603-1003-002		14 255C	1	617-4261-003		13 105	1
603-1010-001		14 255A	1	617-4261-004		13 130	1
604-0317-001		17 6	1	617-4261-005		13 105A	1
604-0317-002		17 5	1	617-4261-006		13 130A	1
605-0145-002		8 35B	1	617-4261-007		13 105B	1
607-0723-001		1 5	1	617-4261-008		13 130B	1
607-0723-002		1 5A	1	617-4261-009		13 116	1
607-0723-006		1 5B	1	617-4261-010		13 141	1
607-1519-001		8 5	1	617-4261-012		13 155	1
607-1519-002		8 5B	1	617-4262-001		14 45	1
607-1519-003		8 5C	1	617-4262-003		14 45A	1
607-1520-001		10 345	1	617-4262-004		14 45B	1
		14 345	1	617-4262-005		14 45C	1
607-1527-001		2 185	1	617-4262-006		14 45D	1
607-1527-002		2 185A	1	617-4422-001		8 25B	1
607-1527-003		2 185B	1			14 1	RF
607-1527-004		2 185C	1	617-4422-003		8 25C	1
607-1644-001		8 5A	1			14 1A	RF
611-0153-002		8 20	1	617-4422-004		8 25D	1
611-0157-001		11 12	0			14 1B	RF
613-1281-001		14 76	1	617-4422-005		8 25E	1
614-0603-001		1 170	1			14 1C	RF
617-3697-001		8 25	1	617-4422-006		8 25F	1
		10 1	RF			14 1D	RF
617-3697-002		8 25A	1	617-4422-007		8 25G	1
		10 1A	RF			14 1E	RF
617-3698-001		8 45	1	617-4422-008		8 25H	1
		11 1	RF			14 1F	RF
617-3698-002		8 45B	1	617-4422-009		8 25I	1
		11 1A	RF			14 1G	RF
617-3716-001		1 190	1	617-4422-010		8 25J	1
		9 1	RF			14 1H	RF
617-3716-002		1 190A	1	617-4422-011		8 25K	1
		9 1A	RF			14 1I	RF
617-3716-005		1 190C	1	617-4422-012		8 25L	1
		9 1C	RF			14 1J	RF
617-3716-006		1 190B	1	617-4422-013		8 25M	1
		9 1B	RF			14 1K	RF
617-3731-001		10 55	1	617-4422-014		8 25N	1
617-3731-003		10 55A	1			14 1L	RF
617-3746-001		9 225	1	617-4422-015		8 25P	1
		16 300	1			14 1M	RF
617-3827-001		11 5	1	617-4422-016		8 25Q	1
617-3829-001		11 25	1			14 1N	RF
617-3831-001		2 5	1	617-4422-017		8 25R	1
		15 5	1			14 1P	RF
617-3831-002		2 5A	1	617-4422-023		8 25S	1
		15 5A	1			14 1Q	RF
617-3837-001		11 90	1	617-4422-024		8 25T	1
617-3837-002		11 90A	1			14 1R	RF

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
617-4422-025		8 25U	1	620-1351-001		1 130B	1
		14 1S	RF	620-1351-002		1 130C	1
617-4422-026		8 25V	1	620-1351-003		1 130A	1
		14 1T	RF	620-1351-004		1 130	1
617-4422-027		8 25W	1	620-1351-007		1 130D	1
		14 1U	RF	620-1351-008		1 130E	1
617-4429-001		8 45A	1	620-1351-009		1 130F	1
		13 1	RF	620-1407-001		1 130G	1
617-4429-002		8 45C	1	620-2461-001		16 100	1
		13 1A	RF	621-1195-001		9 260	1
617-4431-001		14 140	5			16 340	1
		14 140A	3	621-1453-001		9 260A	1
617-4431-002		14 440	5	623-2274-001		5 235C	1
617-4431-003		14 440A	2	623-2275-003		9 135	1
617-4431-004		14 140B	5	623-2275-005		9 135A	1
		14 140C	3	623-2276-002		7 225	1
617-4431-005		14 440B	5	623-2276-003		7 225A	1
617-4431-006		14 440C	2	623-2277-001		4 150	1
617-4438-001		14 245	2	623-2277-002		4 150A	1
617-4438-002		14 245A	2	623-2277-003		4 150B	1
617-4438-003		14 245B	2	623-2283-001		3 260B	1
617-4438-004		14 245C	2	623-2283-002		3 260	1
617-4438-005		14 245D	2	623-2283-003		3 260A	1
617-4438-006		14 245E	2	623-2285-003		12 30	1
617-4439-001		13 5	2	623-2285-005		12 30A	1
		14 300	2	623-2311-002		6 140	1
617-4439-002		14 300D	2	623-2378-001		5 235	1
617-4439-003		13 5A	2	623-2378-002		5 235A	1
		14 300A	2	623-2378-003		5 235B	1
617-4439-004		13 5B	2	623-2825-001		16 210	1
		14 300B	2	623-2825-004		16 210A	1
617-4439-005		14 300E	2	623-3137-001		6 180	1
617-4439-006		13 5C	2	624-0769-001		9 87	1
		14 300C	2			16 150	1
617-4440-001		14 70	2	628-0256-001		8 70	3
617-4440-002		14 70A	2	628-0261-001		2 21	5
617-4440-003		14 70B	2			15 21	5
617-4440-004		14 70C	2	631-0401-001		1 35	4
617-4640-006		1 190D	1	634-1429-004		14 7	0
		16 1	RF	634-1438-001		10 35	1
617-5006-001		14 255B	1	634-1439-001		10 5	1
617-5007-001		14 5B	2	634-1442-001		9 265	1
617-5007-002		14 5C	2	634-1442-002		9 290	1
617-5007-003		14 5D	2	634-1442-003		9 265C	1
617-5412-001		14 485	3	634-1442-004		9 290C	1
		14 485A	5	634-1442-005		9 265A	1
617-5412-002		14 485B	2	634-1442-006		9 290A	1
617-5412-003		14 485C	5	634-1442-007		9 265B	1
		14 485D	3			16 345	1
617-5412-004		14 485E	2	634-1442-008		9 290B	1
617-5489-001		14 75B	1			16 380	1
617-5489-002		14 75C	1	634-1442-009		9 265D	1
620-0723-005		1 127	1	634-1442-010		9 290D	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
634-1443-001		9 160	1	646-0401-001		14 335B	1
634-1443-002		9 160A	1	648-1173-002		1 70	1
634-1443-003		9 160C	1	648-1173-003		1 70A	1
634-1443-004		9 160B	1	648-1177-001		9 88	1
634-1443-006		16 245	1			16 155	1
634-1455-001		9 155	1	651		1 125	1
634-1455-002		9 155A	1	651-8092-001		14 75	1
634-1455-005		9 155C	1	651-8092-002		14 75A	1
634-1455-006		9 155B	1	652		1 25	1
634-1455-008		16 240	1	652-0202-001		9 200	2
634-1463-001		11 40	1			16 280	2
634-1494-001		8 55	1	652-0274-001		14 175	1
634-1494-002		8 55A	1	652-0274-002		14 555	1
634-1615-002		14 6	0	652-0274-003		14 555A	1
634-2003-001		14 4	1	652-2002-001		9 210	2
636-0217-001		8 35	1			16 290	2
636-0217-002		8 35A	1	653-9043-001		1 91	1
636-0220-001		10 265	1	654-0144-001		11 45	1
		14 275	1	654-0144-002		11 45A	1
636-0220-003		14 275A	1	654-0146-001		1 40	2
643-0448-001		1 147	1	654-0174-001		1 40A	2
645-0352-001		10 235	1	654-0222-001		13 50	1
		14 240	1	654-0233-001		13 50A	1
645-5107-001		14 422	AR	65521-136		3 240	1
		14 427	AR			4 130	1
		14 432	AR	65521-436		3 240A	1
645-5107-002		14 423	AR			4 130A	1
		14 428	AR	700-1066-001		1 150	1
		14 433	AR			8 1	RF
645-5107-003		10 197	3	700-1066-002		1 150C	1
		14 421	AR	700-1066-003		1 150D	1
		14 426	AR	700-1066-004		1 150A	1
		14 431	AR			8 1A	RF
645-5125-001		14 261C	AR	700-1066-005		1 150B	1
		14 465	AR			8 1B	RF
646-0323-001		10 340	2	700-1066-006		1 150E	1
		11 35	0			8 1C	RF
646-0323-002		10 340A	2	700-1066-007		1 150F	1
646-0325-001		10 300	7			8 1D	RF
		14 260	1	700-1066-008		1 150G	1
		14 480	1			8 1E	RF
646-0325-003		14 480A	1	700-1066-009		1 150H	1
646-0326-001		10 275	2			8 1F	RF
646-0379-001		13 40	1	700-1066-010		1 150I	1
		14 335	1			8 1G	RF
646-0379-002		13 40A	1	700-1066-011		1 150J	1
		14 335A	1			8 1H	RF
646-0380-001		14 80	1	700-1066-012		1 150K	1
646-0380-003		14 80A	1			8 1I	RF
646-0398-001		14 260A	1	700-1066-013		1 150L	1
646-0399-001		14 180	1			8 1J	RF
		14 560	1	700-1066-014		1 150M	1
646-0399-002		14 180A	1			8 1K	RF

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
704-0487-001		9 180	1	722-2438-012		5 1K	RF
		16 270	1	722-2438-013		1 115L	1
712-0084-001		17 40	1			5 1L	RF
716-0180-001		9 20A	1	722-2438-014		1 115M	1
		17 1A	RF			5 1M	RF
716-0180-002		9 20	1	722-2438-015		1 115N	1
		17 1	RF			5 1N	RF
718-0423-002		5 115	1	722-2438-016		1 115P	1
718-0423-003		5 115A	1			5 1P	RF
718-0423-004		5 115B	1	722-2438-017		1 115Q	1
718-0423-005		5 115C	1			5 1Q	RF
718-0423-007		5 115D	1	722-2438-018		1 115R	1
718-0423-008		5 115E	1			5 1R	RF
718-0423-009		5 115F	1	722-2438-019		1 115S	1
718-0423-010		5 115G	1			5 1S	RF
718-0423-011		5 115H	1	722-2438-020		1 115T	1
718-0423-012		5 115I	1			5 1T	RF
718-0423-013		5 115J	1	722-2438-021		1 115U	1
718-0423-014		5 115K	1			5 1U	RF
720-0123-001		1 95	1	722-2438-022		1 115V	1
		2 1	RF			5 1V	RF
720-0123-002		1 95A	1	722-2438-023		1 115W	1
		2 1A	RF			5 1W	RF
720-0123-003		1 95B	1	722-2438-024		1 115X	1
		2 1B	RF			5 1X	RF
720-0123-004		2 1C	0	722-2438-025		1 115Y	1
720-0167-002		1 95C	1			5 1Y	RF
		15 1	RF	722-2438-026		1 115Z	1
720-0167-004		1 95D	1			5 1Z	RF
		15 1A	RF	722-2438-027		1 116	1
722-2438-001		1 115	1			5 2	RF
		5 1	RF	722-2438-028		1 116A	1
722-2438-002		1 115A	1			5 2A	RF
		5 1A	RF	722-2438-029		1 116B	1
722-2438-003		1 115B	1			5 2B	RF
		5 1B	RF	722-2438-030		1 116C	1
722-2438-004		1 115C	1			5 2C	RF
		5 1C	RF	722-2438-031		1 116D	1
722-2438-005		1 115D	1			5 2D	RF
		5 1D	RF	722-2438-032		1 116E	1
722-2438-006		1 115E	1			5 2E	RF
		5 1E	RF	722-2438-033		1 116F	1
722-2438-007		1 115F	1			5 2F	RF
		5 1F	RF	722-2438-034		1 116G	1
722-2438-008		1 115G	1			5 2G	RF
		5 1G	RF	722-2438-035		1 116H	1
722-2438-009		1 115H	1			5 2H	RF
		5 1H	RF	722-2438-036		1 116I	1
722-2438-010		1 115I	1			5 2I	RF
		5 1I	RF	722-2438-037		1 116J	1
722-2438-011		1 115J	1			5 2J	RF
		5 1J	RF	722-2438-038		1 116K	1
722-2438-012		1 115K	1			5 2K	RF

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ	PART NUMBER	AIRLINE STOCK NUMBER	FIGURE/ ITEM	TTL REQ
722-2438-039		1 116L	1	722-3523-001		6 145	1
		5 2L	RF	7403-09FR06		2 115	7
722-2438-040		1 116M	1			15 120	7
		5 2M	RF	744-0078-001		10 130	1
722-2438-041		1 116N	1			14 130	1
		5 2N	RF	744-0078-002		10 125	1
722-2440-001		2 25	1			14 125	1
		7 1	RF	751-0048-001		6 190	1
722-2440-002		2 25A	1	754-0011-001		10 187	1
		7 1A	RF	754-0011-002		10 185	1
722-2440-003		2 25B	1	7X9-3W06108		10 225	1
		7 1B	RF			14 230	1
		15 25	1	810-2007		1 65A	1
722-2440-004		2 25C	0	810-2007K		1 65B	1
		7 1C	0	810-342		1 50	1
722-2440-005		7 1D	RF	810-371A		1 60	1
		15 25A	1	810-418A		1 65	1
722-2441-001		1 110	1	810-419		1 55	1
		4 1	RF	810-423		1 45	1
722-2441-002		1 110A	1	898-3R1-5K		3 70A	2
		4 1A	RF	9115		10 15	3
722-2441-003		1 110C	1			14 15	3
		4 1C	RF	9535CA0440-0		9 150	4
722-2441-004		1 110B	1			16 225	4
		4 1B	RF	9535CA0440-17		9 150A	4
722-2441-005		1 110D	1			16 225A	4
		4 1D	RF	95C8H		9 30	0
722-2441-006		1 110E	1			16 70	0
		4 1E	RF	9705CA0440-0		9 140	4
722-2441-007		1 110F	1			16 215	4
		4 1F	RF	9705CA115-17		2 20	5
722-2441-008		1 110G	1			15 20	5
		4 1G	RF	973-0031-561		2 150	1
722-2442-001		1 105	1			15 155	1
		3 1	RF	974-0000-574		2 150A	1
722-2442-002		1 105A	1			15 155A	1
		3 1A	RF	980-4100AXUN		1 3C	RF
722-2442-003		1 105B	1	980-4100AXUS		1 3B	RF
		3 1B	RF	980-4100AXXN		1 3A	RF
722-2501-001		1 120	1	980-4100AXXS		1 3	RF
		6 1	RF	980-4100BXUN		1 1E	RF
722-2501-002		1 120A	1	980-4100BXUS		1 1D	RF
		6 1A	RF	980-4100DXUN		1 1C	RF
722-2501-003		1 120B	1	980-4100DXUS		1 1B	RF
		6 1B	RF	980-4100DXXN		1 1A	RF
722-2503-001		9 90	1	980-4100DXXS		1 1	RF
722-2503-002		9 90A	1	980-4101-001		1 2	RF
722-2522-001		10 250	1	980-4101-002		1 2A	RF
		12 1	RF	9LS54LS138J883B		5 85B	1
722-2522-002		12 1A	RF				
		14 270	1				
722-3142-005		16 170	1				
722-3142-009		16 170A	1				

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

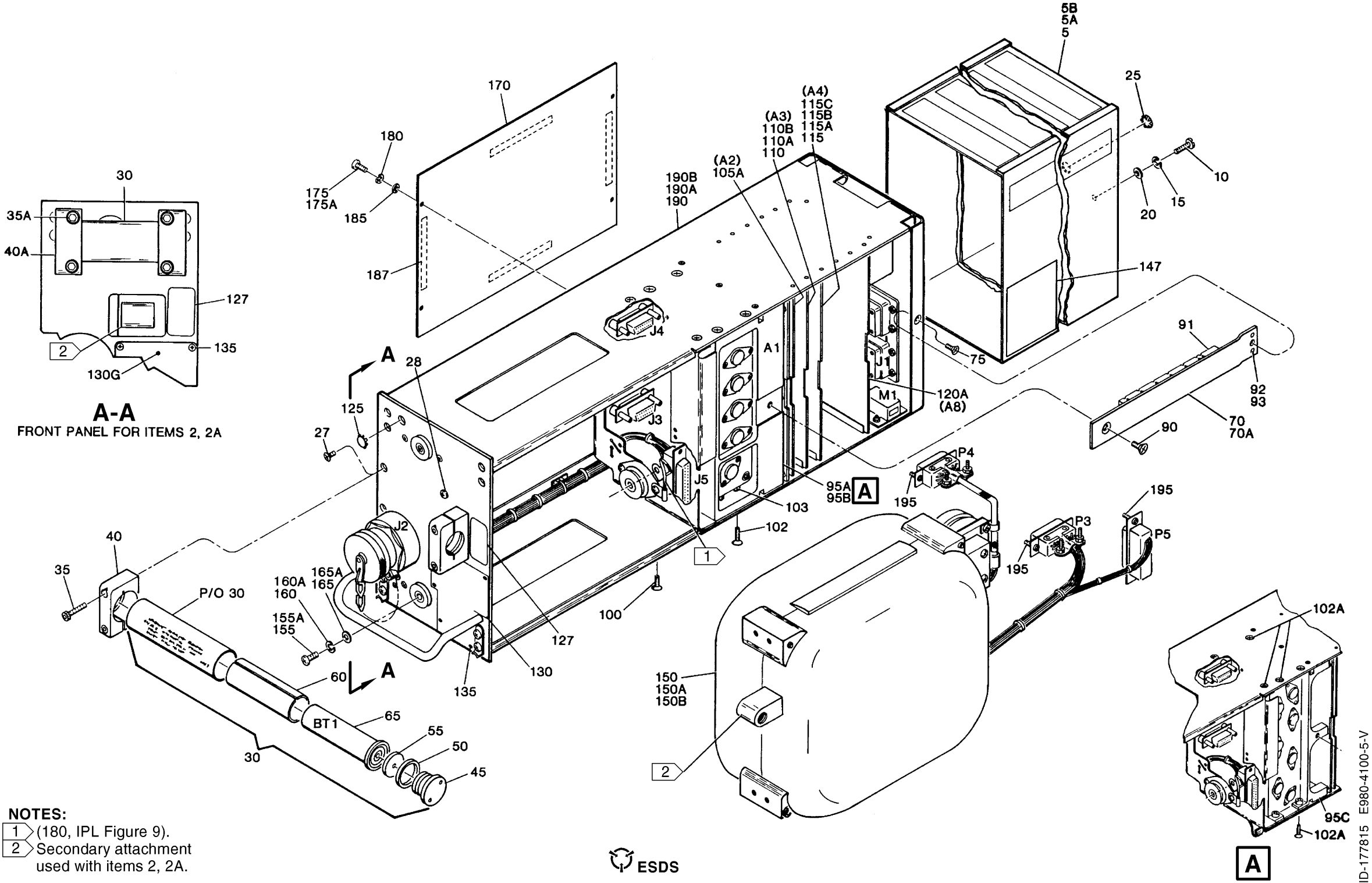
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

6. Parts List

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK



RECORDER, UNIVERSAL FLIGHT DATA
FIGURE 1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
- 1	980-4100DXXS		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-DXXS	A	RF
- 1A	980-4100DXXN		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-DXXN	B	RF
- 1B	980-4100DXUS		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-DXUS	C	RF
- 1C	980-4100DXUN		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-DXUN	D	RF
- 1D	980-4100BXUS		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-BXUS	E	RF
- 1E	980-4100BXUN		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-BXUN	F	RF
- 2	980-4101-001		RECORDER, UNIVERSAL FLIGHT DATA	G	RF
- 2A	980-4101-002		RECORDER, UNIVERSAL FLIGHT DATA	H	RF
- 3	980-4100AXXS		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-AXXS	I	RF
- 3A	980-4100AXXN		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-AXXN	J	RF
- 3B	980-4100AXUS		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-AXUS	K	RF
- 3C	980-4100AXUN		RECORDER, UNIVERSAL FLIGHT DATA TRUE PN IS 980-4100-AXUN	L	RF
5	607-0723-001		.COVER, DUST	E	1
5A	607-0723-002		.COVER, DUST	BDFG	1
5B	607-0723-006		.COVER, DUST	HJL	1
			--- ATTACHING PARTS ---	ACIK	1
10	MS51957-30		.SCREW, PAN HEAD		2
15	MS35338-136		.WASHER, LOCK		2
20	MS15795-805		.WASHER, FLAT		2
			-----*-----		
25	652		..PLUG, METAL (V83330)		1
27	MS51957-40		.SCREW, PAN HEAD	ABHI	5
				J	
28	MS51957-40		.SCREW, PAN HEAD	CDEF	1
				GKL	
30	N15F210B		.UNDERWATER LOCATING DEVICE (ULD) (V94970)	CDEF	1
				GKL	
- 30A	DK100		.UNDERWATER LOCATING DEVICE (V94970) OPTIONAL	CDEF	1
				KL	
- 30B	ELP362D		.UNDERWATER LOCATING DEVICE (V26858) OPTIONAL	CDEF	1
				KL	
- 30C	DK120		.UNDERWATER LOCATING DEVICE (V94970) OPTIONAL	CDEF	1
				KL	
- 30D	ELP362DS		.UNDERWATER LOCATING DEVICE (V26858) OPTIONAL	CDEF	1
				KL	
- 30E	316-6004-003		.UNDERWATER LOCATING DEVICE OPTIONAL (HONEYWELL PN FOR 30A)	CDEF	1
				KL	
- 30F	316-6004-006		.UNDERWATER LOCATING DEVICE OPTIONAL (HONEYWELL PN FOR 30B)	CDEF	1
				KL	
- 30G	316-6004-005		.UNDERWATER LOCATING DEVICE OPTIONAL (HONEYWELL PN FOR 30C)	CDEF	1
				KL	

31-30-32

Page 1037
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
- 30H	316-6004-004		.UNDERWATER LOCATING DEVICE OPTIONAL (HONEYWELL PN FOR 30D)	CDEF KL	1
- 30I	316-6004-001		.UNDERWATER LOCATING DEVICE	G	1
			- - - ATTACHING PARTS - - -		
35	631-0401-001		.SCREW, SOCKET HEAD	CDEF KL	4
35A	LL25U82J28		.SCREW, SOCKET HEAD (V03038)	G	4
- 35B	NAS1352C08-20		.SCREW, SOCKET HEAD OPTIONAL TO 35	CDEF KL	4
- 35C	PHD600-828		.SCREW, SOCKET HEAD OPTIONAL TO 35A	G	4
40	654-0146-001		.MOUNT	CDEF KL	2
40A	654-0174-001		.MOUNT -----*-----	G	2
45	810-423		..CAP, END (V94970)	CDEF GKL	1
50	810-342		..O-RING (V94970)	CDEF GKL	1
55	810-419		..CUSHION, SHOCK (V94970)	CDEF GKL	1
60	810-371A		..CUSHION, SHOCK (V94970)	CDEF GKL	1
65	810-418A		..BATTERY (V94970)	CDEF GKL	1
- 65A	810-2007		..BATTERY (V94970) (USED WITH 30C) (REPLD BY 65B)		1
- 65B	810-2007K		..BATTERY KIT (V94970) (USED WITH 30C) (REPLS -65A)	CDEF GKL	1
- 65C	B362-06192-2		..BATTERY KIT (V26858) (USED WITH 30B)		1
70	648-1173-002		.RETAINER (REPLD BY 70A)		1
70A	648-1173-003		.RETAINER (REPLS 70) - - - ATTACHING PARTS - - -		1
75	MS24693C51		.SCREW, FLAT HEAD		1
- 80	MS21044C08		DELETED		0
- 85	MS15795-805		DELETED		0
90	MS24693C272		.SCREW, FLAT HEAD -----*-----		1
91	653-9043-001		..BUMPER, SILICON RUBBER		1
92	MF1001-08		..NUT, ANCHOR (V15653) - - - ATTACHING PARTS - - -		1
93	MS20426AD3-6		..RIVET -----*-----		2
- 95	720-0123-001		.POWER SUPPLY ASSEMBLY, A1 (ESDS) (REPLD BY 95A) (SEE FIG. 2 FOR BREAKDOWN)		1
95A	720-0123-002		.POWER SUPPLY ASSEMBLY, A1 (ESDS) (REPLS 95) (BUY 95B OR C) (SEE FIG. 2 FOR BREAKDOWN)		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
95B	720-0123-003		.POWER SUPPLY ASSEMBLY, A1 (ESDS) (REPLS 95A) (OPT TO 95C) (SEE FIG. 2 FOR BREAKDOWN)		1
95C	720-0167-002		.POWER SUPPLY ASSEMBLY, A1 (ESDS) (REPLS 95A) (OPT TO 95B,D) (SEE FIG. 15 FOR BREAKDOWN)		1
- 95D	720-0167-004		.POWER SUPPLY ASSEMBLY, A1 (ESDS) (OPT TO 95C) (SEE FIG. 15 FOR BREAKDOWN) - - - ATTACHING PARTS - - -	ABCD GHIJ KL	1
100	MS24693C50		.SCREW, FLAT HEAD (USED WITH 95, 95A, 95B)		4
102	MS24693C51		.SCREW, FLAT HEAD (USED WITH 95, 95A, 95B)		4
102A	MS24693C51		.SCREW, FLAT HEAD (USED WITH 95C,D) (SUPSD BY 102B)		6
- 102B	MS24693C50		.SCREW, FLAT HEAD (SUPSDS 102A)		6
103	MS21044C08		.NUT, LOCKING -----*-----		4
- 105	722-2442-001		.READ/WRITE INTERFACE ASSEMBLY, A2 (PRE SB 1) (REPLD BY 105A) (ESDS) (SEE FIG. 3 FOR BREAKDOWN)		1
105A	722-2442-002		.READ/WRITE INTERFACE ASSEMBLY, A2 (SEE FIG. 3 FOR BREAKDOWN) (REPLS 105) (POST SB 1) (ESDS) (OPT TO 105B, CODES ABCDGH)		1
- 105B	722-2442-003		.READ/WRITE INTERFACE ASSEMBLY, A2 (ESDS) (OPT TO 105A) (SEE FIG. 3 FOR BREAKDOWN)	ABCD GHIJ KL	1
110	722-2441-001		.TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 4 FOR BREAKDOWN) (PRE SB 1) (REPLD BY 110A) (ESDS)		1
- 110A	722-2441-002		.TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 4 FOR BREAKDOWN) (REPLS 110)(REPLD BY 110B) (ESD)(POST SB 1)		1
- 110B	722-2441-004		.TRANSPORT INTERFACE ASSEMBLY, A3 (REPLS 110A)(OPT TO 110C) (ESD)(SEE FIG. 4 FOR BREAK- DOWN)(OPT TO 110D, CODES ABCD GH)(PRE SB 16)(SUPSD BY 110F)		1
- 110C	722-2441-003		.TRANSPORT INTERFACE ASSEMBLY, A3 (OPT TO 110B)(ESD)(SEE FIG. 4 FOR BREAKDOWN)(PRE SB 16) (SUPSD BY 110E)		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
1					
- 110D	722-2441-005		.TRANSPORT INTERFACE ASSEMBLY, A3 (OPT TO 110B)(ESD)(SEE FIG. 4 FOR BREAKDOWN)(PRE SB 16) (SUPSD BY 110G)		1
- 110E	722-2441-006		.TRANSPORT INTERFACE ASSEMBLY, A3 (OPT TO 110F,G)(ESD)(SEE FIG. 4 FOR BREAKDOWN)(SUPSDS 110C)(POST SB 16)		1
- 110F	722-2441-007		.TRANSPORT INTERFACE ASSEMBLY, A3 (OPT TO 110E,G)(ESD)(SEE FIG. 4 FOR BREAKDOWN)(SUPSDS 110B)(POST SB 16)		1
- 110G	722-2441-008		.TRANSPORT INTERFACE ASSEMBLY, A3 (OPT TO 110E,F)(ESD)(SEE FIG. 4 FOR BREAKDOWN)(SUPSDS 110D)(POST SB 16)		1
115	722-2438-001		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (REPLD BY 115A) (PRE SB 1)(SUPSD BY 115C)(ESD) (SEE FIG. 5 FOR BREAKDOWN)		1
- 115A	722-2438-002		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (REPLS 115)(POST SB 1)(ESD)(PRE SB 6)(SUPSD BY 115E)(PRE SB 12)(PRE SB 15) (SUPSD BY 115I)(SEE FIG. 5 FOR BREAKDOWN)	ABCD	1
- 115B	722-2438-003		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (REPLS 115A) (REPLD BY 115C)(ESD)(PRE SB 6) (SUPSD BY 115F)(PRE SB 12) (SUPSD BY 115J)(PRE SB 15) (SEE FIG. 5 FOR BREAKDOWN)	ABCD	1
- 115C	722-2438-004		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (REPLS 115B) (OPT TO 115D)(PRE SB 6)(SUPSD BY 115G)(ESD)(PRE SB 12)(SUPSD BY 115K)(PRE SB 15)(SEE FIG. 5 FOR BREAKDOWN)	ABCD	1
- 115D	722-2438-005		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (REPLS 115B) (OPT TO 115C)(PRE SB 6)(SUPSD BY 115H)(ESD)(PRE SB 12)(SUPSD BY 115L)(PRE SB 15)(SEE FIG. 5 FOR BREAKDOWN)	ABCD	1
- 115E	722-2438-006		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 6, SUPSDS 115A)(PRE SB 12)(SUPSD BY 115I)(PRE SB 15)(PRE SB 16) (SUPSD BY 115R)(SEE FIG. 5 FOR BREAKDOWN)	ABCD	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
- 115F	722-2438-007		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 6, SUPSDS 115B)(PRE SB 12)(SUPSD BY 115J)(PRE SB 15)(PRE SB 16) (SUPSD BY 115S)(SEE FIG. 5 FOR BREAKDOWN)	ABCD	1
- 115G	722-2438-008		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 6, SUPSDS 115C)(PRE SB 12)(SUPSD BY 115K)(PRE SB 15)(PRE SB 16) (SUPSD BY 115T)(SEE FIG. 5 FOR BREAKDOWN)	ABCD	1
- 115H	722-2438-009		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 6, SUPSDS 115D)(PRE SB 12)(SUPSD BY 115L)(PRE SB 15)(PRE SB 16) (SUPSD BY 115U)(SEE FIG. 5 FOR BREAKDOWN)	ABCD	1
- 115I	722-2438-010		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 12, SUPSDS 115A OR 115E)(PRE SB 15)(PRE SB 16)(SUPSD BY 115R) (SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 115Z)	ABCD EF	1
- 115J	722-2438-011		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 12, SUPSDS 115B OR 115F)(PRE SB 15)(PRE SB 16)(SUPSD BY 115S) (SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116)	ABCD EF	1
- 115K	722-2438-012		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 12, SUPSDS 115C OR 115G)(OPT TO 115L)(PRE SB 15)(PRE SB 16) (SUPSD BY 115T)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116A)	ABCD EF	1
- 115L	722-2438-013		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 12, SUPSDS 115D OR 115H)(OPT TO 115K)(PRE SB 15)(PRE SB 16) (SUPSD BY 115U)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116B)	ABCD EF	1
- 115M	722-2438-014		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 15) (PRE SB 16)(SUPSD BY 115V) (SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116C)	IJKL	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
- 115N	722-2438-015		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 15) (PRE SB 16)(SUPSD BY 115W) (SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116D)	IJKL	1
- 115P	722-2438-016		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 15) (PRE SB 16)(SUPSD BY 115X) (SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116E)	IJKL	1
- 115Q	722-2438-017		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 15) (PRE SB 16)(SUPSD BY 115Y) (SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116F)	IJKL	1
- 115R	722-2438-018		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115I)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116G)	ABCD EF	1
- 115S	722-2438-019		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115J)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116H)	ABCD EF	1
- 115T	722-2438-020		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115K)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116I)	ABCD EF	1
- 115U	722-2438-021		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115L)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116J)	ABCD EF	1
- 115V	722-2438-022		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115M)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116K)	IJKL	1
- 115W	722-2438-023		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115N)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116L)	IJKL	1
- 115X	722-2438-024		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115P)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116M)	IJKL	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
1					
- 115Y	722-2438-025		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(POST SB 16) (SUPSDS 115Q)(SEE FIG. 5 FOR BREAKDOWN) (PRE SB 19)(SUPSD BY 116N)	IJKL	1
- 115Z	722-2438-026		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115I)	ABCD EF	1
- 116	722-2438-027		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115J)	ABCD EF	1
- 116A	722-2438-028		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115K)	ABCD EF	1
- 116B	722-2438-029		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115L)	ABCD EF	1
- 116C	722-2438-030		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115M)	IJKL	1
- 116D	722-2438-031		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115N)	IJKL	1
- 116E	722-2438-032		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 SUPSDS 115P)	IJKL	1
- 116F	722-2438-033		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115Q)	IJKL	1
- 116G	722-2438-034		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115R)	ABCD EF	1
- 116H	722-2438-035		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. 5 FOR BREAKDOWN)(POST SB 19, SUPSDS 115S)	ABCD EF	1
- 116I	722-2438-036		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. FOR BREAKDOWN)(POST SB 19, SUPSDS 115T)	ABCD EF	1
- 116J	722-2438-037		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. FOR BREAKDOWN)(POST SB 19, SUPSDS 115U)	ABCD EF	1

31-30-32

Page 1043
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
- 116K	722-2438-038		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. FOR BREAKDOWN)(POST SB 19, SUPSDS 115V)	IJKL	1
- 116L	722-2438-039		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. FOR BREAKDOWN)(POST SB 19, SUPSDS 115W)	IJKL	1
- 116M	722-2438-040		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. FOR BREAKDOWN)(POST SB 19, SUPSDS 115X)	IJKL	1
- 116N	722-2438-041		.RECORDER SYSTEM CONTROLLER ASSEMBLY, A4 (ESD)(SEE FIG. FOR BREAKDOWN)(POST SB 19, SUPSDS 115Y)	IJKL	1
- 120	722-2501-001		.AIRFRAME WIRING INTERFACE, A8 (ESD)(PRE SB 7)(SUPSD BY 120A) (SEE FIG. 6 FOR BREAKDOWN)		1
120A	722-2501-002		.AIRFRAME WIRING INTERFACE, A8 (ESD)(POST SB 7, SUPSDS 120) (OPT TO 120B)(SEE FIG. 6 FOR BREAKDOWN)		1
- 120B	722-2501-003		.AIRFRAME WIRING INTERFACE, A8 (ESD)(OPT TO 120A)(SEE FIG. 6 FOR BREAKDOWN)	ABCD GHIJ KL	1
125	651		.PLUG (V83330)		1
127	620-0723-005		.PLATE, MODIFICATION (POST SB 12)		1
130	620-1351-004		.PLATE, IDENTIFICATION (REPLD BY 130D)	ABCD IJKL	1
- 130A	620-1351-003		.PLATE, IDENTIFICATION (REPLS 130B)(REPLD BY 130E OR 130F)		1
- 130B	620-1351-001		.PLATE, IDENTIFICATION (REPLD BY 130A)	EF	1
- 130C	620-1351-002		.PLATE, IDENTIFICATION (REPLD BY 130D)	ABCD IJKL	1
- 130D	620-1351-007		.PLATE, IDENTIFICATION (REPLS 130/130C)(SEE ITEM 147)(REPLD BY 130E OR 130F)		1
- 130E	620-1351-008		.PLATE, IDENTIFICATION (REPLS 130A AND 130D)	ABCD IJKL	1
- 130F	620-1351-009		.PLATE, IDENTIFICATION (REPLS 130A AND 130D)	EF	1
- 130G	620-1407-001		.PLATE, IDENTIFICATION - - - ATTACHING PARTS - - -	GH	1
135	MS51957-1		.SCREW, PAN HEAD		6
- 140	MS35338-134		DELETED		0
- 145	MS15795-802		DELETED		0
			-----*-----		
147	643-0448-001		.LABEL, PATENT (USE WHENEVER ITEM 130D OR 130E IS USED)		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
150	700-1066-001		.TAPE RECORDER ASSEMBLY (SEE FIG. 8 FOR BREAKDOWN)(REPLD BY 150A)		1
150A	700-1066-004		.TAPE RECORDER ASSEMBLY (REPLS 150/C/D)(POST SB 2) (OPT TO 150E, 150F, 150G) (PRE SB 5)(SUPSD BY 150F) (PRE SB 8)(SUPSD BY 150E) (PRE SB 13)(SUPSD BY 150J) (SEE FIG. 8 FOR BREAKDOWN)		1
150B	700-1066-005		.TAPE RECORDER ASSEMBLY (MFG ON SN 1001 THRU 1250) (OPT TO 150A, 150E, 150F) (PRE SB 5)(SUPSD BY 150F) (PRE SB 8)(SUPSD BY 150E) (PRE SB 13)(SUPSD BY 150J) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150C	700-1066-002		.TAPE RECORDER ASSEMBLY (REPLD BY 150A)		1
- 150D	700-1066-003		.TAPE RECORDER ASSEMBLY (REPLD BY 150A)		1
- 150E	700-1066-006		.TAPE RECORDER ASSEMBLY (OPT TO 150A, 150F, 150G) (POST SB 8)(SUPSDS 150A) (POST SB 8)(SUPSDS 150B) (PRE SB 13)(SUPSD BY 150J) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150F	700-1066-007		.TAPE RECORDER ASSEMBLY (OPT TO 150A, 150E, 150G) (POST SB 5)(SUPSDS 150A) (POST SB 5)(SUPSDS 150B) (PRE SB 11)(SUPSD BY 150I) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150G	700-1066-008		.TAPE RECORDER ASSEMBLY (OPT TO 150A, 150E, 150F)(MAY BE USED ON SN 5999 AND BELOW) (PRE SB 11)(SUPSD BY 150H) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150H	700-1066-009		.TAPE RECORDER ASSEMBLY (OPT TO 150I, 150J) (MFG ON SN 6000 AND ON) (POST SB 11)(SUPSDS 150G) (REPLD BY 150K)(PRE SB 17) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150I	700-1066-010		.TAPE RECORDER ASSEMBLY (OPT TO 150H, 150J)(BUY 150H) (POST SB 11)(SUPSDS 150F) (REPLD BY 150L)(PRE SB 17) (SEE FIG. 8 FOR BREAKDOWN)		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

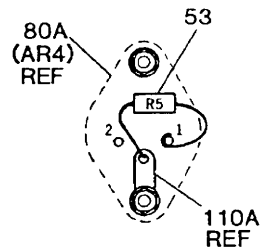
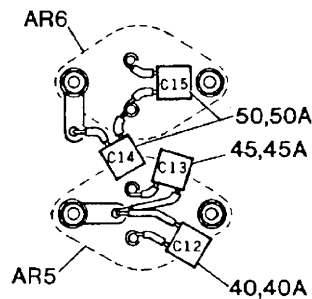
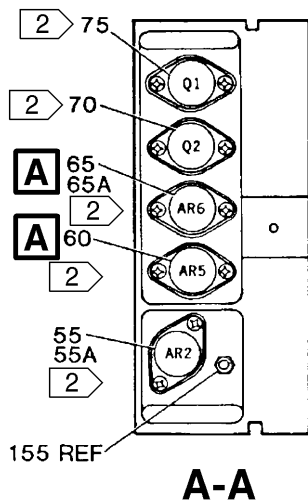
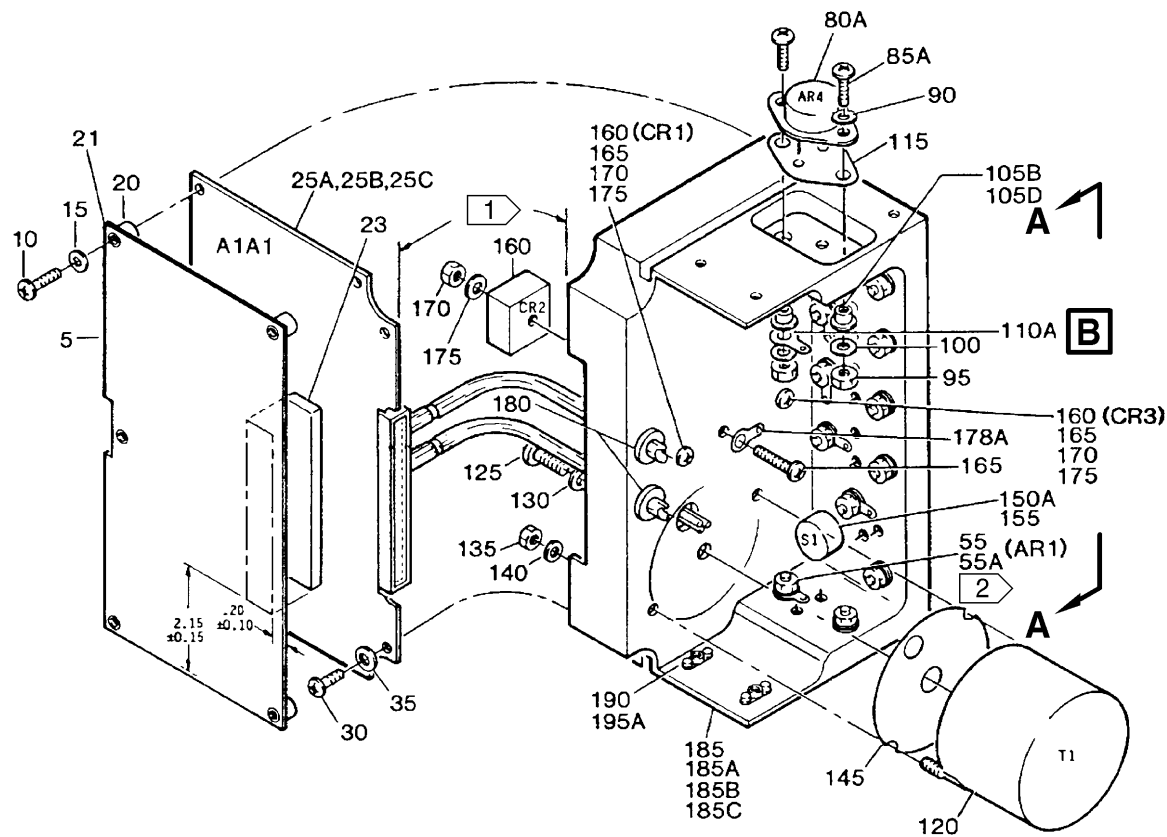
FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
1			1234567		
- 150J	700-1066-011		.TAPE RECORDER ASSEMBLY (OPT TO 150H, 150I)(BUY 150H) (POST SB 13)(SUPSDS 150A) (POST SB 13)(SUPSDS 150B) (POST SB 13)(SUPSDS 150E) (REPLD BY 150M)(PRE SB 17) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150K	700-1066-012		.TAPE RECORDER ASSEMBLY (REPLS 150H)(POST SB 17) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150L	700-1066-013		.TAPE RECORDER ASSEMBLY (REPLS 150I)(POST SB 17) (SEE FIG. 8 FOR BREAKDOWN)		1
- 150M	700-1066-014		.TAPE RECORDER ASSEMBLY (REPLS 150J)(POST SB 17) (SEE FIG. 8 FOR BREAKDOWN)		1
155	MS51957-50		--- ATTACHING PARTS --- .SCREW, PAN HEAD (REPLD BY 155A)		4
155A	NAS623-2-12		.SCREW, PAN HEAD (REPLS 155)		4
160	MS35338-137		.WASHER, LOCK (REPLD BY 160A)		4
160A	MS35338-42		.WASHER, LOCK (REPLS 160)		4
165	MS15795-807		.WASHER, FLAT (OPT TO 165A)		4
165A	AN960-8		.WASHER, FLAT (OPT TO 165) -----*-----		4
170	614-0603-001		.SHIELD --- ATTACHING PARTS ---		1
175	MS51957-14		.SCREW, PAN HEAD (REPLD BY 175A)		4
175A	MS51957-13		.SCREW, PAN HEAD (REPLS 175)		4
180	MS35338-135		.WASHER, LOCK		4
185	MS15795-803		.WASHER, FLAT -----*-----		4
187	600-4143-001		..RUBBER, SPONGE (REPLD WITH MATERIAL, SEE ASSEMBLY SECTION, PARA. 21B)		4
190	617-3716-001		.CHASSIS ASSEMBLY (REPLD BY 190A)(SEE FIG. 9 FOR BREAKDOWN)		1
190A	617-3716-002		.CHASSIS ASSEMBLY (REPLS 190)(BUY 190D)(SEE FIG. 9 FOR BREAKDOWN)		1
190B	617-3716-006		.CHASSIS ASSEMBLY (SUPSDS 190A)(REPLD BY 190D)(SEE FIG. 9 FOR BREAKDOWN)		1
- 190C	617-3716-005		.CHASSIS ASSEMBLY SUPSDS 190B)(REPLD BY 190D)(SEE FIG. 16 FOR BREAKDOWN)		1
- 190D	617-4640-006		.CHASSIS ASSEMBLY (REPLS 190B, 190C)(SEE FIG. 16 FOR BREAKDOWN)		1
195	D20419-21		.SCREWLOCK, PAIR (V71468)		3

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



A-A

NOTES:

- 1 Service loop to be 2.0 ± 0.5 inches when item 25A, 25B or 25C (A1A1) is opened 180 degrees.
- 2 See item 80A (AR4) for hardware.



POWER SUPPLY ASSEMBLY (A1)
FIGURE 2

ID-177816 E720-0123-2-H

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
2					
- 1	720-0123-001		POWER SUPPLY ASSEMBLY, A1 (ESD)(SEE 1-95 FOR NHA)	A	RF
- 1A	720-0123-002		POWER SUPPLY ASSEMBLY, A1 (ESD)(SEE 1-95A FOR NHA)	B	RF
- 1B	720-0123-003		POWER SUPPLY ASSEMBLY, A1 (ESD)(SEE 1-95B FOR NHA)	C	RF
- 1C	720-0123-004		DELETED		0
5	617-3831-001		.EMI SHIELD (REPLD BY 5A)		1
- 5A	617-3831-002		.EMI SHIELD (REPLS 5)		1
			--- ATTACHING PARTS ---		
10	MS35214-16		.SCREW, PAN HEAD		5
15	NAS620B4L		.WASHER, FLAT		5
			-----*-----		
20	9705CA115-17		..STANDOFF (V06540)		5
- 20A	18091BB115-14		..SPACER (V06540)(USED ON 5A)		5
21	628-0261-001		..WASHER, ALUMINUM (USED ON 5)		5
23	600-4143-001		..RUBBER, SPONGE		1
- 25	722-2440-001		.POWER SUPPLY CCA, A1A1 (REPLD BY 25A AND 25B) (SEE FIG. 7 FOR BREAKDOWN)(ESD)		1
25A	722-2440-002		.POWER SUPPLY CCA, A1A1 (REPLS 25) (SEE FIG. 7 FOR BREAKDOWN)(ESD)	AB	1
25B	722-2440-003		.POWER SUPPLY CCA, A1A1 (REPLS 25) (SEE FIG. 7 FOR BREAKDOWN)(ESD)	C	1
25C	722-2440-004		.DELETED --- ATTACHING PARTS ---		0
30	MS35214-14		.SCREW, PAN HEAD		2
35	NAS620B4L		.WASHER, FLAT		2
			-----*-----		
40	M39014-02-1358S		.DELETED		0
40A	M39014-02-1358		.CAPACITOR, 0.33 UF, 50V, 10%		1
45	M39014-02-1350S		.DELETED		0
45A	M39014-02-1350		.CAPACITOR, 0.1 UF, 100V, 10%		1
50	M39014-02-1419S		.DELETED		0
50A	M39014-02-1419		.CAPACITOR, 1.0 UF, 50V, 10%		2
53	RNC55H2430FS		.RESISTOR, 243 OHMS, 1%, 1/10W (USED WITH 25A AND 25B ONLY)		1
55	SG123K883B		.IC, VOLTAGE REGULATOR, +5V (ESD)(V34333)		2
55A	LM123K883B		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V27014) OPTIONAL		2
- 55B	LM123K883C		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V27014) OPTIONAL		2
- 55C	LM123K		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V27014) OPTIONAL		2
- 55D	LM123K		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V64155) OPTIONAL		2
60	UA7812KMQB		.IC, VOLTAGE REGULATOR, +12V (ESD)(V07263)		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
2			1234567		
- 60A	UA7812KM		.IC, VOLTAGE REGULATOR, +12V (ESDS) (V07263) OPTIONAL		1
- 60B	LM140K		.IC, VOLTAGE REGULATOR, +12V (ESDS) (V27014) OPTIONAL		1
- 60C	LM140K12		.IC, VOLTAGE REGULATOR, +12V (ESDS) (V27014) OPTIONAL		1
- 60D	LM140K12		.IC, VOLTAGE REGULATOR, +12V (ESDS) (V04713) OPTIONAL		1
65	UA7912KMQB		.IC, VOLTAGE REGULATOR, -12V (ESD)(V07263)		1
65A	LM120K12-883B		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V27014) OPTIONAL		1
- 65B	UA7912KM		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V07263) OPTIONAL		1
- 65C	LM120K12		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V27014) OPTIONAL		1
- 65D	10408-1		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V54001) OPTIONAL		1
70	JANTX2N3716		.TRANSISTOR, NPN		1
75	JANTX2N3792		.TRANSISTOR, PNP		1
- 80	LM117K883B		.IC, VOLTAGE REGULATOR (ESD) (V27014)		1
80A	SG117K883B		.IC, VOLTAGE REGULATOR (ESDS) (V34333) OPTIONAL		1
- 80B	SG117K883C		.IC, VOLTAGE REGULATOR (ESDS) (V64155) OPTIONAL		1
- 80C	LM117K		.IC, VOLTAGE REGULATOR (ESDS) (V04713) OPTIONAL		1
- 80D	LM117K		.IC, VOLTAGE REGULATOR (ESDS) (V27014) OPTIONAL		1
- 80E	LM117K		.IC, VOLTAGE REGULATOR (ESDS) (V64155) OPTIONAL		1
- 85	MS51957-16		--- ATTACHING PARTS --- .SCREW, PAN HEAD (SUPSD BY 85A)		14
85A	MS51957-17		.SCREW, PAN HEAD (SUPSDS 85)		14
90	MS35333-70		.WASHER, INTERNAL STAR		14
95	MS21044C04		.NUT, LOCK		14
100	MS15795-803		.WASHER, FLAT		14
- 105	5608-9		.DELETED		0
- 105A	NY04-125N		.DELETED		0
105B	NY04-125G		.WASHER, SHOULDER (V08289)		14
- 105C	NY04-125T		.DELETED		0
105D	NY04-90G		.DELETED		0
- 105E	NY04-090G		.DELETED		0
- 105F	NY04-090N		(V08289) OPTIONAL .DELETED (V08289) OPTIONAL		0
- 110	390		-----*----- .LUG (V79963) (SUPSD BY 110A)		7

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
2			1234567		
110A	1412-4		.LUG (V83330) (SUPSDS 110)		7
115	7403-09FR06		.INSULATOR (V55285)		7
120	310-0285-001		.TRANSFORMER		1
			- - - ATTACHING PARTS - - -		
125	MS35216-56		.SCREW, PAN HEAD		1
130	NAS620B10L		.WASHER, FLAT		1
135	MS21044C06		.NUT, LOCK		2
140	MS15795-805		.WASHER, FLAT		2
145	402-3036-001		.INSULATOR		1
			-----*-----		
- 150	973-0031-561		.SWITCH, THERMAL (BUY 150A)		1
150A	974-0000-574		.SWITCH, THERMAL		1
			- - - ATTACHING PARTS - - -		
155	MS21042-06		.NUT, LOCK		1
			-----*-----		
160	JANTX469-1		.RECTIFIER, BRIDGE (ESD)		3
- 160A	M19500-469-01		.RECTIFIER, BRIDGE (ESDS)		3
			OPTIONAL		
			- - - ATTACHING PARTS - - -		
165	MS51957-31		.SCREW, PAN HEAD		3
170	MS21044C06		.NUT, LOCK		3
175	MS15795-805		.WASHER, FLAT		3
			-----*-----		
- 178	390		.LUG, SOLDER (V79963)		1
			(SUPSD BY 178A)		
178A	1488-6		.LUG (V83330)		1
			(SUPSDS 178) (SUPSD BY 178B)		
- 178B	1485-8		.TERMINAL (V83330)		1
			(SUPSDS 178A)		
180	Z427		.GROMMET (V76385)		2
185	607-1527-001		.COVER	A	1
185A	607-1527-002		.COVER	A	1
			OPTIONAL		
185B	607-1527-003		.COVER	BC	1
185C	607-1527-004		.COVER	BC	1
			OPTIONAL		
190	MF1001-08		..NUT, ANCHOR (V15653)		4
			- - - ATTACHING PARTS - - -		
- 195	MS20426A3-4		..RIVET (SUPSD BY 195A)		8
195A	MS20426A3-5		..RIVET (SUPSDS 195)		8
			-----*-----		



READ/WRITE INTERFACE ASSEMBLY (A2)
FIGURE 3

ID-177817 E722-2442-2-G

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
3			1234567		
- 1	722-2442-001		READ/WRITE INTERFACE ASSEMBLY, A2 (SEE FIG. 1-105 FOR NHA) (ESD)	A	RF
- 1A	722-2442-002		READ/WRITE INTERFACE ASSEMBLY, A2 (SEE FIG. 1-105A FOR NHA) (ESD)	B	RF
- 1B	722-2442-003		READ/WRITE INTERFACE ASSEMBLY, A2 (SEE FIG. 1-105B FOR NHA)	C	RF
5	M39014-01-1330		.CAPACITOR, 33 PF, 10%, 200V		2
10	M39014-01-1575		.CAPACITOR, 0.01 UF, 10%, 100V		10
15	HA1-4741-8-883B		.IC, OPERATIONAL AMPLIFIER (ESD)(V34371)	C	5
- 15A	HA1-4741-2		.IC, OPERATIONAL AMPLIFIER (ESDS) (V34371) OPTIONAL CODE C	C	5
- 15B	HA1-4741-2		.IC, OPERATIONAL AMPLIFIER (ESDS) (V07933) OPTIONAL CODE C	C	5
20	CCR06CG332JP		.CAPACITOR, 3300 PF, 5%, 50V		1
25	RNC55H3322FS		.RESISTOR, 33.2K, 1%, 1/10W		18
27	M39014-01-1342		.CAPACITOR, 150 PF, 10%, 200V		2
30	M39014-01-1339		.CAPACITOR, 100 PF, 10%, 200V		17
35	RNC55H1001FS		.RESISTOR, 1K, 1%, 1/10W		2
40	RNC55H1002FS		.RESISTOR, 10K, 1%, 1/10W		8
45	RNC55H1331FS		.RESISTOR, 1.33K, 1%, 1/10W		8
50	M39014-01-1587		.CAPACITOR, 0.047 UF, 10%, 50V		8
55	M39003-01-2064		.CAPACITOR, 6.8 UF, 10%, 35V		2
60	HI1-0509A8		.IC, ANALOG MULTIPLEXER (ESD) (V34371)		2
- 60A	HI1-0509A2		.IC, ANALOG MULTIPLEXER (ESDS) (V34371) OPTIONAL CODE C	C	2
65	M39014-01-1593		.CAPACITOR, 0.1 UF, 10%, 50V		7
70	M8340102K1501FA		.RESISTOR NETWORK, 1.5K		2
- 70A	898-3R1-5K		.RESISTOR NETWORK, 1.5K (V73138) OPTIONAL		2
75	G06D64P4BEBL		.CONNECTOR (V71468)		1
80	MS51957-6		--- ATTACHING PARTS ---		
85	MS21042-02		.SCREW, PAN HEAD		2
90	06/16/5610		.NUT, LOCK		2
			.WASHER, FLAT (V86928) (SUPSD BY 90A)		2
- 90A	06/15/5610		.WASHER, NYLON (V86928) (SUPSDS 90)		2
95	SNJ54LS145J00		-----*-----		
- 95A	SNJ54LS145J		.DECODER (ESD)(V01295)		2
			.DECODER (ESDS) (V01295) OPTIONAL		2
- 95B	S54LS145F883B		.DECODER (ESDS) (V18324) OPTIONAL		2
100	RNC55H2001FS		.RESISTOR, 2K, 1%, 1/10W		8
105	RNC55H3011FS		.RESISTOR, 3.01K, 1%, 1/10W		1
110	M39003-01-2017		.CAPACITOR, 33 UF, 10%, 10V		1
115	LM117H883B		.IC, REGULATOR (ESD)(V27014)		1
120	RCR20G122JS		.RESISTOR, 1.2K, 5%, 1/2W		1

31-30-32

Page 1053
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

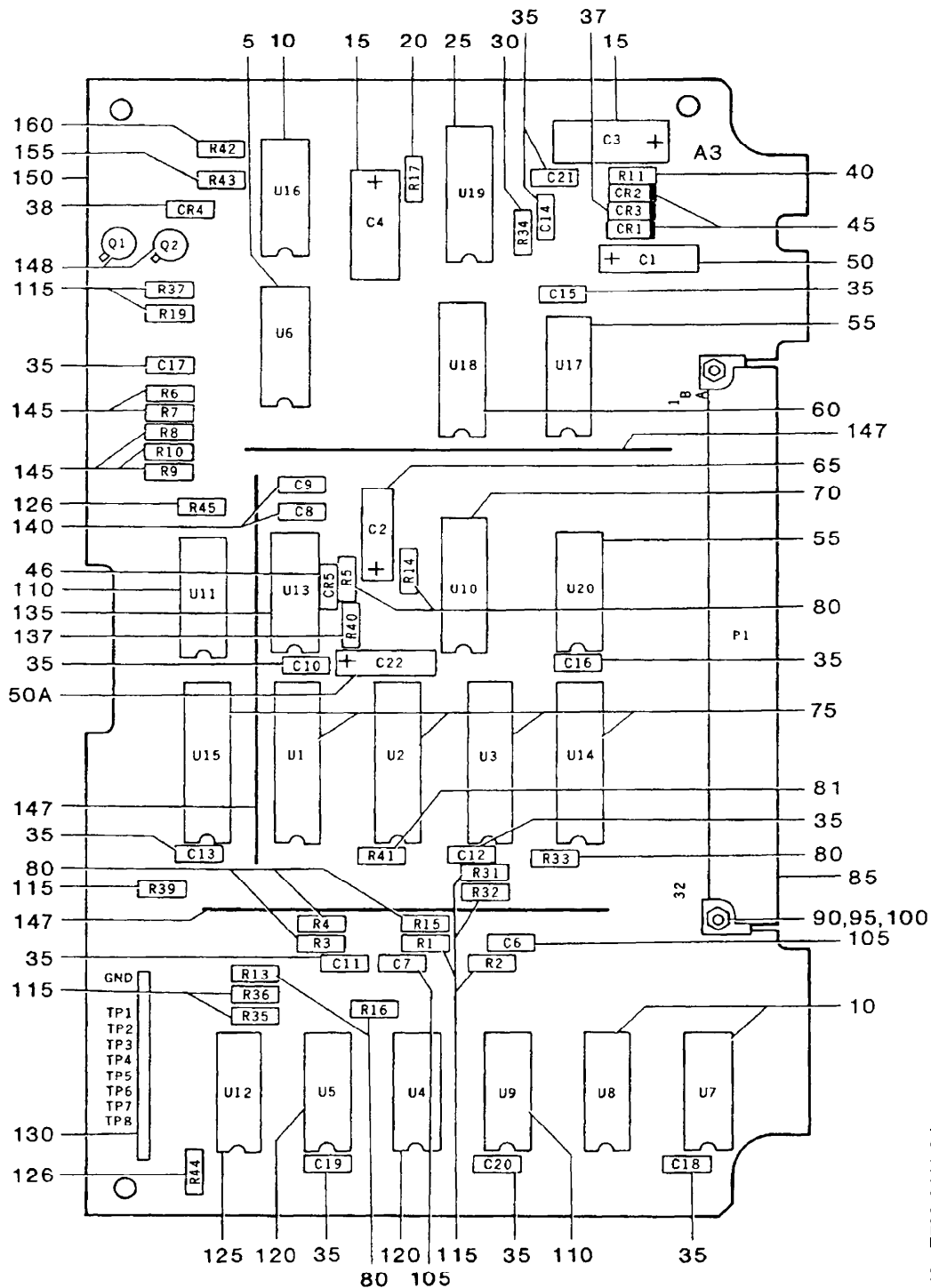
FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
3			1234567		
125	JANTX2N2222A		.TRANSISTOR		1
130	JANTX1N4148-1		.DIODE		18
135	RNC55H2430FS		.RESISTOR, 243 OHMS, 1%, 1/10W		1
140	RNC55H2491FS		.RESISTOR, 2.49K, 1%, 1/10W		2
145	RNC55H4641FS		.RESISTOR, 4.64K, 1%, 1/10W		2
150	RNC55H1002FS		.RESISTOR, 10.0K, 1%, 1/10W		1
			(SELECT FROM THIS OR THE FOLLOWING)		
- 151	RNC55H1072FS		.RESISTOR, 10.7K, 1%, 1/10W		AR
- 152	RNC55H1152FS		.RESISTOR, 11.5K, 1%, 1/10W		AR
- 153	RNC55H1242FS		.RESISTOR, 12.4K, 1%, 1/10W		AR
- 154	RNC55H1332FS		.RESISTOR, 13.3K, 1%, 1/10W		AR
- 155	RNC55H1432FS		.RESISTOR, 14.3K, 1%, 1/10W		AR
- 156	RNC55H1542FS		.RESISTOR, 15.4K, 1%, 1/10W		AR
- 157	RNC55H1652FS		.RESISTOR, 16.5K, 1%, 1/10W		AR
- 158	RNC55H1782FS		.RESISTOR, 17.8K, 1%, 1/10W		AR
- 159	RNC55H1912FS		.RESISTOR, 19.1K, 1%, 1/10W		AR
- 160	RNC55H2052FS		.RESISTOR, 20.5K, 1%, 1/10W		AR
- 161	RNC55H2212FS		.RESISTOR, 22.1K, 1%, 1/10W		AR
- 162	RNC55H2372FS		.RESISTOR, 23.7K, 1%, 1/10W		AR
- 163	RNC55H2552FS		.RESISTOR, 25.5K, 1%, 1/10W		AR
- 164	RNC55H2742FS		.RESISTOR, 27.4K, 1%, 1/10W		AR
- 165	RNC55H2942FS		.RESISTOR, 29.4K, 1%, 1/10W		AR
- 166	RNC55H3162FS		.RESISTOR, 31.6K, 1%, 1/10W		AR
- 167	RNC55H3402FS		.RESISTOR, 34.0K, 1%, 1/10W		AR
- 168	RNC55H3652FS		.RESISTOR, 36.5K, 1%, 1/10W		AR
- 169	RNC55H3922FS		.RESISTOR, 39.2K, 1%, 1/10W		AR
- 170	RNC55H4222FS		.RESISTOR, 42.2K, 1%, 1/10W		AR
- 171	RNC55H4532FS		.RESISTOR, 45.3K, 1%, 1/10W		AR
- 172	RNC55H4872FS		.RESISTOR, 48.7K, 1%, 1/10W		AR
- 173	RNC55H5232FS		.RESISTOR, 52.3K, 1%, 1/10W		AR
- 174	RNC55H5622FS		.RESISTOR, 56.2K, 1%, 1/10W		AR
- 175	RNC55H6042FS		.RESISTOR, 60.4K, 1%, 1/10W		AR
- 176	RNC55H6492FS		.RESISTOR, 64.9K, 1%, 1/10W		AR
- 177	RNC55H6982FS		.RESISTOR, 69.8K, 1%, 1/10W		AR
- 178	RNC55H7502FS		.RESISTOR, 75.0K, 1%, 1/10W		AR
- 179	RNC55H8062FS		.RESISTOR, 80.6K, 1%, 1/10W		AR
- 180	RNC55H8662FS		.RESISTOR, 86.6K, 1%, 1/10W		AR
- 181	RNC55H9312FS		.RESISTOR, 93.1K, 1%, 1/10W		AR
190	M39003-01-2116		.CAPACITOR, 1 UF, 10%, 50V		1
195	SNJ54LS04J00		.IC, INVERTER (ESD)(V01295)		1
- 195A	SNJ54LS04J		.IC, INVERTER (ESDS) (V01295) OPTIONAL		1
- 195B	54LS04BCXJC		.IC, INVERTER (ESDS) (V04713) OPTIONAL		1
- 195C	54LS04BCA		.IC, INVERTER (ESDS) (V18324) OPTIONAL		1
- 195D	DM54LS04J883		.IC, INVERTER (ESDS) (V27014) OPTIONAL		1
200	SNJ54LS02J00		.IC, NOR GATE (ESD)(V01295)		1
- 200A	SNJ54LS02J		.IC, NOR GATE (ESDS) (V01295) OPTIONAL		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
3					
- 200B	54LS02BCXJC		.IC, NOR GATE (ESDS) (V04713) OPTIONAL		1
- 200C	54LS02BCA		.IC, NOR GATE (ESDS) (V18324) OPTIONAL		1
- 200D	DM54LS02J883		.IC, NOR GATE (ESDS) (V27014) OPTIONAL		1
205	SNJ54LS00J00		.IC, NAND GATE (ESD)(V01295)		1
- 205A	SNJ54LS00J		.IC, NAND GATE (ESDS) (V01295) OPTIONAL		1
- 205B	54LS00BCXJC		.IC, NAND GATE (ESDS) (V04713) OPTIONAL		1
- 205C	54LS00BCA		.IC, NAND GATE (ESDS) (V18324) OPTIONAL		1
- 205D	DM54LS00J883		.IC, NAND GATE (ESDS) (V27014) OPTIONAL		1
210	RNC55H3321FS		.RESISTOR, 3.32K, 1%, 1/10W		3
215	LM111H883B		.IC, COMPARATOR (ESD)(V27014)		1
- 215A	LM111H883C		.IC, COMPARATOR (ESDS) (V27014) OPTIONAL		1
217	M39014-02-1419		.CAPACITOR, 1 UF, 10%, 50V		1
220	RNC55H1502FS		.RESISTOR, 15.0K, 1%, 1/10W		1
223	M39014-01-1351		.CAPACITOR, 470 PF, 10%, 200V		1
225	LM136H883B		.IC, VOLTAGE REFERENCE (ESD) (V27014)		1
- 225A	LM136AH883B		.DELETED		0
- 225B	LM136H883C		.IC, VOLTAGE REFERENCE (ESDS) (V27014) OPTIONAL		1
- 225C	LM136AH2-5883C		.IC, VOLTAGE REFERENCE (ESDS) (V27014) OPTIONAL		1
- 225D	LM136AH2-5		.IC, VOLTAGE REFERENCE (ESDS) (V27014) OPTIONAL	AC	1
- 225E	LM136H2-5		.IC, VOLTAGE REFERENCE (ESDS) (V27014) OPTIONAL	AC	1
230	JANTX2N2907A		.TRANSISTOR		1
235	M8340102K1002FA		.RESISTOR NETWORK, 10K		4
240	65521-136		.HEADER (V22526) (SUPSD BY 240A)		1
- 240A	65521-436		.HEADER (V22526) (SUPSDS 240)		1
245	RNC55H1002BS		.RESISTOR, 10.0K, 0.1%, 1/10W		4
250	M39014-01-1356		.CAPACITOR, 820 PF, 10%, 200V		1
255	RNC55H1501FS		.RESISTOR, 1.5K, 1%, 1/10W		2
- 260	623-2283-002		.PRINTED WIRING BOARD	A	1
260A	623-2283-003		.PRINTED WIRING BOARD	BC	1
260B	623-2283-001		.PRINTED WIRING BOARD OPTIONAL	A	1

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101



TRANSPORT INTERFACE ASSEMBLY (A3)
FIGURE 4

ID-177818 E722-2441-2-L

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
4					
- 1	722-2441-001		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110 FOR NHA)(ESD)	A	RF
- 1A	722-2441-002		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110A FOR NHA) (ESD)	B	RF
- 1B	722-2441-004		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110B FOR NHA) (ESD)	C	RF
- 1C	722-2441-003		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110C FOR NHA) (ESD)	D	RF
- 1D	722-2441-005		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110D FOR NHA) (ESD)	E	RF
- 1E	722-2441-006		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110E FOR NHA) (ESD)	F	RF
- 1F	722-2441-007		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110F FOR NHA) (ESD)	G	RF
- 1G	722-2441-008		TRANSPORT INTERFACE ASSEMBLY, A3 (SEE FIG. 1-110G FOR NHA) (ESD)	H	RF
5	SNJ54LS00J00		.IC, NAND GATE (ESD)(V01295)		1
- 5A	SNJ54LS00J		.IC, NAND GATE (ESDS) (V01295) OPTIONAL		1
- 5B	54LS00BCXJC		.IC, NAND GATE (ESDS) (V04713) OPTIONAL		1
- 5C	54LS00BCA		.IC, NAND GATE (ESDS) (V18324) OPTIONAL		1
- 5D	DM54LS00J883		.IC, NAND GATE (ESDS) (V27014) OPTIONAL		1
10	SNJ54LS02J00		.IC, NOR GATE (ESD)(V01295)		3
- 10A	54LS02BCXJC		.IC, NOR GATE (ESDS) (V04713) OPTIONAL		3
- 10B	54LS02BCA		.IC, NOR GATE (ESDS) (V18324) OPTIONAL		3
- 10C	DM54LS02J883		.IC, NOR GATE (ESDS) (V27014) OPTIONAL		3
15	M39003-01-2021		.CAPACITOR, 100 UF, 10%, 10V		2
20	RNC55H133FS		.RESISTOR, 133K, 1%, 1/10W		1
25	SNJ54LS123J00		.IC, ONE SHOT (ESD)(V01295)		1
- 25A	SNJ54LS123J		.IC, ONE SHOT (ESDS) (V01295) OPTIONAL		1
- 25B	54LS123BEXJC		.IC, ONE SHOT (ESDS) (V04713) OPTIONAL		1
- 25C	DM54LS123J883		.IC, ONE SHOT (ESDS) (V27014) OPTIONAL		1
30	RNC55H1003FS		.RESISTOR, 100K, 1%, 1/10W		1
35	M39014-01-1593		.CAPACITOR, 0.1 UF, 10%, 50V		12
37	JANTX1N4148-1		.DIODE		1
38	JANTX1N4148-1		.DIODE	CEGH	1

31-30-32

Page 1057
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
4			1234567		
40	RNC55H1000FS		.RESISTOR, 100 OHMS, 1%, 1/10W		1
45	JANTX1N5711		.DIODE		2
- 45A	JAN1N5711		.DIODE		2
			OPTIONAL		
- 45B	1N5711		.DIODE	ABEH	2
			OPTIONAL		
46	JANTX1N5711		.DIODE	CEGH	1
- 46A	JAN1N5711		.DIODE		1
			OPTIONAL		
- 46B	1N5711		.DIODE	ABEH	1
			OPTIONAL		
50	M39003-01-2017		.CAPACITOR, 33 UF, 10%, 10V		1
50A	M39003-01-2017		.CAPACITOR, 33 UF, 10%, 10V (POST SB 1)	BCDE FGH	1
55	SNJ5417J00		.IC, DRIVER (ESD)(V01295)		2
- 55A	SNJ5417J		.IC, DRIVER (ESDS) (V01295) OPTIONAL		2
60	M8340102K3321FB		.NETWORK, RESISTOR, 3.32K		1
- 60A	M8340102K3301GB		.NETWORK, RESISTOR, 3.32K OPTIONAL		1
65	M39003-01-2064		.CAPACITOR, 6.8 UF, 10%, 35V		1
70	SNJ54LS175J00		.IC, LATCH (ESD)(V01295)		1
- 70A	SNJ54LS175J		.IC, LATCH (ESDS) (V01295) OPTIONAL		1
- 70B	54LS175BEXJC		.IC, LATCH (ESDS) (V04713) OPTIONAL		1
- 70C	S54LS175FBEA		.IC, LATCH (ESDS) (V18324) OPTIONAL		1
- 70D	DM54LS175J883B		.IC, LATCH (ESDS) (V27014) OPTIONAL		1
75	SNJ54LS374J00		.IC, FLIP-FLOP (ESD) (V01295)	ABCE GH	5
- 75A	SNJ54LS374J		.IC, FLIP-FLOP (ESDS) (V01295) OPTIONAL		5
- 75B	54LS374BRAJC		.IC, FLIP-FLOP (ESDS) (V04713) OPTIONAL		5
- 75C	DM54LS374J883B		.IC, FLIP-FLOP (ESDS) (V27014) OPTIONAL		5
- 75D	SN54LS374JB		.IC, FLIP-FLOP (ESDS) (V34335) OPTIONAL		5
76	SNJ54LS374J00		.DELETED		0
77	321-0213-006		.DELETED		0
80	RNC55H3321FS		.RESISTOR, 3.32K, 1%, 1/10W		8
81	RNC55H3321FS		.RESISTOR, 3.32K, 1%, 1/10W	BDF	1
85	G06D64P4BEBL		.CONNECTOR (V71468)		1
			- - - ATTACHING PARTS - - -		
90	MS51957-6		.SCREW, PAN HEAD		2
95	MS21042-02		.NUT, LOCK		2
100	06/16/5610		.WASHER, FLAT (V86928) (SUPSD BY 100A)		2
- 100A	06/15/5610		.WASHER, NYLON (V86928) (SUPSDS 100)		2

-----*-----

31-30-32

Page 1058
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
4					
	105	M39014-01-1575	.CAPACITOR, 0.01 UF, 10%, 100V		2
	110	SNJ54LS14J00	.IC, INVERTER (ESD)(V01295)		2
-	110A	SNJ54LS14J	.IC, INVERTER (ESDS)		2
			(V01295) OPTIONAL		
-	110B	54LS14BCXJC	.IC, INVERTER (ESDS)		2
			(V04713) OPTIONAL		
-	110C	54LS14DMBQ	.IC, INVERTER (ESDS)		2
			(V07263) OPTIONAL		
-	110D	54LS14BCA	.IC, INVERTER (ESDS)		2
			(V18324) OPTIONAL		
-	110E	DM54LS14J883	.IC, INVERTER (ESDS)		2
			(V27014) OPTIONAL		
	115	RNC55H1002FS	.RESISTOR, 10K, 1%, 1/10W		9
	120	SNJ54LS74J00	.IC, FLIP-FLOP (ESD)(V01295)		2
-	120A	SNJ54LS74AJ	.IC, FLIP-FLOP (ESDS)		2
			(V01295) OPTIONAL		
-	120B	54LS74ABCXJC	.IC, FLIP-FLOP (ESDS)		2
			(V04713) OPTIONAL		
-	120C	54LS74ABCA	.IC, FLIP-FLOP (ESDS)		2
			(V18324) OPTIONAL		
-	120D	DM54LS74AJ883	.IC, FLIP-FLOP (ESDS)		2
			(V27014) OPTIONAL		
-	120E	54LS74DMQB	.IC, FLIP-FLOP (ESDS)		2
			(V07263) OPTIONAL		
	125	SNJ54LS10J00	.IC, NAND GATE (ESD)(V01295)		1
-	125A	SNJ54LS10J	.IC, NAND GATE (ESDS)		1
			(V01295) OPTIONAL		
-	125B	54LS10BCXJC	.IC, NAND GATE (ESDS)		1
			(V04713) OPTIONAL		
-	125C	54LS10BCA	.IC, NAND GATE (ESDS)		1
			(V18324) OPTIONAL		
-	125D	DM54LS10J883	.IC, NAND GATE (ESDS)		1
			(V27014) OPTIONAL		
	126	RNC55H3321FS	.RESISTOR, 3.32K, 1%, 1/10W	CEGH	2
	130	65521-136	.HEADER (V22526)		1
			(SUPSD BY 130A)		
-	130A	65521-436	.HEADER (V22526)		1
			(SUPSDS 130)		
	135	MN9102DG	.IC, LATCH (ESD)(V52648)		1
	137	RNC55H78R7FS	.RESISTOR	BCDE	1
			78.7 OHMS, 1%, 1/10W	FGH	
	140	M39014-02-1350	.CAPACITOR, 0.1 UF, 10%, 100V		2
	145	RNC55H3320FS	.RESISTOR, 332 OHMS, 1%, 1/10W		5
	147	C30	.BUSBAR (V99378)		3
			(SUPSD BY 147A)		
-	147A	20000-96	.BUSBAR (V99378)		3
			(SUPSDS 147)		
	148	JANTX2N2222A	.TRANSISTOR	CEGH	2
	150	623-2277-001	.PWB	A	1
-	150A	623-2277-002	.PWB	BDF	1
-	150B	623-2277-003	.PWB	CEGH	1
	155	RNC55H2431FS	.RESISTOR, 2.43K, 1%, 1/10W	CEGH	1

31-30-32

Page 1059
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101

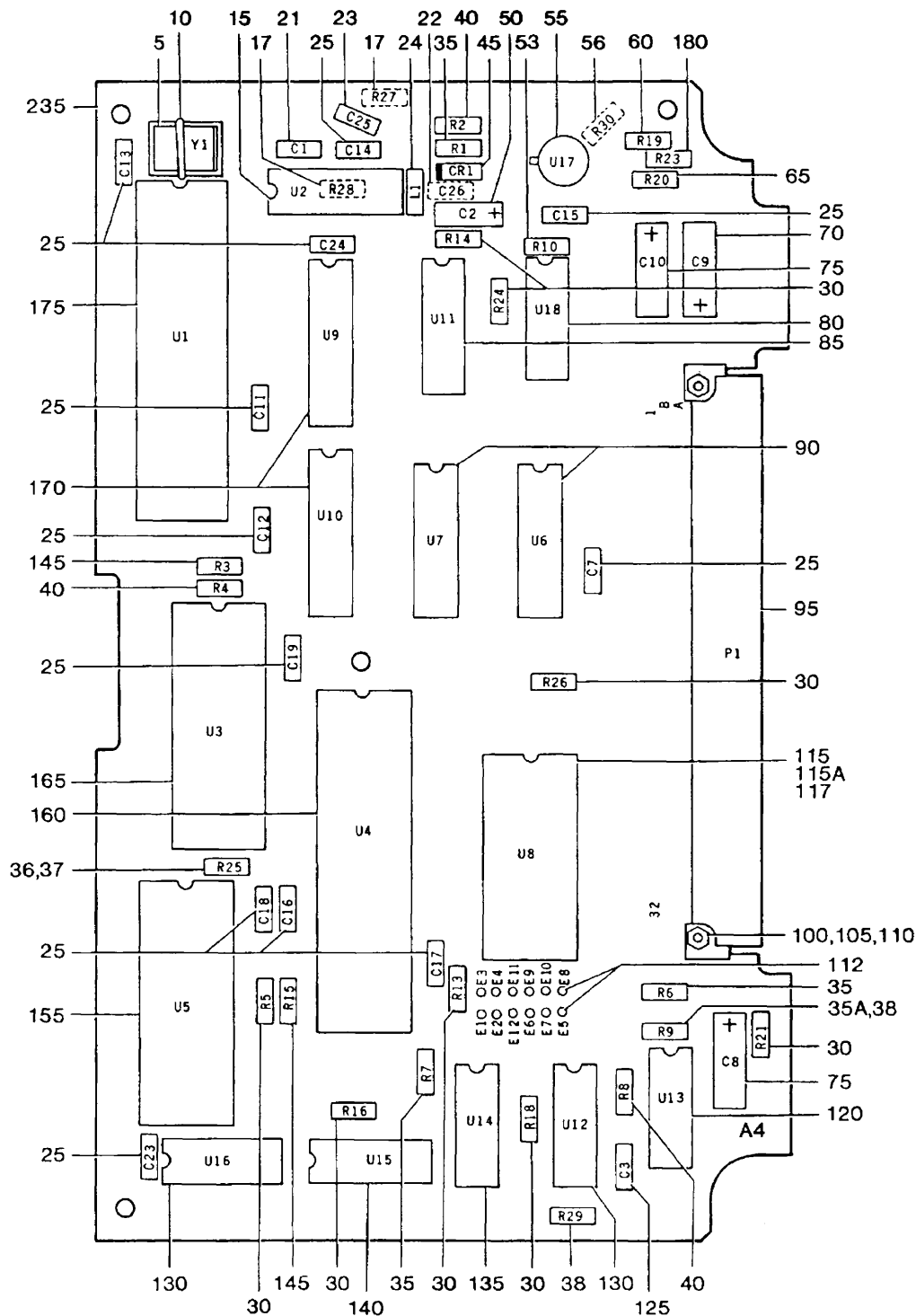
FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
4 160	RNC55H3322FS	1234567	.RESISTOR, 33.2K, 1%, 1/10W	CEGH	1

Honeywell

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



NOTE:

R27, R28, R30, C26 are mounted on circuit side.



RECORDING SYSTEM CONTROLLER ASSEMBLY (A4)
FIGURE 5

D-177819	E722-2438-2-M	E722-2438-4-M	E722-2438-6-M
----------	---------------	---------------	---------------

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
- 1	722-2438-001		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115 FOR NHA)(ESD)	A	RF
- 1A	722-2438-002		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115A FOR NHA)(ESD)	B	RF
- 1B	722-2438-003		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115B FOR NHA)(ESD)	C	RF
- 1C	722-2438-004		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115C AND 1-115D FOR NHA)(ESD)	D	RF
- 1D	722-2438-005		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115D FOR NHA)(ESD)	E	RF
- 1E	722-2438-006		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115E FOR NHA)(ESD)	F	RF
- 1F	722-2438-007		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115F FOR NHA)(ESD)	G	RF
- 1G	722-2438-008		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115G FOR NHA)(ESD)	H	RF
- 1H	722-2438-009		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115H FOR NHA)(ESD)	I	RF
- 1I	722-2438-010		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115I FOR NHA)(ESD)	J	RF
- 1J	722-2438-011		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115J FOR NHA)(ESD)	K	RF
- 1K	722-2438-012		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115K FOR NHA)(ESD)	L	RF
- 1L	722-2438-013		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115L FOR NHA)(ESD)	M	RF
- 1M	722-2438-014		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115M FOR NHA)(ESD)	N	RF
- 1N	722-2438-015		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115N FOR NHA)(ESD)	P	RF
- 1P	722-2438-016		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115P FOR NHA)(ESD)	Q	RF
- 1Q	722-2438-017		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115Q FOR NHA)(ESD)	R	RF

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
5					
- 1R	722-2438-018		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115R FOR NHA)(ESD)	S	RF
- 1S	722-2438-019		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115S FOR NHA)(ESD)	T	RF
- 1T	722-2438-020		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115T FOR NHA)(ESD)	U	RF
- 1U	722-2438-021		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115U FOR NHA)(ESD)	V	RF
- 1V	722-2438-022		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115V FOR NHA)(ESD)	W	RF
- 1W	722-2438-023		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115W FOR NHA)(ESD)	X	RF
- 1X	722-2438-024		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115X FOR NHA)(ESD)	Y	RF
- 1Y	722-2438-025		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115Y FOR NHA)(ESD)	Z	RF
- 1Z	722-2438-026		CCA, RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-115Y FOR NHA)(ESDS)	BA	RF
- 2	722-2438-027		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116 FOR NHA)(ESD)	CA	RF
- 2A	722-2438-028		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116A FOR NHA)(ESD)	DA	RF
- 2B	722-2438-029		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116B FOR NHA)(ESD)	EA	RF
- 2C	722-2438-030		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116C FOR NHA)(ESD)	FA	RF
- 2D	722-2438-031		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116D FOR NHA)(ESD)	GA	RF
- 2E	722-2438-032		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116E FOR NHA)(ESD)	HA	RF
- 2F	722-2438-033		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116F FOR NHA)(ESD)	IA	RF
- 2G	722-2438-034		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116G FOR NHA)(ESD)	JA	RF

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
- 2H	722-2438-035		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116H FOR NHA)(ESD)	KA	RF
- 2I	722-2438-036		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116I FOR NHA)(ESD)	LA	RF
- 2J	722-2438-037		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116J FOR NHA)(ESD)	MA	RF
- 2K	722-2438-038		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116K FOR NHA)(ESD)	NA	RF
- 2L	722-2438-039		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116L FOR NHA)(ESD)	PA	RF
- 2M	722-2438-040		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116M FOR NHA)(ESD)	QA	RF
- 2N	722-2438-041		RECORDING SYSTEM CONTROLLER ASSEMBLY, A4 (SEE FIG. 1-116N FOR NHA)(ESD)	RA	RF
5	M3098-38CR60AU		.CRYSTAL, 18.8744 MEGAHERTZ		1
- 5A	CR60AU18M87440		.CRYSTAL, 18.8744 MEGAHERTZ OPTIONAL		1
- 5B	S1027-2B		.CRYSTAL, 18.8744 MEGAHERTZ (V75378) OPTIONAL		1
10	2829-75-2		--- ATTACHING PARTS --- .STRAP (V98159)		1
15	MD8224B		.IC, CLOCK GENERATOR (ESD) (V34649)(OPT TO 15C)		1
- 15A	D8224		.IC, CLOCK GENERATOR (ESD) (V34649)(OPT TO 15, 15B, 15C, 15D, CODES EIMRVZ)	EIMR VZEA IAMA RA	1
- 15B	MD8224		.IC, CLOCK GENERATOR (ESD) (V34649)(OPT TO 15, 15A, 15C, 15D, CODES EIMRVZ)	EIMR VZEA IAMA RA	1
- 15C	AM8224DMB		.IC, CLOCK GENERATOR (ESD) (V34335)(OPT TO 15)		1
- 15D	AM8224DM		.IC, CLOCK GENERATOR (ESD) (V34335)(OPT TO 15, 15A, 15B, 15C, CODES EIMRVZ)	EIMR VZEA IAMA RA	1
17	RNC55H5110FS		.RESISTOR, 511 OHMS, 1%, 1/10W	A	2
- 20	RNC55H5110FS		.DELETED		0

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
21	M39014-01-1321		.CAPACITOR 10 PF, 10%, 200V (POST SB 1)	BCDE FGHI JKLM STUV WXYZ BACA DAEA JAKA LAMA NAPA QARA	1
22	M39014-01-1324		.CAPACITOR 15 PF, 10%, 200V (POST SB 1)	BCDE FGHI JKLM STUV WXYZ BACA DAEA JAKA LAMA NAPA QARA	1
23	M39014-01-1357		.CAPACITOR 0.001 UF, 10%, 200V (POST SB 1)	BCDE FGHI JKLM STUV WXYZ BACA DAEA JAKA LAMA NAPA QARA	1
24	M75101-3		.INDUCTOR 4.7 UH (POST SB 1) (SUPSD BY 24A)	BCDE FGHI JKLM STUV WXYZ BACA DAEA JAKA LAMA NAPA QARA	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
-	24A	MS75101-3	.INDUCTOR 4.7 UH (SUPSDS 24)	BCDE FGHI JKLM STUV WXYZ BACA DAEA JAKA LAMA NAPA	1
25	M39014-01-1593		.CAPACITOR, 0.1 UF, 10%, 50V		12
30	RNC55H4641FS		.RESISTOR, 4.64K, 1%, 1/10W		8
35	RNC55H1002FS		.RESISTOR, 10K, 1%, 1/10W		3
35A	RNC55H1002FS		.RESISTOR 10K, 1%, 1/10W	ABFJ NSW BAFA JANA	1
36	RNC55H1000FS		.RESISTOR 100 OHMS, 1%, 1/10W	BCDE FGHI JKLM STUV WXYZ BACA DAEA JAKA LAMA NAPA	1
37	RNC55H1002FS		.RESISTOR, 10K, 1%, 1/10W	A	1
38	RNC55H1000FS		.RESISTOR 100 OHMS, 1%, 1/10W	CDEG HIKL MPQT UVX CADA EAGA HAKA LAMA PA	2
40	RNC55H1001FS		.RESISTOR, 1K, 1%, 1/10W		3
45	JANTX1N4148-1		.DIODE		1
50	M39003-01-2014		.CAPACITOR, 4.7 UF, 10%, 10V		1
53	RNC55H1004FS		.RESISTOR, 1 MEGOHM, 1%, 1/10W		1
55	ICL8211MTY883B		.IC, VOLTAGE DETECTOR (ESD) (V32293)		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
- 55A	ICL8211MTY		.IC, VOLTAGE DETECTOR (ESDS) (V32293) OPTIONAL	ABCD EFGI JKMN PQRS TVXY Z BACA EAFA GAHA IAJA KAMA PAQA RA	1
56	RNC55H2004FS		.RESISTOR 2 MEGOHMS, 1%, 1/10W	DEHI LMQR UVYZ DAEA HAIA LAMA QARA	1
60	RNC55H7502FS		.RESISTOR, 75K, 1%, 1/10W		1
65	RNC55H3322FS		.RESISTOR, 33.2K, 1%, 1/10W		1
70	M39003-01-2064		.CAPACITOR, 6.8 UF, 10%, 35V		1
75	M39003-01-2017		.CAPACITOR, 33 UF, 10%, 10V		2
80	SNJ54LS00J00		.IC, NAND GATE (ESD)(V01295)		1
- 80A	SNJ54LS00J		.IC, NAND GATE (ESDS) (V01295) OPTIONAL		1
- 80B	54LS00BCXJC		.IC, NAND GATE (ESDS) (V04713) OPTIONAL		1
- 80C	54LS00BCA		.IC, NAND GATE (ESDS) (V18324) OPTIONAL		1
- 80D	DM54LS00J883		.IC, NAND GATE (ESDS) (V27014) OPTIONAL		1
85	SNJ54LS138J00		.IC, DECODER (ESD)(V01295)		1
- 85A	SNJ54LS138J		.IC, DECODER (ESDS) (V01295) OPTIONAL		1
- 85B	9LS54LS138J883B		.IC, DECODER (ESDS) (V07933) OPTIONAL		1
- 85C	S54LS138F883B		.IC, DECODER (ESDS) (V18324) OPTIONAL		1
- 85D	DM54LS138J883		.IC, DECODER (ESDS) (V27014) OPTIONAL		1
90	1M6561MDN883B		.IC, 1K RAM (ESD)(V32293)		2
- 90A	HM1-6561-8		.IC, 1K RAM (ESDS) (V34371) OPTIONAL		2
- 90B	HM1-6561B8		.IC, 1K RAM (ESDS) (V34371) OPTIONAL		2
- 90C	HM1-6561-2		.IC, 1K RAM (ESDS) (V34371) OPTIONAL		2
- 90D	HM1-6561-883		.IC, 1K RAM (ESDS) (V34371) OPTIONAL		2
95	G06D64P4BEBL		.CONNECTOR (V71468)		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		

5

			--- ATTACHING PARTS ---		
	100	MS51957-6	.SCREW, PAN HEAD		2
	105	MS21042-02	.NUT, LOCK		2
	110	06/16/5610	.WASHER, FLAT (V86928) (SUPSD BY 110A)		2
-	110A	06/15/5610	.WASHER, NYLON (V86928) (SUPSDS 110)		2
			-----*		
	112	CAS01-18B37	.HEADER, STRIP (V52072) (POST SB 6) DELETED	FGHI STUV JAKA LAMA	2
	115	718-0423-002	.IC, PROGRAMMED PROM (ESD)	A	1
	115A	718-0423-003	.IC, PROGRAMMED PROM (ESD) (POST SB 1)(SUPSD BY 115B) (PRE SB 15)	BCDE	1
-	115B	718-0423-004	.IC, PROGRAMMED PROM (ESD) (SUPSDS 115A)(PRE SB 6, SUPSD BY 115C)(PRE SB 12, SUPSD BY 115D)(PRE SB 15)	BCDE	1
-	115C	718-0423-005	.IC, PROGRAMMED PROM (ESD) (POST SB 6, SUPSDS 115B) (PRE SB 15)	FGHI	1
-	115D	718-0423-007	.IC, PROGRAMMED PROM (ESD) (POST SB 12, SUPSDS 115B OR 115C)(PRE SB 15) (PRE SB 19)(SUPSD BY 115J)	JKLM	1
-	115E	718-0423-008	.IC, PROGRAMMED PROM (ESD) (POST SB 15) (PRE SB 19)(SUPSD BY 115K)	NPQR	1
-	115F	718-0423-009	.IC, PROGRAMMED PROM (ESD) (PRE SB 19)(SUPSD BY 115H)	STUV	1
-	115G	718-0423-010	.IC, PROGRAMMED PROM (ESD) (PRE SB 19)(SUPSD BY 115I)	WXYZ	1
-	115H	718-0423-011	.IC, PROGRAMMED PROM (ESD) (POST SB 19, SUPSDS 115F)	JAKA LAMA	1
-	115I	718-0423-012	.IC, PROGRAMMED PROM (ESD) (POST SB 19, SUPSDS 115G)	NAPA QARA	1
-	115J	718-0423-013	.IC, PROGRAMMED PROM (ESD) (POST SB 19, SUPSDS 115D)	BACA DAEA	1
-	115K	718-0423-014	.IC, PROGRAMMED PROM (ESD) (POST SB 19, SUPSDS 115E)	FAGA HAIA	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
117	10624-01-446		.SOCKET (V17117)(POST SB 6)	FGHI JKLM NPQR STUV WXYZ BACA DAEA FAGA HAIA JAKA LAMA NAPA QARA	1
- 117A	IC624SGT		.SOCKET (V55322) OPTIONAL		1
- 117B	ICA246STG30		.SOCKET (V06776) OPTIONAL		1
120	SNJ54LS86J00		.IC, EXCLUSIVE OR GATE (ESD) (V01295)		1
- 120A	SNJ54LS86AJ		.IC, EXCLUSIVE OR GATE (ESDS) (V01295) OPTIONAL		1
- 120B	54LS86BCXJC		.IC, EXCLUSIVE OR GATE (ESDS) (V04713) OPTIONAL		1
- 120C	54LS86BCA		.IC, EXCLUSIVE OR GATE (ESDS) (V18324) OPTIONAL		1
- 120D	DM54LS86J883		.IC, EXCLUSIVE OR GATE (ESDS) (V27014) OPTIONAL		1
125	M39014-01-1569		.CAPACITOR, 0.0047 UF, 10%, 100V		1
130	SNJ54LS14J00		.IC, INVERTER (ESD)(V01295)		2
- 130A	SNJ54LS14J		.IC, INVERTER (ESDS) (V01295) OPTIONAL		2
- 130B	54LS14BCXJC		.IC, INVERTER (ESDS) (V04713) OPTIONAL		2
- 130C	54LS14BCA		.IC, INVERTER (ESDS) (V18324) OPTIONAL		2
- 130D	DM54LS14J883		.IC, INVERTER (ESDS) (V27014) OPTIONAL		2
135	SNJ54LS54J00		.IC, AND OR GATE (ESD)(V01295)		1
- 135A	SNJ54LS54J		.IC, AND OR GATE (ESDS) (V01295) OPTIONAL		1
- 135B	54LS54DMBQ		.IC, AND OR GATE (ESDS) (V27014) OPTIONAL		1
140	SNJ54LS74J00		.IC, FLIP-FLOP (ESD)(V01295)		1
- 140A	SNJ54LS74AJ		.IC, FLIP-FLOP (ESDS) (V01295) OPTIONAL		1
- 140B	54LS74ABCXJC		.IC, FLIP-FLOP (ESDS) (V04713) OPTIONAL		1
- 140C	54LS74DMQB		.IC, FLIP-FLOP (ESDS) (V07263) OPTIONAL		1
- 140D	54LS74ABCA		.IC, FLIP-FLOP (ESDS) (V18324) OPTIONAL		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
- 140E	DM54LS74AJ883		.IC, FLIP-FLOP (ESDS) (V27014) OPTIONAL		1
145	RNC55H1331FS		.RESISTOR, 1.33K, 1%, 1/10W		2
150	NOTUSED		NOT USED		0
155	AM9551DMB		.IC, PROGRAMMABLE INTERFACE (ESD)(V34335)	ABCD EFGH IJKL MNPQ R BACA DAEA FAGA HAIA	1
- 155A	AM9551DI		.IC, PROGRAMMABLE INTERFACE (ESD)(V34335)(OPT TO 155, 155B, CODES EIMR)	EIMR EAIA	1
- 155B	AM9551DIB		.IC, PROGRAMMABLE INTERFACE (ESD)(V34335)(OPT TO 155, 155A, CODES EIMR)	EIMR EAIA	1
- 155C	MD8251A		.IC, PROGRAMMABLE INTERFACE (ESD)(V34649)	STUV WXYZ JAKA LAMA NAPA QARA	1
160	AM9513DMB		.IC, TIMING CONTROLLER (ESD) (V34355) (OPT TO 160B,160E, 160F)		1
- 160A	AM9513SDMB		.DELETED		0
- 160B	AM9513ADMB		.IC, TIMING CONTROLLER (ESD) (V34335)(OPT TO 160,160E,160F)		1
- 160C	AM9513ADI		.DELETED		0
- 160D	AM9513ADIB		.IC, TIMING CONTROLLED (ESD) (V34335)(OPT TO 160, 160B, 160E)	EIMR VZ EAIA MARA	1
- 160E	AM9513DC		.IC, TIMING CONTROLLER (ESD) (V34335)(OPT TO 160, 160B, 160F)		1
- 160F	AM9513-5DMB		.IC, TIMING CONTROLLER (ESD) (V34335)(OPT TO 160, 160B, 160E)		1
165	AM8228DMB883B		.IC, SYSTEM CONTROLLER (ESD) (V34335)(REPLD BY 165A)		1
- 165A	AM8228DMB		.IC, SYSTEM CONTROLLER (ESD) (V34335)(REPLS 165) (REPLD BY 165C)		1
- 165B	MD8228B883C		.IC, SYSTEM CONTROLLER (ESDS) (V34649) OPTIONAL TO 165A		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
- 165C	DP8228MJMIL		.IC, SYSTEM CONTROLLER (ESDS) TRUE PN IS DP8228MJ-MIL (REPLD BY 165D) (V27014) (V97896 322-0223-003)		1
- 165D	IL8228BYA		.IC, SYSTEM CONTROLLER (ESDS) TRUE PN IS IL8228/BYA (REPLS 165C) (VOE3K6) (V97896 322-0223-003)		1
- 165E	IL8228Y1		.IC, SYSTEM CONTROLLER (ESDS) OPTIONAL TO 165D (V0E3K6) (V97896 322-0223-001)		1
170	SNJ54LS244J00		.IC, LINE DRIVER (ESD)(V01295)		2
- 170A	SNJ54LS244J		.IC, LINE DRIVER (ESDS) (V01295) OPTIONAL		2
- 170B	S54LS244F883B		.IC, LINE DRIVER (ESDS) (V18324) OPTIONAL		2
- 170C	54LS244DMQB		.IC, LINE DRIVER (ESDS) (V27014) OPTIONAL		2
- 170D	SN54LS244JB		.IC, LINE DRIVER (ESDS) (V34335) OPTIONAL		2
175	AM9080ADMB883B		.IC, CPU (ESD)(V34335) (REPLD BY 175A)		1
- 175A	MD8080AB		.IC, CPU (ESD)(V34649) (REPLS 175)(OPT TO 175B)		1
- 175B	AM9080ADMB		.IC, CPU (ESD)(V34335) (REPLS 175)(OPT TO 175A)		1
- 175C	ID8080A		.IC, CPU (ESD)(V34335) (OPT TO 175B)	DHLQ UY DAHA LAQA	1
- 175D	AM9080ADI		.IC, CPU (ESD)(V34335) (OPT TO 175A, 175B, FOR CODES EIMRVZ)	EIMR VZ EAIA MARA	1
180	RNC55H5901FS		.RESISTOR, 5.90K, 1%, 1/10W (SELECT FROM THIS OR THE FOLLOWING)		1
- 181	RNC55H6341FS		.RESISTOR, 6.34K, 1%, 1/10W		AR
- 182	RNC55H6811FS		.RESISTOR, 6.81K, 1%, 1/10W		AR
- 183	RNC55H7321FS		.RESISTOR, 7.32K, 1%, 1/10W		AR
- 184	RNC55H7871FS		.RESISTOR, 7.87K, 1%, 1/10W		AR
- 185	RNC55H8451FS		.RESISTOR, 8.45K, 1%, 1/10W		AR
- 186	RNC55H9091FS		.RESISTOR, 9.09K, 1%, 1/10W		AR
- 187	RNC55H9761FS		.RESISTOR, 9.76K, 1%, 1/10W		AR
- 188	RNC55H1052FS		.RESISTOR, 10.5K, 1%, 1/10W		AR
- 189	RNC55H1132FS		.RESISTOR, 11.3K, 1%, 1/10W		AR
- 190	RNC55H1212FS		.RESISTOR, 12.1K, 1%, 1/10W		AR
- 191	RNC55H1302FS		.RESISTOR, 13.0K, 1%, 1/10W		AR
- 192	RNC55H1402FS		.RESISTOR, 14.0K, 1%, 1/10W		AR
- 193	RNC55H1502FS		.RESISTOR, 15.0K, 1%, 1/10W		AR
- 194	RNC55H1622FS		.RESISTOR, 16.2K, 1%, 1/10W		AR
- 195	RNC55H1742FS		.RESISTOR, 17.4K, 1%, 1/10W		AR

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
5					
- 196	RNC55H1872FS		.RESISTOR, 18.7K, 1%, 1/10W		AR
- 197	RNC55H2002FS		.RESISTOR, 20.0K, 1%, 1/10W		AR
- 198	RNC55H2152FS		.RESISTOR, 21.5K, 1%, 1/10W		AR
- 199	RNC55H2322FS		.RESISTOR, 23.2K, 1%, 1/10W		AR
- 200	RNC55H2492FS		.RESISTOR, 24.9K, 1%, 1/10W		AR
- 201	RNC55H2672FS		.RESISTOR, 26.7K, 1%, 1/10W		AR
- 202	RNC55H2872FS		.RESISTOR, 28.7K, 1%, 1/10W		AR
- 203	RNC55H3092FS		.RESISTOR, 30.9K, 1%, 1/10W		AR
- 204	RNC55H3322FS		.RESISTOR, 33.2K, 1%, 1/10W		AR
- 205	RNC55H3572FS		.RESISTOR, 35.7K, 1%, 1/10W		AR
- 206	RNC55H3832FS		.RESISTOR, 38.3K, 1%, 1/10W		AR
- 207	RNC55H4122FS		.RESISTOR, 41.2K, 1%, 1/10W		AR
- 208	RNC55H4422FS		.RESISTOR, 44.2K, 1%, 1/10W		AR
- 209	RNC55H4752FS		.RESISTOR, 47.5K, 1%, 1/10W		AR
- 210	RNC55H5112FS		.RESISTOR, 51.1K, 1%, 1/10W		AR
- 211	RNC55H5492FS		.RESISTOR, 54.9K, 1%, 1/10W		AR
- 212	RNC55H5902FS		.RESISTOR, 59.0K, 1%, 1/10W		AR
- 213	RNC55H6342FS		.RESISTOR, 63.4K, 1%, 1/10W		AR
- 214	RNC55H6812FS		.RESISTOR, 68.1K, 1%, 1/10W		AR
- 215	RNC55H7322FS		.RESISTOR, 73.2K, 1%, 1/10W		AR
- 216	RNC55H7872FS		.RESISTOR, 78.7K, 1%, 1/10W		AR
- 217	RNC55H8452FS		.RESISTOR, 84.5K, 1%, 1/10W		AR
- 218	RNC55H9092FS		.RESISTOR, 90.9K, 1%, 1/10W		AR
- 219	RNC55H9762FS		.RESISTOR, 97.6K, 1%, 1/10W		AR
- 220	RNC55H1053FS		.RESISTOR, 105K, 1%, 1/10W		AR
- 221	RNC55H1133FS		.RESISTOR, 113K, 1%, 1/10W		AR
- 222	RNC55H1213FS		.RESISTOR, 121K, 1%, 1/10W		AR
- 223	RNC55H1303FS		.RESISTOR, 130K, 1%, 1/10W		AR
- 224	RNC55H1403FS		.RESISTOR, 140K, 1%, 1/10W		AR
- 225	RNC55H1503FS		.RESISTOR, 150K, 1%, 1/10W		AR
- 226	RNC55H1623FS		.RESISTOR, 162K, 1%, 1/10W		AR
- 227	RNC55H1743FS		.RESISTOR, 174K, 1%, 1/10W		AR
- 228	RNC55H1873FS		.RESISTOR, 187K, 1%, 1/10W		AR
- 229	RNC55H2003FS		.RESISTOR, 200K, 1%, 1/10W		AR
235	623-2378-001		.PRINTED WIRING BOARD	ABFJ NSW BAFA JANA CDEG	1
- 235A	623-2378-002		.PRINTED WIRING BOARD (OPT TO 235B)	HIKL MPQR TUVX YZ CADA EAGA HAIA KALA MAPA QARA	1

Honeywell

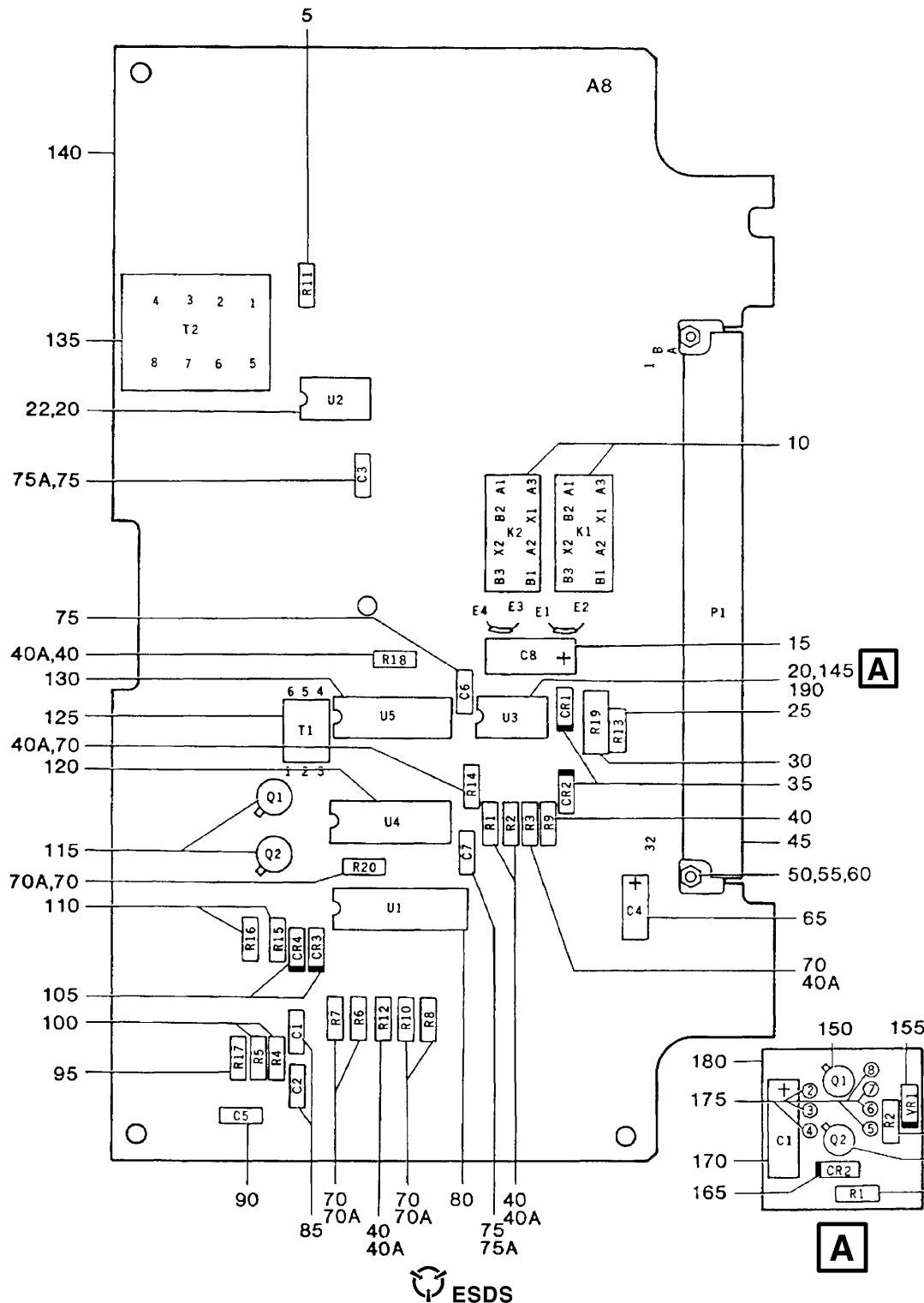
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
5			1234567		
- 235B	623-2378-003		.PRINTED WIRING BOARD (OPT TO 235A)	CDEG HIKL MPQR TUVX YZ CADA EAGA HAIA KALA MAPA QARA	1
- 235C	623-2274-001		.PRINTED WIRING BOARD (OPT TO 235)	ABFJ NSW BAFA JANA	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK



AIRFRAME WIRING INTERFACE ASSEMBLY (A6)
FIGURE 6

ID-177820 E722-2501-2-H

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
6					
- 1	722-2501-001		AIRFRAME WIRING INTERFACE ASSEMBLY, A8 (ESD) (SEE FIG. 1-120 FOR NHA)	A	RF
- 1A	722-2501-002		AIRFRAME WIRING INTERFACE ASSEMBLY, A8 (ESD) (SEE FIG. 1-120A FOR NHA)	B	RF
- 1B	722-2501-003		AIRFRAME WIRING INTERFACE ASSEMBLY, A8 (ESD) (SEE FIG. 1-120B FOR NHA)	C	RF
5	RWR81S51R1FR		.RESISTOR, 51.1 OHMS, 1%, 1W		1
10	3SAV5004D1		.RELAY, 2PDT (V01526)		2
15	M39003-01-2064		.CAPACITOR, 6.8 UF, 10%, 35V		1
- 15A	M39003-01-2304		.CAPACITOR, 6.8 UF, 10%, 35V OPTIONAL		1
20	SNJ55472JG00		.IC, NAND DRIVER (V01295) (ESD)(PRE SB 7, SUPSD BY 145 OR 185, 190, 195 FOR U3)(OPT TO 20A, 20B, 20C, 20D, 20E)	A	2
- 20A	SNJ55472JG883B		.IC, NAND DRIVER (V01295) (ESD)(PRE SB 7, SUPSD BY 145 OR 185, 190, 195 FOR U3)(OPT TO 20, 20B, 20C, 20D, 20E)	A	2
- 20B	SNJ55472JG		.IC, NAND DRIVER (V01295) (ESD)(OPT TO 20, 20A, 20C, 20D, 20E)	A	2
- 20C	SNJ55462JG		.IC, NAND DRIVER (V01295) (ESD)(OPT TO 20, 20A, 20B, 20D, 20E)	A	2
- 20D	SN55462JG		.IC, NAND DRIVER (V01295) (ESD)(OPT TO 20, 20A, 20B, 20C, 20E)	A	2
- 20E	DS55462J8-883		.IC, NAND DRIVER (V27014) (ESD)(OPT TO 20, 20A, 20B, 20C, 20D)	A	2
22	SNJ55472JG		.IC, NAND DRIVER (V01295) (ESD)(OPT TO 22A, 22B, 22C)	BC	1
- 22A	SNJ55462JG		.IC, NAND DRIVER (V01295) (ESD)(OPT TO 22, 22B, 22C)	BC	1
- 22B	SN55462JG		.IC, NAND DRIVER (V01295) (ESD)(OPT TO 22, 22A, 22C)	BC	1
- 22C	DS55462J8-883		.IC, NAND DRIVER (V27014) (ESD)(OPT TO 22, 22A, 22B)	BC	1
25	RWR81S4420FR		.RESISTOR, 442 OHMS, 1%, 1W		1
30	RWR80S10R0FR		.RESISTOR, 10 OHMS, 1%, 2W		1
35	JANTX1N5614		.DIODE		2
- 35A	JAN1N5614		.DIODE OPTIONAL, CODE C		2
40	RNC55H1001FS		.RESISTOR, 1K, 1%, 1/10W	A	5
40A	RNC55H1001FS		.RESISTOR, 1K, 1%, 1/10W (POST SB 7)	BC	6
45	G06D64P4BEBL		.CONNECTOR, 64-PIN (V71468)		1
			- - - ATTACHING PARTS - - -		
50	MS51957-6		.SCREW, PAN HEAD		2

31-30-32

Page 1077
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
6			1234567		
55	MS21042-02		.NUT, LOCK		2
60	06/16/5610		.WASHER, FLAT (V86928)		2
			(SUPSD BY 60A)		
- 60A	06/15/5610		.WASHER, NYLON (V86928)		2
			(SUPSDS 60)		
			-----*-----		
65	M39003-01-2017		.CAPACITOR, 33 UF, 10%, 10V		1
70	RNC55H3321FS		.RESISTOR, 3.32K, 1%, 1/10W	A	7
- 70A	RNC55H3321FS		.RESISTOR, 3.32K, 1%, 1/10W	BC	5
			(POST SB 7)		
75	M39014-01-1593		.CAPACITOR, 0.1 UF, 10%, 50V	A	3
- 75A	M39014-01-1593		.CAPACITOR, 0.1 UF, 10%, 50V	BC	2
			(POST SB 7)		
80	SNJ54LS157J00		.IC, MULTIPLEXER (ESD)(V01295)		1
- 80A	SNJ54LS157J		.IC, MULTIPLEXER (ESDS)		1
			(V01295) OPTIONAL		
- 80B	54LS157BEXJC		.IC, MULTIPLEXER (ESDS)		1
			(V04713) OPTIONAL		
- 80C	54LS157DMQB		.IC, MULTIPLEXER (ESDS)		1
			(V07263) OPTIONAL		
- 80D	54LS157J883B		.IC, MULTIPLEXER (ESDS)		1
			(V07933) OPTIONAL		
- 80E	SN54LS157J883		.IC, MULTIPLEXER (ESDS)		1
			(V27014) OPTIONAL		
85	M39014-01-1569		.CAPACITOR, 4700 PF, 10%, 100V		2
90	M39014-01-1566		.CAPACITOR, 3300 PF, 10%, 100V		1
95	RNC55H3011FS		.RESISTOR, 3.01K, 1%, 1/10W		1
100	RNC55H5901FS		.RESISTOR, 5.90K, 1%, 1/10W		2
105	JANTX1N5711		.DIODE		2
- 105A	JAN1N5711		.DIODE		2
			OPTIONAL		
- 105B	1N5711		.DIODE		2
			OPTIONAL, CODE C		
110	RNC55H3320FS		.RESISTOR, 332 OHMS, 1%, 1/10W		2
115	JANTX2N2222A		.TRANSISTOR		2
120	SNJ5406J00		.IC, BUFFER (ESD)(V01295)		1
- 120A	SNJ5406J		.IC, BUFFER (ESDS)		1
			(V01295) OPTIONAL		
- 120B	DM5406J883		.IC, BUFFER (ESDS)		1
			(V27014) OPTIONAL		
125	SP21		.TRANSFORMER, AUDIO (V81095)		1
- 125A	MET11		.TRANSFORMER, AUDIO		1
			(V63265) OPTIONAL		
130	SNJ54LS00J00		.IC, NAND GATE (ESD)(V01295)		1
- 130A	SNJ54LS00J		.IC, NAND GATE (ESDS)		1
			(V01295) OPTIONAL		
- 130B	54LS00BCXJC		.IC, NAND GATE (ESDS)		1
			(V04713) OPTIONAL		
- 130C	54LS00BCA		.IC, NAND GATE (ESDS)		1
			(V18324) OPTIONAL		
- 130D	DM54LS00J883		.IC, NAND GATE (ESDS)		1
			(V27014) OPTIONAL		

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
6					
	135	M27-165-10	.TRANSFORMER, AUDIO		1
-	135A	310-0501-001	.TRANSFORMER, AUDIO OPTIONAL		1
	140	623-2311-002	.PRINTED WIRING BOARD		1
	145	722-3523-001	.DRIVER, RELAY (POST SB 7, SUPSDS 20, 20A)(OPT TO 185, 190, 195)	BC	1
	150	JANTX2N5582	..TRANSISTOR	BC	2
-	150A	JAN2N5582	..TRANSISTOR OPTIONAL		2
-	150B	2N5582	..TRANSISTOR OPTIONAL		2
	155	JAN1N4615	..DIODE, ZENER	BC	1
	160	RNC55H2321FS	..RESISTOR, 2.32K, 1%, 1/10W	BC	2
	165	JANTX1N4148-1	..DIODE	BC	1
	170	M39003-01-2257	..CAPACITOR, 33 UF, 10%, 10V	BC	1
	175	2002-10TTE	..STRIP, TERMINAL (V19112)	BC	AR
-	175A	TS120TA	..STRIP, TERMINAL (V55322) OPTIONAL		AR
	180	623-3137-001	..BOARD, PRINTED WIRING	BC	1
	185	2829-75-2	..STRAP (V98159)(POST SB 7)	BC	1
	190	751-0048-001	..HEADER ASSEMBLY (POST SB 7)	BC	1
	195	14120-01-446	..SOCKET (V17117)(POST SB 7)	BC	1
-	195A	SBE20STG30	..SOCKET (V06776) OPTIONAL		1
-	195B	SS0120T2	..SOCKET (V55332) OPTIONAL		1

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



ID-177821 E722-2440-2-J

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
7					
- 1	722-2440-001		POWER SUPPLY CCA, A1A1 (ESD)(SEE FIG. 2-25 FOR NHA)	A	RF
- 1A	722-2440-002		POWER SUPPLY CCA, A1A1 (ESD)(SEE FIG. 2-25A FOR NHA)	B	RF
- 1B	722-2440-003		POWER SUPPLY CCA, A1A1 (ESD)(SEE FIG. 2-25B OR 15-25 FOR NHA)	C	RF
- 1C	722-2440-004		DELETED	D	0
- 1D	722-2440-005		POWER SUPPLY CCA, A1A1 (ESD)(SEE FIG. 15-25A FOR NHA)	E	RF
	5	M39018-7-0284M	.CAPACITOR 2300 UF, +75, -10%, 40V - - - ATTACHING PARTS - - -		2
	10	2829-75-2	.STRAP (V98159) -----*-----		4
	12	MS17821-4-9	.CABLE TIE (REPLD BY 12A)		2
- 12A	SST1-5M		.CABLE TIE (V06383)(REPLS 12)		2
	15	RNC60H75R0FS	.RESISTOR 75 OHMS, 1%, 1/4W		2
	20	M39003-01-2116	.CAPACITOR 1 UF, 10%, 50V		1
	25	RNC55H2000FS	.RESISTOR 200 OHMS, 1%, 1/10W		1
	30	RNC55H8252FS	.RESISTOR 82.5K, 1%, 1/10W		1
	35	RWR81S82R5FR	.RESISTOR 82.5 OHMS, 1%, 1W		1
	40	JANTX1N5614	.DIODE		7
- 40A	JAN1N5614		.DIODE OPTIONAL		7
	45	M39014-02-1419	.CAPACITOR 1 UF, 10%, 50V		10
	50	RNC55H8063FS	.RESISTOR , 806K, 1%, 1/10W		1
	55	ICL8211MTY883B	.IC, VOLTAGE DETECTOR (ESD) (V32293)		1
- 55A	ICL8211MTY		.IC, VOLTAGE DETECTOR (ESDS) (V32293) OPTIONAL		1
	56	RNC55H2004FS	.RESISTOR 2 MEGOHMS, 1%, 1/10W	CDE	1
	60	RNC55H1502FS	.RESISTOR, 15K, 1%, 1/10W		1
	65	M39014-02-1350	.CAPACITOR, 0.1 UF, 10%, 100V		5
	70	JANTX1N4970	.DIODE, ZENER, 33V, 5W		1
	75	ULS2003H883B	.IC, TRANSISTOR ARRAY (V56289)		2
- 75A	ULS2003R883C		.IC, TRANSISTOR ARRAY (V56289) OPTIONAL		2
- 75B	SG2003J883B		.IC, TRANSISTOR ARRAY (V34333) OPTIONAL		2
- 75C	SG2003J		.IC, TRANSISTOR ARRAY (V34333) OPTIONAL		2
	80	JANTX1N4148-1	.DIODE		1
	85	RNC55H1001FS	.RESISTOR, 1K, 1%, 1/10W		1

31-30-32

Page 1081
22 Dec 2006

Honeywell

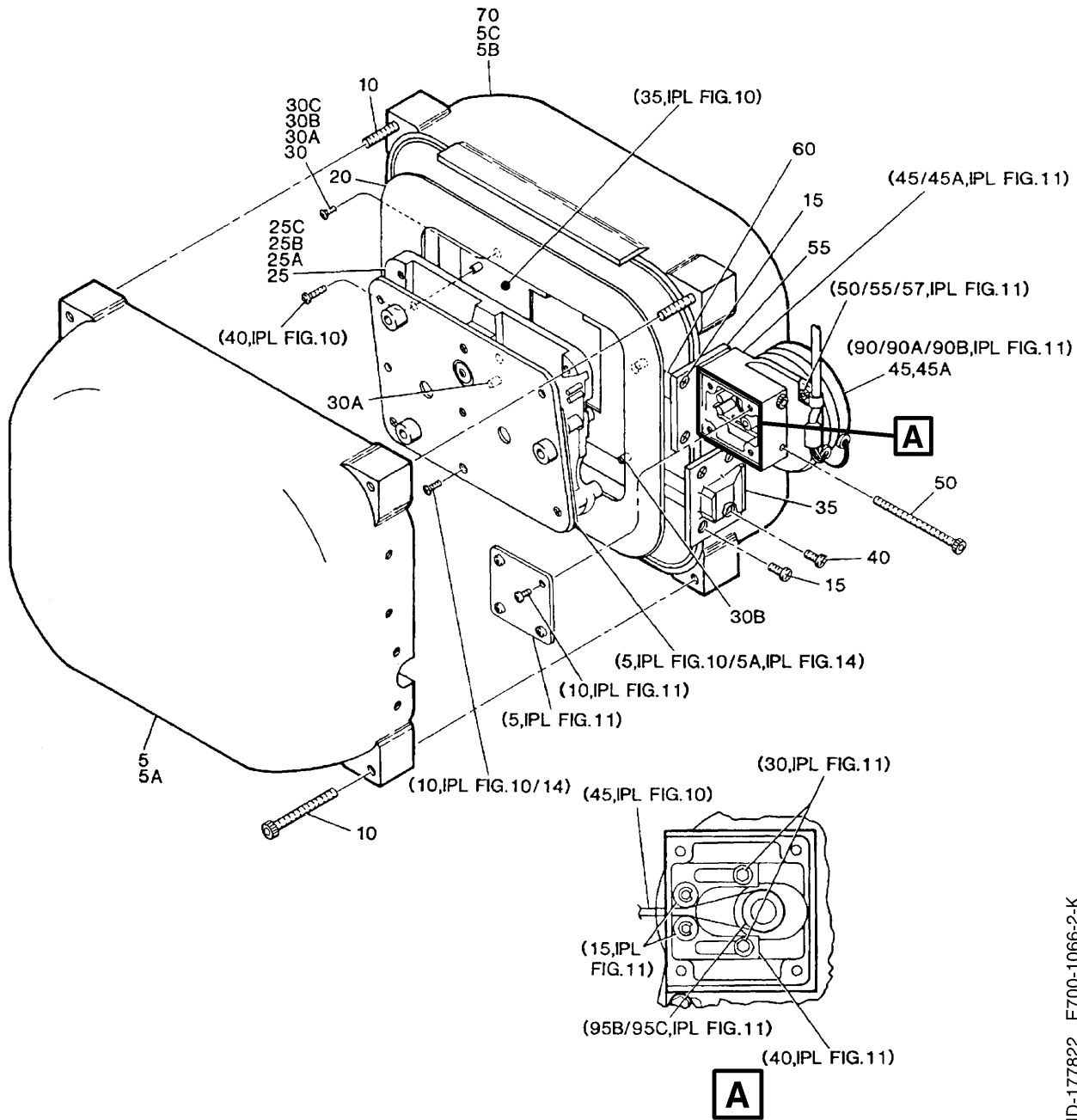
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
7			1234567		
90	RNC55H2801FS		.RESISTOR, 2.8K, 1%, 1/10W		1
95	RNC55H2430FS		.RESISTOR, 243 OHMS, 1%, 1/10W	A	2
95A	RNC55H2430FS		.RESISTOR, 243 OHMS, 1%, 1/10W	BCDE	1
100	JANTX2N2222A		.TRANSISTOR		1
105	G06D64P4BEBL		.CONNECTOR (V71468)		1
			--- ATTACHING PARTS ---		
110	MS51957-6		.SCREW, PAN HEAD		2
115	MS21042-02		.NUT, LOCK		2
120	06/16/5610		.WASHER, FLAT (V86928)		2
			(SUPSD BY 120A)		
- 120A	06/15/5610		.WASHER, NYLONG (V86928)		2
			(SUPSDS 120)		
			-----*-----		
125	UA79M05HMQB		.IC, VOLTAGE REGULATOR (ESD)		1
			(V07263)(OPT TO 125A)		
125A	LM120H5-0-883B		.IC, VOLTAGE REGULATOR (ESD)		1
			(V27014)(OPT TO 125)		
130	M39018-7-0292M		.CAPACITOR, 1000 UF, 10%, 50V		3
			--- ATTACHING PARTS ---		
135	2829-75-2		.STRAP (V98159)		6
137	MS17821-4-9		.CABLE TIE (REPLD BY 137A)		3
- 137A	SST1-5M		.CABLE TIE (V06383)(REPLS 137)		3
			-----*-----		
140	RWR81S10R0FR		.RESISTOR, 10 OHMS, 1%, 1W		1
145	M39003-01-2134		.CAPACITOR, 10 UF, 10%, 50V		2
150	NOTUSED		NOT USED		0
155	M39014-02-1338		.CAPACITOR, 0.01 UF, 10%, 200V		2
160	RWR80SR511FS		.RESISTOR, 0.511 OHM, 1%, 2W		2
165	RNC55H5901FS		.RESISTOR, 5.90K, 1%, 1/10W		1
			(SELECT FROM THIS OR THE FOLLOWING)		
- 166	RNC55H6341FS		.RESISTOR, 6.34K, 1%, 1/10W		AR
- 167	RNC55H6811FS		.RESISTOR, 6.81K, 1%, 1/10W		AR
- 168	RNC55H7321FS		.RESISTOR, 7.32K, 1%, 1/10W		AR
- 169	RNC55H7871FS		.RESISTOR, 7.87K, 1%, 1/10W		AR
- 170	RNC55H8451FS		.RESISTOR, 8.45K, 1%, 1/10W		AR
- 171	RNC55H9091FS		.RESISTOR, 9.09K, 1%, 1/10W		AR
- 172	RNC55H9761FS		.RESISTOR, 9.76K, 1%, 1/10W		AR
- 173	RNC55H1052FS		.RESISTOR, 10.5K, 1%, 1/10W		AR
- 174	RNC55H1132FS		.RESISTOR, 11.3K, 1%, 1/10W		AR
- 175	RNC55H1212FS		.RESISTOR, 12.1K, 1%, 1/10W		AR
- 176	RNC55H1302FS		.RESISTOR, 13.0K, 1%, 1/10W		AR
- 177	RNC55H1402FS		.RESISTOR, 14.0K, 1%, 1/10W		AR
- 178	RNC55H1502FS		.RESISTOR, 15.0K, 1%, 1/10W		AR
- 179	RNC55H1622FS		.RESISTOR, 16.2K, 1%, 1/10W		AR
- 180	RNC55H1742FS		.RESISTOR, 17.4K, 1%, 1/10W		AR
- 181	RNC55H1872FS		.RESISTOR, 18.7K, 1%, 1/10W		AR
- 182	RNC55H2002FS		.RESISTOR, 20.0K, 1%, 1/10W		AR
- 183	RNC55H2152FS		.RESISTOR, 21.5K, 1%, 1/10W		AR
- 184	RNC55H2322FS		.RESISTOR, 23.2K, 1%, 1/10W		AR
- 185	RNC55H2492FS		.RESISTOR, 24.9K, 1%, 1/10W		AR
- 186	RNC55H2672FS		.RESISTOR, 26.7K, 1%, 1/10W		AR

Honeywell

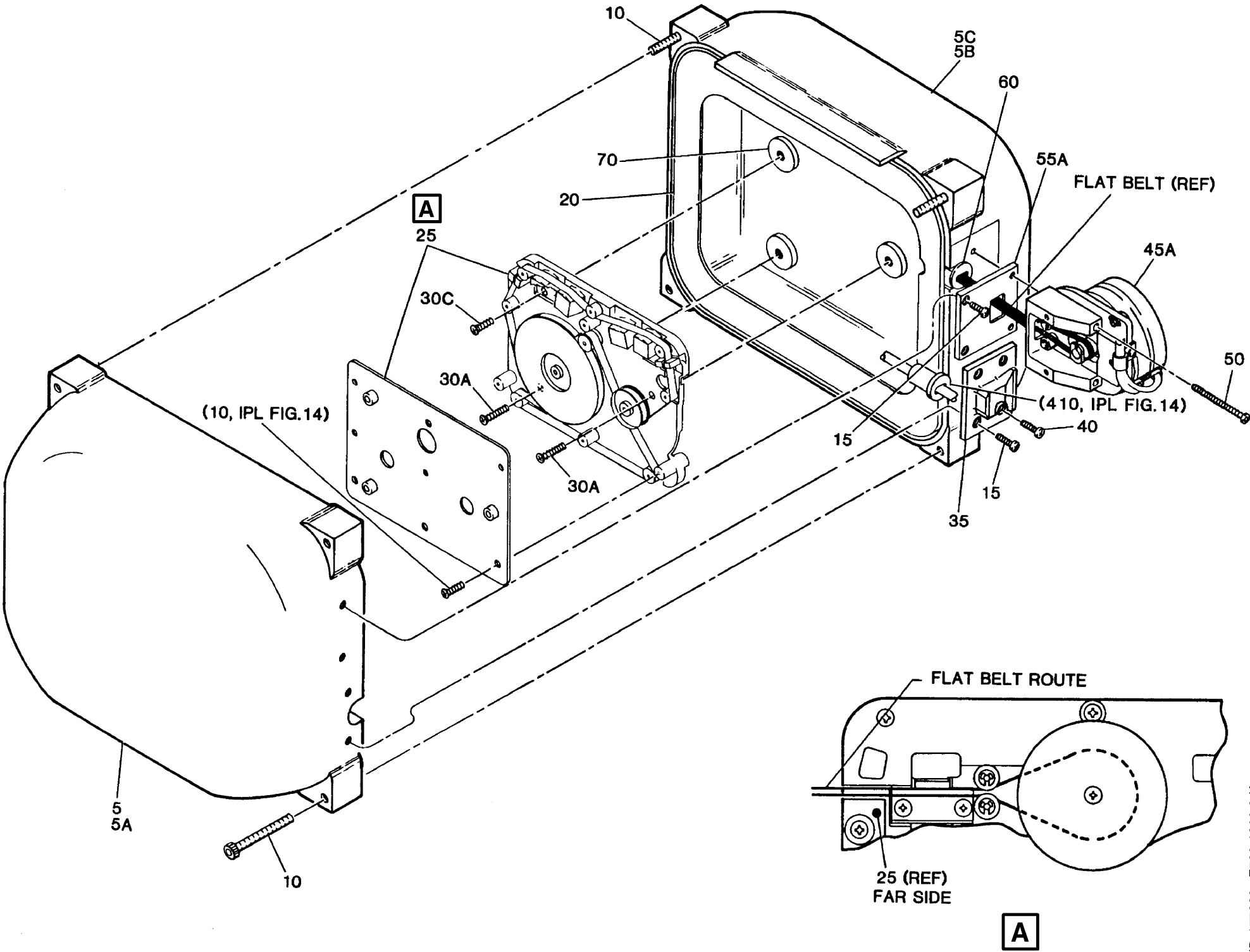
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
7					
- 187	RNC55H2872FS		.RESISTOR, 28.7K, 1%, 1/10W		AR
- 188	RNC55H3092FS		.RESISTOR, 30.9K, 1%, 1/10W		AR
- 189	RNC55H3322FS		.RESISTOR, 33.2K, 1%, 1/10W		AR
- 190	RNC55H3572FS		.RESISTOR, 35.7K, 1%, 1/10W		AR
- 191	RNC55H3832FS		.RESISTOR, 38.3K, 1%, 1/10W		AR
- 192	RNC55H4122FS		.RESISTOR, 41.2K, 1%, 1/10W		AR
- 193	RNC55H4422FS		.RESISTOR, 44.2K, 1%, 1/10W		AR
- 194	RNC55H4752FS		.RESISTOR, 47.5K, 1%, 1/10W		AR
- 195	RNC55H5112FS		.RESISTOR, 51.1K, 1%, 1/10W		AR
- 196	RNC55H5492FS		.RESISTOR, 54.9K, 1%, 1/10W		AR
- 197	RNC55H5902FS		.RESISTOR, 59.0K, 1%, 1/10W		AR
- 198	RNC55H6342FS		.RESISTOR, 63.4K, 1%, 1/10W		AR
- 199	RNC55H6812FS		.RESISTOR, 68.1K, 1%, 1/10W		AR
- 200	RNC55H7322FS		.RESISTOR, 73.2K, 1%, 1/10W		AR
- 201	RNC55H7872FS		.RESISTOR, 78.7K, 1%, 1/10W		AR
- 202	RNC55H8452FS		.RESISTOR, 84.5K, 1%, 1/10W		AR
- 203	RNC55H9092FS		.RESISTOR, 90.9K, 1%, 1/10W		AR
- 204	RNC55H9762FS		.RESISTOR, 97.6K, 1%, 1/10W		AR
- 205	RNC55H1053FS		.RESISTOR, 105K, 1%, 1/10W		AR
- 206	RNC55H1133FS		.RESISTOR, 113K, 1%, 1/10W		AR
- 207	RNC55H1213FS		.RESISTOR, 121K, 1%, 1/10W		AR
- 208	RNC55H1303FS		.RESISTOR, 130K, 1%, 1/10W		AR
- 209	RNC55H1403FS		.RESISTOR, 140K, 1%, 1/10W		AR
- 210	RNC55H1503FS		.RESISTOR, 150K, 1%, 1/10W		AR
- 211	RNC55H1623FS		.RESISTOR, 162K, 1%, 1/10W		AR
- 212	RNC55H1743FS		.RESISTOR, 174K, 1%, 1/10W		AR
- 213	RNC55H1873FS		.RESISTOR, 187K, 1%, 1/10W		AR
- 214	RNC55H2003FS		.RESISTOR, 200K, 1%, 1/10W		AR
220	SG1501AJ883B		.IC, REGULATOR (ESD)(V34333)		1
- 220A	SG1501AJ		.IC, REGULATOR (ESDS) (V34333) OPTIONAL		1
- 220B	MC1568L		.IC, REGULATOR (ESDS) (V04713) OPTIONAL		1
- 220C	MC1568G		.IC, REGULATOR (ESDS) (V04713) OPTIONAL		1
225	623-2276-002		.PRINTED WIRING BOARD	ABCE	1
- 225A	623-2276-003		.PRINTED WIRING BOARD	D	1



ID-177822 E700-1066-2-K

TAPE RECORDER ASSEMBLY
FIGURE 8 (SHEET 1 OF 2)



TAPE RECORDER ASSEMBLY
FIGURE 8 (SHEET 2 OF 2)

ID-177823 E700-1066-3-K

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
8					
- 1	700-1066-001		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150 FOR NHA)	A	RF
- 1A	700-1066-004		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150A FOR NHA)	B	RF
- 1B	700-1066-005		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150B FOR NHA)	C	RF
- 1C	700-1066-006		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150E FOR NHA)	D	RF
- 1D	700-1066-007		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150F FOR NHA)	E	RF
- 1E	700-1066-008		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150G FOR NHA)	F	RF
- 1F	700-1066-009		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150H FOR NHA)	G	RF
- 1G	700-1066-010		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150I FOR NHA)	H	RF
- 1H	700-1066-011		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150J FOR NHA)	I	RF
- 1I	700-1066-012		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150K FOR NHA)	J	RF
- 1J	700-1066-013		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150L FOR NHA)	K	RF
- 1K	700-1066-014		TAPE RECORDER ASSEMBLY (SEE FIG. 1-150M FOR NHA)	L	RF
5	607-1519-001		.COVER, TOP		1
5A	607-1644-001		.COVER, TOP (USED ON FIG.1 ITEM 2, 2A)		1
5B	607-1519-002		.COVER, BOTTOM (OPT TO 5C, CODES BDEFGHI) (BUY 5C)	ABCD EFGH I	1
5C	607-1519-003		.COVER, BOTTOM	BDEF GHIJ KL	1
			--- ATTACHING PARTS ---		
10	56040-4-28		.SCREW, SOCKET HEAD (V71838)		4
- 10A	FS2084-4-28		.SCREW, SOCKET HEAD (V58998) OPTIONAL		4
15	MS51957-43		.SCREW, PAN HEAD -----*-----		4
20	611-0153-002		.GASKET		1
25	617-3697-001		.TAPE TRANSPORT ASSEMBLY (SEE FIG. 10 FOR BREAKDOWN) (SUPSD BY 25A)	A	1
25A	617-3697-002		.TAPE TRANSPORT ASSEMBLY (SEE FIG. 10 FOR BREAKDOWN) (SUPSDS 25)	A	1
25B	617-4422-001		.TAPE TRANSPORT ASSEMBLY (PRE SB 2) (PRE SB 5) (SUPSD BY 25H) (PRE SB 8) (SUPSD BY 25H) (PRE SB 13) (SUPSD BY 25Q) (SEE FIG. 14 FOR BREAKDOWN)	B	1

31-30-32

Page 1087
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
8			1234567		
25C	617-4422-003		.TAPE TRANSPORT ASSEMBLY (POST SB 2) (PRE SB 5) (SUPSD BY 25G) (PRE SB 8) (SUPSD BY 25I) (PRE SB 13) (SUPSD BY 25R) (SEE FIG. 14 FOR BREAKDOWN)	C	1
- 25D	617-4422-004		.TAPE TRANSPORT ASSEMBLY (OPT TO 25F, 25H, 25I) (NOT FOR FIELD USE) (SEE FIG. 14 FOR BREAKDOWN)	DI	1
- 25E	617-4422-005		.TAPE TRANSPORT ASSEMBLY (OPT TO 25G, 25J, 25K) (POST SB 5) (SUPSDS 25B) (PRE SB 11) (SUPSD BY 25N) (SEE FIG. 14 FOR BREAKDOWN)	EH	1
- 25F	617-4422-006		.TAPE TRANSPORT ASSEMBLY (OPT TO 25D, 25H, 25I) NOT FOR FIELD USE (SEE FIG. 14 FOR BREAKDOWN)	DI	1
- 25G	617-4422-007		.TAPE TRANSPORT ASSEMBLY (OPT TO 25E, 25J, 25K) (POST SB 5) (SUPSDS 25C) (PRE SB 11) (SUPSD BY 25P) (SEE FIG. 14 FOR BREAKDOWN)	EH	1
- 25H	617-4422-008		.TAPE TRANSPORT ASSEMBLY (OPT TO 25D, 25F, 25I) (POST SB 8) (SUPSDS 25B) (PRE SB 13) (SUPSD BY 25Q) (SEE FIG. 14 FOR BREAKDOWN)	D	1
- 25I	617-4422-009		.TAPE TRANSPORT ASSEMBLY (OPT TO 25D, 25F, 25H) (POST SB 8) (SUPSDS 25C) (PRE SB 13) (SUPSD BY 25R) (SEE FIG. 14 FOR BREAKDOWN)	D	1
- 25J	617-4422-010		.TAPE TRANSPORT ASSEMBLY (OPT TO 25E, 25G, 25K) (PRE SB 11) (SUPSD BY 25N) (SEE FIG. 14 FOR BREAKDOWN)	E	1
- 25K	617-4422-011		.TAPE TRANSPORT ASSEMBLY (OPT TO 25E, 25G, 25J) (PRE SB 11) (SUPSD BY 25P) (SEE FIG. 14 FOR BREAKDOWN)	E	1
- 25L	617-4422-012		.TAPE TRANSPORT ASSEMBLY (PRE SB 11) (SUPSD BY 25M) (SEE FIG. 14 FOR BREAKDOWN)	F	1
- 25M	617-4422-013		.TAPE TRANSPORT ASSEMBLY (MFG ON SN 6000 AND ON) (POST SB 11) (SUPSDS 25L) (PRE SB 17) (SEE FIG. 14 FOR BREAKDOWN)	G	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
8			1234567		
- 25N	617-4422-014		.TAPE TRANSPORT ASSEMBLY (OPT TO 25P) (BUY 25M) (POST SB 11) (SUPSDS 25E) (POST SB 11) (SUPSDS 25J) (PRE SB 17) (SEE FIG. 14 FOR BREAKDOWN)	H	1
- 25P	617-4422-015		.TAPE TRANSPORT ASSEMBLY (OPT TO 25N) (BUY 25M) (POST SB 11) (SUPSDS 25G) (POST SB 11) (SUPSDS 25K) (PRE SB 17) (SEE FIG. 14 FOR BREAKDOWN)	H	1
- 25Q	617-4422-016		.TAPE TRANSPORT ASSEMBLY (OPT TO 25R) (BUY 25M) (POST SB 13) (SUPSDS 25B) (POST SB 13) (SUPSDS 25H) (PRE SB 17) (SEE FIG. 14 FOR BREAKDOWN)	I	1
- 25R	617-4422-017		.TAPE TRANSPORT ASSEMBLY (OPT TO 25Q) (BUY 25M) (POST SB 13) (SUPSDS 25C) (POST SB 13) (SUPSDS 25I) (PRE SB 17) (SEE FIG. 14 FOR BREAKDOWN)	I	1
- 25S	617-4422-023		.TAPE TRANSPORT ASSEMBLY (REPLS 25M) (POST SB 17) (SEE FIG. 14 FOR BREAKDOWN)	J	1
- 25T	617-4422-024		.TAPE TRANSPORT ASSEMBLY (REPLS 25N,25P) (OPT TO 25U) (POST SB 17) (SEE FIG. 14 FOR BREAKDOWN)	K	1
- 25U	617-4422-025		.TAPE TRANSPORT ASSEMBLY (OPT TO 25T) (POST SB 17) (SEE FIG. 14 FOR BREAKDOWN)	K	1
- 25V	617-4422-026		.TAPE TRANSPORT ASSEMBLY (REPLS 25Q,25R) (OPT TO 25W) (POST SB 17) (SEE FIG. 14 FOR BREAKDOWN)	L	1
- 25W	617-4422-027		.TAPE TRANSPORT ASSEMBLY (OPT TO 25V) (POST SB 17) (SEE FIG. 14 FOR BREAKDOWN)	L	1
			--- ATTACHING PARTS ---		
30	MS24693C3		.SCREW, FLAT HEAD	A	3
30A	MS24693C8		.SCREW, FLAT HEAD	BCDE FGHI JKL	2
30B	MS24693C5		.SCREW, FLAT HEAD	A	4
30C	MS24693C5		.SCREW, FLAT HEAD	BCDE FGHI JKL	1
			-----*		
35	636-0217-001		.SHIELD, HARNESS (OPT TO 35A, 35B)		1
- 35A	636-0217-002		.SHIELD, HARNESS (OPT TO 35, 35B)		1

Honeywell

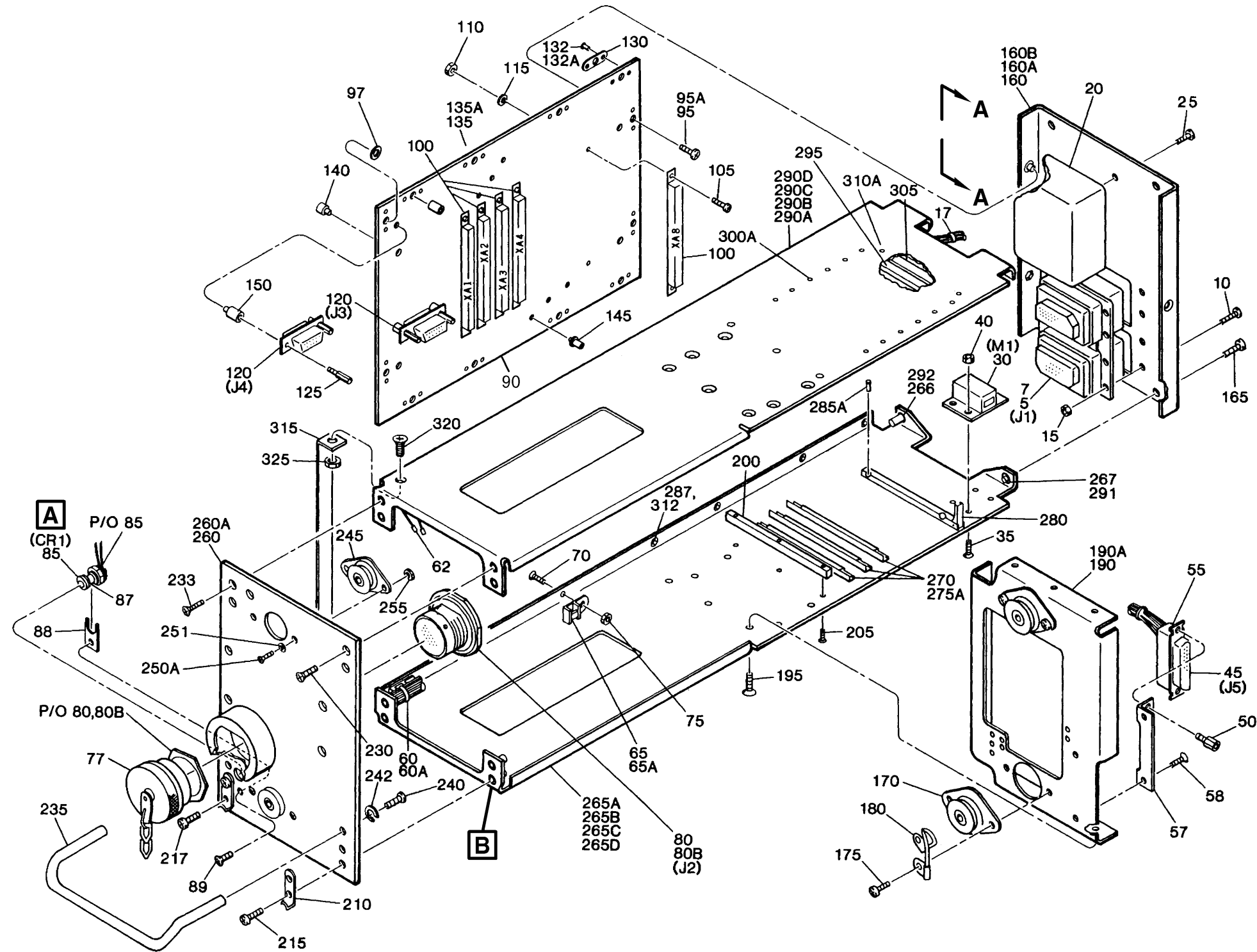
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
8			1234567		
- 35B	605-0145-002		.SHIELD, HARNESS (OPT TO (OPT TO 35, 35A) - - - ATTACHING PARTS - - -		1
40	MS51957-43		.SCREW, PAN HEAD -----*-----		2
45	617-3698-001		.MOTOR DRIVE ASSEMBLY (SEE FIG. 11 FOR BREAKDOWN) (PRE SB 2)	A	1
45A	617-4429-001		.MOTOR DRIVE ASSEMBLY (SEE FIG. 13 FOR BREAKDOWN) (POST SB 2)	BCDE FGHI	1
- 45B	617-3698-002		.MOTOR DRIVE ASSEMBLY (SEE FIG. 11 FOR BREAKDOWN) (POST SB 1)	A	1
- 45C	617-4429-002		.MOTOR DRIVE ASSEMBLY (SEE FIG. 13 FOR BREAKDOWN) - - - ATTACHING PARTS - - -	JKL	1
50	431-0433-001		.SCREW, SOCKET HEAD		2
- 50A	431-0433-002		.DELETED -----*-----		0
55	634-1494-001		.PLATE, MOTOR MOUNT	A	1
- 55A	634-1494-002		.PLATE, MOTOR MOUNT	BCDE FGHI JKL	1
60	600-3964-001		.FEEDTHRU, DRIVE BELT		1
65	NOTUSED		.REDESIGNATED 58		0
70	628-0256-001		.SPACER (USED ONLY WITH 5B) (BUY 5C)	BCDE FGHI	3

Honeywell

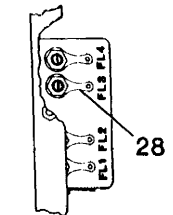
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101

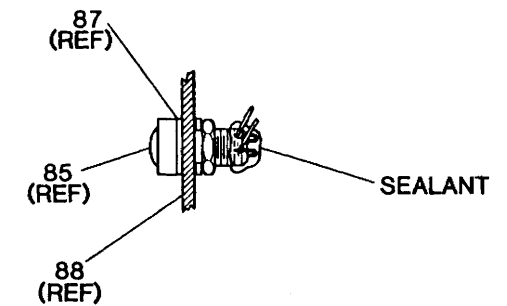


CHASSIS ASSEMBLY

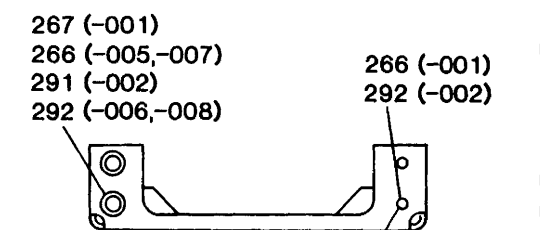
FIGURE 9



A-A



A



267 (-005,-007)
291 (-006,-008)

B

ID-177824 E617-3716-1-F E617-3716-2-F

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
9			1234567		
- 1	617-3716-001		CHASSIS ASSEMBLY (SEE FIG. 1-190 FOR NHA)	A	RF
- 1A	617-3716-002		CHASSIS ASSEMBLY (SEE FIG. 1-190A FOR NHA)	B	RF
- 1B	617-3716-006		CHASSIS ASSEMBLY (SEE FIG. 1-190B FOR NHA)	C	RF
- 1C	617-3716-005		CHASSIS ASSEMBLY (SEE FIG. 1-190C FOR NHA)	D	RF
5	DPX2MA57P00P34-		.CONNECTOR, 57-PIN (V71468) (OPT TO 5A AND 16)		1
- 5A	440-0005-028		.(DPX2MA57P00P34-0001F0 FOR V71468) OPTIONAL - - - ATTACHING PARTS - - -		1
10	MS51957-15		.SCREW, PAN HEAD		8
15	MS21044C04		.NUT, LOCK -----*-----		8
16	030-9081-000		.CONTACT, PIN (V71468) (USED WITH 5A)		57
- 16A	119-2020-080		.CONTACT, PIN (V55104) OPTIONAL		57
17	MS17821-4-9		.TIE, CABLE		1
- 17A	MS3367-4-9		.CABLE TIE OPTIONAL		1
20	716-0180-002		.FILTER ASSEMBLY (SEE FIG. 17 FOR BREAKDOWN)		1
- 20A	716-0180-001		.FILTER ASSEMBLY (SEE FIG. 17 FOR BREAKDOWN) OPTIONAL - - - ATTACHING PARTS - - -		1
25	MS51957-15		.SCREW, PAN HEAD -----*-----		2
28	29		..LUG, SOLDER (V83330) (REPLD BY 28A)		6
- 28A	1410-14		..TERMINAL (V83330)(REPLS 28)		6
30	95C8H		.DELETED - - - ATTACHING PARTS - - -		0
35	MS24693C3		.DELETED		0
40	MS21044C04		.DELETED -----*-----		0
45	DBM25S		.CONNECTOR, 57-PLUG (V71468)		1
- 45A	117DBM25S		.CONNECTOR, 57-PLUG (V02660) OPTIONAL		1
- 45B	MD308G25S1		.CONNECTOR, 57-PLUG (V14283) OPTIONAL - - - ATTACHING PARTS - - -		1
50	D110551		.JACKPOST, PAIR (V71468) -----*-----		1
55	DB50906-1		.SHELL, POTTING (V71468)		1
57	602-2461-001		.BRACKET, CONNECTOR - - - ATTACHING PARTS - - -		1
58	MS24693C2		.SCREW, FLAT HEAD -----*-----		2

31-30-32

Page 1093
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
9			1234567		
	60	MS3367-4-9	.STRAP	AB	4
	60A	MS3367-4-9	.STRAP	CD	6
	65	TC814	.CLAMP, TYWRAP (V59730)	AB	4
	65A	TC814	.CLAMP, TYWRAP (V59730)	CD	6
			- - - ATTACHING PARTS - - -		
	70	MS24693C3	.SCREW, FLAT HEAD		4
	75	MS21044C04	.NUT, LOCK		4
			-----*-----		
	77	MS3181-22N	.CAP, CONNECTOR		1
	80	KPT07A22-55S	.CONNECTOR (V71468)(OPT TO 80A)		1
			(OPT TO 80B, CODE C)		
-	80A	PT07A22-55S	.CONNECTOR (V77820)(OPT TO 80)		1
	80B	KPSE07A22-55S	.CONNECTOR (V71468)	C	1
			(OPT TO 80, CODE C)		
	85	5082-4587	.INDICATOR, LED (V50434)		1
			(OPT TO 85A)		
-	85A	HLMP0454	.LAMP ASSEMBLY (V50434)		1
			(OPT TO 85)		
	87	624-0769-001	.SPACER		1
	88	648-1177-001	.RETAINER		1
			- - - ATTACHING PARTS - - -		
	89	MS24693S4	.SCREW, FLAT HEAD		1
			-----*-----		
	90	722-2503-001	.MOTHERBOARD PWB	AD	1
-	90A	722-2503-002	.MOTHERBOARD, PWB	BC	1
			- - - ATTACHING PARTS - - -		
	95	MS51957-26	.SCREW, PAN HEAD	AD	12
	95A	MS24693C25	.SCREW, FLAT HEAD	BC	12
	97	2162	.WASHER, FIBER (V83330)	AD	1
-	97A	5602-12-62	.WASHER, FIBER		1
			(V86928) OPTIONAL		
			-----*-----		
	100	G06D64P3BBBL	..CONNECTOR, 64-PIN (V71468)		5
			(OPT TO 100A)		
-	100A	440-0507-002	..(20-8457-064-002-025 FOR		5
			V91662) OPTIONAL		
			- - - ATTACHING PARTS - - -		
	105	MS51957-4	..SCREW, PAN HEAD		10
	110	MS20341-2S	..NUT, PLAIN		10
	115	NAS620C2	..WASHER, FLAT (REPD BY 115A)		10
-	115A	06/16/5610	..WASHER, NYLON (V86928)		10
			(REPLS 115)(SUPSD BY 115B)		
-	115B	06/15/5610	..WASHER, NYLON (V86928)		10
			(SUPSDS 115A)		
	117	MS35338-134	..DELETED		0
			-----*-----		
	120	DA15SU	..CONNECTOR, 15-PIN (V71468)		2
			(REPLD BY 120A)		
-	120A	DACK15SBF	..CONNECTOR, 15-PIN (V71468)		2
			(REPLS 120)		
			- - - ATTACHING PARTS - - -		
	125	D110551	..JACKPOST, PAIR (V71468)		2
			-----*-----		

31-30-32

Page 1094
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
9			1234567		
130	MF1001-06		..NUT, ANCHOR (V15653) - - - ATTACHING PARTS - - -	BC	12
132	MS20470A3-6		..RIVET (OPT TO 132A)	BC	24
132A	MS20470A3-5		..RIVET (OPT TO 132) -----*-----	BC	24
135	623-2275-003		..PRINTED WIRING BOARD	AD	1
135A	623-2275-005		..PRINTED WIRING BOARD	BC	1
140	9705CA0440-0		...STANDOFF (V06540)		4
145	1392C1		...TERMINAL (V88245)		7
- 145A	1392C4-5		...TERMINAL (V88245) OPTIONAL		7
- 145B	3531C4-5		...TERMINAL (V16258) OPTIONAL		7
150	9535CA0440-0		...STANDOFF (V06540)		4
- 150A	9535CA0440-17		...STANDOFF (V06540) OPTIONAL		4
- 155	634-1455-001		..CHASSIS	A	1
- 155A	634-1455-002		..CHASSIS	B	1
- 155B	634-1455-006		..CHASSIS	C	1
- 155C	634-1455-005		..CHASSIS	D	1
160	634-1443-001		..PLATE, CONNECTOR	A	1
160A	634-1443-002		..PLATE, CONNECTOR	B	1
160B	634-1443-004		..PLATE, CONNECTOR	C	1
- 160C	634-1443-003		..PLATE, CONNECTOR - - - ATTACHING PARTS - - -	D	1
165	MS51957-43		..SCREW, PAN HEAD -----*-----		4
170	N6550-4		..SHOCKMOUNT (V81860) - - - ATTACHING PARTS - - -		2
175	MS51957-28		..SCREW, PAN HEAD -----*-----		4
180	704-0487-001		..STRAP, GROUND		1
- 185	NOTUSED		NOT USED		1
190	602-2417-001		..BRACKET	AD	1
190A	602-2417-002		..BRACKET - - - ATTACHING PARTS - - -	BC	1
195	MS24693C48		..SCREW, FLAT HEAD -----*-----		6
200	652-0202-001		..GUIDE, POWER SUPPLY - - - ATTACHING PARTS - - -		2
205	MS51959-3		..SCREW, FLAT HEAD -----*-----		6
210	652-2002-001		..HOOK - - - ATTACHING PARTS - - -		2
215	MS51957-45		..SCREW, PAN HEAD		2
217	MS51957-43		..SCREW, PAN HEAD (SUPSD BY 217A)		2
- 217A	MS51957-45		..SCREW, PAN HEAD (SUPSDS 217) -----*-----		2
- 220	NOTUSED		..NOT USED		0
- 225	617-3746-001		..PANEL, FRONT		1

31-30-32

Page 1095
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		

9

			--- ATTACHING PARTS ---		
	230	MS24693C48	..SCREW, FLAT HEAD		2
	233	MS24693C47	..SCREW, FLAT HEAD		2
			(SUPSD BY 233A)		
-	233A	MS24693C48	..SCREW, FLAT HEAD		2
			(SUPSDS 233)		
			-----*-----		
	235	G2525-4	...HANDLE (V04729)		1
-	235A	401-8065-001	...(8296-832A24MIDH2-1-2 FOR		1
			(V55566)) OPTIONAL		
			--- ATTACHING PARTS ---		
	240	MS51957-45	...SCREW, PAN HEAD		2
	242	MS35338-137	...WASHER, LOCK		2
			-----*-----		
	245	N6550-4	...SHOCKMOUNT (V81860)		2
			--- ATTACHING PARTS ---		
-	250	MS24693C25	...SCREW, FLAT HEAD		4
			(REPLD BY 250A)		
	250A	NAS1189E06P5L	...SCREW, FLAT HEAD		4
			(REPLS 250)		
	255	MS35649-264	...NUT, HEX		4
			-----*-----		
	260	621-1195-001	...PANEL		1
	260A	621-1453-001	...PANEL		1
			(USED ON FIG. 1, ITEMS 2, 2A)		
-	265	634-1442-001	..PLATE, CHASSIS	A	1
			(REPLD BY 265C)		
	265A	634-1442-005	..PLATE, CHASSIS	B	1
	265B	634-1442-007	..PLATE, CHASSIS	C	1
	265C	634-1442-003	..PLATE, CHASSIS	A	1
			(REPLS 265)		
-	265D	634-1442-009	..PLATE, CHASSIS	D	1
	266	1032NBSEXRR	...NUT, BLIND PRESS (V86577)		3
-	266A	1032NBSEXRR091	...NUT, BLIND PRESS		3
			(V86577) OPTIONAL		
	267	FE832	...NUT, PRESS (V46384)		3
-	267A	58LNK060	...NUT, PRESS		3
			(V03694) OPTIONAL		
	270	35-1CR2-7	...GUIDE, CARD (V18915)		6
			--- ATTACHING PARTS ---		
-	275	MS20426AD2-4	...RIVET (SUPSD BY 275A)		12
	275A	MS20426AD2-3	...RIVET (SUPSDS 275)		12
			-----*-----		
	280	24SX2-7-3L	...GUIDE, CARD (V18915)		1
			--- ATTACHING PARTS ---		
-	285	MS20426AD2-3	...RIVET (SUPSD BY 285A)		2
	285A	MS20426AD2-4	...RIVET (SUPSDS 285)		2
			-----*-----		
	287	S0S632-8	...NUT, PRESS (V46384)		4
-	287A	46NSEXR8	...NUT, PRESS		4
			(V86577) OPTIONAL		

31-30-32

Page 1096
22 Dec 2006

Honeywell

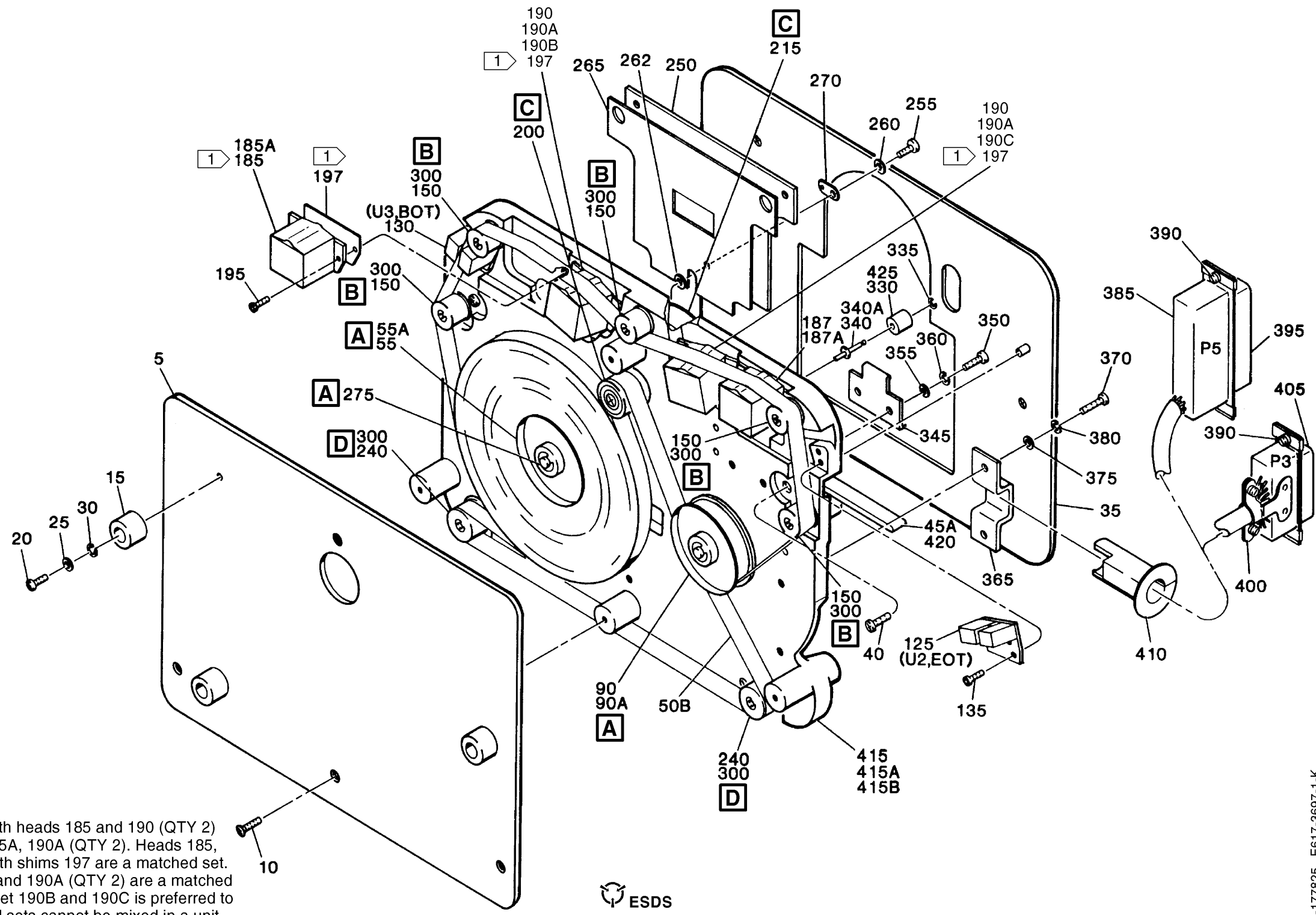
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
9			1234567		
- 290	634-1442-002		..PLATE, CHASSIS (REPLD BY 290C)	A	1
290A	634-1442-006		..PLATE, CHASSIS	B	1
290B	634-1442-008		..PLATE, CHASSIS	C	1
290C	634-1442-004		..PLATE, CHASSIS (REPLS 290)	A	1
290D	634-1442-010		..PLATE, CHASSIS	D	1
291	1032NBSEXRR		...NUT, BLIND PRESS (V86577)		3
- 291A	1032NBSEXRR091		...NUT, BLIND PRESS (V86577) OPTIONAL		3
292	FE832		...NUT, PRESS (V46384)		3
- 292A	58LNK060		...NUT, PRESS (V03694) OPTIONAL		3
295	35-1CR2-7		...GUIDE, CARD (V18915) - - - ATTACHING PARTS - - -		6
- 300	MS20426AD2-4		...RIVET (SUPSD BY 300A)		12
300A	MS20426AD2-3		...RIVET (SUPSDS 300) -----*-----		12
305	24SX2-7-3R		...GUIDE, CARD (V18915) - - - ATTACHING PARTS - - -		1
- 310	MS20426AD2-3		...RIVET (SUPSD BY 310A)		2
310A	MS20426AD2-4		...RIVET (SUPSDS 310) -----*-----		2
312	S0S632-8		...NUT, PRESS (V46384)		4
- 312A	46NSEXR8		...NUT, PRESS (V86577) OPTIONAL		4
315	602-2913-001		..BRACKET, CABLE - - - ATTACHING PARTS - - -	CD	1
320	MS24693C25		..SCREW, FLAT HEAD	CD	2
325	22NKM62		..NUT, NYLON CAP (V72962) -----*-----	CD	2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

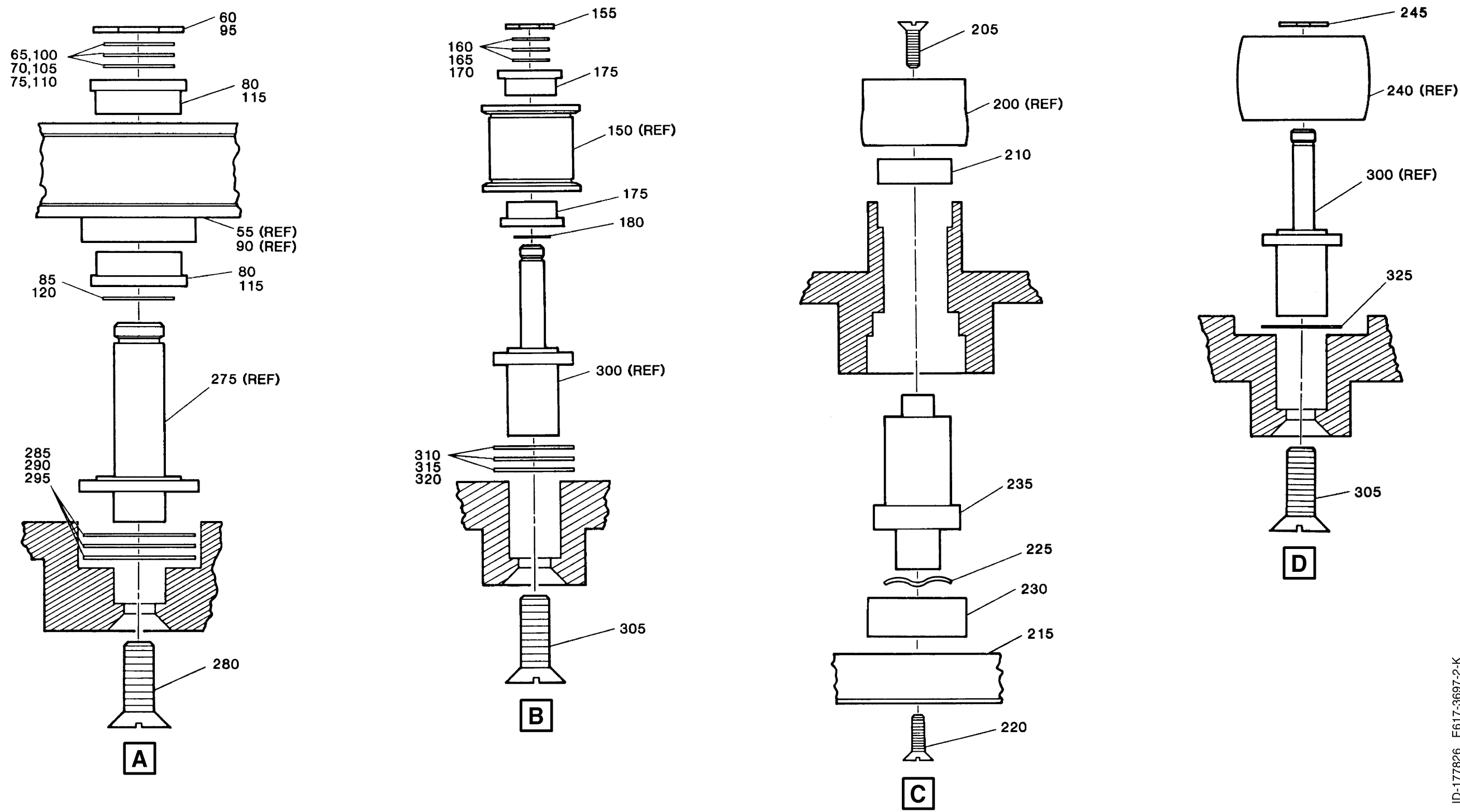
THIS PAGE INTENTIONALLY LEFT BLANK



NOTES:

1 Shim 197 is used with heads 185 and 190 (QTY 2) only and not with 185A, 190A (QTY 2). Heads 185, 187, 190 (QTY 2) with shims 197 are a matched set. Heads 185A, 187A and 190A (QTY 2) are a matched set. Matched head set 190B and 190C is preferred to 190A (QTY 2). Head sets cannot be mixed in a unit.

TAPE TRANSPORT ASSEMBLY
FIGURE 10 (SHEET 1 OF 2)



TAPE TRANSPORT ASSEMBLY
FIGURE 10 (SHEET 2 OF 2)

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
10					
- 1	617-3697-001		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25 FOR NHA)	A	RF
- 1A	617-3697-002		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25A FOR NHA)	B	RF
5	634-1439-001		.PLATE, TOP - - - ATTACHING PARTS - - -		1
10	MS24693C3		.SCREW, FLAT HEAD -----*-----		4
15	9115		.BUMPER, RUBBER (V77969) - - - ATTACHING PARTS - - -		3
20	MS51957-13		.SCREW, PAN HEAD		3
25	MS15795-803		.WASHER, FLAT		3
30	MS35338-135		.WASHER, LOCK -----*-----		3
35	634-1438-001		.PLATE, BOTTOM - - - ATTACHING PARTS - - -	AB	1
40	MS24693C5		.SCREW, FLAT HEAD -----*-----		4
- 45	600-3991-001		.BELT, FLAT (SUPSD BY 45A)		1
45A	600-3991-002		.BELT, FLAT (SUPSDS 45)	AB	1
- 50	406-1508-001		.BELT, PERIPHERAL (SUPSD BY 50A)		1
- 50A	406-1508-002		.BELT, PERIPHERAL (SUPSDS 50) (SUPSD BY 50B)		1
50B	406-1508-003		.BELT, PERIPHERAL (SUPSDS 50A)		1
55	617-3731-001		.REEL, LOADED (SUPSD BY 55A)		1
55A	617-3731-003		.REEL, LOADED (SUPSDS 55)		1
60	5133-18		.RETAINER, E-RING (V79136)		1
65	434-0201-001		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.001 THK		AR
70	434-0201-002		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.003 THK		AR
75	434-0201-003		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.005 THK		AR
80	400-0056-040		.BEARING, BALL		2
85	434-0201-001		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.001 THK		1
90	600-3969-001		.TAPE REEL (SUPSD BY 90A)		1
90A	600-3969-002		.TAPE REEL (SUPSDS 90) - - - ATTACHING PARTS - - -		1
95	5133-18		.RETAINER, E-RING (V79136)		1
100	434-0201-001		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.001 THK		AR
105	434-0201-002		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.003 THK		AR
110	434-0201-003		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.005 THK -----*-----		AR
115	400-0056-040		.BEARING, BALL		2
120	434-0201-001		.SHIM, CRES, 0.190 ID, 0.265 OD, 0.001 THK		1
125	744-0078-002		.SENSOR, EOT		1

31-30-32

Page 1098.5
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
10			1234567		
130	744-0078-001		.SENSOR, BOT - - - ATTACHING PARTS - - -		1
135	MS16995-2		.SCREW, SOCKET HEAD		4
140	NAS620C2		.DELETED		0
145	MS35338-134		.DELETED -----*-----		0
150	600-4108-001		.ROLLER, TAPE GUIDE - - - ATTACHING PARTS - - -		5
155	5133-9H		.RETAINER, E-RING (V79136)		5
160	434-0201-007		.SHIM, CRES, 0.095 ID, 0.142 OD, 0.001 THK		AR
165	434-0201-008		.SHIM, CRES, 0.095 ID, 0.142 OD, 0.003 THK		AR
170	434-0201-009		.SHIM, CRES, 0.095 ID, 0.142 OD, 0.005 THK -----*-----		AR
175	400-0056-041		.BEARING, BALL		10
180	434-0201-007		.SHIM, CRES, 0.095 ID, 0.142 OD, 0.001 THK		5
185	754-0011-002		.HEAD, MAGNETIC, READ/WRITE (SEE NOTE 1)		1
185A	317-1044-002		.HEAD, MAGNETIC, READ/WRITE (SEE NOTE 1)		1
187	754-0011-001		.HEAD, MAGNETIC, READ/WRITE (SEE NOTE 1)		1
187A	317-1044-002		.HEAD, MAGNETIC, READ/WRITE (SEE NOTE 1)		1
190	317-1039-001		.HEAD, MAGNETIC, ERASE (SEE NOTE 1)		2
190A	317-1045-001		.HEAD, MAGNETIC, ERASE (SEE NOTE 1) (OPTIONAL TO MATCH SET 190B AND 190C)		2
190B	317-1045-002		.HEAD, MAGNETIC, ERASE (MATCH WITH 190C) (SEE NOTE 1)		1
190C	317-1045-003		.HEAD, MAGNETIC, ERASE (MATCH WITH 190B) (SEE NOTE 1) - - - ATTACHING PARTS - - -		1
195	MS16995-2		.SCREW, SOCKET HEAD -----*-----		8
197	645-5107-003		.SHIM, HEAD (USED WITH 185,187, 190 ONLY)		3
200	603-1001-001		.PULLEY, TAPE - - - ATTACHING PARTS - - -		1
205	MS24693C3		.SCREW, FLAT HEAD -----*-----		1
210	400-0057-027		.BEARING, BALL		1
215	603-1002-001		.PULLEY, DRIVE - - - ATTACHING PARTS - - -		1
220	MS24693C3		.SCREW, FLAT HEAD		1
225	7X9-3W06108		.WASHER, WAVE (V14519) -----*-----		1
230	400-0057-028		.BEARING, BALL		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
10			1234567		
235	645-0352-001		.SHAFT		1
240	603-1003-001		.ROLLER, BELT		2
			--- ATTACHING PARTS ---		
245	5133-9H		.RETAINER, E-RING (V79136)		2
			-----*-----		
250	722-2522-001		.DISTRIBUTION BOARD (ESD) (SEE FIG. 12 FOR BREAKDOWN)		1
			--- ATTACHING PARTS ---		
255	MS51957-26		.SCREW, PAN HEAD		3
260	NAS620C6L		.WASHER, FLAT		3
262	2254PH141		.WASHER, PHENOLIC (V06540)		AR
			-----*-----		
265	636-0220-001		.SHIELD		1
270	TL56		.LUG, SOLDER (V71218)		1
275	646-0326-001		.PIN, TAPE REEL		2
			--- ATTACHING PARTS ---		
280	MS24693C3		.SCREW, FLAT HEAD		2
285	434-0201-004		.SHIM, CRES, 0.200 ID, 0.420 OD, 0.001 THK		AR
290	434-0201-005		.SHIM, CRES, 0.200 ID, 0.420 OD, 0.003 THK		AR
295	434-0201-006		.SHIM, CRES, 0.200 ID, 0.420 OD, 0.005 THK		AR
			-----*-----		
300	646-0325-001		.PIN, ROLLER		7
			--- ATTACHING PARTS ---		
305	MS24693C3		.SCREW, FLAT HEAD		7
310	434-0201-010		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.001 THK		AR
315	434-0201-011		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.003 THK		AR
320	434-0201-012		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.005 THK		AR
			-----*-----		
325	434-0201-012		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.005 THK		4
330	600-3971-001		.ROLLER, MOTOR DRIVE	AB	2
			--- ATTACHING PARTS ---		
335	5133-9H		.RETAINER, E-RING (V79136)		2
			-----*-----		
340	646-0323-001		.PIN, ROLLER	A	2
340A	646-0323-002		.PIN, ROLLER	B	2
345	607-1520-001		.COVER, HARNESS		1
			--- ATTACHING PARTS ---		
350	MS51957-12		.SCREW, PAN HEAD		2
355	MS15795-803		.WASHER, FLAT		2
360	MS35338-135		.WASHER, LOCK		2
			-----*-----		
365	602-2482-001		.BRACKET, HARNESS		1
			--- ATTACHING PARTS ---		
370	MS51957-13		.SCREW, PAN HEAD		2
375	MS15795-803		.WASHER, FLAT		2

31-30-32

Page 1098.7
22 Dec 2006

Honeywell

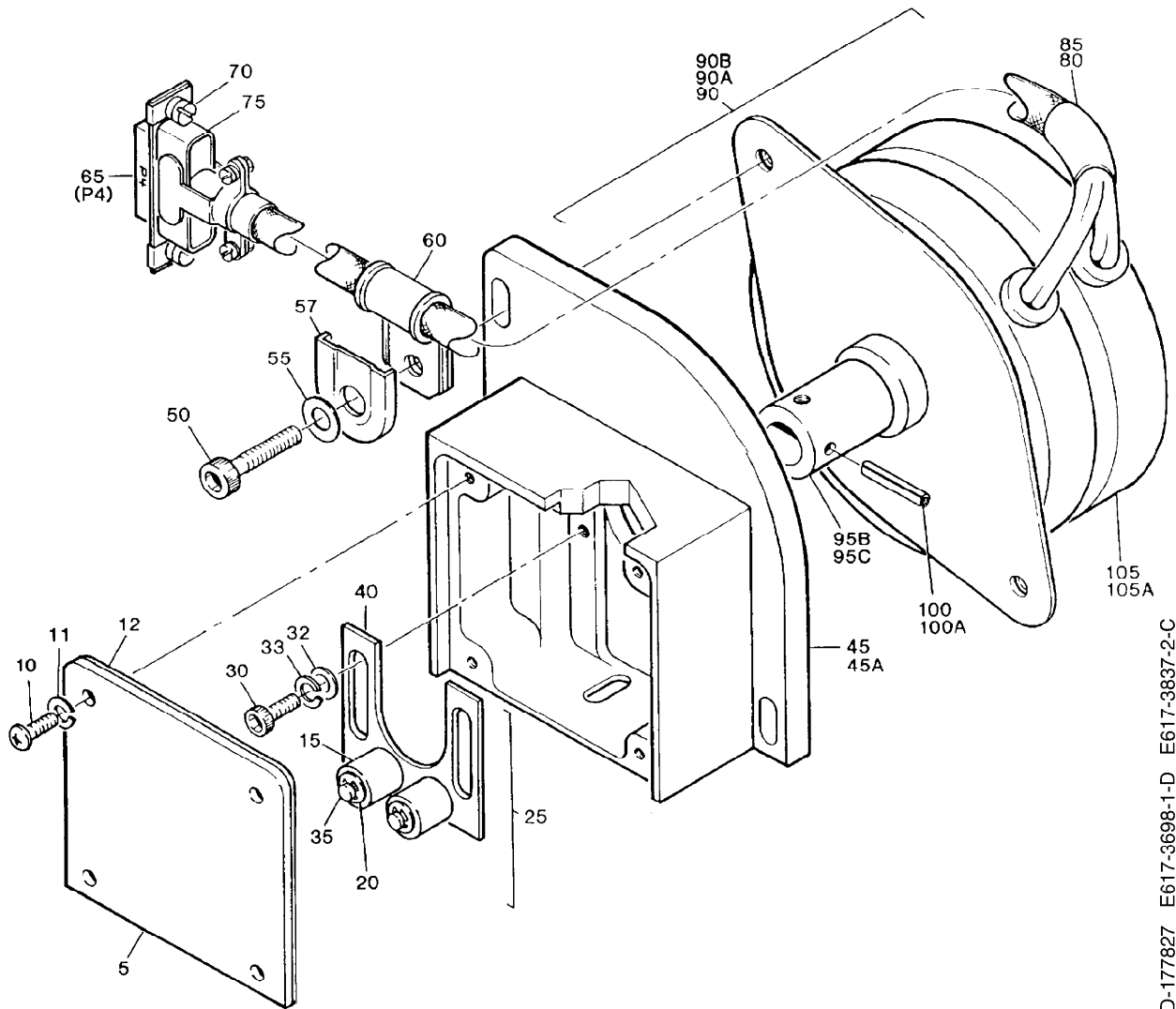
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
10			1234567		
380	MS35338-135		.WASHER, LOCK -----*-----		2
385	600-4103-001		.SHELL, NYLON		1
390	D20419-21		.SCREWLOCK (V71468)		4
395	DBM25P		.CONNECTOR (V71468)		1
400	DA20961		.CLAMP (V71468)		1
405	DAM15P		.CONNECTOR (V71468)		1
410	600-3970-001		.FEEDTHRU		1
415	600-3967-001		.TAPE DECK (OPT TO 415A)	A	1
415A	600-3967-002		.TAPE DECK (OPT TO 415)	A	1
415B	600-3967-003		.TAPE DECK	B	1
420	406-1510-005		.DRIVE BELT MOTOR (POST SB 1)		1
425	600-4674-001		.ROLLER, DRIVE BELT (POST SB 1)		2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK



ID-177827 E617-3698-1-D E617-3837-2-C

MOTOR DRIVE ASSEMBLY
FIGURE 11

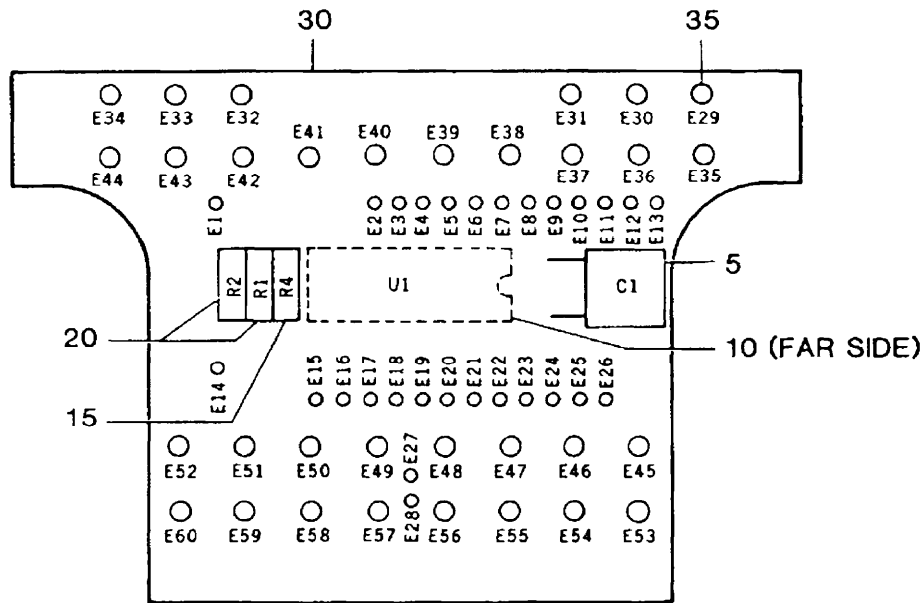
Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

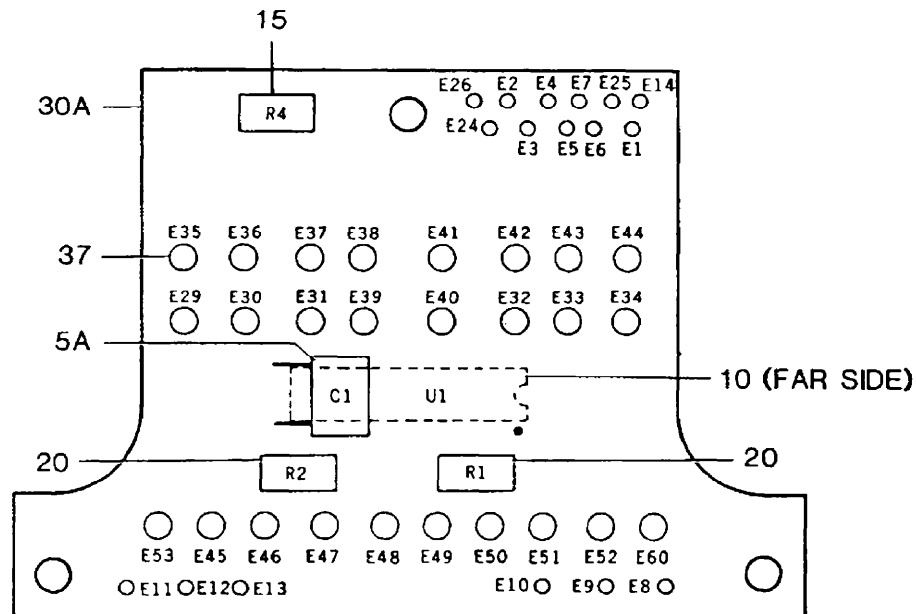
FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
11					
- 1	617-3698-001		MOTOR DRIVE ASSEMBLY (SEE FIG. 8-45 FOR NHA)	A	RF
- 1A	617-3698-002		MOTOR DRIVE ASSEMBLY (SEE FIG. 8-45B FOR NHA)	B	RF
5	617-3827-001		.COVER		1
			--- ATTACHING PARTS ---		
10	MS51954-13		.SCREW, PAN HEAD		4
11	MS35338-135		.WASHER, LOCK SPLIT		4
			-----*-----		
12	611-0157-001		DELETED		0
15	600-3971-001		.ROLLER (PRE SB1)	A	2
- 15A	600-4674-001		.ROLLER, DRIVE BELT (POST SB 1)	B	2
			--- ATTACHING PARTS ---		
20	5133-9H		.RETAINER, E-RING (V79136)		2
			-----*-----		
25	617-3829-001		.PLATE, TENSION		1
			--- ATTACHING PARTS ---		
30	2204-07000CAP		.SCREW, SOCKET HEAD (V71159)		2
32	MS15795-803		.WASHER, FLAT		2
33	MS35338-135		.WASHER, LOCK		2
			-----*-----		
35	646-0323-001		DELETED		0
40	634-1463-001		..PLATE, TENSION		1
45	654-0144-001		.MOUNT, MOTOR		1
45A	654-0144-002		.MOUNT, MOTOR (OPT TO 45)		1
			--- ATTACHING PARTS ---		
50	MS16995-18		.SCREW, SOCKET HEAD		2
55	NAS620C6		.WASHER, FLAT		2
57	D6-140		.WASHER, D (V85480)		1
			-----*-----		
60	01/04/2006		.CLAMP, CABLE (V85480)		1
65	DAM15P		.CONNECTOR (V71468)		1
70	D20419-21		.SCREWLOCK (V71468)		2
75	DA20961		.CLAMP (V71468)		1
80	GSC219		.FERRULE, OUTER (V59730)		1
85	GSC156		.FERRULE, INNER (V59730)		1
90	617-3837-001		.STEPPER MOTOR ASSEMBLY (OPT TO 90A)	A	1
90A	617-3837-002		.STEPPER MOTOR ASSEMBLY	A	1
90B	617-3837-003		.STEPPER MOTOR ASSEMBLY (POST SB 1)	B	1
- 95	600-3975-001		.SUPERSEDED		0
- 95A	600-3975-002		.SUPERSEDED		0
95B	600-3975-003		..PULLEY, DRIVE	A	1
95C	600-3975-004		..PULLEY, DRIVE (POST SB 1)	B	1
100	109-0437MDP		..PIN, SPRING (V00287) (REPLD BY 100A)		1
100A	062-0437MDP		..PIN, SPRING (V00287) (REPLS 100)(POST SB 1)		1
105	360-0082-001		..MOTOR, STEPPER (PRE SB 1)	A	1
105A	360-0082-002		..MOTOR, STEPPER (POST SB 1)	B	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101



722-2522-001



722-2522-002

DISTRIBUTION BOARD
FIGURE 12

ID-177828 E722-2522-1-F E722-2522-2-F

31-30-32

Page 1098.12
22 Dec 2006

Honeywell

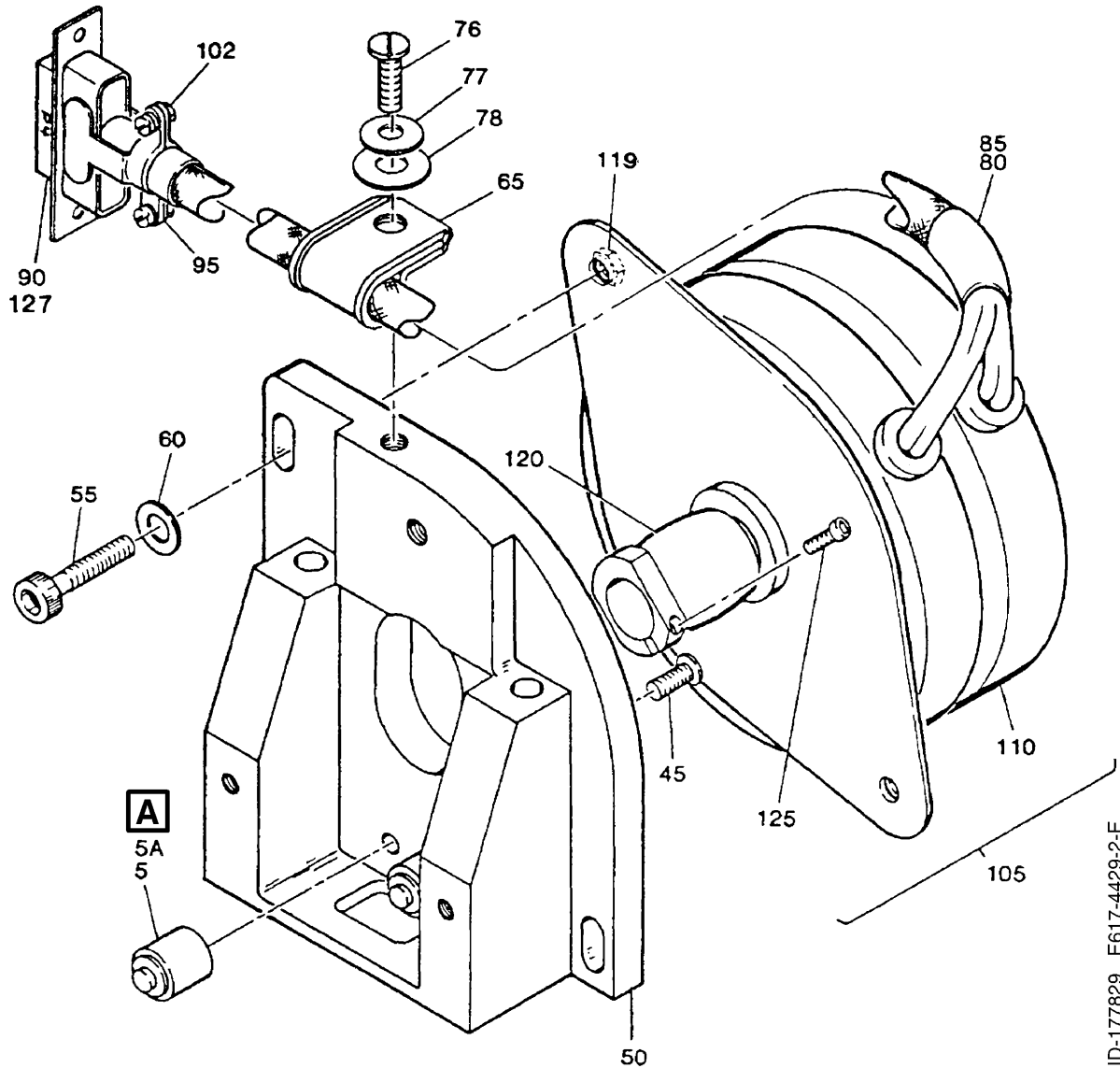
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
12			1234567		
- 1	722-2522-001		DISTRIBUTION BOARD (ESD) (SEE FIG. 10-250 FOR NHA)	A	RF
- 1A	722-2522-002		(SEE DISTRIBUTION BOARD (ESD) FIG. 14-270 FOR NHA)	B	RF
5	M39014-02-1350		.CAPACITOR, 0.1 UF, 10%, 100V	A	1
5A	M39014-01-1593		.CAPACITOR, 0.1 UF, 10%, 50V	B	1
10	SNJ54LS145J00		.IC, DECODER (ESD)(V01295)		1
- 10A	SNJ54LS145J		.DECODER (ESDS) (V01295) OPTIONAL		1
- 10B	SN54LS145F883B		.DECODER (ESDS) (V18324) OPTIONAL		1
15	RNC55H1001FS		.RESISTOR, 1K, 1%, 1/10W		1
20	RWR81S82R5FR		.RESISTOR, 82.5 OHMS, 1%, 1W		2
- 25	NOTUSED		NOT USED		0
30	623-2285-003		.PRINTED WIRING BOARD	A	1
30A	623-2285-005		.PRINTED WIRING BOARD	B	1
35	2002B1		..TERMINAL (V88245)	A	32
- 35A	2002B4-5		..TERMINAL (V88245) OPTIONAL		32
37	2000B4-5		..TERMINAL (V88245)	B	26
- 37A	5075B4-5		..TERMINAL (V16258) OPTIONAL		26

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

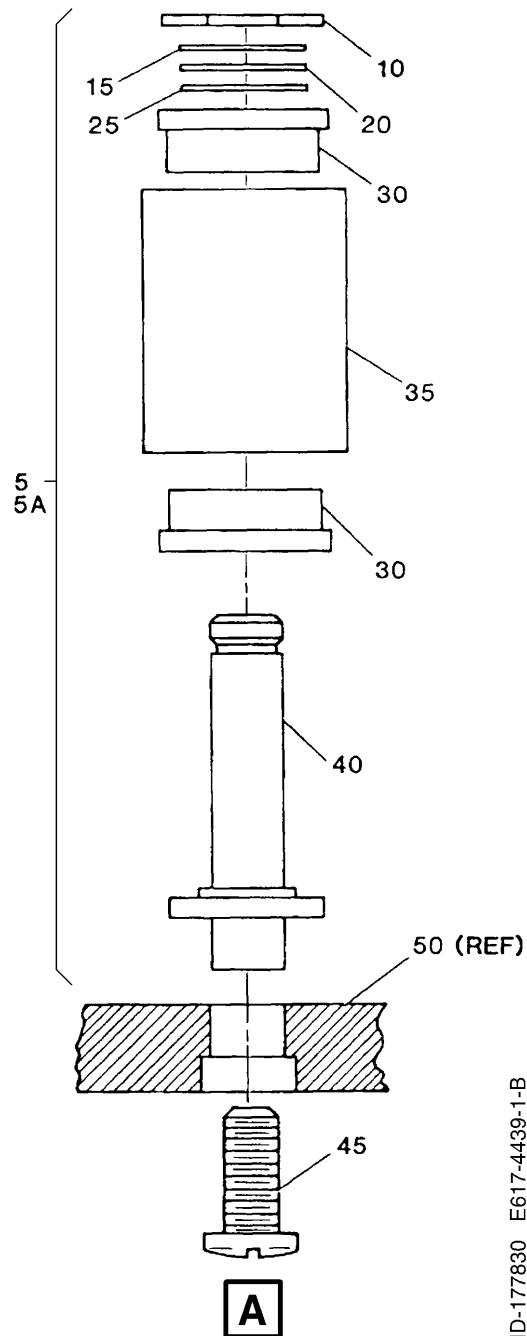


ID-177829 E617-4429-2-E

MOTOR DRIVE ASSEMBLY
FIGURE 13 (SHEET 1 OF 2)

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101



MOTOR DRIVE ASSEMBLY
FIGURE 13 (SHEET 2 OF 2)

31-30-32

Page 1098.16
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
13					
- 1	617-4429-001		MOTOR DRIVE ASSEMBLY (SEE FIG. 8-45A FOR NHA)	A	RF
- 1A	617-4429-002		MOTOR DRIVE ASSEMBLY (SEE FIG. 8-45C FOR NHA)	B	RF
5	617-4439-001		.ROLLER GUIDE ASSEMBLY (OPT TO 5A)(REPLD BY 5B)	A	2
5A	617-4439-003		.ROLLER GUIDE ASSEMBLY (REPLD BY 5C)	A	2
- 5B	617-4439-004		.ROLLER GUIDE ASSEMBLY (REPLS 5)(OPT TO 5C)	B	2
- 5C	617-4439-006		.ROLLER GUIDE ASSEMBLY (REPLS 5A)	B	2
10	5133-9H		..RETAINER, E-RING (V79136)		1
- 10A	MS16633-4009		..RETAINER, E-RING OPTIONAL		1
15	434-0201-007		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.001 THK		AR
20	434-0201-008		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.003 THK		AR
25	434-0201-009		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.005 THK		AR
30	400-0056-042		..BEARING, BALL (REPLD BY 31)	A	2
- 30A	400-0056-041		..BEARING, BALL OPTIONAL	A	2
- 30B	400-0056-052		..BEARING, BALL OPTIONAL	A	2
- 30C	400-0056-053		..BEARING, BALL OPTIONAL	A	2
- 30D	400-0172-001		..BEARING, BALL OPTIONAL	A	2
- 31	400-0056-061		..BEARING, BALL	B	2
- 31A	400-0056-060		..BEARING, BALL (REPLS 30)	B	2
- 31B	400-0056-063		..BEARING, BALL OPTIONAL	B	2
- 31C	400-0056-064		..BEARING, BALL OPTIONAL	B	2
- 31D	400-0056-065		..BEARING, BALL OPTIONAL	B	2
35	600-4681-001		..ROLLER		1
40	646-0379-001		..PIN (USED ON 5, 5B)		1
- 40A	646-0379-002		..PIN (USED ON 5A, 5C) - - - ATTACHING PARTS - - -		1
45	431-0343-014		.SCREW, BIND HEAD		2
- 45A	431-0497-014		.SCREW, BIND HEAD OPTIONAL -----*-----		2
50	654-0222-001		.MOTOR MOUNT (REPLD BY 50A)		1
- 50A	654-0233-001		.MOTOR MOUNT (REPLS 50) - - - ATTACHING PARTS - - -		1
55	MS16995-18		.SCREW, SOCKET HEAD		2

31-30-32

Page 1098.17
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
13			1234567		
60	NAS620C6		.WASHER, FLAT -----*-----		2
65	01/04/2006		.CLAMP (V85480)		1
- 65A	F6NY250-0		.CLAMP, CABLE (V65664) OPTIONAL		1
- 65B	3324BLACK		.CLAMP, CABLE (V28520) OPTIONAL		1
70	D6-140		.DELETED		0
75	MS51957-28		.DELETED		0
			- - - ATTACHING PARTS - - -		
76	MS51957-14		.SCREW, PAN HEAD		1
77	AN960C4L		.WASHER, FLAT		1
78	AN960C6		.WASHER, FLAT		1
- 78A	D6SE140GY		.WASHER, FLAT (V85480) OPTIONAL -----*-----		1
80	GSC219		.FERRULE, OUTER (V59730)		1
85	GSC156		.FERRULE, INNER (V59730)		1
90	DAM15P		.CONNECTOR, 15-PIN (V71468)		1
- 90A	DAM15P		.CONNECTOR (V13556) OPTIONAL		1
95	DA20961		.CLAMP (V71468)		1
100	D20419-21		.DELETED		0
102	MS51957-14		.SCREW, PAN HEAD		2
105	617-4261-003		.STEPPER MOTOR ASSEMBLY (WITHOUT CONNECTOR) (OPT TO 105A, 130, 130A)	A	1
- 105A	617-4261-005		.STEPPER MOTOR ASSEMBLY (WITH CONNECTOR 127)(OPT TO 105,130, 130A)(REPLD BY 105B)	A	1
- 105B	617-4261-007		.STEPPER MOTOR ASSEMBLY (WITH CONNECTOR 127) (REPLS 105A)	B	1
110	360-0082-001		..STEPPER MOTOR	A	1
- 110A	360-0082-003		..MOVED TO ITEM 117		0
- 112	SR1883PPK15-7		...BEARING (V83086)	A	2
- 112A	400-0057-033		...MOVED TO ITEM 118		0
115	LKS632-2		...MOVED TO ITEM 119		0
- 116	617-4261-009		..STEPPER MOTOR ASSEMBLY	B	1
- 117	360-0082-003		...STEPPER MOTOR	B	1
- 118	400-0057-033		...BEARING	B	2
119	LKS632-2		...NUT, PRESS (V46384)	B	2
120	600-4679-001		..PULLEY, DRIVE		1
125	MS16995-2		..SCREW, SOCKET HEAD		1
127	DAM15P		..CONNECTOR, 15-PIN (V71468)		1
- 127A	DAM15P		..CONNECTOR (V13556) OPTIONAL		1
- 130	617-4261-004		.STEPPER MOTOR ASSEMBLY (WITHOUT CONNECTOR) (OPT TO 105, 105A, 130A)	A	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
13			1234567		
- 130A	617-4261-006		..STEPPER MOTOR ASSEMBLY (WITH CONNECTOR 152)(OPT TO 105, 105A, 130)(REPLD BY 130B)	A	1
- 130B	617-4261-008		..STEPPER MOTOR ASSEMBLY (WITH CONNECTOR 152) (REPLS 130A) (OPT TO 105B)	B	1
- 135	360-0082-002		..STEPPER MOTOR	A	1
- 135A	360-0082-004		..MOVED TO ITEM 142		0
- 137	400-0057-032		...MOVED TO ITEM 143		0
- 140	LKS632-1		..MOVED TO ITEM 144		0
- 141	617-4261-010		..STEPPER MOTOR ASSEMBLY	B	1
- 142	360-0082-004		...STEPPER MOTOR	B	1
- 143	400-0057-032		...BEARING	B	2
- 144	LKS632-2		...NUT, PRESS (V46384)	B	2
- 145	600-4679-001		..PULLEY, DRIVE		1
- 150	MS16995-2		..SCREW, SOCKET HEAD		1
- 152	DAM15P		..CONNECTOR, 15-PIN (V71468)		1
- 152A	DAM15P		..CONNECTOR (V13556) OPTIONAL		1
- 155	617-4261-012		..STEPPER MOTOR ASSEMBLY (OPT TO 141)(WITHOUT CONNECTOR)	B	1
- 160	360-0082-006		...STEPPER MOTOR	B	1
- 165	400-0057-033		...BEARING	B	2
- 170	LKS632-1		...NUT, PRESS (V46384)	B	2

Honeywell

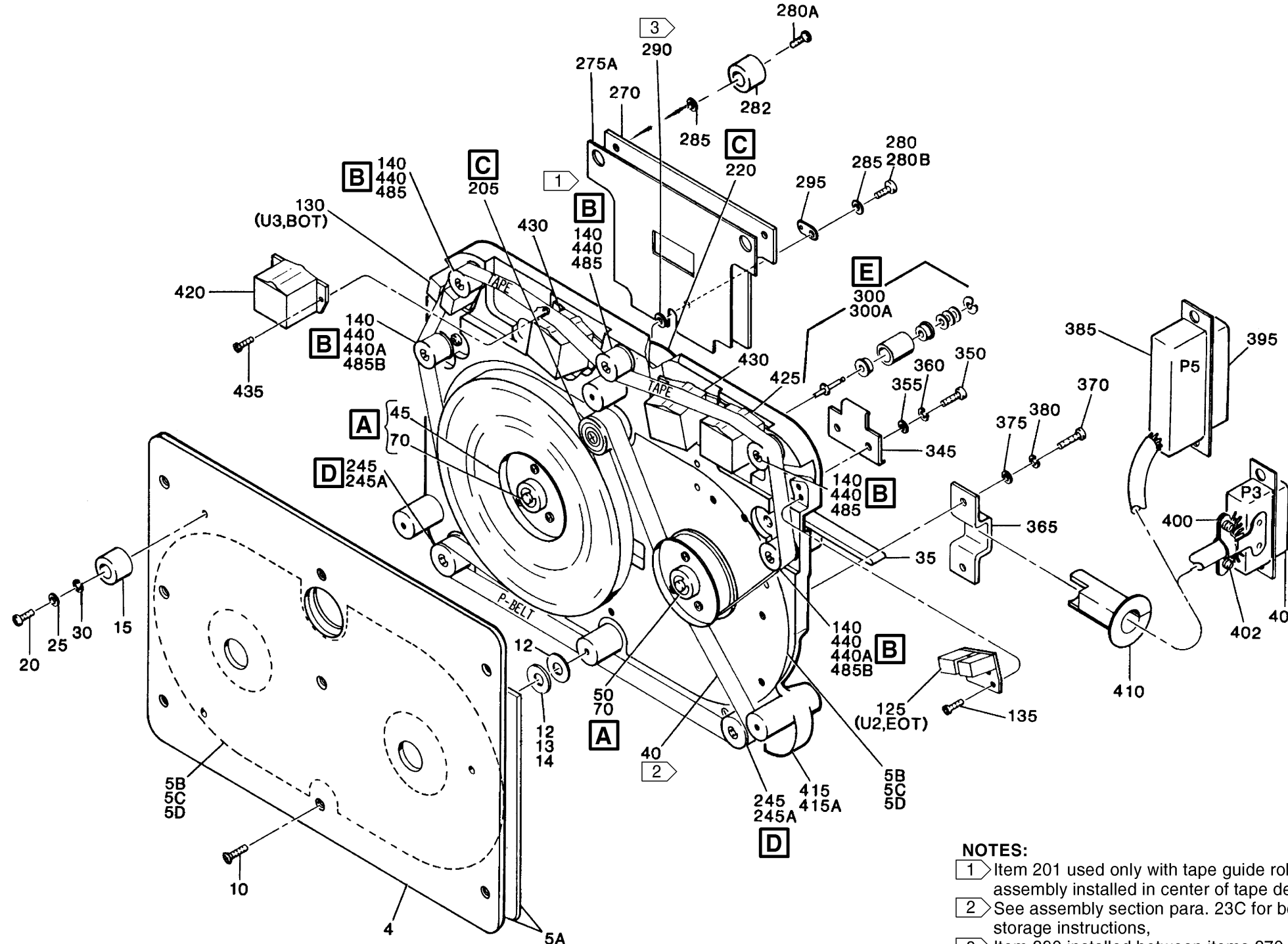
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

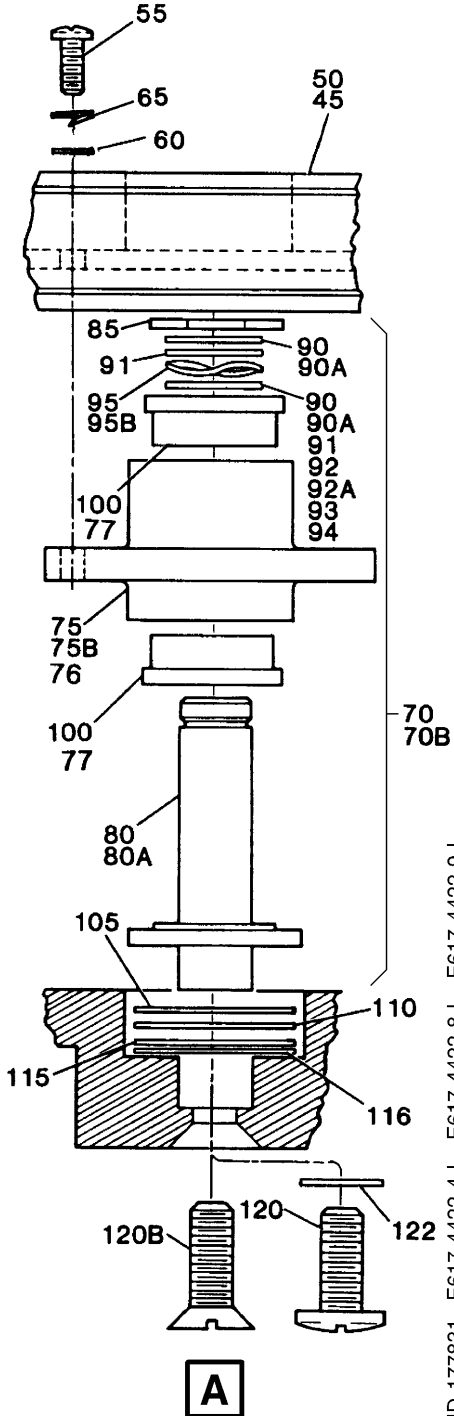
COMPONENT MAINTENANCE MANUAL

Universal Flight Data Recorder / Part No. 980-4100/980-4101



NOTES:

- 1 Item 201 used only with tape guide roller assembly installed in center of tape deck.
- 2 See assembly section para. 23C for belt storage instructions,
- 3 Item 290 installed between items 270 and 275A.

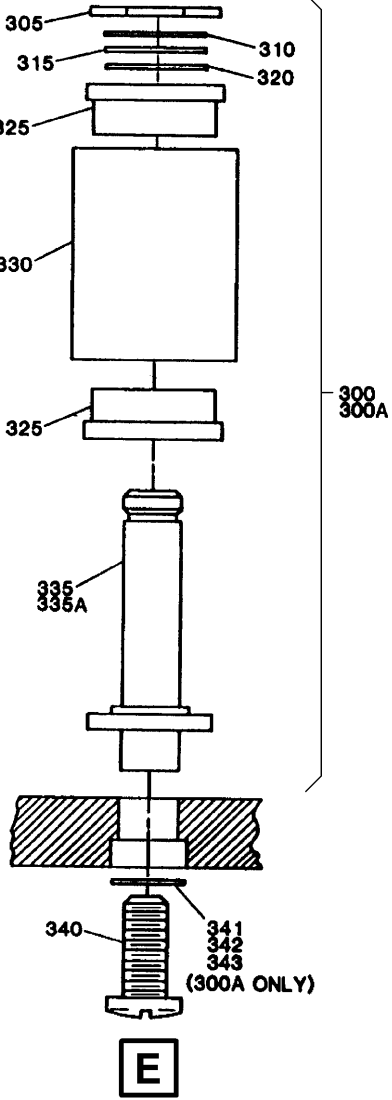
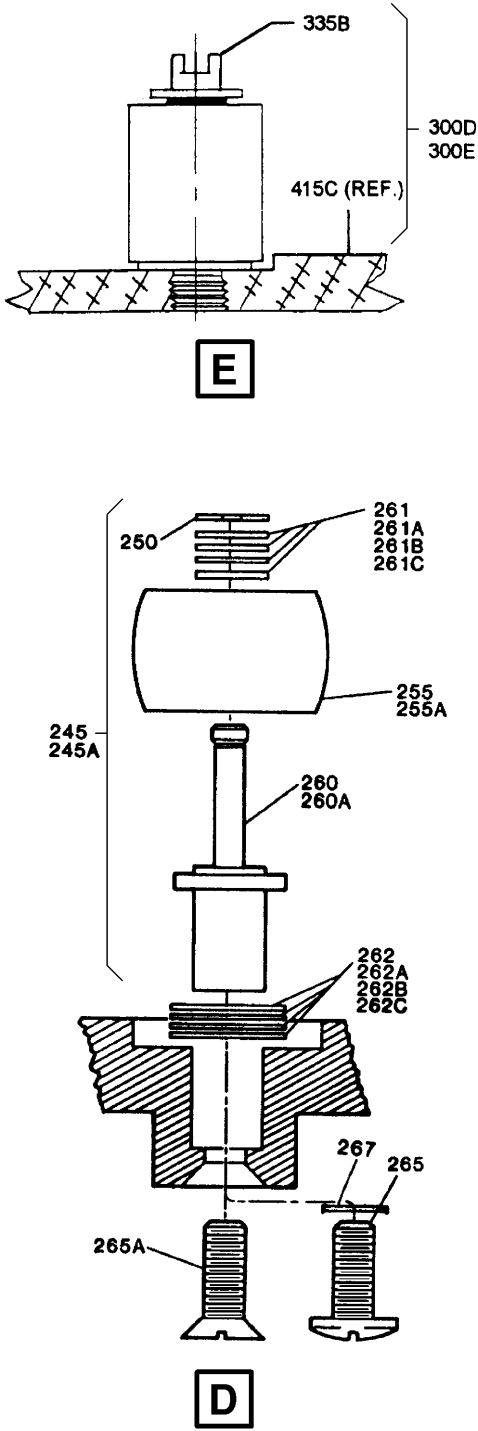
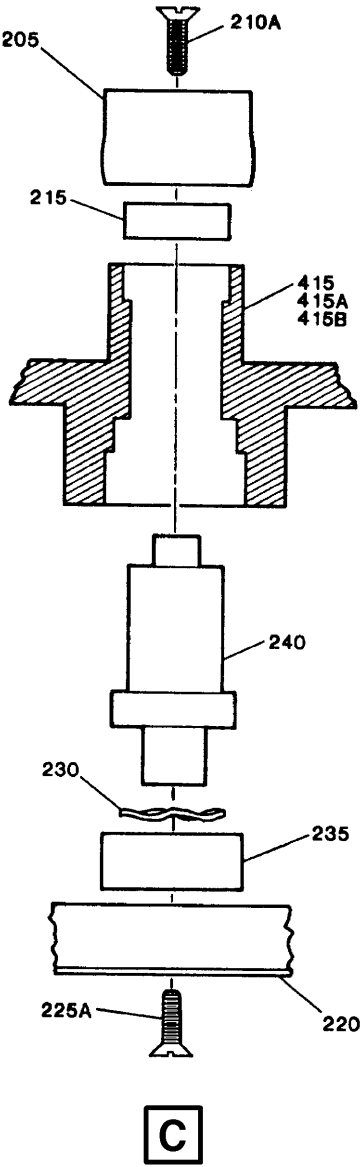
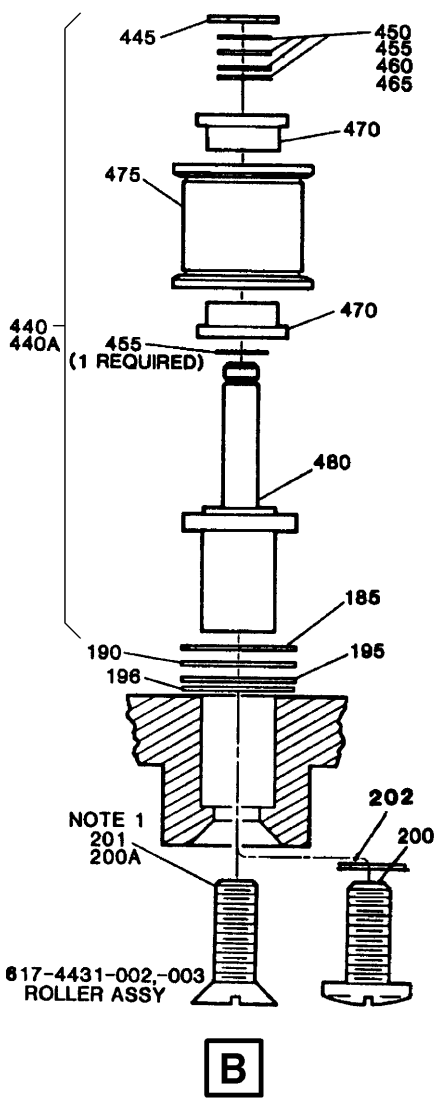
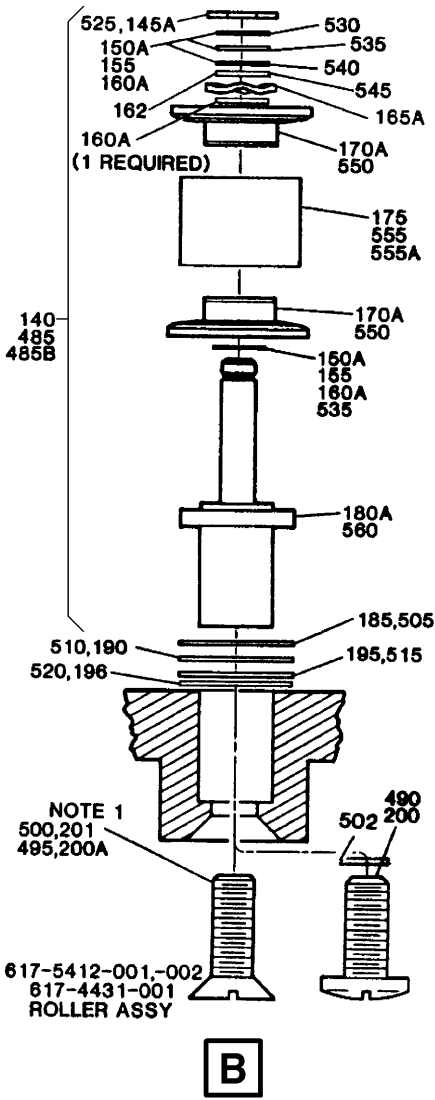


ID-177831 E617-4422-4-L E617-4422-8-L E617-4422-9-L

TAPE TRANSPORT ASSEMBLY
FIGURE 14 (SHEET 1 OF 2)

31-30-32

Page 1098.21/1098.22
22 Dec 2006



TAPE TRANSPORT ASSEMBLY
FIGURE 14 (SHEET 2 OF 2)

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
14					
- 1	617-4422-001		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25B FOR NHA)	A	RF
- 1A	617-4422-003		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25C FOR NHA)	B	RF
- 1B	617-4422-004		DELETED (NOT IN FIELD)		RF
- 1C	617-4422-005		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25E FOR NHA)	D	RF
- 1D	617-4422-006		DELETED (NOT IN FIELD)		RF
- 1E	617-4422-007		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25G FOR NHA)	F	RF
- 1F	617-4422-008		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25H FOR NHA)	G	RF
- 1G	617-4422-009		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25I FOR NHA)	H	RF
- 1H	617-4422-010		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25J FOR NHA)	I	RF
- 1I	617-4422-011		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25K FOR NHA)	J	RF
- 1J	617-4422-012		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25L FOR NHA)	K	RF
- 1K	617-4422-013		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25M FOR NHA)	L	RF
- 1L	617-4422-014		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25N FOR NHA)	M	RF
- 1M	617-4422-015		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25P FOR NHA)	N	RF
- 1N	617-4422-016		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25Q FOR NHA)	P	RF
- 1P	617-4422-017		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25R FOR NHA)	Q	RF
- 1Q	617-4422-023		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25S FOR NHA)	R	RF
- 1R	617-4422-024		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25T FOR NHA)	S	RF
- 1S	617-4422-025		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25U FOR NHA)	T	RF
- 1T	617-4422-026		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25V FOR NHA)	U	RF
- 1U	617-4422-027		TAPE TRANSPORT ASSEMBLY (SEE FIG. 8-25W FOR NHA)	V	RF
4	634-2003-001		.PLATE, TOP (POST SB 11) (SUPSDS 5, 5A)	IJKL MNRS T	1
- 5	617-4205-001		.PLATE ASSEMBLY, TOP (SUPSD BY 5A)(PRE SB 11)(SUPSD BY 4)		1
5A	617-4205-002		.PLATE ASSEMBLY, TOP (SUPSDS 5) (OPT TO 4, CODES IJMN, SN BELOW 5000, OBSOLETE SN 5000 AND ON, EXCEPT CODES PQ) (PRE SB 11)(SUPSD BY 4)	ABCD EFGH IJMN PQUV	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
5B	617-5007-001		.PLATE, CONTROL (OPT TO 5C) (OPT TO 5D, CODES JKLM) (PRE SB 11) (SUPSD BY 5D, CODES MN)	CDEF IJKL MN	2
5C	617-5007-002		.PLATE, CONTROL (OPT TO 5B) (OPT TO 5D, CODES IJKL) (POST SB 5)(PRE SB 11) (SUPSD BY 5D, CODES MN)	KLR	2
5D	617-5007-003		.PLATE, CONTROL (OPT TO 5B, 5C, CODES IJKL)(POST SB 11) (SUPSDS 5B, 5C, CODES MN)	IJMN ST	2
6	634-1615-002		.DELETED		0
7	634-1429-004		.DELETED		0
			- - - ATTACHING PARTS - - -		
10	MS24693C3		.SCREW, FLAT HEAD		4
12	AN960C6L		.WASHER (POST SB 5)(USED WITH 5B, 5C)(OPT TO 13)	CDEF IJKL MNRS T	8
13	5710-23-020		.WASHER (V86928)(USED WITH 5B, 5C)(OPT TO 12)	CDEF IJKL MNRS T	4
14	434-0086-026		.WASHER (USED WITH 5A)	ABCD EFGH IJMN PQUV	4
			-----*-----		
15	9115		.BUMPER, RUBBER (V77969) (OBSOLETE SN 1270 AND ON)		3
			- - - ATTACHING PARTS - - -		
20	MS51957-13		.SCREW, PAN HEAD (OBSOLETE SN 1270 AND ON)		3
25	MS15795-803		.WASHER, FLAT (OBSOLETE SN 1270 AND ON)		3
30	MS35338-135		.WASHER, LOCK (OBSOLETE SN 1270 AND ON)		3
			-----*-----		
35	406-1510-001		.BELT, FLAT		1
40	406-1511-004		.BELT, PERIPHERAL (REPLD BY 40A)		1
-	40A	406-1511-005	.BELT, PERIPHERAL (REPLS 40)		1
45	617-4262-001		.REEL, LOADED	ABGH PQUV	1
-	45A	617-4262-003	.REEL, LOADED (PRE SB 5)	CE	1
-	45B	617-4262-004	.REEL, LOADED (POST SB 5) (OPT TO 45D)	DFIJ KLMN RST	1
-	45C	617-4262-005	.REEL, LOADED (SUPSD BY 45D)	DFIJ KLMN RST	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
- 45D	617-4262-006		..REEL, LOADED (OPT TO 45B) (SUPSDS 45C)	DFIJ KLMN RST	1
50	600-4507-001		..REEL, TAPE (PRE SB 5)	ABGH PQUV	1
- 50A	600-5128-001		..REEL, TAPE (POST SB 5)	CDEF IJKL MNRS T	1
			--- ATTACHING PARTS ---		
55	MS51957-2		..SCREW, PAN HEAD		6
60	NAS620-4		..WASHER, FLAT		6
65	MS35338-39		..WASHER, SPLIT		6
			-----*-----		
70	617-4440-001		..REEL HUB ASSEMBLY (OPT TO 70A)(PRE SB 11, 13)	ABCD EFGH IJK	2
- 70A	617-4440-002		..REEL HUB ASSEMBLY (OPT TO 70)	ABCD EFGH IJK	2
70B	617-4440-003		..REEL HUB ASSEMBLY (POST SB 11, 13)(REPLD BY 70C)	LMNP Q	2
- 70C	617-4440-004		..REEL HUB ASSEMBLY (REPLS 70B)	RSTU V	2
75	651-8092-001		..REEL TAPE HOUSING (USED ON 70)(PRE SB 11,13) (SUPSD BY 75B)	ABCD EFGH IJK	1
- 75A	651-8092-002		..REEL TAPE HOUSING (USED ON 70A)(PRE SB 11,13) (SUPSD BY 75B)	ABCD EFGH IJK	1
75B	617-5489-001		..REEL HUB BEARING ASSEMBLY (USED ON 70B)(POST SB 11,13) (SUPSDS 75,75A,100) (REPLD BY 75C)	LMNP Q	1
- 75C	617-5489-002		..REEL HUB BEARING ASSEMBLY (REPLS 75B)	RSTU V	1
76	613-1281-001		...HOUSING	LMNP QRST UV	1
77	400-0058-004		...BEARING, BALL	LMNP Q	2
- 77A	400-0058-005		...BEARING, BALL	RSTU V	2
80	646-0380-001		..PIN, TAPE REEL (REPLD BY 80A)		1
- 80A	646-0380-003		..PIN, TAPE REEL (REPLS 80)		1
85	5133-18		..RETAINER, E-RING (V79136)		1
90	403-6015-011		..SHIM, CRES, 0.188 ID, 0.250 OD, 0.012 THK (REPLD BY 90A)		AR
90A	B6-5		..SHIM, CRES, 0.188 ID, 0.250 OD, 0.012 THK (V00141) (REPLS 90)		AR

31-30-32

Page 1098.27
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
91	B6-2		..SHIM, CRES, 0.188 ID, 0.250 OD, 0.006 THK (V00141)		AR
92	B6-1		..SHIM, CRES, 0.188 ID, 0.250 OD, 0.004 THK (V00141)	ABCD EFGH IJK	AR
92A	B6-1		..SHIM, CRES, 0.188 ID, 0.250 OD, 0.004 THK (V00141)	LMNP Q	1
93	B6-4		..SHIM, CRES, 0.188 ID, 0.250 OD, 0.010 THK (V00141)	LMNP Q	AR
94	B6-3		..SHIM, CRES, 0.188 ID, 0.250 OD, 0.008 THK (V00141)	LMNP Q	AR
95	434-0171-003		..WASHER, WAVY (REPLD BY 95A)	ABCD EFGH IJK	1
- 95A	W0242-006S		..WASHER, WAVY (V92830)(REPLS 95)(PRE SB 11,13)(SUPSD BY 95B)	ABCD EFGH IJK	1
95B	434-0274-001		..WASHER, WAVY (POST SB 11, 13)(SUPSDS 95A)	LMNP Q	1
100	400-0058-003		..BEARING, BALL (PRE SB 11, 13)(SUPSD BY 75B)	ABCD EFGH	2
105	434-0201-004		--- ATTACHING PARTS --- ..SHIM, CRES, 0.200 ID, 0.420 OD, 0.001 THK		AR
110	434-0201-005		..SHIM, CRES, 0.200 ID, 0.420 OD, 0.003 THK		AR
115	434-0201-006		..SHIM, CRES, 0.200 ID, 0.420 OD, 0.005 THK		AR
116	434-0201-013		..SHIM, CRES, 0.200 ID, 0.420 OD, 0.0015 THK		AR
120	MS51957-14		..SCREW, PAN HEAD	ACDG IKLM PRSU	2
- 120A	MS24693C34		..SCREW (SUPSD BY 120B)	B	2
120B	MS24693C3		..SCREW (SUPSDS 120A)	BEFH JNQT V	2
122	MS35333-70		..WASHER, INTERNAL STAR	ACDG IKLM PRSU	2
125	744-0078-002		-----*----- ..SENSOR, EOT		1
130	744-0078-001		..SENSOR, BOT		1
135	MS16995-2		--- ATTACHING PARTS --- ..SCREW, SOCKET HEAD		4
140	617-4431-001		-----*----- ..ROLLER ASSEMBLY, TAPE GUIDE (SUPSDS 440)(OPT TO 485A) (REPLD BY 140B)	ABCD EFGH IJMN PQ	5

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
14					
- 140A	617-4431-001		..ROLLER ASSEMBLY, TAPE GUIDE (OPT TO 485)(REPLD BY 140C)	KL	3
- 140B	617-4431-004		..ROLLER ASSEMBLY, TAPE GUIDE (REPLS 140)(OPT TO 440B)	ABCD EFGH IJMN STUV	5
- 140C	617-4431-004		..ROLLER ASSEMBLY, TAPE GUIDE (REPLS 140A)(OPT TO 485D)	KLR	3
- 145	5133-9H		..RETAINER, E-RING (V79136) (SUPSD BY 145A)		1
145A	5100-12C		..RETAINER, E-RING (V79136) (SUPSDS 145)(OPT TO 145B)		1
- 145B	MS16624-5012		..RETAINER, E-RING (OPT TO 145A)		1
- 150	434-0086-001		..SHIM, CRES, 0.130 ID, 0.1875 OD, 0.004 THK(SUPSD BY 150A)		AR
150A	434-0086-025		..SHIM, CRES, 0.130 ID, 0.1875 OD, 0.004 THK (SUPSDS 150)		AR
155	434-0086-024		..SHIM, CRES, 0.130 ID, 0.1875 OD, 0.003 THK		AR
- 160	434-0086-029		..SHIM, CRES, 0.130 ID, 0.1875 OD, 0.001 THK(SUPSD BY 160A)		AR
160A	434-0086-023		..SHIM, CRES, 0.130 ID, 0.1875 OD, 0.001 THK (SUPSDS 160)		AR
162	434-0086-136		..SHIM, CRES, 0.130 ID, 0.1875 OD, 0.0015 THK		AR
- 165	434-0071-002		..WASHER, WAVY (REPLD BY 165A)		1
165A	W0367-006S		..WASHER, WAVY (V92830) (REPLS 165)		1
- 170	400-0056-043		..BEARING, BALL (SUPSD BY 170A)		2
170A	400-0056-045		..BEARING, BALL (SUPSDS 170) (USED ON 140, 140A) (REPLD BY 170B)		2
- 170B	400-0056-062		..BEARING, BALL (USED ON 140B, 140C) (REPLS 170A)		2
175	652-0274-001		..ROLLER		1
- 180	646-0399-001		..PIN (SUPSD BY 180A)		1
180A	646-0399-002		..PIN (SUPSDS 180)		1
			- - - ATTACHING PARTS - - -		
185	434-0201-010		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.001 THK		AR
190	434-0201-011		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.003 THK		AR
195	434-0201-012		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.005 THK		AR
196	434-0201-014		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.0015 THK		AR

31-30-32

Page 1098.29
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
200	MS51957-14		.SCREW, PAN HEAD	ACDG IKLM PRSU	4
200A	MS24693C3		.SCREW, FLAT HEAD	BFHJ NQTV	4
201	MS24693C2		.SCREW, FLAT HEAD		1
202	MS35333-70		.WASHER, INTERNAL STAR	ACDG IKLM PRSU	4
			-----*-----		
205	603-1001-001		.PULLEY, TAPE DRIVE (CAPSTAN) (PRE SB 5 OR 8)(PRE SB 13)	ABPU	1
- 205A	603-1001-003		.PULLEY, TAPE DRIVE (CAPSTAN) (POST SB 5 OR 8)(PRE SB 11) (POST SB 13)(OPT TO 205B, CODES IJMNST, SN BELOW 5000, OBSOLETE SN 5000 AND ON)	CDEF IMNS T	1
- 205B	603-1001-005		.PULLEY, TAPE DRIVE (CAPSTAN) (POST SB 11)(OPT TO 205A)	IJMN ST	1
- 205C	603-1001-004		.PULLEY, TAPE DRIVE (CAPSTAN)	GHLK PQRU V	1
			- - - ATTACHING PARTS - - -		
- 210	MS24693C3		.SCREW, FLAT HEAD (SUPSD BY 210A)		1
210A	NAS1189E04P5L		.SCREW, FLAT HEAD, SELF- LOCKING (SUPSDS 210)		1
			-----*-----		
215	400-0057-027		.BEARING, BALL (REPLD BY 215A)		1
- 215A	400-0057-030		.BEARING, BALL (REPLS 215)		1
220	603-1002-001		.PULLEY, TAPE TRANSPORT (FLYWHEEL)		1
			- - - ATTACHING PARTS - - -		
- 225	MS24693C3		.SCREW, FLAT HEAD (SUPSD BY 225A)		1
225A	NAS1189E04P5L		.SCREW, FLAT HEAD, SELF- LOCKING (SUPSDS 225)		1
			-----*-----		
230	7X9-3W06108		.WASHER, WAVY (V14519)		1
235	400-0057-028		.BEARING, BALL (REPLD BY 235A)		1
- 235A	400-0057-031		.BEARING, BALL (REPLS 235)		1
240	645-0352-001		.SHAFT		1
245	617-4438-001		.P-BELT PULLEY ASSEMBLY (SUPSD BY 245B)	ACD	2
245A	617-4438-002		.P-BELT PULLEY ASSEMBLY (SUPSD BY 245C)	B	2
- 245B	617-4438-003		.P-BELT PULLEY ASSEMBLY (SUPSDS 245)(OPT TO 245D)	ACDG IKLM PRSU	2
- 245C	617-4438-004		.P-BELT PULLEY ASSEMBLY (SUPSDS 245A)(OPT TO 245E)	BEFH JNQT V	2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
- 245D	617-4438-005		.P-BELT PULLEY ASSEMBLY (OPT TO 245B)	ACDG IKLM PRSU	2
- 245E	617-4438-006		.P-BELT PULLEY ASSEMBLY (OPT TO 245C)	BEFH JNQT V	2
250	5133-9H		..RETAINER, E-RING (V79136)		1
- 250A	MS16633-4009		..RETAINER, E-RING OPTIONAL		1
255	603-1003-001		..PULLEY, BELT IDLER	ACD	1
255A	603-1010-001		..PULLEY, BELT DRAG IDLER	B	1
- 255B	617-5006-001		..PULLEY, BELT IDLER (USED WITH 245B, 245C)		1
- 255C	603-1003-002		..PULLEY, BELT IDLER (USED WITH 245D, 245E)		1
260	646-0325-001		..PIN, ROLLER	ACDG IKLM PRSU	1
260A	646-0398-001		..PIN, ROLLER, ECCENTRIC	BEFH JNQT V	1
261	434-0201-007		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.001 THK (POST SB 11)		AR
261A	434-0201-008		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.003 THK (POST SB 11)		AR
261B	434-0201-009		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.005 THK (POST SB 11)		AR
261C	645-5125-001		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.0015 THK (POST SB 11)		AR
262	434-0201-012		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.005 THK		AR
262A	434-0201-010		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.001 THK (POST SB 11)		AR
262B	434-0201-011		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.003 THK (POST SB 11)		AR
262C	434-0201-014		.SHIM, CRES, 0.200 ID, 0.300 OD, 0.0015 THK (POST SB 11)		AR
265	MS51957-14		--- ATTACHING PARTS --- .SCREW, PAN HEAD	ACDG IKLM PRSU	2
265A	MS24693C3		.SCREW, FLAT HEAD	BEFH JNQT V	2
267	MS35333-70		.WASHER, INTERNAL STAR	ACDG IKLM PRSU	2
270	722-2522-002		-----*----- .DISTRIBUTION BOARD (SEE FIG. 12 FOR BREAKDOWN)(ESD)		1
- 275	636-0220-001		.SHIELD, DISTRIBUTION BOARD (SUPSD BY 275A)		1

31-30-32

Page 1098.31
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
275A	636-0220-003		.SHIELD, DISTRIBUTION BOARD (SUPSDS 275)		1
280	MS51957-26		--- ATTACHING PARTS --- .SCREW, PAN HEAD	ABCD EFGH IJKL MNPQ	3
280A	MS16995-17		.SCREW (POST SB 17)	RSTU V	2
280B	MS51957-26		.SCREW (POST SB 17)	RSTU V	1
282	NAS43DD1-8		.SPACER (POST SB 17)	RSTU V	2
285	NAS620C6L		.WASHER, FLAT		3
290	2254PH141-1		.WASHER, FIBER (V06540)		3
295	TL56		.LUG (V71218)(OPT TO 295A)		1
- 295A	1488-6		.LUG (V83330)(OPT TO 295) -----*-----		1
300	617-4439-001		.DRIVE BELT ROLLER GUIDE ASSEMBLY (REPLD BY 300B)		2
300A	617-4439-003		.DRIVE BELT ROLLER GUIDE ASSEMBLY (REPLD BY 300C)		2
- 300B	617-4439-004		.DRIVE BELT ROLLER GUIDE ASSEMBLY (REPLS 300)	AIKL MRS	2
- 300C	617-4439-006		.DRIVE BELT ROLLER GUIDE ASSEMBLY (REPLS 300A)	BDFG HJNP QTUV	2
- 300D	617-4439-002		.DRIVE BELT ROLLER GUIDE ASSEMBLY (REPLD BY 300E)	BDFG HJNP TV	2
- 300E	617-4439-005		.DRIVE BELT ROLLER GUIDE ASSEMBLY (REPLS 300D)	BDFG HJNP TV	2
305	5133-9H		..RETAINER, E-RING (V79136) (OPT TO 305A)		1
- 305A	MS16633-4009		..RETAINER, E-RING (OPT TO 305)		1
310	434-0201-007		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.001 THK		AR
315	434-0201-008		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.003 THK		AR
320	434-0201-009		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.005 THK		AR
325	400-0056-042		..BEARING, BALL (REPLD BY 326)		2
- 325A	400-0056-041		..BEARING, BALL OPTIONAL		2
- 325B	400-0056-052		..BEARING, BALL OPTIONAL		2
- 325C	400-0056-053		..BEARING, BALL OPTIONAL		2
- 325D	400-0172-001		..BEARING, BALL OPTIONAL		2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
14					
	326	400-0056-060	..BEARING, BALL (REPLS 325)		2
-	326A	400-0056-061	..BEARING, BALL OPTIONAL		2
-	326B	400-0056-063	..BEARING, BALL OPTIONAL		2
-	326C	400-0056-064	..BEARING, BALL OPTIONAL		2
-	326D	400-0056-065	..BEARING, BALL OPTIONAL		2
	330	600-4681-001	..ROLLER		1
	335	646-0379-001	..PIN (USED ON 300, 300B)		1
	335A	646-0379-002	..PIN (USED ON 300A, 300C)		1
	335B	646-0401-001	..PIN (USED ON 300D, 300E)		1
			--- ATTACHING PARTS ---		
	340	431-0343-013	.SCREW, BIND HEAD		2
	341	434-0201-009	.SHIM, CRES, 0.095 ID, 0.142 OD, 0.005 THK	BEFH JLNQ RTV	AR
	342	07/10/5710	.SHIM, CRES, 0.098 ID, 0.140 OD, 0.010 THK (V86928)	BEFH JLNQ RTV	AR
	343	07/12/5710	.SHIM, CRES, 0.098 ID, 0.140 OD, 0.012 THK (V86928)	BEFH JLNQ RTV	AR
			-----*-----		
	345	607-1520-001	.COVER, HARNESS		1
			--- ATTACHING PARTS ---		
	350	MS51957-12	.SCREW, PAN HEAD		2
	355	MS15795-803	.WASHER, FLAT		2
	360	MS35338-135	.WASHER, LOCK		2
			-----*-----		
	365	602-2482-001	.BRACKET, HARNESS		1
			--- ATTACHING PARTS ---		
	370	MS51957-13	.SCREW, PAN HEAD		2
	375	MS15795-803	.WASHER, FLAT		2
	380	MS35338-135	.WASHER, LOCK		2
			-----*-----		
	385	600-4103-001	.SHELL, NYLON		1
	390	D20419-21	.DELETED		0
	395	DBM25P	.CONNECTOR (V71468)		1
-	395A	117DBM25P	.CONNECTOR (V02660) OPTIONAL		1
-	395B	MD308G25P1	.CONNECTOR (V14283) OPTIONAL		1
	400	DA20961	.CLAMP (V71468)		1
	402	MS51957-14	.SCREW, PAN HEAD		2
	405	DAM15P	.CONNECTOR (V71468)		1
-	405A	DAM15P	.CONNECTOR (V13556) OPTIONAL		1
	410	600-3970-001	.FEEDTHRU		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
415	600-4683-002		.BASEPLATE	ACDG IMPS U	1
415A	600-3967-008		.BASEPLATE	BDFG HJNP QTV	1
- 415B	600-4683-003		.BASEPLATE	KLR	1
- 415C	600-3967-004		.BASEPLATE	BDFG HJNP TV	1
420	317-1044-002		.HEAD, MAGNETIC, READ/WRITE		1
- 420A	317-1044-004		.HEAD, MAGNETIC, READ/WRITE OPTIONAL		1
- 421	645-5107-003		.SHIM, HEAD, 0.005 THK	BEFH IJMN QSTV	AR
- 422	645-5107-001		.SHIM, HEAD, 0.001 THK	IJMN ST	AR
- 423	645-5107-002		.SHIM, HEAD, 0.003 THK	IJKL MNRS T	AR
425	317-1044-001		.HEAD, MAGNETIC, READ/WRITE		1
- 425A	317-1044-003		.HEAD, MAGNETIC, READ/WRITE OPTIONAL		1
- 426	645-5107-003		.SHIM, HEAD, 0.005 THK (POST SB 11)	BEFI JMNS T	AR
- 427	645-5107-001		.SHIM, HEAD, 0.001 THK (POST SB 11)	IJKL MNRS T	AR
- 428	645-5107-002		.SHIM, HEAD, 0.003 THK (POST SB 11)	IJKL MNRS T	AR
430	317-1045-001		.HEAD, MAGNETIC, ERASE		2
- 430A	317-1044-004		.DELETED		0
- 430B	317-1045-004		.HEAD, MAGNETIC, ERASE OPTIONAL		2
- 431	645-5107-003		.SHIM, HEAD, 0.005 THK	BEFI JMNS T	AR
- 432	645-5107-001		.SHIM, HEAD, 0.001 THK	IJKL MNRS T	AR
- 433	645-5107-002		.SHIM, HEAD, 0.003 THK	IJKL MNRS T	AR
435	MS16995-2		--- ATTACHING PARTS --- .SCREW, SOCKET HEAD -----*-----		8

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
440	617-4431-002		..ROLLER ASSEMBLY, TAPE GUIDE (SUPSD BY 485A)(REPLD BY 440B)	ABCD EFGH IJMN PQ KL	5
440A	617-4431-003		..ROLLER ASSEMBLY, TAPE GUIDE (OPT TO 485B)(REPLD BY 440C)		2
- 440B	617-4431-005		..ROLLER ASSEMBLY, TAPE GUIDE (REPLS 440)(OPT TO 140B,485C)	ABCD EFGH IJMN PQST UV KLR	5
- 440C	617-4431-006		..ROLLER ASSEMBLY, TAPE GUIDE (REPLS 440A)(OPT TO 485E)		2
445	5133-9H		..RETAINER, E-RING (V79136) (OPT TO 445A)		1
- 445A	MS16624-5012		..RETAINER, E-RING (OPT TO 445)		1
450	434-0201-009		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.005 THK		AR
455	434-0201-008		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.003 THK		AR
460	434-0201-007		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.001 THK		AR
465	645-5125-001		..SHIM, CRES, 0.095 ID, 0.142 OD, 0.0015 THK		AR
470	400-0056-041		..BEARING, BALL (USED ON 440, 440A)(REPLD BY 471)		2
- 470A	400-0056-042		..BEARING, BALL OPTIONAL		2
- 470B	400-0056-052		..BEARING, BALL OPTIONAL		2
- 470C	400-0056-053		..BEARING, BALL OPTIONAL		2
- 470D	400-0172-001		..BEARING, BALL OPTIONAL		2
- 471	400-0056-060		..BEARING, BALL (USED ON 440B, 440C)(REPLS 470)		2
- 471A	400-0056-061		..BEARING, BALL OPTIONAL		2
- 471B	400-0056-063		..BEARING, BALL OPTIONAL		2
- 471C	400-0056-064		..BEARING, BALL OPTIONAL		2
- 471D	400-0056-065		..BEARING, BALL OPTIONAL		2
475	600-4108-001		..ROLLER (OPT TO 475A) (USED ON 440)		1
- 475A	600-4108-002		..ROLLER (OPT TO 475) (USED ON 440)		1
- 475B	600-4108-003		..ROLLER (USED ON 440A)		1
480	646-0325-001		..PIN, ROLLER		1

31-30-32

Page 1098.35
22 Dec 2006

Honeywell

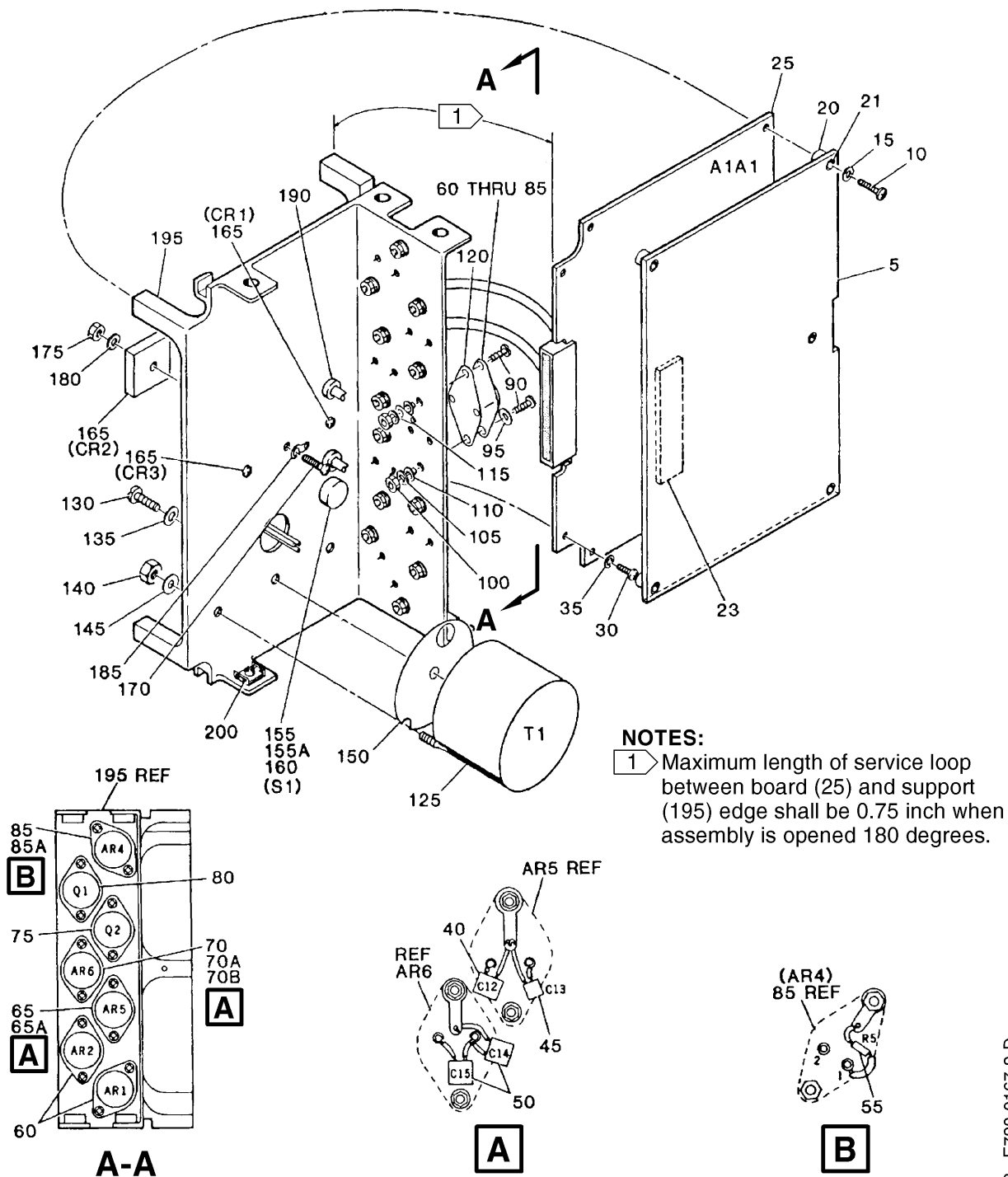
COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
- 480A	646-0325-003		..PIN, ROLLER OPTIONAL		1
485	617-5412-001		..ROLLER ASSEMBLY (OPT TO 140A) (REPLD BY 485C)	KL	3
- 485A	617-5412-001		..ROLLER ASSEMBLY (OPT TO 140) (SUPSDS 440)	ABCD EFGH IJMN PQ	5
485B	617-5412-002		..ROLLER ASSEMBLY (OPT TO 440A) (REPLD BY 485E)	KL	2
- 485C	617-5412-003		..ROLLER ASSEMBLY (REPLS 485A) (OPT TO 140B, 440B)	ABCD EFGH IJMN PQST UV	5
- 485D	617-5412-003		..ROLLER ASSEMBLY (REPLS 485) (OPT TO 140C)	KLR	3
- 485E	617-5412-004		..ROLLER ASSEMBLY (REPLS 485B) (OPT TO 440C) - - - ATTACHING PARTS - - -	KLR	2
490	MS51957-14		..SCREW, PAN HEAD	ACDG IKLM PRSU	4
495	MS24693C3		..SCREW, FLAT HEAD	BFHJ NQTV	4
500	MS24693C2		..SCREW, FLAT HEAD		1
502	MS35333-70		..WASHER, INTERNAL STAR	ACDG IKLM PRSU	4
505	434-0201-010		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.001 THK		AR
510	434-0201-011		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.003 THK		AR
515	434-0201-012		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.005 THK		AR
520	434-0201-014		..SHIM, CRES, 0.200 ID, 0.300 OD, 0.0015 THK -----*-----		AR
525	MS16624-5012		..RING, RETAINING		1
530	434-0086-023		..SHIM, CRES, 0.130 ID, 0.187 OD, 0.001 THK		AR
535	434-0086-024		..SHIM, CRES, 0.130 ID, 0.187 OD, 0.003 THK		AR
540	434-0086-025		..SHIM, CRES, 0.130 ID, 0.187 OD, 0.004 THK		AR
545	434-0086-136		..SHIM, CRES, 0.130 ID, 0.1875 OD, 0.0015 THK		AR
550	400-0056-045		..BEARING, BALL (USED ON 485, 485A, 485B) (REPLD BY 550A)		2

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
14			1234567		
-	550A	400-0056-062	..BEARING, BALL (USED ON 485C, 485D, 485E) (REPLDS 550)		2
	555	652-0274-002	..ROLLER (USED ON 485, 485A)		1
	555A	652-0274-003	..ROLLER (USED ON 485B)		1
	560	646-0399-001	..PIN, ROLLER		1



POWER SUPPLY ASSEMBLY (A1)
FIGURE 15

ID-177833 E720-0167-2-D

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
15			1234567		
- 1	720-0167-002		POWER SUPPLY ASSEMBLY, A1 (ESD) (SEE FIG. 1-95C FOR NHA)	A	RF
- 1A	720-0167-004		POWER SUPPLY ASSEMBLY, A1 (ESD) (SEE FIG. 1-95D FOR NHA)	B	RF
5	617-3831-001		.EMI SHIELD (REPLD BY 5A)		1
- 5A	617-3831-002		.EMI SHIELD (REPLS 5)		1
			--- ATTACHING PARTS ---		
10	MS35214-16		.SCREW, PAN HEAD		5
15	NAS620B4L		.WASHER, FLAT		5
			-----*-----		
20	9705CA115-17		..STANDOFF (V06540)(USED ON 5)		5
- 20A	18091BB115-14		..SPACER (V06540)(USED ON 5A)		5
21	628-0261-001		..WASHER, ALUMINUM (USED ON 5)		5
23	600-4143-001		..RUBBER, SPONGE		1
25	722-2440-003		.POWER SUPPLY CCA, A1A1 (ESD) (SEE FIG. 7 FOR BREAKDOWN)		1
- 25A	722-2440-005		POWER SUPPLY CCA, A1A1 (ESD) (SEE FIG. 7 FOR BREAKDOWN)	B	1
			--- ATTACHING PARTS ---		
30	MS35214-14		.SCREW, PAN HEAD		2
35	NAS620B4L		.WASHER, FLAT		2
			-----*-----		
40	M39014-02-1358		.CAPACITOR, 0.33 UF, 50V, 10%		1
45	M39014-02-1350		.CAPACITOR, 0.1 UF, 100V, 10%		1
50	M39014-02-1419		.CAPACITOR, 1.0 UF, 50V, 10%		2
55	RNC55H2430FS		.RESISTOR, 243 OHMS, 1%, 1/10W		1
60	SG123K883B		.IC, VOLTAGE REGULATOR, +5V (ESD)(V34333)(OPT TO 60A)		2
- 60A	LM123K883B		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V27014) OPTIONAL		2
- 60B	LM123K883C		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V27014) OPTIONAL		2
- 60C	LM123K		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V27014) OPTIONAL	B	2
- 60D	LM123K		.IC, VOLTAGE REGULATOR, +5V (ESDS) (V64155) OPTIONAL	B	2
65	UA7812KMQB		.IC, VOLTAGE REGULATOR, +12V (ESDS)(V07263)		1
65A	LM140K12		.IC, VOLTAGE REGULATOR, +12V (ESDS) (V27014) OPTIONAL		1
- 65B	LM140K12		.IC, VOLTAGE REGULATOR, +12V (ESDS) (V04713) OPTIONAL		1
- 65C	UA7812KM		.IC, VOLTAGE REGULATOR, +12V (ESDS) (V07263) OPTIONAL		1
70	UA7912KMQB		.IC, VOLTAGE REGULATOR, -12V (ESD)(V07263)		1
70A	LM120K12		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V27014) OPTIONAL		1
70B	UA7912KM		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V07263) OPTIONAL		1
- 70C	LM120K12-883		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V27014) OPTIONAL		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		
15					
- 70D	10408-1		.IC, VOLTAGE REGULATOR, -12V (ESDS) (V54001) OPTIONAL		1
75	JNTX2N3716		.TRANSISTOR		1
- 75A	JAN2N3716		.TRANSISTOR OPTIONAL		1
- 75B	JANTXV2N3716		.TRANSISTOR, NPN OPTIONAL (HMN 341-0637-004)		1
80	JANTX2N3792		.TRANSISTOR		1
85	LM117K883B		.IC, VOLTAGE REGULATOR (ESD) (V27014)		1
85A	SG117K883B		.IC, VOLTAGE REGULATOR (ESDS) (V34333) OPTIONAL		1
- 85B	LM117K883C		.IC, VOLTAGE REGULATOR (ESDS) (V64155) OPTIONAL		1
- 85C	LM117K		.IC, VOLTAGE REGULATOR (ESDS) (V04713) OPTIONAL	B	1
- 85D	LM117K		.IC, VOLTAGE REGULATOR (ESDS) (V27014) OPTIONAL	B	1
- 85E	LM117K		.IC, VOLTAGE REGULATOR (ESDS) (V64155) OPTIONAL - - - ATTACHING PARTS - - -	B	1
90	MS51957-17		.SCREW, PAN HEAD		14
95	MS35333-70		.WASHER, INTERNAL STAR		14
100	MS21044C04		.NUT, LOCK		14
105	MS15795-803		.WASHER, FLAT		14
110	NY04-125G		.WASHER, SHOULDER (V08289)		14
- 110A	NY04-090G		.WASHER, SHOULDER (V08289) OPTIONAL		14
- 110B	NY04-090N		.WASHER, SHOULDER (V08289) OPTIONAL -----*-----		14
115	1412-4		.LUG (V83330)		7
120	7403-09FR06		.INSULATOR (V55285)		7
125	310-0285-001		.TRANSFORMER - - - ATTACHING PARTS - - -		1
130	MS35216-56		.SCREW, PAN HEAD		1
135	NAS620B10L		.WASHER, FLAT		1
140	MS21044C06		.NUT, LOCK		2
145	MS15795-805		.WASHER, FLAT -----*-----		2
150	402-3036-001		.INSULATOR		1
155	973-0031-561		.SWITCH, THERMAL (OPT TO 155A)		1
155A	974-0000-574		.SWITCH, THERMAL (OPT TO 155) - - - ATTACHING PARTS - - -		1
160	MS21042-06		.NUT, LOCK -----*-----		1
165	JNTX469-1		.RECTIFIER, BRIDGE		3
- 165A	M19500-469-01		.RECTIFIER, BRIDGE OPTIONAL		3
- 165B	HRSCBH4		.RECTIFIER, BRIDGE (V14099) OPTIONAL		3

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
15			1234567		
- 165C	HRSCBH6		.RECTIFIER, BRIDGE (V14099) OPTIONAL		3
165D	MILS19500-469		.RECTIFIER, BRIDGE TRUE PN IS MIL-S-19500/469 OPTIONAL (V81349)(HMN 340-0056-003)		3
- 165E	MILS19500-469		.RECTIFIER, BRIDGE TRUE PN IS MIL-S-19500/469 OPTIONAL (V81349)(HMN 340-0056-004)		3
			- - - ATTACHING PARTS - - -		
170	MS51957-31		.SCREW, PAN HEAD		3
175	MS21044C06		.NUT, LOCK		3
180	MS15795-805		.WASHER, FLAT -----*-----		3
185	1488-6		.LUG (V83330)(SUPSD BY 185A)		1
- 185A	1485-8		.TERMINAL (V83330)(SUPSDS 185)		1
190	Z427		.GROMMET (V76385)		2
195	602-2955-001		.SUPPORT ASSEMBLY (PRE SB 9, SUPSD BY 195A)		1
- 195A	602-2955-005		.SUPPORT ASSEMBLY (POST SB 9, SUPSDS 195)		1
200	F12NC4284-2-82		..NUT, CLINCH (V72962)		6

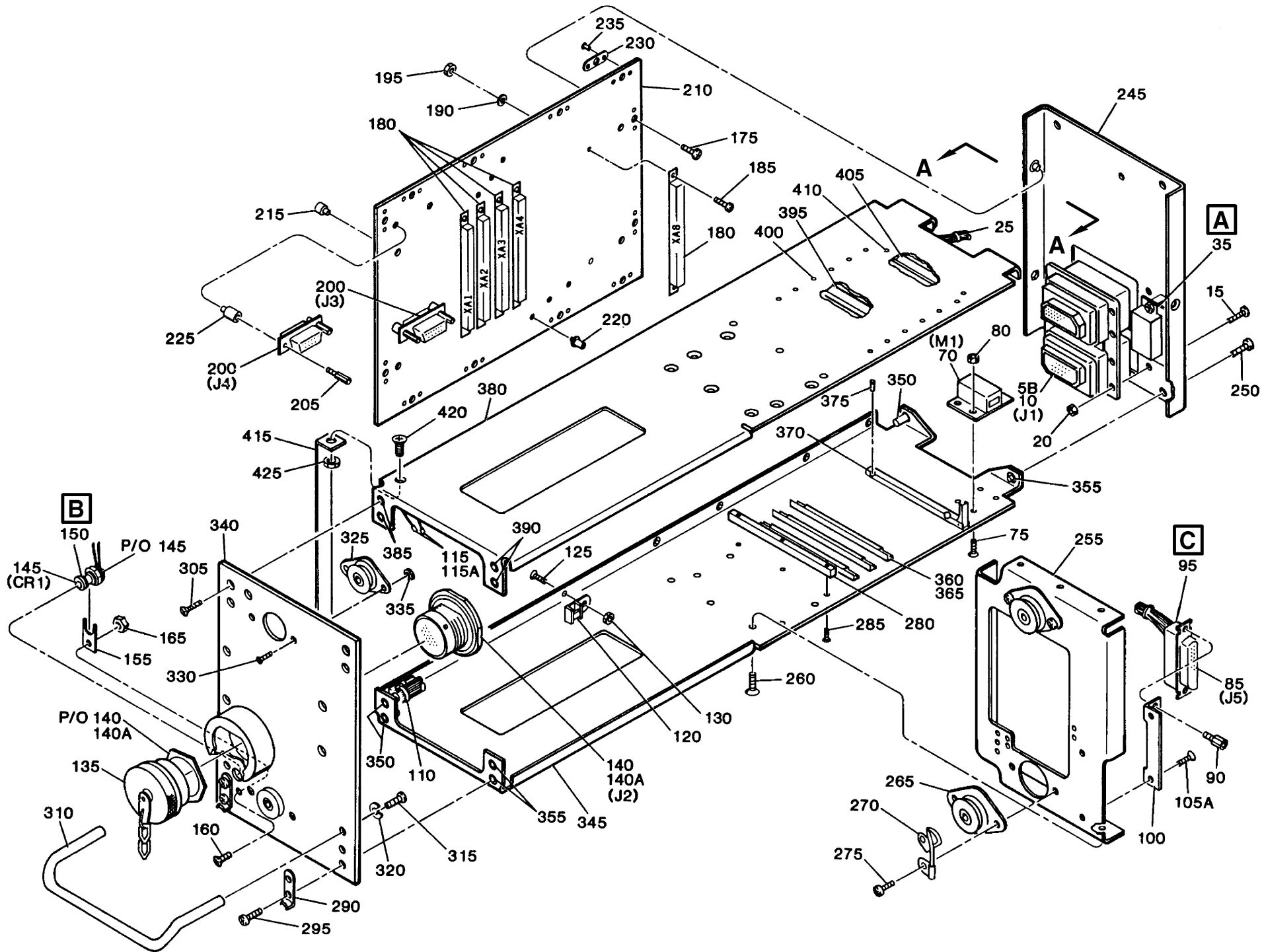
Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

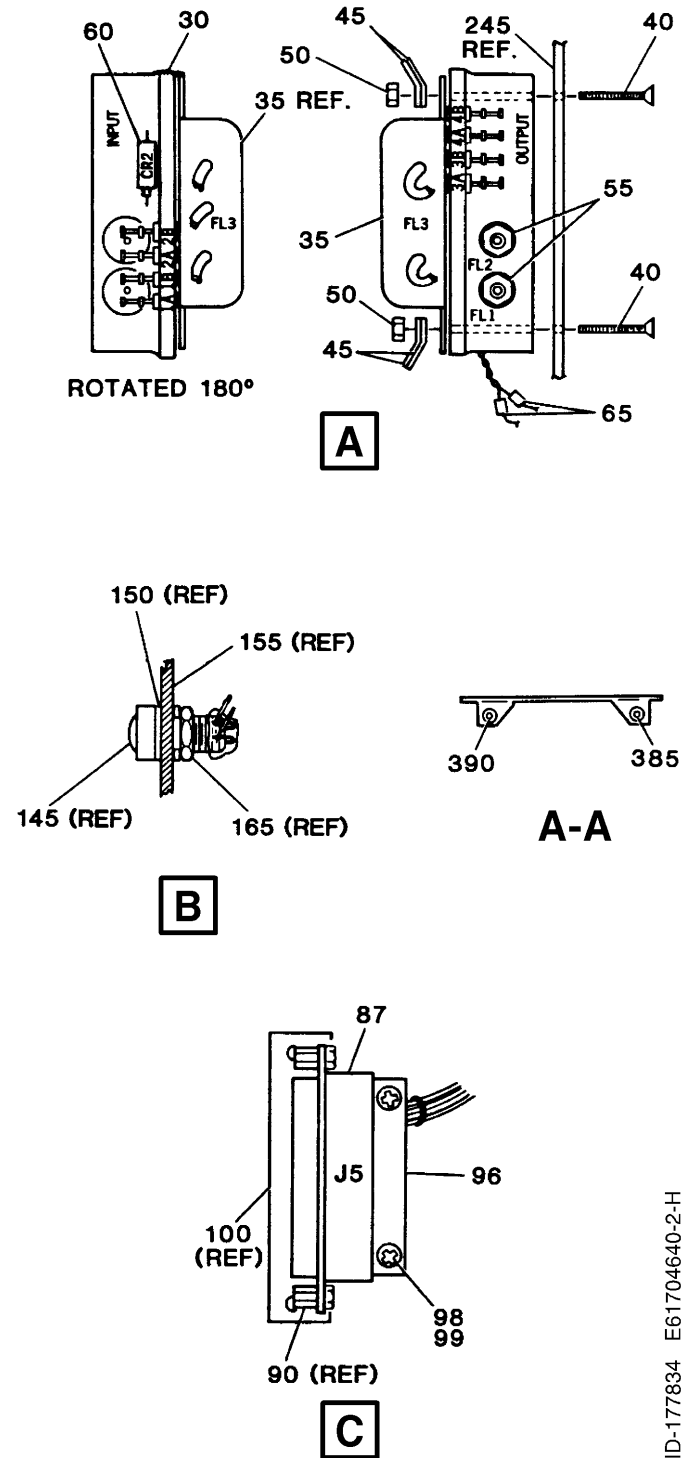
THIS PAGE INTENTIONALLY LEFT BLANK

Honeywell

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



CHASSIS ASSEMBLY
FIGURE 16



ID-177834 E61704640-2-H

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
16			1234567		
- 1	617-4640-006		CHASSIS ASSEMBLY (SEE FIG. 1-190D FOR NHA)		RF
- 5	DPX2MA57P57P34-		.CONNECTOR, 57-PIN (V71468) (SUPSD BY 5B)		1
- 5A	DPX2MA57P57P34-		.CONNECTOR, 57-PIN (V71468) (SUPSD BY 5C)		1
5B	440-0005-009		.CONNECTOR, 57-PIN (DPX2MA57P00P34-0001 FOR V71468)(SUPSDS 5)		1
- 5C	440-0005-028		.CONNECTOR, 57-PIN (DPX2MA57P00P34-0001F0 FOR V71468)(SUPSDS 5A)(OPT TO 5B)		1
10	030-9081-000		.CONTACT, PIN (V71468) (USED WITH 5C)		57
- 10A	119-2020-080		.CONTACT, PIN (V55104) OPTIONAL - - - ATTACHING PARTS - - -		57
15	MS51957-15		.SCREW, PAN HEAD		8
20	MS21044C04		.NUT, LOCK -----*-----		8
25	MS17821-4-9		.TIE, CABLE		1
- 25A	MS3367-4-9		.CABLE TIE OPTIONAL		1
30	602-3139-002		.BRACKET		1
35	GA3A1		.FILTER, EMI (V23880) - - - ATTACHING PARTS - - -		1
40	MS24693C32		.SCREW, FLAT HEAD		2
45	1416-6		.TERMINAL (V83330)		4
50	MS21044C06		.NUT, LOCK -----*-----		2
55	51-717-007		.FILTER, EMI (V33095)		2
- 55A	51-717-021		.FILTER, EMI (V33095) OPTIONAL		2
60	JANTX1N6072A		.DIODE (REPLD BY 60A THRU 60D)		1
- 60A	1-5KE220CA		.DIODE (V04713) (REPLS 60)		1
- 60B	1-5KE220CA		.DIODE (V12954) (REPLS 60)		1
- 60C	1-5KE220CA		.DIODE (V14099) (REPLS 60)		1
- 60D	1-5KE220CA		.DIODE (V14936) (REPLS 60)		1
65	Z673000801		.BEAD, FERRITE (V34899) (SUPSD BY 65A)		2
- 65A	2673000801		.BEAD, FERRITE (V34899) (SUPSDS 65)		2
70	95C8H		.DELETED - - - ATTACHING PARTS - - -		0
75	MS24693C3		.DELETED		0
80	MS21044C04		.DELETED -----*-----		0

31-30-32

Page 1098.45
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
16			1234567		
	85	DBM25S	.CONNECTOR, 25-PIN (V71468) (OPT TO 87, 96, AND 97)		1
-	85A	117DBM25S	.CONNECTOR, 57-PLUG (V02660) OPTIONAL		1
-	85B	MD308G25S1	.CONNECTOR, 57-PLUG (V14283) OPTIONAL		1
	87	M24308-2-283F	.CONNECTOR, 25-PIN (OPT TO 85 AND 95) - - - ATTACHING PARTS - - -		1
	90	D110551	.JACKPOST (PAIR)(V71468) -----*-----		1
	95	DB50906-1	.BACKSHELL, POTTING (V71468) (USED WITH 85)		1
-	96	DB19678-2	.BACKSHELL, STRAIN RELIEF (V71468)(USED WITH 87)		1
-	96A	DB19678-2	.BACKSHELL, STRAIN RELIEF (V71785) OPTIONAL		1
-	97	031-1007-004	.CONTACT, SOCKET (V71468) (USED WITH 87)		24
-	97A	110-2026-057	.CONTACT, SOCKET (V55104) OPTIONAL		24
	98	MS51957-4	.SCREW, PAN HEAD		2
	99	MS21042-02	.NUT, LOCK		2
	100	620-2461-001	.BRACKET, CONNECTOR - - - ATTACHING PARTS - - -		1
-	105	MS51957-15	.SCREW, PAN HEAD (SUPSD BY 105A)		2
	105A	MS24693C3	.SCREW, FLAT HEAD (SUPSDS 105) -----*-----		2
	110	MS3367-4-9	.STRAP		6
-	110A	MS17821-4-9	.CABLE TIE OPTIONAL		6
	115	WMC0LWTB500	.MARKER, WIRE (V85480)		7
	115A	HCPS9-3XXXW	.MARKER, WIRE (V85480) OPTIONAL		7
	120	TC814	.BASE, TYWRAP (V59730) - - - ATTACHING PARTS - - -		6
	125	MS24693C3	.SCREW, FLAT HEAD		6
	130	MS21044C04	.NUT, LOCK -----*-----		6
	135	MS3181-22NA	.CAP, CONNECTOR		1
	140	KPT07A22-55S	.CONNECTOR, 55-PIN (V71468)		1
	140A	KPSE07A22-55S	.CONNECTOR (V71468) OPTIONAL		1
-	140B	PT07A22-55S	.CONNECTOR (V77820) OPTIONAL		1
	145	HLMP0454	.INDICATOR ASSY, LED (V50434)		1
	150	624-0769-001	.SPACER		1
	155	648-1177-001	.RETAINER - - - ATTACHING PARTS - - -		1
	160	MS24693S4	.SCREW, FLAT HEAD		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
16			1234567		
165	MS21044C04		.NUT, LOCK -----*-----		1
- 170	722-3142-005		..MOTHERBOARD PWB		1
- 170A	722-3142-009		..MOTHERBOARD PWB OPTIONAL		1
			--- ATTACHING PARTS ---		
175	MS24693C25		..SCREW, FLAT HEAD -----*-----		12
180	G06D64P3BBBL090		..CONNECTOR, 64-PIN (V71468)		5
- 180A	G06D64P3BBBL		..CONNECTOR, 64-PIN (V71468) OPTIONAL		5
- 180B	440-0507-002		CONNECTOR, 64-PIN (20-8457-064-002-025 FOR V91662) OPTIONAL		5
			--- ATTACHING PARTS ---		
185	MS51957-4		..SCREW, PAN HEAD		10
190	06/16/5610		..WASHER, NYLON (V86928) (SUPSD BY 190A)		10
- 190A	06/15/5610		..WASHER, NYLON (V86928) (SUPSDS 190)		10
195	MS20341-2S		..NUT, LOCK		10
- 195A	MS35649-222		..NUT, PLAIN OPTIONAL -----*-----		10
200	DACK15SBF		..CONNECTOR, 15-PIN (V71468) --- ATTACHING PARTS ---		2
205	D110551		..JACKPOST (PAIR)(V71468) -----*-----		2
210	623-2825-001		..PRINTED WIRING BOARD (USED WITH 170)		1
- 210A	623-2825-004		..PRINTED WIRING BOARD (USED WITH 170A)		1
215	9705CA0440-0		...STANDOFF (V06540)		4
220	1392C1		...TERMINAL (V88245)		7
- 220A	3531C4-5		...TERMINAL (V16258) OPTIONAL		7
225	9535CA0440-0		...STANDOFF (V06540)		4
- 225A	9535CA0440-17		...STANDOFF (V06540) OPTIONAL		4
230	MF1001-06		..NUTPLATE (V15653)		12
235	MS20470A3-5		..RIVET		24
- 240	634-1455-008		..CHASSIS		1
245	634-1443-006		..PLATE, CONNECTOR --- ATTACHING PARTS ---		1
250	MS51957-43		..SCREW, PAN HEAD -----*-----		4
255	602-2417-002		..BRACKET --- ATTACHING PARTS ---		1
260	MS24693C48		..SCREW, FLAT HEAD -----*-----		6
265	N6550-4		..SHOCKMOUNT (V81860)		2
270	704-0487-001		..STRAP, GROUND		1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		

16

			--- ATTACHING PARTS ---		
275	MS51957-28		..SCREW, PAN HEAD		4
			-----*-----		
280	652-0202-001		..GUIDE, POWER SUPPLY		2
			--- ATTACHING PARTS ---		
285	MS51959-3		..SCREW, FLAT HEAD		6
			-----*-----		
290	652-2002-001		..HOOK		2
			--- ATTACHING PARTS ---		
295	MS51957-45		..SCREW, PAN HEAD		4
			-----*-----		
- 300	617-3746-001		..PANEL, FRONT		1
			--- ATTACHING PARTS ---		
305	MS24693C48		..SCREW, FLAT HEAD		4
			-----*-----		
310	G2525-4		...HANDLE (V04729)		1
			--- ATTACHING PARTS ---		
315	MS51957-45		...SCREW, PAN HEAD		2
320	MS35338-137		...WASHER, LOCK		2
			-----*-----		
325	N6550-4		...SHOCKMOUNT (V81860)		2
			--- ATTACHING PARTS ---		
330	NAS1189E06P5L		...SCREW, FLAT HEAD		4
335	MS35649-264		...NUT, HEX		4
			-----*-----		
340	621-1195-001		...PANEL		1
345	634-1442-007		..PLATE, CHASSIS		1
350	1032NBSEXRR		...NUT, BLIND RECESS (V86577)		3
- 350A	1032NBSEXRR091		...NUT, BLIND PRESS		3
			(V86577) OPTIONAL		
355	FE832		...NUT, PRESS (V46384)		3
- 355A	58LNK060		...NUT, PRESS		3
			(V03694) OPTIONAL		
360	35-1CR2-7		...GUIDE, CARD (V18915)		6
			--- ATTACHING PARTS ---		
365	MS20426AD2-3		...RIVET		12
			-----*-----		
370	24SX2-7-3L		...GUIDE, CARD (V18915)		1
			--- ATTACHING PARTS ---		
375	MS20426AD2-4		...RIVET		2
			-----*-----		
380	634-1442-008		..PLATE, CHASSIS		1
385	1032NBSEXRR		...NUT, BLIND RECESS (V86577)		3
- 385A	1032NBSEXRR091		...NUT, BLIND RECESS		3
			(V86577) OPTIONAL		
390	FE832		...NUT, LOCK (V46384)		3
- 390A	58LNK060		...NUT, PRESS		3
			(V03694) OPTIONAL		
395	35-1CR2-7		...GUIDE, CARD (V18915)		6
			--- ATTACHING PARTS ---		
400	MS20426AD2-3		...RIVET		12
			-----*-----		

31-30-32

Page 1098.48
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

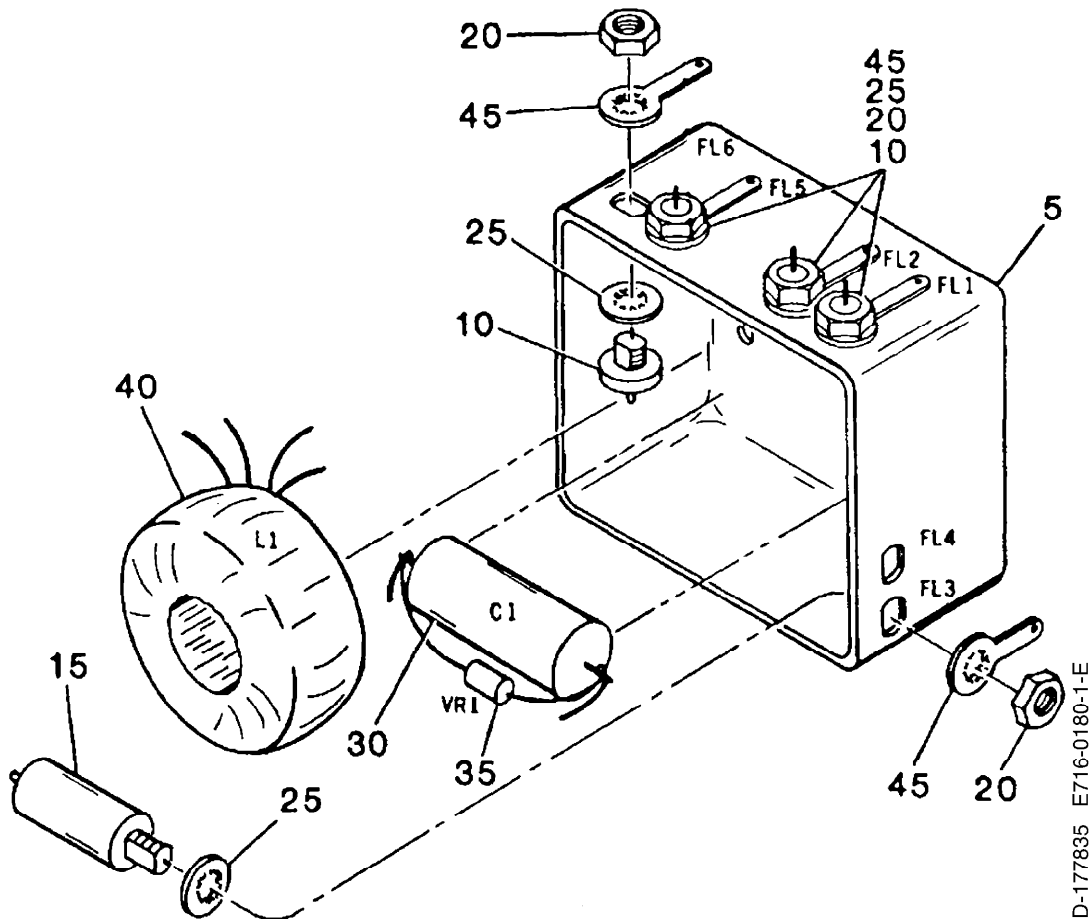
FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
			1234567		

16

405	24SX2-7-3R		...GUIDE, CARD (V18915) - - - ATTACHING PARTS - - -		1
410	MS20426AD2-4		...RIVET -----*-----		2
415	602-2913-001		..BRACKET, CABLE - - - ATTACHING PARTS - - -		1
420	MS24693C25		..SCREW, FLAT HEAD		2
425	22NKM62		..NUT, NYLON CAP (V72962) -----*-----		2

Honeywell

COMPONENT MAINTENANCE MANUAL
Universal Flight Data Recorder / Part No. 980-4100/980-4101



FILTER ASSEMBLY
FIGURE 17

31-30-32

Page 1098.50
22 Dec 2006

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

FIGURE/ITEM	PART NUMBER	AIRLINE STOCK NUMBER	NOMENCLATURE	EFF	UNITS PER ASSY
17			1234567		
- 1	716-0180-002		FILTER ASSEMBLY (SEE FIG. 9-20 FOR NHA)	A	RF
- 1A	716-0180-001		FILTER ASSEMBLY (SEE FIG. 9-20A FOR NHA)	B	RF
5	604-0317-002		.CASE	A	1
- 6	604-0317-001		.CASE	B	1
10	51-717-007		.FILTER, EMI (V33095)		4
- 10A	51-717-021		.FILTER, EMI (V33095) OPTIONAL		4
15	2360-006		.FILTER, EMI (V33095)		2
20	24-0009-1400		- - - ATTACHING PARTS - - - ..NUT (V33095)		6
25	24-0010-0700		..WASHER (V33095)		6
30	M83421-01-5204P		-----*----- .CAPACITOR, 0.33 UF, 10%, 400V		1
35	JANTX1N6072A		.DIODE		1
40	712-0084-001		.INDUCTOR, WINDING		1
45	1410-14		.TERMINAL (V83330)		6
- 50	400-7077-001		.COVER, FILTER ASSEMBLY	B	1

Honeywell

COMPONENT MAINTENANCE MANUAL Universal Flight Data Recorder / Part No. 980-4100/980-4101

THIS PAGE INTENTIONALLY LEFT BLANK